

# Four hundred thirty-nine conjectures in mathematics

Thirty parts across arithmetic, combinatorics, geometry, probability and topology

The Conjecture Engine\*

27 July 2026

## Summary sheet: the four hundred thirty-nine conjectures

**Part I: conjectures from the local–global random model, in importance–novelty order.**

1. **Prime-power contamination calculus (Conjecture 1).** For a balanced pair race  $(n, n + d) \bmod m$ , the surviving prime-square orientations  $(q^2 - d, q^2)$ ,  $(q^2, q^2 + d)$  occupy provably computable residue classes, and the operator  $\mathcal{C}_{d,m}(a; x) = \sum_o \sum_q \Lambda(q^2) \Lambda(\text{prime member})$  determines the drift vector: for twins,  $\mathcal{M}_x(D_1(t) \log^2 t / \sqrt{t}) \rightarrow c_T = \frac{1}{2} C(x, x^2 - 2) \pmod{5}$  and  $\rightarrow 2c_T$  on class 7  $\pmod{8}$ , symmetric differences  $\rightarrow 0$ ; in general each class deficit satisfies  $\mathcal{M}_x(\text{deficit} \cdot \log^2 t / \sqrt{t}) \rightarrow \lim \mathcal{C}_{d,m}(a; x) / \sqrt{x}$ . Convergence of these logarithmic means at drift scale is part of the conjecture (Rubinstein–Sarnak-type structure for pattern races).
2. **Pinned singular-series variance (Conjecture 2).** With  $R(h) = C_4(0, 2, h, h + 2) / \mathfrak{S}(2)^2$  and  $G(H) = \sum_{h \leq H} (1 - h/H)(R(h) - 1)$ : (i) [conditional proposition] the moving-window twin-pair count obeys  $\text{Var} / \mathbb{E} = 1 + \mathfrak{S}(2)(2G(H) - 1) / \log^2 x + o(\log^{-2} x)$ ; (ii)  $-G(H) / \log H \rightarrow \infty$ : pair counts are more sub-Poisson than prime counts.
3. **Least-prime ordering deficit (Conjecture 3).** With  $U(a, q) = \text{Li}(p(a, q)) / \varphi(q)$ : (i)  $U \Rightarrow \text{Exp}(1)$  marginally (attributed), Gumbel maximum under joint independence; (ii) the dyadic averages of  $\Theta(q) = (1 - \mathbb{E}_a[U]) \log q$  over prime  $q \in [Q, 2Q]$  converge to a limit  $\Theta > 0$ : the ordered primes fill residue classes faster than any exchangeable model, in excess of the derived discreteness and injectivity baselines.
4. **Goldbach lane race (Conjecture 4).** For  $n \equiv 2 \pmod{4}$ , prime squares enter ordered Goldbach representations only through the  $(1, 1)$  lane  $\bmod 4$ , so with  $D = R_3 - R_1$  and the explicit weighted census  $D_{\text{sys}}$ :  $\mathbb{E}_x[D - D_{\text{sys}}] = o(\mathbb{E}_x[D_{\text{sys}}])$  under logarithmic sampling; the drift vanishes on the internal null subprogression  $n \equiv 10 \pmod{12}$ ; the sign density of  $D$  tends to  $\frac{1}{2}$ .

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\*This paper was produced by an automated conjecture engine (LLM tools), operated as an exercise in disciplined conjecture-making. All constants, counts, and verification results are reproducible from the accompanying repository (`conjecture-engine`). `run_all.py` regenerates Part I, while the per-part scripts in `verify/` regenerate the later computations. Primality claims are stratified between deterministic verification and Baillie–PSW probable primes. The exact certification boundary and the affected counts are recorded in the computational-check standards section.

5. **Least Goldbach summand (Conjecture 5).** With  $s(n)$  the least prime  $p \nmid n$  with  $n - p$  prime,  $U(n) = \mathfrak{S}(n) \sum_{p \leq s(n), p \nmid n} 1/\log(n - p)$ , and the dyadic-log ensemble  $\mathbb{E}_X$ : (i)  $U \Rightarrow \text{Exp}(1)$ ; (ii)  $\Theta_G(X) = (1 - \mathbb{E}_X U) \log X \rightarrow \Theta_G > 0$ , the Goldbach sibling of the ordering deficit.
6. **Polynomial–exponential entanglement (Conjecture 6).** Primes  $n^2 + 2^n$  force  $n \equiv 3 \pmod{6}$ ; with  $\kappa_S = 3 D_S / \prod_{p \in S} (1 - 1/p)$  built from the CRT-exact survivor density  $D_S$  over the joint period of  $S$  and the lane bonus 3 at the primes 2 and 3:  $\kappa_{S_z}$  converges along the canonical exhaustion  $S_z = \{5 \leq p \leq z\}$  to  $\kappa$ , the counting function is  $\sim \sum_{n \equiv 3(6), n \leq N} \kappa / \log(n^2 + 2^n) \asymp \log N$ ; whether the limit factors exactly (as observed through  $p \leq 19$ ) is an open question.
7. **Multibase Fermat quotients (Conjecture 7).** For  $p \nmid a$ : the Eisenstein–Lerch homomorphism confines multiplicatively dependent bases to a rational subtorus, and (iv) for multiplicatively independent bases  $(q_p(a_1), \dots, q_p(a_r))/p$  equidistributes on the full torus as  $p$  varies; (v) at most finitely many simultaneous Wieferich primes; plus single-base clauses: KS-discrepancy LIL at constant  $1/\sqrt{2}$ , shrinking-target law  $\#\{p \leq x : q_p < K\} \sim \sum \min(1, K/p)$ , Wieferich-count LIL and an almost-sure CLT at the  $du/u$  weighting.
8. **Uniform de Polignac covariance field (Conjecture 8).**  $\pi_d(x) = \mathfrak{S}(d) \text{Li}_2(x)(1 + o(1))$  uniformly for even  $d \leq (\log x)^A$ ; the moving-window residual field has covariance  $K(d, d')H/\log^3 x + (1/\log^4 x) \sum_{|h| \leq H} (H - |h|) K_4(d, d'; h)$ , triple term dominant on the polylogarithmic window class, with Poisson/Gaussian regimes as finite- $x$  corrections to independence.
9. **Race sub-diffusivity (Conjecture 9).** Balanced-race steps at event index have negative autocorrelations obeying  $\rho_k(x) = -(c_k \log \log x + d_k)/\log x (1 + o(1))$ ; under uniformity in  $k$ , tail summability, and  $\sum c_k, \sum |d_k| < \infty$ , the diffusivity is  $\sigma^2(x) = 1 - 2(\sum c_k \log \log x + \sum d_k)/\log x (1 + o(1))$ , those hypotheses being model-layer, not asserted asymptotically; the asymptotic diffusive-vs-rigid dichotomy is registered open, with the rigid branch selected only at Conjecture 1(i-b) for its contaminated races.
10. **Uniform quadratic de Polignac and its constants (Conjecture 10).**  $\#\{n \leq N : n^2 + 1, n^2 + 1 + d \text{ prime}\} = C(d) I_d(N)(1 + o(1))$  uniformly for even  $d \leq (\log N)^B$ ; a separate power-saving rate clause; and the family law: moments of  $C(d)$  converge to derived Euler products (mean 2.7456..., sd 1.6840...), the empirical law converging to the random product with CRT-independent local factors.
11. **First occurrences of prime gaps (Conjecture 11).** On realized gaps:  $\log p(g)/\sqrt{g}$  bounded between positive constants;  $\liminf = \sqrt{e^\gamma/2}$  conditional on the realization hypothesis; and  $\log p(g) = \sqrt{g} + \frac{1}{2} \log g - \frac{1}{2} \log \mathfrak{S}^*(g) + E_g$  with  $\Pr[E_g \leq t] \rightarrow 1 - \exp(-e^{2(t-\mu)})$ , the min-type Gumbel at scale  $\frac{1}{2}$  forced by the slope-2 hazard expansion.
12. **Stern lane race (Conjecture 12).** In  $n = p + 2k^2$ , square contamination  $n = q^2 + 2k^2$  sits in the  $k$ -even lane iff  $n \equiv 1 \pmod{8}$ ,  $k$ -odd iff  $n \equiv 3$ , and is impossible for  $n \equiv 5, 7$  (provable null classes); on contaminated classes the clean-minus-contaminated difference obeys  $\mathbb{E}_X[D - D_{\text{sys}}] = o(\mathbb{E}_X[D_{\text{sys}}])$  with the  $\Lambda(q^2)$ -weighted census  $D_{\text{sys}}$ .
13. **Microscopic variance law (Conjecture 13).** [Conditional on quantitative Hardy–Littlewood] For windows of length  $H = \lambda(\log x)^\alpha$ ,  $\text{Var } X/\mathbb{E}X = 1 - (\log H + \gamma + \log 2\pi -$

- 1)/ $\log x + o(1/\log x)$  uniformly in  $1 \leq \alpha \leq A$ , with Gallagher's Poisson law at the boundary  $\alpha = 1$ .
14. **Null-mechanism race (Conjecture 14).** For  $(n^2 + 1, n^2 + 3)$ , square contamination is algebraically impossible (single bounded exception  $(n, q) = (1, 2)$ ); the race mod 5 is driftless at the contamination scale,  $\mathcal{M}_x(D(t) \log^2 t / \sqrt{t}) \rightarrow 0$ —the negative control for item 1—and the logarithmic occupation of leadership tends to  $\frac{1}{2}$  with Gaussian fluctuations  $\sqrt{\log 2 / \log x}$ , conjectured directly (the event-index invariance principle is a separately falsifiable local hypothesis, which does not imply it).
  15. **Cubic-shift constants (Conjecture 15).** For non-cube  $a$ , the three-state local law gives  $\mathbb{E}_a[\omega(p)] = 1$  exactly, so the limit law of  $C(a)$  has mean exactly 1 (an  $L^2$ -martingale argument), derived sd  $0.2762\dots$ ; counts are uniform over non-cube  $a \leq (\log N)^B$ .
  16. **Hardy–Littlewood's Conjecture F as a family (Conjecture 16).**  $Q_A(N) = C(A)I_A(N)(1 + o(1))$  uniformly over odd  $A \leq (\log N)^B$ ; same-index window covariance  $[C(A, A') - C(A)C(A')]/H/4 \log^2 N$  (negative for locally exclusive pairs), with the off-index pinned sum at the same  $1/4 \log^2 N$  order left open.
  17. **Twin-member Goldbach, orientation-resolved (Conjecture 17).** Under  $n$ -uniform Hardy–Littlewood for the four role systems,  $R_T(n) = [\sum_o \mathfrak{S}_4^{(o)}(n)] \int_5^{n-5} dt / (\log^2 t \log^2(n-t)) (1 + o(1))$ , with the exact identities  $\mathfrak{S}_4^{(\text{lu})} = \mathfrak{S}_4^{(\text{ul})}$  and  $\mathfrak{S}_4^{(\text{uu})}(n) = \mathfrak{S}_4^{(\text{ll})}(n-4)$ .
  18. **Triplet contamination (Conjecture 18).** For  $(n, n+2, n+6) \bmod 5$ , the unique surviving square configuration is the doubly-thinned  $(q^2 - 2, q^2, q^2 + 4)$  (the other two configurations dying for every prime  $q > 3$ ), contaminating class 2:  $\mathcal{M}_x((\pi_3(t; 5, 1) - \pi_3(t; 5, 2)) \log^3 t / \sqrt{t}) \rightarrow c_3$ , the Bateman–Horn constant of  $(q, q^2 - 2, q^2 + 4)$ .
  19. **Sexy-pair contamination matrix (Conjecture 19).** For  $(n, n+6)$  both orientations survive on complementary  $q$ -classes, feeding start classes 3 and 1 (mod 5 and mod 8) with independent constants  $c_A, c_B$  (the  $(q, q^2 \mp 6)$  Bateman–Horn constants) in the drift-scale sense; class 2 (mod 5) and classes 5, 7 (mod 8) are provably clean.
  20. **Fibonacci–Lucas twins (Conjecture 20).** The odd prime-divisor pools of  $F_p$  and  $L_p$  are disjoint (ranks  $p$  vs  $2p$ ;  $\gcd = 2$  iff  $3 \mid p$ ); only finitely many  $p$  have both prime; the naive joint accounting assigns prior mass about  $1.4 \times 10^{-2}$  to the index range beyond  $10^4$ , in which the catalogued index 148091 falls, recorded descriptively and conditionally on the probable-prime status of both  $F_{148091}$  and  $L_{148091}$ , and fatal to any completeness list.
  21. **Factorial twins (Conjecture 21).** Window rigidity:  $n! + a$  is composite for  $2 \leq |a| \leq n$  (theorem), so  $\pm 1$  are the only bounded offsets;  $n = 3$  uniqueness is attributed; the joint-fluctuation clause is a declared-model CLT for  $F_+ - F_-$  under explicit covariance and higher-cumulant hypotheses.
  22. **Boundary trichotomy (Conjecture 22).** For  $\deg F \geq 2$  with positive leading coefficient,  $F(m) - F(j)$  prime forces  $m - j = 1$  outside a bounded region; for  $x^3 + cx$  the boundary lane  $3m^2 - 3m + 1 + c$  is dead-parity for odd  $c$ , dead-3-adic for even  $c \equiv 2 \pmod{3}$ , admissible exactly for  $c \equiv 0, 4 \pmod{6}$ , with uniform Bateman–Horn counts over admissible  $c \leq (\log M)^B$ .

23. **Power-obstruction ladder (Conjecture 23).**  $m^k = p + j^k$  is impossible for composite  $k$  (theorem); for prime  $k$  it forces  $j = m - 1$  and reduces to primality of the irreducible  $D_k(m) = m^k - (m - 1)^k$ , which follows Bateman–Horn uniformly over the prime- $k$  lanes.
24. **Alternating cyclotomic chain (Conjecture 24).** Infinitely many primes  $p$  with  $\Phi_3(p)$  and  $\Phi_6(\Phi_3(p))$  prime, at the derived Bateman–Horn rate; alternation is the unique admissible continuation in the anchored construction space.
25. **Twin cyclotomic bases (Conjecture 25).** Of the  $\varphi(k) = 2$  family, only  $k = 3$  survives ( $k = 4$  parity-dead,  $k = 6$  a translate), and  $\Phi_3(n)$ ,  $\Phi_3(n + 1)$  are simultaneously prime infinitely often with the computed Bateman–Horn constant.

**Part II: structural conjectures, in programme order.**

26. **Connected motif generating functional (Conjecture 26).** A full connected diagram calculus for all joint cumulants of prime motifs.
27. **Complete overlap-renormalization filtration (Conjecture 27).** A classification of every possible covariance scale and rigid eigenspace in a finite motif family.
28. **Regularized mesoscopic local–spectral trace formula (Conjecture 28).** Independently mollified Euler-product and multiple-zero functionals agree after named counterterms, and the limit is regularization independent.
29. **Anchored arithmetic polymer expansion (Conjecture 29).** Palm cluster activities are graded by the number of new prime constraints beyond one anchored motif.
30. **Topological expansion of non-Gaussianity (Conjecture 30).** Each independent cycle in a connected diagram’s overlap graph costs one logarithmic order, so the leading non-Gaussian odd-cumulant term is carried by the tree-like diagrams.
31. **Connected first-arrival functional (Conjecture 31).** The entire least-prime point process has connected kernels obtained from same-class prime correlations.
32. **Tested Gauss-polyspectral reciprocity (Conjecture 32).** Scalar tensor observables have exact Gauss-transport identities under the random-modulus law and dual arithmetic limits.
33. **Rubinstein–Sarnak terminal chaos dichotomy (Conjecture 33).** Terminal extremes are directed by the limiting low-zero chaos measure, with an explicit self-averaging versus Cox criterion.
34. **Nonlinear spectral response calculus (Conjecture 34).** Low-zero and exceptional-zero perturbations pass through first-arrival statistics by universal Volterra response operators.
35. **Local-information capacity of odd class groups (Conjecture 35).** Cohen–Lenstra independence has a sharp two-resource boundary governed by cell entropy and conductor complexity.
36. **Kummer–Haar law on relation tori (Conjecture 36).** Saturated multiplicative relations give the exact horizontal support and Haar law of Fermat-quotient vectors.

37. **Rank-dimensional finite-logarithm large sieve (Conjecture 37).** A positive power range of frequencies obeys a large sieve whose dual dimension is saturated rank.
38. **Mesoscopic-to-lattice shrinking-target transition (Conjecture 38).** Haar universality persists for growing targets, while bounded lattice targets acquire a separate arithmetic regulator intensity.
39. **Global-lattice  $p$ -adic regulator-matrix law (Conjecture 39).** A fixed integral Galois-relation lattice controls horizontal regulator matrices and their determinantal rare events.
40. **Functorial horizontal logarithms on tori (Conjecture 40).** A finitely generated global subgroup of a torus has a functorial matrix-valued finite-logarithm Haar law.
41. **Entropy-conductor profile of arboreal resolution (Conjecture 41).** Exact class-measure entropy and Artin-conductor profiles determine observable depth and finite-index shifts.
42. **Arboreal family large sieve and cumulant independence (Conjecture 42).** Growing preimage quotients satisfy square-root trace bounds and higher connected Frobenius factorization over parameters.
43. **Coloured dynatomic Galois-factor process (Conjecture 43).** Exact-period components generate independent coloured, Frobenius-marked factor processes, with growing-degree support restrictions.
44. **Complexity-uniform primitive valuation process (Conjecture 44).** The complete primitive valuation vector has an adelic product law under an explicit height-discriminant complexity regime and uniform squarefull tails.
45. **Divisor-sensitive dynamical gcd classification (Conjecture 45).** Positive normalized gcd height is exactly eventual entry into a preperiodic subvariety carrying the relevant common divisor.
46. **Frobenius-marked Poisson-Dirichlet process (Conjecture 46).** Macroscopic polynomial-value factors carry fixed-point-size-biased Galois marks.
47. **Three-scale polynomial factorization (Conjecture 47).** Small valuations, mesoscopic scale-invariant factors, and macroscopic marked factors form one conservation-corrected product structure.
48. **Adelic Gibbs gluing for reducible values (Conjecture 48).** Reducible polynomial factor processes are an archimedean- $p$ -adic Gibbs mixture of independent component processes.
49. **Full multivariate adelic saddle law (Conjecture 49).** Joint smoothness is governed by a complete joint Perron action, saddle displacement, and Hessian, not a scalar Euler correction.
50. **Buchstab-Bateman-Horn component flow (Conjecture 50).** Rough prime, semiprime, and higher-almost-prime polynomial values follow universal Buchstab components with Galois marks.

**Part III: six conjectures across fields, in statement order.**

51. **Relative polynomial pretentiousness inverse principle (Conjecture 51).** A persistently nonzero logarithmic correlation of completely multiplicative unimodular functions along multiplicatively independent irreducible polynomials forces one existential block of primitive characters and archimedean parameters with finite relative pretentious distances, the balance identity  $\sum_j t_j \deg P_j = 0$ , and a nonvanishing joint character average at a common modulus.
52. **Monotone threshold purification of vector paths (Conjecture 52).** Every monotone fractional activation schedule of  $m$  vectors of norm at most one in  $\mathbb{R}^d$  admits a rounding to distinct irreversible activation times whose path error is  $O(\sqrt{d})$  uniformly in  $m$ , containing the open Euclidean Steinitz prefix-sum problem and asserting removal of the logarithm from Banaszczyk's  $O(\sqrt{d + \log m})$ .
53. **Entropy dimension of random-free graphons (Conjecture 53).** If the row space of a  $\{0, 1\}$ -valued graphon is Ahlfors  $s$ -regular then  $H(G(n, W))/(n \log n) \rightarrow s$ , the hypothesis class being nonvacuous at every  $s \in (0, 2]$  by Cantor-type threshold constructions.
54. **A Hessian principle for isolated microcanonical graph phases (Conjecture 54).** Under isolated-phase, transversally Lipschitz, and nonemptiness hypotheses, the tangent block averages of a microcanonical coloured graph ensemble fluctuate at scale  $n^{-1}$  around their mean with centred Gaussian limit of covariance  $A^{-1}$ , the inverse of the contracted entropy Hessian.
55. **Sharp stability of entropy idempotence on finite abelian groups (Conjecture 55).** The squared total variation distance from a probability measure to the nearest uniform coset measure is at most  $2/\log 2$  times the convolution entropy defect  $H(\mu * \mu) - H(\mu)$ , sharply, the constant pinned by discretized Gaussians on  $\mathbb{Z}/p\mathbb{Z}$ ; some universal constant already follows from the Green–Manners–Tao entropic inverse theorem, so the sharp constant and the extremal geometry are the content.
56. **Stationary convolution-entropy rigidity (Conjecture 56).** An ergodic stationary process over a finite abelian group preserves entropy under independent self-convolution exactly when its quotient by the translational stabilizer has zero entropy, so in particular every ergodic binary process of intermediate entropy gains entropy under independent XOR.

**Part IV: twenty conjectures on the class-counting polynomial, in statement order.**

57. **Irreducibility and connectivity (Conjecture 57).**  $H_G(U, Y) = C_G(1 + U, Y)$  is irreducible in  $\mathbb{Z}[U, Y]$  exactly when  $G$  is connected, the disconnected direction being immediate from Rossmann's multiplicativity and the connected direction verified by exact factorization for all 996 connected graphs on at most seven vertices.
58. **Recovery of the subgraph-component polynomial (Conjecture 58).** Equality of class-counting polynomials forces equality of Tittmann–Averbouch–Makowsky subgraph-component polynomials, though the two arrays,  $(|S|, |N[S]| - c(G[S]))$  against  $(|S|, c(G[S]))$ , admit no apparent functional relation.



59. **Two-field rigidity (Conjecture 59).** The adjoint-rank distributions of the graphical Lie algebras over  $\mathbb{F}_2$  and  $\mathbb{F}_3$  together determine them over every finite field.
60. **Hybrid rigidity (Conjecture 60).** The  $\mathbb{F}_2$  rank distribution together with the subgraph-component polynomial determines the class-counting polynomial, while the degree sequence in place of  $Q_G$  does not suffice.
61. **Binary edge-deck reconstruction (Conjecture 61).** The multiset of binary specializations  $C_{G-e}(2, Y)$  over the edges determines a graph with at least four edges up to isomorphism, a statement in the orbit of the edge reconstruction conjecture.
62. **Bipartite binary rigidity (Conjecture 62).** Within bipartite graphs the single specialization  $C_G(2, Y)$  determines  $C_G$ .
63. **Connected-pseudoforest binary rigidity (Conjecture 63, resolved false).** The proposed one-field rigidity for trees and connected unicyclic graphs is refuted by an explicit seven-vertex unicyclic pair sharing the binary specialization with distinct degree multisets, the unique violating class through nine vertices, and the tree case survives exhaustively through fourteen vertices.
64. **Chordal binary rigidity (Conjecture 64).** The same one-field rigidity holds within chordal graphs.
65. **Recovery of the domination number (Conjecture 65).** Equality of class-counting polynomials forces equality of domination numbers, a statement strictly between Rossmann's connected-domination theorem and the false total-domination analogue.
66. **Recovery of the independent domination number (Conjecture 66).** Equality of class-counting polynomials forces equality of independent domination numbers.
67. **Global double log-concavity (Conjecture 67).** For every graph the coefficient sequence  $a$  of  $f_G(1 + Z)$  is log-concave and remains log-concave after one application of  $L(a)_i = a_i^2 - a_{i-1}a_{i+1}$ , the third iterate of  $L$  failing for a thirteen-vertex graph.
68. **Strict log-concavity for connected graphs (Conjecture 68).** For connected graphs with at least one edge the inequality  $a_i^2 > a_{i-1}a_{i+1}$  is strict at every interior position of the support.
69. **Strict second-order log-concavity (Conjecture 69).** Under the same hypotheses  $L(a)$  is strictly log-concave at every interior position of its support.
70. **Rank-slice strict log-concavity (Conjecture 70).** For connected graphs every non-monomial rank slice  $[Y^r]F_G(1 + Z, Y)$  has strictly log-concave coefficients at interior positions, the second-order strengthening of the slice statement failing for an eight-vertex graph.
71. **Interval support at fixed rank (Conjecture 71, resolved false).** The proposed gaplessness is refuted by a connected eight-vertex graph whose support sizes attaining rank five skip size two, the unique violating order, while the transposed statement was already false.

72. **Coefficientwise tree extremality (Conjecture 72).** Among  $n$ -vertex trees the shifted polynomial  $f_T(1 + Z)$  is sandwiched coefficientwise between path and star, with equality only at the extremes.
73. **Star rank-majorization (Conjecture 73).** For  $n \geq 5$  and every prime power  $q$  the decreasingly ordered adjoint-rank distribution of the star majorizes that of every  $n$ -vertex tree, the hypothesis being sharp since the star fails to majorize the path at  $(n, q) = (4, 2)$ .
74. **Tree rank-variance extremality (Conjecture 74).** For  $n \geq 4$  and every prime power the variance of  $\text{rank}(\text{ad}_x)$  over a uniformly random element is uniquely minimized by the path and uniquely maximized by the star among  $n$ -vertex trees.
75. **Leaf-count rigidity (Conjecture 75).** Equality in the lemma's bound  $s(T) \geq n - \ell(T) + 2$  holds exactly for paths and stars.
76. **Diameter rigidity (Conjecture 76).** Equality in the lemma's bound  $s(T) \geq \text{diam}(T) + 1$  holds exactly for paths and stars.

**Part V: twenty conjectures on pattern Lie algebras of posets, in statement order.**

77. **Adjoint–Kirillov determination (Conjecture 77).** Equality of adjoint enumerators  $A_{P,q}$  forces equality of Kirillov enumerators  $K_{P,q}$ , the two contractions of the same bracket, verified across 64 adjoint-collision classes.
78. **Lower-central recovery (Conjecture 78).** The adjoint enumerator determines the vector of lower-central factor dimensions of  $L_P$ , which was constant on every adjoint-collision class in testing.
79. **Ordinary Poincaré recovery (Conjecture 79).** The adjoint enumerator is conjectured to determine the ordinary cohomological Poincaré polynomial, tested only for  $\dim H^2$  and  $\dim H^3$  and which rests on the deposited cohomology data.
80. **Derivation recovery (Conjecture 80).** The adjoint enumerator determines  $\dim \text{Der}(L_P)$ , which was constant on every adjoint-collision class.
81. **Adjoint Poincaré recovery (Conjecture 81).** The adjoint enumerator is conjectured to determine every adjoint-cohomology dimension  $\dim H^i(L_P, L_P)$ , tested only for  $\dim H^1$ ,  $\dim H^2$ , and  $\dim H^3$  from the deposited data.
82. **Centroid-dimension recovery (Conjecture 82).** The adjoint enumerator determines  $\dim \text{Cent}(L_P)$ , which was constant on every adjoint-collision class.
83. **Adjoint spectrum characteristic independence (Conjecture 83).** The set of attained adjoint ranks is independent of  $q$ , agreeing across  $\mathbb{F}_2$ ,  $\mathbb{F}_3$ , and  $\mathbb{F}_5$  in every case.
84. **Kirillov spectrum characteristic independence (Conjecture 84).** The set of attained Kirillov ranks is independent of  $q$ , agreeing across  $\mathbb{F}_2$ ,  $\mathbb{F}_3$ , and  $\mathbb{F}_5$  in every case.
85. **Two-field arithmetic rigidity (Conjecture 85).** Equality of adjoint enumerators over  $\mathbb{F}_2$  and  $\mathbb{F}_3$  is conjectured to force equality over every field, tested only at the single holdout  $q = 5$  on 16 three-field collision classes.



86. **Adjoint stochastic field monotonicity (Conjecture 86).** For  $q < q'$  the adjoint-rank distribution over  $\mathbb{F}_{q'}$  first-order stochastically dominates that over  $\mathbb{F}_q$ , held in all 221 field-pair tests, while the likelihood-ratio strengthening is false.
87. **Kirillov likelihood-ratio field monotonicity (Conjecture 87, resolved false).** The proposed monotone likelihood-ratio domination of the normalized even-rank distributions is refuted by a seven-point poset whose likelihood ratio from  $\mathbb{F}_2$  to  $\mathbb{F}_3$  already decreases across the even ranks, while the first-order dominance of Conjecture 86 stands.
88. **Centre-one reverse determination (Conjecture 88).** When  $\dim Z(L_P) = 1$  the Kirillov enumerator determines the adjoint enumerator, verified on all 43 centre-one records.
89. **Centre-one Laplace positivity (Conjecture 89).** When  $\dim Z(L_P) = 1$  the incidence quotient  $R_{P,q}$  has nonnegative integer coefficients, on all 43 centre-one records.
90. **Centre-one quotient shape (Conjecture 90, resolved false).** The proposed unimodality of  $R_{P,q}$  under a line centre is refuted by a seven-point poset whose quotient has a strict interior valley, while the positivity and interval support of Conjecture 89 stand.
91. **Hasse-forest reverse determination (Conjecture 91).** When the Hasse cover graph is a forest the Kirillov enumerator determines the adjoint enumerator, across 44 forest-collision classes.
92. **Hasse-forest coadjoint saturation (Conjecture 92).** On a Hasse forest the Kirillov support is the full even range  $\{0, 2, \dots, b_K\}$ , filled in every case.
93. **Hasse-forest Kirillov unimodality (Conjecture 93).** On a Hasse forest the even-rank sequence of  $K_{P,q}$  is unimodal, strict log-concavity being false.
94. **Hasse-forest breadth-orbit bound (Conjecture 94).** On a Hasse forest  $b_K \leq 2b_A$ , the unrestricted bound failing fourteen times off the forest class.
95. **Unitriangular Kirillov strict log-concavity (Conjecture 95).** For  $\text{ut}_n$  the even-rank sequence of  $K_{L,q}$  is strictly log-concave at every nontrivial interior index, confirmed through  $\text{ut}_6$  over  $\mathbb{F}_2$ .
96. **Adjoint-coadjoint incidence anticorrelation (Conjecture 96).** Sampling on the incidence variety gives negative covariance between adjoint and Kirillov ranks for every nonabelian  $L_P$  and zero exactly in the abelian case.

**Part VI: twenty conjectures on local arithmetic structure, in statement order.**

97. **Local-sieve limit for polygonal compasses (Conjecture 97).** The nearest-prime direction of the polygonal numbers  $P_s$  has a limit equal to a deep local-sieve hazard, whose primes through 47 already predict the order-level bias with correlation 0.896.
98. **Pentagonal and pronic compasses (Conjecture 98).** The pentagonal compass is right-biased in  $(0.56, 0.62)$  and the pronic compass left-biased in  $(0.42, 0.48)$ , opposite directions measured at 0.583 and 0.450.

99. **Opposite polygonal orders (Conjecture 99).** The 60-gonal compass exceeds 0.58 and the 69-gonal compass falls below 0.49, at 0.609 and 0.468 with both signs replicated above  $10^9$ .
100. **Triangular and hexagonal equality (Conjecture 100).** The triangular and hexagonal compass constants are equal, by  $P_6(n) = P_3(2n - 1)$ , agreeing to three places at about 0.536.
101. **Square compass by root class (Conjecture 101).** The nearest-prime direction of  $n^2$  splits sharply by  $n \bmod 6$ , below 0.45 for  $n \equiv 1, 5$  and above 0.54 for  $n \equiv 0, 2, 3, 4$ .
102. **Two-sided order spectrum (Conjecture 102).** As the number of sides grows the polygonal compass straddles one half, so infinitely many orders are left-biased and infinitely many right-biased.
103. **Permanent nonuniformity from length five (Conjecture 103).** Every fixed-length conditional order-pattern law now has a proved limit, while nonuniformity remains open separately for every  $d \geq 5$ , with the length-five chi-square about  $2.1 \times 10^7$ .
104. **Thirteen-fold pattern separation (Conjecture 104).** At window length five the density ratio of the patterns 30241 and 13240 lies in  $(12, 15)$ , measured at 13.31 through  $10^9$ .
105. **Abundancy order constant (Conjecture 105).** Adjacent triples of  $\sigma(n)/n$  are monotone with a density in  $(0.094, 0.098)$ , far below the independent value  $1/3$ , measured at 0.0958.
106. **Totient order constant (Conjecture 106).** Adjacent triples of  $\phi(n)/n$  are monotone with a density in  $(0.019, 0.022)$ , measured at 0.0202.
107. **Five-term totient barrier (Conjecture 107, resolved false).** The proposed barrier is refuted by a theorem of Martin [53], which yields strictly monotone runs of  $\phi(n)/n$  of every length on positive lower density; the runs are astronomically rare, none below  $4 \times 10^9$ , whereas the abundancy analogue already fails at  $n = 36,721,681$ .
108. **Primorial-stride order constant (Conjecture 108).** Along primorial strides the abundancy order density tends to a limit in  $(0.16, 0.19)$ , decreasing from 0.202 at stride 30 toward 0.177.
109. **Dyadic completion overshoot (Conjecture 109).** Raising only the dyadic exponent of an even squarefree six-divisible stride lifts the abundancy order density past  $1/3$  and holds it there, isolated by two controls to the exponents rather than the prime support.
110. **Binomial-slice negative shocks (Conjecture 110, resolved true).** For every  $k \geq 3$  the single-step  $\Omega$ -jump of  $\binom{kn}{n}$  is unbounded below, now proved in Proposition 11 by a Chinese-remainder forcing construction that loads the denominator forms with prime factors while a first-moment bound keeps the numerator forms typical.
111. **Binomial-slice positive shocks (Conjecture 111, resolved true).** For every  $k \geq 2$  the same jump is unbounded above, so the prime-factor count is a two-sided oscillating process; the same forcing construction proves it with the two form families exchanged.

- 112. **Fuss–Catalan two-sided shocks (Conjecture 112, resolved true).** For every  $r \geq 1$  the  $\Omega$ -jump of the Fuss–Catalan numbers is unbounded in both directions, extending the shock phenomenon beyond the binomial coefficient; Proposition 11 proves both directions.
- 113. **Negative shock covariance (Conjecture 113).** In every fixed binomial-slice, central-multinomial, and Fuss–Catalan family successive  $\Omega$ -jumps have a strictly negative lag-one covariance, negative in all 22 tested families; the prime-truncated central-binomial covariance is now exactly  $-\frac{5}{3}$  at every cutoff, the predicted limit for that family.
- 114. **Dyadic multinomial slope (Conjecture 114).** The dyadic  $\Omega$ -jump of  $M_{k,n}$  is conjectured to have slope  $-(k-1)$ , with the  $k=2$  case now proved by cyclotomic factorisation.
- 115. **Eventual sign balance (Conjecture 115, resolved true).** In each binomial-slice and Fuss–Catalan family the positive- and negative-jump densities each tend to  $\frac{1}{2}$  with vanishing zero mass, now proved through the multivariate Erdős–Kac theorem of [173] applied to the exact jump identities.
- 116. **Boundary Euler sum (Conjecture 116).** The mean prime-factor balance of the four composites bracketing a fixed prime gap equals an explicit  $\ell$ -adic Euler sum, matched at correlation 0.998 across 39 gaps.

**Part VII: twenty conjectures on topological invariants of geometric families, in statement order.**

- 117. **Quadric-ladder ratio law (Conjecture 117, resolved true).** The all-quadric Calabi–Yau Euler indices satisfy  $Q_{n+1}/Q_n < 8$  with strictly increasing ratios for  $n \geq 5$ , tending to 8; since resolved true, all three clauses proved with the sharpening  $8 - Q_{n+1}/Q_n \sim 4/n$ , Proposition 16, the limit  $8 = 2^3$  being the  $d=2$  member of the Euler family  $d^{d+1}$ .
- 118. **Normalized merge monotonicity (Conjecture 118).** The degree-normalized Euler index  $E_n/D$  strictly increases under every degree merge, while the raw version fails at  $E_4(2, 2, 4) = 1632 > 1476 = E_4(3, 4)$ , the retained control.
- 119. **Maximal-degree merge monotonicity (Conjecture 119).** Every merge involving a largest degree strictly increases the unnormalized index  $E_n$ , in all 377,356 tested merges through dimension 30.
- 120. **Extremal multidegrees (Conjecture 120).** The all-quadric and hypersurface configurations are the unique Euler extremes at every dimension, likely accessible by coefficient inequalities.
- 121. **Dimension-wise Euler gcd law (Conjecture 121).** The gcd of Calabi–Yau complete-intersection Euler characteristics at dimension  $n$  equals  $24/\gcd(24, n)$ , doubled exactly when  $n \equiv 0, 2 \pmod{8}$ , verified in every dimension 2 to 30.
- 122. **Five-link cap extremum (Conjecture 122).** Among Fano Brieskorn 4-tuples with cap  $A \geq 9$  the middle Betti number is at most  $A-1$ , with equality only at  $(2, 2, A, A)$ , exact through cap 50.
- 123. **Seven-link cap extremum (Conjecture 123).** For 5-tuples with cap  $A \geq 13$  the bound is  $(A-1)(A-2)$ , uniquely at  $(2, 2, A, A, A)$ , the transition at  $A=12$  peaking at 222.

124. **Connected gcd-graph positivity (Conjecture 124).** A connected gcd graph on a Brieskorn 4-tuple forces positive middle homology, a dimension-specific law false at  $N = 5$ , close to prior work.
125. **Common-scale monotonicity (Conjecture 125, resolved true).** Scaling all exponents of a Brieskorn tuple by a common factor never decreases the middle Betti number, now proved in Proposition 18: the scale sequence is a sum of interior Ehrhart counts of integral weighted hypersimplices, with every forward difference through order  $N - 1$  nonnegative.
126. **Scale convexity and log-concavity (Conjecture 126).** The same scale sequences are simultaneously discretely convex and log-concave; the convexity clause is now proved by the Ehrhart model, and the log-concavity clause survives an exact stress of 24,950,000 inequalities.
127. **Hochster interval support (Conjecture 127).** Every aggregated Hochster strand of a flag moment-angle complex has gapless support, though the ordinary Betti sequence of  $K_{2,3}$  is not even unimodal.
128. **Hochster-strand log-concavity (Conjecture 128).** Every strand sequence is log-concave in the subset size, while ultra-log-concavity fails at a connected seven-vertex graph, the retained boundary.
129. **Strand Hurwitz stability (Conjecture 129).** After removing the monomial factor, every strand polynomial has all zeros in the open left half-plane, verified by exact Routh–Hurwitz tests on 50,045 polynomials; real-rootedness is false at four vertices.
130. **Cycle extremality at connectivity two (Conjecture 130).** Among 2-connected graphs the cycle dominates every component-strand coefficient, with coefficientwise equality only for the cycle itself.
131. **Strict component-strand log-concavity (Conjecture 131).** On connected noncomplete graphs the component-strand log-concavity is everywhere strict on positive triples.
132. **Parity-strand unimodality (Conjecture 132).** The even and odd exterior-invariant profiles of every orientation-preserving signed permutation are unimodal, log-concavity failing at dimension 10.
133. **Mapping-torus Betti unimodality (Conjecture 133).** The full Betti sequence of the associated torus mapping torus is unimodal, log-concavity failing already at  $-I_4$  with  $(1, 1, 6, 6, 1, 1)$ .
134. **Sparse-cycle minimizer structure (Conjecture 134).** Some total-Betti minimizer in every dimension  $d \geq 8$  uses pairwise distinct negative cycles and at most the parity-forced positive cycle, as at  $(5, 7)$  for  $d = 12$ .
135. **Square-root-two compression exponent (Conjecture 135, resolved false).** The claimed exponent  $\sqrt{2}$  is refuted by the Landau floor  $M_d \geq 2^d/g(d)$  of Proposition 19: the true exponent is 2, and the square-root fit was a finite-size illusion, since  $\log_2 g(d) \approx d/2$  throughout the tested range.

136. **Torsion budget inequality (Conjecture 136).** The 2-primary torsion generators of these mapping tori never outnumber the total rational Betti number, equivalently negative orbits are at most twice the positive orbits, with equality attained 109 times through dimension 12.

**Part VIII: twenty conjectures on shape laws for enumerative geometric invariants, in statement order.**

137. **Strict dominance law (Conjecture 137).** The signed Euler index of anticanonical hypersurfaces in products of projective spaces strictly increases along every dominance move on the ambient partition, in all 42,691 tested moves through dimension 22.
138. **Strict refinement contraction (Conjecture 138).** Splitting one ambient factor always strictly decreases the signed Euler index, the mirror on ambients of Part VII's merge monotonicity on defining equations.
139. **The 8/9 refinement barrier (Conjecture 139).** The least destructive ambient split asymptotically retains exactly eight ninths of the Euler index, with  $S_{22} = 0.88947$  against  $8/9 = 0.88889$ .
140. **Positive interaction of disjoint refinements (Conjecture 140).** For dimension at least six the log Euler index is strictly supermodular on disjoint split squares, the threshold genuine with four violating squares at dimension five.
141. **Strict Betti log-concavity (Conjecture 141).** The Betti profiles of Göttsche's product are strictly log-concave in the cohomological degree for every seed  $b \geq 3$ , covering every  $\text{Hilb}^n(K3)$ , in 179,200 exact Turán inequalities.
142. **Cohomology-seed total positivity (Conjecture 142).** Adding one seed generator concentrates the normalized profile toward the Lefschetz centre in likelihood-ratio order, in all 53,070 adjacent minors.
143. **Kurtosis trichotomy (Conjecture 143).** Over the proved exact mean  $n$  and variance  $2n/(b+2)$ , the profile kurtosis follows a complete phase diagram in  $b$  with a unique exceptional peak at  $(6, 9)$ , the equality  $\kappa_7(1) = \kappa_7(2) = 9/2$ , and limit  $21/5$ .
144. **Fixed-charge log-concavity (Conjecture 144).** At fixed displacement from middle cohomology the profiles are strictly log-concave in the number of points, making the two-parameter array log-concave along both axes.
145. **Sharp low-dimensional age unimodality (Conjecture 145).** Twisted-sector age histograms of isolated cyclic Calabi–Yau quotients are unimodal in dimensions four and five, sharply:  $\frac{1}{3}(1, 1, 1, 1, 1, 1)$  gives  $(0, 1, 0, 1, 0)$  in dimension six; close to prior work.
146. **Gaussian bulk age law (Conjecture 146).** In the random prime-order model the standardized age distribution tends to a Gaussian, a random-model statement likely accessible to classical equidistribution machinery.
147. **Local central-sector law (Conjecture 147).** Individual age sectors carry lattice-Gaussian mass, with central density  $\sqrt{6/(\pi d)}$ , the local upgrade of the bulk law with the same caveats.

148. **High-sampling restoration of unimodality (Conjecture 148).** Once  $r/d^3 \rightarrow \infty$  the whole histogram is unimodal with probability tending to one, the exponent three deliberately exposed to refinement.
149. **Exterior torsion-entropy log-concavity (Conjecture 149).** The exterior-degree profile of torsion growth rates of hyperbolic toral mapping tori is log-concave, in 81,000 inequalities on 16,200 spectra.
150. **Equality rigidity (Conjecture 150).** Interior equality in the profile forces a two-valued spectrum, in all 222 equality cases of the exact integer corpus.
151. **Sharp binomial upper envelope (Conjecture 151, resolved true).** Convexity on the product of sign simplices proves the sharp upper bound, attained by  $(1, 0, \dots, 0, -1)$ .
152. **Two-level lower envelope (Conjecture 152, resolved true).** Same-sign averaging proves that a two-level spectrum attains every lower envelope.
153. **Nullity-strand interval support (Conjecture 153, resolved false).** The cycle matroid of  $K_{3,3}$  has nullity-one strand  $(9, 0, 6)$  at sizes four to six.
154. **Row-and-column log-concavity (Conjecture 154, resolved false).** The same strand gives the horizontal failure  $0^2 < 9 \cdot 6$ , while the vertical direction remains open separately.
155. **Nearest-rhombus log-concavity (Conjecture 155, resolved false).** A ten-edge graphic matroid has  $H_{2,6} = 2$ ,  $H_{1,5} = 2$ , and  $H_{3,7} = 3$ , giving  $4 < 6$ .
156. **Hurwitz stability of nullity strands (Conjecture 156, resolved false).** The  $K_{3,3}$  strand reduces to  $9 + 6z^2$ , whose zeros lie on the imaginary axis.

**Part IX: twenty conjectures on spectra and filtrations of combinatorial geometries, in statement order.**

157. **Universal negative real roots (Conjecture 157).** The  $h$ -polynomial of every graph associahedron has all zeros real and strictly negative, interpolating the Narayana and Eulerian endpoints, in all 878 deposited root sets and an independent scan of all 771 connected graphs through five vertices; close to prior work.
158. **Connected-deletion interlacing (Conjecture 158).** Deleting a vertex that keeps the graph connected weakly interlaces the  $h$ -zeros, the recursion that would prove the root law, in 3,621 deposited deletion pairs.
159. **Strict Mahler growth under edge addition (Conjecture 159).** Every edge added to a connected graph strictly increases the logarithmic Mahler measure of  $h_G$ , in 3,956 pairs with minimum increment 0.0263.
160. **Edge addition centralizes the spectrum (Conjecture 160).** Every edge addition moves the normalized even-Betti distribution of the toric variety toward middle degree in the symmetric tail order, every tested proper comparison strict.
161. **Path-star Mahler extremality (Conjecture 161).** Among trees the path minimizes and the star maximizes the Mahler measure, each uniquely, the spectral shadow of known coefficientwise orders; close to prior work.



162. **One-crest component law (Conjecture 162).** The component census of the age filtration of an isolated cyclic quotient with distinct weights is unimodal, in 3,135 deposited profiles, distinctness being genuine: a repeated-weight control has census  $(1, 2, 1, 2, 1, 1, 1, 1)$ .
163. **Early component crest (Conjecture 163).** Fragmentation always peaks by half age, so the second half of the filtration only consolidates, by the age duality mechanism.
164. **Half-age connectivity (Conjecture 164).** For  $d \geq 5$  the half-age sublevel complex is already connected, in all 2,905 eligible profiles, with a disconnected repeated-weight control; close to prior work.
165. **One-crest first homology (Conjecture 165).** The  $\mathbf{F}_2$  first-Betti census of the filtration is unimodal, the simplest persistence-type law the object could satisfy.
166. **One-crest second homology (Conjecture 166).** The second-Betti census is unimodal as well, and the degree bound is exact: a distinct-weight quotient with two-crest  $\beta_3$  census is retained as a control.
167. **Positive-temperature log-concavity (Conjecture 167).** Lens-space sine-torsion partition functions are strictly log-concave across the sector index at every positive temperature, in 40,932 triples with minimum curvature 0.0079.
168. **Concavity of mean log torsion (Conjecture 168).** The zero-temperature sector means are strictly concave outright, in 6,822 triples with minimum curvature 0.0770, over the exactly pinned global mean.
169. **The  $\zeta(3)$  central parabola (Conjecture 169).** In the iterated limit the centred sector means follow the parabola  $\frac{3\zeta(3)}{\pi^2}(1-x^2)$ , the constant forced by the proved moment identities.
170. **Age-torsion decoupling (Conjecture 170).** In the random model the age and the log-torsion sum are asymptotically jointly Gaussian and independent, strengthening the annealed age law of Conjecture 146; a random-model claim.
171. **Negative-temperature reversal (Conjecture 171).** On  $-1 \leq s < 0$  the sector shape reverses to strict log-convexity, in 20,466 triples, the window calibrated by an  $s = -2$  control of curvature  $+0.4980$ .
172. **Log-concavity of boundary entropy (Conjecture 172).** The log pseudodeterminants of a simplicial sphere's boundary Laplacians are log-concave across the degree, in 262 triples with minimum margin 51.8.
173. **Entropy per rank decreases (Conjecture 173).** The mean log eigenvalue at each degree weakly decreases with the degree, in all 392 deposited adjacent comparisons.
174. **Simplex rigidity of equality (Conjecture 174).** Any adjacent equality of entropy per rank forces the boundary of a simplex, where all equalities hold, in 130 rigidity tests.
175. **Strict stellar growth (Conjecture 175).** Stellar subdivision of a facet strictly increases every entry of the determinant profile, in 900 comparisons with minimum increment 1.42.
176. **Log-concave stellar increments (Conjecture 176).** The entropy created by a facet subdivision is itself log-concave across the degree, in 450 triples with minimum margin 17.8.

**Part X: twenty conjectures on flux cohomology and protected index laws, in statement order.**

177. **Dense discrete flux saturation (Conjecture 177).** Random sign flux on the torus attains the generic exterior-algebra cohomology profile with probability tending to one, exponentially fast, in 142 of 144 deposited and 71 of 72 independent trials.
178. **Sparse-flux threshold at the coverage scale (Conjecture 178).** Saturation has a coarse threshold at  $(\log n)/n^2$ , the coverage scale of random 3-uniform hypergraphs, the transition observed across 960 deposited trials; the subcritical direction is now proved for every fixed  $c < 2$  at the coverage scale and the supercritical one remains conjectural.
179. **Linear-complexity sparse witnesses (Conjecture 179).** The minimum saturating support grows linearly, now proved from below in the sharp form  $\mu_n \geq \lceil (n-1)/3 \rceil$ , with conjectured linear-hypergraph witnesses of size  $n + C$ , the explicit witnesses using 1, 1, 2, 2, 5, 5, 8, 7 triples for  $n = 3$  to 10.
180. **Coefficient-law universality (Conjecture 180).** Inside the critical window the limiting saturation statistics are the same for every coefficient law, the two deposited laws tracking to 0.2125 against 0.2125; the corresponding statement conditional on the support is false and a five-coordinate witness is retained as a boundary control; a random-model statement.
181. **Finite-field resonance power law (Conjecture 181).** Conditional on the displayed target being the true generic profile, Proposition 29 proves the Frobenius-dependent power law by Lang–Weil. The open content is the generic-profile identification and the geometry of the top resonance components.
182. **Central-binomial law for the complete flux (Conjecture 182, resolved true).** The complete flux has cohomology totals  $F_n$ , a central binomial, over the rationals and  $P_n$  in characteristic two, now proved from a symplectic normal form that reduces the complex to Boolean inclusion matrices.
183. **Prime torsion staircase (Conjecture 183, resolved false).** The staircase clause is now proved for every prime, first  $p$ -torsion being exactly one  $\mathbb{Z}/p$  at  $n = 4p - 1$ , but the elementary-abelian clause is refuted at  $n = 15$  by a summand  $\mathbb{Z}/4$  in degree nine, the least dimension at which one can occur.
184. **Parity-locked torsion count (Conjecture 184).** Saturating sign flux carries exactly  $(P_n - G_n)/2$  even torsion summands, forced by the proved parity calibration together with Conjecture 177 and the now-proved Conjecture 182, while the orders fluctuate, with factors 4, 40, 80 on record; retained as a corollary rather than an independent law.
185. **Coefficientwise endpoint extremality (Conjecture 185).** The all-quadric and hypersurface configurations bound every interior coefficient of the signed  $\chi_y$ -profile, in 7,124 exact comparisons over 680 configurations through dimension 14.
186. **Degree-normalized merge monotonicity (Conjecture 186).** Every coefficient density  $a_p/D$  strictly increases under every equation merge, in 110,777 exact comparisons, extending Conjecture 118’s Euler law to the full protected profile.

187. **Maximal-degree raw monotonicity (Conjecture 187).** Merges through a maximal degree raise every interior coefficient without normalization, in 47,433 exact comparisons.
188. **Real-rooted protected polynomial (Conjecture 188).** The reduced signed  $\chi_y$ -polynomial has only simple strictly negative real zeros, in 369 numerically checked polynomials with smallest normalized separation  $1.276 \times 10^{-3}$ ; resting on numerical evidence.
189. **Strict ultra-log-concavity (Conjecture 189).** The binomial-normalized profile is strictly log-concave, in 6,540 exact inequalities, the first layer of the real-rootedness hierarchy; close to Hodge–Riemann machinery.
190. **Degree-normalized Mahler growth (Conjecture 190).** The Mahler measure per unit degree strictly increases under every merge, in 4,267 numerically checked tests, returning Conjecture 159’s Mahler order on the protected polynomial.
191. **Spectral-radius growth with low-dimensional rigidity (Conjecture 191).** Maximal merges never contract the root radius, with equality exactly in complex dimensions 2, 3, and 5 and strict growth elsewhere, in 2,102 checked tests.
192. **Signature endpoint extremality (Conjecture 192).** The all-quadric and hypersurface configurations bound the normalized signature in every even dimension, with endpoint rigidity, in 8,190 exact configurations through dimension 26.
193. **Degree-normalized signature monotonicity (Conjecture 193).** The signature density  $S/D$  strictly increases under every merge, in 302,130 exact comparisons.
194. **Maximal-degree signature monotonicity (Conjecture 194).** Merges through a maximal degree strictly raise the signature itself, in 80,294 exact tests.
195. **The quadric ratio law with limit 27 (Conjecture 195, resolved true).** Successive all-quadric signatures grow by strictly increasing ratios below 27 converging to 27 with correction  $27/n$ , exact through  $n = 60$  where  $R_{58} = 26.55045876$ ; since resolved true, all three clauses proved from an algebraic generating function, Proposition 31, whose mechanism extends to every repeated block and identifies 27 as  $\cot^6(\pi/6)$ .
196. **Dimension-wise signature gcd law (Conjecture 196).** The gcd of all signatures in even dimension  $n$  is 2 for  $n \equiv 0 \pmod{4}$  and  $2^{\max(4, 1+\nu_2(n+2))}$  for  $n \equiv 2 \pmod{4}$ , the spike 32 at  $n = 14$  confirmed, the signature sibling of Conjecture 121’s Euler gcd law.

**Part XI: twenty conjectures on spectral entropies and multibase carry fields, in statement order.**

197. **All-order path maximality (Conjecture 197).** Among trees of a fixed order the path uniquely maximizes every Rényi entropy of the Laplacian density matrix simultaneously, the order-two and min-entropy cases proved, in 8,775 deposited comparisons through twelve vertices and 3,031 independent ones with smallest margin  $2.23 \times 10^{-4}$ ; close to the star-minimum conjecture.
198. **Leaf-pair closure law (Conjecture 198).** Closing the path between two leaves of a tree into a cycle raises the Laplacian entropy at every order, in 86,982 deposited and 14,252 independent comparisons with smallest margin  $1.66 \times 10^{-3}$ .

199. **Edge-subdivision law (Conjecture 199).** Subdividing any tree edge raises the Laplacian entropy at every order, in 70,126 deposited and 11,249 independent comparisons with the programme's most comfortable margin,  $6.12 \times 10^{-2}$ .
200. **Leaf-compression window (Conjecture 200).** Moving a leaf to a no-smaller neighbour lowers the Laplacian entropy throughout the window from order one half to two, the order-two case proved, in 15,385 deposited window comparisons, while an explicit twelve-vertex tree violates the direction at order 0.1 by  $7.65 \times 10^{-7}$ ; the boundary control is retained.
201. **Unicyclic endpoint law (Conjecture 201).** Among unicyclic graphs the cycle uniquely maximizes and the star-plus-edge uniquely minimizes the Laplacian entropy at every finite order, in 6,230 deposited comparisons; the exact min-entropy tie of the two four-vertex unicyclic graphs excludes the infinite order and is retained as a control.
202. **Tree one-vertex extension law (Conjecture 202).** Pendant attachment and edge subdivision raise the normalized distance signless Laplacian entropy of a tree at every order, in 168,168 deposited comparisons with minimum margin  $4.44 \times 10^{-2}$ , the first entropy law posed for this spectrum.
203. **Low-order extension law (Conjecture 203).** For arbitrary connected graphs the same two extensions raise the metric entropy on the window of orders up to two, in 10,884 deposited and 7,672 independent comparisons, with explicit high-order violations at orders five and ten retained as controls.
204. **Star maximality (Conjecture 204).** The star uniquely maximizes the distance signless Laplacian entropy among trees at every order, reversing the Laplacian programme's path law, in 8,775 deposited and 3,031 independent comparisons with smallest margin  $4.72 \times 10^{-4}$ .
205. **Bipartite path minimality at the Shannon order (Conjecture 205, resolved false).** An interval-certified fourteen-vertex tree refutes the proposed path minimum with gap  $3.15 \times 10^{-5}$ . Exhaustive enumeration of all 3,159 unlabelled trees of that order finds a sharper witness with gap  $9.10 \times 10^{-5}$ .
206. **Balanced-biclique window maximality (Conjecture 206).** The balanced complete bipartite graph maximizes the metric entropy on the window from the Shannon order to order two, in 5,322 deposited and 195 independent comparisons with smallest margin  $1.60 \times 10^{-3}$ .
207. **Unicyclic cycle maximality above the Shannon order (Conjecture 207).** Among unicyclic graphs the cycle uniquely maximizes the metric entropy at every order at least one, in 2,245 deposited comparisons, while a four-vertex noncycle graph wins at order 0.1 by  $6.62 \times 10^{-4}$ ; the boundary control is retained.
208. **Cross-prime carry central limit law (Conjecture 208).** The carry counts of  $n + n$  at finitely many distinct odd primes are asymptotically independent Gaussians, with single-base mean and variance fixed by the proved carry-chain law, marginals and cross-correlations below 0.08 tracked at two million integers, close to the  $q$ -additive different-base theorem.

- 209. **Bounded cross-prime covariance (Conjecture 209).** The covariance of carry counts at two distinct primes stays bounded while each variance grows logarithmically, deposited covariances below 0.24; the part's crispest analytic target.
- 210. **Cross-prime local limit factorization (Conjecture 210).** Joint point probabilities of the carry vector factor into products of Gaussian lattice masses at one-cell resolution.
- 211. **Modular equidistribution of carry vectors (Conjecture 211).** Carry counts at distinct primes jointly equidistribute over arbitrary fixed moduli, deposited discrepancies below  $10^{-3}$  at moduli two and three; the Fourier face of cross-base mixing.
- 212. **Additive large deviations (Conjecture 212).** The vector of carry densities satisfies a large-deviation principle whose rate is the sum of the proved single-base rates, asserted on the interior only, away from the Diophantine boundary where the Graham entanglement lives.
- 213. **Bounded mixed cumulants (Conjecture 213).** Every mixed cumulant of total order at least three across at least two primes stays bounded while marginal cumulants grow, giving a connected-correlation route to the full decoupling.
- 214. **Fixed-multiplier carry vectors (Conjecture 214).** The decoupling survives the richer carry automaton of  $\binom{an}{n}$  for every fixed multiplier, deposited correlations below 0.11 and independent ones below 0.04 at  $a = 3$ .
- 215. **Balanced factorial-ratio vectors (Conjecture 215).** Valuation vectors of every balanced integer factorial ratio decouple across primes, the Landau ratio at  $p = 7, 11$  showing the predicted logarithmic variances with cross-correlation  $-0.016$ .
- 216. **Multibase transducer principle (Conjecture 216).** For bounded finite-state transducers in jointly multiplicatively independent bases, automaton mixing plus the absence of a common periodic factor is asserted to be the complete obstruction list for joint Gaussian decoupling; a programme conjecture.

**Part XII: twenty conjectures on subdivision spectra, spanning-tree correlation, cyclic entropy, and graphon dimension, in statement order.**

- 217. **Critical moment exponent for barycentric vertex Laplacians (Conjecture 217).** The spectral  $\alpha$ -moments of the refined 1-skeleton Laplacian have exponential rate  $\max\{0, \alpha \log d! - \log(d+1)!\}$ , with linear growth at the critical order  $\log(d+1)!/\log d!$ , the exact divergence threshold that weak universality cannot see.
- 218. **Hodge incidence-tail hierarchy (Conjecture 218).** Across Hodge degree  $k$ , the critical moment exponents follow the codimension factorials  $(d-k+1)!$ , with degrees zero and one sharing the tail  $d!$  by supersymmetry and the top degree carrying no exponential tail; the three-dimensional probe pattern is 6, 6, 2, 1.
- 219. **All-degree inclusion-uniform universality (Conjecture 219).** Every inclusion-uniform subdivision rule has universal weak Hodge limits in every degree, the common generalization of Märte's top-degree theorem and Knill's barycentric all-degree theorem; an attribution for that interpolation.

220. **Spectral-tail large deviations from simplex age (Conjecture 220).** Eigenvalues at scale  $B_{d,k}^{\theta m}$  occur at exponential frequency  $-\theta \log(d+1)!$ , the full tail profile whose Legendre data are the critical exponents; the part's boldest statement, with no direct tail support at reachable depth.
221. **Finite-state renormalization for barycentric Hodge limits (Conjecture 221).** A finite vector of boundary Green functions evolves under refinement by a rational map whose attracting fixed point carries all limiting Hodge laws, generalizing the known one-dimensional quadratic decimation; falsifiable by unbounded growth of the minimal boundary closure rank, on which the statement stands or falls.
222. **Complete-split extremality (Conjecture 222).** The spanning-tree correlation of a connected graph is maximized by a complete split graph with mesoscopic core of order root of  $n$  over  $\log n$ , and the maximum is  $n$  minus root of  $n \log n$  to leading order.
223. **Triangle-free bipartite extremality (Conjecture 223).** Under triangle-freeness the maximizer is a complete bipartite graph whose small side migrates at the same mesoscopic scale; the retained control is the crossover from two to three at seventeen vertices that killed the predecessor claim.
224. **Cycle minimum for bridgeless dependence (Conjecture 224).** Every bridgeless graph has spanning-tree correlation at least that of the cycle of the same order, with equality only for the cycle.
225. **Core-periphery stability at the ceiling (Conjecture 225).** Near-extremal families for the correlation ceiling must contain a mesoscopic set joined to almost the whole graph over a near-independent complement, with no structure asserted inside the core, a freedom the order-one split-bipartite gap shows is genuine.
226. **Complete-multipartite reduction (Conjecture 226).** Among complete multipartite graphs the correlation maximizer is always one macroscopic part plus singletons, collapsing the partition optimization to the split family; verified exhaustively through order forty-five.
227. **Localization of near-minimizers on prime cycles (Conjecture 227).** Mesoscopic sequences attaining the proved half-log-two doubling floor asymptotically must, after affine automorphisms, concentrate on cyclic intervals of length little-o of  $p$ : the inverse theorem to the 2026 prime-field entropy-power inequality.
228. **Gaussian rigidity after localization (Conjecture 228).** Under localization and a uniform moment bound, near-minimizing lifts obey a central limit theorem with entropic convergence; the moment hypothesis is load-bearing, since unrestricted continuous rigidity is known to fail.
229. **Heat-kernel wrap crossover (Conjecture 229, resolved false).** The claimed universal heat-kernel wrap limit is refuted by a mesoscopic jump family satisfying every stated hypothesis whose limit carries an extra compound-Poisson factor; a Lindeberg condition at the modular scale is the recorded repair.
230. **Fixed- $m$  entropy-power law on prime cycles (Conjecture 230).** For every fixed number of summands the mesoscopic entropy gain is at least half the logarithm of the



count, with Gaussian rigidity at equality; close to prior work, since the integer-lattice analogue is a recent theorem and only the modular content is new.

- 231. **Uniform pre-wrap entropy monotonicity (Conjecture 231).** Log-concave cyclic laws obey the entropic central-limit increment law uniformly in the number of summands all the way to the wrap scale, with the deposited failure onset near wrap ratio one percent as the retained boundary control.
- 232. **Exact-dimensional entropy identity (Conjecture 232).** For random-free graphons whose row measure is exact-dimensional with local doubling, sampled-graph entropy per  $n \log n$  converges exactly to the dimension.
- 233. **Multifractal average-dimension law (Conjecture 233).** Under an explicit  $L^1$  convergence of local dimensions, the entropy coefficient is the mean pointwise information dimension, neither a Minkowski dimension nor an essential supremum.
- 234. **Second-order graph entropy constant (Conjecture 234).** Smooth row manifolds admit a finite linear-order entropy constant, which the threshold-graphon control proves cannot be the quantization constant of the row space.
- 235. **Equipartition for random-free samples (Conjecture 235).** The sample information content concentrates at the leading scale: almost every sampled graph lies in a typical set of the predicted exponential size, a Shannon–McMillan law for graphon sampling.
- 236. **Sparse-observation dimension law (Conjecture 236).** Under Bernoulli edge-masking with a growing probe count, conditional entropy scales as  $n$  times the logarithm of the probe count times the mean dimension, quantifying information decay under missing data.

**Part XIII: nineteen conjectures on flux torsion, quantum merge monotonicity, and effective string geometry, in statement order.**

- 237. **Odd-degree discrete saturation (Conjecture 237).** Sign-valued random  $r$ -forms attain the generic rational cohomology profile with exponentially small failure probability for every odd degree, lifting the three-form law of Part X to the full family; sixty deposited and twenty-four independent degree-five trials all saturate.
- 238. **Odd coverage-scale threshold (Conjecture 238).** Sparse random odd flux saturates exactly at the isolated-vertex scale of random  $r$ -uniform hypergraphs, with the suspension dimensions excluded by the essentiality hypothesis; the subcritical direction is now proved and the supercritical one remains conjectural.
- 239. **Torsion shape law for the complete flux (Conjecture 239).** The degree-wise prime-torsion multiplicities of the complete flux are strictly log-concave in the interior, with the shifted duality and the interval support, now identified as the full range  $2p+1$  to  $n-2p+2$ , both proved; log-concavity is verified for every prime up to twenty-three through dimension one hundred fifty.
- 240. **Post-onset torsion persistence (Conjecture 240, resolved true).** Once  $p$ -torsion appears at dimension  $4p-1$  its total strictly grows forever, now proved for every prime from the normal form; the two-primary totals run 1, 2, 10, 20, 66, 132.

241. **Normalized mirror-map contraction (Conjecture 241).** In the scale-free normalization by the discriminant scale, every coefficient of the mirror map contracts under equation merges from dimension three on, with the dimension-three order-two equalities as the exact boundary; five hundred eighty-eight independent comparisons pass.
242. **Gopakumar–Vafa amplification (Conjecture 242).** Across every merge edge of the five one-parameter threefolds, the ratio of genus-zero curve counts grows strictly with the degree, an ordering of geometries by quantum growth refined degree by degree.
243. **Monotone conifold approach (Conjecture 243).** The genus-zero invariants of each of the five threefolds are strictly log-convex from degree one, with ratios climbing monotonically to the conifold growth constant.
244. **Merge monotonicity of the conifold constant (Conjecture 244).** The conifold point in the flat coordinate strictly decreases under every merge, so consolidation strictly raises the quantum growth constant, the transcendental image of the proved algebraic ordering.
245. **Conifold-normalized Yukawa monotonicity (Conjecture 245).** At equal fractions of the respective conifold radii, the Yukawa coupling normalized by the classical intersection number strictly increases under merges, with the raw instanton comparison failing everywhere as the retained control.
246. **Quadratic wall complexity (Conjecture 246).** The number of numerical walls meeting a compact region grows at most quadratically in the charge norm. A support-property count is only polynomial and the quadratic exponent requires extra Picard-rank-one geometry.
247. **Sharp Gamma for Brill–Noether stability (Conjecture 247).** The Gamma-constant produced by the Brill–Noether construction of stability conditions is exactly optimal on its ray, its topological floor computed by the proved Chern calibration as five sixths down to one third across the five threefold configurations.
248. **Tame wall atlas (Conjecture 248).** In a fixed effective analytic-exponential presentation, the aggregate wall arrangement has polynomial formula format and a polynomial-size compatible stratification up to every charge cutoff.
249. **Polynomial conditioning of Ricci-flat metrics (Conjecture 249).** All curvature derivatives, the diameter and the regularity scale of Ricci-flat metrics are polynomially controlled in a power-equivalence class of discriminant gauges.
250. **Discriminant-effective balanced convergence (Conjecture 250).** Balanced metrics converge at the classical rate uniformly in the family once the level exceeds a power of the inverse discriminant gauge.
251. **Discriminant-effective Hermitian–Yang–Mills convergence (Conjecture 251).** Balanced-bundle approximations carry a polynomial condition number in the minimum of a discriminant gauge and a quantitative stability margin.
252. **Polynomial matter spectral gap (Conjecture 252).** The first positive bundle-valued Dolbeault eigenvalue is bounded below by a power of the pair gauge wherever the matter cohomology is constant.

- 253. **Certified heterotic Yukawa convergence (Conjecture 253).** Numerically computed physical Yukawa couplings admit a-priori error certificates polynomial in the pair and coupling-zero gauge classes.
- 254. **Saturation-index special-Lagrangian realization (Conjecture 254).** At infinite-distance degenerations, every extremal light charge has a calibrated representative after multiplication by a divisor of the saturation exponent of the monodromy-orbit lattice.
- 255. **LMHS-controlled Laplace towers (Conjecture 255).** The vanishing Ricci-flat Laplace spectrum has finitely many exhaustive towers. Their rational exponents come from the integral limiting mixed Hodge structure and skeleton combinatorics, while the limiting operators may retain metric data.
- 256. **Coordinate monotonicity of normalized complexity (Conjecture 256).** The degree-normalized primitive middle Betti number of a smooth complete intersection strictly increases in every defining degree, growth beyond the trivial multiplicative degree scale.
- 257. **Strict discrete convexity in each degree (Conjecture 257).** Each one-variable restriction of the normalized complexity to the degree lattice is strictly convex, so marginal topological growth accelerates.
- 258. **Additive supermodularity across degrees (Conjecture 258).** Raising one degree strictly increases the absolute marginal value of raising another: distinct defining equations reinforce.
- 259. **Strict log-submodularity (Conjecture 259).** Proportional marginal gains shrink where absolute gains grow, so the logarithm of the normalized complexity is strictly submodular against the supermodularity of the complexity itself.
- 260. **Schur-convexity at fixed degree sum (Conjecture 260).** At fixed dimension, codimension, and adjunction coefficient, majorization of multidegrees raises normalized complexity: balanced degrees minimize it.
- 261. **Adjunction-preserving splits decrease complexity (Conjecture 261).** Splitting one degree  $d$  into a pair summing to  $d + 1$  strictly lowers normalized complexity; the odd-dimensional Calabi–Yau slice is proved via the merge order of Part VII.
- 262. **Strict log-concavity of the nonlinear row (Conjecture 262).** The  $q = 2$  row of every  $\mathbf{P}^1 \times \mathbf{P}^1$  Betti table is strictly log-concave across its support, beyond the published unimodality conjecture; resting on a thirteen-table corpus.
- 263. **Linear row log-concave through its mode (Conjecture 263).** The  $q = 1$  row keeps strict log-concavity on its rising side and at its mode, every observed failure lying in the descending tail.
- 264. **At most one defect in the linear row (Conjecture 264).** The linear row admits at most one log-concavity failure, a nearly totally positive shape with one controlled exception.
- 265. **Defects occur only inside the nonlinear overlap (Conjecture 265).** A linear-row defect can occur only where the nonlinear row is nonzero, so vanishing of the nonlinear syzygies forces local log-concavity of the linear ones.

- 266. **Defect-mode locking (Conjecture 266).** The unique linear-row defect sits at the nonlinear row's mode or one step beyond it, in all eight defect tables; the boldest law of the part.
- 267. **Likelihood-ratio flattening in the degree (Conjecture 267).** Outer primitive Hodge pieces of a smooth hypersurface gain relative weight against inner ones at every degree step, a monotone likelihood-ratio comparison across varieties.
- 268. **Central-window escape (Conjecture 268).** Every central window of the Hodge diamond loses normalized mass monotonically as the degree grows.
- 269. **Convex-order dispersion (Conjecture 269).** Each degree step is a mean-preserving spread of the normalized Hodge profile, so every convex dispersion functional grows.
- 270. **Monotone total-variation convergence (Conjecture 270).** The normalized profiles approach the Eulerian descent law monotonically in total variation, with a single equality at the cubic–quartic threefold step.
- 271. **Parity log-concavity of walk counts (Conjecture 271).** The square-lattice counts satisfy  $c_{n+2}c_{n-2} < c_n^2$  with curvature relaxing monotonically to zero, so two-step ratios descend onto  $\mu^2$ ; the statistic and its asymptotics are Guttman–Jensen's, and the exact strict law is the proposed content.
- 272. **Radial-moment parity log-concavity (Conjecture 272).** Every endpoint-moment sum  $M_q(n)$  obeys the same strict two-step log-concavity at every real order  $q \geq 0$ ; the extreme-order regime is proved.
- 273. **Bridge log-convexity (Conjecture 273).** Bridge counts turn the curvature around:  $b_n^2 < b_{n-1}b_{n+1}$  from  $n = 5$ , sharp at its threshold, so bridge ratios ascend onto  $\mu$ .
- 274. **Bridge height likelihood-ratio growth (Conjecture 274).** The bridge-height array is totally positive of order two in the length, its top corner proved at every length.
- 275. **Endpoint monotone coupling (Conjecture 275).** Two added steps admit a coupling growing both absolute endpoint coordinates simultaneously, strictly implying the open mean-square-displacement monotonicity along parity.
- 276. **Span monotone coupling (Conjecture 276).** The span vector grows in the coordinate-wise stochastic order under two added steps, despite the reweighting of truncation.
- 277. **Span likelihood-ratio expansion (Conjecture 277).** The one-coordinate span grows in likelihood-ratio order along each parity class.
- 278. **Contact stochastic growth (Conjecture 278).** The contact count grows stochastically at every step, so the normalized attractive partition factor is monotone in length.
- 279. **Span negative quadrant dependence (Conjecture 279).** The two spans compete at every joint tail; the same law for absolute endpoint coordinates is false and retained as a control.
- 280. **Radius–atmosphere positive dependence (Conjecture 280).** Far endpoints are locally more extendable at every pair of thresholds.

281. **Radius–contact negative dependence (Conjecture 281).** Folding and displacement oppose at every joint tail, so an infinitesimal attraction shrinks the mean-square displacement.
282. **Contact–atmosphere negative dependence (Conjecture 282).** Contact-rich walks are terminally caged, so attraction immediately throttles the extension ratio.
283. **Collapse trichotomy of partition curvature (Conjecture 283).** The step-two curvature of  $\log Z_n(u)$  scales as  $-\frac{11}{8}n^{-2}$ ,  $-\frac{4}{7}n^{-2}$ , and  $+\sigma(u)n^{-3/2}$  below, at, and above the collapse point; a consolidation of established predictions into one sequence-level law.
284. **Survival monotone in dimension (Conjecture 284).** The probability that a nonbacktracking walk is self-avoiding rises strictly with dimension at every length, proved through length seven in closed form.
285. **Radial moments monotone in dimension (Conjecture 285).** Every positive endpoint moment falls strictly as the dimension rises, the stochastic-order strengthening being false and retained as a control.
286. **Reversible shortcuts amplify winding skewness (Conjecture 286).** A zero-affinity chord added to a driven ring at fixed activity strictly increases the standardized third cumulant of the winding current, against the mixing intuition.
287. **Sequential spectral recruitment (Conjecture 287).** The budgeted-gap frontier of a linear diffusion recruits reversible modes in spectral order: no mode activates before every slower mode.
288. **Convex arcs with marginal-gain resets (Conjecture 288).** Between recruitments the frontier is convex, increasing returns to irreversibility, with downward slope resets at structural thresholds and the trace ceiling reached at finite budget.
289. **Finite two-mode radius (Conjecture 289).** At small budget every global maximizer is supported exactly on the two slowest modes, upgrading the proved infinitesimal selection to an interval.
290. **Rank-two oscillation extremizer (Conjecture 290).** The maximal spectral frequency at fixed entropy production is achieved by a single rotation block, in closed form; closely related to the oscillatory-dissipation decomposition literature.
291. **Cheapest exceptional point is a pair (Conjecture 291).** The minimal entropy production making the drift defective is the pairwise formula, a thermodynamic Wilkinson problem with an elementary answer.
292. **Recruitment under commutator accessibility (Conjecture 292).** Coherent acceleration of a detailed-balance Lindbladian recruits dissipative eigenmodes sparsely and sequentially, with the coupling direction selected by an accessibility-restricted Rayleigh quotient.
293. **Envelope saturation is Lie-algebraic (Conjecture 293).** The classical frontier bounds every coherently driven Lindbladian, with saturation on an interval exactly when the active modes close into  $\mathfrak{su}(2)$  modules.

294. **Exceptional points collapse robust Mpemba basins (Conjecture 294).** Wherever slow-mode cancellation is feasible at all, the measure of preparations achieving it robustly shrinks in inverse proportion to the biorthogonal condition number, proved with prefactor one in the qubit model.
295. **Isoperimetry and symmetry breaking at fixed energy (Conjecture 295).** At fixed total variogram energy the regular simplex uniquely maximizes the extremal coefficient for three variables and at weak dependence; for four or more variables it is proved strictly beaten below that by configurations with fewer active components, and the optimizer is conjectured to climb a recruitment staircase from three active components to all of them.
296. **Pairwise closure at weak dependence (Conjecture 296).** The independence deficit  $m - \theta_m$  is asymptotically the sum of its pairwise deficits: two-body overlaps dominate, and three-body saddles carry strictly larger rates off a generic exceptional set.
297. **Scale elasticity and monotone dependence (Conjecture 297).** The extremal coefficient has logarithmic scale elasticity at most one half and is entrywise monotone, so spectral sparsification distorts it by at most  $((1 + \varepsilon)/(1 - \varepsilon))^{1/2}$ ; closely related to the variogram-level sparsification guarantees.
298. **Positive curvature at complete dependence (Conjecture 298).** The second coefficient of the strong-dependence expansion is nonnegative and vanishes exactly on segments, the finite-point Wills phenomenon; resolved true, both the nonnegativity and the equality classification proved by a support-function Gaussian integration by parts.
299. **Bernoulli-thinning spectroscopy (Conjecture 299).** Thinning commutes with the cluster limit uniformly in the retention probability, so the extremal-index response curve identifies the entire limiting cluster-size law.
300. **Occupancy tomography of cluster supports (Conjecture 300).** Modular occupancy data determines five-mark supports up to congruence, and every occupancy class holds at most two congruence classes; full reconstruction is refuted by Bloom's pair, and occupancy equivalence sits strictly between congruence and homometry.
301. **Consistency of persistent angular homology (Conjecture 301).** The persistence diagrams of top- $k$  angular samples converge to those of the angular support at the rate  $b(U(n/k)) + (\log k/k)^{1/s}$ .
302. **A leakage trichotomy for hidden extremes (Conjecture 302).** After removing the primary angular support, hidden topology is recovered, lost, or crosses over exactly according to whether the expected leaked count tends to zero, to infinity, or to a constant.
303. **The stability barrier of fractional tail tomography (Conjecture 303).** Fractional projection tails identify the angular measure but admit no Hölder inverse; on Hölder classes a logarithmic modulus holds and no polynomial modulus is valid.
304. **A parameter-count threshold for sparse tail tomography (Conjecture 304).** Generic fractional projections identify an  $s$ -atomic angular measure locally at the parameter count  $sd - 1$  and globally one measurement above it; at the count itself, global uniqueness fails on separated instances, recorded in place.



305. **Poissonized blocks remove the universal bias (Conjecture 305).** Poisson block exposure removes the universal  $V^2/(2n)$  correction and the order-1/ $n$  estimator bias at unchanged first-order variance.
306. **The factorial-cumulant design ladder (Conjecture 306).** Integer block-size laws matching  $r$  Poisson factorial cumulants, constructed for every  $r$ , push the finite-block error to order  $n^{-r}$ ; the algebra is proved, the estimator law conjectured.
307. **An adaptation cusp at second-order cancellation (Conjecture 307).** Exact cancellation in a Hall family doubles the oracle exponent inside a window  $|c| \asymp n^{-\beta/(\alpha+4\beta)}$  where no selector adapts at the oracle rate; closely related to the known logarithmic adaptation penalty.
308. **Circle isoperimetry for multiplicative persistence (Conjecture 308).** Among  $n$  points on a Euclidean circle the regular polygon uniquely maximizes the multiplicative persistence ratio of the dominant loop, for every  $n$ ; the unrestricted planar version is refuted by a six-point configuration retained as a control, and the geodesic antecedent is attributed.
309. **Saturation of the stability factor two (Conjecture 309).** On smooth strictly convex curves under farthest-point sampling, the bottleneck error of the landmark diagram is asymptotically exactly twice the Hausdorff gap: the classical stability constant is attained, not slack.
310. **A two-sided projection threshold for loop ratios (Conjecture 310).** Simultaneous preservation of the loop ratios of  $q$  orthogonal hexagonal channels under Gaussian projection has a sharp threshold at dimension of order  $\varepsilon^{-2} \log q$ ; the lower half would be the first matching lower bound for a topological statistic.
311. **Prevalence shrinkage with exponential leakage (Conjecture 311).** An optional feature of prevalence  $q$  appears in the Wasserstein mean at exactly  $q$  times its conditional persistence up to exponentially small matching leakage; the majority-threshold claim is refuted by the flat grouping algebra and retained as a control.
312. **Exponential stability of latent groupings (Conjecture 312).** The probability that any optimal mean grouping betrays the latent grouping is at most  $CNm e^{-c\gamma^2/\sigma^2}$ , and off this event the empirical Fréchet mean is unique, obeys the flat projection formulas, and satisfies featurewise central-limit behavior.
313. **Stable survival tomography of Reeb graphs (Conjecture 313).** Below the separation scale, the phase-averaged Betti curve of a tame Reeb graph is the exact sum of universal kernels  $F_\alpha(\ell_j/h)$  over its spans, and the kink inversion recovering the span spectrum is quantitatively stable away from the collision stratum, the deposited linear inversion failing at coalescing spans, where only a Hölder-2/3 inverse survives.
314. **The dither law survives discretization (Conjecture 314).** Random cover phase makes the Mapper midpoint estimator of a branch critical value exactly unbiased with the dithered-quantizer variance  $s^2/12$ , and discretization at mesh  $\eta$  perturbs bias and variance by mesh-order corrections only.

315. **Three-direction stability and random-slope observability (Conjecture 315).** For separated rectangle-decomposable modules, three fixed generic directions give a locally bi-Lipschitz invariant with polynomial constants, over the proved one-direction completeness, and the direction profile of the matching distance is never needle-like, so uniformly random slopes see near-maximal discrepancy; closely related to the all-line local equivalence and the foliation Lipschitz bounds.
316. **A restricted isometry for persistence sketches (Conjecture 316).** On separated bounded-cardinality diagram classes the persistence-weighted kernel embedding is bi-Lipschitz to the two-Wasserstein metric, and random Fourier sketches at dimension nearly linear in the cardinality preserve the comparison, threading the known non-embeddability obstructions.
317. **Whitney-count evaluations of persistence surfaces (Conjecture 317).** Generic evaluations of the Gaussian persistence surface at the Whitney count  $4m+1$  of points form a complete invariant of separated  $m$ -point diagrams, locally bi-Lipschitz off the discriminant.
318. **Aggregate-landscape moment determinacy (Conjecture 318).** The first  $4m$  moments of the aggregate landscape determine a generic  $m$ -bar diagram, through the exact moment formula proved here; the one-bar case is proved with two moments to spare.
319. **Near-quadratic vineyard complexity in the mean (Conjecture 319).** Random affine motion of  $n$  planar points produces only  $O(n^2 \log n)$  expected changes of the degree-one persistence pairing, far below the quartic worst case that vineyard algorithms are built to survive.
320. **Temporal sampling-efficiency monotonicity (Conjecture 320, resolved false).** The proposed monotone loss of endpoint irreversibility with the sampling lag is refuted by a homogeneous ten-state ring whose efficiency revives by more than two orders of magnitude across an interior lag interval through winding aliasing.
321. **Two-gap-time location bound (Conjecture 321).** The strongest endpoint-level arrow of time occurs within two reversible relaxation times,  $\gamma t_* \leq 2$ .
322. **Spectral integrated-budget bound (Conjecture 322).** The total area under all two-endpoint irreversibility measurements is capped at  $2\sigma/\gamma^2$  once the mixing timescale is fixed.
323. **Irreversibility-gap doubling law (Conjecture 323, resolved true).** The long-time lag KL decays at exactly twice the rate of the slowest dissipative amplitude, now proved in Proposition 61 from the quadratic equivalence of lag KL with the antisymmetric endpoint flux, with the refined power  $m = 2r$  and no genericity.
324. **Generic equality of spectral and irreversibility gaps (Conjecture 324).** The irreversibility gap equals the ordinary spectral gap off a measure-zero exceptional set, where the slowest state-space mode is thermodynamically invisible.
325. **Coarse-grained mode-selection (Conjecture 325, resolved true).** A deterministic partition selects the slowest irreversible mode it fails to annihilate rather than renormalizing the late-time exponent, now proved in Proposition 61 by the same quadratic equivalence applied to the block-summed flux.

326. **Hidden-mode exponential separation (Conjecture 326, resolved false).** The proposed separation of the observed and full arrows of time at exactly the gap between the two irreversibility exponents is refuted by a product of two driven three-cycles: the dominant slow mode has exact zeros, so the coarse-to-full ratio returns to one infinitely often and the liminf separation exponent is zero.
327. **Unicyclic strict efficiency monotonicity (Conjecture 327, resolved false).** The proposed strict decrease of the efficiency at every lag on every driven cycle is refuted by the same homogeneous ring, on which the efficiency revives over an interior lag interval, so the near-decomposable increases are genuine rather than a precision artifact.
328. **Sharp unicyclic amplitude cap, resolved false (Conjecture 328).** The proposed universal cap  $\sup_t \gamma L(t)/\sigma \leq \frac{1}{2}$  is resolved false: an explicit near-decomposable seven-state ring reaches 0.643, so the constant  $\frac{1}{2}$  is not universal.
329. **Wrapped-Brownian irreversibility limit (Conjecture 329, resolved true).** Under weak-affinity diffusive scaling the ring efficiency converges to an explicit wrapped-Gaussian relative-entropy functional, now proved in Proposition 65 by a wrapped local central limit theorem and continuity of the divergence.
330. **Ballistic winding-aliasing horizon (Conjecture 330).** At fixed edge affinity endpoint irreversibility is lost on the ballistic winding timescale  $N/|v|$ , not the diffusive mixing timescale.
331. **Diffusive-to-ballistic Péclet crossover (Conjecture 331, resolved false).** The proposed single universal Péclet crossover curve cannot have the ballistic law as its strong-drift limit for every fixed affinity, because that ballistic coefficient depends on the edge affinity; a one-variable crossover survives only in the weak-edge-bias regime.
332. **OU one-relaxation-time peak horizon (Conjecture 332).** For every stable nonequilibrium Ornstein–Uhlenbeck process the dominant lag-KL peak lies within one drift-relaxation time,  $g_A t_* \leq 1$ .
333. **OU integrated-budget bound (Conjecture 333).** The OU integrated lag KL is capped at  $\sigma_{OU}/g_A^2$ , the Gaussian analogue of the finite-chain budget with no factor-of-two margin.
334. **OU antisymmetric-covariance doubling law (Conjecture 334, resolved true).** The long-time OU lag KL decays at twice the slowest time-antisymmetric covariance mode, now proved in Proposition 68, the Gaussian analogue of the doubling law.
335. **DOZZ spectral log-concavity (Conjecture 335).** The modulus of the symmetric DOZZ structure constant is strictly log-concave in the internal Liouville momentum throughout the interior charge chamber  $Q/3 \leq a < Q/2$ , whose two boundaries each reverse the sign.
336. **Brownian-annulus modular log-concavity (Conjecture 336, resolved false).** The proposed strict log-concavity of the exact modulus density in the logarithmic modulus is refuted at large boundary-length ratio, where the curvature turns positive.
337. **Bulk-entropy spectral front (Conjecture 337).** A fixed bulk quantile of conserved entropy migrates outward only single-exponentially, even where pointwise gradients grow double-exponentially.

- 338. **Super-exponential gradient sparsity (Conjecture 338).** Any super-exponential vorticity-gradient growth forces the near-maximal-gradient set to have vanishing area.
- 339. **Vanishing-viscosity global branch selection (Conjecture 339).** Worst-case 2D enstrophy dissipation  $\Gamma$ -converges at fixed rescaled horizon to a unique global extremal profile modulo symmetry.
- 340. **Reeb-graph spectral desert (Conjecture 340).** The vanishing-viscosity spectrum of a cellular Hamiltonian advection-diffusion splits into an  $O(\nu)$  Reeb-graph band and an  $O(\nu^{1/2})$  band, with a forbidden intermediate range.
- 341. **Hörmander-step enhanced-dissipation universality (Conjecture 341).** The enhanced-dissipation exponent of a one-drift equiregular hypoelliptic flow is  $\nu^{m/(m+2)}$  in the bracket depth  $m$  alone; the exponent is Albritton–Beekie–Novack’s, only the universality is claimed.
- 342. **Frenet-frame completion barrier (Conjecture 342).** A Type-I Euler singularity in a coherent vortex tube forces the total curvature or the scale-invariant torsion variation to diverge, conditional on Type-I.
- 343. **Type-I stretching coherence (Conjecture 343).** A Type-I Euler blow-up concentrates the entire weighted stretching-rate distribution on the intense-vorticity core at the singular clock, conditional on Type-I.
- 344. **Weak-Lorentz projected-Lamb criterion (Conjecture 344).** Finiteness of the projected Lamb vector in  $L_t^2 L_x^{3/2, \infty}$  implies Navier–Stokes regularity, reaching the unresolved weak- $L^3$  velocity endpoint.
- 345. **Helicity-defect stretching law (Conjecture 345).** The large-bandwidth maximal vortex-stretching envelope is suppressed by the universal factor  $(1 - \rho^2)^{3/2}$  in the relative helicity  $\rho$ .
- 346. **Diophantine small-divisor rotation law (Conjecture 346).** The rotational-averaging error of 3D Navier–Stokes is fixed by the sharp arithmetic exponent of the minimum inertial-wave triad detuning.
- 347. **Critical-Richardson viscous log anomaly (Conjecture 347).** At  $\text{Ri} = \frac{1}{4}$  the normalized Kelvin vorticity supremum grows like  $\log(1/\nu)$ , singular exactly at the critical Richardson number.
- 348. **Generalized-SQG neck similarity law (Conjecture 348).** A single self-similar gSQG neck has reciprocal distance/curvature exponents  $(T - t)^{\pm 1/\beta}$  with a nonzero dimensionless product.
- 349. **Boussinesq source-core co-location (Conjecture 349).** An inviscid Boussinesq singularity merges the vorticity maximum and its source  $\partial_x \theta$  at the scalar-gradient scale, with compressive-strain alignment, conditional on blow-up.
- 350. **Minimal-admissible vortex collapse core (Conjecture 350).** The smallest admissible collapsing subsystem generically dominates the planar point-vortex collision set; the three-vortex anchor is Aref’s.

- 351. **High-vorticity non-Beltramization barrier (Conjecture 351).** A Type-I Euler singular core cannot asymptotically Beltramize: its local Lamb-alignment defect stays bounded away from zero, conditional on Type-I.
- 352. **Affine LQG geodesic-star spectrum (Conjecture 352).** The  $D_h$ -dimension of the  $k$ -geodesic-star set is the affine spectrum  $2 + \frac{3-k}{2}(d_\gamma - 2)$ , compressing an entire branching spectrum into  $d_\gamma$ .
- 353. **Exponential scale-mixing of the radial metric residual (Conjecture 353).** The scale-stationary radial LQG metric residual is exponentially mixing with an exponential unit-scale envelope, giving Brownian functional and iterated-logarithm limits.
- 354. **Quantum-cone metric-ball shell FCLT (Conjecture 354).** Logarithmic metric-ball shell increments are exponentially mixing with a Gaussian functional limit, a second-order  $\sigma_\gamma B_{\log(1/r)}$  correction to  $\log V(r)$ .
- 355. **Pointwise sfSLE metric exponent (Conjecture 355).** Space-filling-SLE cells have pointwise metric Hölder exponent exactly  $1/d_\gamma$  at almost every time.
- 356. **Stable-CSBP perimeter universality (Conjecture 356).** Every  $\gamma$ -LQG filled metric ball has a canonical scalar boundary length whose reverse-exploration law is a stable CSBP of index  $1 + 2/d_\gamma$ .
- 357. **Linear confluence response (Conjecture 357).** The geodesic branch defect is exactly linear in the endpoint separation, converging to a universal nondegenerate ratio law.
- 358. **Filled-ball outer boundary dimension (Conjecture 358).** The filled LQG metric-ball outer boundary has  $D_h$ -dimension  $d_\gamma/2$ , the shared boundary-length scalar.
- 359. **fBm distance-profile tangent (Conjecture 359).** The local tangent of the boundary distance profile is fractional Brownian motion of Hurst  $2/d_\gamma$ .
- 360. **Sphere spectral MLR (Conjecture 360).** The Liouville sphere block magnitude is monotone in likelihood ratio in the internal momentum as the cross-ratio moves inward; tested in the global-block limit.
- 361. **Four-point spectral log-concavity (Conjecture 361).** The DOZZ-weighted four-point principal-series magnitude is strictly log-concave in the internal momentum in the interior charge chamber; block part tested in the global limit.
- 362. **Torus thermal spectral MLR (Conjecture 362).** Cooling the torus suppresses higher internal Liouville momentum in the strongest likelihood-ratio sense; tested in the global-torus-block limit.
- 363. **Multipoint joint spectral log-concavity (Conjecture 363).** The log-modulus of the multipoint principal-series sewing kernel has a negative-definite Hessian in the internal momenta; block part tested in the global limit.
- 364. **Quantum-cone nesting area-ratio FCLT (Conjecture 364).** Successive nested-CLE LQG area ratios on the quantum cone are exponentially mixing with a Gaussian functional limit.

- 365. **Density-adaptive Lévy exclusion (Conjecture 365).** An exclusion process whose spatial stable index is an endogenous local-density field converges to a conservative variable-order fractional gain–loss equation.
- 366. **Refreshing heavy traps (Conjecture 366).** Refreshing a heavy-trap environment gives a crossover trichotomy governed by a model-dependent renewal symbol, not a universal tempered-stable law.
- 367. **Critical sparse shortcuts (Conjecture 367).** Adding  $O(N)$  macroscopic shortcut edges to SSEP produces an order-one nonlocal mean-field teleportation term  $\kappa\lambda(\bar{\rho} - \rho)$ .
- 368. **Crowding-renormalized renewal exclusion (Conjecture 368).** Successful-only age reset thins the renewal hazard by the vacancy probability, so exclusion changes the temporal universality exponent to  $\beta(1 - \rho)$ .
- 369. **Activity-fatigued conductances (Conjecture 369).** Bonds that fatigue with use give a coupled degenerate diffusion with finite-time loss of ellipticity at  $p < 1$  and a finite integrated mobility budget.
- 370. **Common heavy-tailed boundary clock (Conjecture 370).** A single shared heavy-tailed boundary clock does not self-average, giving a random Stieltjes boundary law rather than a deterministic fractional Robin condition.
- 371. **Codimension-one nonlinear catalyst (Conjecture 371).** A reactive hypersurface at critical rate  $N$  becomes a singular nonlinear transmission law with derivative jump  $-R(\rho|_{\Sigma})$ .
- 372. **Mixed-clock two-species Maxwell–Stefan (Conjecture 372).** Two species with different temporal universality classes couple through exclusion into a hybrid matrix dispersion relation.
- 373. **Finite crossing-budget membrane (Conjecture 373).** A slow bond with a finite budget of successful crossings switches the transmission condition type at an endogenous stopping time set by integrated local activity.
- 374. **Telegraph-modulated membrane (Conjecture 374).** A shared telegraph noise on a critical slow bond survives the conservative limit as a random Robin PDE; the continuum object is Colantoni’s, a high overlap with prior work.
- 375. **Multi-species chiral odd cross-diffusion (Conjecture 375).** Plaquette-biased multi-color exclusion yields a density-changing odd cross-diffusion sector; the scalar odd case is De Carlo–Gabrielli–Gonçalves, a high overlap with prior work.
- 376. **Perturbative self-generated arrest (Conjecture 376).** Occupation-deepened traps freeze a residual nonconstant profile  $\rho_{\infty} = \bar{\rho} + \epsilon e^{\Delta/(a\bar{\rho})} v_0$ , a sharp perturbative arrest law.
- 377. **Critical-percolation exclusion transfer (Conjecture 377).** SSEP on critical triangular percolation inherits the  $\text{CLE}_6$  Brownian heat flow with no extra transport exponent; the one-particle input is Dankovic–Markering–Miller–Yuan, a high overlap with prior work.



378. **Critically mobile slow membrane (Conjecture 378).** A uniquely matched scaling makes a moving slow bond converge to a Brownian membrane with a surviving random singular interface.
379. **Interior two-sided capacitor membrane (Conjecture 379).** An interior storage node of  $O(N)$  capacity mediates delayed conservative transfer between two SSEP bulks; close to the Franco–Franco–Gonçalves finite-reservoir theorem, a high overlap with prior work.
380. **Escort unexplored-mass alternating-derivative hierarchy (Conjecture 380).** Every moment of the escort unexplored mass has alternating derivatives in the squared circulation  $x = \alpha^2$ ; the natural variable is forced by time-reversal even-analyticity; checked only to finite order.
381. **One-maximum principle for the slow spectral edge (Conjecture 381).** The slow real spectral edge has at most one strict interior local maximum along the circulation ray; its global-unimodality form false and retained as a control.
382. **Unicyclic range stochastic acceleration (Conjecture 382).** On a cycle-rank-one support the whole finite-time range distribution is stochastically accelerated by circulation, false on general graphs and retained as a control.
383. **Pure-cycle stationary-mass coverage acceleration (Conjecture 383).** On a simple cycle the stationary mass of the visited set is stochastically accelerated by circulation, strictly more than range acceleration for nonuniform stationary law.
384. **Local spectral–Smith factorization (Conjecture 384).** The real microscopic spectrum and the fixed-odd-prime Smith cokernel of a symmetric integer Wigner matrix become asymptotically independent, product of the GOE edge and the paired-cokernel law.
385. **Random-integer multiplicative universality (Conjecture 385).** The prime factors of a  $\{0, 1\}$  random determinant obey the Erdős–Kac normal law and a Poisson–Dirichlet  $PD(1)$  law for their large factors in the growing-dimension regime.
386. **Plancherel modular-charge/Airy factorization (Conjecture 386).** The  $e$ -core modular charge of a Plancherel partition is asymptotically independent of its Airy edge, Rostam’s Gaussian charge times the GUE edge.
387. **Stable-spectrum saturation of word measures (Conjecture 387).** The chi-square and KL divergence of an  $S_N$  word measure from uniform decay at the exponent of the best stable Fourier mode, so the exponentially large unstable spectrum does not beat it.
388. **Adelic root-process factorization (Conjecture 388).** The real near-unit root process and the fixed-odd-prime root processes of a random Littlewood polynomial converge to a product law.
389. **Double-scaling tropical shadowing (Conjecture 389).** When  $\beta_N \delta / \log N \rightarrow \infty$ , the zeros of a random Ginibre matrix polynomial at a two-face tropical breakpoint match the zeros of the dominant pencil to  $o(N^{-1/2})$ , so the local law is the branch lift of the complex spherical ensemble.

390. **Kazhdan–Lusztig entropy percolation (Conjecture 390).** The entropy  $n^{-1} \log P_{M(G(n,c/n))}(1)$  converges to a deterministic  $H_{\text{KL}}(c)$ , zero below the Erdős–Rényi threshold and positive above it, with the critical value tending to zero.
391. **Khovanov free energy of random braid closures (Conjecture 391).** The normalized logarithm of the Khovanov bigrading partition function of a random braid closure converges almost surely to a deterministic convex free energy.
392. **Two-sided thin-handle Hodge successive-minima equivalence (Conjecture 392).** For every fixed  $k$ , the first  $k$  Jacobian minima and homologically independent short-geodesic lengths are asymptotically paired in the rare-event window, transferring the Poisson tail constants  $1/(4^k k!)$ .
393. **Floer cancellation saturation (Conjecture 393).** For random Heegaard splittings with  $b_1 = 0$  the excess Floer entropy  $\log \text{rank } \widehat{HF} - \log |H_1|$  is subexponential, so the Floer rank exponent equals the homological torsion exponent.
394. **Quenched random Floer barcode entropy (Conjecture 394).** The exponential growth rate of long finite Floer bars along an i.i.d. Hamiltonian cocycle on a closed surface equals its quenched random topological entropy.
395. **Quenched balanced-cut nullity universality (Conjecture 395).** The empirical nullity law over all balanced cuts of one dense random graph state self-averages to the finite-field random-matrix law.
396. **Dependence-sensitive extreme balanced-cut defect (Conjecture 396).** The largest balanced-cut nullity is asymptotic to  $\sqrt{n \log_q 2}$  despite the quadratic overlap of the cut matrices.
397. **Poisson law for random graph-state distance (Conjecture 397).** Low-weight stabilizers form a Poisson process at the distance threshold, giving an exact extreme law beyond the known first-order constant.
398. **Typical rigidity of labelled cut rank (Conjecture 398).** A dense random graph is, with high probability, determined up to labelled local complementation by its complete bipartite entanglement function.
399. **Positive-rank-tail log-concavity (Conjecture 399).** Cut-rank multiplicities are log-concave from rank two onward; an exact connected 13-vertex certificate shows the rank-one inequality is false.
400. **Full log-concavity for percolated forests (Conjecture 400).** Edge subsets of a forest, counted by their maximum-matching number, form a log-concave sequence.
401. **Hurwitz stability of the percolated-matching polynomial (Conjecture 401).** The matching-number enumerator of all edge subsets of a forest has every zero in the open left half-plane.
402. **Path–star stochastic extremality under bond noise (Conjecture 402).** After independent edge deletion, the star and path stochastically minimize and maximize matching number among trees.

403. **Fixed-leaf broom minimum (Conjecture 403).** At a fixed number of leaves, the subdivided-star broom uniquely minimizes the percolated matching number in stochastic order.
404. **Typical reconstruction from the entanglement enumerator (Conjecture 404).** A uniform labelled tree is typically determined by  $Z_T(x, y)$ , despite a unique collision class at order fifteen.
405. **Random-regular rank-width density (Conjecture 405).** Every fixed-degree random regular graph has a deterministic positive rank-width density.
406. **Cubic branch-width density and kernel transfer (Conjecture 406).** Cubic branch-width has a density that transfers through sparse-kernel repairs and subdivisions to the exact barely-supercritical rank-width constant.
407. **Off-critical grid rank-width surface tension (Conjecture 407).** Bond-percolated square grids have a deterministic linear rank-width constant away from criticality.
408. **Local-complementation orbit entropy saturation (Conjecture 408).** A dense random graph-state orbit reaches the universal  $3^n$  upper bound at exponential scale.
409. **Sparse Erdős–Rényi rank-width density curve (Conjecture 409).** The normalized rank-width has a deterministic curve, zero through the giant-component threshold and positive above it.
410. **Prime continuant law (Conjecture 410).** For almost every real number the number of prime convergent denominators among the first  $N$  is  $(2 \log 2 + o(1)) \log N$ , with a Poisson limit at logarithmic scale; the constant is  $\zeta(2)$  times the reciprocal Lévy constant.
411. **Prime-pair correlations along the convergents (Conjecture 411).** Primality at consecutive indices, and of numerator and denominator, is negatively correlated by exactly  $6/\pi^2$ ; the lag-two factor is a Gauss–Kuzmin product; a typical number has finitely many doubly prime convergents.
412. **Squarefree and totient densities of continuants (Conjecture 412).** Along a typical orbit the squarefree density and the mean of  $\varphi(q_n)/q_n$  both equal  $\prod_p(1 - 1/(p(p+1))) = 0.7044\dots$ , the projective-line analogue of  $6/\pi^2$ .
413. **Erdős–Kac law along the convergents (Conjecture 413).** The number of prime divisors of  $q_n(x)$  is Gaussian with mean and variance  $\log n$  for almost every  $x$ .
414. **Dimension of the prime-continuant sets (Conjecture 414).** Numbers all of whose continuants are prime form a set of Hausdorff dimension one half; numbers with no prime continuant form a set of full dimension.
415. **Digit sets in arithmetic progressions (Conjecture 415).** The dimension of the set with all partial quotients in  $a + m\mathbb{N}_0$  is  $\frac{1}{2} + \frac{1}{2m} - \frac{\log m}{2m^2} - \frac{\psi_\Gamma(a/m)}{2m^2} + O(m^{-3} \log^2 m)$ , with a digamma term separating residue classes of equal density.
416. **Continuant avoidance dimension (Conjecture 416).** The set of numbers none of whose continuants is divisible by  $p$  has dimension  $1 - (6 \log 2/\pi^2)/p + O(p^{-2})$ .

417. **Unit law for the Legendre half-box games (Conjecture 417).** The indifference systems of the Legendre Hankel and Toeplitz half boxes have values  $\mp y/x$  for a unit  $x + y\sqrt{p}$  of norm  $-1$ ; whenever the unit strategy is a probability vector it is optimal, so the game value is a fundamental-unit ratio, proved exactly for forty-one prime instances.
418. **Least odd residue law in the complementary classes (Conjecture 418).** In the two residue classes where the unit strategy fails,  $\sqrt{p}$  times the value is governed by the least odd prime quadratic residue and does not converge.
419. **Brownian shift games (Conjecture 419).** Random Toeplitz and Hankel games, which share their value, have values of order  $m^{-1/2}$  with a universal Brownian limit, one order above full random games.
420. **Hessian big monodromy over finite fields (Conjecture 420).** The point counts of a random quartic and of its Hessian sextic are asymptotically independent symplectic traces, jointly independent of the flex count.
421. **Sato–Tate independence of a curve and its Hessian (Conjecture 421).** For a generic quartic over  $\mathbb{Q}$  the Frobenius traces of the curve and of its Hessian are jointly Sato–Tate equidistributed.
422. **Tangency law and dual-curve point counts (Conjecture 422).** Rational bitangents have rational or conjugate tangency points with independent fair coins, so  $\#C^\vee(\mathbb{F}_q) - \#C(\mathbb{F}_q)$  has an explicit symmetric limit law of variance one.
423. **Positivity of the quarter-box Legendre games (Conjecture 423).** For  $p \equiv 3 \pmod{4}$  the quarter-box Hankel Legendre game always favours the row player, a class-number bias at scale  $m^{-1/2}$ .
424. **Vanishing law for the half box in the class 3 modulo 4 (Conjecture 424).** For  $p \equiv 3 \pmod{4}$  the half-box value is of order  $p^{-3/2}$  and nonpositive when 2 is a residue, while its sign oscillates when 2 is a non-residue.
425. **Finite-rank stable forking (Conjecture 425).** Every supersimple theory of finite  $SU$ -rank has stable forking; the counterexample has infinite rank.
426. **Rank-one stable forking (Conjecture 426).** In a simple theory every forking type of  $SU$ -rank one contains a stable forking formula, and rank one is sharp.
427. **Stable forking for countably categorical simple theories (Conjecture 427).** Countable categoricity restores stable witnesses for forking.
428. **The counterexample is two-dependent (Conjecture 428).** A quantum module is  $k$ -dependent exactly when its graph is, so the stable forking conjecture fails in a 2-dependent supersimple theory.
429. **Transfer of dividing lines from graphs to quantum modules (Conjecture 429).** The module over the quantum algebra of a graph is simple, supersimple, NSOP<sub>1</sub> or stable exactly when the graph is, with  $SU$ -rank  $\omega$  over a rank-one graph.
430. **Stable forking dividing line for quantum modules (Conjecture 430).** The quantum module of a simple graph has stable forking if and only if the graph is stable.

431. **Global linear disjointness is forking in every quantum module (Conjecture 431).** Forking independence is linear disjointness of spans over the full division ring together with independence of the vertex parts of the algebraic closures, with quantifier elimination up to closure.
432. **Where one-basedness fails in the counterexample (Conjecture 432).** The counterexample is not one-based, but canonical bases become bounded over the realisation once finitely many hidden vertices are named, and the theory is CM-trivial.
433. **Failure of stable forking over models (Conjecture 433).** The type of a vector with two hidden vertices over a model and one vector forks without a stable witness, and the weak stable Kim-forking property fails in every quantum module over an unstable NSOP<sub>1</sub> graph.
434. **Decidability and effective forking (Conjecture 434).** The theories over the random and random bipartite graphs are decidable with decidable forking and computable normal forms for types.
435. **The scalar division rings are not simple (Conjecture 435).** The division ring of fractions of the quantum algebra of a graph with an adjacency form of infinite rank modulo two is never simple, in positive characteristic by a theorem of Milliet.
436. **Stable formulas and hidden vertices (Conjecture 436).** One hidden vertex leaves a formula stable, two odd hidden exponents give the order property, and the stable formulas are the Boolean combinations of one-vertex formulas.
437. **Nonforking extensions do not restore stable witnesses (Conjecture 437).** A simple theory has stable forking exactly when every forking type has a nonforking extension with a stable forking formula; in the counterexample no such extension exists.
438. **Definable groups in quantum modules (Conjecture 438).** Every interpretable group is abelian-by-finite and isogenous to a subquotient of a power of the vector sort.
439. **Countable models of the counterexample (Conjecture 439).** The structures of every dimension are elementarily equivalent and exhaust the countable models, giving exactly  $\aleph_0$  countable models of a theory whose vertex sort is countably categorical.

### Abstract

We state four hundred thirty-nine conjectures in thirty parts. The twenty-five conjectures of Part I are each derived from the local-global random model of the primes: Cramér’s model corrected by Hardy–Littlewood singular series, with Bateman–Horn as the organizing framework and Borel–Cantelli accounting for sparse events. Each statement is calibrated to the strongest form the heuristic supports. Every computable constant is computed from its definition, with local admissibility asserted programmatically; every statement was tested against exact counts (probable-prime counts, so labelled, where the integers involved exceed the deterministic certification range), with success measured by the *shape* of agreement (predicted constant, square-root residuals, no drift) rather than by the size of the range searched. The claimed content includes: a prime-power contamination calculus for pattern races—twin races modulo 5 and 8 driven by the pairs  $(q^2 - 2, q^2)$ , cousin, sexy-pair, and triplet predictions derived from the calculus before testing, a Goldbach lane race with a verified internal null lane, a Stern-representation lane race whose null classes are provable by direct congruence, and an

algebraic null control; the measured sub-diffusivity of balanced races (universal negative step correlations tied to the Lenke Oliver–Soundararajan repulsion); canonical ordering deficits for least primes in progressions and for least Goldbach summands; derived covariance kernels for moving-window residual fields, with the diagonal deficit reduced to a pinned singular-series average whose growth already exceeds the single-prime Montgomery–Soundararajan term; distribution laws for the constants of the quadratic-pair and cubic-shift families, with exact mean-value lemmas; an entanglement-aware local-global law for  $n^2 + 2^n$  with an exact-rational CRT factorization; a boundary trichotomy for polynomial ladders; a multibase subtorus law for Fermat quotients; and singular-series waiting-time refinements for first prime gaps. Every statement was checked against the literature by an independent search, with novelty labelled conservatively: statements already known in essence are attributed and kept *inside* the conjecture whose claimed content they calibrate, since those benchmarks are what make the derived constants credible. Every falsifiable-by-instance statement was then stress-tested computationally well beyond its original range, and every constant was recomputed independently from its definition. Part II adds twenty-five structural conjectures in five programmes—connected prime-pattern fields, arithmetic first-arrival fields and class groups, finite logarithms and algebraic tori, arboreal arithmetic dynamics, and adelic factorization processes—each introducing a canonical operator, process, invariant, classification, or phase boundary, with its mechanism, nearest literature boundary, first decisive theorem, and failure mode. Part III adds six conjectures across fields: a relative pretentious inverse principle for polynomial arguments, a threshold purification law for monotone vector paths, an entropy–dimension law for random-free graphons, a Hessian principle for microcanonical graph phases, a sharp stability law for entropy idempotence on finite abelian groups, and a convolution–entropy rigidity for stationary processes. Part IV (Conjectures 57–76) adds twenty conjectures on a single further object, Rossmann’s class-counting polynomial of graphical Lie algebras: rigidity and reconstruction laws for the polynomial and its binary specialization, recovery of the subgraph-component polynomial and of domination invariants, a strict log-concavity ladder with proven failure boundaries, and extremal laws for trees, each tested exhaustively at small order, on adversarial random holdouts, and at every known collision class of the polynomial. Two of the twenty, the pseudoforest one-field rigidity claim and the interval-support claim, have since been refuted, by an explicit seven-vertex pair and a connected eight-vertex graph, and are recorded as resolved false with their counterexamples. Part V (Conjectures 77–96) adds a second algebraic family, the nilpotent pattern Lie algebras of finite posets, and studies two rank-weight enumerators of the bracket tensor, the adjoint and the Kirillov enumerators: determination of structural and coadjoint invariants by the adjoint enumerator, characteristic and field-monotonicity behaviour, a positive unimodal centre-one incidence quotient, and reverse determination, saturation, and a breadth bound for Hasse-forest posets, fifteen of them supported throughout the tested corpus, three stated in greater generality than the computations reach, and two since refuted by explicit seven-point posets and recorded as resolved false, all reverified on an independently generated poset corpus over a proved calibration identity. Part VI (Conjectures 97–116) returns to the local–global aesthetic of Part I on new objects: prime compasses at polynomial centres matched to local-sieve hazards, order patterns of the radical ratio with every fixed conditional density now proved to exist and nonuniformity open beyond length four, order constants of abundancy and totient ratios, unbounded prime-factor shocks in combinatorial sequences resting on seven proved propositions, and a local–global Euler sum for the prime-factor anatomy of prime-gap boundaries, one of the twenty since resolved false by a theorem of G. Martin, the three two-sided shock laws and the sign-balance law since proved, and six stated as measured bands rather than derived constants. Part VII (Conjectures 117–136) crosses into algebraic topology on four canonical geometric families: Euler indices of Calabi–Yau complete intersections ending in an exact dimension-wise gcd law, middle Betti numbers of Brieskorn–Pham links under degree caps and common scaling, Hochster strands of flag moment-angle complexes with interval, log-concavity, and Hurwitz-stability shape laws, and torus mapping



tori with signed monodromy over proved signed-orbit and normalised-merge calibrations, two of the twenty asserting limits beyond the tested range and two close to prior work, the quadric ratio law and normalised merge law since proved in full, the compression-exponent law since resolved false with true exponent two, and the common-scale law since proved by an Ehrhart model. Part VIII (Conjectures 137–156) continues into enumerative topology with five shape-law programmes: Euler indices of anticanonical hypersurfaces in products of projective spaces under dominance and refinement, Betti profiles of Hilbert schemes over a proved pair of exact moment identities, twisted-sector age histograms of cyclic Calabi–Yau quotients, exterior torsion-entropy profiles of hyperbolic toral automorphisms, and bigraded Hochster tables of simple binary matroids, with two limit clauses and three random-model statements recorded and one attribution note recorded, two statements since proved by convexity, and four since refuted by exact graphic certificates. Part IX (Conjectures 157–176) passes to spectra and filtrations of combinatorial geometries: real-rooted  $h$ -polynomials and Mahler orders for graph associahedra, one-crest laws for age-filtered McKay complexes, temperature and shape laws for lens-space sine-torsion sectors over a proved pair of log-sine moment identities, and pseudodeterminant entropy laws for simplicial spheres, with one iterated-limit and one random-model statement recorded and five attribution notes recorded. Part X (Conjectures 177–196) enters string-geometric index theory: saturation, threshold, and torsion-staircase laws for the cohomology of integral three-form flux on tori over a proved coefficient-and-parity calibration, and endpoint, monotonicity, real-rootedness, Mahler, ratio, and gcd laws for the signed  $\chi_y$ -profiles and signatures of Calabi–Yau complete intersections along the merge order, with four random-model statements, one statement beyond the computed range, three numerical-evidence caveats, and three attribution notes recorded, the quadric ratio law and the central-binomial law since proved in full, and the prime-torsion staircase proved for every prime but its elementary-abelian clause resolved false by a summand  $\mathbb{Z}/4$  at dimension fifteen. Part XI (Conjectures 197–216) pairs a spectral programme with an arithmetic one: all-order Rényi entropy monotonicity and extremal laws for the Laplacian density matrices of trees and unicyclic graphs and for the normalized distance signless Laplacian, over a proved degree-and-majorization calibration, and complete cross-prime decoupling laws, Gaussian, local, modular, large-deviation, and cumulant, for the carry fields of central binomial coefficients, over a proved single-base carry-chain law, with the spectral scans recorded as floating-point checks, the nine carry statements limit laws, two attribution notes recorded, and the boundary controls of the entropy windows retained in place, one of the twenty, the bipartite path minimum, since resolved false by an explicit fourteen-vertex tree. Part XII (Conjectures 217–236) runs four programmes at the interface of spectra and information: factorial critical-moment exponents and tail large deviations for iterated barycentric subdivision beyond weak universality, the extremal theory of a new spanning-tree total-correlation invariant with a mesoscopic core-periphery shape principle, the inverse theory of the prime-field entropy-doubling floor through localization, rigidity, and a heat-kernel wrap crossover, and the identification of random-free graphon entropy with the information dimension of the latent row space, over a proved block-additivity and cycle law and a proved threshold-graphon entropy bound, with all scans recorded as floating-point checks, two attribution notes and one evidence gap recorded, and four boundary controls retained in place, one of the twenty, the heat-kernel wrap crossover, since resolved false by an explicit jump family. Part XIII (Conjectures 237–255) extends the string-geometric programmes in five directions: odd-degree flux saturation and thresholds with complete torsion shape laws over the staircase of Part X, strict quantum monotonicity of mirror maps, Gopakumar–Vafa invariants, conifold constants, and normalized Yukawa couplings along the merge order, wall-counting, sharp-constant, and tame-atlas laws for Bridgeland stability, polynomial condition numbers for Ricci-flat metrics, balanced and Hermitian–Yang–Mills convergence, spectral gaps, and certified Yukawa couplings in an algebraic discriminant gauge, and special-Lagrangian realization with LMHS-indexed Laplace towers at infinite distance, over a proved merge-and-Chern calibration, with seven statements, three attribution notes

recorded including one uncounted remark, the raw-instanton reversal retained as an adversarial control, and the post-onset persistence law since proved in full. Part XIV (Conjectures 256–270) adds three programmes on classical projective invariants: a discrete-convexity package for the degree-normalized primitive middle cohomology of complete intersections, controlled-defect laws for the two Betti rows of Segre–Veronese embeddings of  $\mathbf{P}^1 \times \mathbf{P}^1$ , and four monotone-dispersion laws carrying hypersurface Hodge profiles toward the Eulerian descent law, over three proved calibrations, the curve, antidiagonal, and surface propositions, with the syzygy laws resting on a thirteen-table corpus. Part XV (Conjectures 271–285) passes to probability, on the uniform self-avoiding walk: strict parity log-concavity for the walk counts and for every radial-moment sum against strict log-convexity for the bridge counts, likelihood-ratio growth of bridge height and coordinate span with coordinatewise monotone couplings of the absolute endpoint and span vectors, the endpoint coupling strictly implying the open mean-square-displacement monotonicity, four quadrant-dependence signs for spans, radius, atmosphere, and contacts, a partition-curvature trichotomy across the collapse transition whose three constants are the established predictions, and strict dimension monotonicity of nonbacktracking survival and of every positive radial moment, over three proved calibrations, the extreme-order, top-corner, and survival propositions, with one attribution note, one consolidation note, and four negative controls retained in place. Part XVI (Conjectures 286–294) passes to non-equilibrium dynamics, asking how a finite budget of irreversibility is spectrally allocated: a strict amplification law for the standardized winding-current skewness under reversible shortcuts at fixed activity, the complete conjectured architecture of the budgeted-gap frontier of linear diffusions with fixed Gaussian steady state, ordered mode recruitment, convex arcs with marginal-gain resets, exact two-mode support, a rank-two oscillation extremizer, and a pairwise formula for the thermodynamically cheapest exceptional point, the transfer of that architecture to coherently accelerated detailed-balance Lindblad dynamics with an exact Lie-algebraic saturation criterion, and an eigenvector-geometric law for anomalous relaxation, the collapse of robust Mpemba preparation basins in inverse proportion to the biorthogonal condition number at a Liouvillian exceptional point, over three proved calibrations, the pair, envelope, and basin propositions, with one attribution note and four optimizer-level negative findings retained as controls. Part XVII (Conjectures 295–307) passes to extreme-value theory, asking how much of the dependence structure of rare events can be recovered from partial, thinned, or redesigned observation: an isoperimetric recruitment staircase with a proved symmetry-breaking boundary, a pairwise closure law, an elasticity-and-monotonicity law with its dimension-free sparsification consequence, and a positive-curvature law for the Hüsler–Reiss extremal coefficient as a functional on variograms, a thinning-spectroscopy transport law identifying the limiting cluster-size distribution and an occupancy-tomography law placing cluster supports strictly between congruence and homometry, bottleneck-consistency and leakage-trichotomy laws for the persistent homology of extreme directions, a stability barrier and a parameter-count identifiability threshold for fractional tail tomography, and a Poissonization law, a factorial-cumulant design ladder, and an adaptation cusp for randomized block maxima, over four proved calibrations, the tilting, thinning, moment, and design propositions, with two attribution notes and three refutations retained as controls: the simplex isoperimetry claim for four or more variables, the modular-reconstruction claim with Bloom’s homometric pair, and counting-boundary uniqueness for sparse tomography. Part XVIII (Conjectures 308–319) turns to topological data analysis, asking which parts of a persistence computation are laws rather than artifacts: a circle isoperimetry for multiplicative Vietoris–Rips persistence, a saturation law identifying the stability factor two as exactly attained under greedy sampling, and a two-sided Johnson–Lindenstrauss threshold for loop ratios, prevalence-shrinkage and grouping-stability laws making Wasserstein means of diagrams quantitatively Euclidean, survival-tomography and dither laws transferring a proved phase calculus for randomized Mapper to general tame Reeb graphs, a finite-direction stability and random-slope observability law for rectangle-decomposable modules over a proved one-direction completeness theorem, a restricted isometry

for random Fourier persistence sketches, a Whitney-count evaluation law for persistence surfaces, an aggregate-landscape moment-determinacy law, and a near-quadratic average-case law for vineyard events, over proved phase, tomography, and moment calibrations, with three attribution notes and three negative findings retained as controls: unrestricted planar isoperimetry refuted by a six-point configuration, the three-direction completeness claim sharpened to one direction, and the majority-threshold barycenter claim refuted by the flat grouping algebra. Part XIX (Conjectures 320–334) turns to the temporal irreversibility of nonequilibrium dynamics, reading a temporal sampling efficiency off the lag Kullback–Leibler divergence of the two-time endpoint law: monotone efficiency loss, spectral-horizon and integrated-budget bounds, a doubled irreversibility exponent with its generic coincidence with the spectral gap and its coarse-grained mode selection, a wrapped-Brownian scaling limit and a Péclet winding-aliasing crossover on driven cycles, and peak, budget, and doubling laws for Ornstein–Uhlenbeck processes, over a proved layer of endpoint bounds and the exact Gaussian lag divergence, with four long-time and three scaling statements recorded, the efficiency-monotonicity and unicyclic strict-monotonicity laws since resolved false by a homogeneous ring, the sharp unicyclic amplitude cap since resolved false by a near-decomposable ring, and the two Ornstein–Uhlenbeck spectral-abscissa bounds, the peak horizon and the integrated budget, since resolved false by a nonnormal Jordan-block drift. Part XX (Conjectures 335–336) closes with two log-concavity laws for exactly solved objects in random conformal geometry: strict log-concavity of the DOZZ Liouville structure constant in the internal momentum throughout an interior charge chamber whose two boundaries each reverse the sign, and strict log-concavity of the exact Brownian-annulus modulus density in the logarithmic modulus, over a proved layer reducing each curvature to the part of its closed form that carries the sign, both exactly checkable on grids, the DOZZ law requiring continuous curvature control for a proof and the annulus law since resolved false at large boundary ratio. Part XXI (Conjectures 337–351) turns to theoretical fluid dynamics with fifteen sharp singular-behaviour laws: bulk-quantile and gradient-sparsity constraints on two-dimensional Euler small-scale creation and a vanishing-viscosity branch-selection law, a two-band spectral desert and a bracket-depth universality for enhanced dissipation, Frenet-torsion and stretching-coherence barriers and a weak-Lorentz Lamb regularity criterion for three-dimensional Euler and Navier–Stokes, a helicity-defect envelope and a Diophantine rotation law for band-limited and rotating flows, a critical-Richardson viscous log anomaly, a generalized-SQG neck similarity law and a Boussinesq source-core co-location for active scalars, and a minimal-core point-vortex collapse principle with a non-Beltramization barrier, every statement and, with attributions where a sharp exponent or anchor is due to earlier work. Part XXII (Conjectures 352–364) completes the random-conformal-geometry bundle begun in Part XX with thirteen laws in two registers: the metric geometry of Liouville quantum gravity—an affine geodesic-star dimension spectrum, exponential scale-mixing of the radial metric residual, a metric-ball shell functional central limit theorem, a pointwise space-filling-SLE exponent, a stable-branching perimeter universality, a linear confluence response, the filled-ball outer-boundary dimension  $d_\gamma/2$ , and a fractional-Brownian distance-profile tangent—and the spectral convexity of Liouville conformal field theory—monotone likelihood ratio and log-concavity of sphere, four-point, torus, and multipoint principal-series data—together with a conformal-loop-ensemble quantum-area nesting functional central limit theorem, over a proved scaling triangle relating the boundary-length, roughness, and branching exponents, the metric-geometry laws and the field-theory laws recorded where the descendant contribution is tested only in the global-block limit. Part XXIII (Conjectures 365–379) proposes hydrodynamic scaling limits for fifteen interacting particle systems carrying a new structural ingredient: an endogenous local jump index, refreshed or fatigued environments, non-Markovian renewal clocks, singular reaction surfaces, finite crossing budgets, shared common noise, odd multi-species transport, and moving or storing membranes, giving variable-order fractional gain–loss equations, a refreshed-trap crossover trichotomy, a nonlocal teleportation term, a crowding-renormalized temporal exponent, degenerate fatigue diffusion, random Stieltjes and endogenous-stopping-

time boundary laws, hybrid Maxwell–Stefan dispersion, telegraph-random and odd-cross-diffusion hydrodynamics, a perturbative self-arrest residual, a critical-percolation transfer law, and moving and capacitor membranes, every statement a scaling limit, with four high overlaps with closely related 2025–2026 results. Part XXIV (Conjectures 380–383) closes with four fixed-traffic circulation laws for finite-state nonequilibrium Markov chains, in which the stationary law, undirected traffic, and escape rates are held fixed while an irreversible circulation is increased: an alternating-derivative hierarchy in the squared circulation for every moment of the escort unexplored mass, a one-maximum principle for the slow real spectral edge, and stochastic acceleration of the finite-time range on unicyclic supports and of the visited stationary mass on cycles, over a proved layer of endpoint spectral bounds and exact time-reversal even-analyticity that also refutes, by an explicit high-precision counterexample, the natural stronger companion of whole-ray fast-edge monotonicity. Part XXV (Conjectures 384–388) adds five cross-shadow universality laws, each asserting that two limiting observables read off the same discrete randomness decouple by probing different resolution, frequency, or representation sectors: the real spectral edge and the fixed-odd-prime Smith cokernel of a symmetric integer matrix, Erdős–Kac and Poisson–Dirichlet prime-factor laws for bounded-alphabet random determinants, the modular charge and the Airy edge of a Plancherel partition, saturation of a symmetric-group word measure’s complete Fourier spectrum by its best stable mode, and adelic factorization of the real and fixed-prime root processes of a Littlewood polynomial, over a proved layer of anchor identities, every marginal a mature theorem and only the joint law new, each an asymptotic-independence statement whose finite-size diagnostics can only exclude gross obstructions. Part XXVI (Conjectures 389–393) closes with five entropy and rare-event laws for deep invariants on canonical random ensembles: a double-scaling tropical matching of random matrix-polynomial zeros to a two-face spherical-ensemble pencil, the coincidence of the transition to extensive Kazhdan–Lusztig complexity of the random graphic matroid with the Erdős–Rényi giant-cycle threshold, a deterministic convex Khovanov free energy for random braid closures, a two-sided thin-handle equivalence transferring the Weil–Petersson Poisson constants  $1/(4^k k!)$  to every fixed Jacobian successive minimum, and Floer cancellation saturation identifying the Heegaard Floer rank exponent of random Heegaard splittings with the homological torsion exponent, over a proved layer of tropical, block-multiplicativity, convexity, Poisson-tail, and Floer lower-bound anchors, with the recorded gap that the decategorified diagnostics of the two homological laws do not reach their cancellation mechanisms. Part XXVII (Conjecture 394) adds the random Floer-barcode entropy law, over the deterministic surface entropy theorem and an exact lazy Hamiltonian-walk calibration, and strengthens Conjecture 392 to every fixed successive minimum. Part XXVIII (Conjectures 395–409) adds fifteen cut-rank laws for graph states, percolated forests, random regular and sparse graphs, and local-complementation orbits, with exact counterexamples recorded where stronger forms fail. Part XXIX (Conjectures 410–424) adds fifteen statements at the meeting points of number theory, measure theory, game theory and algebraic geometry: prime and squarefree laws for the continuants of a typical real number with the constants  $2 \log 2$ ,  $6/\pi^2$  and  $\prod_p (1 - 1/(p(p+1)))$ , Hausdorff-dimension expansions for digit sets in progressions and for continuant residue avoidance, an exact identification of the values of Legendre-symbol half-box games with coordinate ratios of fundamental units of real quadratic fields together with the least-residue, quarter-box and vanishing laws that accompany it, a Brownian limit for random Toeplitz and Hankel games, and monodromy, Sato–Tate and dual-curve laws for plane quartics and their Hessians. Part XXX (Conjectures 425–439) responds to the refutation of the stable forking conjecture by a module over the division ring of fractions of the quantum algebra of the random bipartite graph: it states the surviving finite-rank, rank-one and countably categorical forms of the conjecture, the two-dependence of the counterexample, a transfer of dividing lines from a graph to its quantum module, the stable forking dichotomy and the forking calculus of these modules, the locus of the failure of one-basedness, failure over models, decidability, non-simplicity of the scalar division rings, the stable formulas, the failure of nonforking extensions to restore stable

witnesses, definable groups, and the countable models, over a proved layer on the quantum algebras of graphs which isolates the mechanism of the counterexample: one hidden vertex leaves a formula stable, two transport the edge relation.

## Contents

|                                                                    |     |
|--------------------------------------------------------------------|-----|
| Summary sheet                                                      | 1   |
| 1 Introduction                                                     | 48  |
| 2 Notation                                                         | 53  |
| I Conjectures from a local–global random model                     | 55  |
| 3 Principal conjectures                                            | 55  |
| 4 An open question                                                 | 68  |
| 5 Family laws and second-order refinements                         | 68  |
| 6 Instances and structural companions                              | 78  |
| II Structural conjectures in five programmes                       | 84  |
| 7 Programme I: connected prime-pattern fields                      | 84  |
| 8 Programme II: arithmetic first-arrival fields and class groups   | 88  |
| 9 Programme III: finite logarithms, regulators, and algebraic tori | 92  |
| 10 Programme IV: arboreal arithmetic dynamics                      | 95  |
| 11 Programme V: adelic factorization and sieve flow                | 99  |
| III Six conjectures across fields                                  | 102 |
| 12 Relative polynomial pretentiousness                             | 103 |
| 13 Monotone threshold purification of vector paths                 | 105 |
| 14 Entropy dimension of random-free graphons                       | 106 |
| 15 A Hessian principle for isolated microcanonical graph phases    | 108 |
| 16 Sharp stability of entropy idempotence on finite abelian groups | 109 |
| 17 Stationary convolution-entropy rigidity                         | 111 |

|      |                                                          |     |
|------|----------------------------------------------------------|-----|
| IV   | The class-counting polynomial of graphical Lie algebras  | 112 |
| 18   | Rigidity and reconstruction                              | 114 |
| 19   | Log-concavity and support                                | 117 |
| 20   | Tree extremality and rigidity                            | 119 |
| V    | Pattern Lie algebras of finite posets                    | 122 |
| 21   | Determination by the adjoint enumerator                  | 123 |
| 22   | Characteristic and field behaviour                       | 125 |
| 23   | Centre-one incidence quotient                            | 127 |
| 24   | Hasse-forest and unitriangular structure                 | 128 |
| VI   | Local structure of primes and arithmetic functions       | 130 |
| 25   | Prime compasses at polynomial centres                    | 133 |
| 26   | Order patterns of the radical ratio                      | 135 |
| 27   | Order laws for multiplicative ratios                     | 136 |
| 28   | Prime-factor shocks in combinatorial sequences           | 137 |
| 29   | The arithmetic anatomy of prime-gap boundaries           | 142 |
| VII  | Topological invariants of geometric families             | 142 |
| 30   | Euler indices of Calabi–Yau complete intersections       | 144 |
| 31   | Middle Betti numbers of Brieskorn–Pham links             | 147 |
| 32   | Hochster strands of flag moment-angle complexes          | 149 |
| 33   | Torus mapping tori with signed monodromy                 | 151 |
| VIII | Shape laws for enumerative geometric invariants          | 153 |
| 34   | Euler indices of anticanonical hypersurfaces in products | 155 |
| 35   | Betti profiles of Hilbert schemes of surfaces            | 156 |
| 36   | Twisted-sector ages of cyclic quotients                  | 157 |



|                                                                               |     |
|-------------------------------------------------------------------------------|-----|
| 37 Exterior torsion entropy of toral automorphisms                            | 158 |
| 38 Matroidal Hochster tables                                                  | 160 |
| IX Spectra and filtrations of combinatorial geometries                        | 162 |
| 39 Toric spectra of graph associahedra                                        | 163 |
| 40 Age-filtered McKay complexes                                               | 165 |
| 41 Sine-torsion sectors of cyclic quotients                                   | 166 |
| 42 Boundary-determinant entropy of simplicial spheres                         | 168 |
| X Flux cohomology and protected index laws                                    | 169 |
| 43 Integral flux complexes on tori                                            | 170 |
| 44 Protected profiles of complete intersections                               | 177 |
| 45 Signature laws for complete intersections                                  | 179 |
| XI Spectral entropies and multibase carry fields                              | 183 |
| 46 Rényi entropies of the Laplacian density matrix                            | 185 |
| 47 Entropy of the distance signless Laplacian                                 | 187 |
| 48 Multibase carry fields                                                     | 189 |
| XII Subdivision spectra, spanning-tree correlation, and information dimension | 192 |
| 49 Spectral tails of barycentric subdivision                                  | 194 |
| 50 Total correlation of uniform spanning trees                                | 196 |
| 51 Prime-cyclic entropy power and the wrap transition                         | 198 |
| 52 Information dimension of random-free graphons                              | 200 |
| XIII Flux torsion, quantum merge monotonicity, and effective string geometry  | 202 |
| 53 Odd flux and torsion shapes                                                | 204 |

|                                                                                 |         |
|---------------------------------------------------------------------------------|---------|
| 54 Quantum monotonicity along the merge order                                   | 206     |
| 55 Stability walls and sharp constants                                          | 207     |
| 56 Effective geometry of Calabi–Yau families                                    | 208     |
| 57 Degenerations and towers                                                     | 210     |
| <br>XIV Convexity, defect, and dispersion laws for projective invariants        | <br>211 |
| 58 Normalized primitive cohomology of complete intersections                    | 214     |
| 59 Controlled defects in Segre–Veronese syzygies                                | 216     |
| 60 Monotone dispersion of hypersurface Hodge profiles                           | 217     |
| <br>XV Stochastic order, curvature, and dependence laws for self-avoiding walks | <br>219 |
| 61 Curvature laws for counts, radial moments, and bridges                       | 222     |
| 62 Monotone transport in the length                                             | 223     |
| 63 Dependence laws at fixed length                                              | 225     |
| 64 Phase and dimension laws                                                     | 226     |
| <br>XVI Budgeted irreversibility and the spectral geometry of relaxation        | <br>227 |
| 65 A current-fluctuation law for equilibrium shortcuts                          | 230     |
| 66 The finite-budget frontier of linear diffusions                              | 231     |
| 67 Coherent acceleration of dissipative quantum dynamics                        | 233     |
| 68 The geometry of anomalous relaxation                                         | 234     |
| <br>XVII Inverse problems and design laws for multivariate extremes             | <br>234 |
| 69 The geometry of Hüsler–Reiss dependence                                      | 238     |
| 70 Cluster inverse problems under thinning                                      | 240     |
| 71 The topology of extreme directions                                           | 242     |
| 72 Tail tomography and its stability                                            | 244     |

|                                                                                |     |
|--------------------------------------------------------------------------------|-----|
| 73 Randomized block designs                                                    | 245 |
| XVIII Extremal, randomized, and inverse laws for persistent topology           | 248 |
| 74 Extremal and sampling laws for Vietoris–Rips persistence                    | 251 |
| 75 The statistical geometry of diagram means                                   | 253 |
| 76 A phase calculus for Mapper                                                 | 254 |
| 77 Rectangle tomography                                                        | 255 |
| 78 Finite summaries and dynamic complexity                                     | 256 |
| XIX Temporal irreversibility and spectral selection in nonequilibrium dynamics | 258 |
| 79 Temporal irreversibility and spectral selection                             | 260 |
| 80 Unicyclic networks and winding aliasing                                     | 263 |
| 81 Nonequilibrium Ornstein–Uhlenbeck dynamics                                  | 266 |
| XX Log-concavity laws for exactly solved random conformal geometry             | 268 |
| 82 Spectral and modular log-concavity                                          | 270 |
| XXI Bold conjectures in theoretical fluid dynamics                             | 271 |
| 83 Small-scale creation in two-dimensional Euler and Navier–Stokes             | 272 |
| 84 Enhanced dissipation and hypoelliptic mixing                                | 273 |
| 85 Geometry and regularity of three-dimensional singularities                  | 274 |
| 86 Band-limited kinematics and rapid rotation                                  | 275 |
| 87 Stratified and active-scalar singularities                                  | 276 |
| 88 Point-vortex collapse and non-Beltramization                                | 278 |
| XXII Metric geometry and spectral convexity in random conformal geometry       | 278 |
| 89 Metric geometry of Liouville quantum gravity                                | 279 |

|                                                                                         |            |
|-----------------------------------------------------------------------------------------|------------|
| 90 Spectral convexity in Liouville conformal field theory                               | 282        |
| 91 Nesting in conformal loop ensembles                                                  | 284        |
| <b>XXIII Hydrodynamic limits of interacting particle systems</b>                        | <b>284</b> |
| 92 Density-adaptive and refreshed exclusion hydrodynamics                               | 285        |
| 93 Renewal exclusion, fatigue, and boundary memory                                      | 286        |
| 94 Catalysts, mixed clocks, and crossing budgets                                        | 288        |
| 95 Common noise, odd transport, and self-generated arrest                               | 289        |
| 96 Critical geometry and moving membranes                                               | 290        |
| <b>XXIV Fixed-traffic circulation laws for nonequilibrium Markov chains</b>             | <b>291</b> |
| 97 Visited-set and spectral circulation laws                                            | 292        |
| <b>XXV Cross-shadow universality</b>                                                    | <b>294</b> |
| <b>XXVI Entropy and rare-event laws for random algebraic and topological structures</b> | <b>298</b> |
| <b>XXVII Random Hamiltonian cocycle entropy</b>                                         | <b>303</b> |
| <b>XXVIII Cut-rank laws for graph states, forests, and random width</b>                 | <b>304</b> |
| <b>XXIX Arithmetic games, prime continuants and Hessian shadows</b>                     | <b>312</b> |
| <b>XXX Forking without stable witnesses: quantum modules over graphs</b>                | <b>323</b> |
| <b>Stress tests and independent recomputation</b>                                       | <b>336</b> |

## 1 Introduction

This collection states four hundred thirty-nine conjectures in thirty parts. Part I (Conjectures 1–25) is a single sustained experiment: the primes, beyond their local structure, are modelled as a random set of density  $1/\log n$ , and the model is pushed to its sharpest consequences.

Part II (Conjectures 26–50) leaves the single model for five structural programmes—connected prime-pattern fields, arithmetic first-arrival fields and class groups, finite logarithms and algebraic tori, arboreal dynamics, and adelic factorization—each contributing one canonical object and its laws. Part III (Conjectures 51–56) leaves number theory itself, carrying the same discipline into multiplicative number theory’s pretentious frontier, discrepancy theory, graph limits, additive information theory, and ergodic theory. Part IV (Conjectures 57–76) turns to a single object on the algebraic side, the class-counting polynomial of graphical Lie algebras, and states for it a rigidity and reconstruction programme, a log-concavity programme, and a tree extremal programme, with the proved layer, an exact rank formula and two lower bounds for trees, separated from the twenty conjectural laws. Part V (Conjectures 77–96) turns to a second algebraic family, the nilpotent pattern Lie algebras of finite posets, whose adjoint and Kirillov rank enumerators carry a determination programme, a characteristic-and-field programme, a centre-one positivity programme, and a Hasse-forest programme over a proved calibration identity, three of the twenty being stated in greater generality than the computations reach and, and two since resolved false with explicit counterexamples. Part VI (Conjectures 97–116) returns to the local–global aesthetic of Part I, reading it off prime compasses at polynomial centres, order patterns of the radical ratio, order constants of multiplicative ratios, prime-factor shocks in combinatorial sequences, and the arithmetic anatomy of prime-gap boundaries, over a proved layer of factorial-ratio jump identities, short-window exchangeability, existence of every fixed radical-pattern density, exact triangular–hexagonal finite-sieve equivalence, and the  $k = 2$  dyadic multinomial slope, one of the twenty since resolved false by a theorem of G. Martin and the shock and sign-balance laws since proved. Part VII (Conjectures 117–136) crosses into algebraic topology, reading exact classical calibrations, the Chern-class formula, the Milnor–Orlik formula, Hochster’s formula, and the Wang sequence, across four canonical families of spaces, over proved signed-orbit and normalised-merge calibrations, with Conjectures 117, 118, and 125 since proved and Conjecture 135 since resolved false. Part VIII (Conjectures 137–156) continues the topological turn with shape laws, order and concavity statements for the exact Euler, Betti, age, entropy, and Hochster arrays of five classical families, over a proved pair of profile moment identities, with Conjectures 151 and 152 since proved and Conjectures 153 to 156 since refuted by exact graphic certificates. Part IX (Conjectures 157–176) passes to spectra and filtrations, reading root location, connectivity, convexity, and determinant laws across four combinatorial geometries, graph associahedra, age-filtered McKay complexes, sine-torsion sectors, and simplicial spheres, over a proved pair of log-sine moment identities. Part X (Conjectures 177–196) enters string-geometric index theory, reading saturation and torsion laws off flux complexes on tori and extremal, monotonicity, and divisibility laws off the protected  $\chi_y$ -profiles and signatures of Calabi–Yau complete intersections, over a proved coefficient-and-parity calibration, its quadric ratio law since proved in full. Part XI (Conjectures 197–216) pairs spectral entropy with carry arithmetic, reading all-order Rényi monotonicity and extremal laws off the Laplacian and distance signless Laplacian density matrices of graphs, over a proved degree-and-majorization calibration, and cross-prime decoupling laws off the carry fields of central binomial coefficients at distinct primes, over a proved single-base carry-chain law. Part XII (Conjectures 217–236) runs four programmes at the interface of spectra and information, reading factorial moment thresholds off barycentric subdivision tails, extremal laws off a new spanning-tree total-correlation invariant, an inverse localization-and-rigidity theory off the prime-field entropy-doubling floor, and dimension identities off random-free graphon entropy, over a proved block-additivity and cycle law and a proved threshold-graphon entropy bound. Part XIII (Conjectures 237–255)

extends the string-geometric programmes, reading odd-degree saturation and torsion shape laws off the flux complexes of Part X, strict quantum monotonicity off the mirror data of the five one-parameter threefolds along the merge order, wall and sharp-constant laws off Bridgeland stability, and effective convergence, spectral-gap, and tower laws off the metric geometry of Calabi–Yau families, over a proved merge-and-Chern calibration. Part XIV (Conjectures 256–270) turns to three classical projective invariants, reading convexity laws off the normalized middle cohomology of complete intersections, controlled-defect laws off the Betti rows of  $\mathbf{P}^1 \times \mathbf{P}^1$ , and monotone-dispersion laws off hypersurface Hodge profiles, over proved curve, antidiagonal, and surface calibrations. Part XV (Conjectures 271–285) passes to probability, reading curvature laws off the count, radial-moment, and bridge sequences of the square-lattice self-avoiding walk, transport and likelihood-ratio laws off its endpoint, span, height, and contact ensembles, quadrant-dependence signs off its joint geometric laws, and phase and dimension laws off the interacting and nonbacktracking ensembles, over proved extreme-order, top-corner, and survival calibrations. Part XVI (Conjectures 286–294) passes to non-equilibrium dynamics, reading a skewness-amplification law off the winding currents of driven rings, recruitment, convexity, oscillation, and cheapest-defectivity laws off the entropy-production-budgeted gap frontier of linear diffusions, accessibility-constrained recruitment and Lie-algebraic saturation laws off coherently driven Lindblad dynamics, and a basin-collapse law off the eigenvector geometry of the Mpemba effect, over proved pair, envelope, and basin calibrations. Part XVII (Conjectures 295–307) passes to extreme-value theory, reading isoperimetric, closure, elasticity, and curvature laws off the Hüsler–Reiss extremal coefficient as a functional on variograms, thinning-spectroscopy and occupancy-tomography laws off the cluster structure of stationary extremes, consistency and leakage laws off the persistent homology of extreme directions, stability barriers and identifiability thresholds off fractional tail tomography, and Poissonization, design-ladder, and adaptation-cusp laws off randomized block maxima, over proved tilting, thinning, moment, and design calibrations. Part XVIII (Conjectures 308–319) turns to topological data analysis, reading isoperimetric, saturation, and projection-threshold laws off degree-one Vietoris–Rips persistence, shrinkage and grouping-stability laws off Wasserstein means of diagrams, survival-tomography and dither laws off the randomized-phase Mapper of tame Reeb graphs, a finite-direction stability law off rectangle-decomposable two-parameter modules, and restricted-isometry, evaluation-count, moment-determinacy, and average-case kinetic laws off finite persistence summaries, over proved phase, tomography, and moment calibrations. Part XIX (Conjectures 320–334) closes in nonequilibrium dynamics, reading monotonicity, spectral-horizon, integrated-budget, and mode-doubling laws off the lag irreversibility of finite Markov chains and Ornstein–Uhlenbeck processes and winding-aliasing scaling laws off driven cycles, over a proved layer of endpoint and Gaussian lag-divergence facts, its efficiency-monotonicity and unicyclic strict-monotonicity laws, its sharp unicyclic amplitude cap, and its two Ornstein–Uhlenbeck spectral-abscissa bounds since resolved false.

Part XXVII (Conjecture 394) adds a random Hamiltonian-cocycle barcode entropy law and strengthens Conjecture 392 to every fixed successive minimum.

Part XXVIII (Conjectures 395–409) adds fifteen laws for the cut-rank entanglement of graph states: quenched and extreme random-matrix laws, forest shape and stability laws, stochastic tree extremals and typical reconstruction, width densities for random regular, sparse, and percolated graphs, and rigidity and orbit-entropy laws under local complementation. Part XXIX (Conjectures 410–424) adds fifteen statements at the meeting points of number theory, measure theory, game theory and algebraic geometry: prime and squarefree laws for the continuants

of a typical real number with the constants  $2 \log 2$ ,  $6/\pi^2$  and  $\prod_p(1 - 1/(p(p+1)))$ , Hausdorff-dimension expansions for digit sets in progressions and for continuant residue avoidance, an exact identification of the values of Legendre-symbol half-box games with coordinate ratios of fundamental units of real quadratic fields together with the least-residue, quarter-box and vanishing laws that accompany it, a Brownian limit for random Toeplitz and Hankel games, and monodromy, Sato–Tate and dual-curve laws for plane quartics and their Hessians. Part XXX (Conjectures 425–439) responds to the refutation of the stable forking conjecture by a module over the division ring of fractions of the quantum algebra of the random bipartite graph: it states the surviving finite-rank, rank-one and countably categorical forms of the conjecture, the two-dependence of the counterexample, a transfer of dividing lines from a graph to its quantum module, the stable forking dichotomy and the forking calculus of these modules, the locus of the failure of one-basedness, failure over models, decidability, non-simplicity of the scalar division rings, the stable formulas, the failure of nonforking extensions to restore stable witnesses, definable groups, and the countable models, over a proved layer on the quantum algebras of graphs which isolates the mechanism of the counterexample: one hidden vertex leaves a formula stable, two transport the edge relation.

One standard governs all thirty parts. A good conjecture is not merely a statement that has resisted counterexample: it is a determinate proposition about a canonical object, with a stated mechanism, and with enough exposed structure to be refuted in pieces. Concretely, every statement here is asked to (i) survive every congruence and size obstruction; (ii) come with a quantitative form whose constants are derived, not fitted; (iii) demand exactly the fluctuations its mechanism earns and no fewer (the Mertens conjecture died of demanding fewer); (iv) sit inside a known hierarchy where one exists (for Part I, Hardy–Littlewood  $k$ -tuples  $\subset$  Schinzel’s Hypothesis  $H \subset$  Bateman–Horn), so that failure would propagate; and (v) expose instance-falsifiable content—congruences, null lanes, exact identities, derived constants—so that computation can refute the derivation even where the asymptotic itself lies beyond finite refutation. Numerical range is the least important column in the ledger: most deep phenomena drift at the rate  $\log \log x$ , and  $\log \log 10^{18} \approx 3.7$ , so verification “to  $10^{18}$ ” is weak evidence by itself. What persuades is the *shape* of the agreement—a count that tracks a derived constant through several decades with residuals that look like noise.

Part I is presented in three sections: the principal conjectures (numbers 1–9), family laws and second-order refinements (numbers 10–17), and instances and structural companions (numbers 18–25). The material they draw on—Bateman–Horn systems, barely-divergent sparse sequences, representation problems with Borel–Cantelli accounting, and statistical laws of the prime sequence—runs through all three. The stress-test section below reports the tests and the independent recomputation of every constant.

One structural lesson organizes several of the statements below. A probability model integrates over density, and is therefore blind to algebraic families of density zero; the admissibility clause, checked at every prime for every system, is what repairs that blindness. The cubes inside the representation problem  $n = p + k^3$  (Theorem 1) are the sharpest instance: they obstruct the problem outright while contributing nothing to any Borel–Cantelli sum taken over all  $n$ . This is the classical lesson of admissibility, and it is why the statements below separate what is provable by congruence or factorization from what a probabilistic accounting supplies. Parts II to V inherit the same separation in a different key: each statement names its canonical object and averaging law, identifies its source, states a first decisive theorem where one exists, and records an honest failure mode.



Three checks stand behind every statement, and the stress-test section reports their outcomes for Part I. First, an independent literature search at neighbourhood depth—defining objects, derived sequences, OEIS comments where relevant, and the abstracts and bodies of near-neighbour papers, read rather than skimmed from search summaries—establishes what is already known. Where a statement exists in essence it is attributed and labelled: some verbatim (Dubner’s twin-sum conjecture, the Stern list for  $p + 2k^2$ , Caldwell–Gallot’s  $e^\gamma \log N$  laws), one with an exact constant catalogued as an OEIS entry (A188596), and the rest attributed accordingly. No such statement stands as a conjecture of its own; each is retained *inside* the conjecture whose claimed content it calibrates, restated in our uniform framework and re-verified at our bounds, and where a statement strengthens a named open problem the containment is stated as its headline. Second, every falsifiable-by-instance statement was verified against exact counts and then re-tested well past its original bound, the computation being designed as an attempt at refutation rather than at confirmation. Third, constants and counts were recomputed from the definitions alone by independent implementations sharing no code with the primary computation. Our novelty policy throughout is conservative: where priority is uncertain we attribute rather than claim.

## Taxonomy: claimed content and its calibration layer

Within Part I, one criterion governs the taxonomy. An unrecorded specialization of Bateman–Horn is analogous to evaluating a classical special function at a new argument—worthwhile as data, but not a new conjecture in the conceptual sense; no conjecture’s claimed content is a bare specialization, and the taxonomy is two-layered *within* each conjecture. The *calibration layer*—the previously-known statements this paper begins from (Dubner’s conjectures, the Stern list, the Caldwell–Gallot laws, Martin’s refined Goldbach, Kourbatov’s record framework, the classical single instances of each family)—appears as attributed remarks inside the conjectures, each with its derived constant re-verified at our bounds; membership in the standard hierarchy of conjectures is itself a virtue, and the calibration layer is what makes the novel layer’s constants credible. The *claimed layer* is graded as follows: mechanism-level content (the contamination calculus 1(v) with its instances 18, 19, 14, 12, 1, 4; the ordering deficits 5, 3; race sub-diffusivity 9; the entanglement law 6); derived second-order and distributional laws (10, 15, 2, 8(ii), 11(iii), 13, 7(iv), 16(ii)); and structural classifications (23, 25, 24, 20(i), 21(i), 22)—with framework attributions stated where an instance lives inside someone else’s theory (Kowalski’s for 15(ii), Montgomery–Soundararajan’s for 2 and 13). Every statement was searched against the literature at neighbourhood depth and verified at the scales reported in the stress-test section. We regard the prime-power contamination calculus (Conjecture 1(v)) and its mechanism family—1 and 4, the fresh cousin-race predictions it generated, and the negative control 14—together with Conjecture 3(ii) and the derived covariance kernels of 8(ii) and 16(ii) as the paper’s core, joined by 13 and 23; Conjectures 20, 6 and 11 carry open modelling questions recorded in place.

One tagging convention runs through every statement below: clauses marked [theorem] or “elementary” are proved; clauses marked “conditional proposition” follow from the hypotheses stated with them; the remaining clauses are conjectures; and open questions are labelled as such and are not counted among the four hundred thirty-nine conjectures. A title tagged “resolved true” or “resolved false” marks a statement settled since its deposit, each carrying a *Resolution* paragraph with the proof or the counterexample.

## Computational-check standards

All counts below are exact (sieves; deterministic Miller–Rabin for  $n < 3.3 \times 10^{24}$ ; Baillie–PSW beyond,). Every comparison reports the Poisson-normalized residual  $z = (\text{obs} - \text{pred})/\sqrt{\text{pred}}$ . Every quoted  $z$ -score is such a residual and is used *descriptively*: where the counts compared are correlated, overlapping, or examined after selection, these residuals are not calibrated tail probabilities, and no significance claim in this paper rests on them. For sparse counts the cumulative comparison is contaminated at small  $x$  by the phantom mass of the main-term integral near its lower limit; in those cases we also report per-decade increments, which are clean. Constants are computed by truncated Euler products with the truncation wobble (difference between cutoffs) reported as an error bar; for conditionally convergent quadratic and cubic products this wobble, not the last digit, is the effective precision.

Primality certification is stratified, and every count inherits the stratum of the integers it involves. Primality of every integer below  $3.3 \times 10^{24}$  is decided *deterministically*, by fixed-base Miller–Rabin over a witness set verified for that range. For larger integers the classification is made by the Baillie–PSW test, and is a *probable-prime* classification, not a proof: no Baillie–PSW pseudoprime is known—and none exists below  $2^{64}$ , where the test is therefore deterministic—but their nonexistence is unproved. Consequently every count involving such integers is a probable-prime count, and is labelled as one wherever it appears. The computations affected are the factorial scans of Conjectures 21 and 2 ( $n! \pm 1$  for  $n$  beyond about 26), the  $n^2 + 2^n$  scan of Conjecture 6 beyond  $n \approx 80$ , the quartic values of the cyclotomic chain of Conjecture 24 beyond  $p \approx 2 \times 10^6$ , and the Fibonacci–Lucas values of Conjecture 20 at all but the smallest indices. Every other count below—sieve censuses, race counts, window fields, and the inputs to every constant—lies entirely inside the deterministic range.

*Data availability.* The accompanying repository (**conjecture-engine**) contains the primary verification programs and the independent reimplementations used for recomputation, the machine-readable results files and exception lists, the Euler-product truncation points used for every constant, and the window-selection rules and random seeds of every sampled computation; `run_all.py` regenerates every number in this paper. Probable-prime classifications are recorded as such in the results files, separately from the deterministically certified ones, so that each count can be read at its own certification stratum.

## 2 Notation

$p, q$  denote primes;  $\gamma$  is Euler’s constant,  $e^\gamma = 1.781072\dots$ ;  $\varphi$  is Euler’s totient;  $\phi = (1 + \sqrt{5})/2$ ;  $F_n$  the Fibonacci numbers;  $p\#$  the primorial. For irreducible  $f_1, \dots, f_k \in \mathbb{Z}[x]$  with positive leading coefficients, the Bateman–Horn singular series is

$$C(f_1, \dots, f_k) = \prod_p \frac{1 - \omega(p)/p}{(1 - 1/p)^k}, \quad \omega(p) = \#\{n \bmod p : p \mid f_1(n) \cdots f_k(n)\},$$

and the corresponding main term is

$$I(N) = \int_2^N \frac{dt}{\prod_i \log f_i(t)},$$

which absorbs the usual  $1/\prod_i \deg f_i$ . The Bateman–Horn conjecture [2] asserts  $\#\{n \leq N : \text{all } f_i(n) \text{ prime}\} \sim C \cdot I(N)$  whenever  $C \neq 0$ ; each Bateman–Horn conjecture below is an instance

with its constant evaluated. For nonlinear systems the product converges only conditionally; throughout, it is taken in the canonical ordering of increasing primes, in which convergence follows from the equidistribution of  $\omega(p)$  about its Chebotarev mean. Our numerical truncation wobble is an error bar for this canonically ordered product. We write  $\text{Li}_k(x) = \int_2^x (\log t)^{-k} dt$ . The twin singular series is  $\mathfrak{S}(d) = 2C_2 \prod_{p|d, p>2} \frac{p-1}{p-2}$  for even  $d$ , with  $2C_2 = 1.3203236\dots$ ; the Goldbach series  $\mathfrak{S}(n)$  is the same product over odd  $p \mid n$ .

Eight of the conjectures below are built over the Bateman–Horn move—choose an admissible system, compute its singular series, state the count—but none is a bare instance of it: 10 and 15 are family laws with derived constant distributions, 23 and 25 carry structural classifications, 24 is a searched chain, and 18, 19, 14 are contamination-calculus predictions (two fresh races and the negative control) whose mechanism is the contamination calculus of Conjecture 1(v). Each conjecture carries its classical calibration instance as an attributed remark. The heuristic is uniform, so we give it once. Model the primality of the values  $f_i(n)$  as independent events of probability  $1/\log f_i(n)$ , corrected by the factor  $(1 - \omega(p)/p)/(1 - 1/p)^k$  at each prime  $p$  to account for the joint local behaviour; multiplying the corrections and integrating gives  $C \cdot I(N)$ .

For locally integrable  $F$  we write  $\mathcal{M}_x(F) = \frac{1}{\log x} \int_2^x F(t) \frac{dt}{t}$  for the logarithmic mean; all race conjectures below state their drift and null clauses through  $\mathcal{M}_x$ , so that “no persistent leader” and “systematic surplus” are claims about a defined functional rather than about pointwise behaviour. Drift clauses are stated at the *drift scale*—for a pair race,  $\mathcal{M}_x(D(t) \log^2 t / \sqrt{t})$  converging to an explicit constant—never as  $\mathcal{M}_x((D - T)/\sqrt{\pi}) \rightarrow 0$ , which is vacuous for the drift coefficient (both  $T/\sqrt{\pi}$  and its multiples have logarithmic mean  $\rightarrow 0$ ). Convergence of  $\mathcal{M}_x$  at drift scale is itself part of each such conjecture: a pure random-walk fluctuation model would make these means divergent in variance, so their convergence encodes the Rubinstein–Sarnak-type almost-periodicity that the zero-oscillation hypothesis names.

Admissibility ( $\omega(p) < p$  for all  $p$ ) is asserted programmatically during every constant computation. The natural-looking pair  $\{n^2 + n + 1, n^2 + n + 3\}$  has  $\omega(3) = 3$  and is rejected; the  $k = 4$  branch of Conjecture 25 and the naive iterated chain behind Conjecture 24 were both rejected by the same machinery, and both rejections became part of the statements.

Novelty labels, from the independent literature search: (a) previously stated in essence; (b) classical family, quantified instance apparently unstated.

**How to read the four hundred thirty-nine.** The count is a numbering convention, not a claim of four hundred thirty-nine independent mechanisms. The contamination calculus is one operator, stated with the twin races and projected onto the Goldbach, Stern, triplet, and sexy-pair patterns, together with one provably null pattern as its control, and those six statements stand or fall largely together. Several conjectures pair a proved classification with a Bateman–Horn lane that is standard once stated, and several keep attributed benchmarks inside them as calibration; the tagging convention above separates these layers, and the novelty labels mark what is claimed as new. Part II is five programmes, each contributing one central object and its laws. Part IV is one polynomial invariant carrying three programmes, and its twenty statements stand or fall in blocks rather than singly, with two, Conjectures 63 and 71, already fallen and recorded as resolved false. Part V is two rank enumerators of one bracket tensor, fifteen conjectures supported throughout the tested corpus, three stated in greater generality than the computations reach, and two resolved false with explicit counterexamples, recorded in place. Part VI is five local-structure programmes in the aesthetic of Part I over a proved layer of jump identities and short-window exchangeability, six of its twenty stating a bias as a measured band, two extrapolating in a parameter rather than the index, one since resolved false, and four since

proved. The corresponding per-part registers for Parts VII through XVII, their programme counts, evidence caveats, retained controls, and resolutions, are carried once by the abstract and the summary sheet rather than repeated here. Depth and independence therefore vary by design across the four hundred thirty-nine, and the summary sheet's ordering, not the raw count, carries the priority claim.

## Part I

# Conjectures from a local–global random model

*Added to the deposit: 28 July 2026.*

## 3 Principal conjectures

The twin-race statement is split into its provable algebraic core and its conjectural clauses, stated through the logarithmic-mean functional  $\mathcal{M}_x$  of the notation section.

**Lemma 1** (Orientation elimination and class assignment). *In  $\psi_2(x) = \sum_{n \leq x} \Lambda(n)\Lambda(n+2)$ , the total contribution of terms in which  $n$  or  $n+2$  is a proper prime power is, apart from  $O(x^{1/3} \log^2 x)$  from cubes and higher powers, carried entirely by the patterns  $(q^2 - 2, q^2)$  with  $q$  prime and  $q^2 - 2$  prime: the mirror pattern  $(q^2, q^2 + 2)$  is annihilated by  $3 \mid q^2 + 2$  for every prime  $q > 3$ . Moreover, for every prime  $q \neq 5$ ,  $q^2 - 2 \equiv 2$  or  $4 \pmod{5}$  (never 1), and for every odd  $q$ ,  $q^2 - 2 \equiv 7 \pmod{8}$ . (Two bounded exceptions: at  $q = 5$  the term  $(23, 25)$  lands outside the twin-start classes altogether, and at  $q = 3$  the mirror term  $(9, 11)$  survives, the annihilation beginning only at  $q > 3$ ; each is a single term, absorbed in the bounded error.)*

The proof is elementary (squares mod 3, 5, 8; the prime-power count). The lemma orients the race: prime-square contamination can enter only specific residue classes, and the model converts that into a drift prediction for the *unweighted* twin counts.

**Conjecture 1** (Twin races modulo 5 and 8, and the prime-power contamination calculus; (b), apparently new). *Twin starts  $p > 5$  lie in classes  $p \equiv 1, 2, 4 \pmod{5}$ ; write  $\pi_t(x; m, a)$  for the number of twin starts at most  $x$  in the class  $a$  modulo  $m$ . Let  $D_1(x) = \pi_t(x; 5, 1) - \frac{1}{2}(\pi_t(x; 5, 2) + \pi_t(x; 5, 4))$  and*

$$T(x) = \frac{1}{2 \log^2 x} \sum_{\substack{q \leq \sqrt{x} \\ q^2 - 2 \text{ prime}}} \log q \log(q^2 - 2).$$

*Then: (i) [drift law, at drift-scale normalization] the clause asserts two sub-claims, labelled separately because they can fail independently. (i-a) [mechanism] class 1 carries the systematic surplus  $T$ : one has the decomposition  $D_1(t) = T(t) + R(t)$  in which the normalized remainder  $R(t) \log^2 t / \sqrt{t}$  possesses a limiting logarithmic distribution with mean zero, so that the entire drift-scale deterministic content of  $D_1$  is the contamination term and its coefficient is thereby identified. (i-b) [averaging] at the drift-scale normalization the remainder averages away:*

$$\mathcal{M}_x \left( \frac{R(t) \log^2 t}{\sqrt{t}} \right) \rightarrow 0, \quad \text{equivalently} \quad \mathcal{M}_x \left( \frac{D_1(t) \log^2 t}{\sqrt{t}} \right) \rightarrow c_T,$$

where

$$c_T := \lim_{x \rightarrow \infty} \frac{T(x) \log^2 x}{\sqrt{x}} = \frac{1}{2} C(x, x^2 - 2)$$

is the half of the Bateman–Horn constant of the pair  $(q, q^2 - 2)$  (the weighted census sum is  $\sim C\sqrt{x}$ , so the limit exists and is explicit). A normalization at the noise scale,  $\mathcal{M}_x((D_1 - T)/\sqrt{\pi_t}) \rightarrow 0$ , would be vacuous for the coefficient of (i-a): since  $T/\sqrt{\pi_t} \asymp 1/\log t$ , one has  $\mathcal{M}_x(T/\sqrt{\pi_t}) \asymp \log \log x / \log x \rightarrow 0$ , so the condition holds with  $T$  replaced by 0,  $2T$ , or any fixed multiple, and does not express “exactly the removed mass.” The statement therefore normalizes at the drift scale, where the claimed limit is a nonzero constant. Convergence at this normalization is a Rubinstein–Sarnak-type regularity assertion: under a pure random-walk model the logarithmic mean would not converge (its variance grows like  $\log x$ ), so (i-b) asserts almost-periodic structure for the twin race and is the precise content of the zero-oscillation hypothesis, while (i-a) is what fixes the limit at  $c_T$  rather than 0 or  $2c_T$ , and is the calculus. The convergence asserted in (i-b) selects the rigid branch of the dichotomy registered at Conjecture 9(iii); the two statements stand or fall together at drift scale; (ii) [zero deterministic drift in the symmetric differences] the symmetric classes carry no deterministic term at the same normalization:  $\mathcal{M}_x((\pi_t(\cdot; 5, 2) - \pi_t(\cdot; 5, 4)) \log^2 t / \sqrt{t}) \rightarrow 0$ , and likewise for each of the three pairwise differences among classes  $\{1, 3, 5\} \pmod{8}$  in clause (iv)—now a statement distinct from (i), since at drift scale “zero” and “ $c_T$ ” are different limits; (iii) [sign density, conditional] conditional on the Gaussian fluctuation model for the normalized remainder in (i)–(ii), the logarithmic density of  $\{D_1 > 0\}$  exists and equals  $\frac{1}{2}$ , approached from above with a finite- $x$  excess of order  $\log \log x / \log x$ : the pointwise drift-to-noise ratio is  $T/\sqrt{\pi_t} \asymp 1/\log t$ , and its logarithmic average  $\mathcal{M}_x(1/\log t) \sim \log \log x / \log x$  carries an extra  $\log \log$  (the scale is not  $1/\log x$ ). In contrast to Chebyshev’s races, the twin race has no persistent leader; (iv) [mod-8 companion] since  $q^2 - 2 \equiv 7 \pmod{8}$  always (Lemma 1), the entire deterministic term falls on the single class  $7 \pmod{8}$ : with  $D_7(x) = \frac{1}{3} \sum_{a \in \{1, 3, 5\}} \pi_t(x; 8, a) - \pi_t(x; 8, 7)$ ,

$$\mathcal{M}_x\left(\frac{D_7(t) \log^2 t}{\sqrt{t}}\right) \rightarrow 2c_T$$

(twice the mod-5 constant, undiluted—at this normalization the factor 2 is a falsifiable prediction, the mechanism and averaging claims separating here exactly as in (i-a)–(i-b)), while classes 1, 3, 5 are mutually symmetric. The mod-5 and mod-8 races are two projections of one mechanism, so their joint behaviour is a family test in the sense of the shape-of-agreement criterion; (v) [contamination calculus for balanced races] for a pattern  $(n, n + d)$  and modulus  $m$ , call the race balanced if the prime-prime singular series is identical across the competing residue classes (as CRT gives for the classes compared in (i)–(iv)) and no mechanism other than prime powers contributes at the  $\sqrt{x}$  scale under the zero-oscillation hypothesis. For balanced races the deterministic drift vector is computed by one operator. Define the contamination operator

$$\mathcal{C}_{d,m}(a; x) = \sum_{\text{orientations } o \in \{(q^2 - d, q^2), (q^2, q^2 + d)\}} \sum_{\substack{q \leq \sqrt{x}, \text{ } o \text{ prime-compatible} \\ \text{start}(o) \equiv a \pmod{m}}} \Lambda(q^2) \Lambda(\text{prime member of } o),$$

where prime-compatible means the non-square member is prime and no fixed prime annihilates the orientation. The provable layer is the orientation census and class assignment (which  $o$  survive, and where their starts land—Lemma 1 and its analogues); the conjectural layer is the transfer: the unweighted class counts are depressed by  $\mathcal{C}_{d,m}(a; x) / \log^2 x$  relative to the clean classes, in the drift-scale limiting-log-mean sense of (i), with the mechanism and averaging claims separated as in (i-a)–(i-b)—for each class  $a$ , the deficit obeys  $\mathcal{M}_x(\text{deficit}_a(t) \log^2 t / \sqrt{t}) \rightarrow c(a) := \lim \mathcal{C}_{d,m}(a; x) / \sqrt{x}$ ,

an explicit Bateman–Horn constant per class, so every coefficient in the vector is identified—i.e. “exactly the removed mass,” which requires the weighted identity, partial summation uniform in classes, higher-power bounds (exponents  $r \geq 3$  contribute  $O(x^{1/3+\epsilon})$  against the operator’s  $\asymp \sqrt{x}$  and are negligible—the one easy layer of the transfer), and the zero-oscillation hypothesis, and is the calculus’s single analytic conjecture rather than a rule. The drift is  $\asymp \sqrt{x}/\log^2 x$  against a count fluctuation  $\asymp \sqrt{x}/\log x$ , i.e. smaller by  $1/\log x$  in standard-deviation units at every height—so the calculus predicts occupation biases under long logarithmic aggregation, never a fixed nonzero standardized mean, and every verification in this family is designed around that fact. (Races with unequal primary constants, or with interfering orientations beyond those enumerated, are outside the calculus until those terms are separated.) In particular, for the cousin pattern  $d = 4$  the calculus makes two predictions with no free choices: the orientation  $(q^2 - 4, q^2)$  is dead apart from a single term ( $q^2 - 4 = (q - 2)(q + 2)$  is composite for  $q > 3$ , while at  $q = 3$  the value 5 is prime: one bounded term, whose start 5 lies outside the mod-5 cousin classes and contributes one bounded term to class 5 mod 8), the orientation  $(q^2, q^2 + 4)$  survives (no prime annihilates it),  $q^2 + 4$  prime forces  $q \equiv \pm 2 \pmod{5}$  for  $q \neq 5$  (at  $q = 5$  the value 29 is prime, but the start  $25 \equiv 0 \pmod{5}$  lies outside the cousin start classes  $\{2, 3, 4\}$ ), so the contaminated cousin-start class is 4 (mod 5) (of  $\{2, 3, 4\}$ ) and—since  $q^2 \equiv 1 \pmod{8}$ —the class 1 (mod 8) (of  $\{1, 3, 5, 7\}$ ): both deficits converge, in the drift-scale limiting-log-mean sense of (i)–(ii), to  $c^c = \lim_{x \rightarrow \infty} T^c(x) \log^2 x / \sqrt{x}$ , the Bateman–Horn constant of the pair  $(q, q^2 + 4)$ , where  $T^c(x) = \frac{1}{\log^2 x} \sum_{q \leq \sqrt{x}, q \neq 5, q^2+4 \text{ prime}} \log q \log(q^2 + 4)$ .

Clause (v) answers the structural question of what general principle the mod-5 and mod-8 races instantiate, and the cousin predictions it yields were derived first and tested second. At  $x = 10^9$  the predicted drift is far inside the noise ( $T^c/\sqrt{\pi_c} \approx 0.09$ , as the  $1/\log x$  law requires), so the sharp statistic is leadership under logarithmic averaging, calibrated against the null of Conjecture 14(ii) (log-occupation  $\frac{1}{2}$  with Gaussian spread  $\sqrt{\log 2/\log x} \approx 0.18$  at  $10^9$ , at the occupation-fraction constant): the class-4 deficit race led on the predicted side with log-density 0.99 (+2.7 null standard deviations), the class-1 mod-8 race with 0.92 (+2.3), both signs agreeing with the prediction, while the control differences among uncontaminated classes stayed within one noise unit of zero in log-mean. This is moderately strong directional evidence—two independent projections each between two and three null standard deviations on the predicted side—though the races share underlying primes and the two statistics are positively correlated, so we do not call it sharp confirmation; the family now has five projections of one mechanism (twin mod 5, twin mod 8, cousin mod 5, cousin mod 8, and the Goldbach lanes of Conjecture 4) plus two algebraic nulls (Conjecture 14 and the dead cousin orientation), which is the shape-of-agreement standard applied to the calculus itself.

A limiting sign density strictly between  $\frac{1}{2}$  and 1 would be inconsistent with the scale analysis: when drift/noise  $\rightarrow 0$ , a Gaussian model forces the sign density to  $\frac{1}{2}$ , and a nontrivial limiting density would require a persistent normalized bias that no mechanism here produces. The fluctuation in (iv) is not written as  $O^*(\sqrt{\pi_t})$ , since  $O$ -notation asserts a provable bound where only a conjectural fluctuation statement is meant. Two model assumptions remain open and are stated as such: the passage from the  $\Lambda$ -weighted count to unweighted twin counts is a partial-summation argument whose error terms need writing out (higher prime powers are disposed of by Lemma 1); and the *zero-oscillation caveat*—the drift law treats the oscillatory explicit-formula contributions to class differences as having vanishing logarithmic mean, the exact analogue of the Rubinstein–Sarnak GRH-plus-linear-independence framework [6], and a conspiracy among hypothetical zeros could in principle mimic or cancel part of the drift; clause (i) is conjectured



on the standard side of that hypothesis.

The mechanism is the twin analogue of the classical explanation of Chebyshev's bias: prime-square terms contaminate the classes that can contain them. (Biases of twin primes in residue classes have been studied before—Sahoo [27] for the mod-4 race, and [28] for prime pairs versus isolated primes—but we could find no prior statement of the mod-5 race, the  $(q^2 - 2, q^2)$  mechanism, or the quantified surplus below.) Mod 3 kills the pattern  $(q^2, q^2 + 2)$  outright, a local accident that halves the mechanism and (as a mod-3 computation shows, overturning the natural first guess of a quadratic-residue bias against class 4) concentrates the surplus entirely on class 1. Data to  $4 \times 10^9$ : class counts (3980017, 3982505, 3981922),  $D_1$  within one noise unit of  $T$  throughout, no persistent leader, and the control race  $\pi_{t,2} - \pi_{t,4}$  wandering like a fair coin. Mod-8 data to  $10^9$ : class counts (856684, 855807, 856046, 855967) for  $a = 1, 3, 5, 7$ ;  $D_7 = +212$  at  $10^9$  against predicted  $T = +254$  with noise  $\sim 1068$ , positive through most of the range (log-density 0.775), and the three pairwise  $\{1, 3, 5\}$  controls all within one noise unit of zero—the two projections of the mechanism behave coherently, none of it provable at present.

**Conjecture 2** (The pair-level Montgomery–Soundararajan reduction; apparently new). *For the window field of Conjecture 8(ii), the diagonal variance of the twin-pair count reduces to a pinned singular-series average. With  $R(h) = C_4(0, 2, h, h + 2)/\mathfrak{S}(2)^2$  for  $h \geq 1$ ,  $C_4$  the Hardy–Littlewood constant of the displayed quadruple, (zero when the offset set is degenerate or inadmissible—all odd  $h$ , and  $h = 2$  since  $\{0, 2, 4\}$  dies at 3; consecutive twin pairs cannot overlap beyond  $(3, 5, 7)$ , so no overlap term exists) and*

$$G(H) = \sum_{1 \leq h \leq H} \left(1 - \frac{h}{H}\right) (R(h) - 1),$$

the derived identity (clause (i)) is

$$\frac{\text{Var}_t \pi_2(t; H)}{\mathbb{E}_t \pi_2(t; H)} = 1 + \frac{\mathfrak{S}(2) (2G(H) - 1)}{\log^2 x} + o\left(\frac{1}{\log^2 x}\right) :$$

with  $p = \mathfrak{S}(2)/\log^2 x$  the per-site intensity, the variance is  $Hp(1 - p) + 2p^2HG(H)$ , so  $\text{Var}/\mathbb{E} = 1 - p + 2pG(H)$ —the Bernoulli diagonal term  $-p = -\mathfrak{S}(2)/\log^2 x$  is of exactly the order of the claimed error and cannot be dropped (it is the same diagonal that, in the single-prime case, merges into Montgomery–Soundararajan's constant  $\gamma + \log 2\pi - 1$ ). With that term retained, the entire deficit left open at Conjecture 8(ii) is the single computable function  $G$  plus the explicit Bernoulli correction—the pair analogue of Montgomery–Soundararajan's reduction of the prime-count variance to  $\sum_{h \leq H} \mathfrak{S}(h)(H - h)$  [19]. Clause (i) is a conditional proposition, not an independent conjecture: given quantitative Hardy–Littlewood for 4-tuples, the reduction follows. The conjectural clause is (ii):  $-G(H)/\log H$  diverges—the pinned average grows strictly faster than the single-prime secondary term  $\frac{1}{2}(\log H + \gamma + \log 2\pi - 1)$ , so pair counts in windows are more sub-Poisson than prime counts—with the caveat that data to  $H = 3000$  cannot distinguish divergence from a pure log with a large constant (nor  $c \log H \log \log H$  from  $(\log H)^\alpha$ ): what the computation establishes is only that the local slope exceeds the single-prime value sevenfold across the computable range; the divergence and its exact form are registered, not evidenced beyond that.

The pinning leaves the local densities in exact balance. At every odd prime  $p$  the relative local factor of  $R(h)$  takes the value  $p/(p - 2)$  for  $h \equiv 0$ , the value  $p(p - 3)/(p - 2)^2$  for  $h \equiv \pm 2$ , and the value  $p(p - 4)/(p - 2)^2$  otherwise, and their exact average over  $h \bmod p$  equals 1, since



$(p-2) + 2(p-3) + (p-3)(p-4) = (p-2)^2$ . Any persistent negativity of  $G(H)$  is therefore not a first-order local-density effect but a global correlation effect across primes, which is what makes its growth rate delicate.

$G$  was computed exactly to  $H = 3000$  (4-tuple constants by Euler products to  $2 \times 10^4$ ):  $G(3000) = -20.9$ , with local logarithmic slope increasing from 3.4 to 3.7 across the range—seven times the single-prime slope  $\frac{1}{2}$ , consistent with (ii) over the computable range while leaving the limiting form undetermined (a quadratic-in-log fit is unstable on subranges). The window measurement brackets the extrapolation: at  $H = 10^5$ ,  $x \in [10^8, 2 \times 10^8]$  the pure-log form predicts  $\text{Var}/\mathbb{E} = 0.97$ , the  $\log^2$  form 0.72, and the observed value is 0.815—intermediate growth, exactly where the computable range points. (Caldwell–Gallot’s factorial-prime law  $\#\{n \leq N : n! + 1 \text{ prime}\} \sim e^\gamma \log N$  [4], the divergent single-sided companion of Conjecture 21, is included as an attributed calibration benchmark: 15 primes to  $n = 700$  against 11.7,  $z = +0.98$ —a probable-prime count,  $n! + 1$  leaving the deterministic range at  $n \approx 26$ .)

*Computational checks.* The pinned singular-series computation and window comparison are reported at the statement; its factorial benchmark stands at 15 primes with  $n \leq 700$  against 11.7 ( $z = +0.98$ ; a probable-prime count for  $n$  beyond about 26), the divergent single-sided companion of Conjecture 21’s convergent side.

**Conjecture 3** (Least primes in progressions; (a) for part (i) [20, 21, 30, 29], (b) for part (ii)). *Let  $p(a, q)$  be the least prime  $\equiv a \pmod{q}$  and  $U(a, q) = \text{Li}(p(a, q))/\varphi(q)$ . Then: (i)  $U \Rightarrow \text{Exp}(1)$  marginally as  $q \rightarrow \infty$ ; and, under the additional hypothesis that the class processes are asymptotically jointly independent in the extreme-value sense (which marginal convergence alone does not supply),  $\max_a U - H_{\varphi(q)}$  converges to a Gumbel law ( $H_n = \sum_{k \leq n} 1/k$ ), the order-statistics form of Wagstaff’s  $\max_a p(a, q) \sim \varphi(q) \log^2 q$ ; (ii) [canonical deficit, dyadic-average form] define, with no reference to any control model,  $\Theta(q) = (1 - \mathbb{E}_a[U(a, q)]) \log q$ . The conjecture is stated for dyadic averages over prime moduli—pointwise convergence along every prime  $q$  may be too strong, since arithmetic fluctuations in  $q$  could persist:*

$$\frac{1}{\#\{q \in [Q, 2Q] \text{ prime}\}} \sum_{\substack{Q \leq q \leq 2Q \\ q \text{ prime}}} \Theta(q) \longrightarrow \Theta > 0 \quad (Q \rightarrow \infty).$$

(The formal claim is model-free: existence and strict positivity of the dyadic-average limit. The stronger reading—that  $\Theta$  exceeds the discreteness level any parity-aware control produces—is the diagnostic layer below, not part of the display, since a control constant has no place in a model-free statement.) The limiting distribution of  $\Theta(q)$  over the dyadic block is left open, and in particular we do not assert  $\text{sd}(\Theta(q)) \rightarrow 0$ : fluctuations driven by the arithmetic of  $q$  and of  $q-1$  (through the singular-series pair terms  $\mathfrak{S}(kq)$  and the class structure mod  $q$ ) may well persist at bounded size, so  $\Theta(q)$  need not concentrate around its dyadic mean.

The decomposition  $\Theta(q) = \theta_{\text{disc}}(q) + \theta_{\text{corr}}(q)$  is a *diagnostic*, not a canonical invariant: its two parts move if the benchmark moves, and clause (ii) is therefore stated through the model-free  $\Theta(q)$  alone. The named benchmark (the parity-aware Cramér–Bernoulli control: candidates  $a, a+q, a+2q, \dots$  independently “prime” with hazard  $h_i = (q/\varphi(q)) \cdot s_i / \log(a+iq)$ ,  $s_i = 2$  for odd candidates and 0 for even, matching density and parity) defines  $\theta_{\text{disc}}(q) = (1 - \mathbb{E}_a[U^{\text{model}}]) \log q$  exactly for each  $q$ , and  $\theta_{\text{corr}} := \Theta - \theta_{\text{disc}}$  is the part of the measured deficit that the control cannot produce; it is this diagnostic that localizes the anomaly in the *ordering* of the primes (Remark 1). Measured on prime moduli  $q \in [1500, 6000]$ :  $\Theta = 1.671 \pm 0.009$ , of which the control accounts for 0.847.

One further deterministic baseline must be subtracted before any of the residue is called an anomaly: for prime  $q$ , the primes  $p < q$  occupy *distinct* classes mod  $q$ —an injective initial phase that no exchangeable allocation reproduces. Its size has a closed form in index time (where  $\text{Li}(p_i) \approx i$ , so  $U$  is the first-hit index over  $\varphi$ ): forcing the first  $\pi(q)$  arrivals to be collision-free and leaving the rest exchangeable gives  $\mathbb{E}[U] = 1 - \pi(q)^2/2\varphi^2 + O(\pi(q)/\varphi^2)$ , hence

$$\Theta_{\text{inj}}(q) = \frac{\pi(q)^2}{2\varphi(q)^2} \log q (1 + o(1)) = \frac{1 + o(1)}{2 \log q},$$

closed form 0.092, 0.082, 0.074 at  $q = 1499, 3001, 5987$ , with Monte Carlo agreeing within sampling error and the exchangeable control at 1.000. The injective phase is therefore real but an order of magnitude too small to be the explanation: it accounts for  $\approx 0.08$  of the measured  $\theta_{\text{corr}} = 0.824 \pm 0.009$ , and carries a  $1/\log q$  decay that the flat measured values do not show. (One might expect the effect to sit at the full deficit scale, since the *affected-class fraction* is  $\pi(q)/\varphi \asymp 1/\log q$ ; but the deficit is the affected fraction times the collision probability an exchangeable model would have suffered in that phase,  $\asymp \pi(q)/2\varphi$ , giving  $1/2 \log^2 q$  in  $\mathbb{E}[U]$ , not  $1/\log q$ .) The injective phase is in any case the  $y < q$  head of the pair-correlation expansion already registered below—collisions require  $q \mid p_2 - p_1$ , impossible before  $y = q$ —so subtracting it is the first term of the stated programme, not a rival explanation. The residue  $\theta_{\text{corr}} - \Theta_{\text{inj}} \approx 0.74$  remains the unexplained ordering anomaly.

A derivation route for the deficit, which we adopt as the conjecture’s stated programme, runs as follows. Let  $V_q(y)$  be the number of reduced classes mod  $q$  not yet visited by a prime  $\leq y$ ; then  $\mathbb{E}_a[U] = \int_0^\infty \overline{V_q(y_\tau)}/\varphi(q) d\tau$  with the Li-time change  $y_\tau$  defined by  $\text{Li}(y_\tau) = \tau\varphi(q)$ , and inclusion–exclusion expands  $V_q(y)$  in prime-tuple correlations *within* a class:

$$\mathbb{E} \binom{N_a}{2} \text{ summed over } a = \#\{p_1 < p_2 \leq y : q \mid p_2 - p_1\} \approx \sum_{1 \leq k \leq y/q} \mathfrak{S}(kq) \int_2^{y-kq} \frac{dt}{\log t \log(t + kq)},$$

so the first correction to the exchangeable occupancy model is carried by the Hardy–Littlewood singular series of gaps divisible by  $q$ . The truncation is structural, not a convenience: a gap  $kq$  requires  $p_1 \leq y - kq$ , so the  $k$ -sum stops at  $y/q$  and the endpoint-shortened integrals supply the triangular weights, while the untruncated sum  $\sum_{k \geq 1} \mathfrak{S}(kq)$  diverges (the singular series has mean one on average over its argument) and the formally factorized expression  $\text{Li}_2(y) \sum_k \mathfrak{S}(kq)$  is meaningless. Evaluating this expansion to second order, and comparing it with the measured  $\Theta$ , is the concrete open computation that would either derive the ordering anomaly from pair correlations or prove it lies deeper; either outcome retires Question 1’s mystery one level.

The stratified experiment behind these numbers separates moduli by factorization type in the matched range  $q \in [1500, 6000]$ : over 40 *prime* moduli,  $\theta_{\text{disc}} = 0.847$  (essentially constant across  $q$ ) and  $\theta_{\text{corr}} = 0.824 \pm 0.009$ ; over 40 *7-smooth* moduli in the same range,  $\theta_{\text{disc}} = 4.32 \pm 0.22$  (the large  $q/\varphi(q)$  hazards make discreteness dominant) while  $\theta_{\text{corr}} = -2.32 \pm 0.22$  reverses sign. The comparison with Leung [29] is thereby given its answer-shaped experiment: Leung derives the exponential law under a uniform Hardy–Littlewood hypothesis and reports discrepancies for *smooth* moduli, and our smooth stratum indeed behaves anomalously—but the positive ordering term survives, with small error bars, exactly where smooth-modulus effects cannot reach. Whether  $\theta_{\text{corr}}$  tends to a constant, and the exact functional form in  $1/\log q$ , remain open: data at  $q \leq 6000$  cannot distinguish  $1/\log q$  from  $(\log \log q)/\log q$ -type refinements—the aggregate  $\Theta \approx 1.5\text{--}1.9$  over  $q \in [10^2, 6 \times 10^3]$ , with  $\mathbb{E}_a[U]$  bottoming near 0.737 around  $q \approx 200$

and recovering through 0.776 by  $q \approx 6000$ , is calibration, not a determination of the form. The averaging family is fixed as: all moduli of the stated type in dyadic ranges of  $q$ , per stratum.

**Remark 1.** *The deficit is reproducible and is not an artefact of the time-change: a Cramér pseudo-prime control (independent Bernoulli “primes” with the correct densities and parity) shows roughly half of it—the discreteness of large per-candidate hazards—while the other half vanishes under any reshuffling of the actual prime residues (iid labels, or a random permutation of the true residue sequence, both restore  $\mathbb{E}[U] = 1.00$ ). The ordered sequence of primes fills residue classes measurably faster than any exchangeable model, and the stratified experiment above shows the effect is not a smooth-modulus artefact: it is largest and cleanest on prime moduli. We measure  $\theta_{\text{corr}}$ ; we cannot yet derive it.*

**Conjecture 4** (The Goldbach lane race; apparently new). *For  $n \equiv 2 \pmod{4}$  let  $R_1(n)$ ,  $R_3(n)$  be the ordered Goldbach representation counts in the lanes  $p \equiv q \equiv 1$  and  $p \equiv q \equiv 3 \pmod{4}$  (the lane convention of the Martin benchmark recorded at Conjecture 5), and  $D(n) = R_3(n) - R_1(n)$ . Prime-square terms  $n = q^2 + p$  ( $q$  odd prime) land exclusively in the (1, 1) lane, since  $q^2 \equiv 1 \pmod{4}$  and then  $n - q^2 \equiv 1 \pmod{4}$  automatically. Consequently the (3, 3) lane leads on average. Define*

$$D_{\text{sys}}(n) = \frac{2}{\bar{\ell}(n)} \sum_{\substack{q \text{ odd prime, } q \leq \sqrt{n} \\ n - q^2 \text{ prime}}} \log q \log(n - q^2), \quad \bar{\ell}(n) = \frac{n - 6}{\int_3^{n-3} \frac{dt}{\log t \log(n - t)}},$$

(the analytic mean of  $\log p \log(n - p)$  under the lane profile; the empirical lane mean is its estimator). Write  $\mathbb{E}_x$  for the average over  $n$  drawn from  $\{n \equiv 2 \pmod{4}, n \leq x\}$  with weight proportional to  $1/n$  (the logarithmic sampling measure, now explicit). Then: (i) [drift law]  $\mathbb{E}_x[D - D_{\text{sys}}] = o(\mathbb{E}_x[D_{\text{sys}}])$  as  $x \rightarrow \infty$ : the (3, 3) lane leads on average by exactly the square contamination; (ii) [weighted mean and internal null lane]  $\mathbb{E}_x[D_{\text{sys}}]$  is given by the Hardy–Littlewood (Conjecture H) local model for  $n - q^2$ , whose local factors vary strongly with  $n$ —most drastically at  $p = 3$ : for  $n \equiv 1 \pmod{3}$ ,  $3 \mid n - q^2$  for every prime  $q > 3$ , so the contamination collapses to a single boundary family, the  $q = 3$  term (when  $n - 9$  is prime): the complementary possibility  $n - q^2 = 3$  is empty on this ensemble, since  $n \equiv 2 \pmod{4}$  forces  $n - 3 \equiv 3 \pmod{4}$ , never a square. The family is negligible in logarithmic mean, so the race carries no drift at the systematic scale on the subprogression  $n \equiv 10 \pmod{12}$ : with the positive comparison scale  $A_G(x) := \mathbb{E}_x^{(\neq 1 \pmod{3})}[D_{\text{sys}}]$ , the logarithmic-ensemble mean of the systematic term over the contaminated lane  $n \not\equiv 1 \pmod{3}$  (a comparison against this subprogression’s own systematic term, which vanishes, would be empty), the claim is  $\mathbb{E}_x[D] = o(A_G(x))$  on  $n \equiv 10 \pmod{12}$ , while the drift concentrates on  $n \not\equiv 1 \pmod{3}$ : an internal null lane, predicted by the same mechanism that predicts the lead; (iii) [sign density] the per- $n$  drift-to-noise ratio  $\kappa(n) = D_{\text{sys}}(n)/\sqrt{\mathfrak{S}(n)J(n)}$  (where  $J(n) = \int_3^{n-3} dt/(\log t \log(n - t))$ , so that  $\mathfrak{S}(n)J(n)$  is the total ordered representation count) satisfies  $\kappa(n) \asymp c(n)/\log n \rightarrow 0$ ; under the Gaussian fluctuation model the logarithmic density of  $\{D > 0\}$  is therefore  $\frac{1}{2}$  in the limit, with the finite- $x$  sign fraction predicted to be  $\mathbb{E}_x[\Phi(\kappa)]$  ( $\Phi$  the standard normal distribution function)—decaying, but far from  $\frac{1}{2}$  at any accessible height.

Derivation status: the identity behind  $D_{\text{sys}}$  is the  $\Lambda$ -weighted lane symmetry minus its prime-power part; the transfer to unweighted counts is a partial-summation argument whose error terms (higher prime powers, which enter only at the  $n^{1/3}$  scale, and the variation of the weight across

the lane) remain to be written out—the same open transfer noted at Conjecture 1—and clause (iii) inherits the zero-oscillation caveat stated there. One further disclosure: the census  $D_{\text{sys}}$  is computed from the primes in the same sample it is compared against, so the level agreement is in-sample; the out-of-sample content of the verification is the null lane and the class assignment, which the mechanism fixes in advance.

This is the Goldbach-partition analogue of Chebyshev’s bias, produced by the same explicit-formula mechanism as Conjecture 1 but through  $q^2 + p$  patterns rather than twin patterns; the single-sign prediction (every square contamination falls into one lane) makes it cleaner than the classical race. We could find no prior statement. The neighbouring phenomenon for consecutive primes is the Lemke Oliver–Soundararajan mod-3 law, whose verification here— $\hat{c}$  drifting  $0.400 \rightarrow 0.370$  over  $10^7 \rightarrow 4 \times 10^9$  with  $(1, 1)/(2, 2)$  symmetry 0.99989—serves as calibration for the present race.

Computational checks. Over 500 log-spaced samples  $n \leq 10^8$ : mean  $D = 76.5$  against mean empirical  $D_{\text{sys}} = 64.5$ , the  $(3, 3)$ -lane lead significant at 5.2 standard errors of the mean. Clause (ii)’s weighted model, computed in presieve-exact form (trial division of each  $n - q^2$  to  $10^3$  with the exact Mertens normalization), reproduces the empirical contamination with no fitted parameter: model mean 66.3 against empirical 67.8 (ratio 0.98) on a 400-sample run. The internal null lane behaves as predicted: on  $n \equiv 1 \pmod{3}$  (123 samples) the model and empirical  $D_{\text{sys}}$  both vanish and the measured lead drops to  $31 \pm 23$  (consistent with zero), while on  $n \not\equiv 1 \pmod{3}$  (277 samples) the lead is  $88 \pm 21$  against predicted contamination 96—the drift lives exactly where the mechanism puts it. Clause (iii): computed  $\mathbb{E}_x[\Phi(\kappa)] = 0.572$  ( $\kappa$  ranging 0–0.64 with mean 0.18) against observed sign fraction 0.588; weak, and quantitatively the weakness the model requires.

**Conjecture 5** (The least Goldbach summand: exponential law and ordering deficit; the time-changed law and deficit constant apparently unstated). *Let  $s(n)$  be the least prime  $p \nmid n$  with  $n - p$  prime (the exclusion  $p \nmid n$  removes a trivial obstruction:  $p \mid n$  forces  $p \mid n - p$ , so  $n - p$  is composite except in the diagonal case  $n = 2p$ , which would otherwise distort the hazard at one atypical point), and time-change by the expected-arrivals clock  $U(n) = \mathfrak{S}(n) \sum_{p \leq s(n), p \nmid n} 1/\log(n - p)$ . Here  $s(n) = +\infty$  when no admissible prime exists, which among even  $n$  happens only at  $n = 4$  and  $n = 6$  and conjecturally at no larger even  $n$ , so the law below runs over the even  $n$  with  $s(n)$  finite. The ensemble is fixed first, since  $U(n)$  is deterministic at each  $n$  and an expectation therefore needs a declared measure: for a scale  $X$ ,*

$$\mathbb{E}_X U = \frac{\sum_{X < n \leq 2X, 2|n} U(n)/n}{\sum_{X < n \leq 2X, 2|n} 1/n}, \quad \Theta_G(X) = (1 - \mathbb{E}_X U) \log X.$$

*Then: (i) [limit law] under  $\mathbb{E}_X$ -sampling,  $U \Rightarrow \text{Exp}(1)$  as  $X \rightarrow \infty$ ; (ii) [canonical ordering deficit]  $\Theta_G(X) \rightarrow \Theta_G > 0$ —the Goldbach sibling of the least-prime deficit of Conjecture 3(ii), produced by the same occupancy mechanism: the ordered primes fill the available representations faster than an exchangeable model, so the first arrival comes early and  $\mathbb{E}_X[U] < 1$ .*

The extremal theory of  $s(n)$  is well developed—Granville, van de Lune and te Riele conjectured  $s(n) = O(\log^2 n \log \log n)$ , and Oliveira e Silva–Herzog–Pardi [12] tabulate first occurrences of minimal Goldbach summands to  $4 \times 10^{18}$  against prime  $k$ -tuple predictions—but the time-changed distributional law (i) and the deficit constant (ii) appear unstated (we could not rule out that the untransformed version of (i) is implicit in the comparisons of [12]). Measured on 2,000 log-sampled  $n \leq 10^8$ :  $\mathbb{E}[U] = 0.796$ , with  $\Theta_G = 3.12 \pm 0.35$  on  $[10^6, 10^7)$  and  $3.42 \pm 0.45$  on  $[10^7, 10^8)$ —positive, stable, and roughly *twice* the least-prime-in-progressions deficit  $\Theta = 1.67$  of Conjecture 3, a

comparison the occupancy expansion there should eventually explain; the Kolmogorov–Smirnov distance to  $\text{Exp}(1)$  (0.088 at this size) is dominated by the deficit displacement itself, as (ii) requires. The derivation route for  $\Theta_G$ , parallel to the occupancy expansion at Conjecture 3: expand the survival probability  $\Pr(s(n) > y) = 1 - \sum_{p \leq y} h_p + \sum_{p_1 < p_2 \leq y} h_{p_1, p_2} - \dots$ , where the pair term requires  $n - p_1$  and  $n - p_2$  simultaneously prime and so carries the Hardy–Littlewood singular series of the shift  $p_2 - p_1$ ; inserting the  $k$ -tuple predictions and integrating against the time change isolates the  $1/\log X$  coefficient—deriving  $\Theta_G$  rather than measuring it is this conjecture’s stated programme.  $\mathfrak{S}(n)$  alone does not encode the between-candidate dependence that the pair term carries. (The (3, 3)-lane refined Goldbach statement—every  $n \equiv 2 \pmod{4}$ ,  $n \geq 6$ , is  $p + q$  with  $p \equiv q \equiv 3 \pmod{4}$ , with the halved singular-series count law, formulated by K. Martin [13]—is included as an attributed calibration benchmark: exhaustively verified to  $10^9$ , count formula matching to 0.2–1.3%; its comparative lane statement is Conjecture 4.)

For sequences whose  $n$ th term has size  $e^{cn}$  the expected number of primes is  $\sum_n \kappa_n/(cn)$ : convergence or divergence of this sum is the entire question, and when it diverges it diverges logarithmically, so the right conjecture has counting function  $\alpha \log x$  with a derived  $\alpha$ .

**Conjecture 6** ( $n^2 + 2^n$ ; sequence is OEIS A064539). *Every prime of the form  $n^2 + 2^n$  with  $n > 1$  has  $n \equiv 3 \pmod{6}$  (elementary), and: (i) infinitely many such primes exist, with counting function  $\asymp \log N$ ; (ii) [candidate law, entanglement-aware form] for a finite set  $S$  of primes  $p \geq 5$  (the primes 2 and 3 are exact in the lane reduction,  $\delta_2 = \delta_3 = 0$ , and  $\text{ord}_2 2$  is undefined) let  $D_S$  be the exact joint density, within the lane  $n \equiv 3 \pmod{6}$ , of  $n$  with  $p \nmid n^2 + 2^n$  for all  $p \in S$ , computed over the full joint period  $\text{lcm}_{p \in S} \text{lcm}(6, p \cdot \text{ord}_p 2)$ , and set  $\kappa_S = 3 D_S / \prod_{p \in S} (1 - 1/p)$ : the factor  $3 = 2 \cdot \frac{3}{2}$  is the exact lane bonus at the primes 2 and 3 (within the lane  $\delta_2 = \delta_3 = 0$ , so each contributes its full  $(1 - 1/p)^{-1}$ ), included in the definition so that  $\kappa_S$  is the complete local constant of the count in (ii-c). The conjecture is stated along the canonical exhaustion  $S_z = \{p \text{ prime} : 5 \leq p \leq z\}$ , and its three layers are logically separate and are asserted separately: (ii-a) the sequence  $\kappa_{S_z}$  converges as  $z \rightarrow \infty$  (a general net over arbitrary finite  $S$  carries an order-of-exhaustion ambiguity we do not need), with limit  $\kappa$ ; (ii-b) whether the limit factors—i.e. whether the exact finite-level factorization observed below persists, so that entanglement is asymptotically absent—is an open question, expressly not counted among this paper’s conjectural assertions, on which (ii-a) takes no position; and (ii-c) the count is  $\sim \sum_{n \leq N, n \equiv 3 \pmod{6}} \kappa / \log(n^2 + 2^n)$ —a genuine further step, since a convergent local density does not automatically govern the global count. The working value  $\kappa^* = 4.2734\dots$  is a hybrid, not a computation of any single  $\kappa_{S_z}$ : the lane factor 3 times the CRT-exact core  $S_{19}$  times independent per-prime factors for  $19 < p \leq 300$ —so  $\kappa^* = \kappa_{S_{300}}$  exactly if and only if the observed exact factorization persists to 300.*

This is not a Bateman–Horn problem: the local conditions at different primes are tied through the orders  $\text{ord}_p 2$ , so the per-prime product  $\prod_p (1 - \delta_p)/(1 - 1/p)$  would tacitly assume their independence. The law is therefore stated through the CRT-exact quantities  $\kappa_S$ , which make no independence assumption. The computation then returned a small surprise in the other direction: through  $p \leq 19$  (joint period 116,396,280) the joint survivor fraction equals the product of the per-prime fractions as an *exact rational identity*, verified in exact arithmetic at every level  $S = \{5 \leq p \leq P\}$ ,  $P \in \{5, 7, 11, 13, 17, 19\}$ . Entanglement is thus absent at all computed levels; whether exact factorization persists for all  $p$ —equivalently, whether the survival events are exactly independent over every joint period, which would itself be a nontrivial equidistribution statement about the orbits of 2—is part of what clause (ii) asks. We separate the robust claim (i) from the candidate law (ii), and note that the eight hits below 6000 are far too few to validate a four-digit constant—the verification below checks consistency, not the constant. The OEIS



records further candidate indices beyond our completed range: 29355, 34653, 57285, 99069, and the probable-prime candidate 1933695. We could not independently re-verify these, but all five satisfy the lane constraint  $n \equiv 3 \pmod{6}$ , and taking them at face value the  $\kappa^*$ -model predicts 2.9 hits in  $(6 \times 10^3, 10^5]$  against 4 reported and 5.9 in  $(6 \times 10^3, 1.94 \times 10^6]$  against 5 ( $z = +0.65$  and  $-0.38$ ): the records extend the consistency check by two decades.

The congruence claim is elementary:  $n$  odd is forced by parity, and for odd  $n$  with  $3 \nmid n$  we have  $n^2 + 2^n \equiv 1 + 2 \equiv 0 \pmod{3}$ . What appears to be a harsh obstruction is, inside the surviving lane, a local bonus:  $\delta_2 = \delta_3 = 0$  contribute the factors  $2 \cdot \frac{3}{2} = 3$  carried explicitly in the definition of  $\kappa_S$ .

*Computational checks* (probable-prime count beyond  $n \approx 80$ ). Hits  $\{3, 9, 15, 21, 33, 2007, 2127, 3759\}$  up to 6000: observed 8, model 8.55,  $z = -0.19$  (over the shorter range  $n \leq 4200$ : 8 vs 8.19,  $z = -0.07$ ); the lane constraint was verified exhaustively (no prime off  $n \equiv 3 \pmod{6}$  to  $n = 3000$ , independently reconfirmed), and the CRT-exact factorization of clause (ii) was verified through  $p \leq 19$  as described at the statement.

**Conjecture 7** (Fermat quotients: the multibase subtorus law and the Wieferich ledger; single-base heuristic is (a) [22], the vertical multibase law apparently new). *With  $q_p(a) = (a^{p-1} - 1)/p \pmod{p}$  (defined for primes  $p \nmid a$ ; all clauses below range over such  $p$  only) and  $q_p = q_p(2)$ , clauses (0)–(v), stated multibase group first: (0) [structure, classical]  $q_p(ab) \equiv q_p(a) + q_p(b) \pmod{p}$  (the Eisenstein–Lerch homomorphism; verified exactly on every tested prime), so for multiplicatively dependent bases the vector  $(q_p(a_1), \dots, q_p(a_r))/p$  is confined to the rational subtorus cut out by the relations; (iv) [multibase equidistribution, the claimed law] for multiplicatively independent bases the vector equidistributes on the full torus as  $p$  varies—the vertical joint law, complementing the fixed- $p$ , varying-base statistics of Ostafe–Shparlinski and Cobeli–Zaharescu—with correlations vanishing and the same LIL calibration as (i) below; (v) [simultaneous Wieferich] the expected count of  $p \leq x$  with  $q_p(2) = q_p(3) = 0$  has convergent sum  $\sum 1/p^2$ : at most finitely many simultaneous Wieferich primes exist (the single-base folklore of this accounting is Conrad’s), and the empirical list is empty to  $10^7$ . Three further clauses on the base-2 quotient alone: (i) [global equidistribution]  $q_p/p$  equidistributes on  $[0, 1)$  with discrepancy at exactly the calibrated random-model scale: in its sharp form,*

$$\limsup_{x \rightarrow \infty} \frac{\sqrt{\pi(x)} D_{KS}(x)}{\sqrt{\log \log \pi(x)}} = \frac{1}{\sqrt{2}},$$

*the Chung–Smirnov law-of-the-iterated-logarithm constant for an i.i.d. uniform sequence—demanding both no more (a bound  $O(\pi(x)^{-1/2})$  would be stronger than the random model supports, the same over-demand that killed the Mertens conjecture) and no less than the model earns: the upper bound alone would be consistent with hidden rigidity, and the matching lower bound asserts the quotients fluctuate like genuine noise; (ii) [shrinking targets] for every fixed  $K \geq 1$ , and uniformly for  $K = K(x) \leq \log x$ ,  $\#\{p \leq x : q_p < K\} \sim \sum_{p \leq x} \min(1, K/p)$ —the fixed- $K$  case down to  $K = 1$  is what governs the zero event and does NOT follow from (i), since  $\{q_p = 0\}$  is a target of shrinking measure  $1/p$ ; (iii) [Wieferich count, fluctuation-calibrated, with the probability space declared]  $W(x) = \#\{\text{Wieferich } p \leq x\}$  is a deterministic function, so its fluctuation clauses are stated in the two legitimate forms, an unqualified CLT for a fixed sequence having no meaning: the deterministic envelope conjecture*

$$\limsup_{x \rightarrow \infty} \frac{W(x) - \log \log x}{\sqrt{2 \log \log x \log \log \log x}} = 1$$

(the iterated-logarithm scale at variance  $V(x) \sim \log \log x$ ; a bound  $O_\varepsilon(V^{1/2+\varepsilon})$  would be weaker than the model's exact prediction), and the almost-sure central limit law, at the correct logarithmic weighting: with  $u = \log \log x$  the variance-time of the count, sample  $u$  with density  $du/u$  on  $[e, L]$ ,  $L = \log \log X$  (equivalently:  $\log u = \log \log \log x$  uniform); then the sampled distribution of  $(W(x) - u)/\sqrt{u}$  converges to  $N(0, 1)$  as  $X \rightarrow \infty$ . The weighting is what makes the clause true, and the two natural alternatives both fail. A block-sampled version on  $[X, X^A]$  fails because only  $\log A = O(1)$  new events enter the block. A uniform-in- $u$  version on  $[1, L]$  fails too, by Brownian scaling: with  $W(e^u) - u \approx B(u)$  and  $u = Ls$ , the self-similarity  $B(Ls)/\sqrt{Ls} \stackrel{d}{=} \tilde{B}(s)/\sqrt{s}$  means the uniform- $u$  empirical distribution converges not to  $\Phi$  but to the random occupation functional  $\int_0^1 \mathbf{1}\{\tilde{B}(s)/\sqrt{s} \leq z\} ds$ —values at proportional times stay correlated, so uniform sampling never averages over enough independent scales. The  $du/u$  weighting is the classical almost-sure-CLT weighting (the continuous analogue of the  $1/k$  weights), for which the model does predict almost-sure convergence of the sampled law to  $\Phi$ . Both clauses are conjectured on the Crandall–Dilcher–Pomerance side of the model choice that clause (iv)'s discussion records.

Two things do not follow. Clause (iii) is not a consequence of (i): a square-root-size error in the aggregate distribution is astronomically larger than  $\log \log x$ , and only the shrinking-target clause (ii) at bounded  $K$  speaks to the zero event. And an  $O(1)$  fluctuation claim would violate this paper's calibration principle of demanding exactly the fluctuations the model earns: with event probabilities  $1/p$  the count has standard deviation  $\sim \sqrt{\log \log x}$ . Nor is any location predicted for a third Wieferich prime: the expected count reaching 3 “at doubly exponential heights” is an expected-count statement whose additive constant is uncalibrated, not a forecast. Consistency data:  $\{1093, 3511\}$  below  $10^8$  against expected  $\approx 2.7$ ;  $\sqrt{n}$  KS = 0.82, 0.82, 0.52, 0.63 at  $x = 10^5, \dots, 10^8$  (bounded, no drift); census at  $K = 100$ : observed 169 against model 161.2.

The multibase clauses were verified at  $10^7$  (664,577 primes): the homomorphism (0) held *exactly* on every spot-checked prime; for the independent pair (2, 3), correlation  $+0.0023 \pm 0.0012$ , a  $20 \times 20$  joint occupancy  $\chi^2$  of 441 on 399 degrees of freedom ( $+1.5\sigma$ ), joint small-quotient census 46 against model 41.2 ( $z = +0.75$ ), and no simultaneous Wieferich prime. Two positions are taken here: the vertical multibase law (iv) sides with the Crandall–Dilcher–Pomerance accounting against Gras's competing probabilistic model (which predicts  $q_p(a) = 0$  *rarer* than  $1/p$ , hence possibly finitely many Wieferich primes even for a single base)—the disagreement is what gives clauses (iii)–(iv) content; and the horizontal literature (fixed  $p$ , base varying: Ostafe–Shparlinski's joint distributions and the Fermat-quotient-matrix statistics of Cobeli–Zaharescu and coauthors) is the attributed neighbour whose vertical counterpart is what this conjecture claims.

**Conjecture 8** (Uniform quantitative de Polignac; (a)-core). (i)  $\pi_d(x) = \#\{p \leq x - d : p, p + d \text{ prime}\} = \mathfrak{S}(d) \text{Li}_2(x) (1 + o(1))$  uniformly for even  $d \leq (\log x)^A$ . (ii) [residual field, with derived covariance kernel] The same primes participate in many pair counts, and the covariance of two of them is carried by triple configurations sharing one prime. For even  $d \neq d'$  define the overlap constant

$$K(d, d') = C_3(0, d, d') + C_3(0, d', d + d') + C_3(0, d, d + d') + C_3(0, |d - d'|, \max(d, d')),$$

where  $C_3$  denotes the Hardy–Littlewood triple constant of the offset set (degenerate or inadmissible triples contributing 0). The field is randomized canonically—covariances of the deterministic cumulative counts are otherwise undefined: for  $t$  uniform on  $[x, 2x]$  and a window length  $H$ , let  $\pi_d(t; H) = \#\{t < n \leq t + H : n, n + d \text{ prime}\}$  and let  $Z_d(t; H)$  be its studentization. The mean



count per window is  $\asymp H/\log^2 x$ , and the regime is set by that ratio (a phase condition: a bare  $H \rightarrow \infty$  does not fix the regime):

$$\text{Cov}_t(\pi_d, \pi_{d'}) \sim \frac{K(d, d') H}{\log^3 x}, \quad \rho(d, d') \sim \frac{K(d, d')}{\sqrt{\mathfrak{S}(d)\mathfrak{S}(d')} \log x},$$

valid for  $\max(d, d') = o(H)$ , which is the standing restriction of this clause and holds throughout the window class fixed below. In general each overlap orientation  $o$ —the in-window displacements  $h_o \in \{0, d, -d', d - d'\}$  and their reflections—enters with multiplicity  $(H - |h_o|)_+$  rather than  $H$ , so that the shared-prime term reads

$$\frac{1}{\log^3 x} \sum_o (H - |h_o|)_+ K_3^{(o)}(d, d'), \quad \sum_o K_3^{(o)}(d, d') = K(d, d'),$$

and an orientation with  $|h_o| > H$  is absent from the window altogether; under  $\max(d, d') = o(H)$  one has  $(H - |h_o|)_+ = H(1 + o(1))$  for every orientation and the displayed form is recovered, so the uniform-in- $d$  statement carries that restriction throughout. Here the shared-prime (triple) term is the first covariance contribution but not the whole of it: the full expansion is

$$\text{Cov}_t(\pi_d, \pi_{d'}) = \frac{K(d, d') H}{\log^3 x} + \frac{1}{\log^4 x} \sum_{|h| \leq H} (H - |h|) K_4(d, d'; h) + \text{error},$$

with  $K_4$  the centred singular series of the off-index four-point sets  $\{0, d, h, h + d'\}$ —the cross analogue of the pinned diagonal average of Conjecture 2—where degenerate offset relations are handled inside the kernels: whenever the four-point set collapses (repeated entries, as at  $h = 0$ ,  $h = d$ ,  $h = -d'$ , or  $d' - d = \pm h$ ) the configuration is a triple or pair and its contribution belongs to the  $K$ -term (or to the mean), so  $K_4$  is defined as the centred singular series of genuinely four-element sets, with special pairs such as  $d' = 2d$  acquiring their extra overlap orientations in  $K(d, d')$ 's census rather than double-counted across kernels. The triple term dominates only where the pinned sum satisfies  $\sum(1 - |h|/H)K_4 = o(\log x)$ ; the dominance range is stated as an explicit function class: for  $H = \lambda(\log x)^\alpha$ , any fixed  $\lambda > 0$  and any  $\alpha \geq 1$  with  $\alpha > A$ —the inequality on the exponents that makes  $\max(d, d') = o(H)$  hold uniformly for all  $d, d' \leq (\log x)^A$ —the per-shift centred average grows only polylogarithmically in  $H$  (the Conjecture-2 scale), so  $\sum(1 - |h|/H)K_4/H = O((\log \log x)^{2+\varepsilon}) = o(\log x)$  and the triple term dominates throughout the polylogarithmic window class; the conjecture asserts the two-term form on exactly that class and takes no position outside it. When  $H/\log^2 x \rightarrow \lambda$  the window counts converge jointly to Poisson laws with these vanishing covariances, and when  $H/\log^2 x \rightarrow \infty$  every fixed finite collection  $\{Z_{d_1}, \dots, Z_{d_r}\}$  is asymptotically jointly normal with independent limiting coordinates: both kernels are finite- $x$  corrections to independence (test below), not nondegenerate limiting kernels. Both distributional conclusions are asserted under an explicit hypothesis beyond the displayed covariances, namely uniform Hardy–Littlewood tuple estimates on the window class fixed above, strong enough to control all higher joint cumulants of the window counts: every joint cumulant of order at least three vanishes in the respective normalizations. The diagonal correction to  $\text{Var}_t(\pi_d)$  is Conjecture 2's pinned average, stated there.

The kernel is verified twice over. Direct window-field measurement ( $x = 10^8$ ,  $H = 10^5$ , 2000 windows, all even  $d \leq 40$ , 190 pairs): the empirical correlation matrix matches the predicted kernel entrywise with correlation 0.86, mean off-diagonal 0.33 observed against 0.37 predicted, the small deficit having the sign of the omitted diagonal corrections, which are also directly

visible as  $\text{Var}/\text{mean} = 0.94$  on average across  $d$ . Second, the deterministic cumulative profile  $z_d(x) = (\pi_d(x) - \mathfrak{S}(d) \text{Li}_2(x)) / \sqrt{\mathfrak{S}(d) \text{Li}_2(x)}$ —the estimator the uniformity verification in (i) uses—shows the same structure at cumulative scale (there  $\rho \sim K \text{Li}_3 / \sqrt{\mathfrak{S}' \text{Li}_2}$ ): mean off-diagonal  $\rho = 0.43$  over all even  $d, d' \leq 60$  at  $x = 10^8$ , hence predicted profile spread  $\sigma^2 \leq 1 - 0.41 = 0.59$  before the diagonal correction, against observed  $\approx 0.27$ , implying a diagonal factor  $\approx 0.5$  of Montgomery–Soundararajan sign and order.

The force of the statement is its uniformity: the singular series takes a thousand different values over  $d \leq 2000$ , spanning a factor of about three between  $d = 2$  and the primorial-rich  $d$ , and the counts must reproduce the entire profile simultaneously with square-root errors. A conspiracy would need to bend a thousand-dimensional vector, not one number. At  $x = 10^8$ : correlation 0.999999, slope 0.99966,  $\max_d |z_d| = 1.80$ ; stressed to  $d \leq 6000$ ,  $\max |z_d| = 2.01$  over three thousand values.

(A further Hardy–Littlewood calibration, on a prime quintuplet constellation, gives observed 15,236 against predicted 15,230.3 at  $4 \times 10^9$ , ratio 1.0004,  $z = +0.05$ .)

**Conjecture 9** (The sub-diffusivity of balanced prime races; apparently new). *Model a balanced race (Conjecture 1(ii)’s symmetric class differences) as the walk of its class-assignment steps, the process defined at event index: enumerate the pattern occurrences  $n_1 < n_2 < \dots$  in the two competing classes, set  $\xi_i = \pm 1$  by class membership of  $n_i$ , and let  $\rho_k(x)$  denote the lag- $k$  sample autocorrelation of  $(\xi_i)$  over the events with  $n_i \leq x$ . Then the walk is measurably sub-diffusive: (i) [decaying-repulsion law] the step autocorrelations are negative at small lags—the race analogue of the Lemke Oliver–Soundararajan consecutive-pattern repulsion [24], which supplies the mechanism—and, since that repulsion is scale-dependent rather than fixed, the conjectured form is a decay law*

$$\rho_k(x) = - \frac{c_k \log \log x + d_k}{\log x} (1 + o(1)),$$

with constants  $c_k, d_k$  in principle derivable from the LOS correlation series (underived here; and the measured values  $\rho_1 \approx -0.037$ ,  $\rho_2 \approx -0.009$  at  $10^9$ —universal across twin and cousin races mod 5 and 8 to within 0.004, each  $\sim 50$  standard errors from zero—are observations at one height, consistent with  $c_1 \approx 0.25$  there, not asymptotic constants); (ii) [long memory, measured] short lags do not exhaust the deficit: the running maxima  $M(x) = \max_{t \leq x} |D(t)| / \sqrt{\text{count}}$  of the four races land at quantiles 0.02–0.17 of the simulated independent-step null (median 2.34 at this span), and still only 0.03–0.23 after correcting for  $\rho_{1,2}$ : the negative correlation persists across lags, so the cumulative diffusivity  $\sigma^2(x) = 1 + 2 \sum_k \rho_k(x)$  sits well below 1 at this height. The model layer is declared:  $(\xi_i)$  is a deterministic, nonstationary family and the  $\rho_k(x)$  are empirical statistics of it, so the Green–Kubo identity  $\sigma^2 = 1 + 2 \sum_k \rho_k$  is invoked under an explicit local-stationarity model hypothesis—the step process restricted to a window  $[x, 2x]$  is modelled as a stationary sequence whose autocorrelations are the measured  $\rho_k(x)$ —and the variance statement can equivalently be taken as a direct definition of  $\sigma^2(x)$  through block averages of  $(S_{n+m} - S_n)^2 / m$  over the events up to  $x$ , a form in which no stationarity is assumed at all. The Green–Kubo sum requires more than the fixed-lag law of (i): the per-lag decay controls each  $\rho_k(x)$  but not the sum unless the decay is uniform in  $k$  with summable tail, since the number of relevant lags may grow with  $x$ . We therefore state the needed hypothesis as part of the clause:  $\rho_k(x)$  obeys the law of (i) uniformly for  $k \leq K(x)$  with  $\sum_{k > K(x)} |\rho_k(x)| = o(\log \log x / \log x)$  for some  $K(x) \rightarrow \infty$ , and—since the tail bound alone leaves the growing head sum uncontrolled—the repulsion constants are summable,  $\sum_k c_k < \infty$  and  $\sum_k |d_k| < \infty$ , so that  $\sigma^2(x) = 1 - 2(\sum_k c_k \log \log x + \sum_k d_k) / \log x (1 + o(1))$  is a genuine consequence rather than an unjustified interchange of limits. The measured correlation

length at  $10^9$  (a few tens of lags, with  $|\rho_k|$  decreasing geometrically in the observable range) is consistent with both hypotheses. Under them,  $\sigma^2(x)$  returns to 1 asymptotically, making sub-diffusivity a finite-height phenomenon of  $\log \log x / \log x$  size, like every other second-order law in this paper; (iii) [registered dichotomy, open] asymptotically the two possibilities are mutually exclusive, and which of them holds is the open question. Either the fixed-lag expansion of (i) extends uniformly in  $k$  with summable constants—in which case the hypotheses of (ii) hold, (ii) applies, and the race is asymptotically diffusive with  $\sigma^2(x) \rightarrow 1$  (running maxima  $\sim \sqrt{2 \log \log x}$ , Darling–Erdős class)—or that expansion fails at long lags, and only in that case can the race inherit the almost-periodic rigidity that Rubinstein–Sarnak-type structure [6] forces on classical races (maxima  $\asymp (\log \log \log x)^A$ , the Montgomery–Ng class). Rigidity therefore requires a failure of the hypotheses of (ii) at long lags, and cannot coexist with them. No explicit formula is known for twin patterns, the two branches differ only beyond any computable height, and we register the dichotomy without choosing in this conjecture. What is conjectured here is therefore the fixed-lag law (i) outright, and the variance law of (ii) conditionally on its stated uniformity and summability hypotheses, which belong to the finite-height model layer and are not asserted asymptotically: asserting them would silently select the diffusive branch of (iii). The pattern-specific selection is made elsewhere—Conjecture 1 asserts the rigid branch for its races, through (i-b) for the contaminated combinations and through the drift-scale nulls of its clause (ii) for the symmetric differences—and consistency between the two conjectures then requires exactly that the hypotheses of (ii) fail at long lags for those races.

The repulsion constants  $\rho_1, \rho_2$  are pattern-independent across our four races—a universality the LOS correlation series should predict quantitatively, and the conjecture’s stated derivation programme—and the suppressed maxima are the first (to our knowledge) direct measurement of race sub-diffusivity, an effect invisible to endpoint statistics. (The twin-gap record benchmark— $G_t(x) \asymp \log^3 x$  with working constant  $1/(2C_2)$ , inside Kourbatov’s extreme-value framework [15]—is included as an attributed calibration benchmark: records to  $4 \times 10^9$  wander in  $[0.41, 0.56] \log^3 x$ , approaching from below on a log log clock, largest observed twin-to-twin gap 5292 after the twin at 2,466,641,069; the Cramér–Granville fragility of such constants [17] is why the constant was always treated as first-order.)

## 4 An open question

**Question 1.** Determine  $\Theta(q)$  of Conjecture 3: derive, from the Hardy–Littlewood correlations or otherwise, the first-order deficit of  $\mathbb{E}_a \text{Li}(p(a, q)) / \varphi(q)$  below its random-model value 1—equivalently, explain quantitatively why the primes in their natural order occupy residue classes faster than any exchangeable resampling of themselves.

## 5 Family laws and second-order refinements

**Conjecture 10** (Uniform quadratic de Polignac; family law apparently unstated, the  $d = 2$  instance is (a), cf. OEIS A080149, [32]). For even  $d \geq 2$  let  $\pi_d^q(N) = \#\{n \leq N : n^2 + 1, n^2 + 1 + d \text{ both prime}\}$  and  $C(d) = C(x^2 + 1, x^2 + 1 + d)$ , with  $I_d(N)$  the integral  $I(N)$  of this pair. Then: (i) [uniform family law] for every fixed  $B > 0$ ,  $\pi_d^q(N) = C(d) I_d(N)(1 + o(1))$  uniformly

over even  $d \leq (\log N)^B$ ; (ii) [registered rate, strictly stronger] there is  $\eta_B > 0$  with

$$\sup_{\substack{2 \leq d \leq (\log N)^B \\ 2 \mid d}} \left| \frac{\pi_d^q(N)}{C(d) I_d(N)} - 1 \right| \ll_B (\log N)^{-\eta_B}$$

—an error clause that no version of Bateman–Horn supplies, registered separately from (i) since its analytic content is independent; (iii) [family statistic: the law of the constants] each local factor of  $C(d)$  depends only on  $d \bmod p$ , and the factors at any finite set of primes are exactly independent as  $d$  varies (CRT), with the passage to the infinite product controlled by the convergent tail-variance sum—which supplies the uniform integrability (via  $L^2$ -boundedness of the normalized partial products) that the moment identities require, since almost-sure convergence alone would not carry means through the limit; so the moments of  $C(d)$  over even  $d \leq D$  converge, as  $D \rightarrow \infty$ , to derived Euler products; in particular

$$\frac{1}{\#\{2 \mid d \leq D\}} \sum_{2 \mid d \leq D} C(d) \rightarrow \bar{C} = 2.7456 \dots, \quad \text{sd}(C(d)) \rightarrow 1.6840 \dots,$$

and the empirical distribution of  $C(d)$  converges to the law of the random product  $\prod_p f_p(U_p)$  with independent uniform residues  $U_p$ . In particular ( $d = 2$ , the classical instance):  $\#\{n \leq N : n^2 + 1, n^2 + 3 \text{ both prime}\} \sim C(2) I(N)$ ,  $C(2) = 2.954014 \dots$

This is the quadratic analogue of the uniform de Polignac law of Conjecture 8: not one count but the whole profile of constants must be matched at once, and the family, unlike its linear parent, has no known prior statement. Every even shift is admissible (for  $p \geq 5$ ,  $\omega(p) \leq 4 < p$ ; the classes mod 3 and 5 vary with  $d$ , which is what makes the  $C(d)$ -profile nonconstant, spanning a factor of about twenty over  $d \leq 300$ , from 0.47 at  $d = 298$  to 9.06 at  $d = 162$ ). At  $d = 2$  the local analysis reads:  $n$  even,  $n \not\equiv 0 \pmod{3}$ ,  $\omega(p) = 2 + \left(\frac{-1}{p}\right) + \left(\frac{-3}{p}\right)$  for  $p \geq 5$ , and the conjecture implies infinitely many twin pairs  $(m+1, m+3)$  with  $m$  a perfect square. Profile verification at  $N = 10^6$  over all 150 even shifts  $d \leq 300$ : correlation 0.99976 between observed and predicted counts, regression slope 1.0022, residual mean  $z = +0.24$  with spread 0.84 and  $\max_d |z| = 2.28$ —the shape-of-agreement standard applied across the family. Clause (iii) was verified by computing both sides independently: the derived Euler-product moments give mean 2.7456 and standard deviation 1.6840, against 2.7434 and 1.6726 measured over the 150 constants with  $d \leq 300$  (ratios 0.9992 and 0.9932)—a family-level statistic whose constants are derived, not fitted. Clause (iii), unlike (i) and (ii), may prove accessible unconditionally by the exact-CRT and martingale route recorded at the cubic family below; the hard content of the conjecture is the uniform law and the rate clause. One question the family raises is recorded as open rather than conjectured: the maximal range  $D(N)$  for which the uniform law can hold (the analogue of the Elliott–Halberstam question for this family—whether there is a sharp transition at some power scale  $N^\delta$  rather than a logarithmic one).

*Computational checks (the classical single instance).* At  $d = 2$  and bound  $10^7$ : observed 32,898 against predicted 32,862.7, ratio 1.0011,  $z = +0.19$ ; the family-profile verification across the 150 shifts  $d \leq 300$  is reported above.

**Conjecture 11** (First occurrences of prime gaps, liminf form; (b)). *Let  $\mathcal{G}$  be the set of even  $g$  that occur as gaps between consecutive primes (all even  $g$ , under Polignac’s conjecture), and for  $g \in \mathcal{G}$  let  $p(g)$  be the prime starting the first such gap. Then: (i)  $\log p(g)/\sqrt{g}$  is bounded*

between positive constants on  $\mathcal{G}$ ; (ii) [liminf law, stated as the dual of the maximal-gap limsup, conditional on the realization hypothesis below]

$$\liminf_{\substack{g \rightarrow \infty \\ g \in \mathcal{G}}} \frac{\log p(g)}{\sqrt{g}} = \sqrt{e^\gamma/2} = 0.943682\dots$$

The inversion of Granville's  $\limsup G(X)/\log^2 X = 2e^{-\gamma}$  into first-occurrence coordinates is not automatic: the limsup controls how large maximal gaps get, not which gap sizes are realized as first occurrences near the envelope. Clause (ii) is therefore conditional on the realization hypothesis: infinitely many  $g \in \mathcal{G}$  first occur inside a maximal-gap event at the Cramér–Granville envelope scale, i.e. with  $g = (2e^{-\gamma} + o(1)) \log^2 p(g)$ . Under that hypothesis the displayed value is forced by algebra; without it only the lower bound  $\liminf \log p(g)/\sqrt{g} \geq \sqrt{e^\gamma/2}$  follows from the envelope. The value is registered as that dual (against the slope-1 laws), not claimed as an independent phenomenon; (iii) [second-order, singular-series dependence] sampling  $g$  uniformly from the realized gaps  $\mathcal{G} \cap [G, 2G]$  and letting  $G \rightarrow \infty$ ,

$$\log p(g) = \sqrt{g} + \frac{1}{2} \log g - \frac{1}{2} \log \mathfrak{S}^*(g) + E_g, \quad \mathfrak{S}^*(g) = \prod_{\substack{p|g \\ p>2}} \frac{p-1}{p-2},$$

where the error  $E_g$  converges in distribution: for the sampled  $g$ ,

$$\Pr[E_g \leq t] \longrightarrow 1 - \exp(-e^{2(t-\mu)}) \quad (G \rightarrow \infty)$$

for some centring constant  $\mu$ —the min-type (reversed) Gumbel law at scale  $\frac{1}{2}$ , i.e. the law of  $\frac{1}{2} \log W$  for  $W$  a unit exponential. The scale is forced by the hazard computation: the cumulative intensity of gap- $g$  events is  $\Lambda_g(y) \asymp (\mathfrak{S}(g)/y^2) \exp(y - g/y)$  in  $y = \log x$  (the factor  $e^{-g/y}$  is the no-intervening-prime probability), so  $\frac{d}{dy} \log \Lambda_g = 1 + g/y^2 - 2/y = 2 + o(1)$  at the centring point  $y_0 \approx \sqrt{g}$ , and the first-passage identity  $\Lambda_g(y) = W$  gives  $E_g = y - y_0 = \frac{1}{2} \log W + o(1)$ . The scale is  $\frac{1}{2}$  and not 1 precisely because the slope of the log-intensity at  $y_0$  is 2 (both the  $y$ -term and the  $g/y$ -term contribute), and the same slope-2 expansion is what produces the  $\frac{1}{2}$  coefficients of the centring. (Gumbel-type modelling of gap extremes and first occurrences is Kourbatov–Wolf territory [15, 25]; the claim of this clause is only the  $\mathfrak{S}^*$ -centring and the scale- $\frac{1}{2}$  first-passage form, not the extreme-value framework. Sampling uniformly over the realized gaps in  $[G, 2G]$  conditions on realization, which the independent first-passage model does not represent; the clause conjectures that this selection does not shift the limiting law, and that assumption is part of what a refutation would refute.) Smooth gaps, whose tuple constant is large, appear earlier by exactly half a logarithm of that constant—the waiting-time first-passage computation makes the coefficient  $-\frac{1}{2}$ , not a free parameter, while  $\mu$  is left as the one unresolved constant of the clause. (The restriction to  $\mathcal{G}$  is needed:  $p(g)$  is not known to be defined for every even  $g$ , and quantifying over all  $g$  would smuggle Polignac's conjecture into the statement. The  $\mathfrak{S}^*$ -clause records that the Hardy–Littlewood factor of  $g$  materially shifts waiting times; it does not move the liminf constant, since  $\log \mathfrak{S}^*(g) = O(\log \log g)$ .)

Clause (iii) was tested on the full first-occurrence table to  $10^9$ : over the 100 even gaps realized in  $[60, 500]$ , regressing  $\log p(g) - \sqrt{g} - \frac{1}{2} \log g$  on  $\log \mathfrak{S}^*(g)$  (with a  $\sqrt{g}$  trend term absorbing the liminf-versus-typical drift) gives coefficient  $-0.466$  against the predicted  $-\frac{1}{2}$ , partial correlation  $-0.24$ : the singular series of the gap is visible in when the gap first appears, at the predicted strength.



The full limit does not follow by inverting Granville’s  $\limsup G(x)/\log^2 x = 2e^{-\gamma}$ : that inversion is invalid, since a limsup envelope for maximal gaps controls only where gaps *can first appear at the earliest*, i.e. a lower envelope for  $p(g)$ —hence a liminf claim, not a limit. Whether  $\log p(g)/\sqrt{g}$  converges at all, and if so whether the limit is Shanks’s classical 1 [16] (as Wolf’s refinement  $p(g) \sim \sqrt{g} e^{\sqrt{g}}$  and the Kourbatov–Wolf residue-class laws [25] also assert) or the envelope value, involves at least three distinct phenomena—whether a given  $g$  occurs near the extreme scale at all, the distribution of exact gap lengths near records, and the difference between record gaps and non-record first occurrences—and is left open here. The two candidate constants differ by 6%, beneath the resolution of any feasible computation: our regression slope over  $g \geq 100$  falls from 1.307 at  $10^9$  to 1.297 at  $4 \times 10^9$ , far above both and decreasing, as it must be for either.

**Conjecture 12** (The Stern lane race; apparently new). *In Stern’s representation problem  $n = p + 2k^2$  ( $n$  odd), race the  $k$ -even and  $k$ -odd lanes. Prime-square contamination of the  $\Lambda$ -weighted lane counts comes from  $n = q^2 + 2k^2$ —the norm form  $x^2 + 2y^2$  of  $\mathbb{Q}(\sqrt{-2})$  with prime  $x$ —and since  $q^2 \equiv 1 \pmod{8}$  while  $2k^2 \equiv 0$  or  $2 \pmod{8}$  by the parity of  $k$ : (i) [class assignment, direct congruence] contamination sits in the  $k$ -even lane iff  $n \equiv 1 \pmod{8}$ , in the  $k$ -odd lane iff  $n \equiv 3 \pmod{8}$ , and for  $n \equiv 5, 7 \pmod{8}$  no square contamination exists at all:  $q^2 + 2k^2 \equiv 1$  or  $3 \pmod{8}$  always, a two-line congruence (genus theory enters only for the converse question of which  $n$  are represented), so the null classes are provable, not conjectured; (ii) [drift law, in the ensemble form of Conjecture 4] Fix the ensemble: for a scale  $X$ , let  $\mathbb{E}_X$  average over odd  $n \in (X, 2X]$  of the given residue class mod 8 with weight proportional to  $1/n$  (the logarithmic sampling measure, as at Conjectures 5 and 4). Define, per class, with  $R_o(n)$  and  $R_e(n)$  the numbers of representations  $n = p + 2k^2$  with  $k$  odd and  $k$  even respectively, the clean-minus-contaminated difference with explicit sign:  $D(n) = R_o - R_e$  for  $n \equiv 1 \pmod{8}$  (even lane contaminated),  $D(n) = R_e - R_o$  for  $n \equiv 3 \pmod{8}$ , and  $D(n) = R_e - R_o$  (sign immaterial) on the null classes  $n \equiv 5, 7 \pmod{8}$ . Set  $D_{\text{sys}}(n) = [\sum_{(q,k): q^2+2k^2=n, q \text{ prime}, k \geq 1} \log q] / \lambda_{\text{an}}(n)$ —the weight is  $\Lambda(q^2) = \log q$ , each representation counted once in the  $k \geq 1$  convention (the ordered-representation factor 2 of Conjecture 4 has no counterpart here, and doubling would be a normalization error)—where  $\lambda_{\text{an}}(n)$  is the analytic lane mean of  $\log p$  over the candidate profile, defined with no reference to which candidates are prime:*

$$\lambda_{\text{an}}(n) = \frac{\int_0^{\sqrt{(n-2)/2}} du}{\int_0^{\sqrt{(n-2)/2}} \frac{du}{\log(n-2u^2)}},$$

the mean of  $\log(n-2u^2)$  under the density proportional to  $1/\log(n-2u^2)$  on the candidate interval (the upper limit keeps  $n-2u^2 \geq 2$ , the convention of  $I(N)$ , avoiding the nonintegrable point  $n-2u^2 = 1$ ), which is the profile a Bateman–Horn heuristic assigns to the prime representations. The definition is out-of-sample: the verification’s empirical lane mean over the observed prime representations is an estimator of  $\lambda_{\text{an}}$ , used as a consistency check, not part of the definition. The law, coefficient-identifying as at Conjecture 4(i): on the contaminated classes

$$\mathbb{E}_X[D - D_{\text{sys}}] = o(\mathbb{E}_X[D_{\text{sys}}]) \quad (X \rightarrow \infty),$$

and on the null classes—where  $D_{\text{sys}}$  vanishes identically, so that a comparison against it would be empty—the statement is made against the positive comparison scale

$$A(X) := \mathbb{E}_X^{(1)}[D_{\text{sys}}],$$

the logarithmic-ensemble mean of the systematic census taken over the contaminated class  $n \equiv 1 \pmod{8}$ : writing  $D_{\text{null}}$  for the difference  $D$  on a null class  $n \equiv 5$  or  $7 \pmod{8}$ , the claim is  $\mathbb{E}_X[D_{\text{null}}] = o(A(X))$ .

This is the calculus applied to a *representation* problem with norm-form contamination—the drift weight is carried by the arithmetic of  $\mathbb{Q}(\sqrt{-2})$  rather than by a polynomial lane—and it comes with two provable null classes. On depth: an *asymptotic* for the contaminating count  $\#\{(q, k) : q^2 + 2k^2 = n, q \text{ prime}\}$ , averaged over  $n$ , is a norm-form-with-prime-argument problem of genuine analytic difficulty (the same species as primes represented with restricted variables);  $D_{\text{sys}}(n)$  is *computed exactly per sample*, so the verification tests the transfer conjecture against the true census, not against an unproved averaged asymptotic. Verified on 2,500 log-sampled odd  $n \leq 10^8$  (mean  $\sim 330$  representations each): contaminated strata pooled  $D = +0.27 \pm 0.33$  against predicted  $D_{\text{sys}} = +0.15$  (at the  $\Lambda(q^2)$  weight); null strata pooled  $-0.22 \pm 0.33$ , consistent with zero (one of the four strata sits at  $-2\sigma$  individually, within a four-test family). (Stern’s exceptional list—the ten known odd integers that are not  $p + 2k^2$ , the largest 5993, with the seven prime members the Stern primes; completeness of the list is conjectural, OEIS A060003—is included as the attributed calibration benchmark anchoring this conjecture: no further exception occurs below  $10^9$  in our computation.)

**Conjecture 13** (Microscopic variance law; (b); the constant is Montgomery–Soundararajan’s [19]). *Fix  $\lambda > 0$  and  $c \in (0, 1]$ , and let  $X$  count primes in  $[t, t + \lambda \log x]$  for  $t$  uniform on  $[x, (1+c)x]$ . Then, under the Hardy–Littlewood  $k$ -tuple conjectures,  $X \Rightarrow \text{Poisson}(\lambda)$  (Gallagher’s argument [18], which is conditional on exactly those conjectures), and—conditional on the quantitative hypothesis spelled out after the statement (uniform Hardy–Littlewood with  $o(h)$  control of the averaged singular series), without which the display below is a registered extrapolation rather than a consequence—at finite  $x$*

$$\frac{\text{Var } X}{\mathbb{E} X} = 1 - \frac{\log(\lambda \log x) + \gamma + \log 2\pi - 1}{\log x} + o\left(\frac{1}{\log x}\right),$$

with  $\gamma + \log 2\pi - 1 = 1.41509\dots$ , the leading correction being independent of the sampling constant  $c$ . [Interpolation clause] More generally, for windows of length  $H = \lambda(\log x)^\alpha$  the same formula with  $\log H$  in place of  $\log(\lambda \log x)$ ,

$$\frac{\text{Var } X}{\mathbb{E} X} = 1 - \frac{\log H + \gamma + \log 2\pi - 1}{\log x} + o\left(\frac{1}{\log x}\right),$$

holds uniformly for  $1 \leq \alpha \leq A$  (any fixed  $A$  with  $H \leq x^{o(1)}$ ): one expression interpolates from Gallagher’s Poisson boundary ( $\alpha = 1$ , deficit  $\asymp \log \log x / \log x$ ) through the mesoscopic Montgomery–Soundararajan regime (deficit  $\asymp \alpha \log \log x / \log x$ ), with no transition other than the slow growth of  $\log H$ .

The deficit arises by inserting Montgomery’s singular-series average  $\sum_{d \leq h} \mathfrak{S}(d)(h - d) = \frac{h^2}{2} - \frac{h}{2}(\log h + \gamma + \log 2\pi - 1) + o(h)$  into the second factorial moment at the microscopic window  $h = \lambda \log x$ . The constant  $\gamma + \log 2\pi - 1$  is Montgomery–Soundararajan’s, was identified in their framework long before this paper, and the statement is a *microscopic extrapolation and numerical test* of their second-moment formula at Gallagher’s boundary—not a newly discovered second-order law. The hypothesis must also be quantitative: what is required is not the qualitative  $k$ -tuple conjectures but uniform Hardy–Littlewood estimates for all shifts up to  $O(\log x)$ , control



of the average singular series at the  $o(h)$  level, and bounds for prime powers, endpoint weights, and the variation of  $\log t$  across the sampling range. Three further qualifications are built into the statement: Gallagher’s Poisson law is conditional, not a theorem about primes; the sampling measure for  $t$  is specified, since different weightings shift lower-order terms; and the error is claimed only as  $o(1/\log x)$ . The naive claim  $\text{Var}/\mathbb{E} = 1$  fails at every accessible scale by 18–28%; the corrected law matches with no fitted parameter, and the observed residual  $\approx +0.003$  is consistent with (but not evidence for) a genuine  $1/\log^2 x$  term. (Numerical precedent: Sanchis-Lozano [26] fitted the Montgomery–Soundararajan variance shape at  $x \sim 10^8$  in the mesoscopic regime  $h \gg \log N$ ; the microscopic statement and test appear to be new.)

| $\lambda$ | $x = 10^9$ |           | $x = 4 \times 10^9$ |           |
|-----------|------------|-----------|---------------------|-----------|
|           | observed   | predicted | observed            | predicted |
| 1/2       | 0.8241     | 0.8189    | —                   | —         |
| 1         | 0.7910     | 0.7854    | 0.7998              | 0.7960    |
| 2         | 0.7527     | 0.7520    | 0.7670              | 0.7646    |
| 4         | 0.7201     | 0.7185    | —                   | —         |

The residual  $\approx +0.003$  sits at the scale of a  $1/\log^2 x$  term, consistent with the hedged reading at the statement.

**Conjecture 14** (The null-mechanism race; apparently new). *For the quadratic twin pairs of Conjecture 10 at  $d = 2$ , square contamination is algebraically impossible:  $n^2 + 1 = q^2$  has no solution with  $n \geq 1$ , and  $n^2 + 3 = q^2$  only at  $(n, q) = (1, 2)$ . The surviving classes of  $n \bmod 5$  are  $\{0, 1, 4\}$  (the single escape  $n = 2$ , the pair  $(5, 7)$  in the killed class 2, is one bounded term), and the classes 1 and 4 have identical singular series (elementary, via CRT). Let  $\pi_a^q(x)$  count solutions with  $n \equiv a \pmod{5}$ ,  $D(x) = \pi_1^q(x) - \pi_4^q(x)$ , and define the normalized race process  $Y(x) = D(x)/\sqrt{\pi_1^q(x) + \pi_4^q(x)}$ . Then, in contrast to the twin race of Conjecture 1, this race is driftless: (i)  $\mathcal{M}_x(D(t) \log^2 t / \sqrt{t}) \rightarrow 0$  at the drift-scale normalization of Conjecture 1(i)—the scale at which the contaminated races converge to nonzero constants—so this is the genuine negation of Conjecture 1(i) for this pair (the weaker normalization  $\mathcal{M}_x(Y) \rightarrow 0$  with  $Y = D/\sqrt{\pi^q}$  would not exclude a drift at the contamination scale, hence would not make this race a control; the form stated excludes exactly that); (ii) [occupation law, conjectured directly, with a separately falsifiable local hypothesis] the process is defined precisely at event index: let  $n_1 < n_2 < \dots$  enumerate the  $n$  with  $n^2 + 1$  and  $n^2 + 3$  both prime and  $n \equiv 1$  or  $4 \pmod{5}$ , set  $\xi_i = +1$  if  $n_i \equiv 1$  and  $-1$  if  $n_i \equiv 4$ , and let  $S_N = \xi_1 + \dots + \xi_N$  (so  $D(x) = S_{N(x)}$  with  $N(x)$  the number of such events to  $x$ ). The sequence  $(\xi_i)$  is deterministic, so the hypothesis is stated for a randomly sited window rather than for  $S_N$  itself. Fix a growth scale  $N_0$  and a window length  $N = N(N_0)$  with  $N = o(N_0)$ , and choose the starting index  $M$  of the window according to the logarithmic measure on  $[1, N_0]$ :*

$$\Pr[M = m] = \left( \sum_{1 \leq m' \leq N_0} \frac{1}{m'} \right)^{-1} \frac{1}{m}, \quad 1 \leq m \leq N_0.$$

*The hypothesis is an invariance principle of ASIP type, stated in triangular-array form, for the induced random path  $(S_{M+j} - S_M)_{0 \leq j \leq N}$  under that sampling law: on a probability space carrying  $M$  and a standard Brownian motion  $B$ ,*

$$S_{M+j} - S_M = B(j) + O(N^{1/2-\delta}) \quad \text{uniformly for } 0 \leq j \leq N,$$

with probability tending to one as  $N_0 \rightarrow \infty$ , for some  $\delta > 0$ ; the sampling law of  $M$  changes with  $N_0$ , so the coupling is asserted per  $N_0$  with vanishing exceptional probability rather than almost surely on a single space. The logarithmic sampling law is the same clock in which the occupation statement below is read, and randomizing the window start is what supplies the deterministic step sequence with a probability space. The weak functional limit theorem— $S_{\lfloor N\tau \rfloor} / \sqrt{N}$  converging weakly to standard Brownian motion on  $\tau \in [0, 1]$ —is retained as the weaker, separately falsifiable consequence of the windowed ASIP. The occupation law itself, however, is not a consequence of the windowed hypothesis, and is conjectured directly. The windowed path  $S_{M+j} - S_M$  is recentred at the window start, and recentring erases exactly the absolute level that occupation measures: a deterministic bias  $b(n) = \sqrt{n}/\log n$  added to  $S_n$  shifts the level by  $\sqrt{n}/\log n$  while contributing window increments of size  $\sim j/(2\sqrt{n} \log n) \ll \sqrt{j}$ , invisible to the windowed ASIP at every window scale, yet it would drive the occupation fraction to 1. A global coupling  $D(t) = B(N(t)) + o(\sqrt{N(t)})$  with  $N(t) \asymp t/\log^2 t$  would suffice, but for the deterministic sequence  $(\xi_i)$  no randomization is available that makes such a coupling well posed, so we do not hypothesize it. The Brownian computation is instead the motivation that identifies the limit and the constant: for a coupled walk, in logarithmic time  $u = \log t$  the Lamperti reduction  $Z(s) = e^{-s/2} B(e^s)$  is a stationary, ergodic Ornstein–Uhlenbeck process, and the logarithmic occupation measure of leadership converges almost surely to  $\frac{1}{2}$ . The conjectured law is that conclusion itself:

$$\frac{1}{\log x} \int_2^x \mathbf{1}_{\{D(t) > 0\}} \frac{dt}{t} \rightarrow \frac{1}{2},$$

with fluctuations of the Gaussian scale  $\sqrt{\log 2 / \log x}$  (the occupation-indicator covariance  $\frac{1}{2\pi} \arcsin e^{-u/2}$  integrates to  $\frac{1}{2} \log 2$ ; note that the integrated covariance for the sign average  $2\mathbf{1} - 1$  is four times this, its standard deviation twice, the occupation fraction itself carrying  $\sqrt{\log 2 / \log x}$ ); the classical arcsine spread survives only in the natural event clock, i.e. for occupation fractions of  $\{B(m) > 0, m \leq N(x)\}$ . A persistent systematic leader—a log-occupation stuck near 0 or 1 beyond the stated Gaussian scale—is the refutation.

The clock matters here. An arcsine-distributed limit for the logarithmic occupation itself is incompatible with the change of clock: the arcsine law lives in the natural time of the walk, whereas under  $dt/t$  averaging the Lamperti transform is ergodic and forces the limit  $\frac{1}{2}$ . The distinction is what makes the statistic informative: the observed log-occupation 0.21 at  $10^7$  sits 1.4 null standard deviations below  $\frac{1}{2}$  (scale  $\sqrt{\log 2 / \log x} \approx 0.21$  there)—still consistent with the driftless null—whereas read as an arcsine draw the same number would carry no information at all.

This is the negative control that completes the mechanism family of Conjectures 1 and 4: those races carry a predicted drift *because* prime squares can infiltrate specific residue classes of specific patterns; here the quadratic form blocks squares identically, so the same model predicts *nothing*—a falsifiable contrast. If a persistent drift were found in this race, the mechanism story behind the family would be wrong no matter how well the positive cases fit. The absence of *this* drift source does not by itself prove the race driftless—oscillatory zero terms, higher prime powers, and endpoint corrections are other conceivable sources—so clause (i) is a conjecture under the same zero-oscillation hypothesis as Conjecture 1, not a consequence of the square-blocking algebra alone. What the algebra supplies is the *contrast*: whatever residual model one adopts, it must produce drift in Conjectures 1 and 4 and none here. Computational checks at  $10^7$ : 32,898 pairs, class counts 10,917 against 10,981 ( $D = -64$ , i.e.  $-0.43$  noise units), log-mean normalized drift  $-0.46$  (consistent with 0 at one noise unit), log-occupation of leadership 0.21—1.4 null standard

deviations below  $\frac{1}{2}$  on the clock and constant of clause (ii), within the fair race's 95% band. (The quadratic triple  $n^2 + \{1, 3, 7\}$ ,  $C = 10.64599 \pm 0.006$ , is included as a calibration benchmark—observed 3,963 against predicted 3,998.6 at  $10^7$ ,  $z = -0.56$ , independently recomputed to  $3 \times 10^7$ ; the longer pattern  $n^2 + \{1, 3, 7, 9, 13\}$  has been recorded in the OEIS since 2000, so the tuple family has prior presence and only its singular-series evaluation is plausibly new.)

*Computational checks.* The race data are reported at the statement above; independent recomputation of this conjecture's calibration member, the quadratic triple, gave 10.647706, with primes to  $10^9$ —inside our stated wobble—and an independent count to  $3 \times 10^7$  with final ratio 1.0059.

**Conjecture 15** (The cubic shift family: the distribution of its constants; instance-level law apparently unstated). *For non-cube  $a \geq 1$  let  $C(a) = C(x^3 + a)$ . Only  $p \equiv 1 \pmod{3}$  moves the product, and the local root count there takes three values as  $a$  varies mod  $p$ :  $\omega = 3$  with probability  $(p-1)/3p$  ( $-a$  a nonzero cube),  $\omega = 1$  with probability  $1/p$  (the case  $p \mid a$ , easy to overlook and essential to the mean-value identity below), and  $\omega = 0$  with probability  $2(p-1)/3p$ ; for  $p \equiv 2 \pmod{3}$  the cube map is a bijection and  $\omega = 1$  identically. Then: (i) [mean-value one, elementary lemma]  $\mathbb{E}_a[\omega(p)] = 3 \cdot \frac{p-1}{3p} + \frac{1}{p} = 1$  exactly at every  $p$ , so the derived mean of  $C(a)$  is exactly 1—the cubic analogue of the Korevaar–te Riele mean-value-one theorem for linear prime-pair constants, here a one-line computation at each finite level; passing the mean through the infinite product needs more than almost-sure convergence, and the needed ingredient is on hand: the convergent variance sum makes the partial products an  $L^2$ -bounded martingale (each factor has mean 1 and is independent of the earlier ones), so they are uniformly integrable and the limit law has mean exactly 1, not merely limit-of-means 1; (ii) [limit law] the empirical law of  $\{C(a) : a \leq A, a \text{ a non-cube}\}$  converges, as  $A \rightarrow \infty$ , to the law of the random Euler product  $\prod_{p \equiv 1(3)} f_p(\xi_p)$  with the three-state local variables  $\xi_p$  above—independent at any finite set of primes exactly, by CRT, with the infinite product's tail controlled by the convergent variance sum (unlike the quadratic family of Conjecture 10(iii), no normalization is needed)—with derived standard deviation  $0.2762\dots$  computed from the full three-state law; (iii) [uniformity]  $\#\{n \leq N : n^3 + a \text{ prime}\} = C(a)I_a(N)(1 + o(1))$  uniformly over non-cube  $a \leq (\log N)^B$ .*

The framework here is Kowalski's theory of averages of Euler products and singular-series distributions [11], which proves such limit laws for  $k$ -tuple singular series; the one-parameter cubic shift family, its exact mean-one lemma, and the short-range uniformity appear unstated, and we present (ii) as an instance-level law inside that framework rather than a new mechanism. Computational checks: derived moments (mean exactly 1, sd 0.2762) against the 294 non-cube shifts  $a \leq 300$ : empirical mean 1.0215 (+1.5 profile standard errors), sd 0.2481 (finite-family tail correlations account for the sd gap: for  $p > a_{\max}$  the residues of  $a$  are not yet equidistributed); uniformity profile over 57 shifts at  $N = 2 \times 10^5$ : mean  $z = +0.09$ , spread 0.72, max  $|z| = 2.31$ . (The single instance  $a = 2$ ,  $C = 1.298435 \pm 0.0003$ , verified at  $10^7$  with  $z = +0.32$ , is included as the calibration member; no statement of Bunyakovsky type is proven for any cubic, the nearest theorem being Heath-Brown's  $x^3 + 2y^3$  [3], which is what keeps the whole family conjectural.)

*Computational checks (the classical single instance).* At  $a = 2$  and bound  $10^7$ : observed 287,956 against predicted 287,784.8, ratio 1.0006,  $z = +0.32$ ; the family-profile verification across the 57 shifts is reported above.

**Conjecture 16** (Hardy–Littlewood's Conjecture F as a family; (a), [1], cf. [23]). *For odd  $A$  let  $Q_A(N) = \#\{n \leq N : n^2 + n + A \text{ prime}\}$  and  $C(A) = C(x^2 + x + A)$ . Then: (i) for every fixed  $B > 0$ ,  $Q_A(N) = C(A)I_A(N)(1 + o(1))$  uniformly over odd  $1 \leq A \leq (\log N)^B$ ; (ii) [window*

field] the residual field obeys the analogue of Conjecture 8(ii), kernel derived and probability space specified: for  $t$  uniform on  $[N, 2N]$  and window length  $H \leq N^{o(1)}$ , let  $Q_A(t; H) = \#\{t < n \leq t + H : n^2 + n + A \text{ prime}\}$  (mean  $\asymp H/2 \log N$ , which sets the regime). Two members share the index window, so with  $C(A, A') = C(x^2 + x + A, x^2 + x + A')$ ,

$$\text{Cov}_t(Q_A, Q_{A'}) \sim [C(A, A') - C(A)C(A')] \frac{H}{4 \log^2 N},$$

with  $C(A, A') = 0$  (hence negative correlation) for the locally exclusive pairs. This is the same-index contribution only: the full covariance adds the off-index pinned sum

$$\frac{1}{4 \log^2 N} \sum_{0 < |h| \leq H} (H - |h|) [C(f_A(x), f_{A'}(x + h)) - C(A)C(A')],$$

the two-parameter analogue of Conjecture 2's  $G(H)$ , whose evaluation is this family's open component. (The order of this sum matters: the joint event here is two single prime values at distinct indices, of probability  $\asymp 1/4 \log^2 N$ , unlike Conjecture 8's pair-of-pairs events at  $1/\log^4 x$ . At that order the off-index sum, which grows like a pinned  $G$ -average over  $\asymp H$  shifts against the same-index term's single  $H$ , is generically the larger component, so the numerical cancellation of the same-index kernel establishes nothing about the full cross-member covariance, and the localization of the observed profile deficit to the diagonal is an open question rather than an inference.) For  $H/\log N \rightarrow \lambda$  the counts are jointly Poisson with local exclusions, and for  $H/\log N \rightarrow \infty$  each fixed finite collection of studentized counts is asymptotically jointly normal with independent limiting coordinates, all correlations of size  $\asymp 1/\log N$  making both kernels finite- $N$  corrections to independence. As at Conjecture 8(ii), both distributional conclusions are asserted under an explicit hypothesis beyond the displayed covariances—uniform Hardy–Littlewood tuple estimates over the family, strong enough that every joint cumulant of order at least three of the studentized counts vanishes in the respective normalizations—since the covariance kernels alone do not determine the limit laws. (Uniformity over a range growing with  $N$  is the substantive claim; uniformity over any fixed finite set of  $A$  is logically automatic from the individual asymptotics.) Among odd  $A \leq 199$ , Euler's  $A = 41$  has the largest constant,  $C(41) = 6.6395463 \dots$  (Cohen's high-precision value [31]).

Numerical evaluation of the kernel (all odd  $A, A' \leq 99$  at  $N = 10^6$ : 1,225 pairs, of which 289 locally exclusive) returned a structurally clean answer: the positive correlations of admissible pairs and the negative correlations of exclusive pairs *cancel*, mean off-diagonal  $\rho = -0.003$ , so the derived *same-index* cross-member kernel is null. With the off-index sum at its correct order (larger, not smaller, than the same-index term), this cancellation does not license the conclusion that the members are asymptotically independent or that the observed profile variance  $\approx 0.37$  must be diagonal; the natural reading remains that each  $\text{Var}(Q_A)$  is sub-Poisson through within-sequence pair correlations, the quadratic-family analogue of the Montgomery–Soundararajan deficit (Conjecture 13), but the off-index cross terms are an unevaluated competitor—and part of the spread is truncation noise in the constants themselves (the  $\pm 10^{-3}$  wobble moves each  $z_A$  by  $\sim 0.3$ ). Deriving the diagonal law and the off-index pinned sums is the family's remaining open component, and the more important one; the numerical cancellation of the *same-index* cross-member kernel is a finite computation over  $A \leq 99$ , which localizes the deficit off the same-index diagonal without proving that the averaged kernel or the off-index contributions vanish—an analytic classification of the sign of  $C(A, A') - C(A)C(A')$ , and the evaluation of the

two-parameter pinned sums, remain open. At  $N = 10^6$ : mean ratio 1.0006, standard deviation 0.0027,  $\max |z| = 1.92$ , rank correlation 0.99988; for  $A = 41$ , observed 261,080 against predicted 261,017.6 ( $z = +0.12$ , using Cohen’s constant). On precision: a truncated Euler product for  $C(41)$  at cutoff  $2 \times 10^5$  reads 6.64092—wrong in the fourth significant decimal—because these conditionally convergent products drift at the  $10^{-3}$  scale over practical cutoffs. Quoted digits for all quadratic-family constants in this paper are limited by the stated truncation wobble, not by the last printed digit. The entire  $10^8$ -bit primality dataset compresses to one product formula per  $A$ .

**Conjecture 17** (Twin-member Goldbach, orientation-resolved; the basis conjecture and the integral kernel are Dubner’s [9]; the orientation decomposition apparently new). *For even  $n$  let  $R_T(n)$  count ordered  $(a, b)$ ,  $a + b = n$ , with  $a$  and  $b$  both members of twin pairs, each ordered role (lower/upper) counted. Each of the four role assignments is a 4-form linear system in  $t = a$  with root set  $\{0, \pm 2\} \cup \{n, n \pm 2\}$ -type mod each  $p$ , hence its own  $n$ -dependent singular series  $\mathfrak{S}_4^{(o)}(n)$ , and*

$$R_T(n) = \left[ \sum_o \mathfrak{S}_4^{(o)}(n) \right] \int_5^{n-5} \frac{dt}{\log^2 t \log^2(n-t)} (1 + o(1)),$$

*the four constants computed exactly per  $n$  from the coincidence pattern of the root sets (which of  $n, n \pm 2, n \pm 4$  each small prime divides). The error term is stated as  $(1 + o(1))$ , not  $(1 + O(1/\log n))$ : our own measurements show a second-order deficit whose  $1/\log n$  coefficient is not yet pinned down (below). Two precision notes: the asymptotic runs over varying  $n$ , so what is assumed is a Hardy–Littlewood estimate for 4-form systems uniform in the  $n$ -dependent coefficients—strictly stronger than the fixed-system conjecture—and the number 5, the unique prime that is simultaneously a lower twin member (of  $(5, 7)$ ) and an upper one (of  $(3, 5)$ ), is double-counted consistently by the ordered-role convention, affecting finitely many representations per  $n$  and no asymptotic.*

Dubner’s paper already contains the integral kernel (his quotient  $\mu_4$ ) and the qualitative basis conjecture; the content claimed here is the orientation-resolved singular series and its  $n$ -profile. Two exact identities compress the four orientations: the substitution  $t \mapsto n - t$  exchanges the two summand roles, forcing  $\mathfrak{S}_4^{(\text{lu})}(n) = \mathfrak{S}_4^{(\text{ul})}(n)$  identically (each of the (ll) and (uu) systems being self-dual under it), and the affine substitution  $t \mapsto t + 2$  gives  $\mathfrak{S}_4^{(\text{uu})}(n) = \mathfrak{S}_4^{(\text{ll})}(n - 4)$ . So only one function of  $n$  per symmetry class is free, and the profile is really two-dimensional; determining the average of  $\sum_o \mathfrak{S}_4^{(o)}(n)$  over  $n$  in closed form is the conjecture’s registered identity-hunting programme. Verified on 150 log-sampled  $n \leq 10^8$ : the profile *shape* is confirmed at log-log correlation 0.9993 across two decades, while the *level* runs at 0.87 of prediction on  $[10^6, 10^7]$  and 0.81 on  $[10^7, 10^8]$ —a systematic deficit of second-order size ( $\approx 3.4/\log n$  at the top range, the same order as the 18–28% finite-height Poisson deficit of Conjecture 13), which we record as its open component rather than absorb into a fitted constant: either the 4-tuple second-order correction accounts for it, or the orientation model needs repair, and the two are distinguishable at  $10^{10}$ . The *direction* of the trend is itself the anomaly: a genuine  $1 - c/\log n$  correction shrinks as  $n$  grows, so a ratio moving *away* from 1 across the decades indicates either a different error shape or a normalization problem, and settling which is part of what the open component must resolve. (The attributed benchmark stands inside the statement: every even  $n \geq 4210$  is a sum of two twin members, with the conjectured complete list of 35 exceptions (OEIS A007534)—verified to  $10^8$ , re-swept to  $10^9$ , independently re-verified by a different algorithm, the same 35 exceptions each time.)



## 6 Instances and structural companions

**Conjecture 18** (Contamination in prime triplets; apparently new). *The contamination calculus of Conjecture 1(v) extends to  $k$ -tuples, and the triplet pattern  $(n, n+2, n+6)$  is its first instance beyond pairs. Of the three configurations placing a prime square inside the pattern, two die algebraically for  $q > 3$ — $(q^2, q^2+2, q^2+6)$  by  $3 \mid q^2+2$ , and  $(q^2-6, q^2-4, q^2)$  because  $q^2-4 = (q-2)(q+2)$  is composite (at  $q=3$  the factorization is trivial,  $q^2-4=5$  is prime, and the single bounded exceptional configuration  $(3, 5, 9)$  occurs—one term, harmless to every asymptotic)—leaving the single doubly-thinned survivor  $(q^2-2, q^2, q^2+4)$ , which requires  $q^2-2$  and  $q^2+4$  simultaneously prime and forces  $q \equiv \pm 2 \pmod{5}$  for  $q \neq 5$  (else  $5 \mid q^2+4$ ; at  $q=5$  all three conditions do hold, the configuration being  $(23, 25, 29)$ , but its start  $23 \equiv 3 \pmod{5}$  lies outside the triplet start classes  $\{1, 2\}$ , so the single term is harmless). Triplet starts lie in classes  $n \equiv 1, 2 \pmod{5}$  (single bounded exception  $(5, 7, 11)$ , start  $\equiv 0$ ), and the surviving configuration has start  $q^2-2 \equiv 2$ : class 2 is contaminated and class 1 leads, in the drift-scale form of Conjecture 1(i):*

$$\mathcal{M}_x \left( \frac{(\pi_3(t; 5, 1) - \pi_3(t; 5, 2)) \log^3 t}{\sqrt{t}} \right) \rightarrow c_3 := \lim_{x \rightarrow \infty} \frac{T_3(x) \log^3 x}{\sqrt{x}}, \quad T_3(x) = \frac{1}{\log^3 x} \sum_{\substack{q \leq \sqrt{x}, q \neq 5 \\ q^2-2, q^2+4 \text{ prime}}} \log(q^2-2) \log q \log$$

( $c_3$  exists and is the Bateman–Horn constant of the triple  $(q, q^2-2, q^2+4)$ , since the census sum is  $\sim c_3 \sqrt{x}$ ; convergence of the logarithmic mean at this normalization is the same Rubinstein–Sarnak-type structure hypothesis as at Conjecture 1(i); a normalization at the noise scale,  $\mathcal{M}_x((D - T_3)/\sqrt{\pi_3})$  with  $D$  the class difference displayed above and  $\pi_3$  the total triplet count to  $t$ , would be vacuous for the coefficient, exactly as at the pair races), with drift  $\asymp \sqrt{x}/\log^3 x$ , one logarithm weaker than the pair races, and drift-to-noise  $\asymp \log^{-3/2} x$  against their  $\log^{-1}$ , a scale the calculus itself predicts and the verification must respect. The mechanism clause and the averaging clause are asserted separately, as at Conjecture 1(i).

This is the calculus applied at tuple level: the configuration census, the double thinning, and the mod-5 class assignment are all forced, and the prediction was derived before the data were taken. At  $10^9$ : classes 189,837 against 189,670 ( $D = +167$  on the predicted side;  $T_3 = +25$  against noise 616, so the endpoint is uninformative exactly as the  $\log^{-3/2}$  law requires), leadership log-density 0.65 (+0.8 null standard deviations at the occupation constant of Conjecture 14(ii)—directionally consistent, sharpness unavailable at this height by the calculus’s own accounting). (The Chernick chain  $\{p, 2p-1, 3p-2\}$ —the population behind the universal-form Carmichael numbers, quantified by Dubner [10]—is included as an attributed calibration benchmark:  $C = 2.858249$ , observed 125,379 against predicted 125,429.4 at  $3 \times 10^8$ ,  $z = -0.14$ .)

*Computational checks.* The triplet-race data and the Chernick-chain calibration count are reported at the statement above.

**Conjecture 19** (The contamination matrix: sexy pairs with two surviving orientations; apparently new). *For the pattern  $(n, n+6)$ , both prime-square orientations survive, on complementary classes of  $q \pmod{5}$ —the first matrix instance of the calculus (every pair race so far had exactly one): orientation  $A = (q^2-6, q^2)$  requires  $q^2-6$  prime, forcing  $q \equiv \pm 2 \pmod{5}$  for  $q \neq 5$  (at  $q=5$  the value 19 is prime, but its start  $19 \equiv 4 \pmod{5}$  lies outside the sexy start classes  $\{1, 2, 3\}$ ), and lands in start class 3 ( $\pmod{5}$ ), class 3 ( $\pmod{8}$ ); orientation  $B = (q^2, q^2+6)$  requires  $q^2+6$  prime, forcing  $q \equiv \pm 1 \pmod{5}$  (the term  $q=5$  falls outside the start classes), and lands in class*



1 (mod 5), class 1 (mod 8). Sexy starts occupy classes  $\{1, 2, 3\} \bmod 5$  and  $\{1, 3, 5, 7\} \bmod 8$ ; with  $T_A(x) = \frac{1}{\log^2 x} \sum_{q^2-6 \text{ prime}, q \neq 5} \log q \log(q^2 - 6)$ ,  $T_B(x) = \frac{1}{\log^2 x} \sum_{q^2+6 \text{ prime}, q \neq 5} \log q \log(q^2 + 6)$ , and  $c_A, c_B$  their drift-scale limits ( $c_A = \lim T_A \log^2 x / \sqrt{x}$ , the Bateman–Horn constant of  $(q, q^2 - 6)$ , likewise  $c_B$  for  $(q, q^2 + 6)$ ), the predicted drift vector, in the drift-scale limiting-log-mean sense of Conjecture 1(i) (an  $\mathcal{M}_x$ -over- $\sqrt{\pi}$  normalization would be coefficient-blind), is, writing  $\pi(a)$  for the number of sexy starts up to  $t$  in class  $a$  of the modulus in question: mod 5:  $\mathcal{M}_x((\pi(2) - \pi(3)) \log^2 t / \sqrt{t}) \rightarrow c_A$ ,  $\mathcal{M}_x((\pi(2) - \pi(1)) \log^2 t / \sqrt{t}) \rightarrow c_B$  (class 2 clean); mod 8:  $\mathcal{M}_x((\frac{1}{2}(\pi(5) + \pi(7)) - \pi(3)) \log^2 t / \sqrt{t}) \rightarrow c_A$ ,  $\mathcal{M}_x((\frac{1}{2}(\pi(5) + \pi(7)) - \pi(1)) \log^2 t / \sqrt{t}) \rightarrow c_B$  (classes 5, 7 clean, mutually symmetric, with  $\mathcal{M}_x((\pi(5) - \pi(7)) \log^2 t / \sqrt{t}) \rightarrow 0$ ). Two independently computable drift constants, one certified null class per modulus. The mechanism clause and the averaging clause are asserted separately, as at Conjecture 1(i).

The matrix structure—two orientations feeding disjoint classes on complementary  $q$ -classes, with an interior null—is what distinguishes this from every single-orientation race, and it tests the calculus beyond its balanced one-mechanism cases; the class assignments were verified independently before the data were taken. At  $10^9$  the predicted components ( $T_A = 191$ ,  $T_B = 110$ ) sit far inside the noise ( $\approx 2,100$ ), as the  $1/\log x$  law forces; all four measured components and the 5–7 control lie within one noise unit of their predictions (trivially, at this height), and the four leadership log-densities (0.48, 0.56, 0.53, 0.73) sit at  $(-0.1, +0.3, +0.1, +1.2)$  null standard deviations (occupation constant 0.18): three at the null, the mod-8  $B$ -component mildly positive on the predicted side. The matrix is *registered and consistent*, with sharpness unavailable below  $\sim 10^{14}$  by the calculus’s own accounting. (The pair  $\{p, p^2 - 2\}$ , OEIS A062326,  $C = 3.383216$ , the input that sets Conjecture 1’s drift scale, is included as an attributed calibration benchmark: verified at  $10^7$ ,  $z = -1.31$ , independently recomputed to  $10^9$ ; a refutation of its infinitude would remove the twin race’s predicted mechanism.)

*Computational checks.* The sexy-pair race data are reported at the statement above; independent recomputation of this conjecture’s calibration member, the pair  $\{p, p^2 - 2\}$ , gave 3.383227, with primes to  $10^9$ —inside our stated wobble—and an independent count to  $3 \times 10^7$  with final ratio 0.9965.

**Conjecture 20** (Fibonacci–Lucas twins: the convergent side stress-tested; the object is OEIS A080327). (i) [rank-disjointness, elementary, with its exception stated] every odd prime factor  $r$  of  $L_p$  has rank of apparition exactly  $2p$  (from  $z(r) \mid 2p$ ,  $z(r) \nmid p$ , and  $z(r) = 2$  being impossible), and every prime factor of  $F_p$  ( $p \neq 5$  prime) has rank exactly  $p$ : the odd divisor pools of the two numbers are disjoint. The prime 2 is the unique exception: 2 divides both  $F_p$  and  $L_p$  precisely when  $p = 3$  ( $F_3 = 2$ ,  $L_3 = 4$ ;  $\gcd(F_p, L_p) = 2$  iff  $3 \mid p$ )—the restriction to odd divisors is essential, the clause being false at  $p = 3$  without it. Disjoint pools remove shared-prime correlation but not all correlation: dependence can still enter through the order structure of the recurrence, which is why (iii) matters; (ii) [finiteness] only finitely many primes  $p$  have  $F_p$  and  $L_p$  simultaneously prime; (iii) [calibration clause, quantitative] the naive joint accounting—probability  $(e^\gamma \log p / (p \log \phi))^2$  per index—assigns prior mass about  $1.4 \times 10^{-2}$  to the whole index range beyond  $10^4$ , in which the catalogued index  $p = 148091$  falls (OEIS A080327; both  $F_p$  and  $L_p$  are probable primes—numbers of roughly 30,950 digits, with Lucas numbers proved prime only up to index 56003—so everything recorded here is conditional on both probable-prime classifications, for which no Baillie–PSW pseudoprime is known: no unconditional refutation is claimed from uncertified numbers). That rare-event datum is recorded descriptively, with no significance claim attached to it: the model was examined after the event was known, and no test protocol was

*fixed in advance. A single surprising event is not a strict refutation, and it does not by itself estimate the size of the correction: an underestimated constant, genuine cross-dependence, and a mis-calibrated finite-range tail are competing explanations that one event cannot separate. What the datum does bear on (conditionally as above) is not the model but the list: it is fatal to any Goldilocks completeness list; the conjectural content of (ii) is that even the corrected accounting has convergent sum.*

This conjecture is the collection’s designated stress test of the convergent Borel–Cantelli template that Conjecture 21 exemplifies. Our scan to  $p \leq 10^4$  found exactly the joint indices  $\{5, 7, 11, 13, 17, 47\}$  with expected further mass  $1.4 \times 10^{-2}$ , while the catalogued index 148091, at which  $F_p$  and  $L_p$  are both probable primes, lies beyond that range. We therefore conjecture finiteness *without* a completeness clause: the Goldilocks-maximal list that is defensible for factorial twins is indefensible here. The lesson, in the language of Grantham–Granville [7], is that recurrence-sequence constants need their local corrections *before* tail masses are trusted. The single-sided calibration (probable-prime counts:  $F_p$  and  $L_p$  leave the deterministic range at very small  $p$ ): 25 Fibonacci prime indices  $\leq 10^4$  against the naive screening prediction 29.2 (the finite sum of the declared per-index hazard over primes to  $10^4$ ) ( $z = -0.8$ , a mild deficit, not significant on its own); 29 Lucas prime indices  $\leq 10^4$ ; the corrected  $c_F$  needs a Granville-type patch, and clause (iii) quantifies the same need at the joint level.

*Computational checks.* Joint scan and single-sided calibration reported at the statement (Fibonacci prime indices  $\{3, 5, 7, \dots, 9311, 9677\}$ , 25 of predicted 29.2,  $z = -0.78$ ; Lucas indices 29; joint  $\{5, 7, 11, 13, 17, 47\}$  to  $10^4$ )—probable-prime counts at all but the smallest indices, the Fibonacci and Lucas values leaving the deterministic range almost at once.

**Conjecture 21** (Factorial twins; the uniqueness core is (a), OEIS A088054; clauses (i) and (iii) are ours). (i) [*window rigidity, elementary*] for  $n \geq 4$  and  $2 \leq |a| \leq n$  the number  $n! + a$  is composite (any prime factor  $p$  of  $a$  satisfies  $p \leq n$ , so  $p \mid n! + a$ , and  $n! + a > n \geq p$ ); the two small anomalies  $2! - 2 = 0$  and  $3! - 3 = 3$  are the only exceptions, which is why the clause is restricted to  $n \geq 4$ . Hence among all offsets of bounded size the only candidates for primality near  $n!$  are  $a = \pm 1$ : the twin question below is the unique bounded-offset constellation question at  $n!$ , and every admissible offset set is a subset of  $\{-1, +1\}$ . (ii) [*uniqueness; (a)*] Only finitely many  $n$  have  $n! - 1$  and  $n! + 1$  simultaneously prime, and the complete list is  $n = 3$ : i.e.  $6 = 3! = 3\#$  is the unique factorial lying between twin primes. (iii) [*joint fluctuation model*] the two single-sided counts  $F_{\pm}(N) = \#\{n \leq N : n! \pm 1 \text{ prime}\}$  are each  $\sim e^{\gamma} \log N$  and, under the independent-indices model stated below, asymptotically independent, with  $F_+ - F_-$  normalized by  $\sqrt{2e^{\gamma} \log N}$  asymptotically standard normal. The probability space is declared: the clause is about a random model in which the events  $E_n^{\pm} = \{n! \pm 1 \text{ prime}\}$  carry the Caldwell–Gallot marginal hazards  $e^{\gamma}/n$  with an unspecified dependence structure, and the Gaussian limit is asserted under three explicit hypotheses on that structure: the same-index bound  $\sum_{n \leq N} |\text{Cov}(\mathbf{1}_{E_n^+}, \mathbf{1}_{E_n^-})| = o(\log N)$ , which pins the variance at  $2e^{\gamma} \log N (1 + o(1))$  and which the joint law  $\sim (e^{\gamma}/n)^2$  derived below supplies with a convergent sum—without it the remaining hypotheses leave the second cumulant free, and a model with  $\Pr(E_n^+ \cap E_n^-) = e^{\gamma}/2n$  halves the limiting variance; the cross-index covariance bound  $\sum_{m < n \leq N} |\text{Cov}(\mathbf{1}_{E_m}, \mathbf{1}_{E_n})| = o(\log N)$  over the four sign pairs (the variance itself is  $\asymp \log N$ ), and—since pairwise control alone cannot preclude higher-order dependence—the higher-cumulant condition  $\kappa_r(F_+ - F_-) = o((\log N)^{r/2})$  for each fixed  $r \geq 3$ , the method-of-moments sufficient condition. None of the three is imported from an independence model; whether they hold for the arithmetic dependence across factorial indices is the open component of this clause.

Each event separately has Caldwell–Gallot probability  $\sim e^\gamma/n$  [4] and divergent sum (the  $n! + 1$  side is the classical law retained as calibration at Conjecture 2); the *joint* event has probability  $\sim (e^\gamma/n)^2$ , whose sum converges. This is the suite’s exemplar of the convergent side of the Borel–Cantelli dichotomy—the accounting that predicts finitude rather than infinitude. The at-a-common-index independence in (iii) has a derivation: a generic twin pair near  $n!$  would carry the Hardy–Littlewood twin factor  $2C_2 \prod_{2 < p \leq n} \frac{p-2}{p-1} / \prod_{2 < p \leq n} (\frac{p-1}{p})^2$ -type couplings, but here the screening (coprimality to all  $p \leq n$ ) is *deterministic*—both neighbours always pass it—and the twin coupling factor and the reciprocal sieve factors cancel to first order, leaving the joint probability  $\sim (e^\gamma/n)^2$  with no singular-series correlation; what remains open, and is recorded in (iii), is dependence *across* distinct indices  $n$ , whose divisibility structures are nested. That open component has a concrete first computation, which we register as the conjecture’s programme: for  $m < n$  and a prime  $p > n$ , the residues  $m! \bmod p$  and  $n! \bmod p$  are deterministically linked by  $n! = m! \cdot (m+1) \cdots n$ , so the covariance of the events  $p \mid m! \pm 1$  and  $p \mid n! \pm 1$  is an explicit character-sum average over the multiplier  $(m+1) \cdots n$ ; summing it over  $p$  either bounds the cross-index dependence at  $o(1/n)$ —vindicating the independent-indices model—or exhibits genuine coupling, and the computation is finite at each height. Verified for  $n \leq 700$  ( $F_+ = 15$ ,  $F_- = 16$ : difference  $-1$  against noise scale  $\approx 4.8$ ); the model-expected number of further twin examples beyond 700 is  $e^{2\gamma}/700 \approx 4.5 \times 10^{-3}$  (computed tail). The defensible core is finiteness; the exact one-element list is the Goldilocks-maximal form—strictly stronger than the tail estimate can guarantee. On priority: the exact uniqueness conjecture is on the public record, OEIS A088054 stating that 3 is conjecturally the intersection of A002981 and A002982, so clause (ii) is attributed there and the conjecture’s claimed content is clauses (i) and (iii).

The primorial analogue— $p\# \pm 1$  both prime only for  $p \in \{3, 5, 11\}$ —is due to Lillie [5], whose abstract gives both the  $O(n^{-2})$  joint probability and the prediction that there are three instances in total; it is included here as an attributed calibration benchmark, independently verified to  $p \leq 4000$ .

*Computational checks* (probable-prime count for  $n$  beyond about 26). Exhaustive to  $n \leq 700$ : the only factorial twin is  $n = 3$ ; individually,  $n! + 1$  is prime for  $n \in \{2, 3, 11, 27, 37, 41, 73, 77, 116, 154, 320, 340, 399, 4$  and  $n! - 1$  for  $n \in \{3, 4, 6, 7, 12, 14, 30, 32, 33, 38, 94, 166, 324, 379, 469, 546\}$ —two divergent single-sided laws straddling their shared constant while the joint count stops at one, the convergent accounting in action. (The primorial companion, attributed to Lillie [5]: exhaustive to  $p \leq 4000$ , twins exactly  $\{3, 5, 11\}$ , with  $p\# + 1$  prime for eleven  $p$  against the Caldwell–Gallot prediction 12.4,  $z = -0.40$ ; again a probable-prime count above the deterministic range.)

**Theorem 1** (Cube obstruction). *For  $k \geq 2$ , the cube  $k^3$  is representable as  $p + j^3$  with  $p$  prime,  $j \geq 1$ , if and only if  $3k^2 - 3k + 1$  is prime. Consequently the set of integers not representable as  $p + k^3$  is infinite: the non-representable cubes alone have counting function  $\sim x^{1/3}$ . (That the full exceptional set is  $\sim x^{1/3}$  follows only when combined with Conjecture 22(iii), and is conjectural.)*

*Proof.*  $k^3 - j^3 = (k - j)(k^2 + kj + j^2)$ ; for  $1 \leq j \leq k - 2$  both factors exceed 1, so the difference is composite. The only candidate is  $j = k - 1$ , giving  $3k^2 - 3k + 1$ . Since  $3k^2 - 3k + 1$  is composite for a density-one set of  $k$  (all but  $O(K/\log K)$  of  $k \leq K$ , by an upper-bound sieve of Brun or Selberg type applied to the quadratic’s prime values), all but a vanishing proportion of cubes are unrepresentable.  $\square$

The restriction to non-cubes in the next conjecture is therefore essential: the exception set of  $n = p + k^3$  over *all*  $n$  is infinite, and the cubes are exactly what makes it so. The effect is visible immediately in the data—all 412 exceptions found in  $[10^8, 10^9]$  are perfect cubes—and it

is invisible to a Borel–Cantelli sum taken over all  $n$ , since the cubes have density zero. The cube lane is not lost, only relocated: by Theorem 1 it is governed by the primality of  $3k^2 - 3k + 1$ , which is a Bateman–Horn family in its own right and appears as such in Conjecture 22.

**Conjecture 22** (The boundary trichotomy for polynomial ladders; the divisibility principle is classical and its cubic case is Cunningham’s cuban observation (OEIS A002407); the family classification is apparently new). *For  $F \in \mathbb{Z}[x]$  with  $\deg F \geq 2$  and positive leading coefficient, and  $m > j \geq 1$ ,  $(m - j) \mid F(m) - F(j)$  (textbook), and the divided difference  $(F(m) - F(j))/(m - j)$ —whose leading form is then positive on the cone  $m > j \geq 1$ —exceeds 1 outside an effectively bounded region; so  $F(m) - F(j)$  prime forces  $m - j = 1$  apart from finitely many exceptional pairs (for the family below the exceptional set is provably empty). Both hypotheses are necessary: for  $\deg F = 1$  the cofactor can be identically 1 and the collapse fails, and for negative leading coefficient the divided difference tends to  $-\infty$  on the cone, so the “exceeds 1” clause would be false as stated (one may equivalently keep general sign and read the claim through  $|F(m) - F(j)|$ ). Every representation problem  $F(m) = p + F(j)$  in this range thus collapses to the boundary polynomial  $D_F(m) = F(m) - F(m - 1)$ . The content is the classification of the boundary lanes. For  $F = x^3 + cx$ ,  $c \geq 0$ : (i) [trichotomy, elementary]  $D_F = 3m^2 - 3m + 1 + c$  and the lane is dead-parity for  $c$  odd ( $D_F$  always even), dead-3-adic for even  $c \equiv 2 \pmod{3}$  ( $3 \mid D_F$  always), and admissible exactly for  $c \equiv 0, 4 \pmod{6}$ ; (ii) [uniform boundary law] over the admissible lanes,  $\#\{m \leq M : D_F(m) \text{ prime}\} \sim C(D_F) I(M)$  uniformly for  $c \leq (\log M)^B$ —a Bateman–Horn family whose members are indexed by the ladder classification; (iii) [attributed core] the case  $c = 0$  is the cuban ladder: Cunningham’s classical observation that a prime difference of cubes forces consecutive arguments, with the non-cube exceptional set of  $n = p + k^3$  finite (Hardy–Littlewood’s  $E_3(X) = O(1)$  [1]).*

The reducible-boundary phenomenon is part of the same classification:  $F = x^4$  has  $D_F = (2m - 1)(2m^2 - 2m + 1)$ —the boundary lane itself factors, which is precisely the composite- $k$  obstruction of Conjecture 23 seen from the ladder side, and  $F = x^3 + x^2$  gives  $D_F = m(3m - 1)$ , dead by reducibility. Verified: the collapse is algebraically empty for  $x^3 + cx$  ( $c \geq 0$ ), the trichotomy checked to  $m = 5000$  on all  $c \leq 12$ , and the two smallest admissible new lanes counted at  $M = 10^6$ :  $c = 4$  ( $C = 2.12956$ ,  $z = +0.27$ ) and  $c = 6$  ( $C = 2.68954$ ,  $z = -0.92$ ). (Empirically the non-cube exception census of (iii) by decade is 4, 27, 168, 763, 2011, 2808, 1181, 88 up to  $10^8$ , peaking at the sixth decade and collapsing thereafter, largest found 78,526,384.)

**Conjecture 23** (The power-obstruction ladder: an elementary structural proposition with a Bateman–Horn corollary; apparently new as a family). *For  $k \geq 2$  and  $m \geq 2$ , write  $D_k(m) = m^k - (m - 1)^k$ , and call  $m^k$  representable if  $m^k = p + j^k$  for some prime  $p$  and  $j \geq 1$ . Then: (i) [theorem] for composite  $k$ , no  $k$ -th power is representable: if  $r$  is a proper divisor of  $k$  with  $r > 1$  (such  $r$  exists precisely because  $k$  is composite; the excluded value  $r = 1$  would allow only the useless factor  $m - j = 1$  at  $j = m - 1$ ), then for every  $1 \leq j < m$  the difference  $m^k - j^k$  has the proper factor  $m^r - j^r > 1$  with cofactor  $> 1$ ; (ii) [elementary] for prime  $k$  (including  $k = 2$ ),  $m^k$  is representable if and only if  $D_k(m)$  is prime; and  $D_k$  is irreducible for prime  $k$  (a root  $\alpha$  has  $(\alpha/(\alpha - 1))^k = 1$  with  $\alpha/(\alpha - 1) \neq 1$ , so  $\zeta = \alpha/(\alpha - 1)$  is a primitive  $k$ -th root of unity and  $\mathbb{Q}(\alpha) = \mathbb{Q}(\zeta)$  has degree  $k - 1 = \deg D_k$ , forcing irreducibility; explicitly  $D_k(x) = (x - 1)^{k-1} \Phi_k(x/(x - 1))$ . The forcing  $j = m - 1$  needs only  $k \geq 2$ ); (iii) [conjecture] for each prime  $k \geq 3$  the representable lane follows Bateman–Horn ( $k = 2$  is a theorem, the prime number theorem on the lane  $2m - 1$ ):  $\#\{m \leq M : D_k(m) \text{ prime}\} \sim C_k \int_2^M dt / \log D_k(t)$ .*

The dividing line is prime versus composite  $k$ , not odd versus even: for *odd* composite  $k = rs$  the factorization of clause (i) applies verbatim, so that  $D_9(m)$ , for instance, is *never* prime. This was checked directly for  $k = 9, 15, 21, 25, 27, 33$ —no prime value of  $D_k$  occurs up to  $m = 2000$ , and the predicted divisor  $D_r(m) \mid D_k(m)$  is present identically—and it is the same algebraic mechanism, seen from the ladder side, that Theorem 1 exhibits for cubes. The family generalizes the cubic case  $k = 3$  of Conjecture 22.

*Computational checks.* The even-rung theorem was checked numerically to  $10^8$  (no fourth power is  $p + j^4$ ); the three verifiable Bateman–Horn lanes  $k = 2, 3, 5$  (constants 2 exactly, 3.36181, and  $3.67770 \pm 0.007$ , the quartic product computed by brute-force root counts to  $10^5$ ) all sit within  $|z| < 0.6$ : at bound  $10^7$  the lane  $k = 2$  gives observed 1,270,606 against predicted 1,270,902.8 (ratio 0.9998,  $z = -0.26$ ); at  $10^6$  the lane  $k = 3$  gives 126,826 against 126,641.6 (ratio 1.0015,  $z = +0.52$ ); and at  $8 \times 10^5$  the lane  $k = 5$  gives 56,925 against 57,021.3 (ratio 0.9983,  $z = -0.40$ ).

**Conjecture 24** (The alternating cyclotomic chain; apparently new). *There are infinitely many primes  $p$  such that, writing  $u = \Phi_3(p) = p^2 + p + 1$ , the three integers*

$$p, \quad u = p^2 + p + 1, \quad \Phi_6(u) = u^2 - u + 1 = p^4 + 2p^3 + 2p^2 + p + 1$$

*are all prime, with  $\#\{p \leq x\} \sim C_6^{\text{ch}} I(x)$  for the Bateman–Horn system  $\{x, x^2 + x + 1, x^4 + 2x^3 + 2x^2 + x + 1\}$ ,  $C_6^{\text{ch}} = 3.6143 \pm 0.011$ . The first chains are (2, 7, 43) and (3, 13, 157).*

This is the repunit analogue of a Cunningham chain: a prime base  $p$ , its length-3 repunit  $u = (p^3 - 1)/(p - 1)$ , then  $(u^3 + 1)/(u + 1)$ . The chain must *alternate* between  $\Phi_3$  and  $\Phi_6$ : the naive iterate  $\{p, \Phi_3(p), \Phi_3(\Phi_3(p))\}$  is *inadmissible*—if  $p \equiv 2 \pmod{3}$  then  $u \equiv 1 \pmod{3}$ , so  $3 \mid \Phi_3(u)$  always, while  $p \equiv 1 \pmod{3}$  forces  $3 \mid u$ , so the admissibility check rejects the naive chain outright. The uniqueness claim is made over a specified construction space: among length-3 chains  $\{x, \Phi_j(x), \Phi_k(\Phi_j(x))\}$  with  $j, k \in \{3, 6\}$  *extending the repunit sub-chain* (i.e.  $j = 3$ ), the mod-3 arithmetic above forces  $k = 6$ : alternation is the unique admissible continuation. (Dropping the repunit anchor admits one sibling, the pure- $\Phi_6$  chain  $\{x, \Phi_6(x), \Phi_6(\Phi_6(x))\}$  on the complementary lane  $x \equiv 1 \pmod{3}$ ; since  $\Phi_6(x) = \Phi_3(x - 1)$ , its second element is again a repunit value, and we regard the classification of all admissible words in the two-letter alphabet  $\{\Phi_3, \Phi_6\}$ —which words are obstruction-free at every prime, and whether they form a regular language—as a natural open programme.) The length-2 sub-chain  $\{p, p^2 + p + 1\}$  is the classical cyclotomic Sophie Germain statement, included here as an attributed calibration benchmark ( $C = 1.521661 \pm 0.0005$ , catalogue value 1.5217315..., OEIS A188596; verified at  $z = -1.06$  at  $10^7$ ): infinitely many prime bases have prime repunits of length 3, the base-independent home of the phenomenon whose base-2 shadow is the Mersenne sequence. The length-3 chain stated here is, as far as we can find, unstated, and its quartic layer makes it the highest-degree system in the suite.

*Computational checks.* At bound  $10^7$ : observed 1,362 against predicted 1,357.2, ratio 1.0035,  $z = +0.13$ . The chain census by decade is 4, 11, 41, 224, 1362 at  $N = 10^3, \dots, 10^7$  (a probable-prime count for the quartic layer beyond  $p \approx 2 \times 10^6$ ;  $z$  from +0.40 to +0.13, no drift), the quartic layer’s singular series computed by brute-force root counts to  $10^5$ ; the retained sub-chain benchmark  $\{p, p^2 + p + 1\}$  stands at 33,661 against 33,855.5 at  $10^7$  ( $z = -1.06$ ). Independent recomputation of that calibration member, the Sophie Germain sub-chain, gave 1.521730, with primes to  $10^9$ —inside our stated wobble—and an independent count to  $3 \times 10^7$  with final ratio 0.9962.



**Conjecture 25** (Twin cyclotomic bases: the complete quadratic family; no prior trace found). *For the three cyclotomic polynomials of degree 2 (indices  $k$  with  $\varphi(k) = 2$ , i.e.  $k \in \{3, 4, 6\}$ ), consider  $\Phi_k(n)$  and  $\Phi_k(n+1)$  simultaneously prime. Then: (i) [parity obstruction, elementary] for  $k = 4$  one member of the pair  $(n^2 + 1, n^2 + 2n + 2)$  is even for every  $n$ , so the only prime pair is  $n = 1$ , giving  $(2, 5)$ —the same obstruction that forbids consecutive  $n^2 + 1$  primes; (ii) [collapse, elementary]  $\Phi_6(x) = \Phi_3(x - 1)$  identically, so the  $k = 6$  pair at index  $n$  is the  $k = 3$  pair at index  $n - 1$ : the family contains exactly one nontrivial statement; (iii) [conjecture, the live instance] there are infinitely many  $n$  for which the length-3 repunits in bases  $n$  and  $n + 1$  are simultaneously prime:  $\#\{n \leq N : \Phi_3(n) = n^2 + n + 1 \text{ and } \Phi_3(n + 1) = n^2 + 3n + 3 \text{ both prime}\} \sim C_5 I(N)$ , with  $C_5 = C(x^2 + x + 1, x^2 + 3x + 3)$ .*

Both forms in (iii) are odd for every  $n$  (each is  $\Phi_3$  of an integer), and mod 3 each vanishes on exactly one class ( $n \equiv 1$  and  $n \equiv 0$  respectively), so the pair is admissible with  $\omega(3) = 2$ . Unlike an arbitrary shifted pair, the companion here is structurally forced: it is the same cyclotomic value at the consecutive base, making this the natural “twin” statement inside the cyclotomic-chain family of Conjecture 24. The family analysis (i)–(ii) is what a family *lift* of the statement uncovers, and both clauses were found by the admissibility computation itself: the  $k = 4$  branch was rejected as inadmissible at  $p = 2$  at singular-series time (both-odd count to  $10^6$ : zero, prime pairs:  $n = 1$  only), and the  $k = 6$  run returned counts *identical* to  $k = 3$ , which is the numerical shadow of the polynomial identity in (ii).

*Computational checks.*  $C_5 = 2.964239 \pm 0.002$ ; at bound  $10^7$ , observed 33,274 against predicted 32,976.7, ratio 1.0090,  $z = +1.64$ .

## Part II

# Structural conjectures in five programmes

*Added to the deposit: 3 August 2026.*

The twenty-five conjectures of this part are organized by mechanism rather than by the contamination calculus of Part I. Each names a canonical object and averaging law, identifies an arithmetic, geometric, spectral, or adelic source, and states a first decisive theorem and a failure mode.

## 7 Programme I: connected prime-pattern fields

Let  $H \subset \mathbb{Z}$  be finite and admissible, and write

$$\Lambda_H(n) = \prod_{h \in H} \Lambda(n + h), \quad Y_H(x, L) = \sum_n w\left(\frac{n - x}{L}\right) (\Lambda_H(n) - \mathfrak{S}(H)),$$

where  $w$  is a fixed smooth compactly supported function and  $\mathfrak{S}(H)$  is the Hardy–Littlewood singular series  $\prod_p (1 - \nu_H(p)/p)(1 - 1/p)^{-|H|}$ , with  $\nu_H(p)$  the number of residue classes occupied by  $H$  modulo  $p$ . The window position  $x$  is sampled from  $[X, 2X]$  with density  $dx/(x \log 2)$  and  $X^\varepsilon \leq L \leq X^{1-\varepsilon}$ . For translated motifs  $H_i + t_i$ , their *exact overlap type* records every equality among shifted prime constraints. The connected singular series  $\mathfrak{S}^c(K_1, \dots, K_r)$  is obtained by Möbius inversion over set partitions of the indexed tuple  $(K_1, \dots, K_r)$ .



**Conjecture 26** (Connected motif generating functional). *Fix admissible motifs  $H_1, \dots, H_m$  and put*

$$\mathcal{Z}_{X,L}(\mathbf{z}) = \mathbb{E}_x \exp \left( \sum_{i=1}^m z_i Y_{H_i}(x, L) \right).$$

*For every fixed total degree  $R$ , the degree- $\leq R$  Taylor polynomial of  $\log \mathcal{Z}_{X,L}$  has a uniform asymptotic expansion indexed by connected exact-overlap diagrams of at most  $R$  translated motifs. A diagram whose coincidence hypergraph connects the  $r$  translated sets  $H_{i_1} + t_1, \dots, H_{i_r} + t_r$  contributes, at the leading order of its stratum,*

$$W_\tau(L; \mathbf{t}) \mathfrak{S} \left( \bigcup_{j=1}^r (H_{i_j} + t_j) \right) \prod_v (\log X)^{m_v-1},$$

*where  $W_\tau(L; \mathbf{t}) = L \int_{\mathbb{R}} \prod_{j=1}^r w(u + t_j/L) du$  is the archimedean window-overlap weight of the diagram and  $m_v$  is the multiplicity of the prime constraint at  $v$ : any set partition that separates a coincidence loses at least one factor of  $\log X$ , so the Möbius alternation leaves the single-block moment as the leading term of a connected diagram. The same alternation cancels the leading terms of disconnected diagrams, whose contribution is carried by the connected singular series of their disjoint blocks at strictly smaller logarithmic order, reducing for the fully disjoint type to  $\mathfrak{S}^c(H_{i_1} + t_1, \dots, H_{i_r} + t_r)$  with no logarithmic enhancement. The log powers are therefore partition dependent and are never factored across the alternation, and every prime-power counterterm is included in the local weight attached to its multiplicity. Ordering terms lexicographically by powers of  $L$  and  $\log X$ , the remainder after any fixed truncation is smaller than the last retained scale, uniformly in the mesoscopic range.*

**Significance.** This is a multitype connected field theory for prime constellations. It simultaneously organizes covariance, odd moments, mixed motifs, overlap singularities, and lower-order prime-power effects. Montgomery–Soundararajan and Kuperberg provide the closest one-field moment frameworks, while constrained singular-series sums provide the nearest local input [19, 34, 35]. The new content is the complete connected multitype functional. The first decisive theorem is the full third-cumulant formula for two distinct pair motifs. A failure would reveal a term of  $\log \mathcal{Z}_{X,L}$  at a scale indexed by no connected exact-overlap diagram, a correlation source among the motif fields  $Y_{H_i}(x, L)$  invisible to the Möbius alternation over set partitions.

**Conjecture 27** (Complete overlap-renormalization filtration). *Fix a finite motif family  $H_1, \dots, H_m$ . Apply the degree-two specialization of Conjecture 26 and order all of its connected covariance strata by their asymptotic scales  $s_1 \succ s_2 \succ \dots$  (lexicographically in powers of  $L$ ,  $\log X$ , and the renormalized disjoint scales). In the mesoscopic range scales sharing the same total power of  $L$  and of logarithms differ only by powers of  $\theta = \log L / \log X$  and merge into a single stratum whose limiting form is polynomial in  $\theta$ . The filtration, the graded convergence, and the eigenvalue asymptotics below are asserted for every fixed  $\theta \in (\varepsilon, 1 - \varepsilon)$ . Recursively subtract the earlier strata and let  $A_\nu = A_\nu(\theta)$  be the limiting covariance form at scale  $s_\nu$ , and put*

$$\mathcal{W}_\nu = \bigcap_{\mu < \nu} \ker A_\mu.$$

*Each  $A_\nu$  is asserted to induce a nondegenerate form on the graded quotient  $\mathcal{W}_\nu / \mathcal{W}_{\nu+1}$  for every such  $\theta$ , so that  $\mathcal{W}_{\nu+1} = \mathcal{W}_\nu \cap \ker A_\nu$  and the filtration does not collapse. For every fixed truncation*

of this ordered diagram expansion, the spectral projections of the covariance matrix converge to the associated graded spaces  $\mathcal{W}_\nu/\mathcal{W}_{\nu+1}$ , and every eigenvalue whose first nonzero term occurs within the truncation is asymptotic to  $s_\nu$  times a positive eigenvalue of the induced form  $A_\nu$ . Vectors annihilating all retained forms pass canonically to the next connected diagram scales of Conjecture 26.

Consequently every asymptotic variance scale of a finite motif statistic is generated by a connected overlap or renormalized disjoint diagram from Conjecture 26, and no scale external to that diagrammatic expansion occurs. The filtration is invariant under replacing the motif family by one with isomorphic exact-overlap incidence algebra and identical local Euler weights.

**Significance.** The claim classifies all mechanisms by which a finite linear combination of prime motifs can become rigid. It predicts a canonical renormalization filtration, not merely one variance formula, and gives an algorithm for constructing observables that expose deeper arithmetic strata. A counterexample would identify a genuinely new source of rigidity outside overlap and disjoint correlation. The first theorem is a three-motif example in which one overlap form has a rational kernel and the next form is nondegenerate on it.

**Conjecture 28** (Regularized mesoscopic local–spectral trace formula). *Fix motifs  $H_i, H_j$ , a Schwartz function  $\phi$  with  $\phi(0) = 0$ , an even cutoff  $\eta \in C_c^\infty(\mathbb{R})$  with  $\eta = 1$  near the origin, and an even mollifier  $\kappa \in C_c^\infty(\mathbb{R})$  of integral one. Let  $Q_{ij}^{\text{loc}}(\phi; L)$  be the exact-overlap-renormalized singular-series form*

$$\sum_{t \in \mathbb{Z}} \left[ L \int_{\mathbb{R}} w(u) w(u + t/L) du \right] \phi(t/L) \mathfrak{S}^c(H_i, H_j + t).$$

Let  $\Lambda_{T,\eta}^\sharp$  be the Weil explicit-formula distribution in the logarithmic variable, with each nontrivial zero of ordinate  $\gamma$  multiplied by  $\eta(\gamma/T)$  and with the pole, trivial-zero, archimedean, and prime-power terms kept as separately named summands. For  $\delta > 0$ , put

$$\Lambda_{T,\eta,\delta}^\sharp = \Lambda_{T,\eta}^\sharp *_{\log} \kappa_\delta, \quad \kappa_\delta(u) = \delta^{-1} \kappa(u/\delta).$$

This is a smooth function, so products of its shifted copies are unambiguous. Define

$$Q_{ij}^{\text{spec}}[\eta, \kappa] = Q_{ij}^{\text{spec}}(\phi; X, L; T, \delta, \eta, \kappa)$$

by inserting these mollified copies into the smoothed motif covariance, applying the same exact-overlap Möbius projection as on the local side, and subtracting the explicitly specified pole, diagonal, trivial-zero, archimedean, and prime-power counterterms before the mollifier is removed. In normal form: expand each inserted copy into its separately named summands together with the mollified zero sum and expand the covariance multilinearly;  $Q_{ij}^{\text{spec}}$  is then the zero–zero block of that expansion with its diagonal term removed, the subtracted counterterms being exactly the cross terms containing a pole, trivial-zero, archimedean, or prime-power summand, together with the diagonal term of the zero–zero block, so that two readers form the same object.

For every sufficiently large fixed  $A, B$ , take

$$T = (X/L)(\log X)^A, \quad \delta = (L/X)(\log X)^{-B}.$$

Then

$$Q_{ij}^{\text{spec}}(\phi; X, L; T, \delta, \eta, \kappa) - Q_{ij}^{\text{loc}}(\phi; L) = o(V_{ij}(X, L))$$

uniformly for  $X^\varepsilon \leq L \leq X^{1-\varepsilon}$  and for  $\phi$  in bounded Schwartz sets, where  $V_{ij}$  is the first nonzero disjoint covariance scale. The same limit holds jointly for every fixed matrix of motifs and tests. Changing  $A, B, \eta$ , or  $\kappa$  within the stated admissible class changes the renormalized spectral form by  $o(V_{ij})$ , so the limiting trace functional is canonical.

**Significance.** The spectral side is a finite smooth expression before any products are taken, and regularization independence is part of the conjecture rather than an implicit convention. The statement extends the Goldston–Montgomery and Chan prime/zero equivalence from one prime-counting field to products of shifted von Mangoldt functions [55, 33]. Failure can occur in two mathematically different ways: a missing mesoscopic covariance functional, or a genuine regularization anomaly. The first decisive theorem is the pair-motif covariance with one factor unshifted and two independent admissible mollifiers.

**Conjecture 29** (Anchored arithmetic polymer expansion for prime motifs). *Fix an admissible motif  $H$  of size  $k$ . A finite set  $A \subset \mathbb{Z}$  of occurrence starts is a polymer when  $0 \in A$ , the overlap graph on  $A$  is connected, and  $U_A = \bigcup_{t \in A} (H + t)$  is admissible. Define its anchored codimension*

$$d_H(A) = |U_A| - |H|;$$

*this is the number of new prime constraints beyond the occurrence anchored at 0. In the regularized spectral model of Conjecture 28, let  $Z$  be the limiting low-mode vector (the joint limit, asserted as part of this conjecture, of the mode integrals of the Conjecture 28 functional against frequencies below a fixed cutoff  $K$  on the  $L$ -scale, the Conjecture 28 limits taken first and  $K \rightarrow \infty$  last), let  $\lambda(t \mid Z)$  be the conditional one-point occurrence intensity, and set*

$$\lambda_H^0 = \frac{\mathfrak{S}(H)}{(\log X)^k}, \quad \tilde{\lambda}(t \mid Z) = \lambda(t \mid Z) / \lambda_H^0.$$

*For every fixed polymer  $A$ , uniformly for bounded diameter,*

$$\frac{(\log X)^{d_H(A)} \rho^{(|A|)}(A \mid Z)}{\lambda(0 \mid Z) \prod_{t \in A \setminus \{0\}} \tilde{\lambda}(t \mid Z)} \rightarrow \frac{\mathfrak{S}(U_A)}{\mathfrak{S}(H)}$$

*in probability in the successive limits defining  $Z$ . Thus the conditional environment supplies only the one-point intensity, while every connected Palm activity is graded by  $(\log X)^{-d_H(A)}$  and weighted by the complete-union singular series.*

*If admissible overlapping translates exist, put*

$$\delta(H) = \min_{\substack{t \neq 0 \\ H \cup (H+t) \text{ admissible}}} (|H \cup (H+t)| - |H|), \quad \mathcal{T}_H = \{t \neq 0 : H \cup (H+t) \text{ admissible}, |H \cup (H+t)| - |H| = \delta(H)\}$$

*Then the cluster-start fraction has the first nontrivial expansion*

$$\theta_X(H) = 1 - \frac{c_H}{(\log X)^{\delta(H)}} + o((\log X)^{-\delta(H)}), \quad c_H = \sum_{\substack{t \in \mathcal{T}_H \\ t < 0}} \frac{\mathfrak{S}(H \cup (H+t))}{\mathfrak{S}(H)}.$$

*The set  $\mathcal{T}_H$  is symmetric under  $t \mapsto -t$ , and the cluster-start fraction counts, for each occurrence, only whether it has an overlapping predecessor, so the sum is restricted to  $t < 0$  and equals half the symmetric sum over  $\mathcal{T}_H$ . If no admissible overlap exists,  $\theta_X(H) = 1 + o((\log X)^{-M})$  for every fixed  $M$ . Higher corrections are the connected Ursell sums (the signed connected sums of cluster-expansion theory) of anchored polymers ordered by  $d_H(A)$ . These finite- $X$  Gibbs expansions are projectively consistent and subcritical, and after division by the low-zero intensity the cluster-start process is Poisson to each fixed order of this triangular expansion.*

**Significance.** The relevant rarity is the number of new constraints after one occurrence is given, not the enhancement relative to a product of all one-point intensities. For two translates sharing  $s$  of the  $k$  prime positions, the Palm scale is  $(\log X)^{-(k-s)}$ . A universal  $1/\log X$  correction therefore holds only for special motifs, such as pair motifs that admit an overlapping translate. The law keeps polymers of every size, supplies a canonical extremal-index exponent  $\delta(H)$ , and links Hardy–Littlewood constants, low-zero environments, Palm theory, and extreme motif gaps. For  $H = \{0, 6\}$  one has  $\delta(H) = 1$  and the pair-motif coefficient is recovered. The first decisive theorem is the two-element polymer law for one pair motif, the Palm limit at  $A = \{0, t\}$  with  $t \in \mathcal{T}_H$  together with the expansion of  $\theta_X(H)$  at order  $(\log X)^{-\delta(H)}$ . A failure would show that the low-mode environment  $Z$  supplies more than the one-point intensity, so connected Palm activities would not be graded by  $(\log X)^{-d_H(A)}$  and weighted by  $\mathfrak{S}(U_A)$  alone.

**Conjecture 30** (Topological expansion of non-Gaussian prime cumulants). *For every connected exact-overlap diagram in Conjecture 26, expand its connected singular series over the linking primes and form the linkage graph of each term, whose vertices are the motif occurrences and whose edges join two occurrences sharing a linking prime in that term. Let  $g$  be the first Betti number of the linkage graph. Once all larger coincidence strata have been subtracted, the total contribution of the terms of Betti number  $g$  is smaller than that of the spanning-tree terms with the same external motifs by at least  $(\log L)^{-g}$ . The complete leading coefficient at each logarithmic order is the sum of the linked-term weights with the corresponding Betti number. In particular, the first nonzero odd cumulant of a balanced motif statistic, a combination  $Y_c$  whose coefficient vector annihilates every coincidence-stratum form of Conjecture 27, is carried by incidence trees. For the single-prime field the fully spread connected part  $\kappa_{2m+1}^{\text{sp}}$ , defined by subtracting every coincidence stratum as in Conjecture 27, obeys*

$$\kappa_{2m+1}^{\text{sp}} = (c_m + o(1))L^m(\log L)^{m+1},$$

where  $c_m$  is the total connected tree weight. The raw cumulant adds the explicit coincidence diagrams of Conjecture 26, which carry powers of  $\log X$  and are computable term by term, so the tree law is a statement about the spread diagrams, exactly the regime of Kuperberg’s singular-series sums.

**Significance.** This predicts a topological genus expansion for prime statistics: every independent cycle costs a logarithmic order. Kuperberg’s odd-moment conjecture is the one-field calibration, and the incidence-homology filtration with its motif-dependent coefficients is the new mechanism [34]. The first theorem is the case  $m = 1$  for the single-prime field, the spread third-cumulant tree law  $\kappa_3^{\text{sp}} = (c_1 + o(1))L(\log L)^2$  with its one-cycle terms suppressed by  $(\log L)^{-1}$ . A failure would show that logarithmic suppression is controlled by an arithmetic invariant not visible in diagram topology.

## 8 Programme II: arithmetic first-arrival fields and class groups

For a prime modulus  $q$  and  $a \in \mathbb{F}_q^\times$ , let  $p(q, a)$  (argument order reversed from Part I’s  $p(a, q)$ ) be the least prime in the class  $a$  and put

$$T_q(a) = \frac{\text{Li}(p(q, a))}{q-1}, \quad \mathcal{N}_{q,a} = \sum_{p \equiv a \pmod{q}} \delta_{\text{Li}(p)/(q-1)}.$$

The probability space chooses  $q$  uniformly among primes in  $[Q, 2Q]$  and then chooses  $a$  uniformly from  $\mathbb{F}_q^\times$ .

**Conjecture 31** (Connected first-arrival functional). *Let  $w_{r,Q}^c$  be the connected factorial-cumulant measure of the random point process  $\mathcal{N}_{q,a}$ . For every fixed  $r \geq 2$  there is a signed measure  $K_r$ , locally finite on the off-diagonal region  $\{u \in \mathbb{R}_+^r : u_i \text{ distinct}\}$ , such that, on that region (the diagonal strata carry strictly smaller normalizations, as in the stratification of Conjecture 26, and are excluded here),*

$$(\log Q)^{r-1} w_{r,Q}^c \implies K_r$$

*vaguely, and  $K_1$  denotes the limiting intensity measure, which in this clock is Lebesgue measure on  $\mathbb{R}_+$ . The measure  $K_r$  is obtained by averaging, over the modulus aspect, the connected Hardy–Littlewood densities of  $r$  primes constrained to one residue class. For every nonnegative compactly supported  $f$  in a fixed uniform ball on which the Laplace functional is analytic,*

$$\log \mathbb{E} \exp(-\langle f, \mathcal{N}_{q,a} \rangle) = \sum_{r \geq 1} \frac{(-1)^r}{r!} \int \prod_{j=1}^r (1 - e^{-f(u_j)}) dw_{r,Q}^c(\mathbf{u}),$$

*and the asymptotic expansion obtained by inserting the limits  $K_r$  is uniform to every fixed connected order. In particular the entire survival function, not merely its first correction, is determined by the hierarchy  $(K_r)_{r \geq 2}$ .*

**Significance.** This is an arithmetic point-process theory of least primes. The coupon-collector model of Li–Pratt–Shakan supplies the nearest extremal analogy, but not a connected Laplace functional [36]. Truth would determine waiting times, occupancy covariance, cover times, and terminal clustering from one hierarchy. The first theorem is the existence and explicit evaluation of  $K_2$  on compact off-diagonal sets. A failure would show that least primes in residue classes cluster through a mechanism not captured by modulus-averaged connected Hardy–Littlewood densities, a connected correlation of  $\mathcal{N}_{q,a}$  living outside the hierarchy  $(K_r)_{r \geq 2}$ .

**Conjecture 32** (Tested Gauss-polyspectral reciprocity for the least-prime field). *Set  $T_q(0) = \text{Li}(q)/(q-1)$  and centre the resulting field  $f_q$  on all of  $\mathbb{F}_q$ . Let  $\mathbf{A}_q$  and  $\mathbf{M}_q$  be its complete additive and multiplicative Fourier coefficient vectors, including the explicit rank-one coordinate needed to record the value at 0. There is then an explicit invertible Gauss matrix  $U_q$  with*

$$\mathbf{M}_q = U_q \mathbf{A}_q.$$

*Choose  $q$  uniformly among primes in  $[Q, 2Q]$ . For every finite family of degrees  $r_1, \dots, r_m$  and uniformly bounded fibrewise test tensors  $\Phi_{j,q}$ , define the scalar additive observables*

$$X_{j,Q}(q) = \langle \Phi_{j,q}, \mathbf{A}_q^{\otimes r_j} \rangle$$

*and the transported multiplicative observables*

$$Y_{j,Q}(q) = \left\langle (U_q^{-1})^{*\otimes r_j} \Phi_{j,q}, \mathbf{M}_q^{\otimes r_j} \right\rangle.$$

*For every modulus,  $X_{j,Q}(q) = Y_{j,Q}(q)$ , so all joint cumulants under the random-modulus law satisfy the exact finite identity*

$$\text{Cum}_Q(X_{1,Q}, \dots, X_{m,Q}) = \text{Cum}_Q(Y_{1,Q}, \dots, Y_{m,Q}).$$

This is the probability-space-correct form of Gauss transport. It makes no reference to a cumulant of a deterministic coefficient vector.

The conjectural content is that, after normalization by the first nonzero connected scale, tests supported on additive zero-sum tensors converge to contractions of the Fourier transforms of the arrival kernels  $K_r$  from Conjecture 31. Transporting those tests fibrewise by the exact Gauss matrices and only then taking the modulus average gives the same limits as the corresponding multiple explicit-formula correlations of Dirichlet  $L$ -function zeros. The two descriptions agree test by test: every uniformly bounded fibrewise family of test tensors yields the same normalized limits along both routes, so the two limiting multilinear functionals coincide on the tested class itself, no completion being invoked.

**Significance.** The elementary input is the finite Gauss matrix. The new claim is a complete higher-order tomography principle for the random-modulus least-prime field. Every scalar statistic is compared on a genuine probability space, while the varying finite fields are handled fibrewise before averaging. Failure would isolate a tensor class on which local same-class prime correlations and character-zero correlations disagree even though the underlying finite transforms are exact. The first theorem is the quadratic identity for tests localized to one additive lag and its transported character kernel.

**Conjecture 33** (Rubinstein–Sarnak terminal chaos dichotomy). *Assume GRH and a  $q$ -aspect linear-independence/random-phase hypothesis for the low zeros of primitive Dirichlet  $L$ -functions. Let  $x_q$  satisfy  $\text{Li}(x_q)/(q-1) = \log(q-1)$  and let  $G_{q,A}(a)$  be the centred, variance-normalized low-zero field obtained from characters  $\chi \neq \chi_0$  and zeros  $|\gamma| \leq (\log q)^A$ , with the standard compensator  $-\frac{1}{2} \text{Var } G_{q,A}(a)$ . Define the random empirical chaos measure*

$$\mathcal{M}_{q,A}(dg) = \frac{1}{q-1} \sum_{a \in \mathbb{F}_q^\times} \exp\left(G_{q,A}(a) - \frac{1}{2} \text{Var } G_{q,A}(a)\right) \delta_{G_{q,A}(a)}(dg).$$

Let  $b_Q$  be the smallest level with  $\mathbb{E}_q \#\{a : T_q(a) > b_Q\} \leq 1$ . In the successive limits  $Q \rightarrow \infty$  and then  $A \rightarrow \infty$ , the pair consisting of  $\mathcal{M}_{q,A}$  and

$$\sum_a \delta_{(G_{q,A}(a), T_q(a) - b_Q)}$$

converges jointly. Conditionally on the limiting chaos measure  $\mathcal{M}$ , the terminal cluster-start process is Poisson with directing measure

$$e^{-x} \mathcal{M}(dg) dx.$$

At finite modulus the cluster-start thinning enters through the terminal extremal index  $\theta_{\text{occ}}(q)$ , the analogue for the last classes of the cluster-start fraction  $\theta_X(H)$  of Conjecture 29, together with the terminal connected decorations supplied by the hierarchy of Conjecture 31. Both corrections enter at the first nonzero connected scale and vanish in the successive limit, so  $\theta_{\text{occ}}(q) \rightarrow 1$  and the limiting directing measure carries no cluster factor.

The nature of the directing measure is governed by the covariance-energy statistic

$$\mathfrak{E}_{q,A} = \frac{1}{(q-1)^2} \sum_{a,b \in \mathbb{F}_q^\times} \text{Cov}(G_{q,A}(a), G_{q,A}(b))^2.$$



If  $\mathfrak{E}_{q,A} \rightarrow 0$  along the successive limit,  $\mathcal{M}$  self-averages to the deterministic exponentially tilted Gaussian law and the terminal process is an ordinary decorated Poisson process. If  $\liminf \mathfrak{E}_{q,A} > 0$  and the Gaussian-chaos second moments are uniformly integrable,  $\mathcal{M}$  is nondegenerate and the limit is a genuine Cox process. No nondegeneracy is asserted without this criterion.

**Significance.** Rubinstein–Sarnak theory supplies the fixed-modulus prime-race environment [6]. The new object is its terminal first-arrival chaos measure in the modulus aspect. The statement now treats self-averaging as a serious competing model rather than assuming a Cox limit by terminology. It predicts the complete last class process, its prime-race marks, and an explicit spectral criterion separating a deterministic environment from persistent random mixing. The first theorem is the self-averaging branch at one fixed cutoff  $A$ , where  $\mathfrak{E}_{q,A} \rightarrow 0$  forces the deterministic exponentially tilted Gaussian limit of  $\mathcal{M}_{q,A}$  and an ordinary decorated Poisson law for the terminal cluster-start process.

**Conjecture 34** (Nonlinear spectral response calculus for arithmetic first arrivals). *Let a sequence of modulus ensembles have a small explicit-formula perturbation of the arrival intensity*

$$V_Q(a, t) = \sum_{\chi \in \mathcal{C}_Q} v_{Q,\chi}(t) \chi(a), \quad \|V_Q\|_T \rightarrow 0,$$

where  $\mathcal{C}_Q$  has uniformly bounded cardinality and, on every fixed interval  $[0, T]$ , the canonical norm is  $\|V\|_T = \sup_{a, 0 \leq t \leq T} |V(a, t)|$ . Assume that the connected hierarchy of Conjecture 31 satisfies an exponential cluster bound there, so its Laplace functional is analytic in a fixed  $\|\cdot\|_T$ -ball. If  $S_Q(a, t)$  and  $S_Q^{(0)}(t)$  are the perturbed and ordinary survival functions, then for every fixed  $R$

$$\log \frac{S_Q(a, t)}{S_Q^{(0)}(t)} = \sum_{m=1}^R \mathcal{R}_m[V_Q^{\otimes m}](a, t) + o(\|V_Q\|_T^R)$$

uniformly on compact  $t$ -ranges. The causal Volterra operator  $\mathcal{R}_m$  is the universal linked-cluster contraction of the full hierarchy  $(K_r)_{r \geq 1}$  with  $m$  marked perturbation insertions, and truncating the connected hierarchy at order  $J$  gives a convergent approximation as  $J \rightarrow \infty$ .

If the input character support is  $\mathcal{C}_Q$ , the  $m$ th response has Fourier support contained in the products of at most  $m$  input characters, with the reverse inclusion possibly failing through cancellation or the trivial character. In particular the linear response preserves character rank, while any new harmonics first appear at quadratic order with coefficients determined by connected three-point arrival correlations.

**Significance.** This is a nonlinear transfer calculus from zeros to first-hit statistics. A Siegel zero, a cluster of low zeros, and an ordinary prime-race bias become different inputs to the same response operators. The exceptional-zero rank-one statement is a special case, while the second-order harmonics provide a direct falsification test. The first theorem is the Fréchet derivative  $\mathcal{R}_1$  from the two-point occupancy kernel.

**Conjecture 35** (Local-information capacity of odd class groups). *Fix an odd prime  $\ell$ . Sample negative fundamental discriminants  $D \in [-2Y, -Y]$ , conditioned on a fixed coarse two-primary invariant. For a set of odd primes  $S_Y$ , put  $M_Y = \prod_{p \in S_Y} p$ , let  $\Sigma_Y(D) = ((D/p))_{p \in S_Y}$  be the attainable sign vector, and let  $H_Y$  be the logarithm of the number of attainable cells. Let  $\delta_Y(\sigma)$  be the exact CRT density of the sign cell after imposing quadratic reciprocity and the fixed*

two-primary data (for a single odd unramified prime the density of each sign is  $p/(2(p+1))$ ), and an unconditioned cell density is the corresponding product), and let  $\mathcal{D}_Y$  be the ambient discriminant family. Write

$$\Delta_Y(S_Y) = \sup_{\sigma} \left| \frac{\#\{D \in \mathcal{D}_Y : \Sigma_Y(D) = \sigma\}}{|\mathcal{D}_Y| \delta_Y(\sigma)} - 1 \right|$$

over attainable cells with  $|\mathcal{D}_Y| \delta_Y(\sigma) \geq \log Y$ .

There is a sequence  $\varepsilon_Y \rightarrow 0$ , independent of  $S_Y$  in the subcritical range, such that uniformly over those cells

$$d_{\text{TV}}(\mathcal{L}(\text{Cl}(D)[\ell^\infty] \mid \Sigma_Y = \sigma, \mathcal{Z}\text{-data}), \mathcal{L}_{\text{CL}, \ell}) \leq \varepsilon_Y + O(\Delta_Y(S_Y)),$$

and the analogous bound holds for every fixed Cohen–Lenstra moment. In particular, for every  $\varepsilon > 0$ , the conclusion is uniform whenever

$$H_Y \leq (1 - \varepsilon) \log Y, \quad \log M_Y \leq (\tfrac{1}{2} - \varepsilon) \log Y, \quad \Delta_Y(S_Y) = o(1).$$

Since every odd prime contributes at most  $\log 3 \leq \log p$  of cell entropy,  $H_Y \leq \log M_Y$  throughout, so the conductor constraint is the binding one in the displayed range, while the entropy governs the impossibility clause below. At  $H_Y \geq (1 + \varepsilon) \log Y$  no uniform statement is possible. Up to the conductor scale at which  $\Delta_Y$  becomes macroscopic, every first-order failure of local independence factors through this discriminant-cell discrepancy, and there is no earlier odd-class-group obstruction.

**Significance.** Wood’s framework treats fixed local conditions [37]. The new statement proposes the full information capacity of an odd class group under a growing local sigma-algebra, with the entropy governing impossibility and the conductor governing equidistribution. It would transfer directly to Selmer groups, ray class groups, and unramified extension statistics. The first theorem is uniformity for  $|S_Y| \rightarrow \infty$  with  $M_Y = Y^{o(1)}$ . The discrepancy half is not the obstruction: for cells of macroscopic mass the equidistribution of discriminants is elementary for  $M_Y \leq Y^{1/2-\varepsilon}$ , and squarefree-in-progressions technology carries it to  $Y^{2/3-\varepsilon}$ , so the conjectural content lies entirely in the class-group law over the growing cell family.

## 9 Programme III: finite logarithms, regulators, and algebraic tori

For  $p \nmid a$ , write

$$q_p(a) = \frac{a^{p-1} - 1}{p} \pmod{p}.$$

For a generating tuple  $g = (g_1, \dots, g_s)$ , let  $R(g)$  be its multiplicative relation lattice and let  $\mathbb{T}_g \subset (\mathbb{R}/\mathbb{Z})^s$  be the identity component of its annihilator. Its dimension is the saturated multiplicative rank. Write  $e(x) = e^{2\pi i x}$ .

**Conjecture 36** (Kummer–Haar law on relation tori). *Let  $\Gamma \leq \mathbb{Q}^\times$  be finitely generated and torsion-free, saturated modulo torsion (if  $x^n \in \Gamma \cdot \{\pm 1\}$  for some  $n \geq 1$  then  $x \in \Gamma \cdot \{\pm 1\}$ ), and*

let  $g$  be any ordered generating tuple. As  $p$  varies in every fixed compatible cyclotomic–Kummer Chebotarev class, the vectors

$$\frac{(q_p(g_1), \dots, q_p(g_s))}{p} \in (\mathbb{R}/\mathbb{Z})^s$$

equidistribute with Haar measure on  $\mathbb{T}_g$ . More strongly, for every finite collection of compatible Chebotarev restrictions, the conditional empirical measures converge to the Haar disintegration on the corresponding components, and the support is contained in no proper closed coset of  $\mathbb{T}_g$ . The law is invariant under changing generators and depends on  $\Gamma$  only through its saturation and the fixed Kummer conditioning data.

**Significance.** Katz formulated horizontal Wierferich equidistribution principles for algebraic groups, and Shparlinski proved estimates in different horizontal and averaged regimes [38, 39]. The new content here is the exact saturated relation torus, full generator invariance, and disintegration under fixed Kummer classes. It turns multiplicative rank into a geometric support theorem rather than a heuristic count of coordinates. The first decisive theorem is the Haar law on  $\mathbb{T}_g$  for one rank-one saturated  $\Gamma$  presented by two dependent generators, restricted to a single fixed compatible cyclotomic–Kummer Chebotarev class. A failure would show that the vectors  $q_p(g_i)/p$  remember more of  $\Gamma$  than its saturation, concentrating on a proper closed coset of  $\mathbb{T}_g$  that no relation in  $R(g)$  predicts.

**Conjecture 37** (Rank-dimensional large sieve for finite logarithms). *Choose a Smith-normal-form basis of the character lattice  $X^*(\mathbb{T}_g) = \mathbb{Z}^s/R(g)^{\text{sat}}$ . There is a constant  $\delta_\Gamma > 0$  such that, for every  $\varepsilon > 0$ , every  $M \leq X^{\delta_\Gamma}$ , and all complex weights  $a_p$  supported on  $X < p \leq 2X$ ,*

$$\sum_{\substack{m \in X^*(\mathbb{T}_g) \\ \|m\|_\infty \leq M}} \left| \sum_{X < p \leq 2X} a_p e\left(\frac{\langle m, q_p(g) \rangle}{p}\right) \right|^2 \ll_{\Gamma, \varepsilon} (M^{\text{rank } \Gamma} + \pi(X)) X^\varepsilon \sum_p |a_p|^2.$$

The same inequality holds after every fixed compatible cyclotomic–Kummer restriction, and for matrix-valued weights with the Hilbert–Schmidt norm. The exponent  $\delta_\Gamma$  is invariant under changing the generating tuple.

**Significance.** This is the quantitative mechanism behind Conjecture 36. It predicts that saturated rank is not only the support dimension but the exact dual dimension in a prime-varying large sieve. A proof for any positive power range would control genuine shrinking targets and would be far beyond currently available fixed-frequency tests. Failure would reveal an unrecognized spacing or energy obstruction among finite-logarithm vectors.

**Conjecture 38** (Mesoscopic-to-lattice shrinking-target transition). *Let  $\Gamma \leq \mathbb{Q}^\times$  be torsion-free of rank  $r$  and saturated modulo torsion, and choose a basis of  $\Gamma$ , identifying its relation torus with  $(\mathbb{R}/\mathbb{Z})^r$ . For integers  $w_p$  with  $1 \leq w_p = o(p)$ , let  $E_p(w)$  be the event that every coordinate of the Fermat-quotient vector has a representative of absolute value at most  $w_p$ .*

Mesoscopic universality. *If  $w_p \rightarrow \infty$ , then in every range in which the Haar mass*

$$\lambda(X) = \sum_{p \leq X} \left( \frac{2w_p + 1}{p} \right)^r$$

tends to infinity, the count of  $E_p(w)$  is asymptotic to  $\lambda(X)$ . On prime blocks with bounded Haar mass, the randomly translated event process converges to a Poisson process with that mass.

**Lattice transition.** For every fixed finite target set  $B \subset \mathbb{Z}^r$ , there is an arithmetic regulator intensity  $\kappa_{\Gamma,B} \geq 0$  such that

$$\#\{p \leq X : q_p|_{\Gamma} \in B\} = (\kappa_{\Gamma,B} + o(1)) \sum_{p \leq X} p^{-r}$$

whenever the sum diverges, while only finitely many such primes occur when  $r \geq 2$ . As a separately falsifiable genericity clause, for Kummer-generic  $\Gamma$  (the Kummer fields  $\mathbb{Q}(\zeta_m, \Gamma^{1/m})$  have the maximal degree  $\varphi(m)m^r$  for every  $m$ ) and  $B = \{0\}$  one has  $\kappa_{\Gamma,B} = 1$ . The transition from Haar universality to the arithmetic intensity occurs only when  $w_p$  remains bounded.

**Significance.** This separates two universality classes that a single undifferentiated model conflates. Growing targets are controlled by Haar geometry, while exact Wieferich targets can carry a genuinely arithmetic regulator intensity. Gras's much rarer fixed-base model and the Haar model therefore disagree precisely at the lattice boundary [40, 41]. The first theorem is mesoscopic universality for  $w_p = p^\eta$  with one rank-one base.

For a  $p$ -adic unit  $a$ , write  $\langle a \rangle_p$  for the Teichmüller lift of its residue, and for a prime  $\mathfrak{p} \mid p$  of a number field write  $\tau_{\mathfrak{p}}$  for the Teichmüller lift in the completion at  $\mathfrak{p}$ . Define  $q_{p,k}(a) = p^{-1} \log(a \langle a \rangle_p^{-1}) \pmod{p^k}$ . The matrix entries below are the  $\mathfrak{p}$ -adic analogues of  $q_{p,k}$ , and on  $\mathbb{Q}$  with  $k = 1$  one has  $q_{p,1}(a) \equiv -q_p(a) \pmod{p}$ , the Fermat quotient up to sign.

**Conjecture 39** (Global-lattice horizontal  $p$ -adic regulator-matrix law). *Let  $K/\mathbb{Q}$  be a fixed Galois number field with group  $G$ , and let  $\Gamma \leq \mathcal{O}_K^\times$  be torsion-free and saturated modulo  $\mu(K)$ , with ordered basis  $\epsilon_1, \dots, \epsilon_s$ . Define the fixed global relation lattice*

$$\mathcal{R}_{\Gamma} = \left\{ (c_{\sigma j}) \in \mathbb{Z}^{G \times s} : \prod_{\sigma \in G} \prod_{j=1}^s \sigma(\epsilon_j)^{c_{\sigma j}} \in \mu(K) \right\}.$$

Let  $\mathbb{T}_{\Gamma} \subset (\mathbb{R}/\mathbb{Z})^{G \times s}$  be its annihilator. This lattice contains the norm and Galois relations and is independent of the chosen basis up to the natural integral change of coordinates.

For a rational prime  $p$  splitting completely in  $K$ , label the primes above  $p$  by  $G$ , use the Teichmüller lift in each completion, and form the depth- $k$  matrix

$$L_{p,k}(\Gamma) = \left( p^{-1} \log_{\mathfrak{p}}(\epsilon_j \tau_{\mathfrak{p}}(\bar{\epsilon}_j)^{-1}) \pmod{p^k} \right)_{\mathfrak{p} \mid p, 1 \leq j \leq s}.$$

It lies in the annihilator of  $\mathcal{R}_{\Gamma} \otimes \mathbb{Z}/p^k \mathbb{Z}$ . As  $p$  varies through any fixed compatible Frobenius class,  $L_{p,1}/p$  equidistributes with Haar measure on  $\mathbb{T}_{\Gamma}$ , and the depth- $k$  matrices are Haar at every fixed depth  $k$ , with the depth- $(k+1)$  law pushing forward to the depth- $k$  law under reduction (the moduli vary with  $p$ , so this is a per-depth statement with compatible marginals, not Haar on one fixed group).

Let  $\mathcal{D}$  be a basis-invariant determinantal or Smith stratum in this projective module, and let  $\mu_{p,k}(\mathcal{D})$  be its exact local Haar mass. The primes for which  $L_{p,k} \in \mathcal{D}$  have counting law governed by  $\sum_{p \leq X} \mu_{p,k}(\mathcal{D})$ . For a determinantal or Smith stratum of codimension  $c$ , whose point counts are polynomial in  $p$ ,  $\mu_{p,1}(\mathcal{D}) = p^{-c} + O(p^{-c-1})$ , and codimension one has a Poisson point process in  $\log \log p$  after uniform random translation, while codimension at least two has a summable large-prime tail together with the exact small-prime atoms.

**Significance.** The horizontal state space is one fixed global torus rather than a relation module that changes with  $p$ . The conjecture turns mod- $p$  Leopoldt defects into a Galois-equivariant random-matrix problem, with exact Smith strata retaining higher depth. Existing mod- $p$  Leopoldt and random-matrix theories provide neighbouring languages [41, 42, 43]. Recent number-field Wieferich results strengthen the arithmetic motivation without supplying this horizontal matrix law [58, 59]. The first theorem is the rank-one split-prime law for a real quadratic field. A failure would show that the matrices  $L_{p,k}(\Gamma)$  favour some Smith stratum beyond its exact local Haar mass, so mod- $p$  Leopoldt defects would be governed by an arithmetic invariant outside the fixed global torus  $\mathbb{T}_\Gamma$ .

**Conjecture 40** (Functorial horizontal finite logarithms on algebraic tori). *Let  $T/\mathbb{Q}$  be an algebraic torus, let  $\mathcal{T}$  be its smooth integral model away from a finite set, and let  $\Gamma \leq T(\mathbb{Q})$  be finitely generated. Let  $S$  be the identity component of the Zariski closure of  $\Gamma$ . At a good unramified prime  $p$ , let  $\tau_p(\bar{P})$  be the canonical prime-to- $p$  torsion lift of the reduction of  $P \in \Gamma$  and define*

$$\ell_p(P) = p^{-1} \log_T(P \tau_p(\bar{P})^{-1}) \pmod{p} \in \mathrm{Lie}(S)(\mathbb{F}_p).$$

*Here  $\mathrm{Lie}(S)(\mathbb{F}_p)$  is identified with  $(\mathbb{Z}/p)^{\dim S}$  through a fixed integral basis of the Lie algebra of  $\mathcal{T}$ , with representatives in  $[0, p)$ ; a different basis changes the matrix by a fixed element of  $\mathrm{GL}_{\dim S}(\mathbb{Z})$ , which preserves the Haar law and the shrinking-target intensities. For any generators  $P_1, \dots, P_s$  of  $\Gamma$ , the normalized matrix  $(\ell_p(P_i)/p)_i$  equidistributes, in every compatible Frobenius class, with Haar measure on the real relation subtorus cut out jointly by the algebraic subgroup  $S$  and the integral relations among the  $P_i$ . For every morphism of tori  $\phi : T \rightarrow T'$ ,*

$$\ell_p(\phi(P)) = d\phi(\ell_p(P)),$$

*and the horizontal Haar measures push forward under the induced map. Isogenies preserve the shrinking-target intensities after the explicit finite kernel correction.*

**Significance.** Katz already proposed Wieferich equidistribution beyond  $\mathbb{G}_m$  [38]. The additional content is a matrix-valued law for a finitely generated global subgroup, relation-subtorus support, fixed-Frobenius disintegration, and exact functoriality under morphisms and isogenies. It turns horizontal finite logarithms into a functor on the category of tori with arithmetic subgroups. The first decisive theorem is the Haar law for the group generated by a single point of a one-dimensional nonsplit torus  $T$ , in a single fixed compatible Frobenius class. A failure would show that the matrices  $(\ell_p(P_i)/p)_i$  concentrate on a proper closed subset of the relation subtorus, an arithmetic constraint beyond the algebraic subgroup  $S$  and the integral relations among the  $P_i$ .

## 10 Programme IV: arboreal arithmetic dynamics

For a degree- $d$  map  $f$  and basepoint  $a$ , write  $K_n = \mathbb{Q}(f^{-n}(a))$ ,  $G_n = \mathrm{Gal}(K_n/\mathbb{Q})$ , and let  $\mu_n(C) = |C|/|G_n|$  on conjugacy classes. For primes  $p \leq X$  unramified in  $K_n$ , let  $\hat{\mu}_{n,X}$  be the empirical Frobenius measure.

**Conjecture 41** (Entropy–conductor profile of arboreal resolution). *For a degree- $d$  map  $f$  and level  $n$ , let  $G_{n,f}$  be the arboreal quotient and  $\mu_{n,f}$  its Haar measure on conjugacy classes. For  $N$  independent samples from  $\mu_{n,f}$  define the exact sampling obstruction*

$$\mathcal{S}_{n,f}(N) = \mathbb{E} d_{\mathrm{TV}} \left( N^{-1} \sum_{j=1}^N \delta_{Z_j}, \mu_{n,f} \right).$$

For each nontrivial irreducible Artin representation  $\rho$  of  $G_{n,f}$ , let  $q(\rho)$  be its Artin conductor and define the GRH conductor energy

$$\mathcal{A}_{n,f}(X) = \frac{\log X}{\sqrt{X}} \left( \sum_{\rho \neq 1} (\dim \rho)^2 (\log q(\rho) + \dim(\rho) \log X)^2 \right)^{1/2}.$$

This is a canonical character-theoretic replacement for a single discriminant proxy. For a parameter family  $\mathcal{F}(B)$  put

$$n_{\mathcal{F}}^*(B, X; \eta) = \max \left\{ n : \mathbb{E}_{f \in \mathcal{F}(B)} [\mathcal{S}_{n,f}(\pi(X)) + \mathcal{A}_{n,f}(X)] \leq \eta \right\}.$$

Assume the generic arboreal image is open, exceptional parameters have density zero, Artin holomorphy and GRH hold at the levels considered, and the family is in the large-sieve range of Conjecture 42 below. Then the actual empirical resolution depth, defined by the same total-variation threshold with prime Frobenius samples, differs from  $n_{\mathcal{F}}^*(B, X; \eta)$  by  $O_{\mathcal{F}, \eta}(1)$ .

More precisely, suppose, uniformly for a density-one set of parameters, that the class measures have a limiting complete Rényi profile

$$\psi_{\mathcal{F}}(\alpha) = \lim_{n \rightarrow \infty} d^{-n} \log \sum_C \mu_{n,f}(C)^\alpha$$

uniformly for  $\alpha$  in a fixed interval containing  $[1/2, 2]$ , and the corresponding Artin-conductor energies have finite positive exponential profiles on the same  $d^n$  scale. Then

$$n_{\mathcal{F}}^*(B, X; \eta) = \log_d \log X + O_{\mathcal{F}, \eta}(1).$$

Two families of the same degree with boundedly different complete Rényi and conductor profiles have resolution depths differing by  $O(1)$ , and in particular a fixed-index open subgroup changes only the bounded shift.

**Significance.** The observable depth is defined through the exact nonuniform sampling law and a representation-by-representation conductor energy. The  $\log_d \log X$  scale follows from explicit Rényi and conductor profiles on the natural  $d^n$  complexity scale. Large arboreal images and effective Chebotarev are inputs [44, 45, 46]. Recent work on fixed-point profiles shows why the full class-measure tail, rather than group order alone, matters [60]. The first theorem is bounded-shift stability for one explicit index-two subgroup of the binary wreath tower. A failure would show that resolution depth responds to a family invariant beyond the complete Rényi profile  $\psi_{\mathcal{F}}(\alpha)$  and the conductor energy  $\mathcal{A}_{n,f}(X)$ , so two families with matching profiles could separate by an unbounded shift.

**Conjecture 42** (Arboreal family large sieve and cumulant independence). *Let  $\mathcal{F} \rightarrow \mathbb{A}^1$  be a generically finite-index arboreal family. For each level  $n$  and irreducible representation  $\rho$  of the generic quotient  $G_n$ , let  $\mathcal{V}_{n,\rho}$  be the associated middle-extension sheaf on the good parameter locus (the canonical extension of the corresponding local system across the bad parameters). Define the geometric conductor budgets*

$$\mathfrak{C}_{n,r} = \sum_{\rho_1, \dots, \rho_r} \left( \prod_{j=1}^r \dim \rho_j \right)^2 \prod_{j=1}^r \text{cond}(\mathcal{V}_{n,\rho_j})^2, \quad \mathfrak{C}_n = \mathfrak{C}_{n,1},$$



where each conductor includes rank, the number of singular points, and the tame-drop and Swan terms (the tame and wild ramification invariants at those points); for  $r \geq 2$  the budget is attached to the product trace function on the CRT parameter set, whose factors live over distinct characteristics, so its complexity is by definition this product of local conductors rather than the conductor of an external tensor product over a single base.

Square-root tier. For every  $\varepsilon > 0$  and all complex coefficients  $\alpha_{p,\rho}$ ,

$$\sum_{|c| \leq B} \left| \sum_{p \leq X} \sum_{\rho \neq 1} \alpha_{p,\rho} \operatorname{tr} \rho(\operatorname{Frob}_{p,c}) \right|^2 \ll_{\varepsilon, \mathcal{F}} (B + X^2 \mathfrak{C}_n^{1+\varepsilon}) \sum_{p,\rho} |\alpha_{p,\rho}|^2,$$

uniformly when  $X^2 \mathfrak{C}_n^{1+\varepsilon} \leq B^{1-\varepsilon}$ , after deleting the singular fibres.

Connected tier. For every fixed  $r \geq 3$ , centred trace functions at distinct primes have family-averaged connected cumulants with square-root cancellation in the parameter-family count, uniformly in the range  $X^r \mathfrak{C}_{n,r}^{1+\varepsilon} \leq B^{1-\varepsilon}$ , and equivalently, after normalization, every fixed connected Frobenius cumulant tends to zero. An exponential connected-moment bound makes this convergence uniform in  $r$  over the range needed for occupancy functionals.

Consequently, whenever the analytic Chebotarev error is  $o(1)$  but the iid sampling functional of  $\mu_n$  is nonvanishing, the complete conjugacy-cylinder count vector, averaged over parameters, has the nonuniform multinomial occupancy law of independent samples from  $\mu_n$ , including its missing-mass, collision, and total-variation profile. The multinomial conclusion is derived from the connected tier, not from the  $L^2$  bound alone.

**Significance.** This separates the two analytic inputs that a single-tier statement compresses into one claim. The square-root tier is a growing-monodromy large sieve in the spirit of Kowalski [47]. The connected tier is the genuinely stronger cross-prime independence principle required for an occupancy limit. The conductor budget is attached to the actual sheaves and their product trace functions rather than to an undefined scalar conductor of a cover. The first theorem is the connected tier at  $r = 3$  for one fixed level  $n$ , square-root cancellation of the family-averaged third connected cumulant within the range governed by  $\mathfrak{C}_{n,3}$ . A failure identifies either geometric complexity missed by the sheaf conductors or arithmetic entanglement surviving all fixed-order trace tests.

**Conjecture 43** (Coloured dynatomic Galois-factor process). For  $f_c(x) = x^d + c$ , let the critical exact-period polynomial factor as  $\Psi_n(c) = \prod_{j=1}^{s_n} F_{n,j}(c)$  over  $\mathbb{Z}$ . Sample  $c$  from a dyadic height interval away from postcritically finite, discriminant, and fixed-resultant loci.

For every fixed  $n$ , condition on any fixed finite local valuation state and normalize the prime factors of each  $F_{n,j}(c)$  by that component's residual logarithmic mass. The component-coloured factor processes converge jointly to independent Frobenius-marked Poisson–Dirichlet PD(1) processes, with the mark in colour  $j$  given by the fixed-point-size-biased conjugacy law in the splitting field of  $F_{n,j}$ . The total logarithmic mass of ramified factors tends to zero.

Uniformly for  $n = n(H)$  with  $\deg \Psi_n = o(\log H)$ , every fixed correlation functional whose support lies inside the available level of distribution converges to the corresponding restricted-support correlation of these coloured marked processes. No full-process claim is made at growing degree without level-one distribution.

**Significance.** Exact-period components are genuine geometric colours, and their Galois marks distinguish dynamical families with similar unmarked factor sizes. The statement deliberately sep-

arates the full fixed-degree process from the restricted-support growing-degree law, in accordance with the distribution thresholds in Bharadwaj–Rodgers [51]. The first theorem is a two-colour correlation law at one fixed period. A failure would show that distinct exact-period components are entangled, with the splitting fields of two factors  $F_{n,j}$  sharing Frobenius information that survives conditioning on the fixed finite local valuation state.

**Conjecture 44** (Complexity-uniform primitive valuation process for dynatomic values). *For the critical exact-period polynomial  $\Psi_n(c)$ , remove its canonical greatest common divisor in  $\mathbb{Q}[c]$  with the earlier critical-orbit product  $\prod_{m < n} f_c^m(0)$ , in the notation of Conjecture 43, and denote the result by  $\Psi_n^{\text{new}}$ . Define the logarithmic arithmetic complexity*

$$\mathfrak{H}_n = \deg \Psi_n^{\text{new}} + \log^+ H_{\text{coeff}}(\Psi_n^{\text{new}}) + \log^+ |\text{Disc}(\Psi_n^{\text{new}})| + \sum_{m < n} \log^+ |\text{Res}(\Psi_n^{\text{new}}, f_c^m(0))|,$$

where zero resultants have already been removed by the canonical gcd. For each prime  $p$ , let  $\nu_{n,p}$  be the exact Haar law on  $\mathbb{Z}_p$  of the primitive valuation

$$V_{n,p}(c) = v_p(\Psi_n^{\text{new}}(c)) \mathbf{1}_{p \nmid f_c^m(0) \ \forall m < n}.$$

All discriminant, resultant, and collision primes remain inside these local laws.

For every control function  $\omega(H) = o(\log H)$ , uniformly over levels with  $\mathfrak{H}_n \leq \omega(H)$ , the finite-prime valuation vectors for  $c$  in a dyadic height interval converge to the corresponding marginals of  $\bigotimes_p \nu_{n,p}$ . In addition, for every  $\eta > 0$ ,

$$\lim_{z \rightarrow \infty} \limsup_{H \rightarrow \infty} \sup_{n: \mathfrak{H}_n \leq \omega(H)} \mathbb{P} \left( \sum_{p > z} (\log p)(V_{n,p} - 1)_+ > \eta \log |\Psi_n^{\text{new}}(c)| \right) = 0,$$

and the analogous unnormalized tail  $\sum_{p > z} \mathbb{P}(V_{n,p} \geq 2)$  tends to zero whenever the local squarefree product is positive. These two clauses give convergence in the topology generated by bounded cylinder functions and the logarithmic squarefull-mass functional.

Consequently

$$\mathbb{P}(\Psi_n^{\text{new}}(c) \text{ is squarefree}) = \prod_p \tilde{\nu}_{n,p}(\{0, 1\}) + o(1)$$

uniformly in the stated complexity range, where  $\tilde{\nu}_{n,p} = \nu_{n,p}$  except at the finitely many collision primes dividing one of the resultants  $\text{Res}(\Psi_n^{\text{new}}, f_c^m(0))$ , at which  $\tilde{\nu}_{n,p}$  is the law of the raw valuation  $v_p(\Psi_n^{\text{new}}(c))$ , and the logarithmic squarefull mass converges to the law determined by the complete local product.

**Significance.** Degree alone does not control a growing polynomial family. The controlling range includes coefficient height, discriminant, and resultants, while the infinite-product conclusion is supported by an explicit uniform squarefull-tail assertion. The local process still retains every bad-prime and collision state and remains level dependent, as primitive hits at a fixed prime occur by bounded time modulo  $p^k$ . For the unicritical families  $x^d + c$  the resultants at the first levels are units, so the stated complexity range is nonempty there. The first decisive theorem is the primitive valuation law with its uniform squarefull tail at one fixed level of the unicritical family  $x^d + c$ , giving the squarefree asymptotic for that single  $\Psi_n^{\text{new}}$ . A failure would show that  $\mathfrak{H}_n$  is not the right complexity gauge, with some growing-level family carrying squarefull mass or valuation correlations that degree, coefficient height, discriminant, and resultants together do not see. The nearest literature treats primitive divisors and fixed-polynomial squarefree values, not this complexity-uniform adelic process [49, 48].

**Conjecture 45** (Divisor-sensitive classification of dynamical gcd height). *Let  $F = (f, g)$  be a split map on  $(\mathbb{P}^1)^2$ , with  $f, g$  disintegrated (in the sense of Medvedev and Scanlon [57], so neither is linearly conjugate to a power map, a Chebyshev map, or a close relative of these) of the same degree, and let  $P = (a, b)$  have positive canonical heights. Let  $D_1, D_2$  be the two coordinate-zero divisors and let  $E$  be the exceptional divisor of the blow-up at  $D_1 \cap D_2$ . Then*

$$\limsup_{n \rightarrow \infty} \frac{h_E(F^n(P))}{\max\{h(f^n(a)), h(g^n(b))\}} > 0$$

*if and only if there exist  $m \geq 0$  and an irreducible  $F$ -periodic curve  $Z$  containing  $F^m(P)$ , with  $Z$  not contained in  $D_1 \cup D_2$  (automatic when both coordinates have positive canonical height), such that, on the normalization of  $Z$ , the pullbacks of  $D_1$  and  $D_2$  have a common nonzero effective component. In the absence of such eventual entry into a divisor-compatible periodic curve, the normalized gcd height tends to zero.*

*The converse is unconditional in every setting where the required Vojta inequality for the blow-up is known, and otherwise is explicitly conditional on that inequality. The same criterion extends to split maps on  $(\mathbb{P}^1)^r$  by replacing a common component with positive codimension-one intersection multiplicity among the pulled-back divisors.*

**Significance.** The obstruction is not merely a common dynamical quotient. It is a periodic geometric relation carrying the actual divisors measured by the gcd height. Eventual entry is the right notion, since a preperiodic tail can carry the divisor relation before periodicity begins. Recent work resolves several set-theoretic dynamical-GCD problems, while this height-theoretic divisor classification is the additional boundary [50]. The first theorem is the positive-limsup direction, that eventual entry of the orbit into a divisor-compatible periodic curve  $Z$  forces  $\limsup h_E(F^n(P)) / \max\{h(f^n(a)), h(g^n(b))\} > 0$ . Failure would point to a new source of macroscopic Diophantine proximity outside invariant geometry.

## 11 Programme V: adelic factorization and sieve flow

Throughout this programme  $P^+(m)$  and  $P^-(m)$  denote the largest and smallest prime factors of  $m$ , the constant  $\gamma$  is Euler's,  $\rho_f(p)$  counts the roots of  $f$  modulo  $p$ , and every sample  $n \in [N, 2N]$  is drawn uniformly.

**Conjecture 46** (Frobenius-marked Poisson–Dirichlet process). *Let  $f \in \mathbb{Z}[x]$  be irreducible with splitting field  $L$  and transitive Galois group  $G$ . Sample  $n \in [N, 2N]$ . Mark each unramified prime factor  $p$  of  $f(n)$  by its conjugacy class  $C \subset G$  and give it mass  $u = \log p / \log |f(n)|$ . The macroscopic marked process converges to a marked PD(1) process with conditional mark law*

$$\mathbb{P}(C \mid u) = \frac{|C|}{|G|} \text{fix}(C), \quad \text{fix}(C) \text{ the number of fixed roots of an element of } C,$$

*independent of  $u$ . Given the masses and every fixed finite local valuation state, the marks of finitely many macroscopic factors are asymptotically independent. Under a normal quotient equipped with the induced permutation representation, the marked process pushes forward functorially.*

**Significance.** Chebotarev contributes  $|C|/|G|$ , divisibility by a polynomial value size-biases by the number of fixed roots, and Burnside’s lemma normalizes the result. Bharadwaj–Rodgers supply the unmarked factor-process framework [51]. Simultaneous Galois marking, conditional independence, and functorial pushforward are the new content. It connects Galois theory to probabilistic factorization directly. The first theorem is the mark law for the largest prime factor of one irreducible quadratic  $f$ , the two-class case where the size bias  $\text{fix}(C)$  already separates the marked law from the plain Chebotarev weight  $|C|/|G|$ . A failure would show that divisibility size-biases Frobenius by more than the fixed-root count, a coupling between factor masses  $u$  and Galois marks that survives conditioning on the finite local valuation state.

**Conjecture 47** (Three-scale factorization of polynomial values). *Let  $f \in \mathbb{Z}[x]$  be irreducible, sample  $n \in [N, 2N]$ , choose  $y = y(N) \rightarrow \infty$  with  $\log y = o(\log N)$ , and write  $R_y = f(n)/S_y(f(n))$ , where  $S_y(m) = \prod_{p \leq y} p^{v_p(m)}$  is the  $y$ -smooth part. Jointly with the complete small-prime valuation field  $(v_p(f(n)))_{p \leq y}$ , the residual logarithmic factor process*

$$\sum_{p|R_y} v_p(R_y) \delta_{(\log p / \log |R_y|, \text{Frob}_p)}$$

*converges to the following conservation-corrected product object: the small valuations have their polynomial Kubilius law (independent exact local limit laws for the small-prime valuations), and conditional on their consumed logarithmic mass the residual process is a scale-invariant Poisson process of intensity  $du/u$ , marked by Conjecture 46, conditioned to have total mass one. The convergence is in total variation for the complete valuation field, the range  $\log y = o(\log N)$  being the classical Kubilius range for the field at  $f(x) = x$ , and in the sense of finite-dimensional distributions for the residual process jointly with the field. The small field and the unconditioned residual Poisson process are asymptotically independent, and all leading dependence after conditioning is the single residual-mass constraint.*

*For disjoint normalized size bands bounded away from zero, the unconditioned residual process has independent increments. Its mesoscopic restriction and its ranked macroscopic atoms yield respectively the scale-invariant factor process and the Frobenius-marked PD(1) partition. Thus the local Kubilius field is the boundary variable fixing the available mass, while the mesoscopic and macroscopic laws are two projections of one residual process, and no additional cross-scale coupling survives.*

**Significance.** Raw small–large independence is false because small factors consume mass. This conjecture identifies the complete conservation-corrected product structure and adds the mesoscopic scale that is absent from a two-scale statement. It would justify, in one limit theorem, the local sieve model, the scale-invariant factor process, and the macroscopic partition. The first decisive theorem is the conservation-corrected joint law at one fixed prime  $p$  and one normalized size band, where the only surviving dependence is the residual-mass constraint. A failure would show a cross-scale coupling between the small-prime valuation field and the residual process beyond the single residual-mass constraint, so smooth-part consumption would not exhaust the dependence between the sieve scale and the factor process.

**Conjecture 48** (Adelic Gibbs gluing for reducible polynomial values). *Let  $f = \prod_{i=1}^s g_i$  be squarefree with irreducible components, sample  $n \in [N, 2N]$ , and put  $t = n/N$ . For each prime  $p$ , let  $\nu_p$  be the exact Haar law on  $\mathbb{Z}_p$  of the valuation vector  $(v_p(g_1(x)), \dots, v_p(g_s(x)))$ . For  $y \rightarrow \infty$  with  $\log y = o(\log N)$ , consider stable convergence jointly with the archimedean*

coordinate  $t$  and condition on  $\mathcal{L}_y = (v_p(g_i(n)))_{p \leq y, i \leq s}$ . After normalizing each component by its own residual logarithmic mass, the coloured Frobenius-marked factor processes converge to conditionally independent copies of the three-scale law of Conjecture 47, with the marks of Conjecture 46.

Unconditionally, the joint process is the projective Gibbs mixture obtained by integrating those conditional product laws against Lebesgue measure in  $t$  and the adelic local law  $\bigotimes_p \nu_p$ , in the iterated order  $N \rightarrow \infty$  and then  $y \rightarrow \infty$ . Every bounded finite-dimensional Laplace functional is determined by

$$Z_p(\mathbf{z}) = \int_{\mathbb{Z}_p} \prod_i z_i^{v_p(g_i(x))} dx$$

together with the archimedean size profile  $(\log |g_i(Nt)|)_i$ . No additional colour coupling survives after this adelic–archimedean mixture.

**Significance.** A scalar Euler factor cannot specify a point process. This statement gives the full local-to-global gluing rule and includes the archimedean variable that can couple component sizes. It is functorial under multiplying, splitting, or regrouping components. The first theorem is the case of two linear components, where  $Z_p(\mathbf{z})$  is computable in closed form and the mixture over  $t$  and  $\bigotimes_p \nu_p$  already yields the full two-colour gluing law. A persistent medium-prime or archimedean coupling after the stated mixture would identify a new nonadelic invariant.

**Conjecture 49** (Full multivariate adelic saddle law for joint smoothness). *Let  $f_1, \dots, f_s \in \mathbb{Z}[x]$  be pairwise coprime, primitive, and jointly admissible, meaning that for every prime  $p$  some residue  $x$  has  $p \nmid \prod_i f_i(x)$ , sample  $n \in [N, 2N]$ , and put  $t = n/N$ . For complex variables  $\mathbf{s}$  and smoothness bounds  $\mathbf{y}$ , define*

$$\mathcal{Z}_{p,\mathbf{y}}(\mathbf{s}) = \int_{\mathbb{Z}_p} p^{-\sum_{i:p \leq y_i} (s_i-1)v_p(f_i(x))} dx, \quad \widehat{\mathcal{Z}}_{p,\mathbf{y}}(\mathbf{s}) = \frac{\mathcal{Z}_{p,\mathbf{y}}(\mathbf{s})}{\mathcal{Z}_{p,\mathbf{y}}(+\infty)}.$$

The denominator is the exact Haar probability that  $p$  divides none of the relevant  $f_i(x)$ . Put

$$\Phi_t(\mathbf{s}) = \sum_p \log \widehat{\mathcal{Z}}_{p,\mathbf{y}}(\mathbf{s}) + \sum_i (s_i - 1) \log |f_i(Nt)|.$$

For  $f(x) = x$ , this gives the usual truncated zeta factor  $(1 - p^{-s})^{-1}$ . Define the joint local probability by the Perron-calibrated multivariate inverse Mellin integral

$$\mathcal{P}_t^{\text{joint}} = \frac{1}{(2\pi i)^s} \int_{\boldsymbol{\sigma}_t + i\mathbb{R}^s} e^{\Phi_t(\mathbf{s})} \prod_{i=1}^s \frac{ds_i}{s_i},$$

with the standard truncated-contour limit, where  $\boldsymbol{\sigma}_t$  is the unique real saddle in the admissible region. The factors  $s_i^{-1}$  are part of the canonical cumulative smoothness event, and for one variable they recover the Hildebrand–Tenenbaum Perron normalization [56].

Uniformly when every  $u_i(t) = \log |f_i(Nt)| / \log y_i$  lies in a fixed compact saddle-point region,

$$\mathbb{P}(P^+(f_i(n)) \leq y_i \ \forall i) \sim \int_1^2 \mathcal{P}_t^{\text{joint}} dt.$$

The integral has the full multivariate saddle expansion. Writing  $H_t = \nabla^2 \Phi_t(\boldsymbol{\sigma}_t)$ , its leading term includes the joint action, the amplitude  $\prod_i \sigma_{i,t}^{-1}$ , and  $(\det H_t)^{-1/2}$ , with the usual lattice correction when the valuation span is proper.

If  $\mathcal{P}_{i,t}^{\text{marg}}$  are the identically calibrated one-polynomial probabilities, then  $\mathcal{P}_t^{\text{joint}} / \prod_i \mathcal{P}_{i,t}^{\text{marg}}$  is the complete ratio of joint and marginal actions, Perron amplitudes, Hessian determinants, and lattice corrections. Its Euler-product component is the ratio of the exact  $p$ -adic partition functions evaluated at the joint and marginal saddles. None of these factors may be replaced by a scalar Euler correction at the marginal saddle.

**Significance.** The statement is calibrated to the cumulative smoothness probability. It combines exact higher  $p$ -adic valuations, moving archimedean sizes, the joint saddle displacement, Perron amplitudes, mixed curvature, and possible lattice effects. Even one-polynomial smooth-value asymptotics are difficult [52, 54]. A complete multivariate adelic saddle law would be a major new local–global bridge. A failure would show a joint smoothness dependence not produced by the exact  $p$ -adic partition functions and the moving archimedean sizes, an arithmetic coupling among the  $f_i$  outside the saddle displacement, the Hessian  $H_t$ , and the lattice correction.

**Conjecture 50** (Buchstab–Bateman–Horn component flow). *Let  $f \in \mathbb{Z}[x]$  be primitive, irreducible, and admissible ( $\rho_f(p) < p$  for every prime  $p$ ), and sample  $n \in [N, 2N]$ . For  $y$  with  $u_n = \log |f(n)| / \log y$  in a fixed compact subset of  $(1, \infty)$ , let  $\Omega_y(f(n))$  be the number of prime factors of  $f(n)$  exceeding  $y$ , counted with multiplicity, on the event  $P^-(f(n)) > y$ . Put*

$$V_f(y) = \prod_{p \leq y} \left(1 - \frac{\rho_f(p)}{p}\right).$$

Define the Buchstab component functions by

$$\omega_1(u) = \frac{1}{u}, \quad \omega_j(u) = \frac{1}{u(j-1)!} \int_{\substack{t_1, \dots, t_{j-1} \geq 1 \\ t_1 + \dots + t_{j-1} \leq u-1}} \frac{dt_1 \cdots dt_{j-1}}{t_1 \cdots t_{j-1}} \quad (j \geq 2),$$

so that  $\sum_{j \geq 1} \omega_j(u) = \omega(u)$ . Then for every fixed  $j$ ,

$$\mathbb{P}(P^-(f(n)) > y, \Omega_y(f(n)) = j) = V_f(y) e^\gamma \mathbb{E}_{n \in [N, 2N]} \omega_j(u_n) + o(V_f(y)),$$

uniformly on compact  $u$ -ranges. Conditional on the ordered logarithmic sizes of the  $j$  remaining factors, their unramified Frobenius marks are independent with the fixed-point-size-biased law of Conjecture 46. Summing over  $j$  gives the rough-value Buchstab law,  $j = 1$  gives the Bateman–Horn prime density, and  $j = 2$  gives the semiprime transition. Whenever  $\max_{N \leq n \leq 2N} |f(n)| < y^2$ , every rough value has exactly one prime factor above  $y$ , so the  $j = 1$  component exhausts the rough set, while the density formula itself remains conjectural.

**Significance.** The conjecture resolves a rough polynomial value into its complete finite- $u$  factor count and Galois mark content. Primes and semiprimes are not inserted as separate heuristics. They are components of one universal flow. Generalized Buchstab equations are the random-integer antecedent [61], and the fixed-polynomial, Galois-marked component law with its local factors is the new statement. It is strongly falsifiable even when total roughness follows Buchstab.

## Part III

# Six conjectures across fields

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The six conjectures below leave the prime model. They range over multiplicative number theory, discrepancy theory, graph limits, additive information theory, and ergodic theory, and each is held to the same standard as the rest of the collection: a canonical object, a stated mechanism, the nearest literature boundary, a first decisive theorem where one exists, and an honest failure mode.

## 12 Relative polynomial pretentiousness

Let  $P \in \mathbb{Z}[x]$  be primitive (the greatest common divisor of its coefficients is 1) and irreducible, with positive leading coefficient. For a prime  $p$  write

$$\rho_P(p) = \#\{a \in \mathbb{Z}/p\mathbb{Z} : P(a) \equiv 0 \pmod{p}\}.$$

The weight  $\rho_P(p)/p$  is the first-order frequency with which  $p$  divides a polynomial value  $P(n)$ . A function  $f$  on the positive integers is completely multiplicative if  $f(mn) = f(m)f(n)$  for all  $m, n \geq 1$ . For completely multiplicative  $f, g$  with values in the closed complex unit disc, define the truncated  $P$ -relative pretentious distance

$$\mathbb{D}_P(f, g; X)^2 = \sum_{p \leq X} \frac{\rho_P(p)}{p} (1 - \operatorname{Re}(f(p)\overline{g(p)})), \quad \mathbb{D}_P(f, g)^2 = \lim_{X \rightarrow \infty} \mathbb{D}_P(f, g; X)^2 \in [0, \infty].$$

Every summand is nonnegative, so the truncated distance is nondecreasing in  $X$  and the limit exists. Throughout, a primitive Dirichlet character  $\chi$  of conductor  $c$  is regarded as the  $c$ -periodic completely multiplicative function on  $\mathbb{Z}$  that vanishes on integers not coprime to  $c$ , so that expressions such as  $\chi(P(a))$  are defined for every residue  $a$  modulo any multiple of  $c$ . For  $t \in \mathbb{R}$ , the comparison function  $n \mapsto \chi(n)n^{it}$  is completely multiplicative with values in the closed unit disc.

Let  $P_1, \dots, P_m \in \mathbb{Z}[x]$  be pairwise distinct primitive irreducible polynomials with positive leading coefficients and no fixed prime divisor, meaning  $\rho_{P_j}(p) < p$  for every  $j$  and every prime  $p$ . Assume multiplicative independence modulo constants:

$$\prod_{j=1}^m P_j(x)^{e_j} \in \mathbb{Q}^\times, \quad e_j \in \mathbb{Z} \implies e_1 = \dots = e_m = 0.$$

Write  $\mathbb{S}^1$  for the unit circle in the complex plane.

**Conjecture 51** (Relative polynomial pretentiousness inverse principle). *Let  $f_1, \dots, f_m : \{1, 2, 3, \dots\} \rightarrow \mathbb{S}^1$  be completely multiplicative. If*

$$\limsup_{X \rightarrow \infty} \left| \frac{1}{\log X} \sum_{\substack{n \leq X \\ P_j(n) > 0 \ \forall j}} \frac{\prod_{j=1}^m f_j(P_j(n))}{n} \right| > 0,$$

*then there exist primitive Dirichlet characters  $\chi_1, \dots, \chi_m$  and real numbers  $t_1, \dots, t_m$  such that the following three conditions hold simultaneously:*

- (i)  $\mathbb{D}_{P_j}(f_j, \chi_j(n)n^{it_j}) < \infty$  for every  $1 \leq j \leq m$ ,
- (ii)  $\sum_{j=1}^m t_j \deg P_j = 0$ ,

(iii) for some common multiple  $q$  of the conductors of  $\chi_1, \dots, \chi_m$ ,

$$\frac{1}{q} \sum_{a \bmod q} \prod_{j=1}^m \chi_j(P_j(a)) \neq 0,$$

the characters being extended by 0 on nonunits as above.

**Significance.** The mechanism is that a persistent logarithmic correlation should have exactly two sources: an archimedean resonance carried by the factors  $n^{it_j}$  and a finite local resonance carried by the characters  $\chi_j$ . The balance condition (ii) is forced by the expansion  $P_j(n)^{it_j} = c_j^{it_j} n^{it_j \deg P_j} (1 + o(1))$ , with  $c_j$  the leading coefficient, because logarithmic averaging annihilates any nonzero total power of  $n$ , and (iii) asserts that the finite-modulus resonances are jointly compatible, so that the model correlation does not vanish identically for local reasons. Packaging the conclusion as one existential block matters: the characters and archimedean parameters cannot be chosen independently for each  $j$ , and it is the joint system (i)–(iii) that reproduces the correlation. The decisive-theorem ledger for the linear case is as follows. For linear polynomials the two-point case is Tao’s logarithmically averaged two-point Elliott theorem [75]. The odd-order cases are the theorems of Tao–Teräväinen, and the shape of the conclusion here, one existential block coupling characters, archimedean parameters, and a nonvanishing local average, is the template of their structure theorem for logarithmically averaged correlations [76]. The general even-order linear case is precisely the logarithmically averaged Elliott conjecture. All of the novelty of the present statement therefore sits at  $\max_j \deg P_j \geq 2$ , where even the logarithmic mean value of  $f(n^2 + 1)$  for a single nonpretentious completely multiplicative  $f$  is open, and a logarithmic mean value theorem for  $f(n^2 + 1)$  with a conclusion of the stated relative-pretentious type would be the first decisive theorem. The failure mode is a system of functions nonpretentious in every relative metric whose logarithmic correlation has positive limsup, which would reveal a correlation mechanism beyond the archimedean and finite local resonances.

**Remark 2.** *The restriction to completely multiplicative  $f_j$  is not cosmetic. For 1-bounded multiplicative functions that are not completely multiplicative, the fixed-parameter form of the conclusion is false: Matomäki–Radziwiłł–Tao [71] constructed a 1-bounded multiplicative function that pretends to be  $n^{it_k}$  for different parameters  $t_k$  on different scales, refuting the original fixed-parameter Elliott conjecture, and the corrected conjecture must allow the comparison parameter to vary with the scale  $X$ . For the unimodular completely multiplicative  $f_j$  of the present statement the fixed-parameter form survives, and by a rigidity argument rather than by monotonicity alone. Fix a member  $f_j$  of the system, a primitive Dirichlet character  $\chi$ , and scales  $X_k \leq X_{k+1}$ , the scale index  $k$  being unrelated to the function index  $j$ . If  $\chi n^{is_k}$  and  $\chi n^{is_{k+1}}$  are within bounded truncated distance of  $f_j$  at the scales  $X_k$  and  $X_{k+1}$  respectively, monotonicity of the truncated distance in the scale places both within bounded distance of  $f_j$  at the common scale  $X_k$ , the triangle inequality places them within bounded distance of each other, and the growth in  $X$  of  $\sum_{p \leq X} (1 - \cos(\delta \log p))/p$  for fixed  $\delta \neq 0$  then forces  $|s_{k+1} - s_k| = O(1/\log X_k)$ . After passing to a geometrically growing subsequence of the scales,  $\log X_{k+1} \geq 2 \log X_k$  say, which monotonicity permits without weakening the conclusion, the increments are summable, the parameters converge, and monotonicity converts boundedness along the scales into finiteness of the full distance at the limit parameter. Unimodularity at the primes is consumed at the triangle-inequality step: the passage through the middle function  $f_j$  rests on the identity  $\chi n^{is_k} \overline{\chi n^{is_{k+1}}} = (\chi n^{is_k} \bar{f}_j)(f_j \overline{\chi n^{is_{k+1}}})$ , valid at the primes exactly when  $|f_j(p)| = 1$ , and for values of modulus less than one the two*

comparisons decouple on the primes where the modulus drops, which is exactly the room the scale-switching construction occupies.

**Remark 3.** For each  $j$ , the finitely many primes dividing  $\text{disc}(P_j)\text{lc}(P_j)$ , the discriminant times the leading coefficient, may be weighted arbitrarily: replacing the weights  $\rho_{P_j}(p)/p$  at those primes by any bounded values changes  $\mathbb{D}_{P_j}(f, g)^2$  by a bounded amount and leaves the finiteness condition (i) unchanged. The relative metric is therefore canonical up to the bad primes, and the conjecture is insensitive to that choice.

### 13 Monotone threshold purification of vector paths

Let  $d, m \geq 1$ . Let  $v_1, \dots, v_m \in \mathbb{R}^d$  satisfy  $\|v_i\|_2 \leq 1$ , where  $\|\cdot\|_2$  is the Euclidean norm. Let  $a_i : [0, 1] \rightarrow [0, 1]$  be continuous nondecreasing functions with  $a_i(0) = 0$  and  $a_i(1) = 1$ , called a monotone activation schedule, and define the fractional path

$$F(t) = \sum_{i=1}^m a_i(t) v_i.$$

A threshold rounding of  $F$  is a choice of pairwise distinct  $\tau_1, \dots, \tau_m \in (0, 1]$  together with the pure path

$$G(t) = \sum_{i: \tau_i \leq t} v_i.$$

**Conjecture 52** (Monotone threshold purification of vector paths). *There exists a universal constant  $C < \infty$  such that for every  $d, m \geq 1$ , every  $(v_i)_{i \leq m}$ , and every monotone activation schedule  $(a_i)_{i \leq m}$  as above, some threshold rounding satisfies*

$$\sup_{0 \leq t \leq 1} \|G(t) - F(t)\|_2 \leq C\sqrt{d}.$$

**Significance.** The headline attribution comes first. The special case  $a_i(t) = t$  for all  $i$  with  $\sum_i v_i = 0$  is exactly the open prefix-sum form of the Euclidean Steinitz problem at the scale  $O(\sqrt{d})$ : there  $F \equiv 0$ , and ordering the vectors by their thresholds turns the display into the assertion that some permutation  $\pi$  has all prefix sums bounded,  $\max_{k \leq m} \|\sum_{i \leq k} v_{\pi(i)}\|_2 \leq C\sqrt{d}$ . The distinctness of the thresholds is what makes this reduction exact: allowing coincident thresholds would admit the rounding with every  $\tau_i = 1$ , whose pure path is identically zero in this special case and exposes no prefix sum. Grinberg–Sevast’yanov [66] proved the bound  $d$  in every norm, the best known Euclidean bound is Banaszczyk’s [62, 63]  $O(\sqrt{d + \log m})$ , now matched constructively in the high-dimensional regime  $d \geq \text{polylog}(m)$  [174], and so already in this special case the conjecture asserts the removal of the logarithmic term uniformly in  $m$ . For the general threshold form, even the existence of any bound  $C(d)$  depending on  $d$  alone and uniform in  $m$  appears to be open. The mechanism is that  $F$  is a monotone trajectory in the zonotope generated by the  $v_i$ , the set of sums  $\sum_i \lambda_i v_i$  with  $\lambda_i \in [0, 1]$ , and the conjecture asserts that every monotone zonotope path admits an integral shadow at the optimal Euclidean discrepancy scale. The order  $\sqrt{d}$  is verified sharp: take  $m = d$ ,  $v_i = e_i$  the standard basis, and  $a_i(t) = t$ , so that at  $t = 1/2$  every coordinate of  $G(1/2) - F(1/2)$  equals  $\pm 1/2$  whatever the thresholds, giving  $\|G(1/2) - F(1/2)\|_2 = \sqrt{d}/2$ . No decisive theorem exists: the first would be any  $m$ -uniform bound  $C(d)$  for the threshold form, even with a  $d$ -dependence far worse than  $\sqrt{d}$ .

A falsification honesty note: since even the existence of a finite bound at each fixed  $d$  is open, the conjecture can fail in two ways, at fixed dimension, through paths with  $d$  fixed and  $m \rightarrow \infty$  whose best threshold roundings have unbounded discrepancy, or asymptotically, through an infinite family of monotone paths with  $d \rightarrow \infty$  every threshold rounding of which has discrepancy  $\omega(\sqrt{d})$ . Small- $d$  computer searches at bounded  $m$  cannot decide either failure mode.

## 14 Entropy dimension of random-free graphons

A graphon is a symmetric measurable function  $W : [0, 1]^2 \rightarrow [0, 1]$ , and  $W$  is random-free if it takes values in  $\{0, 1\}$  almost everywhere. Define the row pseudometric

$$d_W(x, y) = \int_0^1 |W(x, z) - W(y, z)| dz.$$

Identifying points at zero distance and completing yields a metric-measure space  $(\Omega_W, d_W, \rho_W)$ , the row space, where  $\rho_W$  is the pushforward of Lebesgue measure. The row space is Ahlfors  $s$ -regular if it is compact and there exist  $c, C, r_0 > 0$  such that

$$cr^s \leq \rho_W(B(x, r)) \leq Cr^s$$

for every  $x \in \text{supp } \rho_W$  and every  $0 < r < r_0$ . Let  $G(n, W)$  be the random labelled graph on  $[n] = \{1, \dots, n\}$  obtained from independent uniform  $U_1, \dots, U_n$  on  $[0, 1]$  by declaring  $ij$  an edge exactly when  $W(U_i, U_j) = 1$ , and let  $H(G(n, W))$  be its Shannon entropy with natural logarithms.

**Conjecture 53** (Entropy dimension of random-free graphons).

(i) [nonvacuity] *For every  $s \in (0, 2]$  there exists a random-free graphon whose row space is Ahlfors  $s$ -regular.*

(ii) [refined entropy-dimension law] *If the row space of a random-free graphon  $W$  is Ahlfors  $s$ -regular for some  $s > 0$ , then*

$$H(G(n, W)) = sn \log n + O_W(n).$$

**Proposition 1** (Entropy-dimension limit law). *If the row space of a random-free graphon  $W$  is Ahlfors  $s$ -regular for some  $s > 0$ , then*

$$H(G(n, W)) = sn \log n + O_W(n \log \log n),$$

and in particular  $\lim_{n \rightarrow \infty} H(G(n, W)) / (n \log n) = s$ .

*Proof.* Write  $d = d_W$  and  $\rho = \rho_W$ , and use natural logarithms. For the lower bound, set  $k = \lfloor n / \log n \rfloor$ , fix the labelling, and designate the last  $k$  vertices as tests, with latent points  $Z = (Z_1, \dots, Z_k)$  and remaining latent points  $X_1, \dots, X_{n-k}$ . The cross matrix  $M = (W(X_i, Z_j))_{i,j}$  is a function of the labelled graph, and conditioning cannot increase entropy, so  $H(G(n, W)) \geq H(M) \geq H(M \mid Z)$ . Conditionally on  $Z$  the signature rows  $S_Z(X_i) = (W(X_i, Z_1), \dots, W(X_i, Z_k))$  are independent with a common law, whence  $H(M \mid Z) = (n - k) \mathbb{E}_Z H(S_Z(X))$  with  $X$  an independent uniform point. Let  $C_Z$  be the collision probability of the signature law. Shannon entropy dominates the Rényi entropy of order two, so  $H(S_Z(X)) \geq -\log C_Z$ , and convexity of  $-\log$  gives  $\mathbb{E}_Z H(S_Z(X)) \geq -\log \mathbb{E}_Z C_Z$ . For independent uniform  $X$  and  $Y$  the tests are

independent uniform points, two rows agree at a test with probability  $1 - d(X, Y)$ , and Fubini gives the exact identity  $\mathbb{E}_Z C_Z = \mathbb{E}_{X,Y} (1 - d(X, Y))^k$ . Since  $(1 - u)^k \leq e^{-ku}$  and the upper mass bound gives  $\rho(B(x, r)) \leq Cr^s$ , a dyadic shell decomposition yields, uniformly in  $x$ ,

$$\int e^{-k d(x,y)} d\rho(y) \leq \rho(B(x, 1/k)) + \sum_{m \geq 0} e^{-2^m} \rho(B(x, 2^{m+1}/k)) = O_W(k^{-s}).$$

Hence  $\mathbb{E}_Z C_Z = O_W(k^{-s})$ , so  $\mathbb{E}_Z H(S_Z(X)) \geq s \log k - O_W(1)$  and  $H(G(n, W)) \geq (n - k)(s \log k - O_W(1)) = s n \log n - O_W(n \log \log n)$ .

For the upper bound, let  $\delta \in (0, \frac{1}{2})$ . The lower mass bound makes the balls of radius  $\delta/2$  around a maximal  $\delta$ -separated set disjoint, each of mass at least  $c(\delta/2)^s$ , so the row space admits a Borel partition into  $N_\delta = O_W(\delta^{-s})$  cells of diameter at most  $2\delta$ . Let  $Q_i$  be the cell of the  $i$ th latent row. Subadditivity and conditioning give  $H(G(n, W)) \leq n \log N_\delta + \binom{n}{2} \mathbb{E}_{(a,b)} h(p_{ab})$ , where  $(a, b)$  is the cell pair of two independent rows,  $p_{ab}$  is the conditional edge probability, and  $h$  is the binary entropy. For a  $\{0, 1\}$ -valued  $W$ , draw independent latent points  $X, X'$  with rows in the cell  $a$  and independent latent points  $Y, Y'$  with rows in the cell  $b$ , so that  $W(X, Y)$  and  $W(X', Y')$  are independent indicators of mean  $p_{ab}$  and  $p_{ab}(1 - p_{ab}) = \frac{1}{2} \mathbb{P}(W(X, Y) \neq W(X', Y'))$ , while the union bound gives  $\mathbb{P}(W(X, Y) \neq W(X', Y')) \leq \mathbb{P}(W(X, Y) \neq W(X', Y)) + \mathbb{P}(W(X', Y) \neq W(X', Y'))$ . Averaging each term over the opposite cell with its mass removes that cell restriction, so  $\mathbb{E}_{(a,b)} [p_{ab}(1 - p_{ab})] \leq \frac{1}{2} (\mathbb{E}_a \mathbb{E}_{X, X' \in a} d(X, X') + \mathbb{E}_b \mathbb{E}_{Y, Y' \in b} d(Y, Y')) \leq 2\delta$ . The bound  $h(p) \leq \varphi(p(1 - p))$  with the concave increasing  $\varphi(u) = 2u(1 + \log(1/u))$  and Jensen give  $\mathbb{E}_{(a,b)} h(p_{ab}) \leq \varphi(2\delta) = O(\delta \log(1/\delta))$ . Therefore  $H(G(n, W)) \leq s n \log(1/\delta) + O_W(n) + O(n^2 \delta \log(1/\delta))$ , and the choice  $\delta = (n \log^2 n)^{-1}$  gives  $H(G(n, W)) \leq s n \log n + O_W(n \log \log n)$ .  $\square$

**Remark 4.** *The conjecture is the refinement of the error term of proposition 1 to  $O_W(n)$ , which lies beyond both bounds of the proof: locating the mispredicted edges of a resolution-1/n code costs order  $n \log n$  if done naively, and the two-scale correction that codes coarse types first stops at  $O(n \log \log n)$ .*

**Significance.** The mechanism is metric distinguishability of rows: two row types at  $d_W$ -distance  $\varepsilon$  disagree on an  $\varepsilon$ -fraction of potential neighbours and become statistically distinguishable against  $n$  sampled vertices at a scale comparable to  $1/n$ , an Ahlfors  $s$ -regular row space has about  $n^s$  distinguishable types at that scale, and the latent type of each of the  $n$  labelled vertices then carries  $s \log n + O(1)$  nats. Clause (i) certifies that the hypothesis class is nonvacuous at every fractional  $s$ : it is expected to be provable by Cantor-type threshold constructions, in which  $W$  is the indicator of a threshold event read along a self-similar Cantor set, and it is included so that clause (ii) quantifies over a genuinely rich family. The nearest literature boundary is Hatami–Norine [143]: a graphon is random-free exactly when  $H(G(n, W)) = o(n^2)$ , and every subquadratic growth order occurs for suitable random-free graphons, so no hypothesis short of a metric-measure condition on the row space can pin the coefficient of  $n \log n$ , and Ahlfors regularity is exactly the condition that does. The decisive theorem half-exists: the limit law is provable from regularity by the collision-entropy and quantization bounds of proposition 1, so the first theorem for the conjecture proper is the linear error term for one explicit self-similar threshold graphon of noninteger  $s$ , together with a proof of clause (i). The failure mode is a genuinely growing correction: an Ahlfors regular random-free graphon whose entropy exceeds  $s n \log n + C n$  for every  $C$  would show that the resolution-1/n coding loss is intrinsic rather than an artefact of the two-scale method.

## 15 A Hessian principle for isolated microcanonical graph phases

Fix a partition

$$[0, 1] = I_1 \sqcup \cdots \sqcup I_q, \quad |I_a| = \alpha_a > 0.$$

A coloured graphon is a symmetric measurable  $U : [0, 1]^2 \rightarrow [0, 1]$  considered together with this fixed colour partition. Let  $p(U) \in [0, 1]^D$  with  $D = q(q+1)/2$  denote the block-average vector

$$p_{ab}(U) = \frac{1}{|I_a||I_b|} \int_{I_a \times I_b} U(x, y) \, dx \, dy, \quad 1 \leq a \leq b \leq q.$$

A coloured finite graph is a finite simple graph  $F$  with a colour  $c(v) \in \{1, \dots, q\}$  attached to each vertex, and its coloured homomorphism density in  $U$  is

$$t(F, U) = \int_{[0, 1]^{V(F)}} \prod_{v \in V(F)} \alpha_{c(v)}^{-1} \mathbf{1}_{I_{c(v)}}(x_v) \prod_{uv \in E(F)} U(x_u, x_v) \prod_{v \in V(F)} dx_v.$$

Fix coloured finite graphs  $F_1, \dots, F_k$  and put  $\mathbf{t}(U) = (t(F_1, U), \dots, t(F_k, U))$ . Define the graphon entropy

$$\mathcal{S}(U) = \frac{1}{2} \int_{[0, 1]^2} [-U \log U - (1 - U) \log(1 - U)] \, dx \, dy,$$

with  $0 \log 0 = 0$ , and for a feasible constraint vector  $\mathbf{a}$  the contracted entropy

$$\Psi_{\mathbf{a}}(p) = \sup\{\mathcal{S}(U) : \mathbf{t}(U) = \mathbf{a}, p(U) = p\}.$$

The finite graphs are coloured with pinned classes. For each  $n$  set  $n_a = \lfloor \alpha_a n \rfloor$  and  $N_n = \sum_a n_a$ , and fix a partition of the vertex set  $[N_n]$  into colour classes  $V_1, \dots, V_q$  with  $|V_a| = \lfloor \alpha_a n \rfloor$  exactly, so that  $N_n = n + O(q)$ . For a graph  $G$  on  $[N_n]$  with these colour classes, partition each  $I_a$  into  $n_a$  consecutive intervals of length  $|I_a|/n_a$  assigned to the vertices of  $V_a$ , let  $U_G$  be the  $\{0, 1\}$ -valued coloured graphon equal to 1 on a product of two vertex intervals exactly when the corresponding pair is an edge, and put  $\mathbf{t}(G) = \mathbf{t}(U_G)$  and  $\hat{p}(G) = p(U_G)$ .

Fix  $0 < \eta < 1$  and assume the following regularity hypotheses.

(H1) The supremum of  $\mathcal{S}(U)$  over  $\mathbf{t}(U) = \mathbf{a}$  is attained at a unique coloured graphon  $U_*$ , up to measure-preserving maps preserving each colour class.

(H2) Writing  $p_* = p(U_*)$ , the feasible block-average set  $\{p(U) : \mathbf{t}(U) = \mathbf{a}\}$  is, near  $p_*$ , a  $C^3$  embedded manifold  $M \subset \mathbb{R}^D$ .

(H3) The restriction  $\Psi_{\mathbf{a}}|_M$  is  $C^3$  near  $p_*$  and has there a unique local maximum.

(H4) The quadratic form  $A = -D_M^2 \Psi_{\mathbf{a}}(p_*)$  is positive definite on the tangent space  $T = T_{p_*} M$ .

(H5) [transversally Lipschitz constraint correspondence] There exist  $\delta_0 > 0$  and  $L < \infty$  such that every coloured graphon  $U$  with  $\|p(U) - p_*\| \leq \delta_0$  and  $\|\mathbf{t}(U) - \mathbf{a}\|_{\infty} = \delta \leq \delta_0$  satisfies  $\text{dist}(p(U), M) \leq L\delta$ . In particular the conditioning window below confines  $\hat{p}_n$ , on the event  $\|\hat{p}_n - p_*\| \leq \delta_0$ , to an  $o(n^{-1})$  thickening of  $M$ .

(H6) [nonemptiness] For every sufficiently large  $n$  the conditioning set  $\{G : \|\mathbf{t}(G) - \mathbf{a}\|_{\infty} \leq n^{-1-\eta}\}$  over graphs  $G$  on  $[N_n]$  with the pinned colour classes is nonempty.

Let  $G_n$  be uniform on the conditioning set of (H6), let  $\hat{p}_n = \hat{p}(G_n)$ , and let  $\Pi_T$  be orthogonal projection of  $\mathbb{R}^D$  onto  $T$ .

**Conjecture 54** (Hessian principle for an isolated microcanonical phase). *Under hypotheses (H1)–(H6),*

$$n \Pi_T(\hat{p}_n - \mathbb{E} \hat{p}_n) \xrightarrow{d} N_T(0, A^{-1}),$$

where  $N_T(0, A^{-1})$  denotes the centred Gaussian law on  $T$  whose precision form is  $A$ .



**Significance.** The mechanism is a Laplace expansion at the dense-graph entropy exponent  $n^2$ : on the feasible manifold,  $\Psi_{\mathbf{a}}(p_* + u) = \Psi_{\mathbf{a}}(p_*) - \frac{1}{2}\langle Au, u \rangle + O(\|u\|^3)$ , so the fluctuation scale is  $u \asymp n^{-1}$  and the predicted covariance on the tangent space is  $A^{-1}$ , while (H5) suppresses the transversal directions below the fluctuation scale, which is exactly why the conclusion is stated only after projection by  $\Pi_T$  and centring at  $\mathbb{E}\hat{p}_n$ . The nearest literature boundary is the Chatterjee–Varadhan [64] large deviation principle, which supplies the  $n^2$  normalization and the variational description of conditioned dense graphs, together with the multipodal structure theory of Kenyon–Radin–Ren–Sadun [67] and Neeman–Radin–Sadun [72] (constrained entropy maximizers are step-function graphons with finitely many blocks), which makes hypotheses (H1)–(H4) verifiable in concrete constrained ensembles. A scale check disposes of one suspect: a single edge toggle moves each coordinate of  $\mathbf{t}(G)$  by  $O(n^{-2})$ , so the lattice of attainable constraint values has spacing of order  $n^{-2}$ , far below the fluctuation scale  $n^{-1}$ , and the genuine concern is therefore Laplace-ratio control, namely uniform comparison of microcanonical counts with their Laplace approximations across windows of width  $o(n^{-1})$ , not arithmetic oscillation. Nonemptiness (H6) is a genuine hypothesis and may fail along subsequences of  $n$  for arithmetic reasons, in which case the statement is asserted only along the subsequence where it holds. No decisive theorem exists at this generality: the natural first one is the case  $q = 1$  with the edge-triangle constraint pair in a region where the entropy maximizer is bipodal, a two-block step function, and (H1)–(H5) have been verified. The failure mode is an isolated nondegenerate phase whose tangent fluctuations are non-Gaussian or live at a different scale, which would necessarily come from a failure of Laplace-ratio control. The conjectural content is exactly that (H1)–(H6) already force this local regularity: a subexponential prefactor in the microcanonical counts varying across the conditioning window would leave every hypothesis intact while tilting the limit away from  $N_T(0, A^{-1})$ , and the conjecture asserts that isolated nondegenerate phases exclude it.

## 16 Sharp stability of entropy idempotence on finite abelian groups

Let  $G$  be a finite abelian group and  $\mu$  a probability measure on  $G$ . Write  $H(\mu)$  for Shannon entropy with natural logarithms,  $\mu * \mu$  for the convolution, the law of  $X + Y$  with  $X, Y$  independent of law  $\mu$ , and

$$\Delta(\mu) = H(\mu * \mu) - H(\mu)$$

for the convolution entropy defect. If  $H \leq G$  is a subgroup and  $a \in G$ , let  $u_{a+H}$  be the uniform measure on the coset  $a + H$ . Total variation distance is

$$\|\mu - \nu\|_{\text{TV}} = \frac{1}{2} \sum_{x \in G} |\mu(x) - \nu(x)|.$$

**Conjecture 55** (Sharp stability of entropy idempotence). *For every finite abelian group  $G$  and every probability measure  $\mu$  on  $G$ , with entropies in natural logarithms,*

$$\inf_{\substack{H \leq G \\ a \in G}} \|\mu - u_{a+H}\|_{\text{TV}}^2 \leq \frac{2}{\log 2} (H(\mu * \mu) - H(\mu)),$$

*and the constant  $2/\log 2$  is sharp.*

**Remark 5.** *The equality case is verified. With  $X, Y$  independent of law  $\mu$ ,*

$$\Delta(\mu) = H(X + Y) - H(X) = H(X + Y) - H(X + Y | Y) = I(X + Y; Y) \geq 0,$$

*where  $I$  denotes mutual information. If  $\Delta(\mu) = 0$  then  $X + Y$  is independent of  $Y$ , which forces the translates of  $\mu$  by the elements of its support to coincide, and hence forces  $\mu$  to be uniform on a coset of a subgroup. The right-hand side therefore vanishes exactly on the proposed extremizers.*

**Significance.** The mechanism is that the defect is the mutual information created by one self-convolution, so the conjecture is a dimension-free quantitative classification of near-idempotent measures with cosets as the exact equality case. The sharpness discussion identifies the extremal geometry. Coset perturbations  $\mu = (1 + \varepsilon f)u_H$  with  $\sum_{x \in H} f(x) = 0$  give, as  $\varepsilon \rightarrow 0$ , the ratio

$$\frac{\|\mu - u_H\|_{\text{TV}}^2}{\Delta(\mu)} \rightarrow \frac{(\mathbb{E}|f|)^2}{2 \mathbb{E}f^2} \leq \frac{1}{2}$$

by Cauchy–Schwarz, so perturbative examples certify at most the constant  $1/2$ . The near-extremizers are instead discretized Gaussians at intermediate scale on  $\mathbb{Z}/p\mathbb{Z}$ : for  $1 \ll \sigma(p) \ll p$ , the discretization  $\mu_p$  of a centred Gaussian of standard deviation  $\sigma(p)$  has  $\Delta(\mu_p) \rightarrow \frac{1}{2} \log 2$ , the entropy-power constant recording the entropy gain of doubling a Gaussian variance, while its total variation distance to every  $u_{a+H}$  tends to 1, since the only subgroups of  $\mathbb{Z}/p\mathbb{Z}$  are trivial and full and  $\mu_p$  is asymptotically singular to every point mass and to the uniform measure. Hence no constant below  $2/\log 2$  can serve, the sharpness half of the statement. The positioning against the literature is as follows: the entropic inverse theorems of Tao’s sumset entropy theory [74] and the entropic polynomial Freiman–Ruzsa theorem of Gowers–Green–Manners–Tao [65] conclude in Ruzsa distance, an entropic closeness of  $X$  to a subgroup-uniform variable, not in total variation; the subgroup-form inverse theorem of Green–Manners–Tao [87], supplying for small defect a finite subgroup within entropic Ruzsa distance a bounded multiple of  $\Delta(\mu)$ , upgrades to total variation by a Pinsker argument, and with the trivial bound in the large-defect regime this settles the existence of some finite universal constant, so the sharp value is what the statement adds. On  $\mathbb{Z}/p\mathbb{Z}$  there are no proper nontrivial subgroups, so the conjecture specializes to a discrete entropy-power-type lower bound for spread measures: any  $\mu$  far in total variation from every point mass and from the uniform measure must create a definite amount of entropy under self-convolution. The natural first decisive theorem is that  $\mathbb{Z}/p\mathbb{Z}$  specialization with the sharp constant. The defect also has the exact form  $H(\mu * \mu) - H(\mu) = \mathbb{E}_Y D_{\text{KL}}(\tau_Y \mu \| \mu * \mu)$ , with  $\tau_y$  translation by  $y$ , so a small defect makes typical support-translates of  $\mu$  close to  $\mu * \mu$  in relative entropy, and Pinsker’s inequality converts that closeness to total variation: the route to a universal constant is to upgrade approximate invariance under many translations to proximity to a coset uniform. The failure mode is a sequence  $\mu_n$  with  $\Delta(\mu_n) \rightarrow 0$  whose coset distance is much larger than  $\sqrt{\Delta(\mu_n)}$ , and the natural hunting ground is measures straddling several subgroup scales inside groups with rich subgroup lattices, a family for which computational search through  $\mathbb{Z}/2^8\mathbb{Z}$  finds no ratio  $\|\mu - u_{a+H}\|_{\text{TV}}^2/\Delta(\mu)$  above the discretized-Gaussian value  $2/\log 2$ . Of the original assertion’s three layers, the existence of some universal constant is settled by the inverse theorem as noted, and the conjecture is the remaining two: the sharp value  $2/\log 2$  and the classification of near-extremizers as the discretized Gaussians at intermediate scale.

## 17 Stationary convolution-entropy rigidity

Let  $K$  be a finite abelian group and  $\Omega = K^{\mathbb{Z}}$  with coordinatewise addition and the left shift  $\sigma$ . Let  $\mu$  be a  $\sigma$ -invariant ergodic Borel probability measure on  $\Omega$ , and write  $h_\sigma(\mu)$  for its Kolmogorov–Sinai entropy. Define the translational stabilizer

$$H_\mu = \{h \in K^{\mathbb{Z}} : (\tau_h)_*\mu = \mu\}, \quad \tau_h(x) = x + h,$$

a closed shift-invariant subgroup of  $K^{\mathbb{Z}}$ , and let  $\pi_\mu : K^{\mathbb{Z}} \rightarrow K^{\mathbb{Z}}/H_\mu$  be the quotient homomorphism. Let  $\mu * \mu$  be the law of  $X + Y$  for independent  $X, Y \sim \mu$ .

**Conjecture 56** (Stationary convolution-entropy rigidity). *For every shift-ergodic  $\mu$ ,*

$$h_\sigma(\mu * \mu) = h_\sigma(\mu) \iff h_\sigma((\pi_\mu)_*\mu) = 0.$$

**Remark 6.** *The binary specialization deserves prominence. Take  $K = \mathbb{Z}/2\mathbb{Z}$ . By Kitchens’ structure theory of group shifts [68], every closed shift-invariant subgroup of  $(\mathbb{Z}/2\mathbb{Z})^{\mathbb{Z}}$  is a group shift of finite type, and every proper one is the solution set of a nontrivial linear recurrence and hence finite. If  $H_\mu$  is finite the quotient map has finite fibres and  $h_\sigma((\pi_\mu)_*\mu) = h_\sigma(\mu)$ , while if  $H_\mu$  is everything then  $\mu$  is the Haar measure, the uniform Bernoulli process, with entropy  $\log 2$ . Since  $h_\sigma(\mu * \mu) \geq h_\sigma(\mu)$  always, by conditioning on one summand, the conjecture therefore asserts: every ergodic binary process with  $0 < h_\sigma(\mu) < \log 2$  gains entropy under independent XOR, the coordinatewise sum modulo 2 with an independent copy.*

**Proposition 2** (Easy direction, conditional). Suppose  $h_\sigma((\pi_\mu)_*\mu) = 0$ . Assume the following two ingredients.

(a) The Abramov–Rokhlin entropy addition formula holds for the Haar-fibred compact group extension  $\pi_\mu : (\Omega, \mu, \sigma) \rightarrow (\Omega/H_\mu, (\pi_\mu)_*\mu, \sigma)$  and for the corresponding extension of  $\mu * \mu$ , so that each total entropy splits as the quotient entropy plus the common Haar fibre entropy, the conditional measures of  $\mu$  and of  $\mu * \mu$  over the quotient being Haar measures on cosets of  $H_\mu$ .

(b) The entropy of a factor of an independent joining is bounded by the sum of the entropies of the factors, so that  $(\pi_\mu)_*(\mu * \mu) = ((\pi_\mu)_*\mu) * ((\pi_\mu)_*\mu)$  has entropy at most  $2h_\sigma((\pi_\mu)_*\mu) = 0$ .

Then  $h_\sigma(\mu * \mu) = h_\sigma(\mu)$ .

**Significance.** The mechanism is the stationary analogue of entropy idempotence on a finite group: equality of entropy under independent self-convolution should force all positive entropy to be Haar-type randomness along a translation-invariant subgroup process, modulo a quotient that may retain arbitrarily rich zero-entropy dynamics such as rotations, substitutions, or Toeplitz systems. The easy direction is recorded above as a conditional proposition with its two ingredients named, the Abramov–Rokhlin formula for Haar-fibred compact group extensions and the factor-of-joining entropy bound, and the difficult direction is the rigidity claim that no other equality mechanism exists. The nearest literature boundary consists of Kułaga-Przymus–Lemańczyk [69] on the entropy of products of independent stationary processes and Lindenstrauss–Meiri–Peres [70] on the entropy of convolutions, neither of which decides the binary question of remark 6. No decisive theorem appears to be on record even for Markov processes, and strict XOR entropy gain for mixing two-state Markov chains would be the first. The failure-mode honesty note is this: the natural counterexample hunting ground is the Ornstein–Weiss [73] bilaterally deterministic processes, positive entropy processes whose distant past and distant future jointly determine the whole trajectory, together with  $T, T^{-1}$ -type constructions (skew products driven by a random

walk, in the style of Kalikow), and no search has yet probed either family, so the negation of the conjecture has an unexplored natural habitat. The gap the rigidity must close is quantitative: equality of entropy rates asserts only that the block mutual information  $I(X_{1:n} + Y_{1:n}; Y_{1:n})$  is  $o(n)$ , not that it is bounded, so sublinear information spread through arbitrarily long-range structure is the resource a counterexample would exploit. The natural staircase of evidence runs from independent letters through Markov and mixing processes to completely positive entropy.

## Part IV

# The class-counting polynomial of graphical Lie algebras

*Added to the deposit: 5 August 2026.*

The twenty conjectures below concern a single canonical object, the class-counting polynomial of a finite graph, approached through the two-step nilpotent Lie algebras that realize it as a rank generating function. They form three programmes. A rigidity and reconstruction programme (Conjectures 57 to 66) asserts that coarse specializations of the polynomial already determine it and that the polynomial in turn determines classical invariants. A log-concavity programme (Conjectures 67 to 71) locates the exact strength of positivity that the coefficient arrays support, with the boundary calibrated by explicit counterexamples one level higher. A tree extremal programme (Conjectures 72 to 76) places the path and the star at the two ends of every ordering the polynomial induces on trees. The proved layer is separated from the conjectural one: the rank formula (Proposition 3) and the two lower bounds on the number of rank values in a tree (Lemma 2) are theorems with full proofs, joined by the attainment of those bounds by the path and the star (Proposition 4) and the reduction of the star majorization to a single anti-concentration bound (Proposition 5), and every conjecture is stated against that floor. Each statement is held to the standard of the rest of the collection: a canonical object, a stated mechanism, the nearest literature boundary, a first decisive theorem where one exists, and an honest failure mode. Two statements, Conjectures 63 and 71, have since been resolved false by explicit counterexamples recorded with them, a seven-vertex unicyclic pair and a connected eight-vertex graph respectively, and both are retained with their resolutions rather than renumbered.

For a finite simple graph  $G$  on vertex set  $V(G)$  with edge set  $E(G)$  and a field  $F$ , the *graphical Lie algebra*  $L_G(F)$  is the two-step nilpotent Lie algebra over  $F$  with basis  $\{v_i : i \in V(G)\}$  together with the central basis  $\{z_e : e \in E(G)\}$ , the bracket being  $[v_a, v_b] = z_{ab}$  for each edge  $\{a, b\}$  with  $a < b$ ,  $[v_b, v_a] = -z_{ab}$ , and all other brackets of basis vectors zero. For  $S \subseteq V(G)$  write  $N_G[S]$  for the closed neighbourhood of  $S$ , the set of vertices lying in  $S$  or adjacent to it, and  $c(G[S])$  for the number of connected components of the induced subgraph  $G[S]$ , and set

$$\rho_G(S) = |N_G[S]| - c(G[S]), \quad \rho_G(\emptyset) = 0.$$

**Proposition 3** (Rank formula). Let  $G$  be a graph on  $n$  vertices, let  $F$  be a field, and let  $x \in L_G(F)$  have vertex part supported on  $S$ , all coordinates  $x_i$  with  $i \in S$  being nonzero. Then

$$\text{rank}(\text{ad}_x) = \rho_G(S),$$

independently of the field and of the nonzero values.

*Proof.* The map  $\text{ad}_x$  kills the centre and sends  $v_j$  to  $\sum_{i \in S, i \sim j} \pm x_i z_{ij}$ , so  $\text{rank}(\text{ad}_x)$  is the rank of the linear map  $y \mapsto [x, y]$  on the vertex space, whose coefficient at  $z_{ab}$  is  $x_a y_b - x_b y_a$ . A vector  $y$  lies in the kernel if and only if  $x_a y_b = x_b y_a$  for every edge  $\{a, b\}$ . If both endpoints lie in  $S$  this forces the ratio  $y_i/x_i$  to be constant along edges of  $G[S]$ , one free parameter per connected component of  $G[S]$ . If exactly one endpoint  $a$  lies in  $S$  it forces  $y_b = 0$ , for every  $b \in N(S) \setminus S$ . Edges with neither endpoint in  $S$  impose nothing, so  $y_j$  is free for every  $j$  outside  $N[S]$ . Hence  $\dim \ker = (n - |N[S]|) + c(G[S])$ , and the rank is  $|N[S]| - c(G[S])$ .  $\square$

The formula was verified before it was proved: on eighteen thousand random pairs of a graph and an element, the rank of  $\text{ad}_x$  computed by direct linear algebra over  $\mathbb{F}_2$ ,  $\mathbb{F}_3$ , and  $\mathbb{F}_5$  agreed with  $\rho_G(S)$  in every case, and the proof above was written only afterwards.

The *class-counting polynomial* of  $G$  is

$$C_G(X, Y) = \sum_{S \subseteq V(G)} (X - 1)^{|S|} Y^{\rho_G(S)}.$$

The polynomial is Rossmann's [77]: he introduced it, proved that its specializations enumerate the conjugacy classes of graphical groups over  $\mathbb{F}_q$  by size, proved the multiplicativity  $C_{G \sqcup H} = C_G C_H$  over disjoint unions, and showed that the cardinalities of the connected dominating sets of  $G$  can be read off from it. Set

$$H_G(U, Y) = C_G(1 + U, Y), \quad F_G(X, Y) = X^{|E(G)|} C_G(X, X^{-1}Y), \quad f_G(X) = F_G(X, 1),$$

so that the coefficient of  $U^k Y^r$  in  $H_G$  counts the sets  $S$  with  $|S| = k$  and  $\rho_G(S) = r$ . At  $X = q$  the polynomial records the distribution of  $\text{rank}(\text{ad}_x)$  over  $L_G(\mathbb{F}_q)$ , by Proposition 3. The *subgraph-component polynomial* of Tittmann–Averbouch–Makowsky [78] is

$$Q_G(U, W) = \sum_{S \subseteq V(G)} U^{|S|} W^{c(G[S])}.$$

For a tree  $T$  on  $n$  vertices let  $s(T)$  be the number of distinct values taken by  $\rho_T$ , let  $\ell(T)$  be the number of leaves, and let  $\text{diam}(T)$  be the diameter.

**Lemma 2** (Two lower bounds on the number of rank values). *For every tree  $T$  on  $n \geq 2$  vertices,*

$$s(T) \geq n - \ell(T) + 2 \quad \text{and} \quad s(T) \geq \text{diam}(T) + 1.$$

*Proof.* For the first bound, when  $n = 2$  the values 0 and 1 occur. For  $n \geq 3$  the internal vertices, those that are not leaves, induce a subtree with  $m = n - \ell(T) \geq 1$  vertices. Order them  $v_1, \dots, v_m$  by breadth-first search from an internal root, so that every prefix  $S_j = \{v_1, \dots, v_j\}$  is connected and  $\rho(S_j) = |N[S_j]| - 1$ . Each  $v_{j+1}$  has degree at least two in  $T$  and its parent lies in the prefix, so it has a neighbour  $c$  on the far side, whose only neighbour inside  $N[S_j]$  could be  $v_{j+1}$  itself, which is not in  $S_j$ . Hence  $c$  lies in  $N[S_{j+1}]$  but not in  $N[S_j]$ , the sizes strictly increase, and the chain realizes  $m$  distinct values, all at least  $\deg(v_1) \geq 2$ . Together with  $\rho(\emptyset) = 0$  and the value 1 at a single leaf this gives at least  $m + 2 = n - \ell(T) + 2$  values. For the second bound take a diametral path  $v_0, v_1, \dots, v_d$  with  $d = \text{diam}(T)$ , whose endpoint  $v_0$  is a leaf. For the prefixes  $S_j = \{v_0, \dots, v_{j-1}\}$  with  $1 \leq j \leq d$ , the vertex  $v_{j+1}$  witnesses strict growth of  $|N[S_j]|$  as long as it exists, since a tree has no edge from  $v_{j+1}$  back into  $\{v_0, \dots, v_{j-1}\}$ . The chain realizes  $d$  distinct values starting from  $\rho(S_1) = 1$ , and  $\rho(\emptyset) = 0$  gives  $d + 1$  in total.  $\square$

**Proposition 4** (Attainment of the bounds: path and star). The path  $P_n$  has  $s(P_n) = n$  for every  $n \geq 2$ , and the star  $K_{1,n-1}$  has  $s(K_{1,n-1}) = 3$  for every  $n \geq 3$ . Both trees attain equality in each bound of Lemma 2.

*Proof.* For a connected graph on  $n$  vertices and a nonempty subset  $S$  one has  $1 \leq \rho(S) \leq n - 1$ , the lower bound because a single vertex gives  $\rho(\{v\}) = \deg(v) \geq 1$  and the upper because  $|N[S]| \leq n$  and  $c(G[S]) \geq 1$ . With  $\rho(\emptyset) = 0$  this leaves at most  $n$  distinct values, so  $s(T) \leq n$  for every tree  $T$ . The path has  $\text{diam}(P_n) = n - 1$ , so Lemma 2 gives  $s(P_n) \geq n$ , forcing  $s(P_n) = n$ , and with  $\ell(P_n) = 2$  both  $n - \ell(P_n) + 2$  and  $\text{diam}(P_n) + 1$  equal  $n$ . For the star with  $n \geq 3$ , Proposition 3 gives attained ranks  $\{0, 1, n - 1\}$ , so  $s(K_{1,n-1}) = 3$ , and with  $\ell(K_{1,n-1}) = n - 1$  and  $\text{diam}(K_{1,n-1}) = 2$  both  $n - \ell + 2$  and  $\text{diam} + 1$  equal 3.  $\square$

## 18 Rigidity and reconstruction

**Conjecture 57** (Irreducibility and connectivity). *For every graph  $G$  with at least one vertex,  $H_G(U, Y)$  is irreducible in  $\mathbb{Z}[U, Y]$  if and only if  $G$  is connected.*

**Significance.** Rossmann’s multiplicativity [77] makes the disconnected direction immediate, each factor being nonconstant since  $H_{K_1} = 1 + U$ , so the content is the connected direction. The mechanism is that a factorization of  $H_G$  would express the joint array of  $(|S|, \rho_G(S))$  over all vertex subsets as a product of two smaller arrays, which is exactly what a separation of  $G$  into two vertex-disjoint parts produces, and the conjecture asserts that no other source of such a product structure exists: connectivity of the graph should be readable as algebraic indecomposability of the polynomial. The nearest literature boundary is the multiplicativity theorem itself, which supplies one direction and gives the statement its shape. The verification is exact rather than probabilistic: all 996 connected graphs on at most seven vertices have irreducible  $H_G$ , by exact factorization over  $\mathbb{Z}$ . A first decisive theorem would be irreducibility of  $H_{P_n}$  for the paths, a one-parameter family at the bottom of the connectivity hierarchy. The failure mode is a connected graph whose polynomial factors, which would reveal a product structure in the rank array not induced by any decomposition of the graph and would make the polynomial blind to connectivity.

**Conjecture 58** (Recovery of the subgraph-component polynomial). *For all graphs  $G$  and  $H$ ,  $C_G = C_H$  implies  $Q_G = Q_H$ .*

**Significance.** The two polynomials record transverse marginals of the induced-subgraph structure:  $C_G$  records the joint distribution of  $(|S|, |N_G[S]| - c(G[S]))$  over all vertex subsets and  $Q_G$  the joint distribution of  $(|S|, c(G[S]))$ , and no functional relation between the two arrays is apparent, so the mechanism must be rigidity of the pairs of arrays that actually arise from graphs. The nearest literature boundary pairs the two sources: the class-counting polynomial and its enumerative theorems are Rossmann’s [77], the subgraph-component polynomial is Tittmann–Averbouch–Makowsky’s [78], and no bridge between them is on record. The polynomial  $C_G$  does not determine the graph, and every known collision class was tested: the 119 collision classes among the 1252 graphs on at most seven vertices, a randomized eight-vertex corpus, the unique collision class among all 3159 trees on fourteen vertices (there is none through thirteen, and the class was refound independently by exhaustive search), and the first unicyclic collision at ten vertices. In every class  $Q_G$  agrees, as do the domination invariants of Conjectures 65 and 66.



A first decisive theorem would be a derivation of  $Q_T$  from  $C_T$  for trees. The failure mode is a  $C$ -collision pair with distinct  $Q$ , which would show that the closed-neighbourhood array and the component array are genuinely independent invariants and would break the entire recovery programme at its first joint.

**Conjecture 59** (Two-field rigidity). *If  $C_G(2, Y) = C_H(2, Y)$  and  $C_G(3, Y) = C_H(3, Y)$ , then  $C_G = C_H$ . Equivalently, the adjoint-rank distributions of the graphical Lie algebras over  $\mathbb{F}_2$  and  $\mathbb{F}_3$  determine them over every finite field.*

**Significance.** The two hypotheses are the unweighted and the  $2^{|S|}$ -weighted rank statistics,  $C_G(2, Y) = \sum_S Y^{\rho_G(S)}$  and  $C_G(3, Y) = \sum_S 2^{|S|} Y^{\rho_G(S)}$ , and recovering a nonnegative integer array from these two evaluations is impossible for general arrays, so the mechanism is rigidity of the arrays that graphs realize rather than linear algebra. The equivalence with the finite-field form is Proposition 3, and the nearest literature boundary is Rossmann’s enumeration theorem [77], through which the conjecture reads: two conjugacy-class censuses, over  $\mathbb{F}_2$  and  $\mathbb{F}_3$ , determine the census over every finite field. The verification is exhaustive on at most seven vertices and covers six thousand random eight-vertex graphs. A first decisive theorem would be two-field rigidity for trees. The failure mode is a pair of graphs agreeing over  $\mathbb{F}_2$  and  $\mathbb{F}_3$  yet separated over some larger field, which would reveal genuinely field-dependent information in the adjoint-rank distribution and would locate it at a specific weight  $(q - 1)^{|S|}$ .

**Conjecture 60** (Hybrid rigidity). *If  $C_G(2, Y) = C_H(2, Y)$  and  $Q_G = Q_H$ , then  $C_G = C_H$ .*

**Significance.** The mechanism is complementarity of marginals: the binary specialization carries the rank distribution and  $Q_G$  carries the joint size-and-component array, and the conjecture asserts that these two projections of the induced-subgraph data jointly pin the full class-counting polynomial. The pairing is calibrated against a cheaper substitute, since  $C_G(2, Y)$  together with the degree sequence does not determine  $C_G$ , so the component information in  $Q_G$  is doing work that no degree data can do. The nearest literature boundary is again the pair Rossmann [77] and Tittmann–Averbouch–Makowsky [78], whose polynomials the hypothesis mixes. The verification is exhaustive on at most seven vertices. A first decisive theorem would be the tree case. The failure mode is a pair agreeing in both hypotheses but not in  $C$ , which would show that the joint array holds correlation information invisible to this pair of its natural projections.

**Conjecture 61** (Binary edge-deck reconstruction). *For graphs  $G$  and  $H$  with at least four edges each, if the multisets  $\{C_{G-e}(2, Y) : e \in E(G)\}$  and  $\{C_{H-e}(2, Y) : e \in E(H)\}$  coincide, then  $G$  and  $H$  are isomorphic.*

**Significance.** This lies in the orbit of the edge reconstruction conjecture, for which Bondy–Hemming [79] is the survey. Each card here retains only the binary rank distribution of an edge-deleted subgraph rather than its isomorphism type, so the deck is a drastic coarsening of the classical edge deck, and the mechanism is that rank distributions vary so tightly under single-edge deletion that the multiset of shadows should already separate graphs. The nearest literature boundary is the reconstruction literature surveyed in [79], against which the present statement trades weaker cards for the same conclusion. Since each binary card is computed from the corresponding ordinary card, the ordinary edge deck determines the binary deck, so the statement implies the edge reconstruction conjecture on this range and is strictly the bolder

claim. The verification is exhaustive on at most seven vertices among graphs with at least four edges. A first decisive theorem would be reconstruction of trees from the binary deck. The failure mode is a nonisomorphic pair with equal binary decks, which would measure exactly how much the rank distribution forgets under edge deletion, and a counterexample here would be the least surprising in the part, which is why the statement is the least-supported in the part.

**Conjecture 62** (Bipartite binary rigidity). *For bipartite graphs  $G$  and  $H$ ,  $C_G(2, Y) = C_H(2, Y)$  implies  $C_G = C_H$ .*

**Significance.** A single specialization cannot determine the polynomial in general, since the implication fails already for connected unicyclic graphs on seven vertices (the resolution of Conjecture 63), so the mechanism is that a hereditary class restriction removes the room in which the unweighted rank distribution and the size-weighted array can decouple. The nearest literature boundary is Rossmann’s specialization theorem [77], which gives the hypothesis its meaning as the  $\mathbb{F}_2$  census. The verification is a six-thousand-graph random holdout of bipartite graphs on nine to twelve vertices, spanning 2747 distinct binary specializations, with no violation. A first decisive theorem would be the tree case, shared with the surviving tree layer of Conjecture 63. The failure mode is a bipartite pair with equal binary specialization and distinct  $C$ , which would locate the boundary of one-field rigidity strictly inside the bipartite world and would redirect the class-restriction programme towards sparser families.

**Conjecture 63** (Connected-pseudoforest binary rigidity, resolved false). *For  $G$  and  $H$  each a tree or a connected unicyclic graph,  $C_G(2, Y) = C_H(2, Y)$  implies  $C_G = C_H$ .*

**Remark 7** (Resolution). *The conjecture is false as stated. The connected unicyclic graphs on vertex set  $\{0, \dots, 6\}$  with edge sets  $\{\{0, 3\}, \{0, 4\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{3, 6\}, \{4, 5\}\}$  and  $\{\{0, 1\}, \{0, 3\}, \{0, 4\}, \{1, 2\}, \{2, 3\}, \{2, 4\}, \{3, 6\}, \{4, 5\}\}$  share the binary specialization  $C(2, Y) = 1 + 3Y + 7Y^2 + 29Y^3 + 32Y^4 + 40Y^5 + 16Y^6$  while their class-counting polynomials differ already in the singleton stratum, whose coefficients record the degree multisets  $\{1, 1, 1, 2, 3, 3, 3\}$  and  $\{1, 1, 2, 2, 2, 3, 3\}$ . An exhaustive scan shows the pair is the unique violating class among all trees and connected unicyclic graphs through nine vertices. One member is bipartite and not chordal and the other chordal and not bipartite, and the ternary specializations differ, so Conjectures 62, 64, and 59 are untouched. The tree case survives exhaustively through fourteen vertices and remains open. Connectedness was in any case not the failure boundary, since the implication also fails for disconnected pseudoforests.*

**Significance.** The statement is retained with its resolution because the failure is itself informative. The refuting pair shows that on unicyclic graphs the binary rank distribution can forget the degree multiset, the coarsest datum beyond the vertex and edge counts, so one cycle already leaves enough freedom to decouple the unweighted rank census from the size-weighted array. The failure calibrates the neighbouring statements rather than undermining them: the pair straddles the classes of Conjectures 62 and 64, one member bipartite and one chordal, so the bipartite and chordal rigidity claims stand on graphs the counterexample cannot reach, and the two-field claim of Conjecture 59 absorbs it, since the ternary specializations differ. The surviving positive content is the tree case, exhaustively unrefuted through fourteen vertices, which is exactly the first decisive theorem the original statement named.

**Conjecture 64** (Chordal binary rigidity). *For chordal graphs  $G$  and  $H$ ,  $C_G(2, Y) = C_H(2, Y)$  implies  $C_G = C_H$ .*

**Significance.** Chordal graphs, those in which every cycle of length at least four has a chord, form the natural dense counterpart to the sparse classes of Conjectures 62 and 63, and the mechanism is the same class rigidity tested against a family with unbounded clique structure rather than bounded cycle structure. The nearest literature boundary is once more the specialization theorem [77] that interprets the hypothesis over  $\mathbb{F}_2$ . The verification is a six-thousand-graph random holdout of chordal graphs on nine to twelve vertices, spanning 5598 distinct binary specializations, with no violation. A first decisive theorem would be the case of trees, which are chordal, followed by block graphs. The failure mode is a chordal pair separating the binary specialization from the polynomial. The resolution of Conjecture 63 sharpens rather than settles the question here, since the refuting pair there has exactly one chordal member, so chordal rigidity survives it and its status is genuinely open.

**Conjecture 65** (Recovery of the domination number). *For all graphs  $G$  and  $H$ ,  $C_G = C_H$  implies  $\gamma(G) = \gamma(H)$ , where  $\gamma$  is the domination number.*

**Significance.** A set is dominating exactly when its closed neighbourhood is the whole vertex set, and the array behind  $C_G$  records  $|N_G[S]|$  only through the difference  $\rho_G(S) = |N_G[S]| - c(G[S])$ , so the dominating sets are entangled with component counts and the mechanism is that the entanglement can be undone at the extreme where  $\gamma$  lives. The conjecture sits strictly between a theorem and a false analogue, which is the nearest literature boundary: the cardinalities of connected dominating sets are readable from the coefficients, by Rossmann [77], while the total domination number is not determined by  $C_G$ . Every known collision class of  $C$ , the corpus recorded at Conjecture 58, has agreeing domination number. A first decisive theorem would be the tree case, where the unique fourteen-vertex collision class is the only known test. The failure mode is a collision class with distinct domination numbers, which would place  $\gamma$  on the undetermined side of the boundary alongside total domination and would show that the connected-domination theorem is the exact limit of what the polynomial sees.

**Conjecture 66** (Recovery of the independent domination number). *For all graphs  $G$  and  $H$ ,  $C_G = C_H$  implies  $i(G) = i(H)$ , where  $i$  is the independent domination number, the least cardinality of a set that is simultaneously independent and dominating.*

**Significance.** Independent dominating sets impose the two conditions the array treats most asymmetrically, domination through  $|N_G[S]|$  and independence through  $c(G[S]) = |S|$ , and the mechanism is that their interaction leaves a trace in the joint distribution of  $(|S|, \rho_G(S))$  even though neither condition is separately readable. The nearest literature boundary is the same theorem-versus-false-analogue frame as Conjecture 65: connected domination is readable [77], total domination is not determined, and  $i$  is conjectured to fall on the determined side. Every known collision class of  $C$ , the corpus recorded at Conjecture 58, has agreeing independent domination number. A first decisive theorem would be recovery of  $i$  for trees. The failure mode is a collision class with distinct  $i$ , which would split the domination invariants into determined and undetermined families along a line that no present structural explanation predicts.

## 19 Log-concavity and support

**Conjecture 67** (Global double log-concavity). *For every graph  $G$  the coefficient sequence  $a$  of  $f_G(1 + Z)$  is log-concave, and so is its transform  $L(a)$  defined by  $L(a)_i = a_i^2 - a_{i-1}a_{i+1}$ ,*

*out-of-range terms being zero.*

**Significance.** The expansion point is canonical: at  $X = 1 + Z$  the polynomial is expanded about the degenerate point  $X = 1$ , where the weights  $(X - 1)^{|S|}$  vanish for every nonempty set, and  $f_G$  folds the rank variable into the size variable at the same point, so the sequence  $a$  is the most compressed image of the whole array. The statement is calibrated to the strongest form the data support: log-concavity is asserted, its once-iterated form is asserted, and the calibration is exact because the third iterate of  $L$  fails for a thirteen-vertex graph. The nearest literature boundary is Rossmann’s paper [77], whose theorems about this polynomial are enumerative, the specialization, multiplicativity, and dominating-set results, so the positivity layer opens a new face of the object. The verification covers the full atlas of graphs on at most seven vertices and twenty-five hundred random graphs on eight to ten vertices. A first decisive theorem would be log-concavity of  $f_T(1 + Z)$  for trees. The failure mode is graded: a failure of the second level would compress the true boundary into the narrow band between one and three iterates, while a failure of bare log-concavity on some large graph would show that the seven-vertex atlas sits below the threshold where the array’s geometry turns.

**Conjecture 68** (Strict log-concavity for connected graphs). *For connected  $G$  with at least one edge,  $a_i^2 > a_{i-1}a_{i+1}$  at every interior position of the support of  $a$ , where  $a$  is the coefficient sequence of  $f_G(1 + Z)$ .*

**Significance.** The mechanism is that equality cases of log-concavity should be confined to the degenerate stratum, and in this part the degenerate stratum is disconnectedness, exactly as in Conjecture 57: multiplicativity [77] builds the polynomials of disconnected graphs as products, and products are where interior coincidences are manufactured. The nearest literature boundary is that same multiplicativity theorem, which marks the hypothesis as necessary in spirit. The verification is the corpus of Conjecture 67, the full seven-vertex atlas and twenty-five hundred random graphs on eight to ten vertices, with strictness observed throughout the connected stratum. A first decisive theorem would be strictness for paths and stars, the two ends of the tree order of Conjecture 72. The failure mode is a connected graph with an interior equality, an exact coincidence  $a_i^2 = a_{i-1}a_{i+1}$ , which would demand an algebraic explanation and would show that connectivity does not exhaust the sources of degeneracy.

**Conjecture 69** (Strict second-order log-concavity). *For connected  $G$  with at least one edge, the transform  $L(a)$  of Conjecture 67 is strictly log-concave at every interior position of its support.*

**Significance.** The statement pushes the strictness mechanism of Conjecture 68 one level up the  $L$ -hierarchy, and its position is the sharpest the evidence permits, since the third iterate of  $L$  fails already for a thirteen-vertex graph and the second-order slice strengthening fails for an eight-vertex graph (Conjecture 70), so the conjecture threads a boundary with counterexamples visible on two sides. The nearest literature boundary is as in Conjecture 67: the polynomial’s literature [77] is enumerative and the positivity hierarchy is uncharted. The verification is the same corpus, the full seven-vertex atlas and twenty-five hundred random graphs on eight to ten vertices. A first decisive theorem would be second-order strictness for paths. The failure mode is a connected graph whose  $L(a)$  has an interior equality or reversal, which by the two flanking counterexamples would pin the exact rung of the hierarchy at which graphs stop being log-concave.

**Conjecture 70** (Rank-slice strict log-concavity). *For connected  $G$  and every attainable rank  $r$ , the slice  $A_{G,r}(Z) = [Y^r] F_G(1 + Z, Y)$ , when it is not a monomial, has strictly log-concave coefficients at every interior position of its support.*

**Significance.** The mechanism refines Conjecture 68 from the folded sequence to each rank slice separately: the sets of a fixed rank  $r$  are counted by size, and strict log-concavity asserts that each of these level sets is unimodal in the strong multiplicative sense, so that positivity holds fibrewise and not only in aggregate. The calibration is exact here too, since the second-order strengthening of this slice statement fails for an eight-vertex graph, a counterexample reconfirmed by the independent implementation, so the slice hierarchy stops one level below the global hierarchy of Conjecture 69. The nearest literature boundary is Rossmann’s [77] reading of the slices as class data over finite fields. The verification is the full seven-vertex atlas and twenty-five hundred random graphs on eight to ten vertices. A first decisive theorem would be strict log-concavity of every slice for stars. The failure mode is a connected graph with a non-log-concave slice, which would show that aggregate log-concavity, if it survives in Conjecture 67, is an artefact of summation rather than a fibrewise phenomenon.

**Conjecture 71** (Interval support at fixed rank, resolved false). *For every graph  $G$  and every attainable rank  $r$ , the set  $K_r(G) = \{|S| : \rho_G(S) = r\}$  is an interval of integers.*

**Remark 8** (Resolution). *The conjecture is false as stated. The connected graph on vertex set  $\{0, \dots, 7\}$  with edge set  $\{\{0, 3\}, \{0, 6\}, \{0, 7\}, \{1, 4\}, \{1, 6\}, \{1, 7\}, \{2, 5\}, \{2, 6\}, \{3, 6\}, \{3, 7\}, \{4, 6\}, \{4, 7\}, \{5, 7\}\}$  attains adjoint rank five at support sizes one and three but at no size two: the degree-five vertices 6 and 7 give rank five at size one, the independent set  $\{0, 1, 2\}$  has closed neighbourhood all eight vertices and so gives  $8 - 3 = 5$  at size three, while every two-vertex set has rank in  $\{3, 4, 6\}$ . Hence  $K_5(G) = \{1, 3, 4, 5, 6\}$  has an interior gap at size two. An exhaustive scan of all 12,346 graphs on eight vertices finds exactly two violations, this graph and one companion, both connected, and none on seven or fewer vertices.*

**Significance.** The statement is retained with its resolution because the failure is informative and sharply located. A connected eight-vertex graph attains rank five at the support sizes  $\{1, 3, 4, 5, 6\}$ , skipping size two, so the size direction of the array is not gapless in general, and since the transposed statement was already false, neither direction carries interval structure. The failure does not reach the slice positivity beneath which the support claim was placed: on the two refuting graphs every rank slice  $[Y^r] F_G(1 + Z, Y)$  of Conjecture 70 stays strictly log-concave, since each slice weights the raw size counts by a full binomial row that closes the gap, so gapless raw support is not after all necessary for the slice statement. The nearest literature boundary is unchanged, Rossmann’s [77] reading of one extreme stratum of the array. The surviving positive content is the interval property for all graphs through seven vertices and, conjecturally, for trees, which the eight-vertex counterexample leaves open.

## 20 Tree extremality and rigidity

**Conjecture 72** (Coefficientwise tree extremality). *For every tree  $T$  on  $n$  vertices,*

$$f_{P_n}(1 + Z) \leq f_T(1 + Z) \leq f_{K_{1,n-1}}(1 + Z)$$

*coefficientwise, with equality only for the path and the star respectively.*

**Significance.** The mechanism is visible in Lemma 2: the rank landscape of a tree is driven by its internal vertices and its diameter, and the path and the star are the two trees that extremize both statistics simultaneously, the path with two leaves and maximal diameter, the star with maximal leaf count and diameter two. The conjecture asserts that this extremality is not merely a matter of the range of  $\rho_T$  but holds coefficient by coefficient in the shifted polynomial, the strongest ordering the object admits. The nearest literature boundary is Rossmann’s enumeration [77], which interprets each coefficientwise inequality as a family of inequalities between class censuses over every finite field at once. The verification is exhaustive over all trees through fourteen vertices. A first decisive theorem would be either inequality for one nontrivial family, say the star bound for caterpillars. The failure mode is a tree escaping the sandwich at a single coefficient, which would show that the coefficientwise order on trees is not anchored at the degree extremes and would force the extremal programme to work slice by slice instead.

**Conjecture 73** (Star rank-majorization). *For every  $n \geq 5$  and every prime power  $q$ , the decreasingly ordered adjoint-rank distribution of  $L_{K_{1,n-1}}(\mathbb{F}_q)$  majorizes that of  $L_T(\mathbb{F}_q)$  for every  $n$ -vertex tree  $T$ .*

**Remark 9.** *The hypothesis  $n \geq 5$  cannot be dropped: at  $n = 4$  and  $q = 2$  the star fails to majorize the path. The dual assertion, that the path is majorization-minimal among  $n$ -vertex trees, is false. The failure at  $(n, q) = (4, 2)$  was found in the verification reported below.*

**Proposition 5** (Reduction of the star majorization). Let  $T$  be a tree on  $n \geq 3$  vertices and  $q$  a prime power, and write  $\pi_{T,q}(r) = q^{-n} \sum_{\rho_T(S)=r} (q-1)^{|S|}$  for the adjoint-rank distribution of a uniformly random element of  $L_T(\mathbb{F}_q)$ . The decreasingly ordered distribution of the star  $K_{1,n-1}$  majorizes that of  $T$  if and only if

$$\max_r \pi_{T,q}(r) \leq 1 - \frac{1}{q}.$$

*Proof.* By Proposition 3 the star attains ranks 0, 1, and  $n-1$ , from the empty support, the nonempty sets of leaves, and the supports meeting the centre, with masses  $1 - q^{-1}$ ,  $q^{-1} - q^{-n}$ , and  $q^{-n}$  in decreasing order for every  $q \geq 2$ . For any tree  $\rho_T(S) = 0$  holds only at  $S = \emptyset$ , so  $\pi_{T,q}(0) = q^{-n}$ , and this is the least atom, every nonempty rank carrying mass a sum of terms  $q^{-n}(q-1)^{|S|} \geq q^{-n}$ . A tree on  $n \geq 3$  vertices has  $\text{diam} \geq 2$ , so Lemma 2 gives at least three attained ranks. Majorization of the three-atom star vector over the padded tree vector is the family of partial-sum inequalities on decreasingly ordered atoms, both sides totalling 1. The first reads  $1 - q^{-1} \geq \max_r \pi_{T,q}(r)$ , the displayed bound. The second reads  $1 - q^{-n} \geq \pi^{(1)} + \pi^{(2)}$  for the two largest tree atoms, and holds automatically, since removing them leaves at least one atom of mass at least  $q^{-n}$ . Every later star partial sum equals 1. Hence the whole family holds if and only if its first member does.  $\square$

**Significance.** Majorization compares two probability vectors through all partial sums of their decreasingly ordered entries, so the conjecture asserts that the star’s adjoint-rank distribution is the most concentrated among trees uniformly across the whole partial-sum hierarchy and across every finite field at once. The mechanism is concentration: by Proposition 3 the rank of a random element is a statistic of its vertex support, and the star compresses the possible values harder than any other tree. The remark shows the inequality system is genuinely tight, which is the honest register for a majorization claim: at  $(n, q) = (4, 2)$  one partial sum reverses, and the dual minimality of the path fails outright, so the surviving statement is exactly the one



the data leave standing. The nearest literature boundary is Rossmann's [77] reading of these distributions as class data of graphical groups. The verification is exhaustive for  $q \in \{2, 3, 4, 5\}$  over all trees through twelve vertices, for  $q \in \{2, 3\}$  at thirteen, and for  $q = 2$  at fourteen. Proposition 5 reduces the whole majorization, across every partial sum and every field, to the single anti-concentration bound  $\max_r \pi_{T,q}(r) \leq 1 - q^{-1}$ , whose extremal value is the star's own top atom, so the failure at  $(n, q) = (4, 2)$  is exactly a violation of that bound. A first decisive theorem would be the reduced bound for all  $n \geq 5$  trees, of which the comparison of star against path is the tightest case. The failure mode is a tree and a prime power breaking one partial sum at some  $n \geq 5$ , which would show that the small- $n$  anomaly is the visible end of a persistent phenomenon rather than a boundary effect.

**Conjecture 74** (Tree rank-variance extremality). *For every  $n \geq 4$  and every prime power  $q$ , among  $n$ -vertex trees the variance of  $\text{rank}(\text{ad}_x)$  over uniformly random  $x \in L_T(\mathbb{F}_q)$  is uniquely minimized by the path and uniquely maximized by the star.*

**Significance.** The two tree orderings of this programme are not redundant: majorization compares the ordered probabilities while the variance measures the spread of the rank values themselves, and the star sits at the top of both scales, most concentrated in the sense of Conjecture 73 and most spread in the sense of variance, with the path now appearing as the genuine minimum. The mechanism is again Proposition 3, which makes the variance a purely combinatorial functional of the array  $(|S|, \rho_T(S))$  weighted by  $(q - 1)^{|S|}$ . The nearest literature boundary is the enumerative frame of Rossmann [77], in which the variance is a second moment of the class census. The verification matches Conjecture 73: exhaustive for  $q \in \{2, 3, 4, 5\}$  through twelve vertices, for  $q \in \{2, 3\}$  at thirteen, and for  $q = 2$  at fourteen. A first decisive theorem would be the uniqueness of the star maximum at  $q = 2$ . The failure mode is a tree overtaking an extreme at some  $(n, q)$ , and since the statement begins at  $n = 4$  where the majorization statement already fails, a violation would separate the variance order from the majorization order and show that the two concentration scales genuinely diverge on trees.

**Conjecture 75** (Leaf-count rigidity). *For a tree  $T$  on  $n \geq 2$  vertices, equality holds in the bound  $s(T) \geq n - \ell(T) + 2$  of Lemma 2 exactly when  $T$  is a path or a star.*

**Significance.** The lemma's proof realizes  $n - \ell(T) + 2$  rank values along a breadth-first chain of internal vertices, and equality demands that every subset of vertices lands on one of the chain's values, so the mechanism is rigidity of the equality case: only the two extreme trees should leave no room for excess values off the chain. The nearest literature boundary is internal, the lemma itself, whose bound the conjecture upgrades to a characterization, in the same relation as an isoperimetric equality case stands to its inequality. The verification is exhaustive over all trees through fourteen vertices. The forward half, that the path and the star do achieve equality, is proved in Proposition 4, leaving uniqueness, that no third tree is equally tight, as the open content. The failure mode is a third tree achieving equality, which would show that the leaf count fails to see some genuinely different extremal shape and would refute the rigidity reading of Lemma 2 while leaving the lemma intact.

**Conjecture 76** (Diameter rigidity). *For a tree  $T$  on  $n \geq 2$  vertices, equality holds in the bound  $s(T) \geq \text{diam}(T) + 1$  of Lemma 2 exactly when  $T$  is a path or a star.*

**Significance.** The diametral chain of the lemma realizes  $\text{diam}(T) + 1$  values, and equality requires the whole subset lattice to produce nothing beyond the chain, so the mechanism parallels Conjecture 75 with the diameter replacing the internal vertex count, and the two rigidity statements pin the same extremal pair from two independent directions, each serving as a cross-check on the other. The nearest literature boundary is again Lemma 2 itself. The verification is exhaustive over all trees through fourteen vertices. The forward half, that the path and the star do achieve equality, is proved in Proposition 4, so only uniqueness remains open. The failure mode is a tree of small diameter and few rank values outside the pair, which would show the diameter bound to be loose in a structured way and would decouple the two rigidity statements that the exhaustive search through fourteen vertices keeps locked together.

## Part V

# Pattern Lie algebras of finite posets

*Added to the deposit: 6 August 2026.*

The twenty conjectures below concern the pattern Lie algebras of finite posets, studied through two rank-weight enumerators that record how the bracket contracts against an element and against a functional. They form four programmes. A determination programme (Conjectures 77 to 82) asserts that the adjoint enumerator alone determines the coadjoint enumerator and a list of classical invariants. A characteristic-and-field programme (Conjectures 83 to 87) identifies what in these distributions is independent of the ground field and how the field affects them when it does. A centre-one programme (Conjectures 88 to 90) studies the incidence quotient that measures the difference between the two enumerators when the centre is a line. A Hasse-forest and unitriangular programme (Conjectures 91 to 96) relates coadjoint structure to the shape of the cover graph and treats the unitriangular chain as a separate extremal case. The proved results are kept separate from the conjectural ones: the incidence identity (Proposition 6) is a theorem with a full proof, the standard adjoint–coadjoint duality of this family, included for calibration and not claimed as new, and every conjecture is stated relative to it. Fifteen of the twenty are supported throughout the tested corpus. Three more, Conjectures 79, 81, and 85, are stated in greater generality than the computations reach, and each carries a remark giving the restricted case its evidence covers. Two, Conjectures 87 and 90, have since been refuted by explicit seven-point posets and are recorded as resolved false with their counterexamples.

For a finite poset  $P$  on a set of points and a prime power  $q$ , the *pattern Lie algebra* over  $\mathbb{F}_q$  is

$$L_P(\mathbb{F}_q) = \text{span}\{e_{ij} : i <_P j\}$$

inside the strictly upper triangular matrices under a fixed linear extension of  $P$ , with the matrix-unit bracket

$$[e_{ij}, e_{kl}] = \delta_{jk} e_{il} - \delta_{li} e_{kj}.$$

It is nilpotent, and  $\dim L_P$  is the number of strict order relations of  $P$ . Write  $\text{ad}_x(y) = [x, y]$  for the adjoint map of  $x$ , and for a functional  $f \in L_P(\mathbb{F}_q)^*$  let  $B_f(u, v) = f([u, v])$  be the alternating Kirillov form. The two natural contractions of the bracket tensor give two rank-weight enumerators over a single finite field,

$$A_{P,q}(T) = \sum_{x \in L_P(\mathbb{F}_q)} T^{\text{rank}(\text{ad}_x)}, \quad K_{P,q}(T) = \sum_{f \in L_P(\mathbb{F}_q)^*} T^{\text{rank}(B_f)},$$

the *adjoint enumerator* and the *Kirillov enumerator*, each a distribution over one finite field. Write  $b_A(P, q) = \max_x \text{rank}(\text{ad}_x)$  for the *adjoint breadth* and  $b_K(P, q) = \max_f \text{rank}(B_f)$  for the *maximal Kirillov rank*, the largest coadjoint-orbit rank. Write  $Z(L_P)$ ,  $\text{Cent}(L_P)$ , and  $\text{Der}(L_P)$  for the centre, the centroid (the linear maps commuting with every  $\text{ad}_x$ ), and the derivation algebra, and  $\gamma_i(L_P)$  for the lower central series. Say  $P$  is a *Hasse forest* if the undirected Hasse cover graph of  $P$  is a forest. The chain on  $n$  points gives  $L = \text{ut}_n(\mathbb{F}_q)$ , the strictly upper triangular Lie algebra.

**Proposition 6** (Incidence identity and calibration). Let  $L = L_P(\mathbb{F}_q)$  and let

$$I_L = \{(x, f) \in L \times L^* : f([x, L]) = 0\}$$

be the incidence variety. Then

$$|I_L| = q^{\dim L} A_{L,q}(q^{-1}) = q^{\dim L} K_{L,q}(q^{-1}).$$

Consequently  $A_{L,q}(1) = K_{L,q}(1) = q^{\dim L}$  and  $A_{L,q}(q^{-1}) = K_{L,q}(q^{-1})$ , so  $(1 - T)(1 - qT)$  divides  $A_{L,q}(T) - K_{L,q}(T)$  in  $\mathbb{Z}[T]$ .

*Proof.* Count  $I_L$  two ways. For fixed  $x$  the condition  $f([x, L]) = 0$  says  $f$  vanishes on the image  $[x, L] = \text{im}(\text{ad}_x)$ , a subspace of dimension  $\text{rank}(\text{ad}_x)$ , so the number of admissible  $f$  is  $q^{\dim L - \text{rank}(\text{ad}_x)}$ . Summing over  $x$  gives

$$|I_L| = q^{\dim L} \sum_x q^{-\text{rank}(\text{ad}_x)} = q^{\dim L} A_{L,q}(q^{-1}).$$

For fixed  $f$  the condition says  $x$  lies in the radical of the alternating form  $B_f$ , of dimension  $\dim L - \text{rank}(B_f)$ , giving  $q^{\dim L - \text{rank}(B_f)}$  admissible  $x$ , so  $|I_L| = q^{\dim L} K_{L,q}(q^{-1})$ . Equating the two counts gives  $A_{L,q}(q^{-1}) = K_{L,q}(q^{-1})$ . Taking every element and every functional, the rank-0 terms included, gives  $A_{L,q}(1) = K_{L,q}(1) = q^{\dim L}$ . The two evaluations exhibit  $T = 1$  and  $T = 1/q$  as common roots of  $A_{L,q} - K_{L,q}$ , and since  $q > 1$  these are distinct, so  $(1 - T)(1 - qT)$  divides  $A_{L,q} - K_{L,q}$ .  $\square$

The identity is the standard adjoint–coadjoint duality of this family, a Knuth-type contraction of one bracket tensor, and is recorded here as calibration rather than as a new result. For the centre-one conjectures define

$$R_{P,q}(T) = \frac{K_{P,q}(T) - A_{P,q}(T)}{(1 - T)(1 - qT)},$$

a polynomial in  $\mathbb{Z}[T]$  by Proposition 6.

## 21 Determination by the adjoint enumerator

**Conjecture 77** (Adjoint–Kirillov determination). *For all finite posets  $P$  and  $Q$  and every prime power  $q$ , if  $A_{P,q}(T) = A_{Q,q}(T)$  then  $K_{P,q}(T) = K_{Q,q}(T)$ .*

**Significance.** The adjoint and Kirillov enumerators are the two contractions of the same bracket, one in the element variable and one in the functional variable, and Proposition 6 forces them to agree at  $T = 1$  and  $T = 1/q$ . The conjecture asserts that the adjoint enumerator determines the Kirillov enumerator at every value of  $T$ , not only at those two points. The nearest related result is Delsarte’s [85] analysis of the rank distributions of alternating bilinear forms over a finite field, which describes the Kirillov side alone but does not connect it to the adjoint side. A tractable special case is the chain  $L = \text{ut}_n(\mathbb{F}_q)$ , where the divisibility of Proposition 6 already relates the two enumerators. In tests on the corpus described in the stress-test section, the adjoint enumerator determined the Kirillov enumerator across 64 adjoint-collision classes with no exception. A counterexample would be an adjoint-collision pair with distinct Kirillov enumerators, which would show that  $K$  is not determined by  $A$  beyond the two points where the proposition equates them.

**Conjecture 78** (Lower-central recovery). *For every finite poset  $P$  and every prime power  $q$ , the enumerator  $A_{P,q}(T)$  determines the vector of lower-central factor dimensions  $(\dim \gamma_i(L_P)/\gamma_{i+1}(L_P))_i$ .*

**Significance.** The rank of  $\text{ad}_x$  is the dimension of the image  $[x, L_P]$ , and the lower central series is built from these images as  $x$  ranges over the algebra. The conjecture asserts that the distribution of adjoint ranks determines the graded dimensions  $\dim \gamma_i/\gamma_{i+1}$ . The nearest related result is the breadth theory of Cameron–Coll–Mayers–Russoniello [80], in which the maximal adjoint rank  $b_A$  is one value of the distribution that the full enumerator records. A tractable special case is the chain, where  $\text{ut}_n$  has a known lower central series matched to its adjoint spectrum. In tests on the corpus described in the stress-test section, the lower-central factor vector was constant on every adjoint-collision class. A counterexample would be an adjoint-collision class with two distinct factor vectors, which would show that the lower-central factors are not determined by  $A$ .

**Conjecture 79** (Ordinary Poincaré recovery). *For every finite poset  $P$  and every prime power  $q$ , the enumerator  $A_{P,q}(T)$  determines the ordinary cohomological Poincaré polynomial  $\sum_i \dim H^i(L_P, \mathbb{F}_q) U^i$ .*

**Remark 10.** *The evidence tests only the equality of  $\dim H^2(L_P, \mathbb{F}_q)$  and  $\dim H^3(L_P, \mathbb{F}_q)$  across adjoint-collision classes, and it rests on the deposited cohomology computation rather than on an independent recomputation of Lie-algebra cohomology. The universal statement over all degrees is more tentative than the other conjectures of this part, the remaining cases being untested.*

**Significance.** The Chevalley–Eilenberg differential of  $L_P$  is built from the same structure constants that the adjoint maps record, so the adjoint-rank distribution constrains the cohomological Betti numbers through the ranks of those differentials. The nearest related result is the poset Lie cohomology of Lampret–Vavpetič [82], the closest computation of  $H^i(L_P, \mathbb{F}_q)$  for this family. No decisive special case is proposed, and the tested range is the two degrees recorded in the remark. A counterexample would be an adjoint-collision class with distinct ordinary Betti numbers in some degree, which would show that cohomology is not determined by the adjoint enumerator.

**Conjecture 80** (Derivation recovery). *For every finite poset  $P$  and every prime power  $q$ , the enumerator  $A_{P,q}(T)$  determines  $\dim \text{Der}(L_P)$ .*

**Significance.** A derivation is constrained at each point by its interaction with the adjoint action, so the derivation algebra is determined by the adjoint representation, of which the adjoint-rank distribution is a coarse record. The conjecture asserts that this distribution determines the dimension of the space of derivations. The nearest related result is again the breadth theory of Cameron–Coll–Mayers–Russoniello [80], which reads structural invariants of  $L_P$  off the same adjoint data. A tractable special case is the chain, where  $\dim \operatorname{Der}(\operatorname{ut}_n)$  is classical. In tests on the corpus described in the stress-test section,  $\dim \operatorname{Der}(L_P)$  was constant on every adjoint-collision class. A counterexample would be an adjoint-collision class with distinct derivation dimensions, which would show that  $\dim \operatorname{Der}(L_P)$  is not determined by the adjoint distribution.

**Conjecture 81** (Adjoint Poincaré recovery). *For every finite poset  $P$  and every prime power  $q$ , the enumerator  $A_{P,q}(T)$  determines the adjoint-cohomology dimensions  $\dim H^i(L_P, L_P)$  for every  $i$ .*

**Remark 11.** *The evidence tests only the equality of  $\dim H^1(L_P, L_P)$ ,  $\dim H^2(L_P, L_P)$ , and  $\dim H^3(L_P, L_P)$  across adjoint-collision classes, and it rests on the deposited cohomology computation rather than on an independent recomputation. The universal statement over all degrees is more tentative than the other conjectures of this part, the remaining cases being untested.*

**Significance.** Cohomology with coefficients in the adjoint module governs the deformation theory of  $L_P$ , and its differentials are built from the same brackets whose ranks the adjoint maps record, so the situation parallels Conjecture 79 with the algebra acting on itself. The nearest related result is the poset Lie cohomology of Lampret–Vavpetič [82]. No decisive special case is proposed, and the tested range is the three degrees recorded in the remark. A counterexample would be an adjoint-collision class with distinct adjoint Betti numbers in some degree, which would show that the deformation theory is not determined by the adjoint enumerator.

**Conjecture 82** (Centroid-dimension recovery). *For every finite poset  $P$  and every prime power  $q$ , the enumerator  $A_{P,q}(T)$  determines  $\dim \operatorname{Cent}(L_P)$ .*

**Significance.** The centroid is by definition the space of linear maps commuting with every  $\operatorname{ad}_x$ , so it is an invariant of the adjoint representation, of which the adjoint-rank distribution is a coarse record. The conjecture asserts that this distribution determines its dimension. The nearest related result is again the breadth theory of Cameron–Coll–Mayers–Russoniello [80], whose invariants are defined on the same adjoint data. A tractable special case is the chain, where  $\dim \operatorname{Cent}(\operatorname{ut}_n) = n$  for  $n \geq 3$ . In tests on the corpus described in the stress-test section,  $\dim \operatorname{Cent}(L_P)$  was constant on every adjoint-collision class. A counterexample would be an adjoint-collision class with distinct centroid dimensions, which would show that the centroid dimension is not determined by the adjoint-rank distribution.

## 22 Characteristic and field behaviour

**Conjecture 83** (Adjoint spectrum characteristic independence). *For every finite poset  $P$ , the set  $\{\operatorname{rank}(\operatorname{ad}_x) : x \in L_P(\mathbb{F}_q)\}$  of attained adjoint ranks is independent of  $q$ .*

**Significance.** The attained adjoint ranks are governed by which support patterns of  $x$  can produce a given image dimension, a combinatorial condition on  $P$ . The conjecture asserts that

the set of attained values depends only on the poset, while the multiplicities depend on the field. The nearest related result is Halasi–Pálffy [83], whose theorem that pattern-group class counts are not polynomial in  $q$  shows that the counts depend on the field. The invariance of the support is a stronger statement about a smaller object. A tractable special case is the chain, where the attained ranks of  $\text{ut}_n$  form an interval determined by  $n$ . In tests on the corpus described in the stress-test section, the attained-rank sets agreed across  $\mathbb{F}_2$ ,  $\mathbb{F}_3$ , and  $\mathbb{F}_5$  in every case. A counterexample would be a poset with an adjoint rank attained over one field but not another, which would show that the support depends on the characteristic.

**Conjecture 84** (Kirillov spectrum characteristic independence). *For every finite poset  $P$ , the set  $\{\text{rank}(B_f) : f \in L_P(\mathbb{F}_q)^*\}$  of attained Kirillov ranks is independent of  $q$ .*

**Significance.** The attained Kirillov ranks are the even numbers realized as ranks of the alternating forms  $B_f$ , and which of these occur is a combinatorial condition on the coadjoint action of  $P$ . The conjecture is the coadjoint analogue of Conjecture 83. The nearest related results are Delsarte [85] for the rank distributions of alternating forms and Halasi–Pálffy [83] for the field dependence of the associated class counts, against which the invariance of the support is the finer statement. A tractable special case is the chain. In tests on the corpus described in the stress-test section, the attained Kirillov-rank sets agreed across  $\mathbb{F}_2$ ,  $\mathbb{F}_3$ , and  $\mathbb{F}_5$  in every case. A counterexample would be a poset with a Kirillov rank attained over one field but not another, which would show field dependence in the coadjoint support.

**Conjecture 85** (Two-field arithmetic rigidity). *For all finite posets  $P$  and  $Q$ , if  $A_{P,2} = A_{Q,2}$  and  $A_{P,3} = A_{Q,3}$  then  $A_{P,q} = A_{Q,q}$  for every prime power  $q$ .*

**Remark 12.** *Among the posets computed over all three fields, the only field beyond  $q = 2$  and  $q = 3$  at which equality was actually tested is the single holdout  $q = 5$ . The universal statement over every prime power is more tentative than the other conjectures of this part, the remaining cases being untested.*

**Significance.** The two adjoint enumerators over the smallest fields are two weighted counts drawn from the poset’s rank array. The conjecture asserts that the arrays realized by posets are rigid enough that no third field carries information absent from the first two. The nearest related result is Halasi–Pálffy [83], whose non-polynomiality theorem shows that larger fields can behave differently in general, against which the two-field claim asserts that this family is nonetheless determined. No decisive special case is proposed, and the tested range is the single holdout recorded in the remark. Equality of  $A$  at  $q = 2$  and  $q = 3$  forced equality at  $q = 5$  across 16 three-field collision classes. A counterexample would be a pair agreeing over  $\mathbb{F}_2$  and  $\mathbb{F}_3$  but differing over some larger field.

**Conjecture 86** (Adjoint stochastic field monotonicity). *For every finite poset  $P$  and prime powers  $q < q'$ , the distribution of  $\text{rank}(\text{ad}_x)$  for uniformly random  $x \in L_P(\mathbb{F}_{q'})$  first-order stochastically dominates its distribution over  $\mathbb{F}_q$ .*

**Significance.** First-order stochastic dominance compares two distributions across all of their tail probabilities at once, so the conjecture asserts that raising the field can only move the adjoint rank upward in this sense. The reason is that a larger field provides more elements of high rank



at each support pattern. The nearest related result is the breadth theory of Cameron–Coll–Mayers–Russoniello [80], in which the top rank  $b_A$  is stable while the rest of the distribution shifts. The statement is calibrated to the strongest form that holds. The monotone-likelihood-ratio strengthening is false, failing three times already from  $q = 2$  to  $q = 3$ , so first-order dominance is the appropriate form. In tests on the corpus described in the stress-test section, first-order dominance held in all 221 field-pair tests, while the likelihood-ratio strengthening failed as recorded. A counterexample would be a poset and a field pair reversing one tail probability, which would show that first-order dominance is too strong and that the field can move rank downward.

**Conjecture 87** (Kirillov likelihood-ratio field monotonicity, resolved false). *For every finite poset  $P$  and prime powers  $q < q'$ , the normalized even-rank distribution defined by  $K_{P,q}$  dominates that defined by  $K_{P,q'}$  in monotone likelihood-ratio order.*

**Remark 13** (Resolution). *The conjecture is false as stated. The seven-point poset with cover relations  $1 \prec 2, 2 \prec 3, 2 \prec 4, 1 \prec 5, 5 \prec 7, 1 \prec 6, 6 \prec 7$  gives a ten-dimensional pattern algebra whose Kirillov enumerator has even-rank coefficients 128, 384, 128, 384 at ranks 0, 2, 4, 6 over  $\mathbb{F}_2$  and 2187, 17496, 4374, 34992 over  $\mathbb{F}_3$ . The likelihood ratio  $[T^{2j}]K_{P,3}/[T^{2j}]K_{P,2}$  then takes the values 17.1, 45.6, 34.2, 91.1 across the four even ranks, decreasing from rank 2 to rank 4, so the monotone likelihood-ratio order already fails at the pair  $q = 2, q' = 3$ . The first-order dominance of Conjecture 86 is untouched, being the weaker order the data do support.*

**Significance.** The statement is retained with its resolution because the failure locates the boundary precisely. Monotone likelihood-ratio order is strictly stronger than first-order dominance and compares the two distributions ratio by ratio across the even ranks, and the refuting poset shows that the coadjoint side does not satisfy this strengthening any more than the adjoint side does in Conjecture 86. The nearest related result is Delsarte [85], whose rank distributions of alternating forms are the model for the even-rank sequence. The surviving positive content is the first-order Kirillov dominance that the same data support, the coadjoint analogue of Conjecture 86, which the counterexample leaves standing.

## 23 Centre-one incidence quotient

**Conjecture 88** (Centre-one reverse determination). *For every finite poset  $P$  with  $\dim Z(L_P) = 1$  and every prime power  $q$ , the enumerator  $K_{P,q}(T)$  determines  $A_{P,q}(T)$ .*

**Significance.** Conjecture 77 states the implication from the adjoint side to the coadjoint side, and this is the reverse implication under the hypothesis that the centre is a line. The reason is that a one-dimensional centre makes the incidence quotient of Proposition 6 rigid enough to invert. The nearest related result is Delsarte [85], whose alternating-form rank distributions are what  $K_{P,q}$  records. A tractable special case is the chain  $ut_n$ , whose centre is the single corner entry. In tests on the corpus described in the stress-test section, the Kirillov enumerator determined the adjoint enumerator on all 43 centre-one records. A counterexample would be a centre-one pair with equal  $K$  and distinct  $A$ , which would show that  $A$  is not determined by  $K$  even when the centre is a line.

**Conjecture 89** (Centre-one Laplace positivity). *For every finite poset  $P$  with  $\dim Z(L_P) = 1$  and every prime power  $q$ , the polynomial  $R_{P,q}(T)$  has nonnegative integer coefficients.*

**Significance.** The quotient  $R_{P,q}$  measures the difference between the coadjoint and adjoint enumerators after the two common roots of Proposition 6 are divided out. The conjecture asserts that under a line centre this difference is a counting polynomial, so its coefficients are nonnegative integers. The nearest related result is Delsarte [85], against whose alternating-form model the positivity of  $R_{P,q}$  constrains how far  $K$  can exceed  $A$ . A tractable special case is the chain, where  $R$  is computable in closed form. In tests on the corpus described in the stress-test section,  $R_{P,q}$  had nonnegative integer coefficients on all 43 centre-one records. A counterexample would be a centre-one poset with a negative coefficient in  $R_{P,q}$ , which would show that  $R_{P,q}$  is not a counting polynomial and would remove the basis for the shape statement of Conjecture 90.

**Conjecture 90** (Centre-one quotient shape, resolved false). *For every finite poset  $P$  with  $\dim Z(L_P) = 1$  and every prime power  $q$ , the coefficient sequence of  $R_{P,q}(T)$  has interval support and is unimodal.*

**Remark 14** (Resolution). *The conjecture is false as stated. The seven-point poset with cover relations  $1 \prec 2, 2 \prec 3, 2 \prec 4, 3 \prec 6, 4 \prec 5, 5 \prec 6, 6 \prec 7$  has a one-dimensional centre and a nineteen-dimensional pattern algebra over  $\mathbb{F}_2$ , and its incidence quotient  $R_{P,2}$  has nonnegative integer coefficients with interval support but is not unimodal: the coefficients at degrees 8, 9, and 10 are 206336, 198656, 225280, a strict interior valley between two higher values. The positivity of Conjecture 89 and the interval support both hold on this poset, so the unimodality alone fails.*

**Significance.** The statement is retained with its resolution because the failure sharpens the neighbouring positivity result. Given the positivity of Conjecture 89, the natural next question was the shape of the coefficient sequence, and the refuting poset shows that positivity does not force a single peak, since the quotient can carry a strict interior valley while keeping nonnegative coefficients and gapless support. The nearest related results are Delsarte [85] for the alternating-form arithmetic and Coll–Mayers [81] for the coadjoint-orbit reading of  $K$ . The surviving positive content is exactly the positivity and interval support of Conjecture 89, which the counterexample leaves intact, so the regularity of  $R_{P,q}$  under a line centre stops at nonnegativity and gapless support and does not extend to unimodality.

## 24 Hasse-forest and unitriangular structure

**Conjecture 91** (Hasse-forest reverse determination). *For every finite poset  $P$  whose Hasse cover graph is a forest and every prime power  $q$ , the enumerator  $K_{P,q}(T)$  determines  $A_{P,q}(T)$ .*

**Significance.** This is the reverse determination of Conjecture 77 under a structural hypothesis complementary to the centre-one hypothesis of Conjecture 88. The reason is that a forest cover graph builds the coadjoint form from independent covers, which makes the incidence quotient rigid enough to invert. The nearest related result is Coll–Mayers [81], whose index computations for Lie poset algebras are sharpest on tree-shaped posets. A tractable special case is the chain, a forest on one branch. In tests on the corpus described in the stress-test section, the Kirillov enumerator determined the adjoint enumerator across 44 forest-collision classes. A counterexample would be a forest pair with equal  $K$  and distinct  $A$ , which would show that the forest hypothesis is not the right setting for reverse determination.

**Conjecture 92** (Hasse-forest coadjoint saturation). *For every finite poset  $P$  whose Hasse cover graph is a forest and every prime power  $q$ , with  $b_K = b_K(P, q)$  the maximal Kirillov rank,*

$$\text{supp } K_{P,q} = \{0, 2, 4, \dots, b_K\},$$

*so every even rank up to  $b_K$  is attained.*

**Significance.** An alternating form has even rank, and the conjecture asserts that on a forest every even rank up to the top is realized by some functional. The reason is that the independent covers of a forest allow the coadjoint rank to be adjusted one pair at a time. The nearest related results are Coll–Mayers [81], whose index is the codimension complementary to  $b_K$ , and Delsarte [85], whose alternating-form model predicts which even ranks can occur. A tractable special case is the chain, where  $K_{\text{ut}_n}$  fills its even range. In tests on the corpus described in the stress-test section, the forest Kirillov support filled every even rank up to  $b_K$  in each case. A counterexample would be a forest with a gap in its Kirillov support, which would show that the coadjoint ranks of tree-shaped posets can skip values and would remove the support hypothesis used in Conjecture 93.

**Conjecture 93** (Hasse-forest Kirillov unimodality). *For every finite poset  $P$  whose Hasse cover graph is a forest and every prime power  $q$ , the even-rank sequence  $([T^{2j}]K_{P,q}(T))_j$  is unimodal.*

**Significance.** On the gapless support of Conjecture 92 the next question is the shape of the even-rank counts. The conjecture asserts that the coadjoint rank distribution of a forest rises to a single peak and falls, with no interior dip. The nearest related results are Delsarte [85] for the alternating-form counts and Coll–Mayers [81] for the coadjoint reading. The statement is calibrated to the strongest form that holds. Strict log-concavity is false, so unimodality is the appropriate form. In tests on the corpus described in the stress-test section, the forest even-rank sequence was unimodal throughout. A counterexample would be a forest whose even-rank sequence has two peaks, which would show that the coadjoint distribution of tree-shaped posets need not be concentrated and would confine the shape statement to the unitriangular chain of Conjecture 95.

**Conjecture 94** (Hasse-forest breadth–orbit bound). *For every finite poset  $P$  whose Hasse cover graph is a forest and every prime power  $q$ ,*

$$b_K(P, q) \leq 2b_A(P, q).$$

**Significance.** The bound relates the maximal coadjoint-orbit rank to the adjoint breadth. The reason is that on a forest the largest alternating form cannot exceed twice the largest adjoint image, since both come from the same covers. The nearest related results are Cameron–Coll–Mayers–Russoniello [80] for the breadth  $b_A$  and Coll–Mayers [81] for the index that controls  $b_K$ . The forest hypothesis is essential. The unrestricted inequality is false, with many counterexamples off the forest class. In tests on the corpus described in the stress-test section, the forest bound held everywhere, and the unrestricted bound failed fourteen times off the forest class. A counterexample would be a forest violating  $b_K \leq 2b_A$ , which would show that a tree-shaped poset can carry a coadjoint orbit large relative to its adjoint breadth.

**Conjecture 95** (Unitriangular Kirillov strict log-concavity). *For  $L = \text{ut}_n(\mathbb{F}_q)$  and every prime power  $q$ , the even-rank sequence  $([T^{2j}]K_{L,q}(T))_j$  is strictly log-concave at every nontrivial interior index.*

**Significance.** The chain is the extremal forest, and here the unimodality of Conjecture 93 strengthens to strict log-concavity. The conjecture asserts that the coadjoint rank distribution of  $ut_n$  is as concentrated as the family allows. The nearest related result is Marberg [84], whose character and orbit enumeration for the unitriangular group is the closest account of the coadjoint data of  $ut_n$ . A tractable special case would be strict log-concavity for a single infinite family of indices, such as the central ones. In tests on the corpus described in the stress-test section, strict log-concavity was confirmed for  $ut_4$ ,  $ut_5$ , and  $ut_6$  over  $\mathbb{F}_2$ , for  $ut_4$  and  $ut_5$  over  $\mathbb{F}_3$ , and for  $ut_4$  over  $\mathbb{F}_5$ . A counterexample would be an interior index of some  $ut_n$  where the strict inequality fails, which would show that even the chain falls short of strict concentration.

**Conjecture 96** (Adjoint–coadjoint incidence anticorrelation). *For every nonabelian pattern Lie algebra  $L = L_P$  and every prime power  $q$ , sampling  $(x, f)$  uniformly on the incidence variety  $I_L = \{(x, f) : f([x, L]) = 0\}$  gives*

$$\text{Cov}(\text{rank}(\text{ad}_x), \text{rank}(B_f)) < 0,$$

*with covariance zero exactly when  $L_P$  is abelian.*

**Significance.** The incidence variety of Proposition 6 pairs an element with a functional annihilating its adjoint image. The conjecture asserts that a high adjoint rank leaves little room for the coadjoint form to be large, so the two ranks are negatively correlated on  $I_L$ . The nearest related result is Coll–Mayers [81], whose coadjoint-orbit analysis of Lie poset algebras is the setting in which  $\text{rank}(B_f)$  is an orbit dimension. A tractable special case is the abelian case, where both ranks are zero and the covariance vanishes, giving the equality clause. In tests on the corpus described in the stress-test section, the covariance was negative on every nonabelian poset tested and exactly zero on all sixteen abelian ones. The statement is calibrated to the strongest form that holds. The stronger conditional stochastic-order and conditional-mean forms are false, so the sign of the covariance is the surviving claim. A counterexample would be a nonabelian poset with nonnegative incidence covariance, which would decouple the two ranks on the variety that defines their duality.

## Part VI

# Local structure of primes and arithmetic functions

*Added to the deposit: 10 August 2026.*

The twenty conjectures below return to the local–global aesthetic of Part I, but read it off objects other than residue classes of consecutive primes. Each takes a familiar arithmetic quantity, looks at it through a local lens that ought to have looked symmetric, and takes the residual asymmetry seriously. They form five programmes. A compass programme (Conjectures 97 to 102) measures the direction of the nearest prime to a polynomial centre and matches it to a local-sieve hazard. An order-pattern programme (Conjectures 103 and 104) studies the order patterns of  $\text{rad}(n)/n$  on short windows. An order programme (Conjectures 105 to 109) measures how often abundancy and totient ratios move monotonically. A shock programme (Conjectures 110 to 115)

tracks the prime-factor count of combinatorial sequences that drop many prime factors in a single step, a *shock*. A boundary programme (Conjecture 116) reads the prime-factor anatomy of the four integers bracketing a prime gap. The proved layer is kept separate. Proposition 7 gives the factorial-ratio jump identities, Proposition 8 gives short-window exchangeability, Proposition 9 proves that every fixed radical-pattern density exists, Proposition 10 proves exact finite-sieve equivalence for triangular and hexagonal centres, Proposition 14 proves the  $k = 2$  case of the dyadic multinomial slope, Proposition 11 settles the three shock laws, Conjectures 110 to 112, by an elementary forcing construction, and Proposition 12 evaluates the prime-truncated lag-one covariance of the central-binomial slice at exactly  $-\frac{5}{3}$ , and Proposition 13 shows the lag-one shock covariance converges to a rational number in every family. Six of the twenty state a bias as a numerical band rather than a derived constant, where they occur, the compass and order constants of Conjectures 98, 99, 100, 105, 106, and 108. Two, Conjectures 102 and 113, extrapolate in a parameter rather than in the index and are the boldest.

For a composite integer  $a$  let  $p^-(a) < a < p^+(a)$  be its neighbouring primes and set  $L(a) = a - p^-(a)$  and  $U(a) = p^+(a) - a$ . The *prime compass* of  $a$  points right when  $U(a) < L(a)$ , left when  $L(a) < U(a)$ , and is tied when the two distances agree. For a sequence  $A_1 < A_2 < \dots$  of composite centres write

$$\theta_A(x) = \frac{\#\{n : A_n \leq x, U(A_n) < L(A_n)\}}{\#\{n : A_n \leq x, U(A_n) \neq L(A_n)\}}$$

for the right-closer frequency among the untied centres, and  $\theta_A = \lim_{x \rightarrow \infty} \theta_A(x)$  when it exists; we call  $\theta_A$  the *compass constant* of  $A$ . The polygonal numbers are  $P_s(n) = \frac{1}{2}((s-2)n^2 - (s-4)n)$  for  $s \geq 3$ , the pronic numbers are  $Q(n) = n(n+1)$ , and the squares are  $n^2$ . For the radical  $\text{rad}(n) = \prod_{p|n} p$  write  $R(n) = \text{rad}(n)/n$ , and for a window of length  $d$  call the permutation  $\pi_0 \dots \pi_{d-1}$  of  $\{0, \dots, d-1\}$  the *order pattern*—the ordinal pattern of the window—at  $n$  when  $R(n+\pi_0) < \dots < R(n+\pi_{d-1})$ , counting only windows with no ties. Write  $\delta_\pi$  for the density of the order pattern  $\pi$  among the untied windows, the limiting ratio of the two window counts, so that  $\sum_\pi \delta_\pi = 1$ ; ties have positive density, adjacent squarefree integers already tying at  $R = 1$ , so an unconditional normalization would be deficient. For the abundancy index  $\sigma(n)/n$  and the totient ratio  $\phi(n)/n$  call an adjacent triple *monotone* when it is strictly increasing or strictly decreasing, and for a stride  $W$  let  $\mu(W)$  be the density of  $n$  for which  $\sigma(n)/n, \sigma(n+W)/(n+W), \sigma(n+2W)/(n+2W)$  is monotone. Write  $\Omega(m)$  for the number of prime factors of  $m$  with multiplicity, and set the binomial slice  $A_{k,n} = \binom{kn}{n}$ , the central multinomial  $M_{k,n} = (kn)!/(n!)^k$ , and the Fuss–Catalan number  $F_{r,n} = \frac{1}{rn+1} \binom{(r+1)n}{n}$ , with single-step jump  $J_{k,n} = \Omega(A_{k,n+1}) - \Omega(A_{k,n})$ . Finally, for consecutive odd primes  $p < q$  set

$$D(p, q) = \Omega(p+1) + \Omega(q-1) - \Omega(p-1) - \Omega(q+1),$$

the inward-minus-outward prime-factor balance of the four composites bracketing the gap.

**Proposition 7** (Factorial-ratio jump identities). For all  $k \geq 2$  and  $n \geq 1$ ,

$$J_{k,n} = \sum_{j=1}^k \Omega(kn+j) - \Omega(n+1) - \sum_{j=1}^{k-1} \Omega((k-1)n+j).$$

In particular  $\Omega\binom{2n+2}{n+1} - \Omega\binom{2n}{n} = 1 + \Omega(2n+1) - \Omega(n+1)$ , and for the Catalan numbers  $C_n = \frac{1}{n+1} \binom{2n}{n}$ ,  $\Omega(C_{n+1}) - \Omega(C_n) = 1 + \Omega(2n+1) - \Omega(n+2)$ . Consequently the central binomial jump  $J_{2,n}$ ,

the Catalan jump  $\Omega(C_{n+1}) - \Omega(C_n)$ , and the central multinomial jump  $\Omega(M_{k,n+1}) - \Omega(M_{k,n})$  for every  $k \geq 2$  are unbounded below, while the binomial slice  $J_{k,n}$  for  $k \geq 3$  is settled by the forcing construction of Proposition 11.

*Proof.* The ratio  $A_{k,n+1}/A_{k,n} = \binom{k(n+1)}{n+1} / \binom{kn}{n}$  is the rational number  $\prod_{j=1}^k (kn + j) / ((n+1) \prod_{j=1}^{k-1} ((k-1)n + j))$ , an exact identity of factorials in which every factor is a positive integer, so  $\Omega$  is additive across it and the displayed formula follows, the two specializations being the cases  $k = 2$  for the central binomial coefficient and its Catalan quotient. For the unboundedness we use the elementary bound that an odd integer  $m$  has  $\Omega(m) \leq \log_3 m$ , since each of its prime factors is at least 3. Taking  $k = 2$  and  $n+1 = 2^t$  gives  $J_{2,n} = 1 + \Omega(2^{t+1} - 1) - t$  with  $2^{t+1} - 1$  odd, so  $J_{2,n} < 1 + (t+1) \log_3 2 - t \rightarrow -\infty$ ; the Catalan jump  $1 + \Omega(2n+1) - \Omega(n+2)$  at  $n+2 = 2^t$  equals  $1 + \Omega(2^{t+1} - 3) - t \rightarrow -\infty$  by the same bound. For the central multinomial  $M_{k,n+1}/M_{k,n} = \prod_{j=1}^k (kn + j) / (n+1)^k$ , so  $\Omega(M_{k,n+1}) - \Omega(M_{k,n}) = \sum_{j=1}^k \Omega(kn + j) - k \Omega(n+1)$ ; at  $n+1 = 2^t$  the top factor  $k \cdot 2^t$  contributes  $t + \Omega(k)$ , while each of the  $k-1$  remaining factors  $k \cdot 2^t - i$  with  $1 \leq i \leq k-1$  has  $v_2(k \cdot 2^t - i) = v_2(i) \leq \log_2 k$  and odd part below  $k \cdot 2^t$ , so  $\Omega(k \cdot 2^t - i) \leq \log_2 k + t \log_3 2 + O(1)$  and the jump is at most  $-(1 - \log_3 2)(k-1)t + O(1) \rightarrow -\infty$ . For the binomial slice with  $k \geq 3$  the extra block  $-\sum_{j=1}^{k-1} \Omega((k-1)n + j)$  need not be large, so this argument does not force  $J_{k,n} \rightarrow -\infty$ ; that case needs the forcing construction of Proposition 11.  $\square$

**Proposition 8** (Short-window radical exchangeability). For every window length  $d \leq 4$  the strict order-pattern densities of  $R(n) = \text{rad}(n)/n$  exist and are uniform, each equal to  $1/d!$ .

*Proof.* The value  $R(n) = \text{rad}(n)/n$  is the reciprocal of the powerful part of  $n$ , determined by the prime squares dividing  $n$ :  $1/R(n) = \prod_{p^2|n} p^{v_p(n)-1}$ . In a window  $n, n+1, \dots, n+d-1$  of length  $d \leq 4$  no prime square  $p^2$  can divide two distinct entries, since consecutive multiples of  $p^2$  differ by  $p^2 \geq 4 \geq d$ , with the single boundary case  $p = 2, d = 4$  excluded because 4 divides only one of any four consecutive integers. Hence the powerful parts of the  $d$  entries are supported on disjoint sets of primes, and by the Chinese remainder theorem the vector  $(R(n), \dots, R(n+d-1))$  is asymptotically exchangeable, so every strict order is equally likely.  $\square$

**Proposition 9** (Existence of radical-pattern densities). For every fixed window length  $d$ , each unconditional strict order-pattern count for  $R(n) = \text{rad}(n)/n$  has a natural density. The untied windows have positive density. Consequently every conditional density  $\delta_\pi$  exists.

*Proof.* Fix  $d$ . For  $Y \geq 2$  and  $K \geq 3$ , truncate  $R(m)$  by retaining only primes  $p \leq Y$  and replacing each valuation by  $\min(v_p(m), K-1)$ . The resulting value is periodic modulo  $\prod_{p \leq Y} p^K$ . The truncated vector on  $m, m+1, \dots, m+d-1$  can differ from the true vector only if  $p^2 \mid m+i$  for some  $p > Y$ , or if  $p^K \mid m+i$  for some  $p \leq Y$ . The upper density of this exceptional set is at most

$$d \sum_{p > Y} \frac{1}{p^2} + d \sum_{p \leq Y} \frac{1}{p^K}.$$

It tends to zero by first taking  $Y$  and then  $K$  large. Every strict-pattern indicator for the truncated vector is periodic. Its densities form a Cauchy net under refinement, so every unconditional strict-pattern density exists.

It remains to show that their sum is positive. For each prime  $p \leq d$ , choose an exponent  $a_p$  with  $p^{a_p} > d$  and a residue class modulo  $p^{a_p}$  on which none of the  $d$  entries is divisible by  $p^{a_p}$ . This



fixes every small-prime valuation in the window. Next choose distinct primes  $q_0, \dots, q_{d-1} > d$  and exponents  $e_i \geq 2$ . The Chinese remainder theorem permits

$$m + i \equiv q_i^{e_i} \pmod{q_i^{e_i+1}}$$

for every  $i$ . No  $q_i$  divides another entry of the window. The exponents can be chosen so that the resulting  $d$  powerful parts, including the fixed small-prime contributions, are pairwise distinct. In this progression, require every remaining prime  $p$  to divide no entry to order two. For  $p > d$  this excludes at most  $d < p^2$  residue classes modulo  $p^2$ . The elementary squarefree sieve therefore gives the positive density

$$\prod_{p \nmid Q} \left(1 - \frac{\nu(p)}{p^2}\right) > 0,$$

where  $Q$  is the fixed modulus and  $0 \leq \nu(p) \leq d$ . On this positive-density set the powerful parts, and hence the values of  $R$ , are pairwise distinct. The density of untied windows is therefore positive, so division by it gives every  $\delta_\pi$ .  $\square$

**Proposition 10** (Triangular–hexagonal finite-sieve equivalence). *For every modulus  $m \geq 1$  and residue  $c \bmod m$ ,*

$$\frac{1}{2m} \#\{n \bmod 2m : P_3(n) \equiv c \pmod{m}\} = \frac{1}{m} \#\{k \bmod m : P_6(k) \equiv c \pmod{m}\}.$$

Consequently the triangular and hexagonal local compass frequencies are exactly equal for every finite set of sieving primes.

*Proof.* The involution  $n \mapsto -n - 1$  on  $\mathbb{Z}/2m\mathbb{Z}$  reverses parity and preserves  $P_3(n) = n(n+1)/2$  modulo  $m$ . It therefore pairs every even index with an odd index having the same triangular residue. The odd indices are  $n = 2k - 1$  with  $k \bmod m$ , and  $P_3(2k - 1) = P_6(k)$ . This proves the displayed equality. At any finite sieve depth the first surviving offsets on both sides of a centre depend only on its residue modulo the product of the sieving primes, so the equality of residue laws gives equality of the local compass frequencies.  $\square$

## 25 Prime compasses at polynomial centres

**Conjecture 97** (Local-sieve limit for polygonal compasses). *For every fixed  $s \geq 3$  the limit  $\theta_s = \theta_{P_s}$  exists. Moreover, if the nearest-prime race is replaced by the first offset on each side surviving every prime at most  $y$ , the resulting local compass frequency tends to  $\theta_s$  as  $y \rightarrow \infty$ .*

**Significance.** The direction of the nearest prime to a centre  $a$  is decided by two competing waiting times, and for a polynomial centre the admissible offsets on each side are filtered by the residue pattern of  $P_s$  modulo the small primes, so the compass should be governed by a local-sieve hazard rather than by the size of  $P_s$ . The conjecture asserts both that the frequency settles and that the deep-sieve local model is its limit, which is the content beyond a mere claim of bias. The nearest related result is the Cramér–Granville model of gaps between primes [17], in which such local hazards are the leading correction to the exponential law. A first decisive theorem would be the existence of  $\theta_3$  for the triangular numbers. In tests on the corpus described in the stress-test section, the sieve using only the primes through 47 predicted the order-level bias with correlation 0.896 across  $s \leq 100$ . A counterexample would be a polygonal order whose frequency fails to settle, or whose deep local model does not approach the observed value.

**Conjecture 98** (Pentagonal and pronic compasses). *The pentagonal compass satisfies  $0.56 < \theta_{P_5} < 0.62$ , while the pronic compass satisfies  $0.42 < \theta_Q < 0.48$ .*

**Significance.** The pentagonal and pronic sequences are the two simplest quadratic centres of opposite local character, and the conjecture records that their compasses point in opposite directions, one right-biased and one left-biased, which rules out any explanation by the size of the centres alone. The bands are stated as numerical intervals rather than derived constants, the honest register for a measured bias whose local model of Conjecture 97 explains its sign but not yet its value. The nearest related result is again the local hazard reading of prime gaps [17]. In tests on the corpus described in the stress-test section, the frequencies were 0.583 over 51,630 pentagonal centres and 0.450 over 63,236 pronic centres, with the signs replicated on a holdout above  $10^9$ . A counterexample would be either frequency settling outside its band, or the two sequences sharing a direction.

**Conjecture 99** (Opposite polygonal orders). *The 60-gonal compass satisfies  $\theta_{60} > 0.58$  and the 69-gonal compass satisfies  $\theta_{69} < 0.49$ .*

**Significance.** Two nearby polygonal orders can carry opposite biases because the coefficients of  $P_s$  traverse different residue patterns as  $s$  varies, so the compass is a function of the order and not a monotone drift in it. The conjecture fixes the two most extreme observed orders as a right-biased and a left-biased witness, calibrating the spectrum of Conjecture 102. The nearest related result is the local-sieve reading of Conjecture 97, of which this is a two-point instance. In tests on the corpus described in the stress-test section, the frequencies were 0.609 and 0.468, each replicated above  $10^9$ . A counterexample would be either order settling on the wrong side of its stated cut.

**Conjecture 100** (Triangular and hexagonal equality). *The triangular and hexagonal compass constants are equal,  $\theta_{P_3} = \theta_{P_6} \in (0.52, 0.56)$ .*

**Significance.** The identity  $P_6(n) = P_3(2n - 1)$  realizes the hexagonal numbers as the odd-index triangular subsequence, and the conjecture asserts that this subsequence carries the same compass limit as the whole triangular sequence, so the bias is insensitive to the arithmetic-progression thinning of the index. Proposition 10 now proves the exact finite-sieve content. The two centre sequences have the same residue law for every modulus and hence the same compass frequency at every finite sieve depth. What remains open is the transfer from those local models to the actual nearest primes in Conjecture 97, together with the common numerical band. In tests on the corpus described in the stress-test section, the two constants agreed to three places, 0.53595 against 0.53572. A counterexample would be a persistent gap between the two frequencies at large centres.

**Conjecture 101** (Square compass by root class). *For each residue  $r \bmod 6$  the conditional square compass  $\theta_{\square, r}$  over roots  $n \equiv r$  exists, with  $\theta_{\square, r} < 0.45$  for  $r \in \{1, 5\}$  and  $\theta_{\square, r} > 0.54$  for  $r \in \{0, 2, 3, 4\}$ .*

**Significance.** The overall imbalance of the nearest prime to a square is a coarse average, and the conjecture is the finer claim that the bias splits sharply and stably by the residue of the root modulo six, a fingerprint that survives conditioning where the pooled rate does not. The

mechanism is that  $n \bmod 6$  fixes the residues of  $n^2 \pm 1$  and  $n^2 \pm 2$  at the primes 2 and 3, forcing which side carries the small forced factors. The nearest related result is the local hazard model of Conjecture 97 specialized to  $s$  with a square centre. In tests on the corpus described in the stress-test section, the six classes gave frequencies from 0.419 to 0.572 over about 10,540 roots each, with the same signs below  $10^9$ . A counterexample would be a root class settling on the wrong side of 0.5 from the stated sign.

**Conjecture 102** (Two-sided order spectrum). *The polygonal compass spectrum straddles one half in the limit of many sides,*

$$\liminf_{s \rightarrow \infty} \theta_s < 0.49 < 0.58 < \limsup_{s \rightarrow \infty} \theta_s,$$

*so infinitely many polygonal orders are left-biased and infinitely many are right-biased.*

**Significance.** Where Conjectures 98 to 101 extrapolate in the index  $n$  at fixed order, this statement extrapolates in the order  $s$  itself, asserting that the coefficients of  $P_s$  keep producing both directions as  $s$  grows, and it is the boldest of the compass programme for exactly that reason. The mechanism is that the residue pattern of  $P_s$  modulo a fixed prime set runs through a full cycle as  $s$  varies, so no eventual sign can set in. The nearest related result is the local hazard model of Conjecture 97, which makes the two-sidedness plausible without proving it. In tests on the corpus described in the stress-test section, the observed orders  $3 \leq s \leq 100$  ranged from 0.468 at  $s = 69$  to 0.609 at  $s = 60$ . A counterexample would be an eventual one-sided bias, an order beyond which every  $\theta_s$  lies on one side of one half.

## 26 Order patterns of the radical ratio

**Conjecture 103** (Permanent nonuniformity from length five). *For every fixed window length  $d \geq 5$  the strict order-pattern densities of  $R(n) = \text{rad}(n)/n$  exist, and their distribution is not uniform.*

**Significance.** The first coupling event,  $4 \mid n$  and  $4 \mid n+4$ , biases the endpoints of the length-five window. Proposition 8 makes the law exactly uniform through length four, while Proposition 9 now proves that every fixed-length conditional law exists by periodic approximation and a positive-density squarefree sieve. The open content is nonuniformity at each length  $d \geq 5$ . Nonuniformity at length five alone does not settle a longer length because the longer law conditions on the whole longer window being untied. The nearest related result is the distribution theory of powerful numbers [86], which governs the density of the coupling events. In tests on the corpus described in the stress-test section, the chi-square against uniformity jumped from 0.45 at length four to about  $2.1 \times 10^7$  at length five through  $10^9$ . A counterexample would be a length  $d \geq 5$  at which every strict order pattern has density  $1/d!$ .

**Conjecture 104** (Thirteen-fold pattern separation). *At window length five the strict densities satisfy  $12 < \delta_{30241}/\delta_{13240} < 15$ , with 30241 in the dominant class and 13240 in the suppressed class.*

**Significance.** The length-five order-pattern law is not merely nonuniform but sharply structured, and the conjecture pins the ratio of its most common to one of its least common patterns as a bounded constant strictly above one, so the asymmetry is a definite feature and not a vanishing edge effect. The mechanism is the same endpoint coupling of Proposition 8, which favours patterns placing the coupled endpoints coherently and suppresses those placing them against the coupling. The nearest related result is again the powerful-number density [86] that sets the coupling strength. In tests on the corpus described in the stress-test section, the ratio fell from 14.06 at  $2 \times 10^7$  to 13.31 at  $10^9$  over 23,154,595 strict windows, with the length-four and length-five cell counts reproduced by an independent implementation. A counterexample would be the ratio settling outside the band, or the two patterns exchanging dominance.

## 27 Order laws for multiplicative ratios

**Conjecture 105** (Abundancy order constant). *The density of  $n$  for which  $\sigma(n)/n, \sigma(n+1)/(n+1), \sigma(n+2)/(n+2)$  is monotone exists and lies in  $(0.094, 0.098)$ .*

**Significance.** For three independent continuous samples the monotone density is  $2/3! = 1/3$ , and the conjecture records that adjacent abundancy triples are far from independent, monotone only about a tenth of the time, with the middle value an extremum of the triple more than two thirds of the time. The band is stated numerically, the honest register for a measured order constant whose value the mechanism does not yet supply. The nearest related result is the classical distribution function of  $\sigma(n)/n$  [86], a one-point law that says nothing about the joint order of a triple. In tests on the corpus described in the stress-test section, the pooled rate was 0.09584 over 19,999,991 strict triples, stable across seven disjoint blocks. A counterexample would be the density settling outside the band or failing to converge.

**Conjecture 106** (Totient order constant). *The density of  $n$  for which  $\phi(n)/n, \phi(n+1)/(n+1), \phi(n+2)/(n+2)$  is monotone exists and lies in  $(0.019, 0.022)$ .*

**Significance.** The totient ratio is even more strongly anti-monotone than the abundancy index, its adjacent triples monotone only about one part in fifty, and the conjecture records this second measured order constant as a companion to Conjecture 105 on the reciprocal-flavoured multiplicative function. The mechanism is that  $\phi(n)/n = \prod_{p|n}(1 - 1/p)$  and  $\sigma(n)/n$  move oppositely with the small prime divisors of  $n$ , so the two order constants probe the same local dependence from two sides. The nearest related result is the classical distribution function of  $\phi(n)/n$  [86]. In tests on the corpus described in the stress-test section, the pooled rate was 0.02025 over 19,999,990 strict triples. A counterexample would be the density settling outside the band.

**Conjecture 107** (Five-term totient barrier, resolved false). *No five consecutive values of  $\phi(n)/n$  are strictly increasing, and no five are strictly decreasing.*

**Remark 15** (Resolution). *The conjecture is false. A theorem of Martin [53] states that for every  $k$  and every prescribed strict ordering of  $\phi(n+1), \dots, \phi(n+k)$  with each successive ratio exceeding any fixed constant  $C$ , the set of  $n$  realizing it has positive lower density. Apply it with  $k = 5$ . Since  $\phi(m+1)/(m+1) > \phi(m)/m$  exactly when  $\phi(m+1)/\phi(m) > 1 + 1/m$ , prescribing  $\phi(n+i+1)/\phi(n+i) > 2$  for  $i = 1, \dots, 4$  forces  $\phi(n+i)/(n+i)$  strictly increasing across the*

five consecutive integers; prescribing  $\phi(n+i)/\phi(n+i+1) > 2$  forces a strictly decreasing run in the same way. Hence strictly monotone runs of  $\phi(n)/n$  of length five—indeed of every length, in both directions—occur on a set of positive lower density. That density is astronomically small, which is why no run appears below  $4 \times 10^9$ , but it is positive, so the barrier does not hold.

**Significance.** The statement is retained with its resolution because the contrast it draws survives the refutation. Long strictly monotone runs of  $\phi(n)/n$  do exist, by Martin’s theorem, but only on a set of astronomically small positive density, with none below  $4 \times 10^9$ , whereas the abundancy analogue already fails at a findable height: a strictly increasing five-term run of  $\sigma(n)/n$  begins at  $n = 36,721,681$ . The two functions therefore differ not in whether long monotone runs occur, since both admit them, but in how far one must search to meet one, a quantitative separation the original barrier misread as an absolute prohibition for the totient. The surviving content is this gap and its mechanism, that  $\phi(m+1)/\phi(m)$  must clear  $1 + 1/m$  to advance the ratio, which is exactly the simultaneous-inequality construction of [53] pushed to a fixed length.

**Conjecture 108** (Primorial-stride order constant). *Along primorial strides  $W_y = \prod_{p \leq y} p$  the abundancy order density  $\mu(W_y)$  has a limit as  $y \rightarrow \infty$ , and  $\lim_{y \rightarrow \infty} \mu(W_y) \in (0.16, 0.19)$ .*

**Significance.** Widening the stride to a primorial makes the three sampled arguments share their small prime structure in a controlled way, and the conjecture records that the monotone density then rises from the adjacent value of Conjecture 105 toward a distinct limit, so the order constant is a function of the stride’s prime support. The band is again a measured interval. The nearest related result is the distribution function of  $\sigma(n)/n$  [86], here conditioned along an arithmetic progression of common modulus. In tests on the corpus described in the stress-test section, the rate fell from 0.202 at  $W = 30$  to 0.1768 at  $W = 30030$ , the later blocks decreasing toward a limit below the pooled value. A counterexample would be  $\mu(W_y)$  failing to converge or landing outside the band.

**Conjecture 109** (Dyadic completion overshoot). *For every even squarefree  $W$  divisible by six the limit  $\lim_{e \rightarrow \infty} \mu(2^e W)$  exists and exceeds  $1/3$ , so  $\mu(2^e W) > 1/3$  for all large  $e$ .*

**Significance.** Raising only the dyadic exponent of a fixed squarefree stride, rather than its prime support, is the sharp control, and the conjecture records that this raises the monotone density past the independent value  $1/3$  and holds it there, the opposite direction from the primorial depression of Conjecture 108. The mechanism is isolated by two controls: replacing  $\sigma(n)/n$  by the squarefree abundancy or by  $n/\phi(n)$  removes most of the effect, so it is carried by the exponents, especially  $v_2$ , and not by the set of prime divisors. The nearest related result is the distribution theory of  $\sigma(n)/n$  [86]. In tests on the corpus described in the stress-test section, three stride ladders rose monotonically toward values above  $1/3$ , and an adversarial scan of 73 further squarefree supports had every final level above  $1/3$ , the smallest being 0.3341. A counterexample would be a stride whose completed density stayed at or below  $1/3$ .

## 28 Prime-factor shocks in combinatorial sequences

**Conjecture 110** (Binomial-slice negative shocks, resolved true). *For every fixed  $k \geq 3$  the jump  $J_{k,n} = \Omega\binom{k(n+1)}{n+1} - \Omega\binom{kn}{n}$  is unbounded below.*

**Significance.** The binomial slice grows super-exponentially while its prime-factor count can fall, because a step can replace many small prime powers by fewer large primes, and the conjecture asserts arbitrarily deep single-step losses for every slice past the central one. The mechanism is the exact identity of Proposition 7, in which the negative term  $-\Omega(n+1)$  can be made large by a power-of-two argument. The nearest related result is Proposition 7 itself, which proves the case at hand for  $k=2$ ; the general  $k$  is not reduced to it, since the slice carries a second negative block that the power-of-two argument does not control, and stays open. A first decisive theorem is the  $k=2$  case, already proved. In tests on the corpus described in the stress-test section, the minima reached  $-21$  through  $-23$  for  $k=3$  to  $8$  over millions of steps each. A counterexample would be a slice with a finite floor on its downward jumps. *Resolution.* The conjecture is true: Proposition 11 forces arbitrarily deep negative jumps for every  $k \geq 3$  by a Chinese-remainder construction that loads the denominator forms  $(k-1)n+j$  with many prime factors while a first-moment bound keeps the numerator forms typical.

**Conjecture 111** (Binomial-slice positive shocks, resolved true). *For every fixed  $k \geq 2$  the jump  $J_{k,n}$  is unbounded above.*

**Significance.** The upward companion of Conjecture 110 asserts that a single step can also gain arbitrarily many prime factors, so the jump sequence is two-sided and not merely floored, and the two together make the slice's prime-factor count a genuinely oscillating additive process. The mechanism is again the identity of Proposition 7, now with the positive block  $\sum_j \Omega(kn+j)$  made large by choosing  $n$  so that the numerator forms carry high prime powers. The nearest related result is the theory of the normal and large values of  $\Omega$  on linear forms [86]. In tests on the corpus described in the stress-test section, the maxima reached  $13$  through  $26$  across  $k=2$  to  $8$ . A counterexample would be a slice whose upward jumps are bounded. *Resolution.* The conjecture is true: Proposition 11 applies the same forcing with the two form families exchanged, loading the numerator forms  $kn+j$ ; the second negative block that resisted the power-of-two argument is controlled on average rather than pointwise.

**Conjecture 112** (Fuss–Catalan two-sided shocks, resolved true). *For every fixed  $r \geq 1$  the jump  $\Omega(F_{r,n+1}) - \Omega(F_{r,n})$  is unbounded both below and above.*

**Significance.** The Fuss–Catalan numbers are the natural one-parameter generalization of the Catalan sequence, and the conjecture extends the two-sided shock phenomenon to them, so the effect is a feature of factorial-ratio families rather than of the binomial coefficient alone. The mechanism is the exact consecutive-ratio identity for  $F_{r,n}$ , derived like Proposition 7 without factoring the enormous integer itself. The nearest related result is Proposition 7, whose Catalan case is the instance  $r=1$ . In tests on the corpus described in the stress-test section, orders  $1$  to  $8$  gave minima from  $-20$  to  $-25$  and maxima from  $13$  to  $24$ . A counterexample would be an order with a one-sided bound on its jumps. *Resolution.* The conjecture is true in both directions for every  $r \geq 1$ : Proposition 11 applies to the Fuss–Catalan ratio's two families of forms, which are pairwise nonproportional since  $(r+1)j' = rj$  has no admissible solution.

**Proposition 11** (Two-sided shock forcing). *For every fixed  $k \geq 2$  the jump  $J_{k,n}$  is unbounded below and unbounded above, and for every fixed  $r \geq 1$  the Fuss–Catalan jump  $\Omega(F_{r,n+1}) - \Omega(F_{r,n})$  is unbounded below and unbounded above. In particular Conjectures 110, 111, and 112 are true.*

*Proof.* Proposition 7 and  $\Omega(kn+k) = \Omega(k) + \Omega(n+1)$  give  $J_{k,n} = \Omega(k) + \sum_{j=1}^{k-1} (\Omega(kn+j) - \Omega((k-1)n+j))$ , and cancelling  $rn+1$  against the denominator block and  $(r+1)(n+1)$  against  $n+1$  in the



exact ratio  $F_{r,n+1}/F_{r,n}$  gives  $\Omega(F_{r,n+1}) - \Omega(F_{r,n}) = \Omega(r+1) + \sum_{j=1}^r \Omega((r+1)n+j) - \sum_{j=2}^{r+1} \Omega(rn+j)$ . Each claim is therefore an instance of the following: for affine forms  $L_i(n) = a_i n + b_i$ ,  $1 \leq i \leq s$ , and  $L'_j(n) = a'_j n + b'_j$ ,  $1 \leq j \leq s$ , with positive integer coefficients and no  $L_i$  proportional to any  $L'_j$ , the difference  $\sum_i \Omega(L_i(n)) - \sum_j \Omega(L'_j(n))$  is unbounded above. Nonproportionality holds in all four applications:  $kj' = (k-1)j$  forces  $k \mid j$  with  $1 \leq j \leq k-1$ , and  $(r+1)j' = rj$  forces  $r+1 \mid j$  with  $1 \leq j \leq r$ , both impossible.

For the claim, extract the contents  $\gcd(a'_j, b'_j)$ , whose contribution to  $\Omega(L'_j)$  is constant, so the  $L'_j$  may be taken primitive; let  $R$  bound the coefficients and the cross-resultants  $|a_i b'_j - a'_j b_i|$ , fix  $m$ , and choose  $sm$  distinct primes exceeding  $R$ , one block  $Q_i$  of  $m$  primes for each  $L_i$ . Each congruence  $L_i(n) \equiv 0 \pmod{Q_i}$  is a single residue class since the primes exceed  $a_i$ , and the classes combine by the Chinese remainder theorem into  $n \equiv \rho \pmod{Q}$  with  $Q = Q_1 \cdots Q_s$ , on which  $\Omega(L_i(n)) \geq m$  for every  $i$ . No forced prime  $p$  divides any  $L'_j(n)$ : together with  $p \mid L_i(n)$  it would divide  $a_i L'_j(n) - a'_j L_i(n) = a_i b'_j - a'_j b_i$ , a nonzero integer of absolute value below  $p$ . Now let  $n$  run over  $\rho + tQ$  for  $1 \leq t \leq Q^3$ , so that every value  $L'_j(n)$  is at most a constant times  $Q^4$ . A prime power  $p^v \leq Q^3$  with  $p \nmid Q$  divides  $L'_j(n)$  along at most  $Q^3/p^v + 1$  values of  $t$ , primes dividing  $a'_j$  divide no value of the primitive form, and summing  $1/p^v$  over prime powers gives an average of  $\log \log Q + O(1)$  per form, the  $+1$  terms totalling  $O(Q^3/\log Q)$ . Prime powers exceeding  $Q^3$  add  $O(1)$  per form on average: a prime above  $Q^3$  divides a value at most once and to the first power, since  $(Q^3)^2$  exceeds every value, while prime powers  $p^v > Q^3$  with  $v \geq 2$  are squarefull, number  $O(Q^2)$  below the value bound, and each catches at most two values of  $t$ . By Markov some  $t \leq Q^3$  has  $\sum_j \Omega(L'_j(n)) \leq 2s(\log \log Q + O(1))$ , whence  $\sum_i \Omega(L_i(n)) - \sum_j \Omega(L'_j(n)) \geq sm - 2s \log \log Q - O(s) \rightarrow \infty$  as  $m \rightarrow \infty$ , since  $\log \log Q = O(\log(sm \log m))$ . Explicit witnesses are cheap: six primes per form already force  $J_{3,n} \leq -20$  and  $J_{4,n} \leq -30$  at heights near  $10^{21}$  and  $10^{35}$ , and  $J_{3,n} \geq 21$  on the positive side, beyond every scanned extreme.  $\square$

**Conjecture 113** (Negative shock covariance). *In every fixed binomial-slice ( $k \geq 2$ ), central-multinomial ( $k \geq 2$ ), and Fuss–Catalan ( $r \geq 1$ ) family the lag-one covariance of successive jumps has a finite strictly negative limit.*

**Significance.** Beyond their range, the jumps carry a local temporal structure, and the conjecture asserts that consecutive jumps are negatively associated, a large loss tending to be followed by a partial recovery, which is the signature of a mean-reverting additive process rather than an independent one. The statement is on the covariance and not the correlation, since the slowly growing variance can send the correlation toward zero while a definite local covariance survives, and it is exposed because it asserts a sign uniformly across families. The nearest related result is the theory of correlations of additive functions on nearby arguments [86]. In tests on the corpus described in the stress-test section, all 22 tested families had negative covariance, from about  $-1.15$  to  $-59.5$ . A counterexample would be a family with nonnegative or divergent limiting covariance. The central-binomial slice now carries an exact local constant: Proposition 12 evaluates the prime-truncated covariance at exactly  $-\frac{5}{3}$  for every cutoff at least 3, so the limit for that family should equal  $-\frac{5}{3}$  outright, the surviving step being that primes beyond any cutoff contribute nothing in the limit; the full covariance stands at  $-1.47$  by  $2 \times 10^6$  and drifts toward the constant. That the limit exists at all, and is rational, now follows in every family from Proposition 13: each jump is a signed sum of  $\Omega$  over affine forms whose signed coefficients cancel, so all but finitely many primes contribute nothing to the lag-one covariance and the remainder is a finite sum of local rational densities; only its uniform negativity across families stays conjectural.

**Proposition 12** (Exact truncated covariance for the central-binomial slice). For a cutoff  $y \geq 2$  write  $\Omega_y(m) = \sum_{p \leq y} v_p(m)$  and  $J_n^{(y)} = 1 + \Omega_y(2n+1) - \Omega_y(n+1)$ . The lag-one covariance of the truncated central-binomial jump over  $n \leq N$  converges as  $N \rightarrow \infty$ , to  $-1$  at  $y = 2$  and to

$$\lim_{N \rightarrow \infty} \text{Cov}(J_n^{(y)}, J_{n+1}^{(y)}) = -\frac{5}{3} \quad \text{for every } y \geq 3.$$

*Proof.* Write  $A = 2n+1$ ,  $B = n+1$ ,  $C = 2n+3$ ,  $D = n+2$ , so that  $J_n^{(y)} - 1 = \sum_{p \leq y} (v_p(A) - v_p(B))$  and  $J_{n+1}^{(y)} - 1 = \sum_{p \leq y} (v_p(C) - v_p(D))$ . For  $n$  uniform on  $\{1, \dots, N\}$  the joint residues modulo any fixed prime powers equidistribute as  $N \rightarrow \infty$ , valuations at distinct primes decouple in the limit, and all limiting expectations exist by the geometric decay of  $\Pr[p^r \mid an+b]$ , so the limiting covariance is the sum over  $p \leq y$  of the local covariance  $\text{Cov}(v_p(A) - v_p(B), v_p(C) - v_p(D))$ . For a form  $an+b$  with  $p \nmid a$  the limiting mean of  $v_p$  is  $\sum_{r \geq 1} p^{-r} = 1/(p-1)$ , and two such forms whose roots differ modulo  $p$  are never simultaneously divisible by  $p$ , so their valuations have covariance  $-1/(p-1)^2$ . For  $p \geq 5$  all four pairs  $(A, C)$ ,  $(A, D)$ ,  $(B, C)$ ,  $(B, D)$  have distinct roots, the cross-resultants being 2, 3, 1, 1, and the four equal covariances cancel in the signed sum. For  $p = 2$ ,  $A$  and  $C$  are odd while  $B$  and  $D$  are consecutive, so  $v_2(B)v_2(D) = 0$  identically, each mean is 1, and the local contribution is  $\text{Cov}(v_2(B), v_2(D)) = -1$ . For  $p = 3$  the exceptional pair is  $(A, D)$ , with  $A = 2D - 3$ : divisibility by 3 is simultaneous, along the single class  $n \equiv 1 \pmod{3}$ , while  $9 \mid A$  and  $9 \mid D$  are incompatible since  $9 \nmid 3$ , so  $\mathbf{E}[v_3(A)v_3(D)] = \frac{1}{3} + 2 \sum_{r \geq 2} 3^{-r} = \frac{2}{3}$  and  $\text{Cov}(v_3(A), v_3(D)) = \frac{2}{3} - \frac{1}{4} = \frac{5}{12}$ ; the other three pairs have distinct roots modulo 3 and covariance  $-\frac{1}{4}$  each, so the local contribution is  $-\frac{1}{4} - \frac{5}{12} + \frac{1}{4} - \frac{1}{4} = -\frac{2}{3}$ . Summing the local contributions gives  $-1$  at  $y = 2$  and  $-1 - \frac{2}{3} = -\frac{5}{3}$  for every  $y \geq 3$ .  $\square$

**Proposition 13** (Existence of the shock covariance). In each fixed binomial-slice ( $k \geq 2$ ), central-multinomial ( $k \geq 2$ ), and Fuss–Catalan ( $r \geq 1$ ) family, the lag-one covariance of successive  $\Omega$ -jumps over  $n \leq N$  converges as  $N \rightarrow \infty$  to a rational number.

*Proof.* By Proposition 7 each jump equals a constant plus a signed sum  $\sum_i s_i \Omega(L_i(n))$  over affine forms  $L_i(n) = a_i n + b_i$  with positive integer coefficients, the numerator and denominator carrying equally many forms, so  $\sum_i s_i = 0$ . Writing  $\Omega = \sum_p v_p$  and using that the valuations at distinct primes decouple as  $N \rightarrow \infty$ , with all mixed moments existing by the geometric decay of  $\Pr[p^m \mid an+b]$ , the limiting lag-one covariance is  $\sum_p C_p$  with

$$C_p = \sum_{i,j} s_i s_j \text{Cov}(v_p(L_i(n)), v_p(L_j(n+1))),$$

a sum over the forms  $L_i(n)$  and their unit shifts  $L_j(n+1) = a_j n + (a_j + b_j)$ . No  $L_i(n)$  is proportional to any  $L_j(n+1)$ , so the pairwise resultants  $a_i(a_j + b_j) - a_j b_i$  are nonzero integers; let  $p$  divide none of them and none of the leading coefficients  $a_i$ . Then no such pair shares a root modulo  $p$ , so the two forms are never simultaneously divisible by  $p$ ; each valuation has limiting mean  $1/(p-1)$  and every pairwise covariance equals  $-1/(p-1)^2$ , whence  $C_p = -(p-1)^{-2} (\sum_i s_i)^2 = 0$ . Only the finitely many primes dividing a leading coefficient or one of the resultants contribute, and each such  $C_p$  is a rational number fixed by the  $p$ -adic densities of the forms. The covariance is therefore a finite sum of rationals, hence rational.  $\square$

**Conjecture 114** (Dyadic multinomial slope). For every fixed  $k \geq 2$  the dyadic jump of the central multinomial obeys

$$\frac{\Omega(M_{k,2^m}) - \Omega(M_{k,2^{m-1}})}{m} \longrightarrow -(k-1).$$

**Significance.** Proposition 7 gives a negative linear bound for the dyadic multinomial jump, and the conjecture sharpens the constant to  $-(k-1)$ . At  $n = 2^m - 1$  the statement is equivalent to  $\Omega(k2^m - c) = o(m)$  for every fixed  $1 \leq c < k$ . Proposition 14 proves this when  $k = 2$  by separating the cyclotomic factors of  $2^{m+1} - 1$  into small and large orders. The remaining fixed- $k$  cases are prime-factor questions for the exponential shifts  $k2^m - c$ . In tests on the corpus described in the stress-test section, the observed jumps at  $m = 21$  ran from about  $-16$  for  $k = 2$  to  $-118$  for  $k = 8$ , in the predicted direction. A counterexample would be a fixed  $k$  with a nonzero linear contribution from one of the shifted terms.

**Proposition 14** (The dyadic multinomial slope for  $k = 2$ ). The conclusion of Conjecture 114 holds for  $k = 2$ .

*Proof.* At  $n = 2^m - 1$ , the jump identity gives

$$\Omega(M_{2,2^m}) - \Omega(M_{2,2^m-1}) = 1 - m + \Omega(2^{m+1} - 1).$$

Put  $N = m + 1$  and factor  $2^N - 1 = \prod_{d|N} \Phi_d(2)$ . Uniformly in  $d$ ,

$$\log \Phi_d(2) = \varphi(d) \log 2 + O(1).$$

This follows from the Möbius product for  $\Phi_d(2)$  and the convergence of  $\sum_{r \geq 1} |\log(1 - 2^{-r})|$ . A prime factor  $p$  of  $\Phi_d(2)$  either has multiplicative order  $d$  modulo  $p$ , in which case  $p \equiv 1 \pmod{d}$ , or is the largest prime factor of  $d$ . In the exceptional case it occurs to the first power, by the standard order lemma and lifting the exponent.

For divisors  $d \leq \sqrt{N}$ , the total number of prime factors with multiplicity is at most

$$O \left( \sum_{\substack{d|N \\ d \leq \sqrt{N}}} \varphi(d) \right) = N^{1/2+o(1)}.$$

For  $d > \sqrt{N}$ , every nonexceptional prime factor is larger than  $d$ , so these factors contribute at most

$$O \left( \sum_{\substack{d|N \\ d > \sqrt{N}}} \frac{\varphi(d)}{\log d} \right) = O \left( \frac{N}{\log N} \right).$$

The exceptional factors contribute at most  $\tau(N) = N^{o(1)}$ . Here we used  $\sum_{d|N} \varphi(d) = N$  and the standard divisor bound. Hence  $\Omega(2^N - 1) = o(N)$ . Dividing the displayed jump by  $m$  gives the limit  $-1$ .  $\square$

**Conjecture 115** (Eventual sign balance, resolved true). *For every fixed binomial-slice and Fuss–Catalan family the positive-jump and negative-jump densities each tend to  $\frac{1}{2}$ , while the zero-jump density tends to 0.*

**Significance.** The jump has a small positive mean and a broadening variance, and the conjecture asserts that the sign nonetheless balances in the limit, an Erdős–Kac-type prediction that the fluctuations swamp the drift so that up-steps and down-steps become equally frequent. The

mechanism is that the jump is a difference of additive functions on linear forms whose variance grows without bound, so the centred sign is asymptotically symmetric. The nearest related result is the Erdős–Kac theorem and its descendants for additive functions [86]. In tests on the corpus described in the stress-test section, the negative fractions stood at 0.356 to 0.435 and rose across prefix blocks as the variance widened. A counterexample would be a family whose sign densities settled away from one half. *Resolution.* The conjecture is true. By the identities of Propositions 7 and 11, each jump is, up to a bounded term, a signed sum of  $\Omega$  over  $2(k-1)$ , respectively  $2r$ , pairwise nonproportional affine forms, whose primitive parts are pairwise coprime linear polynomials. The multivariate Erdős–Kac theorem of El-Baz, Loughran, and Sofos [173], whose one-variable case recovers Tanaka’s theorem for pairwise coprime irreducible polynomials, makes the vector of  $\omega$ -values asymptotically Gaussian with identity covariance, so the signed sum, normalized by  $\sqrt{2(k-1)\log\log n}$ , is asymptotically standard normal; replacing  $\omega$  by  $\Omega$  shifts each coordinate by a quantity of bounded mean, negligible at this scale. The diverging variance swamps the bounded centring, giving sign densities  $\frac{1}{2}$ , and the zero-jump density vanishes because the limit law assigns every shrinking central window vanishing mass.

## 29 The arithmetic anatomy of prime-gap boundaries

**Conjecture 116** (Boundary Euler sum). *For each prime  $\ell$  let  $r$  be uniform on  $\mathbb{Z}/\ell^a\mathbb{Z}$  conditioned on  $r(r+g) \not\equiv 0 \pmod{\ell}$ , and set*

$$m_{\ell,a}(g) = \mathbb{E}[v_\ell(r+1) + v_\ell(r+g-1) - v_\ell(r-1) - v_\ell(r+g+1)].$$

*Then for every fixed even  $g$  occurring infinitely often as a consecutive-prime gap,*

$$\lim_{x \rightarrow \infty} \mathbb{E}[D(p, q) \mid p \leq x, q - p = g] = \sum_{\ell} \lim_{a \rightarrow \infty} m_{\ell,a}(g).$$

**Significance.** The prime-factor balance of the four composites bracketing a gap oscillates strongly with the gap, and the conjecture is a local–global law identifying its mean with an explicit Euler sum of  $\ell$ -adic expectations, the random variable being the factor anatomy of the neighbours rather than a residue transition of the prime endpoints. The mechanism is elementary at the small primes: for  $g \equiv 2 \pmod{6}$  both inward neighbours are forced multiples of three, for  $g \equiv 4 \pmod{6}$  both outward neighbours are, and when  $6 \mid g$  the three-adic contribution cancels, with the higher primes superimposing smaller waves. The nearest related result is the Hardy–Littlewood correlation heuristic for tuples inside a gap [1], sharing the calculus’s aesthetic of the Lemke Oliver–Soundararajan biases [24] while measuring a different object. A first decisive theorem would be the law for  $g = 2$  from a Hardy–Littlewood model with the consecutive-gap exclusion built in. In tests on the corpus described in the stress-test section, truncating the sum at  $\ell = 47$  gave correlation 0.998 and root-mean-square error 0.24 across the 39 gaps with at least 500 samples. A counterexample would be a well-sampled gap whose mean disagreed with the truncated Euler sum beyond the truncation error.

## Part VII

# Topological invariants of geometric families

*Added to the deposit: 12 August 2026.*

The twenty conjectures below leave arithmetic for algebraic topology while keeping the same discipline. Each takes a canonical family of spaces, computes an exact classical invariant across the whole family, and takes the resulting shape law seriously. They form four programmes. A Calabi–Yau programme (Conjectures 117 to 121) studies the Euler indices of smooth complete intersections in projective space, ending in an exact dimension-wise gcd law. A link programme (Conjectures 122 to 126) studies the middle Betti numbers of Brieskorn–Pham links under degree caps and common scaling. A moment-angle programme (Conjectures 127 to 131) studies the Hochster strands of flag moment-angle complexes, their support, shape, and zero locations. A monodromy programme (Conjectures 132 to 136) studies torus mapping tori with signed-permutation monodromy over the signed-orbit calibration proved below. The calibration layer is classical and exact: the Chern-class formula for complete intersections, the Milnor–Orlik formula for Brieskorn links, Hochster’s formula for bigraded Betti numbers, and the Wang sequence for mapping tori. Six adversarial controls are retained in place, each refuting a tempting strengthening. Two of the twenty, the limit clause of Conjecture 117 and Conjecture 135, are the strongest statements of the part. Two more, Conjectures 120 and 124, lie close to classical methods. Since deposit, Conjectures 117 and 118 have been resolved true by Propositions 16 and 17, and Conjecture 135 has been resolved false by the Landau floor of Proposition 19, which fixes the compression exponent at 2. Conjecture 125 and the convexity clause of Conjecture 126 have since been proved by the Ehrhart model of Proposition 18.

For  $n \geq 2$  and a multidegree  $\mathbf{d} = (d_1, \dots, d_r)$  with every  $d_i \geq 2$  and  $\sum_i d_i = n + r + 1$ , let  $X_{\mathbf{d}}^{(n)} \subset \mathbf{CP}^{n+r}$  be a smooth complete intersection of multidegree  $\mathbf{d}$ , a Calabi–Yau  $n$ -fold, with Euler characteristic computed exactly by  $c(TX) = (1 + H)^{n+r+1} / \prod_i (1 + d_i H)$  and  $\chi = (\prod_i d_i) [H^n] c(TX)$ ; write  $E_n(\mathbf{d}) = (-1)^n \chi(X_{\mathbf{d}}^{(n)})$ ,  $D(\mathbf{d}) = \prod_i d_i$ , and let the *merge*  $M_{ij}\mathbf{d}$  replace  $d_i, d_j$  by  $d_i + d_j - 1$ , preserving both  $n$  and the Calabi–Yau condition. Write  $Q_n = E_n(2, \dots, 2)$  for the all-quadric configuration of  $n + 1$  quadrics. For  $\mathbf{a} = (a_1, \dots, a_N)$  with  $a_i \geq 2$ , let  $L(\mathbf{a})$  be the Brieskorn–Pham link  $\{z_1^{a_1} + \dots + z_N^{a_N} = 0\} \cap S^{2N-1}$ , of real dimension  $2N - 3$ , with middle Betti number  $\beta(\mathbf{a}) = b_{N-2}(L(\mathbf{a}); \mathbf{Q})$  given exactly by the Milnor–Orlik formula [88]

$$\beta(\mathbf{a}) = (-1)^N + \sum_{\emptyset \neq J \subseteq [N]} (-1)^{N-|J|} \frac{\prod_{j \in J} a_j}{\text{lcm}_{j \in J} a_j},$$

the *Fano* tuples being those with  $\sum_i 1/a_i > 1$ , with cap  $A = \max_i a_i$  and scale sequence  $\beta_{\mathbf{a}}(k) = \beta(ka_1, \dots, ka_N)$ . For a finite simple graph  $G$  on  $m$  vertices with clique complex  $K = \text{Cl}(G)$ , define the Hochster strands

$$h_{r,s}(G) = \sum_{\substack{I \subseteq V(G) \\ |I|=s}} \dim_{\mathbf{F}_2} \tilde{H}_r(K_I; \mathbf{F}_2), \quad H_{G,r}(z) = \sum_s h_{r,s}(G) z^s,$$

the bigraded Betti sectors of the Stanley–Reisner ring and of the moment-angle complex  $\mathcal{Z}_K$  by Hochster’s formula [89, 90]. Finally, for a signed permutation matrix  $A \in SL(d, \mathbb{Z})$  let  $M_A$  be the mapping torus of the induced automorphism of the torus  $T^d$ , set  $f_k(A) = \dim_{\mathbf{Q}} \ker(\wedge^k A - I)$ , so that the Wang sequence gives  $b_k(M_A; \mathbf{Q}) = f_k(A) + f_{k-1}(A)$  [97, 96], and let  $\tau(A) = \sum_k d(\text{Tor } H_k(M_A; \mathbb{Z}))$  count minimal torsion generators. The cap  $A$  of the link programme and the matrix  $A$  of the monodromy programme are unrelated symbols, each local to its own sections.

**Proposition 15** (Signed-orbit calibration). Let  $A$  be a signed permutation matrix on  $\mathbb{Z}^d$ . Then  $\wedge^k A$  permutes the wedge basis up to sign, and every orbit of the induced permutation of  $k$ -subsets

carries a holonomy sign, the product of the signs collected around one period. If  $p_k$  and  $n_k$  are the numbers of positive and negative orbits at level  $k$ , then

$$f_k(A) = p_k, \quad \text{coker}(\wedge^k A - I) \cong \mathbb{Z}^{p_k} \oplus (\mathbb{Z}/2)^{n_k},$$

and consequently  $b_k(M_A; \mathbf{Q}) = p_k + p_{k-1}$ ,  $\text{Tor } H_k(M_A; \mathbb{Z}) \cong (\mathbb{Z}/2)^{n_k}$ , and  $\tau(A) = \sum_k n_k$ , while  $\sum_k b_k(M_A; \mathbf{Q}) = 2 \sum_k p_k$ .

*Proof.* On the wedge basis  $e_S$ ,  $S$  a  $k$ -subset,  $\wedge^k A$  acts by  $e_S \mapsto \pm e_{\sigma(S)}$  with  $\sigma$  the induced permutation, so  $\wedge^k A - I$  is block-diagonal over the orbits of  $\sigma$ . On an orbit of length  $L$ , successive substitutions  $e_i \mapsto \pm e_i$  absorb all but one sign into the basis; such changes alter individual signs but preserve their product around the orbit, the holonomy  $\varepsilon$ , which is therefore invariant. The block becomes  $C_\varepsilon - I$  with  $C_\varepsilon$  the cyclic shift carrying a single sign  $\varepsilon$  on its closing edge. Eliminating  $x_{i+1} = x_i$  along the open edges reduces the block to the single relation  $(1 - \varepsilon)x = 0$ , so its cokernel is  $\mathbb{Z}$  and its rational kernel one-dimensional when  $\varepsilon = +1$ , and its cokernel is  $\mathbb{Z}/2$  and its rational kernel zero when  $\varepsilon = -1$ . Summing over orbits gives  $f_k = p_k$  and the displayed cokernel. The Wang sequence of the fibration  $T^d \rightarrow M_A \rightarrow S^1$  reads  $0 \rightarrow \text{coker}(\wedge^k A - I) \rightarrow H_k(M_A; \mathbb{Z}) \rightarrow \ker(\wedge^{k-1} A - I) \rightarrow 0$ , and the kernel is free, so the sequence splits, giving  $b_k = p_k + p_{k-1}$  and  $\text{Tor } H_k \cong (\mathbb{Z}/2)^{n_k}$ . The two sums follow by telescoping.  $\square$

### 30 Euler indices of Calabi–Yau complete intersections

**Conjecture 117** (Quadric-ladder ratio law, resolved true). *For every  $n \geq 5$ ,*

$$\frac{Q_{n+1}}{Q_n} < 8, \quad \frac{Q_{n+2}}{Q_{n+1}} > \frac{Q_{n+1}}{Q_n}, \quad \text{and} \quad \lim_{n \rightarrow \infty} \frac{Q_{n+1}}{Q_n} = 8.$$

**Significance.** The all-quadric configuration is the maximal-codimension endpoint of the family, and the conjecture asserts that its Euler index grows with a definite exponential ratio approached monotonically from below, a rare closed growth law for a topological index across Calabi–Yau dimension. The small-dimensional boundary is genuine: the values run 24, 128, 960, 6912 at  $n = 2$  to 5, the ratio rising to 7.5 and dipping to 7.2 before the monotone regime sets in. The limit clause is an extrapolation. The nearest related result is the exact Chern-class evaluation itself, surveyed with the complete-intersection landscape in [95]. A first decisive theorem would be the strict bound  $Q_{n+1}/Q_n < 8$  from the coefficient recursion. In tests on the corpus described in the stress-test section, all finite clauses held exactly through  $n = 200$ , the ratio reaching 7.9337 by  $n = 60$ . A counterexample would be a ratio at least 8 or a convexity break beyond  $n = 5$ . Since deposit the conjecture has been resolved true, by proof rather than by computation: Proposition 16 shows the ladder’s generating function is algebraic and derives an exact recurrence whose interval induction pins every ratio inside the corridor  $(8 - 4/(n+1), 8 - 4/(n+2))$  from dimension five onward, proving all three clauses with the previously unstated first correction  $8 - Q_{n+1}/Q_n \sim 4/n$ ; the recurrence also reproduces the deposited ladder through  $n = 200$ . The limit  $8 = 2^3$  is the  $d = 2$  member of the equal-degree Euler family  $d^{d+1}$  of Proposition 32.

**Proposition 16** (Resolution of the quadric-ladder ratio law). Conjecture 117 holds in full, with the sharpening  $8 - Q_{n+1}/Q_n \sim 4/n$ . The generating function  $A(z) = \sum_{n \geq 0} Q_n z^n$ , with  $Q_0 = 2$  and  $Q_1 = 0$ , is algebraic,

$$A = \frac{2(1-y)^3}{1-3y}, \quad z = \frac{y(1-2y)}{2(1-y)^2}, \quad y(0) = 0,$$



and the coefficients satisfy  $(n+1)(n+2)Q_{n+2} = (n+1)(7n+9)Q_{n+1} + 4(n+2)(2n+3)Q_n$ .

*Proof.* The Chern-class formula gives  $Q_n = [t^n]\varphi(t)^{n+1}$  for  $\varphi(t) = 2(1-t)^2/(1-2t)$ : indeed  $E_n(2, \dots, 2) = (-1)^n 2^{n+1} [H^n](1+H)^{2n+2}(1+2H)^{-n-1}$ , and the substitution  $t = -H$  converts one extraction into the other. For each small  $z$  the equation  $y = z\varphi(y)$  has exactly one root  $y(z)$  near the origin, and the residue form of Lagrange inversion, summing the geometric series  $\sum_n (z\varphi(t)/t)^n$  on a small contour and picking up the simple pole at  $y(z)$ , gives

$$A(z) = \sum_{n \geq 0} [t^n]\varphi(t)^{n+1} z^n = \frac{\varphi(y)}{1 - z\varphi'(y)};$$

inverting  $y = z\varphi(y)$  gives  $z = y(1-2y)/(2(1-y)^2)$ , and the residue value simplifies to  $2(1-y)^3/(1-3y)$ , the stated parametrization. Differentiating along it shows that  $(1+z)(1-8z)A'' - 9(1+4z)A' - 24A$  is a rational function of  $y$  that vanishes identically, a finite identity checked by exact expansion, and extracting the coefficient of  $z^n$  from this differential equation yields the recurrence.

Now the three clauses. Set  $q_n = Q_{n+1}/Q_n$ , so the recurrence reads  $q_{n+1} = \frac{7n+9}{n+2} + \frac{4(2n+3)}{(n+1)q_n}$ , strictly decreasing as a function of  $q_n$ . Write  $\delta_n = 8 - q_n$ , and call the two-sided bound  $4/(n+2) < \delta_n < 4/(n+1)$  the corridor at index  $n$ . The corridor holds at the base  $n = 5$ :  $Q_5 = 6912$  and  $Q_6 = 51,072$  give  $\delta_5 = 11/18 \in (4/7, 4/6)$ . For the step, the map carries the corridor at index  $n$  into the interval bounded by the images of its two walls, since it is decreasing, and the two inequalities placing that image inside the corridor at index  $n+1$  clear denominators to polynomials with positive coefficients in  $n-5$ ; so the corridor propagates to every  $n \geq 5$ . The corridor gives  $q_n < 8$ ; it gives  $q_{n+1} > 8 - 4/(n+2) > q_n$ , the strict increase; and both walls are asymptotic to  $4/n$ , so  $q_n \rightarrow 8$  with  $8 - q_n \sim 4/n$ . For  $2 \leq n \leq 4$  the ratios are  $16/3$ ,  $15/2$ , and  $36/5$ , all below 8.  $\square$

**Conjecture 118** (Normalized merge monotonicity, resolved true). *For every  $n \geq 3$ , every multidegree  $\mathbf{d}$  with  $r \geq 2$ , and every pair  $i \neq j$ ,*

$$\frac{E_n(M_{ij}\mathbf{d})}{D(M_{ij}\mathbf{d})} > \frac{E_n(\mathbf{d})}{D(\mathbf{d})}.$$

**Significance.** A merge lowers the codimension by one at fixed dimension, and the conjecture asserts that the degree-normalized Euler index strictly increases along every merge path from the all-quadric floor to the hypersurface ceiling, so the normalized index is a strict order on the merge poset. The normalization is essential and calibrated: the raw strengthening is false, since  $E_4(2, 2, 4) = 1632 > 1476 = E_4(3, 4)$  although  $(2, 2, 4) \mapsto (3, 4)$  is a merge, and this counterexample is retained. The nearest related result is the complete-intersection Euler data of [95]. A first decisive theorem would be the two-equation case  $r = 2$ . In tests on the corpus described in the stress-test section, all 1,568,731 merges through dimension 30 satisfied the strict inequality, reverified independently through dimension 18. A counterexample would be one merge with a nonincreasing normalized index. Since deposit the conjecture has been resolved true. Proposition 17 expresses every normalised merge increment as a coefficient whose factors have nonnegative coefficients and whose target coefficient is strictly positive.

**Proposition 17** (Resolution of normalised merge monotonicity). Put  $a_\ell = d_\ell - 1$ , and let  $a = a_i$  and  $b = a_j$  for the entries being merged. Then

$$\frac{E_n(M_{ij}\mathbf{d})}{D(M_{ij}\mathbf{d})} - \frac{E_n(\mathbf{d})}{D(\mathbf{d})} = ab[u^{n-2}] \frac{1}{(1+u)^2(1-au)(1-bu)(1-(a+b)u)} \prod_{\ell \neq i,j} \frac{1}{1-a_\ell u} > 0.$$

Consequently Conjecture 118 holds in full. The same inequality also holds in dimension  $n = 2$ .

*Proof.* The Calabi–Yau condition gives  $\sum_{\ell} a_{\ell} = n + 1$ . The Chern-class formula and the coefficient substitution  $t = u/(1 + u)$  give

$$\frac{E_n(\mathbf{d})}{D(\mathbf{d})} = [t^n] \frac{(1 - t)^{n+r+1}}{\prod_{\ell} (1 - d_{\ell} t)} = [u^n] \frac{1}{(1 + u)^2} \prod_{\ell} \frac{1}{1 - a_{\ell} u}.$$

The merge replaces  $a, b$  by  $a + b$ . Subtracting the two rational series and using

$$\frac{1}{1 - (a + b)u} - \frac{1}{(1 - au)(1 - bu)} = \frac{abu^2}{(1 - au)(1 - bu)(1 - (a + b)u)}$$

gives the displayed coefficient formula. For every integer  $c \geq 1$  and  $q \geq 0$ ,

$$[u^q] \frac{1}{(1 + u)(1 - cu)} = \frac{c^{q+1} + (-1)^q}{c + 1} \geq 0.$$

Pair the two factors  $(1 + u)^{-1}$  with  $(1 - au)^{-1}$  and  $(1 - bu)^{-1}$ . Every remaining factor is a geometric series with nonnegative coefficients. The coefficient of degree  $n - 2$  is strictly positive because  $(1 - (a + b)u)^{-1}$  supplies the term  $(a + b)^{n-2} u^{n-2}$  while all other factors supply their constant terms. Since  $ab > 0$ , the merge increment is positive.  $\square$

**Conjecture 119** (Maximal-degree merge monotonicity). *For every  $n \geq 3$  and every  $\mathbf{d}$  with  $r \geq 2$ , if  $d_i = \max_j d_j$  then  $E_n(M_{ik}\mathbf{d}) > E_n(\mathbf{d})$  for every  $k \neq i$ .*

**Significance.** Raw merges can lower the Euler index, but the conjecture isolates the direction that survives: merging any degree into a largest degree always raises the unnormalized index, so the failure of raw monotonicity is confined to merges among small degrees. The retained counterexample of Conjecture 118 merges the two smallest degrees, and the conjecture asserts that this is the only kind of failure. The nearest related result is again the Euler census of [95]. A first decisive theorem would be the hypersurface-forming merges, where the target is a single degree. In tests on the corpus described in the stress-test section, all 377,356 maximal-degree merges through dimension 30 increased the index strictly. A counterexample would be a maximal-degree merge that fails to increase  $E_n$ .

**Conjecture 120** (Extremal multidegrees). *For every  $n \geq 3$  and every multidegree  $\mathbf{d}$ ,*

$$E_n(\underbrace{2, \dots, 2}_{n+1}) \leq E_n(\mathbf{d}) \leq E_n(n + 2),$$

*with equality only at the all-quadric and hypersurface configurations.*

**Significance.** The statement pins the two ends of the family as the unique extremes of the Euler index at every dimension, turning the visually extreme configurations into a precise rigidity claim. It is close to prior work: the bounds are likely accessible by elementary coefficient inequalities, and the retained content is the equality rigidity together with its calibrating role for Conjectures 118 and 119, whose merge order it bounds. The nearest related result is the complete-intersection census of [95]. A first decisive theorem would be either inequality with its equality case. In tests on the corpus described in the stress-test section, all 35,467 configurations through dimension 30 obeyed both bounds with equality only at the endpoints. A counterexample would be an interior configuration matching an extreme.

**Conjecture 121** (Dimension-wise Euler gcd law). *Let  $g_n$  be the greatest common divisor of  $\chi(X_{\mathbf{d}}^{(n)})$  over all multidegrees at dimension  $n$ . Then*

$$g_n = \frac{24}{\gcd(24, n)} \times \begin{cases} 2, & n \equiv 0, 2 \pmod{8}, \\ 1, & \text{otherwise.} \end{cases}$$

**Significance.** The conjecture gives a closed dimension-wise arithmetic law for the whole family, and the mod-8 doubling is the surprise: the natural guess  $24/\gcd(24, n)$  misses exactly the residues 0 and 2 modulo 8, where every Euler characteristic in the family carries one further factor of two. At  $n = 2$  the law reproduces  $\chi = 24$  for every Calabi–Yau complete-intersection surface, the K3 case. The mechanism should be a congruence for the coefficient family, and a proof programme runs through the image of these complete intersections in complex cobordism against Todd-genus and elliptic-genus congruences. The nearest related result is the divisibility role of Euler numbers in quotient constructions surveyed in [95]. A first decisive theorem would be the divisibility half, that the displayed integer divides every  $\chi$  at dimension  $n$ . In tests on the corpus described in the stress-test section, the formula matched the exact gcd in every dimension 2 to 30, reverified independently through dimension 26. A counterexample would be a dimension whose gcd differs.

### 31 Middle Betti numbers of Brieskorn–Pham links

**Conjecture 122** (Five-link cap extremum). *For  $N = 4$  and every cap  $A \geq 9$ , every sorted Fano tuple  $\mathbf{a}$  with  $\max_i a_i = A$  satisfies  $b_2(L(\mathbf{a})) \leq A - 1$ , with equality only for  $\mathbf{a} = (2, 2, A, A)$ .*

**Significance.** Among five-dimensional Fano Brieskorn links with a fixed largest exponent, the conjecture identifies the exact maximal middle Betti number and its unique maximizer, a linear cap law with a rigidity clause. The threshold is genuine: at  $A = 3$  the tuple  $(3, 3, 3, 3)$  has  $b_2 = 6$ , defeating the eventual formula, so the linear branch begins only after the low-cap oscillation dies out. The nearest related results are the Milnor–Orlik computation [88] and the role of these links in positive Sasakian geometry [93, 94]. A first decisive theorem would be the bound for tuples containing two entries equal to 2. In tests on the corpus described in the stress-test section, the bound and its rigidity held for every sorted Fano 4-tuple at every exact cap through 50, reverified independently through 25. A counterexample would be a tuple beating the cap or a second maximizer.

**Conjecture 123** (Seven-link cap extremum). *For  $N = 5$  and every cap  $A \geq 13$ , every sorted Fano tuple with  $\max_i a_i = A$  satisfies  $b_3(L(\mathbf{a})) \leq (A - 1)(A - 2)$ , with equality only for  $(2, 2, A, A, A)$ .*

**Significance.** One dimension up, the cap law becomes quadratic with the analogous unique maximizer, and the pair of statements suggests a general polynomial cap law in the link dimension. The transition is sharper than at five dimensions: at  $A = 12$  the maximizer is  $(2, 3, 12, 12, 12)$  with  $b_3 = 222$ , well above the eventual  $(A - 1)(A - 2)$ , so the stable branch begins only at  $A = 13$ . The nearest related results are again [88, 93]. A first decisive theorem would be the asymptotic order  $b_3 \ll A^2$  under the Fano condition. In tests on the corpus described in the stress-test section, the complete exact-cap census through  $A = 40$  found no exception, the transition at  $A = 12$  reverified independently. A counterexample would be a tuple above the quadratic cap or a second maximizer at some cap.

**Conjecture 124** (Connected gcd-graph positivity). *For  $N = 4$ , if the graph on  $\{1, 2, 3, 4\}$  joining  $i \sim j$  when  $\gcd(a_i, a_j) > 1$  is connected, then  $b_2(L(\mathbf{a})) \geq 1$ .*

**Significance.** The conjecture reads rational middle homology off an arithmetic graph: pairwise entanglement of the exponents, propagated along a connected graph, forces the five-dimensional link to carry homology. It is dimension-specific and calibrated: the five-tuple  $(2, 2, 2, 2, 2)$  has a connected gcd graph and vanishing middle Betti number, so no naive extension to all  $N$  holds. It lies close to the classical graph-theoretic criteria for Brieskorn homology spheres [88, 93]. Those criteria detect integral homology spheres, and a vanishing rational middle homology can still coexist with finite torsion, so they do not settle the Betti statement directly; what they do supply is the exact Milnor–Orlik count  $b_2 = \#\{(k_1, \dots, k_4) : 1 \leq k_i \leq a_i - 1, \sum_i k_i/a_i \in \mathbb{Z}\}$ , and the conjecture is that a connected gcd graph forces this count to be positive, a claim confirmed on the whole tested corpus and calibrating the scaling laws. A proof from first principles would organise the four-variable inclusion-exclusion by connected graph type. A first decisive theorem would be the case of a gcd-path. In tests on the corpus described in the stress-test section, all 137,440 connected tuples with entries at most 60 had  $b_2 \geq 1$ , reverified independently through entries at most 40. A counterexample would be a connected tuple with vanishing middle homology.

**Conjecture 125** (Common-scale monotonicity, resolved true). *For every  $N \geq 4$  and every base tuple  $\mathbf{a}$ , the scale sequence satisfies  $\beta_{\mathbf{a}}(k+1) \geq \beta_{\mathbf{a}}(k)$  for every  $k \geq 1$ .*

**Significance.** Scaling every exponent by a common factor multiplies the ambient weights without changing their ratios, and the conjecture asserts that the middle Betti number can only grow along this ray. The scale formula is an alternating polynomial in  $k$ , so monotonicity is not coefficientwise obvious and encodes a positivity of the finite differences of the Milnor–Orlik sum. The nearest related result is the formula itself [88]. A first decisive theorem would be monotonicity for tuples of equal entries, where the polynomial is explicit. In tests on the corpus described in the stress-test section, all 52,593 base tuples of lengths 4 to 6 were monotone at every scale through 10, reverified independently on random tuples through scale 11. A counterexample would be one base tuple with a descending scale step. *Resolution.* The conjecture is true: Proposition 18 identifies the scale sequence as a sum of interior Ehrhart counting functions of  $N - 1$  integral weighted hypersimplices, whose forward differences of every order through  $N - 1$  are nonnegative, strictly more than monotonicity.

**Proposition 18** (Ehrhart model for common scaling). *For a base tuple  $\mathbf{a}$  and  $1 \leq m \leq N - 1$  let  $P_m(\mathbf{a}) = \{x \in \mathbf{R}^N : 0 \leq x_i \leq a_i, \sum_i x_i/a_i = m\}$ , a lattice polytope of dimension  $N - 1$ . Then*

$$\beta_{\mathbf{a}}(k) = \sum_{m=1}^{N-1} \#(\text{relint}(kP_m(\mathbf{a})) \cap \mathbb{Z}^N),$$

and consequently every forward difference of the scale sequence of order  $r \leq N - 1$  is nonnegative at every  $k \geq 1$ . In particular Conjecture 125 is true, and the discrete-convexity clause of Conjecture 126 is true.

*Proof.* The multiplication map  $\prod_{j \in J} \mu_{a_j} \rightarrow \mu_{\text{lcm}(a_j)}$  on roots of unity is surjective with kernel of order  $\prod_{j \in J} a_j / \text{lcm}_{j \in J} a_j$ , so that quantity counts the tuples over  $J$  with product 1; by inclusion–exclusion over the coordinates forced to equal 1, the Milnor–Orlik sum counts the tuples  $(\zeta_1, \dots, \zeta_N)$  with each  $\zeta_i$  a nontrivial  $a_i$ -th root of unity and  $\prod_i \zeta_i = 1$ , the empty

set contributing the term  $(-1)^N$ . For the scaled tuple  $k\mathbf{a}$  write  $\zeta_i = \exp(2\pi i r_i / (ka_i))$  with  $0 < r_i < ka_i$ ; the product is 1 exactly when  $\sum_i r_i / a_i = km$  for an integer  $m$ , and since each summand lies strictly between 0 and  $k$ , the integer satisfies  $1 \leq m \leq N - 1$ . These conditions say precisely that  $(r_1, \dots, r_N)$  is a lattice point of the relative interior of  $kP_m(\mathbf{a})$ , proving the displayed identity. Each  $P_m(\mathbf{a})$  is integral: the substitution  $y_i = x_i / a_i$  carries it to the hypersimplex  $\{0 \leq y_i \leq 1, \sum_i y_i = m\}$ , whose vertices are 0–1 vectors, so every vertex of  $P_m(\mathbf{a})$  has coordinates 0 or  $a_i$ . For a lattice polytope  $P$  of dimension  $d$  with  $h^*$ -vector  $(h_0^*, \dots, h_d^*)$ , Ehrhart reciprocity [175] turns the expansion  $L_P(k) = \sum_i h_i^* \binom{k+d-i}{d}$  into  $L_{P^\circ}(k) = \sum_i h_i^* \binom{k+i-1}{d}$ , and the  $h_i^*$  are nonnegative integers [176]; applying the forward difference  $r$  times gives  $\Delta^r L_{P^\circ}(k) = \sum_i h_i^* \binom{k+i-1}{d-r} \geq 0$  for  $0 \leq r \leq d$ . Summing over the  $N - 1$  polytopes  $P_m(\mathbf{a})$ , all of dimension  $N - 1$ , yields  $\Delta^r \beta_{\mathbf{a}}(k) \geq 0$  for every  $k \geq 1$  and  $r \leq N - 1$ ; the case  $r = 1$  is Conjecture 125, and the case  $r = 2$  is the convexity clause of Conjecture 126.  $\square$

**Conjecture 126** (Scale convexity and log-concavity). *For every  $N \geq 4$  and every base tuple, the scale sequence is simultaneously discretely convex and log-concave:*

$$\beta_{\mathbf{a}}(k+1) - 2\beta_{\mathbf{a}}(k) + \beta_{\mathbf{a}}(k-1) \geq 0, \quad \beta_{\mathbf{a}}(k)^2 \geq \beta_{\mathbf{a}}(k-1)\beta_{\mathbf{a}}(k+1).$$

**Significance.** The coexistence is unusually rigid: convexity says the absolute increments grow, log-concavity says their relative growth slows, and together they confine the scale sequence to a narrow polynomial-like corridor. A proof might come from total positivity of the Milnor–Orlik polynomial after finite differencing, or from a representation as a positive mixture, and either route would explain Conjecture 125 as a corollary. The nearest related result is [88]. A first decisive theorem would be either inequality for equal-entry tuples. In tests on the corpus described in the stress-test section, both inequalities held across the entire scale corpus of Conjecture 125. A counterexample would be a base tuple violating either inequality at some scale. *Resolution of the convexity clause.* The first inequality is true, the case  $r = 2$  of Proposition 18, whose positive-mixture representation is exactly the route anticipated above. Nonnegative  $h^*$ -coefficients do not imply the log-concavity clause, which remains open; a further exact stress of 24,950,000 scale inequalities on 25,000 random tuples with entries through 120 and scales through 501 found no violation of either clause.

## 32 Hochster strands of flag moment-angle complexes

**Conjecture 127** (Hochster interval support). *For every finite simple graph  $G$  and every  $r \geq 0$ , the support  $\{s : h_{r,s}(G) > 0\}$  is an interval of integers.*

**Significance.** Adding a vertex can kill the homology of an individual induced subcomplex, so no monotone mechanism forces gapless support; the conjecture asserts that aggregation over all vertex subsets nevertheless removes every gap in every homological degree. The sector grading is genuinely finer than the ordinary one: for  $G = K_{2,3}$  the ordinary moment-angle Betti sequence is  $(1, 0, 0, 4, 2, 0, 3, 2, 0, 0, 0)$ , not unimodal, while every strand support in the tested corpus is an interval. The nearest related results are Hochster’s formula and the bigraded theory of moment-angle complexes [89, 90], and, closest of all, Whieldon’s gaplessness conjecture for the nonlinear strands of edge ideals [91], of which the present statement is the  $\mathbf{F}_2$  Stanley–Reisner form: although Abedelfatah–Nevo [92] disproved the field version for strands beyond the third,

the universal-coefficient bound  $\dim_{\mathbf{F}_2} \tilde{H}_r \geq \dim_{\mathbf{Q}} \tilde{H}_r$  shows that 2-torsion can only fill a support gap and never open one, so their construction leaves the  $\mathbf{F}_2$  statement standing. A first decisive theorem would be interval support for the component strand  $r = 0$ . In tests on the corpus described in the stress-test section, all 50,454 strands over 34,867 graphs had interval support, reverified independently on an exhaustive labeled corpus through five vertices and random holdouts at six and seven. A counterexample would be one graph, one degree, and one internal zero.

**Conjecture 128** (Hochster-strand log-concavity). *For every  $G$ ,  $r$ , and  $s$ ,*

$$h_{r,s}(G)^2 \geq h_{r,s-1}(G) h_{r,s+1}(G).$$

**Significance.** The strand sequences are conjectured log-concave in the subset size, a shape law for aggregated homology ranks across all induced subcomplexes. The strengthening that would place it in the Lorentzian-polynomial orbit is false and retained: ultra-log-concavity, normalized by binomial coefficients, already fails for a connected seven-vertex graph whose component strand is  $(10, 19, 14, 5, 1)$  at sizes two through six. Ordinary log-concavity is thus the calibrated boundary. The nearest related results are [89, 90]; the  $\mathbf{F}_2$  coefficient caveat recorded for Conjecture 127 applies here as well, a support gap being what would break the inequality. A first decisive theorem would be log-concavity of the component strand for trees. In tests on the corpus described in the stress-test section, no failure appeared in 50,454 strand sequences, reverified independently through five vertices exhaustively. A counterexample would be one graph and one degree with a convex triple.

**Conjecture 129** (Strand Hurwitz stability). *For every  $G$  and  $r$  with  $H_{G,r}$  nonconstant, every zero of  $H_{G,r}(z)/z^{\min \text{supp}}$ , the strand polynomial divided by its lowest monomial, satisfies  $\operatorname{Re} z < 0$ .*

**Significance.** Real-rootedness, the strongest classical shape property, is already false at four vertices: the empty graph has component polynomial  $6 + 8z + 3z^2$  after removing the monomial factor, with nonreal zeros. Those zeros nevertheless lie in the open left half-plane, and the conjecture asserts that this Hurwitz stability is the correct universal zero law for all Hochster strands, strictly between log-concavity and real-rootedness. The nearest related results are the bigraded computations of [90]; as for Conjecture 127, the  $\mathbf{F}_2$  coefficient field is what keeps the field counterexample of [92] from applying. A first decisive theorem would be stability for the empty graphs, whose strand polynomials have closed form. In tests on the corpus described in the stress-test section, all 50,045 nonconstant strand polynomials passed exact Routh–Hurwitz tests with integer determinants, reverified independently with an exact fraction-free implementation. A counterexample would be one strand polynomial with a zero in the closed right half-plane.

**Conjecture 130** (Cycle extremality at connectivity two). *If  $G$  is 2-vertex-connected on  $m \geq 4$  vertices, then  $h_{0,s}(G) \leq h_{0,s}(C_m)$  for every  $s$ , with equality in every coefficient only for the cycle  $C_m$ .*

**Significance.** The component strand measures aggregate disconnection of induced subgraphs, and the cycle, the sparsest 2-connected topology, is conjectured to disconnect the most, coefficient by coefficient, among all 2-connected graphs of its order. Every added chord can only help induced subgraphs stay connected, which is the mechanism, but the coefficientwise uniformity across all  $s$  is the content. The nearest related results are [89, 90]. A first decisive theorem would



be the top coefficient, the number of disconnecting vertex pairs. In tests on the corpus described in the stress-test section, all 12,185 two-connected graphs of the corpus, exhaustive through six vertices with seeded holdouts through ten, obeyed the domination, reverified independently on all 11,616 two-connected labeled graphs through six vertices. A counterexample would be a 2-connected graph beating the cycle in one coefficient.

**Conjecture 131** (Strict component-strand log-concavity). *For every connected noncomplete graph, every triple of positive consecutive component-strand coefficients is strictly log-concave:  $h_{0,s}^2 > h_{0,s-1}h_{0,s+1}$ .*

**Significance.** On the component strand of connected noncomplete graphs the log-concavity of Conjecture 128 is conjectured to be everywhere strict, so equality cases concentrate entirely in higher strands or degenerate graphs; complete graphs are excluded because their reduced component strand vanishes identically. Strictness pins the shape law against the boundary and makes every near-equality a finite certificate worth retaining. The nearest related results are [89, 90]. A first decisive theorem would be strictness for paths. In tests on the corpus described in the stress-test section, no equality occurred among positive triples across the corpus, reverified independently through five vertices exhaustively. A counterexample would be one connected noncomplete graph with an exactly geometric triple.

### 33 Torus mapping tori with signed monodromy

**Conjecture 132** (Parity-strand unimodality). *For every orientation-preserving signed permutation  $A$ , the sequences  $(f_0(A), f_2(A), f_4(A), \dots)$  and  $(f_1(A), f_3(A), f_5(A), \dots)$  are unimodal.*

**Significance.** By Proposition 15 the invariants  $f_k$  count positive orbits of the signed action on  $k$ -subsets, and the conjecture asserts that each parity class of this orbit-count profile rises and falls once. Log-concavity is too strong and the failure is retained: at  $d = 10$  the type with one positive 8-cycle and one positive 2-cycle has even strand  $(1, 5, 26, 26, 5, 1)$ , where  $5^2 < 1 \cdot 26$ . Unimodality is thus the calibrated boundary. The nearest related result is the Wang calibration [96] through Proposition 15. A first decisive theorem would be unimodality for a single signed cycle. In tests on the corpus described in the stress-test section, all 57,758 orientation-preserving signed cycle types through dimension 21 passed, reverified independently through dimension 11. A counterexample would be a type with a two-peaked parity strand.

**Conjecture 133** (Mapping-torus Betti unimodality). *For every orientation-preserving signed permutation  $A$  in dimension  $d$ , the Betti sequence  $(b_0(M_A), \dots, b_{d+1}(M_A))$  is unimodal.*

**Significance.** Poincaré duality makes the Betti sequence symmetric but not unimodal, so the claim is a genuine constraint on the exterior invariant profile, obtained from Conjecture 132's strands through the Wang sum  $b_k = f_k + f_{k-1}$ . Log-concavity fails already at  $A = -I_4$ , whose Betti sequence is  $(1, 1, 6, 6, 1, 1)$ , and the failure is retained, so unimodality is again the calibrated boundary. The nearest related result is [96] with Proposition 15. A first decisive theorem would be the reduction of Conjecture 133 to Conjecture 132 by a two-strand interleaving argument. In tests on the corpus described in the stress-test section, every type through dimension 21 passed, reverified independently through dimension 11. A counterexample would be a type with a two-peaked Betti sequence.

**Conjecture 134** (Sparse-cycle minimizer structure). *For every  $d \geq 8$ , some minimizer of the total Betti number  $\sum_k b_k(M_A; \mathbf{Q})$  over orientation-preserving signed permutations has no positive signed cycle when  $d$  is even, exactly one when  $d$  is odd, and pairwise distinct negative-cycle lengths.*

**Significance.** Total cohomology is minimized by destroying invariant vectors, and the conjecture asserts the structure of an optimal destroyer: negative cycles of pairwise distinct lengths, with at most the parity-forced positive cycle. Repeated cycle lengths create coincident eigenvalue charges and hence extra zero-sum relations, which is the mechanism. The observed minimizing negative partitions include (5, 7) at  $d = 12$ , (5, 6, 7) at  $d = 18$ , and (5, 7, 8) at  $d = 20$ . The nearest related result is Proposition 15, which turns the question into orbit counting. A first decisive theorem would be that repeating a negative-cycle length never decreases the total below the distinct-length optimum. In tests on the corpus described in the stress-test section, every exact minimum through dimension 21 had the stated form, reverified independently through dimension 12. A counterexample would be a dimension whose every minimizer violates the pattern.

**Conjecture 135** (Square-root-two compression exponent, resolved false). *Let  $M_d$  be the minimum of  $\sum_k b_k(M_A; \mathbf{Q})$  over orientation-preserving signed permutations in dimension  $d$ . Then  $M_d^{1/d} \rightarrow \sqrt{2}$  as  $d \rightarrow \infty$ .*

**Significance.** The conjecture fixes the exponential compression rate of the minimal total cohomology of this finite monodromy sector, and it is the boldest statement of the part: it asserts a limit, and the finite roots approach it non-monotonically. The tempting finite law is false and retained:  $M_d = 2^{\lfloor (d+3)/2 \rfloor}$  holds through  $d = 11$  and fails at  $d = 12$ , where the true minimum is 160, attained by negative cycles of lengths 7 and 5, not 128. The diagnostic  $(\log M_d - (d/2) \log 2) / \sqrt{d}$  stays below 0.42 for  $8 \leq d \leq 21$ , consistent with a subexponential correction. The nearest related result is Proposition 15. A first decisive theorem would be the lower bound  $M_d \geq 2^{d/2}$  up to subexponential factors. A counterexample would be a divergent or different limit.

**Proposition 19** (Landau floor for the minimal total cohomology). *For every signed permutation  $A$  on  $\mathbb{Z}^d$ ,*

$$\sum_k b_k(M_A; \mathbf{Q}) \geq \frac{2^{d+1}}{\text{ord}(A)} \geq \frac{2^d}{g(d)},$$

where  $g(d)$  is Landau's function, the maximal order of a permutation of  $d$  letters. Consequently  $2^d/g(d) \leq M_d \leq 2^{d+1}$ , and since  $\log g(d) \sim \sqrt{d \log d}$ , the minimal total cohomology satisfies  $M_d^{1/d} \rightarrow 2$ .

*Proof.* By Proposition 15,  $\sum_k b_k(M_A; \mathbf{Q}) = 2P$  with  $P = \sum_k p_k = \sum_k \dim_{\mathbf{Q}} \ker(\wedge^k A - I)$ , the dimension of the fixed space of  $A$  on the full exterior algebra  $\Lambda^\bullet(\mathbf{Q}^d)$ . Since  $A$  has finite order  $m$ , that fixed space is the invariant subspace of the cyclic group generated by  $A$ , and averaging characters gives  $P = \frac{1}{m} \sum_{j=0}^{m-1} \text{tr} \Lambda^\bullet(A^j) = \frac{1}{m} \sum_{j=0}^{m-1} \det(I + A^j)$ . Every term is nonnegative:  $A^j$  is again a signed permutation, and a signed cycle of length  $L$  with holonomy  $\varepsilon$  contributes  $\prod_{\zeta^L = \varepsilon} (1 + \zeta) = 1 - (-1)^L \varepsilon \in \{0, 2\}$  to the determinant. The term  $j = 0$  alone is  $2^d$ , so  $P \geq 2^d/m$ . If the underlying permutation has cycle lengths  $\ell_1, \dots, \ell_r$ , then  $A^{\text{lcm}(\ell_i)}$  is diagonal with entries  $\pm 1$ , so  $m \leq 2 \text{lcm}(\ell_1, \dots, \ell_r) \leq 2g(d)$ , giving the displayed floor. The identity monodromy gives

$M_d \leq 2 \cdot 2^d$ , and Landau’s asymptotic  $\log g(d) \sim \sqrt{d \log d}$  [172] makes  $(2^d/g(d))^{1/d} \rightarrow 2$ , so the two-sided bound forces  $M_d^{1/d} \rightarrow 2$ .  $\square$

**Remark 16** (Resolution). *The conjecture is false: by Proposition 19 the compression exponent is 2, not  $\sqrt{2}$ , with the whole deviation from  $2^{d+1}$  confined to a subexponential factor at most  $2g(d)$ . The finite scans could not separate the two laws, because  $\log_2 g(d) \approx d/2$  throughout the tested range— $\log_2 g(21) \approx 9.7$  against  $d/2 = 10.5$ —so the floor  $2^d/g(d)$  is numerically indistinguishable from  $2^{d/2}$  there, and the square-root fit was a finite-size illusion. The observed minima respect the floor with room to spare,  $M_{12} = 160$  against  $2^{12}/g(12) = 68$ , and the surviving question is the exact order of the subexponential correction  $\log_2(2^{d+1}/M_d)$ , now pinned between 0 and  $\log_2(2g(d)) = O(\sqrt{d \log d})$ . The torsion budget of Conjecture 136 is untouched: it is consistent with the true growth, and no incompatibility remains once the exponent is corrected.*

**Conjecture 136** (Torsion budget inequality). *For every signed permutation  $A \in SL(d, \mathbb{Z})$ ,*

$$\tau(A) \leq \sum_k b_k(M_A; \mathbf{Q}), \quad \text{equivalently} \quad \sum_k n_k \leq 2 \sum_k p_k$$

*in the orbit counts of Proposition 15.*

**Significance.** By Proposition 15 the integral torsion of these mapping tori is elementary 2-primary with one generator per negative orbit, so the conjecture is a purely combinatorial inequality: negative orbits of the signed action on the full exterior algebra never outnumber twice the positive orbits. It is not a universal-coefficient generality, and equality is attained repeatedly, at 109 of the 1,580 deposited signed types through dimension 12, so the inequality is sharp and a proof likely needs an explicit two-to-one map from negative to positive orbits. The nearest related result is Proposition 15. A first decisive theorem would be the inequality for a single signed cycle, where both sides are divisor sums. In tests on the corpus described in the stress-test section, every type through dimension 12 passed, reverified independently with the equality cases recounted. A counterexample would be a type with torsion exceeding the rational budget.

## Part VIII

# Shape laws for enumerative geometric invariants

*Added to the deposit: 12 August 2026.*

The twenty conjectures below continue the topological turn of Part VII with a second discipline: take a classical enumerative invariant of a geometric family, compute it exactly across the family, and conjecture the shape of the resulting array. They form five programmes. A hypersurface programme (Conjectures 137 to 140) studies the Euler indices of anticanonical hypersurfaces in products of projective spaces under an order dual to Part VII’s: where Part VII merged defining equations at fixed ambient, these statements split ambient factors at fixed dimension. A Hilbert-scheme programme (Conjectures 141 to 144) studies the Betti profiles of Göttsche’s product over a proved pair of exact moment identities. An orbifold programme (Conjectures 145 to 148) studies the twisted-sector age histograms of cyclic Calabi–Yau quotients.

An entropy programme (Conjectures 149 to 152) studies the exterior torsion-entropy profile of hyperbolic toral automorphisms. A matroid programme (Conjectures 153 to 156) lifts the Hochster strand laws of Part VII from graphs to simple binary matroids. The calibration layer is exact throughout: the Chern-class evaluation, Göttsche’s product, the age grading of the McKay correspondence, the eigenvalue evaluation of torsion growth rates, and Hochster’s formula. Retained controls again mark every calibrated boundary. The limit clauses of Conjectures 139 and 143 assert limits; Conjectures 146 to 148 are random-model statements resting on Monte Carlo evidence, with Conjectures 146 and 147 likely within reach of classical equidistribution machinery; Conjecture 145 lies close to classical methods. Since deposit, Conjectures 151 and 152 have been proved by convexity, while Conjectures 153 to 156 have been refuted by two explicit graphic matroids.

For a partition  $\lambda = (\lambda_1, \dots, \lambda_r)$  of  $d+1$  let  $A_\lambda = \prod_i \mathbf{CP}^{\lambda_i}$  and let  $X_\lambda \in |-K_{A_\lambda}|$  be a smooth anticanonical hypersurface, a Calabi–Yau  $d$ -fold; its Euler characteristic is evaluated exactly from  $c(TA_\lambda) = \prod_i (1 + H_i)^{\lambda_i+1}$  and  $[X_\lambda] = \sum_i (\lambda_i + 1)H_i$ , and we write  $E_d(\lambda) = (-1)^d \chi(X_\lambda)$ . A *split*  $S_{m,a}\lambda$  replaces one part  $m$  by  $a$  and  $m - a$ ; a *dominance move* transfers one unit from a smaller positive part to a weakly larger part. For  $b \geq 1$  define  $h_b(n, k)$  by

$$\sum_{n,k \geq 0} h_b(n, k) q^n z^k = \prod_{m \geq 1} \frac{1}{(1 - z^{m-1} q^m)(1 - z^m q^m)^b (1 - z^{m+1} q^m)},$$

so that  $h_{22}(n, k) = b_{2k}(\text{Hilb}^n(\mathbf{K}3))$  by Göttsche’s theorem [98]. For  $r \geq 3$  and weights  $a_1, \dots, a_d \in (\mathbb{Z}/r)^\times$  with  $\sum_i a_i \equiv 0 \pmod{r}$ , the cyclic quotient  $\frac{1}{r}(a_1, \dots, a_d)$  has ages  $\text{age}(j) = \frac{1}{r} \sum_i (ja_i \bmod r)$  for  $1 \leq j < r$ , integers between 1 and  $d-1$ , with histogram  $N_k = \#\{j : \text{age}(j) = k\}$  grading twisted sectors in the McKay correspondence [101]. For a hyperbolic  $A \in SL(d, \mathbb{Z})$  with eigenvalue log-moduli  $x_1, \dots, x_d$  summing to zero, set

$$h_k(x) = \sum_{|S|=k} \left( \sum_{i \in S} x_i \right)_+, \quad 1 \leq k \leq d-1;$$

away from exterior resonances  $h_k$  is the exponential growth rate of  $|\det(\wedge^k A^n - I)|$ , the degree- $k$  torsion order of the mapping torus of  $A^n$ , since each factor  $|\lambda_S^n - 1|$  grows like  $|\lambda_S|^n$  or stays bounded [102, 105]. Finally, for a simple binary matroid  $M$  on ground set  $E$  with independence complex  $\Delta(M)$ , put  $\beta(A) = \dim \tilde{H}_{r(A)-1}(\Delta(M|_A); \mathbf{F}_2)$  and

$$H_{c,s}(M) = \sum_{\substack{A \subseteq E, |A|=s \\ |A|-r(A)=c}} \beta(A),$$

the aggregated multigraded Betti numbers of the Stanley–Reisner ring by Hochster’s formula [89], graded by restriction size and nullity, with strand polynomial  $\sum_s H_{c,s}(M) z^s$  at each fixed nullity  $c$ . The profile  $h_b(n, k)$  and the spectral functional  $h_k(x)$  are unrelated symbols, each local to its own programme.

**Proposition 20** (Profile moment identities). For every  $b \geq 1$  and  $n \geq 1$ , the probability measure on  $\{0, \dots, 2n\}$  proportional to  $h_b(n, \cdot)$  has mean exactly  $n$  and variance exactly  $2n/(b+2)$ .

*Proof.* Write  $F(q, z)$  for the generating product and substitute  $(q, z) \mapsto (qe^{-t}, e^t)$ : the factor arguments  $z^{m-1}q^m, z^m q^m, z^{m+1}q^m$  become  $q^m e^{-t}, q^m, q^m e^t$ , so

$$\Phi(t) = \log F(qe^{-t}, e^t) = - \sum_{m \geq 1} [\log(1 - q^m e^{-t}) + b \log(1 - q^m) + \log(1 - q^m e^t)],$$

and  $[q^n] F(qe^{-t}, e^t) = \sum_k h_b(n, k) e^{t(k-n)}$  is the moment generating function of the centred profile. The two  $t$ -dependent families are exchanged by  $t \mapsto -t$ , so  $\Phi'(0) = 0$  and the mean is  $n$ . Differentiating twice,  $\Phi''(0) = 2 \sum_m q^m / (1 - q^m)^2$ , so with  $P_n = \sum_k h_b(n, k)$  and  $G_n = \sum_k (k - n)^2 h_b(n, k)$ ,

$$\sum_n G_n q^n = 2F(q, 1) \sum_m \frac{q^m}{(1 - q^m)^2}.$$

Both Lambert series  $\sum_m q^m / (1 - q^m)^2$  and  $\sum_m m q^m / (1 - q^m)$  expand to  $\sum_N \sigma(N) q^N$ , hence are equal, while  $F(q, 1) = \prod_m (1 - q^m)^{-(b+2)}$  gives  $q \partial_q \log F(q, 1) = (b+2) \sum_m m q^m / (1 - q^m)$ . Therefore  $\sum_n G_n q^n = \frac{2}{b+2} q \partial_q F(q, 1) = \frac{2}{b+2} \sum_n n P_n q^n$ , that is,  $G_n / P_n = 2n / (b+2)$  for every  $n$ .  $\square$

### 34 Euler indices of anticanonical hypersurfaces in products

**Conjecture 137** (Strict dominance law). *For every  $d \geq 3$  and every dominance move  $\lambda \rightarrow \mu$  on partitions of  $d+1$ ,*

$$E_d(\mu) > E_d(\lambda).$$

**Significance.** Concentrating ambient dimension into fewer projective factors strictly increases the signed Euler index, so  $E_d$  is strictly increasing along the dominance order on partitions, from the split ambient  $(\mathbf{CP}^1)^{d+1}$  to the single factor  $\mathbf{CP}^{d+1}$ . The Chern coefficient is an alternating expression, and the conjectured order asks for a hidden positive basis for the dominance differences. The nearest related results are the CICY constructions and censuses [95, 104], which enumerate these geometries without ordering them. A first decisive theorem would be the two-factor case. In tests on the corpus described in the stress-test section, all 42,691 elementary moves for  $3 \leq d \leq 22$  passed, reverified independently through  $d = 18$ . A counterexample would be one reversed or equal move.

**Conjecture 138** (Strict refinement contraction). *For every  $d \geq 3$ , every  $\lambda$ , and every nontrivial split of one part,*

$$E_d(S_{m,a}\lambda) < E_d(\lambda).$$

**Significance.** Factoring one ambient projective direction always strictly reduces the signed Euler index, the exact mirror of Part VII's merge monotonicity read on the ambient rather than the defining equations. The sign calibration is essential: without the factor  $(-1)^d$  the statement is false in odd dimension, and the two statements together make  $E_d$  a strict order antitone in refinement. The nearest related results are [95, 104]. A first decisive theorem would be splits of the one-part partition, refining a single hypersurface ambient. In tests on the corpus described in the stress-test section, all 42,861 splits through  $d = 22$  passed, reverified independently through  $d = 18$ . A counterexample would be one split that fails to contract.

**Conjecture 139** (The 8/9 refinement barrier). *Let  $S_d$  be the maximum of  $E_d(S_{m,a}\lambda) / E_d(\lambda)$  over all  $\lambda$  and all one-part splits. Then  $S_d \rightarrow 8/9$  as  $d \rightarrow \infty$ .*

**Significance.** The mechanism should be a closed-form ratio for the extremal split, which the data identify as splitting a single unit off the dominant part. The nearest related results are [95, 104]. A first decisive theorem would be the exact evaluation of the split ratio along the

one-part family. In tests on the corpus described in the stress-test section, the exact maxima decrease toward the limit from dimension four on,  $S_{22} = 0.8894748$  against  $8/9 = 0.8888889$ , the independent recomputation giving  $S_{18} = 0.8897556$ . A counterexample would be a different limit or two separated accumulation points.

**Conjecture 140** (Positive interaction of disjoint refinements). *For  $d \geq 6$  and splits  $S, T$  acting on distinct parts of  $\lambda$ ,*

$$E_d(\lambda) E_d(ST\lambda) > E_d(S\lambda) E_d(T\lambda).$$

**Significance.** The logarithm of the signed Euler index is strictly supermodular on disjoint refinement squares: two independent factorizations lose less together than the product of their separate losses. The threshold is calibrated and retained: the unqualified statement is false in dimension five, where the exact scan finds four violating squares, so the hypothesis  $d \geq 6$  is a genuine boundary rather than caution. The nearest related results are [95, 104]. A first decisive theorem would be the two-part ambient case. In tests on the corpus described in the stress-test section, all 24,680 split squares for  $6 \leq d \leq 18$  passed, reverified independently for  $6 \leq d \leq 12$ . A counterexample would be one square with submodular interaction.

## 35 Betti profiles of Hilbert schemes of surfaces

**Conjecture 141** (Strict Betti log-concavity). *For every  $b \geq 3$ ,  $n \geq 1$ , and  $0 < k < 2n$ ,*

$$h_b(n, k)^2 > h_b(n, k-1) h_b(n, k+1).$$

**Significance.** The Betti profile of Göttsche’s product is conjectured strictly log-concave in the cohomological degree, a multiplicative curvature condition strictly above the unimodality that the bivariate strict-unimodality machinery of [99] supplies for products of this type; at  $b = 22$  the statement covers every  $\text{Hilb}^n(\mathbb{K}^3)$ . Ultra-log-concavity is deliberately not asserted, matching the calibration pattern of the strand programmes. The nearest related results are Göttsche’s formula [98] and the bivariate unimodality criterion of [99], stated there for Laurent-series products rather than for these profiles. A first decisive theorem would be the middle three degrees for large  $n$  from the asymptotic profile. In tests on the corpus described in the stress-test section, 179,200 exact Turán inequalities passed for  $3 \leq b \leq 30$  and  $n \leq 80$ , reverified independently for  $b \leq 12$ ,  $n \leq 40$ . A counterexample would be one nonpositive Turán determinant.

**Conjecture 142** (Cohomology-seed total positivity). *For every  $b \geq 1$ ,  $n \geq 1$ , and  $0 \leq k < n$ ,*

$$h_{b+1}(n, k+1) h_b(n, k) \geq h_{b+1}(n, k) h_b(n, k+1).$$

**Significance.** Adding one seed cohomology generator concentrates the normalized profile toward the Lefschetz centre in monotone likelihood-ratio order, a total-positivity statement across the seed parameter transverse to the shape law of Conjecture 141. The mechanism suggested by the data is a planar network or a stable multivariate refinement of the product. The nearest related result is [98]. A first decisive theorem would be the boundary minors  $k = 0$ . In tests on the corpus described in the stress-test section, all 53,070 adjacent two-by-two minors for  $b \leq 30$ ,  $n \leq 60$  passed, reverified independently for  $b \leq 12$ . A counterexample would be one negative minor.



**Conjecture 143** (Kurtosis trichotomy). *Let  $\kappa_b(n)$  be the standardized fourth moment of the profile of Proposition 20. Then: for  $b = 3, 4, 5$ ,  $\kappa_b(n)$  increases strictly to  $21/5$ ; for  $b = 6$  it has a unique maximum at  $n = 9$  and decreases thereafter to  $21/5$ ; for  $b = 7$ ,  $\kappa_7(1) = \kappa_7(2)$  and it decreases strictly thereafter; for  $b \geq 8$  it decreases strictly to  $21/5$ .*

**Significance.** Over the exact mean and variance of Proposition 20, the profile's kurtosis follows a complete finite- $n$  phase diagram with a single exceptional peak at  $(b, n) = (6, 9)$  and a logistic-type limit  $21/5$ , the limit clauses beyond finite check. The initially stronger assertion of strict decrease for  $b = 7$  was refuted by the exact equality  $\kappa_7(1) = \kappa_7(2) = 9/2$  and the corrected boundary is retained. The nearest related result is the asymptotic  $\chi_y$ -profile analysis [100], whose neighbourhood contains the limit but not the finite phase diagram. A first decisive theorem would be the closed form of  $\kappa_b(1)$ . In tests on the corpus described in the stress-test section, every stated comparison passed for  $3 \leq b \leq 22$  and  $n \leq 80$ , the full diagram reverified independently through  $n = 50$ . A counterexample would be a violated comparison or a different limit.

**Conjecture 144** (Fixed-charge log-concavity). *For every  $b \geq 3$  and fixed  $j \geq 0$ , the sequence  $n \mapsto h_b(n, n + j)$ ,  $n \geq \max(1, j)$ , is strictly log-concave.*

**Significance.** Holding the displacement from middle cohomology fixed and varying the number of points gives the transverse direction to Conjecture 141, and the conjecture asserts strict log-concavity there as well, so the whole two-parameter array is log-concave along both natural axes. The nearest related result is [98]. A first decisive theorem would be the central diagonal  $j = 0$  at  $b = 22$ , the middle Betti numbers of  $\text{Hilb}^n(\text{K3})$ . In tests on the corpus described in the stress-test section, 48,384 exact diagonal Turán inequalities passed for  $b \leq 30$ ,  $j \leq 25$ ,  $n \leq 80$ , reverified independently for  $b \leq 12$ ,  $j \leq 14$ ,  $n \leq 40$ . A counterexample would be one convex diagonal triple.

## 36 Twisted-sector ages of cyclic quotients

**Conjecture 145** (Sharp low-dimensional age unimodality). *For  $d = 4$  and  $d = 5$ , the age histogram  $(N_1, \dots, N_{d-1})$  is unimodal for every isolated cyclic Calabi–Yau quotient  $\frac{1}{r}(a_1, \dots, a_d)$ .*

**Significance.** The age duality  $\text{age}(j) + \text{age}(-j) = d$  makes the histogram symmetric in mass but not in shape, and the conjecture asserts single-peakedness in exactly the dimensions where it can hold: in dimension six the quotient  $\frac{1}{3}(1, 1, 1, 1, 1, 1)$  has histogram  $(0, 1, 0, 1, 0)$ , so the range is sharp. It is close to prior work, since symmetry leaves few independent entries in these dimensions and the statement may be a short unrecognized lemma of the quotient-singularity literature [101]. A first decisive theorem would be the single inequality  $N_2 \geq N_1$  at  $d = 4$ . In tests on the corpus described in the stress-test section, all 450,999 admissible weight multisets with  $r \leq 60$  passed, reverified independently for  $r \leq 40$  on 58,393 multisets. A counterexample would be one two-peaked histogram.

**Conjecture 146** (Gaussian bulk age law). *Let  $r$  be prime and let distinct nonzero weights  $a_1, \dots, a_d$  be uniform conditioned on zero sum. If  $d, r \rightarrow \infty$  with  $d = o(r)$ , then in probability the empirical law of  $(\text{age}(j) - d/2)/\sqrt{d/12}$  over  $1 \leq j < r$  converges weakly to the standard Gaussian.*

**Significance.** Each age is a sum of  $d$  fractional rotations whose joint behaviour over the group orbit is quasi-independent, and the conjecture is the resulting finite-population central limit law for twisted-sector gradings. It carries two caveats: as a random-model statement resting on Monte Carlo evidence, and as likely within reach of classical equidistribution and discrepancy machinery, so its value is as a calibration target rather than a deep unknown. The nearest related result is the age grading itself [101]. A first decisive theorem would be convergence of the second moment. In tests on the corpus described in the stress-test section, the mean lattice-corrected Kolmogorov distance fell from 0.0398 at  $(d, r) = (10, 101)$  to 0.0162 at  $(40, 1009)$ , the independent recomputation giving 0.0344 and 0.0178. A counterexample would be a nonvanishing limiting distance.

**Conjecture 147** (Local central-sector law). *Under the sampling law of Conjecture 146 with  $d$  even,  $N_{d/2}/(r-1) \sim \sqrt{6/(\pi d)}$ , and uniformly for  $|k - d/2| = O(\sqrt{d \log d})$  the ratio  $N_k/(r-1)$  is asymptotic to the corresponding lattice Gaussian mass.*

**Significance.** The weak law of Conjecture 146 is upgraded to a local limit law: individual age sectors, not just their bulk distribution, carry Gaussian mass, which is what an application to sector counts actually needs. It shares both caveats of Conjecture 146, and a proof requires a local central limit theorem for the correlated residue orbit rather than larger simulations. The nearest related result is [101]. A first decisive theorem would be the central sector alone. In tests on the corpus described in the stress-test section, the ratio of observed central mass to the prediction was 1.034, 1.048, 1.025, 1.006 across the four sampled regimes. A counterexample would be a sector with non-Gaussian limiting mass.

**Conjecture 148** (High-sampling restoration of unimodality). *In the random prime-order model, if  $r/d^3 \rightarrow \infty$  then the entire age histogram is unimodal with probability tending to one.*

**Significance.** Deterministic unimodality fails from dimension six onward, but enough group elements suppress histogram noise and restore the Gaussian shape generically, and the conjecture pins the sampling threshold at the third power of the dimension. The exponent is deliberately exposed: replacing  $d^3$  by a smaller power is the cleanest refinement or refutation, and the statement carries the random-model caveat. The nearest related result is Conjecture 147, whose local law at the fluctuation scale predicts exactly this exponent. A first decisive theorem would be unimodality with probability tending to one at  $r \geq d^4$ . In tests on the corpus described in the stress-test section, no failure occurred in 180 trials each at  $(6, 1009)$ ,  $(8, 2003)$ ,  $(10, 4001)$ , while visibly lower ratios  $r/d^3$  produced failure rates reaching 14.6%. A counterexample would be a sequence with  $r/d^3 \rightarrow \infty$  and non-vanishing failure probability.

## 37 Exterior torsion entropy of toral automorphisms

**Conjecture 149** (Exterior torsion-entropy log-concavity). *For every nonzero zero-sum real spectrum  $x$  and  $1 < k < d-1$ ,*

$$h_k(x)^2 \geq h_{k-1}(x) h_{k+1}(x).$$

**Significance.** The torsion growth rates of a hyperbolic toral mapping torus, read across the exterior degree, form a log-concave profile: a Hodge-shaped inequality for a piecewise-linear

functional of subset sums rather than for an intersection form. The nearest related results are the torsion-growth programme of [102], whose setting is arithmetic groups rather than mapping tori, and the toral determinant calculus of [105], which computes the rates without ordering them. A first decisive theorem would be  $d = 4$ , where a finite chamber decomposition suffices. In tests on the corpus described in the stress-test section, all 81,000 inequalities on 16,200 random zero-sum spectra through  $d = 12$  passed, with an exact integer check alongside, reverified independently on 3,000 random spectra and the full integer corpus of Conjecture 150. A counterexample would be one convex triple of rates.

**Conjecture 150** (Equality rigidity). *If an interior equality  $h_k(x)^2 = h_{k-1}(x)h_{k+1}(x)$  occurs, then the multiset  $\{x_1, \dots, x_d\}$  has exactly two distinct values.*

**Significance.** Equality in the profile inequality is conjectured to force a two-level spectrum, the only configuration whose subset-sum chambers align enough to flatten a Turán determinant; the converse is deliberately not asserted. Rigidity turns Conjecture 149 into a strict law away from an explicit degenerate locus. The nearest related results are [102, 105]. A first decisive theorem would be the case of integer spectra with entries in  $\{-1, 0, 1\}$ . In tests on the corpus described in the stress-test section, all 222 equality cases among 48,512 exact integer spectra with entries in  $[-2, 2]$  were two-valued, the independent exhaustive recount through  $d = 7$  finding 122 equality cases, all two-valued. A counterexample would be one three-valued equality.

**Conjecture 151** (Sharp binomial upper envelope, resolved true). *For every zero-sum spectrum and every  $1 \leq k \leq d - 1$ ,*

$$h_k(x) \leq \binom{d-2}{k-1} h_1(x),$$

*with equality in every degree for spectra proportional to  $(1, 0, \dots, 0, -1)$ .*

**Significance.** Every exterior torsion rate is controlled by the ordinary topological entropy  $h_1$  with the sharp binomial constant. Proposition 21 proves the statement by normalising  $h_1 = 1$  and maximising a convex function on the product of the positive and negative sign simplices. A vertex has one coordinate 1, one coordinate  $-1$ , and all others zero, which gives exactly  $\binom{d-2}{k-1}$  positive subsets. The nearest related results remain [102, 105]. The former finite tests now serve as regression checks for the proved inequality.

**Proposition 21** (Resolution of the sharp binomial upper envelope). Conjecture 151 holds in full.

*Proof.* If  $h_1(x) = 0$ , then the zero-sum spectrum is zero and the result is immediate. Otherwise scale so that  $h_1(x) = 1$ . Fix the sets  $P$  and  $N$  of positive and negative coordinates. The positive coordinates lie in the simplex  $\sum_{i \in P} x_i = 1$ , while the magnitudes of the negative coordinates lie in the simplex  $\sum_{j \in N} (-x_j) = 1$ . On the closure of their product,

$$h_k(x) = \sum_{|S|=k} \left( \sum_{i \in S} x_i \right)_+$$

is a convex function, since it is a sum of positive parts of linear forms. A convex function on a compact polytope attains a maximum at a vertex. At such a vertex exactly one coordinate is 1,

exactly one is  $-1$ , and every other coordinate is zero. A  $k$ -subset contributes 1 precisely when it contains the positive coordinate and avoids the negative coordinate. There are  $\binom{d-2}{k-1}$  such subsets. Rescaling proves the inequality, and the vertex calculation proves the asserted equality for spectra proportional to  $(1, 0, \dots, 0, -1)$ .  $\square$

**Conjecture 152** (Two-level lower envelope, resolved true). *Normalize  $h_1(x) = 1$ . For every  $d$  and  $k$ , the minimum of  $h_k$  over zero-sum spectra is attained by a two-level spectrum  $x^{p,q}$  with  $p$  entries  $1/p$ ,  $q$  entries  $-1/q$ , and  $d - p - q$  zeros.*

**Significance.** Together with Conjecture 151, this result places every exterior torsion-entropy profile in a sharp two-sided corridor governed by elementary two-level spectra. Proposition 22 proves the reduction by averaging coordinates within each sign class. Convexity and symmetry show that every such averaging step can only decrease  $h_k$ , so equal positive coordinates and equal negative coordinates attain the minimum. The nearest related results remain [102, 105]. The former random and exact comparisons now serve as regression checks for the finite envelope.

**Proposition 22** (Resolution of the two-level lower envelope). Conjecture 152 holds in full.

*Proof.* Let  $x$  be a zero-sum spectrum with  $h_1(x) = 1$ , and let  $P$  and  $N$  be its positive and negative index sets with sizes  $p$  and  $q$ . The function  $h_k$  is convex and invariant under permutations within  $P$  and within  $N$ . Replace two positive coordinates by their average. The new spectrum is the midpoint of the original spectrum and the spectrum obtained by swapping those coordinates, so convexity and symmetry show that  $h_k$  cannot increase. Repeating this operation makes every positive coordinate equal to  $1/p$ . The same argument applied to the negative magnitudes makes every negative coordinate equal to  $-1/q$ . The zero coordinates remain zero. Thus

$$h_k(x^{p,q}) \leq h_k(x).$$

There are only finitely many pairs  $p, q \geq 1$  with  $p + q \leq d$ , and every  $x^{p,q}$  is admissible. The global minimum is therefore attained by one of these two-level spectra.  $\square$

## 38 Matroidal Hochster tables

**Conjecture 153** (Nullity-strand interval support, resolved false). *For every simple binary matroid  $M$  and every nullity  $c$ , the support  $\{s : H_{c,s}(M) > 0\}$  is an interval of integers.*

**Proposition 23** (A graphic nullity-strand certificate). Let  $M = M(K_{3,3})$  be the cycle matroid of the complete bipartite graph. Then

$$H_{1,4}(M) = 9, \quad H_{1,5}(M) = 0, \quad H_{1,6}(M) = 6.$$

The matroid  $M$  is simple, binary, and 3-connected.

*Proof.* In a nullity-one graphic restriction, the top reduced homology is nonzero exactly when the restricted edge set is a circuit. Indeed, any edge outside the unique circuit is a coloop, which makes the independence complex a cone. The circuits of  $M(K_{3,3})$  are the cycles of  $K_{3,3}$ . There are  $\binom{3}{2}^2 = 9$  cycles of length four, no cycles of length five by bipartiteness, and six Hamilton cycles of length six. This gives the displayed strand. The graph  $K_{3,3}$  is simple and 3-vertex-connected, so its cycle matroid is simple, binary, and 3-connected.  $\square$

**Remark 17** (Resolution). *Conjecture 153 is false. Proposition 23 gives nullity-one support  $\{4, 6\}$ , with an internal gap at size five.*

**Significance.** The statement is retained with its resolution because the obstruction is elementary and survives strong connectivity. The cycle matroid of  $K_{3,3}$  is simple, binary, and 3-connected, yet its nullity-one strand is supported at sizes four and six with a zero between them. Simplicity and connectivity therefore do not repair interval support. The nearest related results remain Hochster’s formula [89] and the homology of matroid independence complexes [106]. A surviving problem is to identify a structural class, narrower than simple binary matroids, on which interval support does hold.

**Conjecture 154** (Row-and-column log-concavity, resolved false). *Both coordinate directions of the table are log-concave:  $H_{c,s}^2 \geq H_{c,s-1}H_{c,s+1}$  and  $H_{c,s}^2 \geq H_{c-1,s}H_{c+1,s}$ .*

**Remark 18** (Resolution). *Conjecture 154 is false. For the matroid of Proposition 23, the horizontal inequality at  $(c, s) = (1, 5)$  reads  $0^2 \geq 9 \cdot 6$ .*

**Significance.** The horizontal direction already fails on the nullity-one circuit census of  $K_{3,3}$ , where the gap at size five forces the strongest possible reversal. The vertical direction is not refuted by this example and remains a separate question. The nearest related results are the Hodge-theoretic log-concavity of matroid invariants [103] and the shellability framework [106], neither of which controls this bigraded array. Any repaired row statement must exclude basic circuit gaps rather than rely on simplicity or 3-connectivity.

**Conjecture 155** (Nearest-rhombus log-concavity, resolved false). *Both diagonal directions are log-concave:  $H_{c,s}^2 \geq H_{c-1,s-1}H_{c+1,s+1}$  and  $H_{c,s}^2 \geq H_{c-1,s+1}H_{c+1,s-1}$ .*

**Proposition 24** (A graphic rhombus certificate). Let  $G$  have vertex set  $\{0, \dots, 6\}$  and edge set

$$\{03, 04, 05, 13, 14, 23, 24, 26, 34, 56\},$$

where  $ij$  denotes the edge  $\{i, j\}$ . For its cycle matroid  $M(G)$ ,

$$H_{1,5} = 2, \quad H_{2,6} = 2, \quad H_{3,7} = 3.$$

The graph  $G$  is simple and 2-connected.

*Proof.* For a restriction  $A$ , its contribution is the absolute reduced Euler characteristic of its independence complex,

$$\beta(A) = \left| -1 + \sum_{\substack{\emptyset \neq I \subseteq A \\ I \text{ independent}}} (-1)^{|I|-1} \right|.$$

Deletion and contraction on the ten displayed edges gives the following complete list of nonzero contributions to the three cells. At  $(1, 5)$  the two five-cycles  $\{03, 05, 23, 26, 56\}$  and  $\{04, 05, 24, 26, 56\}$  contribute one each. At  $(2, 6)$  the restriction  $\{03, 04, 13, 14, 23, 24\}$  contributes two. At  $(3, 7)$  that restriction together with 34 contributes three. Every other restriction of the required size and nullity has zero top reduced homology. Summing gives the displayed values. Deleting any one vertex leaves  $G$  connected, so  $G$  is 2-connected.  $\square$

**Remark 19** (Resolution). *Conjecture 155 is false. Proposition 24 gives*

$$H_{2,6}^2 = 4 < 6 = H_{1,5}H_{3,7}.$$

**Significance.** The first nearest-diagonal inequality fails on a small simple graphic matroid, so the table is not the unrestricted shadow of a Lorentzian or strongly log-concave bivariate polynomial. The certificate is 2-connected and uses only ten edges. The opposite diagonal direction remains a separate question. The nearest related circle is still the Hodge-theoretic theory around [103], but any connection must pass through a more restrictive class or a modified array.

**Conjecture 156** (Hurwitz stability of nullity strands, resolved false). *For every simple binary matroid and every nonconstant nullity strand, every zero of the strand polynomial  $\sum_s H_{c,s}(M) z^{s-s_{\min}}$ , with  $s_{\min}$  the least size in its support, has strictly negative real part.*

**Remark 20** (Resolution). *Conjecture 156 is false. The nullity-one polynomial of Proposition 23, after removing its lowest monomial, is  $9 + 6z^2$ . Its zeros are  $\pm i\sqrt{3}/2$  and have real part zero.*

**Significance.** Strict Hurwitz stability fails at the boundary of the left half-plane. The  $K_{3,3}$  strand has a missing middle coefficient, and its two zeros lie on the imaginary axis. This links the failure directly to the support gap that refutes Conjectures 153 and 154. The nearest related results remain [89, 90] and the parallel graph-strand law of Part VII. Closed-left-half-plane stability is a possible weakening, but it is not established here.

## Part IX

# Spectra and filtrations of combinatorial geometries

*Added to the deposit: 12 August 2026.*

The twenty conjectures below extend the geometric turn of Parts VII and VIII to spectra and filtrations. They form four programmes. A toric programme (Conjectures 157 to 161) studies the  $h$ -polynomials of graph associahedra, the even-Betti spectra of their smooth toric varieties, through root location, interlacing, and a Mahler-measure order. A McKay programme (Conjectures 162 to 166) filters the Cayley graph of an isolated cyclic Calabi–Yau quotient by orbifold age and conjectures one-crest laws for the low homology of the resulting flag complexes. A torsion programme (Conjectures 167 to 171) studies microcanonical ensembles of lens-space sine factors, their temperature-dependent sector shapes, and a  $\zeta(3)$  central parabola over the log-sine moment identities proved below. An entropy programme (Conjectures 172 to 176) orders the boundary-map pseudodeterminants of simplicial spheres across degree and under stellar refinement. The calibration layer is exact where the objects are: tubing  $f$ -vectors,  $\mathbf{F}_2$  homology, and coefficient convolutions are integer computations, while root locations and pseudodeterminants are checked spectrally in floating point and. Four retained controls mark calibrated boundaries. Conjectures 169 and 170 assert limits in a random or iterated-limit model and are the boldest; Conjectures 157, 161, 164, 169, and 170 sit closest to classical machinery; and Conjecture 170 deliberately strengthens the annealed Gaussian age law drawn from Conjecture 146 to a joint decoupling law.

For a connected simple graph  $G$  on  $n \geq 2$  vertices, a *tube* is a proper connected induced subgraph, two tubes being compatible when nested or disjoint without adjacency, and a *tubing* is



a set of pairwise compatible tubes; the graph associahedron  $P_G$  is the simple  $(n-1)$ -polytope whose codimension- $k$  faces are the tubings with  $k$  tubes [107]. Write  $h_G(t) = \sum_i h_i(G)t^i$  for its  $h$ -polynomial, so that  $h_i(G) = b_{2i}(X_G)$  for the smooth projective toric variety of the normal fan, a positive palindromic sequence; put  $\mathcal{M}(G) = \sum_{h_G(\rho)=0} \log \max(1, |\rho|)$ , the logarithmic Mahler measure, and  $\mu_G(i) = h_i(G)/h_G(1)$ . For an odd prime  $p$  and pairwise distinct weights  $a = (a_1, \dots, a_d) \in (\mathbb{F}_p^\times)^d$  with  $\sum_i a_i = 0$ , the quotient  $\frac{1}{p}(a_1, \dots, a_d)$  acts on the vertex set  $\mathbb{Z}/p$ , including 0, through the Cayley graph  $x \sim y$  when  $y - x \in \{\pm a_1, \dots, \pm a_d\}$ ; with  $\text{age}(j) = \frac{1}{p} \sum_i [ja_i]_p$  and  $\text{age}(0) = 0$ , let  $K_k$  be the flag complex of the induced graph on  $\{j : \text{age}(j) \leq k\}$  for  $0 \leq k \leq d-1$ , and write  $c_k$  and  $b_{q,k}$  for its number of components and its  $q$ -th  $\mathbf{F}_2$  Betti number. For a prime  $p \geq 5$  let  $\Omega_{p,d,m}$  be the tuples  $x \in \{1, \dots, p-1\}^d$  with  $\sum_i x_i = mp$ , let  $H_p(x) = \sum_i \log(2 \sin(\pi x_i/p))$ , the lens-space log-torsion energy [108], and set  $Z_{p,d,m}(s) = \mathbb{E}[e^{sH_p}]$  and  $\Phi_{p,d,m} = \mathbb{E}[H_p]$  over uniform  $\Omega_{p,d,m}$ . Finally, for a finite simplicial  $d$ -sphere  $K$  with boundary maps  $\partial_k$ , put  $D_k(K) = \text{pdet}(\partial_k \partial_k^*)$ , the product of nonzero eigenvalues,  $E_k(K) = \log D_k(K)$ ,  $r_k(K) = \text{rank } \partial_k$ , and  $A_k(K) = E_k(K)/r_k(K)$ , the entropy per boundary rank, with higher matrix-tree readings of these determinants in [109, 110].

The sector expectations  $Z_{p,d,m}$  and  $\Phi_{p,d,m}$  are defined only when  $\Omega_{p,d,m}$  is nonempty. This holds exactly when  $d \leq mp \leq d(p-1)$ .

**Proposition 25** (Log-sine moment identities). For every integer  $p \geq 2$ ,

$$\prod_{j=1}^{p-1} 2 \sin \frac{\pi j}{p} = p, \quad \text{and} \quad \int_0^1 \left(u - \frac{1}{2}\right)^2 \log(2 \sin \pi u) du = -\frac{\zeta(3)}{2\pi^2}.$$

*Proof.* Evaluating  $\prod_{j=1}^{p-1} (z - \zeta^j) = (z^p - 1)/(z - 1)$  at  $z = 1$  for  $\zeta = e^{2\pi i/p}$  gives  $\prod_j (1 - \zeta^j) = p$ ; taking absolute values and  $|1 - \zeta^j| = 2 \sin(\pi j/p)$  gives the product identity, so the sector energies of the definitions have exact mean  $\log p/(p-1)$  per coordinate. For the integral, expand  $\log(2 \sin \pi u) = -\sum_{k \geq 1} \cos(2\pi k u)/k$  on  $(0, 1)$  and  $(u - \frac{1}{2})^2 = \frac{1}{12} + B_2(u)$  with  $B_2(u) = \pi^{-2} \sum_{k \geq 1} \cos(2\pi k u)/k^2$ . The constant term contributes nothing, since  $\int_0^1 \log(2 \sin \pi u) du = 0$  by the vanishing mean of every cosine, and orthogonality of the cosines gives  $\int_0^1 B_2(u) \cos(2\pi k u) du = 1/(2\pi^2 k^2)$ , so the integral equals  $-\sum_{k \geq 1} k^{-1} \cdot (2\pi^2 k^2)^{-1} = -\zeta(3)/(2\pi^2)$ .  $\square$

### 39 Toric spectra of graph associahedra

**Conjecture 157** (Universal negative real roots). For every connected simple graph  $G$ , every zero of  $h_G(t)$  is real and strictly negative.

**Significance.** The two endpoint families are classical: the path gives the associahedron with Narayana  $h$ -polynomial and the complete graph the permutohedron with Eulerian  $h$ -polynomial, both real-rooted, and the conjecture asserts that every graph associahedron between them shares the property. It is close to prior work as a natural question in an active neighbourhood, and it is sharply family-specific: the analogous real-rootedness for all flag simple polytopes fails in high dimension, so a proof must use the graphical structure, most plausibly a stable refinement of the descent enumerator of [107]. A first decisive theorem would be the chordal case. In tests on the corpus described in the stress-test section, all 878 deposited root sets and an independent recomputation over all 771 labelled connected graphs through five vertices passed. A counterexample would be one graph with a nonreal or nonnegative zero.

**Conjecture 158** (Connected-deletion interlacing). *If  $v$  is a vertex of  $G$  with  $G - v$  connected, the zeros of  $h_{G-v}$  weakly interlace the zeros of  $h_G$ .*

**Significance.** Interlacing is the inductive engine of real-rootedness, and the conjecture asserts that vertex deletion is its compatible recursion for graph associahedra, so a deletion recurrence in the interlacing family framework would prove Conjecture 157 at the same stroke. The restriction to connected deletions is structural, since  $P_{G-v}$  requires a connected graph in the same convention. The nearest related results are the coefficientwise tree-shift inequalities of [111, 112], which order coefficients rather than roots. A first decisive theorem would be leaf deletion for trees. In tests on the corpus described in the stress-test section, all 3,621 deposited connected-deletion pairs and 2,971 independently recomputed pairs passed. A counterexample would be one pair with a crossing root.

**Conjecture 159** (Strict Mahler growth under edge addition). *For every missing edge  $e$  of a connected graph  $G$ ,  $\mathcal{M}(G + e) > \mathcal{M}(G)$ .*

**Significance.** Adding an edge coarsens the tube structure and enlarges the  $h$ -vector, and the conjecture asserts that the multiplicative root radius responds strictly monotonically, making the Mahler measure a strict connectivity monotone on the graphs of one vertex set, from the path's minimum toward the complete graph. The observable does not occur in the graph-associahedron literature located by our search [107, 111]. A first decisive theorem would be edges completing a triangle on the path. In tests on the corpus described in the stress-test section, all 3,956 deposited graph-edge pairs passed with minimum increment 0.0263, reverified independently on 3,227 pairs. A counterexample would be one edge with nonincreasing Mahler measure.

**Conjecture 160** (Edge addition centralizes the spectrum). *For every missing edge  $e$  of a connected graph  $G$  and every symmetric outer tail that is nonempty and proper,  $\sum_{|i-c|\geq r} \mu_{G+e}(i) \leq \sum_{|i-c|\geq r} \mu_G(i)$ , where  $c = (n-1)/2$ .*

**Significance.** The normalized even-Betti distribution of the toric variety is conjectured to become more centrally peaked under every edge addition, in the symmetric tail order that implies monotonicity for every increasing function of the distance from middle degree, a concentration companion to the outward root motion of Conjecture 159. The nearest related results are the coefficient bounds of [111], which compare unnormalized coefficients. A first decisive theorem would be the outermost tail, the ratio of end coefficients to total mass. In tests on the corpus described in the stress-test section, all deposited tail cross-products were nonnegative with minimum integer margin 2, so every tested proper comparison was strict, and the independent recomputation found no violation. A counterexample would be one edge and one tail with mass moving outward. In the gamma basis  $h_G(t) = \sum_j \gamma_j(G) t^j (1+t)^{d-2j}$ , nonnegative for these flag polytopes [111], the normalized coefficient distribution is the law of  $J + \text{Bin}(d - 2J, \frac{1}{2})$  with  $J$  drawn proportionally to  $\gamma_J(G) 4^{-J}$ , and the conditional symmetric tails shrink as  $J$  grows, by binomial unimodality; the statement therefore follows from first-order stochastic dominance of the gamma-index law under edge addition. That dominance holds across all 450 edge additions among connected graphs through six vertices, counted up to isomorphism of the pair, while its likelihood-ratio strengthening already fails at five vertices, with gamma vectors  $(1, 13, 8)$  and  $(1, 17, 10)$ , so the dominance form is the calibrated sufficient condition.

**Conjecture 161** (Path–star Mahler extremality). *For every tree  $T$  on  $n \geq 4$  vertices,  $\mathcal{M}(P_n) \leq \mathcal{M}(T) \leq \mathcal{M}(K_{1,n-1})$ , with equality on the left only for the path and on the right only for the star.*

**Significance.** The path and the star are extremal for several coefficientwise orders on tree associahedra [111, 112], and the conjecture proposes the spectral strengthening: the same pair bounds the nonlinear Mahler functional of all roots, with equality rigidity. It is close to prior work as the spectral shadow of known extremal orders. A first decisive theorem would be either bound along Aisbett’s tree shifts, which would chain the inequality from path to star. In tests on the corpus described in the stress-test section, all 1,857 deposited bound-and-rigidity tests and an independent scan of every labelled tree through six vertices passed. A counterexample would be a third tree attaining either bound or a tree escaping the corridor.

## 40 Age-filtered McKay complexes

**Conjecture 162** (One-crest component law). *For every isolated cyclic quotient with pairwise distinct weights, the component sequence  $(c_0, \dots, c_{d-1})$  is unimodal.*

**Significance.** As the age threshold rises, components are born and merge with no general monotonicity, and the conjecture asserts that the census nevertheless has a single crest. Pairwise distinctness is a calibrated boundary, not caution: the repeated-weight quotient  $\frac{1}{17}(2, 3, 3, 3, 10, 10, 10, 10)$  has component profile  $(1, 2, 1, 2, 1, 1, 1, 1)$ , refuting the unrestricted statement, and the control is retained. The nearest related results are the age grading of the McKay correspondence [101] and quiver connectivity, neither of which filters the Cayley graph by age. A first decisive theorem would be unimodality for  $d = 3$ . In tests on the corpus described in the stress-test section, all 3,135 deposited profiles and an independent exhaustive recomputation at  $p \in \{7, 11, 13\}$  passed. A counterexample would be one distinct-weight quotient with a two-crest census.

**Conjecture 163** (Early component crest). *The first index attaining  $\max_k c_k$  is at most  $\lfloor d/2 \rfloor$ .*

**Significance.** Unimodality says nothing about location, and the conjecture adds it: fragmentation peaks no later than half age, so the second half of the filtration only consolidates. The mechanism is the age duality  $\text{age}(j) + \text{age}(-j) = d$ , which pairs late sublevels with complements of early ones. The nearest related result is [101]. A first decisive theorem would be the crest location for  $d = 4$ . In tests on the corpus described in the stress-test section, all deposited and independently recomputed profiles peaked by half age. A counterexample would be a quotient whose fragmentation peaks late.

**Conjecture 164** (Half-age connectivity). *For  $d \geq 5$ , the sublevel complex  $K_{\lceil d/2 \rceil}$  is connected.*

**Significance.** The full Cayley graph is connected for elementary reasons; the content is that connectivity is already achieved by the middle age threshold, a quantitative connectivity law for the filtration. It is close to prior work, close to standard Cayley-graph connectivity questions, and its distinctness hypothesis is again a genuine boundary: the repeated-weight quotient  $\frac{1}{11}(1, 1, 5, 5, 5, 5)$  has two components at its half-age level, and the control is retained. A first decisive theorem would be connectivity of  $K_{\lceil d/2 \rceil}$  when some weight equals 1. In tests on the corpus described in the stress-test section, all 2,905 deposited profiles with  $d \geq 5$  and

the independent recomputation passed. A counterexample would be a distinct-weight quotient disconnected at half age.

**Conjecture 165** (One-crest first homology). *The sequence  $(b_{1,0}, \dots, b_{1,d-1})$  is unimodal.*

**Significance.** Cycles are created by new vertices and killed by flag two-cells in no fixed order, and the conjecture asserts a one-crest shadow for the ungraded first Betti census of the filtration, the simplest persistence-type law the object could satisfy. The nearest related results are Floer-theoretic age filtrations, which grade different complexes. A first decisive theorem would be unimodality for  $d = 4$ . In tests on the corpus described in the stress-test section, all deposited and independently recomputed  $\mathbf{F}_2$  profiles passed. A counterexample would be one quotient with a two-crest  $b_1$  census.

**Conjecture 166** (One-crest second homology). *The sequence  $(b_{2,0}, \dots, b_{2,d-1})$  is unimodal.*

**Significance.** The degree bound is genuine and calibrated: the distinct-weight quotient  $\frac{1}{31}(2, 8, 16, 17, 18, 20, 24, 26, 27, 28)$  has  $\beta_3$  profile  $(0, 0, 0, 0, 0, 4, 3, 4, 4, 4)$ , not unimodal, so the one-crest law stops at the second homology and the retained control marks the boundary. Together with Conjectures 162 and 165 this confines the low-dimensional shape of the whole filtration. The nearest related result is [101]. A first decisive theorem would be  $d = 5$ . In tests on the corpus described in the stress-test section, all deposited and independently recomputed profiles passed. A counterexample would be a two-crest  $b_2$  census.

## 41 Sine-torsion sectors of cyclic quotients

**Conjecture 167** (Positive-temperature log-concavity). *For every  $s > 0$  and  $2 \leq m \leq d - 2$  such that  $\Omega_{p,d,q}$  is nonempty for  $q \in \{m - 1, m, m + 1\}$ ,  $Z_{p,d,m}(s)^2 > Z_{p,d,m-1}(s) Z_{p,d,m+1}(s)$ .*

**Significance.** The sector index  $m$  is the microcanonical charge of the age construction, and the conjecture asserts that at every positive temperature the sector profile of the torsion partition function is strictly log-concave, an ordering of lens-space character factors [108] by their orbifold charge. The non-emptiness qualification is the exact domain on which all three expectations are defined. A proof route is total positivity of the coefficient kernel of powers of a positive polynomial. A first decisive theorem would be fixed small  $(p, d)$  by exact convolution. In tests on the corpus described in the stress-test section, all 40,932 deposited triples passed with minimum curvature 0.0079, reverified independently at five  $(p, d)$  pairs. A counterexample would be one sector triple with the wrong curvature sign.

**Conjecture 168** (Concavity of mean log torsion). *For  $2 \leq m \leq d - 2$  such that  $\Omega_{p,d,q}$  is nonempty for  $q \in \{m - 1, m, m + 1\}$ ,  $2\Phi_{p,d,m} > \Phi_{p,d,m-1} + \Phi_{p,d,m+1}$ .*

**Significance.** This is the zero-temperature shadow of Conjecture 167, stated separately because strict inequalities need not survive differentiation at a symmetry point: the mean log torsion is conjectured strictly concave in the sector index outright. By Proposition 25 the global mean over all sectors is pinned exactly, so the concavity is a genuine shape statement about the conditional means. A first decisive theorem would be  $d = 4$ , three sectors and one inequality. In tests on the corpus described in the stress-test section, all 6,822 deposited triples passed with minimum

curvature 0.0770, reverified independently. A counterexample would be one convex triple of sector means.

**Conjecture 169** (The  $\zeta(3)$  central parabola). *Let  $m_d$  satisfy  $(m_d - d/2)/\sqrt{d/12} \rightarrow x$ . Then, taking  $p \rightarrow \infty$  through primes first and  $d \rightarrow \infty$  second,*

$$\lim_{d \rightarrow \infty} \lim_{p \rightarrow \infty} \left( \Phi_{p,d,m_d} - \frac{d \log p}{p-1} \right) = \frac{3\zeta(3)}{\pi^2} (1 - x^2).$$

**Significance.** The centering is exact by Proposition 25, and the constant is forced by the proposition's second identity through a conditional local-limit expansion, so the conjectural content is precisely the uniform interchange of the singular log-sine observable with the conditioning. The statement is an iterated-limit claim, its machinery classical. A first decisive theorem would be the central value  $x = 0$ . In tests on the corpus described in the stress-test section, four exact dynamic-programming regimes matched the parabola with central-window root-mean-square errors 0.027 to 0.038. A counterexample would be a different limiting profile.

**Conjecture 170** (Age–torsion decoupling). *With weights uniform on distinct nonzero zero-sum tuples and  $J$  uniform on  $\mathbb{F}_p^\times$ , the standardized pair of  $\text{age}_A(J)$  and of the centred log-torsion sum converges to a standard bivariate Gaussian as  $p, d \rightarrow \infty$  with  $d = o(p)$ .*

**Significance.** The involution  $J \mapsto -J$  forces exact zero covariance, and the conjecture asserts far more: the singular torsion statistic is itself asymptotically Gaussian and asymptotically independent of the age. It strengthens the annealed corollary of Conjecture 146, the Gaussian law of a single uniformly drawn age, by the torsion coordinate and the joint law, and it carries the same random-model and accessibility caveats. A first decisive theorem would be the torsion marginal alone. In tests on the corpus described in the stress-test section, three seeded ensembles of up to 60,480 character samples had empirical correlations below  $5 \times 10^{-16}$ , a consistency check of the exact symmetry rather than evidence of independence, while the informative diagnostics, the variances and fourth moments, moved toward Gaussian values; sharper tests would target mixed fourth cumulants or conditional torsion laws in narrow age windows. A counterexample would be a persistent dependence or a non-Gaussian marginal.

**Conjecture 171** (Negative-temperature reversal). *For  $-1 \leq s < 0$  and  $2 \leq m \leq d-2$  such that  $\Omega_{p,d,q}$  is nonempty for  $q \in \{m-1, m, m+1\}$ ,  $Z_{p,d,m}(s)^2 < Z_{p,d,m-1}(s) Z_{p,d,m+1}(s)$ .*

**Significance.** Cooling through zero reverses the sector shape: strict log-concavity at positive temperature becomes strict log-convexity on the first negative-temperature window, and the window is a calibrated boundary, since at  $(p, d, s) = (11, 4, -2)$  the central curvature has the wrong sign, +0.4980, a retained control reproduced independently to six decimals. The pair of Conjectures 167 and 171 is thus one slice of a sharper phase diagram whose critical temperature is unknown. A first decisive theorem would be the reversal at  $s = -1$  for  $d = 4$ . In tests on the corpus described in the stress-test section, all 20,466 deposited triples passed with curvature at most  $-0.0076$ , reverified independently. A counterexample would be a log-concave triple inside the window.

## 42 Boundary-determinant entropy of simplicial spheres

**Conjecture 172** (Log-concavity of boundary entropy). *For every simplicial  $d$ -sphere and  $2 \leq k \leq d-1$ ,  $E_k(K)^2 \geq E_{k-1}(K) E_{k+1}(K)$ .*

**Significance.** The log pseudodeterminants of the boundary Laplacians, the higher spanning-tree entropies of the matrix-tree readings [109, 110], are conjectured log-concave across the degree, a second-order organization of the whole determinant profile of a sphere. A proof route is an Alexandrov–Fenchel-type injection between adjacent-degree tree counts. A first decisive theorem would be cross-polytope boundaries, where every eigenvalue is explicit. In tests on the corpus described in the stress-test section, all 262 deposited triples passed with minimum margin 51.8, reverified independently on spheres through dimension four. A counterexample would be one sphere and one degree with a convex triple.

**Conjecture 173** (Entropy per rank decreases). *For every simplicial  $d$ -sphere,  $A_1(K) \geq A_2(K) \geq \dots \geq A_d(K)$ .*

**Significance.** Normalizing each entropy by its boundary rank gives the mean log eigenvalue at each degree, and the conjecture asserts this spectral density weakens monotonically with the degree, so lower-dimensional skeleta carry the densest determinant mass. Poincaré duality for the sphere chain complex pairs the two ends, which is the suggested mechanism. The nearest related results are [109, 110]. A first decisive theorem would be the first step  $A_1 \geq A_2$  for polytopal spheres. In tests on the corpus described in the stress-test section, all 392 deposited adjacent comparisons passed, reverified independently. A counterexample would be one increasing adjacent pair.

**Conjecture 174** (Simplex rigidity of equality). *An adjacent equality  $A_k(K) = A_{k+1}(K)$  occurs for some  $k$  if and only if  $K$  is the boundary of a simplex, in which case all adjacent equalities occur.*

**Significance.** The simplex boundary is the unique sphere whose entropy-per-rank chain is flat, so Conjecture 173 is strict at every step for every other sphere, an equality-rigidity statement in the register of the collection’s other extremal characterizations. The mechanism is that the simplex boundary’s Laplacian spectra at all degrees are powers of a single integer. A first decisive theorem would be strictness for cross-polytopes. In tests on the corpus described in the stress-test section, all 130 deposited rigidity tests passed, the simplex equalities holding to numerical zero, reverified independently. A counterexample would be a nonsimplicial sphere with one flat step.

**Conjecture 175** (Strict stellar growth). *If  $K'$  is obtained from  $K$  by stellar subdivision of a facet, then  $E_k(K') > E_k(K)$  for every  $1 \leq k \leq d$ .*

**Significance.** The most elementary local refinement of a triangulated sphere is conjectured to increase every entry of the determinant profile strictly, making the profile a complexity monotone for stellar moves. A facet subdivision is a block update of the boundary maps, so Schur complements and the matrix determinant lemma are the natural proof route and would give an explicit increment formula. The nearest related results are [109, 110]. A first decisive theorem would be the top degree, where the update is rank-one-like. In tests on the corpus described in the stress-test section, all 900 deposited comparisons over 225 subdivision pairs passed with



minimum increment 1.42, reverified independently through dimension four. A counterexample would be one degree that fails to grow.

**Conjecture 176** (Log-concave stellar increments). *With  $\delta_k = E_k(K') - E_k(K)$  as in Conjecture 175,  $\delta_k^2 \geq \delta_{k-1}\delta_{k+1}$  for  $2 \leq k \leq d-1$ .*

**Significance.** Not only does every degree grow under a facet subdivision, but the created entropy is conjectured to have a single log-concave degree profile, so local refinement injects spectral complexity in an organized shape rather than arbitrarily. Together with Conjecture 172 this asserts log-concavity for the profile and for its most elementary increments simultaneously. The nearest related results are [109, 110]. A first decisive theorem would be stacked spheres, where the increments are explicit. In tests on the corpus described in the stress-test section, all 450 deposited triples passed with minimum margin 17.8, reverified independently. A counterexample would be one subdivision with a convex increment triple.

## Part X

# Flux cohomology and protected index laws

*Added to the deposit: 12 August 2026.*

The twenty conjectures below carry the collection into string-geometric index theory. They form three programmes. A flux programme (Conjectures 177 to 184) studies the cohomology of wedge multiplication by an integral three-form on a torus, the invariant-forms model of  $H$ -twisted cohomology, through saturation laws for random flux, sparse witnesses, and an arithmetic torsion staircase for the complete flux. A profile programme (Conjectures 185 to 191) studies the signed  $\chi_y$ -coefficient profile of projective Calabi–Yau complete intersections along the merge order of Part VII, extending the proved Euler monotonicity of Conjecture 118 to every protected coefficient, with root, log-concavity, and Mahler laws. A signature programme (Conjectures 192 to 196) does the same for the Hirzebruch signature, ending in a quadric ratio law with limit 27 and a dimension-wise gcd law that is the signature sibling of Conjecture 121. The calibration layer is exact where the objects are: finite-field cohomology, Smith normal forms,  $\chi_y$ -coefficients, signatures, merge inequalities, ratios, and gcds are integer or rational computations, while characteristic-zero ranks beyond dimension nine are dual-prime proxies and the root, Mahler, and radius laws of Conjectures 188, 190, and 191 are checked spectrally in floating point and. Two retained controls mark calibrated boundaries, a falsified all-prime torsion precursor and a falsified elementary-torsion strengthening. Conjectures 177, 178, 180, and 184 are stated in a random model and Conjecture 181 is stated in greater generality than the computations reach, the part’s boldest statement; Conjectures 184, 189, and 195 sit closest to classical machinery; and Conjecture 190 returns the Mahler order of Conjecture 159 on a different polynomial. Since deposit, three of the twenty have been resolved. Conjecture 195 is resolved true: Proposition 31 proves all three of its clauses from an algebraic generating function. Conjecture 182 is resolved true and Conjecture 183 resolved false: Proposition 30 puts the complete flux in a symplectic normal form, proving both cohomology totals and the whole torsion staircase, and exhibiting at  $n = 15$  a summand  $\mathbb{Z}/4$  that refutes the elementary-abelian clause. Proposition 28 has since proved the subcritical direction of Conjecture 178 and the linear lower bound of Conjecture 179, and has repaired Conjecture 180, whose support-conditional reading is false and is retained as a

control; Proposition 27 sharpens the subcritical direction to every fixed  $c < 2$  at the coverage scale itself.

For  $n \geq 3$  and a commutative ring  $R$  let  $E_n(R) = \Lambda^* R^n$ , and for an integral three-form  $H \in \Lambda^3 \mathbb{Z}^n$  let  $d_H$  be wedge multiplication by  $H$ , a differential because  $H$  has odd degree; write  $b_k(H; R)$  for the dimensions of the cohomology of  $(E_n(R), d_H)$ ,  $B(H; R) = \sum_k b_k(H; R)$ , and  $t_p(H)$  for the number of  $p$ -primary cyclic summands of the integral cohomology. This complex is the invariant-forms model of  $H$ -twisted cohomology [113], whose K-theoretic refinement classifies Ramond–Ramond charges in a topologically nontrivial B-field [114]. The *generic target*  $g_n$  is the cohomology profile on the Zariski-open locus of maximal wedge ranks, that of a generic principal ideal generated in degree three: for  $n = 2m$  it is  $(3^{m-1}, 2 \cdot 3^{m-1}, 3^{m-1})$  in degrees  $m-1, m, m+1$ ; for  $n \equiv 3 \pmod{4}$  it is  $3^{(n-1)/2}$  in each of the two middle degrees; for  $n \equiv 1 \pmod{4}$  it has inner pair  $3^{(n-1)/2}$  and outer pair 1, except that the outer pair is 3 at  $n = 9$ ; its total is  $G_n$ . Lundqvist and Nicklasson prove a lower bound for the generic Hilbert series in every odd degree, disproving the Moreno-Socias–Snellman conjecture above degree three, and establish equality only in special cases, leaving the minimal series open in general and in degree three in particular [115]; the displayed profile is therefore the expected one and not a cited theorem, and the exceptional outer pair at  $n = 9$  is the four-dimensional kernel computed there. The *complete flux* is  $H_\Delta = \sum_{i < j < k} e_i \wedge e_j \wedge e_k$ , and the target totals are  $F_n = \binom{n+1}{(n+1)/2}$  for odd  $n$ ,  $F_n = 2\binom{n}{n/2}$  for even  $n$ , and  $P_n = 2^{n-1} + 2^{(n-1)/2}$  for odd  $n$ ,  $P_n = 2(2^{n-2} + 2^{(n-2)/2})$  for even  $n$ . Write  $t_{p,k}(n)$  for the number of  $p$ -primary cyclic summands of the integral cohomology of  $H_\Delta$  in degree  $k$ , and  $T_p(n) = \sum_k t_{p,k}(n)$  for their total. On the geometric side,  $X_{\mathbf{d}}^n \subset \mathbb{CP}^{n+r}$  is the Calabi–Yau complete intersection of multidegree  $\mathbf{d} = (d_1, \dots, d_r)$ ,  $d_i \geq 2$ ,  $\sum_i d_i = n + r + 1$ , with degree  $D(\mathbf{d}) = \prod_i d_i$ ; merging replaces  $d_i, d_j$  by  $d_i + d_j - 1$ , preserving dimension and the Calabi–Yau condition, exactly as in Part VII. Writing  $\chi_y = \sum_p \chi^p y^p$  for the Hirzebruch genus [117], put  $a_p(\mathbf{d}) = (-1)^{n-p} \chi^p(X_{\mathbf{d}}^n)$  and  $A_{\mathbf{d}}(y) = \sum_p a_p(\mathbf{d}) y^p$ , a palindromic integer polynomial with nonnegative coefficients throughout the tested range; for even  $n$  put  $S(\mathbf{d}) = (-1)^{n/2} \sigma(X_{\mathbf{d}}^n)$ . Let  $Q = (2, \dots, 2)$  be the all-quadric and  $Y = (n+2)$  the hypersurface configuration.

**Proposition 26** (Coefficient and parity calibrations). For every  $H \in \Lambda^3 \mathbb{Z}^n$  and every prime  $p$ ,  $B(H; \mathbb{F}_p) = B(H; \mathbb{Q}) + 2t_p(H)$ . If every coefficient of  $H$  is odd, then the complexes of  $H$  and of  $H_\Delta$  over  $\mathbf{F}_2$  coincide, so  $B(H; \mathbb{F}_2) = B(H_\Delta; \mathbb{F}_2)$ .

*Proof.* The complex  $(E_n(\mathbb{Z}), d_H)$  is a bounded complex of finitely generated free abelian groups, so the universal coefficient theorem gives  $H^k(E_n(\mathbb{F}_p), d_H) \cong H^k(E_n(\mathbb{Z}), d_H) \otimes_{\mathbb{F}_p} \oplus \text{Tor}(H^{k+3}(E_n(\mathbb{Z}), d_H), \mathbb{F}_p)$ . A free summand contributes one dimension in its own degree; a  $p$ -primary cyclic summand contributes one dimension in its own degree through the tensor factor and one three degrees below through the torsion factor, the differential having degree three; a cyclic summand of order prime to  $p$  contributes nothing. Summing over  $k$  gives the identity. For the second claim, the matrix of  $d_H$  over  $\mathbf{F}_2$  depends only on the coefficients of  $H$  modulo 2, and a three-form with all coefficients odd reduces modulo 2 to  $H_\Delta$ .  $\square$

## 43 Integral flux complexes on tori

**Conjecture 177** (Dense discrete flux saturation). Fix a finite symmetric set  $V \subset \mathbb{Z} \setminus \{0\}$ , for example  $\{\pm 1\}$ , and let every coefficient of  $H_n \in \Lambda^3 \mathbb{Z}^n$  be independent and uniform on  $V$ . Then  $\Pr[(b_k(H_n; \mathbb{Q}))_k = g_n] \rightarrow 1$ , and the failure probability decays exponentially in  $n$ .

**Significance.** Over an infinite field a generic three-form attains the generic profile; the conjecture asserts that the smallest possible discrete ensembles already do so with overwhelming probability, turning the generic principal-ideal profile of [115] into a probabilistic law for discrete flux backgrounds and compressing the ordinary  $2^n$  torus cohomology to a  $3^{n/2}$ -scale protected sector. The mechanism is that the discrete cube must avoid every determinantal degeneracy locus of the wedge-rank strata. A first decisive theorem would be a uniform positive lower bound on the saturation probability. In tests on the corpus described in the stress-test section, 142 of 144 deposited dense trials and 71 of 72 independent trials hit  $g_n$  for  $3 \leq n \leq 11$ . A counterexample would be a law and a subsequence with failure probability bounded away from zero.

**Conjecture 178** (Sparse-flux threshold at the coverage scale). *Include each coordinate triple independently with probability  $p_n$  and give included triples independent coefficients from a fixed finite symmetric set of nonzero integers. If  $p_n n^2 / \log n \rightarrow \infty$  then  $\Pr[(b_k; \mathbb{Q})_k = g_n] \rightarrow 1$ .*

**Significance.** The scale  $(\log n)/n^2$  is the isolated-vertex scale of 3-uniform random hypergraphs, while uncovered pairs persist to the far larger scale  $(\log n)/n$ , so the conjectured last obstruction to generic wedge ranks is the uncovered vertex alone, a correction, in the spirit of hitting-time thresholds in random graph theory; even that obstruction is parity-sensitive, since for  $n \equiv 0 \pmod{4}$  the deposited profiles satisfy  $g_n = (1+t)g_{n-1}$ , so a form ignoring one coordinate can still saturate, and the vertex mechanism itself would place the critical constant near 2 outside those dimensions; a sharper form should have a critical window  $p = (c_* \log n + O(\log \log n))/n^2$ , but no value of  $c_*$  is claimed and the statement is confined to the coarse threshold. The zero direction is now proved in Proposition 28, a promotion: below the scale unused coordinates proliferate, and the deposited profile admits at most one of them, so the surviving content is the supercritical direction, which extrapolates, with the random model. Proposition 27 sharpens the zero direction to every fixed  $c < 2$  at the critical scale, by a second-moment count of unused coordinates, so the coverage constant 2 named above is now a rigorous lower-side bound. In tests on the corpus described in the stress-test section, 960 deposited trials at  $p = c \log n / n^2$  rose from hit rate 0 at  $c = 1$  to 0.875 and 0.9125 at  $c = 10$  under the two laws, and an independent sweep rose from 0/36 through 5/36 to 34/36 at  $c = 1, 4, 10$ . A counterexample would be a sparse sequence saturating below the scale or a dense one failing above it.

**Proposition 27** (Coverage floor for the sparse threshold). If  $p_n = (c + o(1)) \log n / n^2$  with a fixed  $c < 2$ , then the saturation probability of Conjecture 178 tends to zero.

*Proof.* Let  $U_n$  count coordinates contained in no included triple. A fixed coordinate lies in  $\binom{n-1}{2}$  triples, so  $\Pr[i \text{ unused}] = (1 - p_n)^{\binom{n-1}{2}} = n^{-c/2+o(1)}$  and  $\mathbf{E}U_n = n^{1-c/2+o(1)} \rightarrow \infty$ . Two coordinates share exactly  $n-2$  incident triples, so  $\Pr[i, j \text{ both unused}] / \Pr[i \text{ unused}]^2 = (1 - p_n)^{-(n-2)} = 1 + o(1)$ , whence  $\text{Var } U_n = o((\mathbf{E}U_n)^2)$  and  $U_n \geq 2$  with probability tending to one. By the support obstruction of Proposition 28, a form attaining the deposited profile leaves at most one coordinate unused, so saturation fails.  $\square$

**Conjecture 179** (Linear-complexity sparse witnesses). *Let  $\mu_n$  be the minimum support size of a  $\{\pm 1\}$ -valued three-form on  $\mathbb{Z}^n$  with rational profile  $g_n$ . Then for all sufficiently large  $n$  a saturating support of at most  $n + C$  triples can be chosen to form a linear 3-uniform hypergraph, for an absolute constant  $C$ ; in particular  $\mu_n = \Theta(n)$ .*

**Significance.** Between the dense ensemble of Conjecture 177 and the threshold of Conjecture 178 sits the deterministic minimum, and the conjecture pins it at linear growth with witnesses of the simplest combinatorial type, pairwise intersections at most one vertex, giving sparse flux backgrounds with generic protected topology and transparent charge combinatorics. The linear lower bound is now proved, and in the sharp form  $\mu_n \geq \lceil (n-1)/3 \rceil$ : by Proposition 28 a saturating form leaves at most one coordinate unused, while a support of  $s$  triples meets at most  $3s$  of them. The content of the conjecture is therefore the matching upper construction. Explicit witnesses use 1, 1, 2, 2, 5, 5, 8, 7 triples for  $n = 3$  to 10, the exceptional dimension 9 being the one nonlinear support, matching the exceptional outer pair of  $g_9$ ; a proof route is partial Steiner systems with block-decomposed rank counts. In tests on the corpus described in the stress-test section, all eight deposited witnesses were reproduced independently, and exhaustive scans at  $n = 5, 6$  confirmed  $\mu_5 = \mu_6 = 2$ . A counterexample would be sublinear supports or forced nonlinearity infinitely often.

**Conjecture 180** (Coefficient-law universality in the sparse window). *Fix  $c > 0$  and let the support be sampled at  $p = c \log n / n^2$  as in Conjecture 178. For every finite symmetric coefficient set  $V \subset \mathbb{Z} \setminus \{0\}$ , the limiting saturation probability and the limiting joint law of the excess vector  $(b_k(H; \mathbb{Q}) - g_{n,k})_k$  exist and depend only on  $c$ , not on  $V$ .*

**Significance.** Thresholds located by a support process should not remember the coefficient sampler, and the conjecture asserts exactly this universality, making sparse flux-topology statistics a property of the hypergraph ensemble rather than an artifact of a particular discrete law; no arithmetic exclusion is needed at the rational level, since a common divisor of all coefficients scales the differential invertibly over  $\mathbb{Q}$  and changes no rank, so the deposited form's exclusion of such laws has been removed, with the window and the excess vector now fixed explicitly in the statement. The mechanism is rank universality for structured sparse random matrices, but not conditionally on the support: the profile is genuinely coefficient-sensitive, and the smallest witness, retained here as a boundary control, is the four-triple support  $\{012, 013, 024, 034\}$  on five coordinates, where all triples meet in one coordinate and the form is  $e_0 \wedge \omega$  for a two-form  $\omega$  on the other four. It is minimal: on four coordinates  $\Lambda^3(\mathbb{Q}^4) \cong (\mathbb{Q}^4)^* \otimes \det$  has a single  $GL_4$ -orbit of nonzero forms, so no coefficient moves the profile, while a coefficient-sensitive support needs an even cover, which three triples cannot give since their nine incidences are odd. The profile is  $(0, 3, 9, 9, 3, 0)$  when the Pfaffian of  $\omega$  vanishes and  $(0, 1, 9, 9, 1, 0)$  when it does not, and the chance of the degenerate branch is  $\frac{1}{2}$  for  $\{\pm 1\}$  against  $\frac{3}{16}$  for  $\{\pm 1, \pm 2\}$ , so the law given the support does depend on  $V$ . By Proposition 28(iii) coefficient sensitivity needs a subsupport covering every coordinate evenly, hence with at most  $\frac{3}{2}|S'|$  coordinates, and each fixed such circuit has expected multiplicity  $O(n^{-|S'|/2}(\log n)^{|S'|})$  in the window and disappears from it, which is what localizes the mechanism above. That is as far as the argument reaches, and it is not a proof of the conjecture, a limitation: the window carries  $|S| \sim \frac{\varepsilon}{6} n \log n$  triples on  $n$  coordinates, so  $\text{rank}_{\mathbb{F}_2} A \leq n < |S|$  and the left kernel has dimension at least  $|S| - n = \Theta(n \log n)$ ; circuits are forced rather than rare, and a first-moment estimate for one fixed circuit cannot be summed over the at least  $2^{|S|-n}$  of them. Controlling the large circuits is the surviving problem. The statement above is annealed, averaging over the random support; the quenched strengthening instead fixes the support and asks that the conditional laws  $\mathcal{L}_V(\cdot | S)$  and  $\mathcal{L}_W(\cdot | S)$  converge to one another in total variation. Exact equality of those conditional laws fails at every finite  $n$ , by the four-triple control above, but the convergence itself is untouched by it and is open. Both the random model and the limit clause are. A first decisive theorem would be two-point universality,

$\{\pm 1\}$  against  $\{\pm 1, \pm 2, \pm 3\}$ . In tests on the corpus described in the stress-test section, the two deposited laws tracked each other across 960 trials, with saturation rates 0.2125 against 0.2125 at  $c = 4$  and 0.500 against 0.525 at  $c = 6$ . A counterexample would be two admissible laws with separated limiting curves.

**Proposition 28** (Support obstructions and coefficient rescaling). Let  $H \in \Lambda^3 \mathbb{Z}^n$  have support  $S$ , let  $u$  be the number of coordinates lying in no triple of  $S$ , and let  $A$  be the  $|S| \times n$  incidence matrix of  $S$ . (i) The Betti polynomial of  $(E_n(\mathbb{Q}), d_H)$  is  $(1+t)^u$  times that of the form read in the used coordinates, and  $H$  attains the generic profile only if  $g_n = (1+t)^u g_{n-u}$ ; for the deposited profile this forces  $u \leq 1$ , with  $u = 1$  possible only when  $n \equiv 0 \pmod{4}$ . (ii) Consequently  $\mu_n \geq \lceil (n-1)/3 \rceil$ , and if  $p_n n^2 / \log n \rightarrow 0$  then the saturation probability of Conjecture 178 tends to zero. (iii) If  $\text{rank}_{\mathbf{F}_2} A = |S|$  then the rational profile of  $H$  depends only on  $S$  and not on the coefficients. Coefficient dependence therefore requires a nonempty  $S' \subseteq S$  covering every coordinate an even number of times, and then  $|S'|$  is even,  $|S'| \geq 4$ , and  $|V(S')| \leq \frac{3}{2}|S'|$ . Since  $A$  has  $n$  columns the criterion requires  $|S| \leq n$ , so it is vacuous in the window of Conjecture 180, where  $|S| = \Theta(n \log n)$  and the left kernel of  $A$  has dimension at least  $|S| - n$ .

*Proof.* If a coordinate  $e$  lies in no triple of  $S$  then  $\Lambda^* \mathbb{Q}^n = \Lambda^* V_0 \otimes \Lambda^* \langle e \rangle$  with  $d_H$  acting as  $d_H \otimes 1$ , so the Betti polynomial gains a factor  $1+t$ , and iterating gives  $(1+t)^u$ . Since  $b_k = \binom{n}{k} - \text{rank}_k - \text{rank}_{k-3}$  and both ranks are generically maximal, the generic profile minimizes every  $b_k$  separately, so the profile  $h$  of the form read in the  $n-u$  used coordinates satisfies  $h_k \geq g_{n-u,k}$  for all  $k$ . If  $H$  saturates then  $g_n = (1+t)^u h$  and  $G_n = 2^u \sum_k h_k \geq 2^u G_{n-u}$ , while suspending a generic  $(n-u)$ -form is admissible and gives  $G_n \leq 2^u G_{n-u}$ ; equality forces  $h = g_{n-u}$ , hence  $g_n = (1+t)^u g_{n-u}$ . For the deposited profile, comparing the displayed values gives  $g_n = (1+t)g_{n-1}$  exactly when  $n \equiv 0 \pmod{4}$ , and never  $g_n = (1+t)^2 g_{n-2}$ , since that would require the same identity at  $n$  and at  $n-1$ . This is (i).

A support of  $s$  triples meets at most  $3s$  coordinates, so  $u \geq n - 3s$ , and (i) gives  $n - 3s \leq 1$ , which is the stated bound on  $\mu_n$ , integrality giving the ceiling. For the threshold, a fixed coordinate lies in no chosen triple with probability  $(1-p_n)^{\binom{n-1}{2}} = \exp(-\frac{1}{2}p_n n^2(1+o(1)))$ , which is  $n^{-o(1)}$  when  $p_n n^2 = o(\log n)$ , so the expected number of unused coordinates is  $n^{1-o(1)}$ ; two coordinates lie in only  $n-2$  common triples and  $p_n n = o(1)$ , so their indicators are asymptotically uncorrelated and a second-moment estimate gives  $u \rightarrow \infty$  in probability. By (i) the saturation probability tends to zero, which is (ii).

For (iii), the substitution  $e_i \mapsto \lambda_i e_i$  with  $\lambda \in (\mathbb{Q}^\times)^n$  is an algebra automorphism carrying  $d_H$  to the differential of the form with coefficients  $c_t \prod_{i \in t} \lambda_i$ , so it leaves the profile unchanged. Writing  $\lambda_i = \varepsilon_i \exp(x_i)$ , a prescribed coefficient vector is reached exactly when  $Ax = \log |c|$  over  $\mathbb{Q}$  and  $A\varepsilon = \text{sign}(c)$  over  $\mathbf{F}_2$ ; full row rank over  $\mathbf{F}_2$  supplies both, since it forces full row rank over  $\mathbb{Q}$ , so every coefficient vector is equivalent to  $(1, \dots, 1)$ . If instead the rows are dependent over  $\mathbf{F}_2$ , a nonzero relation  $\sum_t \mu_t \chi_t = 0$  has support  $S'$  covering every coordinate an even number of times, so  $3|S'| = \sum_{t \in S'} |t| \geq 2|V(S')|$ ; the same parity forces  $3|S'|$  even, hence  $|S'|$  even, and  $|S'| = 2$  is impossible for distinct triples, so  $|S'| \geq 4$ . Finally  $\text{rank}_{\mathbf{F}_2} A \leq n$ , so the criterion can hold only when  $|S| \leq n$ .  $\square$

**Conjecture 181** (Finite-field resonance power law). Fix  $n \geq 5$ . There are a positive integer  $\delta_n$ , a finite Galois extension  $K_n/\mathbb{Q}$ , and nonnegative integers  $\kappa_{n,\sigma}$  indexed by the Frobenius conjugacy classes  $\sigma$  of  $K_n/\mathbb{Q}$ , positive for at least one class, such that, for  $H$  uniform on  $\Lambda^3 \mathbb{F}_p^n$ , the probability of the resonance event  $R_{n,p}$ , that the profile  $(b_k(H; \mathbb{F}_p))_k$  exceeds  $g_n$  in some degree,



satisfies

$$\Pr(R_{n,p}) = \kappa_{n,\text{Frob}_p} p^{-\delta_n} + O_n(p^{-\delta_n-1/2}) \quad \text{as } p \rightarrow \infty.$$

**Proposition 29** (Conditional Lang–Weil resonance law). Fix  $n \geq 5$  and suppose that the displayed  $g_n$  is the characteristic-zero profile on the Zariski-open locus of maximal wedge ranks. Then the conclusion of Conjecture 181 holds for every sufficiently large prime of good reduction. More precisely,  $\delta_n$  is the codimension of the largest geometric component of the resonance scheme, and  $\kappa_{n,\text{Frob}_p}$  is the number of its largest geometric components fixed by Frobenius.

*Proof.* Put  $M = \binom{n}{3}$ . The coefficients of a three-form give the affine space  $\mathbb{A}_{\mathbb{Z}}^M$ . Under the hypothesis on  $g_n$ , the event that some wedge map has rank below its generic rank is the set of  $\mathbb{F}_p$ -points of a proper determinantal subscheme  $X_n \subset \mathbb{A}_{\mathbb{Z}}^M$  for every sufficiently large prime. Let  $d$  be the greatest dimension of a geometric irreducible component of its characteristic-zero fibre and set  $\delta_n = M - d$ . Choose a finite Galois extension over which all components of dimension  $d$  are defined. Away from finitely many ramified primes and primes of bad reduction, componentwise Lang–Weil [171] gives

$$\#X_n(\mathbb{F}_p) = \kappa_{n,\text{Frob}_p} p^d + O_n(p^{d-1/2}).$$

Intersections of distinct components of dimension  $d$  and every smaller component contribute  $O_n(p^{d-1})$ . Dividing by  $p^M$  proves the stated probability law. The Galois action on the finite set of largest components shows that the leading coefficient depends only on the Frobenius conjugacy class.  $\square$

**Significance.** Proposition 29 separates the formal counting law from the open geometry. Once  $g_n$  is the true characteristic-zero generic profile, Lang–Weil gives the displayed asymptotic. The remaining content is to prove that identification, determine  $\delta_n$ , describe the largest components and their fields of definition, and compute the Frobenius classes that fix them. Frobenius can permute these components, as for  $x^2 + 1$  over  $\mathbb{F}_p$ , so neither a prime-independent constant nor a congruence-only description is justified without further geometry. The deposited screen of 2,800 exact finite-field samples at  $5 \leq n \leq 8$  detects concentration at small characteristics but is too small to estimate  $\delta_n$ . A first decisive theorem would determine the top resonance components for one  $n$ . A failure of the expected generic profile would invalidate the unconditional form while leaving the proposition intact.

**Conjecture 182** (Central-binomial law for the complete flux, resolved true). *For every  $n \geq 4$ ,  $B(H_{\Delta}; \mathbb{Q}) = F_n$  and  $B(H_{\Delta}; \mathbb{F}_2) = P_n$ .*

**Significance.** The most symmetric integral flux has closed cohomology totals, a central binomial over the rationals and a near-power of two in characteristic two, giving an exact protected-state count for a maximally symmetric background; the binomial shape calls for a decomposition of the complex, the natural proof route and the reason this is rated the most approachable flux law. In tests on the corpus described in the stress-test section, both formulas held for every  $4 \leq n \leq 13$ , the characteristic-two totals exactly and the rational totals by exact Smith forms through  $n = 9$  and dual large-prime ranks beyond, reproduced independently throughout. A counterexample would be one dimension violating either formula. *Resolution.* The conjecture is true: Proposition 30 proves both formulas, reducing wedge multiplication by  $H_{\Delta} = \alpha \wedge \omega$  to Boolean inclusion matrices, whose rational ranks are full and whose characteristic-two complex is exact. The rational total is the binomial transform  $\sum_s \binom{g}{s} 2^s \binom{g-s}{\lfloor (g-s)/2 \rfloor} = \binom{2g+1}{g}$  and the



characteristic-two total is  $4^g + 2^g$ ,  $g = \lfloor (n-1)/2 \rfloor$ ; the symmetric-group route anticipated at deposit is replaced by a symplectic normal form.

**Conjecture 183** (Prime torsion staircase for the complete flux, resolved false). *For every prime  $p$ , the integral cohomology of  $(E_n(\mathbb{Z}), d_{H_\Delta})$  has no  $p$ -primary torsion for  $n \leq 4p-2$  and exactly one  $p$ -primary cyclic summand, isomorphic to  $\mathbb{Z}/p$ , at  $n = 4p-1$ . Moreover, for every  $n$  the 2-primary subgroup is elementary abelian of rank  $t_2(H_\Delta) = (P_n - F_n)/2$ .*

**Significance.** Arithmetic torsion in the flux charge group appears along a staircase indexed by the primes, and the boundary is calibrated by a falsified precursor, retained in place: the claim that all torsion is elementary 2-torsion held through  $n = 9$  in exact Smith form and died at  $(p, n) = (3, 11)$ , where the characteristic-three total exceeds  $F_{11} = 924$  by exactly 2, one  $\mathbb{Z}/3$  summand; the excesses grow to 4 and 28 at  $n = 12, 13$ , and the next decisive prediction is first 5-torsion at  $n = 19$ . By Proposition 26 the rank clause is equivalent to the two totals of Conjecture 182, and the exponent-two clause was the boldest part of the statement. A proof route would be a modular filtration whose semisimplicity first fails at  $n = 4p-1$ , a mechanism that is a target for explanation rather than a consequence of general theory, since nothing elementary singles out  $4p-1$ . In tests on the corpus described in the stress-test section, the staircase and rank clause were verified in exact Smith form through  $n = 9$  and the  $(3, 11)$ ,  $(3, 12)$ ,  $(3, 13)$  excesses reproduced independently, with no resonance at  $p = 5, 7$  through  $n = 13$ . A counterexample would be torsion before a step, a different onset group, or a divisor exceeding two in the 2-primary part. *Resolution.* The staircase clause is true and the exponent clause is false, both by Proposition 30: the onset at  $n = 4p-1$  with a single  $\mathbb{Z}/p$  holds for every prime, while at  $n = 15$  the singleton-free sector carries the inclusion matrix  $U_{7,3}$ , whose Smith normal form  $1^{20}, 2^9, 6^5, 12$  contributes a summand  $\mathbb{Z}/4$  in degree nine. This is exactly the divisor exceeding two that the statement named as its own refutation, and  $n = 15$  is the least dimension at which it can occur, so the exhaustive range through  $n = 13$  could not have seen it; the surviving question is the exact 2-adic invariant factors, whose first departure from exponent one is now located.

**Proposition 30** (Complete-flux normal form). Write  $\alpha = e_1 + \cdots + e_n$  and  $\omega = \sum_{2 \leq i < j \leq n} e_i \wedge e_j$ , and put  $g = \lfloor (n-1)/2 \rfloor$ . Then  $H_\Delta = \alpha \wedge \omega$ , and the integral cohomology of  $(E_n(\mathbb{Z}), d_{H_\Delta})$  satisfies: (i)  $B(H_\Delta; \mathbb{Q}) = F_n$  and  $B(H_\Delta; \mathbb{F}_2) = P_n$ ; (ii) for every prime  $p$  there is no  $p$ -primary torsion for  $n \leq 4p-2$ , and exactly one  $p$ -primary cyclic summand, isomorphic to  $\mathbb{Z}/p$ , at  $n = 4p-1$ ; (iii) whenever  $p$ -primary torsion is present, the positive support of  $k \mapsto t_{p,k}(n)$  is the full interval  $2p+1 \leq k \leq n-2p+2$ ; (iv)  $T_p(n+1) > T_p(n)$  for every  $n \geq 4p-1$ ; (v) at  $n = 15$  the 2-primary subgroup is not elementary abelian: it contains a summand  $\mathbb{Z}/4$  in degree 9.

*Proof.* Comparing coefficients,  $\alpha \wedge \omega$  produces each triple meeting  $\{1\}$  once, and each triple  $\{a < b < c\} \subset \{2, \dots, n\}$  three times, from  $e_a \wedge (e_b \wedge e_c)$ ,  $e_b \wedge (e_a \wedge e_c)$  and  $e_c \wedge (e_a \wedge e_b)$  with signs  $+$ ,  $-$ ,  $+$ ; so  $H_\Delta = \alpha \wedge \omega$ . The substitution  $(e_1, \dots, e_n) \mapsto (\alpha, e_2, \dots, e_n)$  is unimodular, so with  $W = \langle e_2, \dots, e_n \rangle$  we may write  $E_n(\mathbb{Z}) = \Lambda^* W \oplus \alpha \wedge \Lambda^* W$ . Since  $\alpha \wedge \alpha = 0$  the differential annihilates  $\alpha \wedge \Lambda^* W$  and carries  $x \in \Lambda^* W$  to  $\alpha \wedge Lx$ , where  $L = \omega \wedge -$  acts on  $\Lambda^* W$ ; as  $\alpha \wedge -$  is injective on  $\Lambda^* W$ , the cohomology is  $\ker L \oplus \alpha \wedge \operatorname{coker} L$ , so it is  $\operatorname{coker} L$  that carries every torsion summand.

The alternating form  $\omega$  has all elementary divisors equal to 1, so an integral change of basis of  $W$  writes it as  $x_1 \wedge y_1 + \cdots + x_g \wedge y_g$ , together with one null basis vector when  $n$  is even. Split  $\Lambda^* W$  according to the state of each symplectic pair, empty,  $x_i$ ,  $y_i$ , or  $x_i y_i$ . Then  $L$  annihilates every pair held in a singleton state, and on the remaining  $r = g - s$  pairs it sends the set of doubly

occupied pairs to the sum of its one-element extensions; that is, on a sector with  $s$  singletons it is the 0–1 inclusion matrix  $U_{r,j}$  from  $j$ -subsets to  $(j+1)$ -subsets of an  $r$ -set. There are  $\binom{g}{s}2^s$  such sectors, doubled by the null vector when  $n$  is even, and a monomial with  $s$  singletons and  $j+1$  doubly occupied pairs sits in cohomological degree  $s+2j+3$ .

Wilson’s diagonal form for  $t$ -subset versus  $k$ -subset inclusion matrices [116], applied to  $U_{r,j}$  when  $2j+1 \leq r$  and to its transpose otherwise, gives diagonal entries  $h-i$  with multiplicity  $\binom{r}{i} - \binom{r}{i-1}$  for  $0 \leq i < h$ , where  $h = \min(j+1, r-j)$ . These number  $\binom{r}{h-1} = \min(\binom{r}{j}, \binom{r}{j+1})$ , so over  $\mathbb{Q}$  every  $U_{r,j}$  has full possible rank, a sector of  $r$  active pairs contributes  $2^{\lfloor r/2 \rfloor}$  to the total cohomology, and summing over sectors gives  $\sum_s \binom{g}{s} 2^s \cdot 2^{\lfloor (g-s)/2 \rfloor} = 2^{\binom{g+1}{2}}$ , which is  $F_n$  for odd  $n$  and half of it for even  $n$  before the null vector doubles it back. Over  $\mathbf{F}_2$  the operator is multiplication by  $u_1 + \dots + u_r$  in  $\mathbf{F}_2[u_1, \dots, u_r]/(u_i^2)$ , hence conjugate under a linear change of variables to multiplication by  $u_1$  and therefore exact, so its ranks sum to  $2^{r-1}$  and such a sector contributes  $2^r$ , or 2 when  $r=0$ ; summing gives  $4^g + 2^g$ , again doubled by the null vector in even dimension, which is  $P_n$ . This is (i).

A diagonal entry divisible by  $p$  forces  $h \geq p$  and hence  $r \geq 2p-1$ ; since  $r \leq g$ , no  $p$ -torsion occurs for  $n \leq 4p-2$ . At  $n = 4p-1$  we have  $g = 2p-1$ , the only qualifying sector is  $s=0$ ,  $r=2p-1$ ,  $j=p-1$ , and its diagonal contains the entry  $p$  exactly once, giving a single summand  $\mathbb{Z}/p$ ; this is (ii). In a sector with  $r$  active pairs,  $p$  divides a diagonal entry of  $U_{r,j}$  exactly when  $p-1 \leq j \leq r-p$ , so that sector realises the degrees  $s+2j+3$  for  $p-1 \leq j \leq g-s-p$ , together with their shifts by one when  $n$  is even. The sectors  $s=0$  and  $s=1$  already supply the two parities, running over  $2p+1, 2p+3, \dots, 2g-2p+3$  and  $2p+2, 2p+4, \dots, 2g-2p+2$  respectively, and the intervals for larger  $s$  are nested inside these; the union is therefore the single interval  $2p+1 \leq k \leq n-2p+2$ . This is (iii).

For (iv) let  $A_r$  count the  $p$ -primary summands contributed by one pure  $r$ -pair sector and put  $T_g = \sum_r \binom{g}{r} 2^{g-r} A_r$ , so that  $T_p(2g+1) = T_g$  while the null vector gives  $T_p(2g+2) = 2T_g$ . Pascal’s rule yields  $T_{g+1} - 2T_g = \sum_{r \geq 1} \binom{g}{r-1} 2^{g+1-r} A_r$ , which is positive as soon as some  $A_r$  is, that is, after onset.

Finally let  $n = 15$ , so  $g = 7$ . The sector  $s=0$ ,  $r=7$ ,  $j=3$  has  $h=4$ , so Wilson’s diagonal carries the entry 4: the Smith normal form of the  $35 \times 35$  matrix  $U_{7,3}$  is  $1^{20}, 2^9, 6^5, 12$ , its cokernel is  $(\mathbb{Z}/2)^{14} \oplus (\mathbb{Z}/3)^6 \oplus \mathbb{Z}/4$ , and the corresponding degree is  $s+2j+3 = 9$ . This is (v).  $\square$

**Conjecture 184** (Parity-locked torsion count for sign flux). *For dense  $\{\pm 1\}$  flux as in Conjecture 177, conditional on the rational profile equalling  $g_n$ , the number of 2-primary cyclic summands is exactly  $t_2(H_n) = (P_n - G_n)/2$ ; unconditionally, this count holds with probability tending to one.*

**Significance.** A random flux ensemble is conjectured to carry a deterministic number of torsion charge directions even while their orders fluctuate: by Proposition 26, every sign flux has the characteristic-two cohomology of the complete flux, so the count is forced by Conjectures 177 and 182 together and the statement is essentially their corollary, numbered separately for the exact count it pins rather than as an independent law, and close to prior work since the mechanism is the classical universal coefficient theorem. The elementary-torsion strengthening is false and the control is retained: deposited Smith forms contain cyclic factors of orders 4, 40, and 80. A first decisive theorem would be the conditional identity, which needs only the mod-two clause of Conjecture 182. In tests on the corpus described in the stress-test section, all sixteen deposited exact Smith trials at  $5 \leq n \leq 8$  had the predicted count, reproduced independently.

A counterexample would be a saturating sign flux with the wrong number of even summands. *Resolution.* With Conjecture 182 proved, the conditional clause is a theorem: Proposition 26 gives  $t_2(H) = (B(H; \mathbb{F}_2) - B(H; \mathbb{Q}))/2$  for every sign flux, Proposition 30 gives  $B(H; \mathbb{F}_2) = P_n$ , and the conditioning supplies  $B(H; \mathbb{Q}) = G_n$ . The unconditional clause is precisely the saturation event of Conjecture 177, so the statement is retained as a corollary of that conjecture rather than as an independent law.

## 44 Protected profiles of complete intersections

**Conjecture 185** (Coefficientwise endpoint extremality). *For  $n \geq 3$  and every interior index  $1 \leq p \leq n - 1$ ,  $a_p(Q) \leq a_p(\mathbf{d}) \leq a_p(Y)$ , and equality across all interior indices holds only for the endpoint configuration itself. Dimension 2 is the equality calibration: every configuration has the K3 profile  $(2, 20, 2)$ .*

**Significance.** The all-quadric and hypersurface geometries are conjectured to bound every coefficient of the protected profile throughout the complete-intersection landscape [95], refining the Euler extremality aesthetic of Part VII from one number to the full  $\chi_y$ -vector. The mechanism is connectivity of the configuration poset under merges, with a correction: Conjecture 186 controls only the degree-normalized coefficients, and since the total degree strictly decreases under every merge, normalized monotonicity does not transfer to the raw corridor; the upper endpoint should instead flow from the maximal-merge law of Conjecture 187 along maximal chains, and the all-quadric lower comparison, which needs merges away from the maximum, is the harder half. A first decisive theorem would be the extremality of  $Y$  from Conjecture 187 along maximal-merge chains. In tests on the corpus described in the stress-test section, all 7,124 deposited comparisons over 680 configurations through dimension 14 passed in exact arithmetic, reverified independently together with the rigidity clause. A counterexample would be one configuration and one index outside the corridor.

**Conjecture 186** (Degree-normalized merge monotonicity). *If  $\mathbf{d}'$  is obtained from  $\mathbf{d}$  by merging two entries, then  $a_p(\mathbf{d}')/D(\mathbf{d}') > a_p(\mathbf{d})/D(\mathbf{d})$  for every  $1 \leq p \leq n - 1$ .*

**Significance.** Dividing by the degree, the natural volume normalization, every protected coefficient density strictly increases each time two defining equations are consolidated, a monotone coordinate on Calabi–Yau presentations that extends Conjecture 118’s Euler law coefficientwise. The proof route is positivity of the one-merge difference in the classical generating series for  $\chi_y$  of complete intersections [117]. A first decisive theorem would be one interior coefficient, say  $p = 1$ , for all merges. In tests on the corpus described in the stress-test section, all 110,777 deposited exact comparisons through dimension 14 passed, with an independent recomputation of 25,921 deduplicated merge tests finding no violation. A counterexample would be one merge and one index with nonincreasing density.

**Conjecture 187** (Maximal-degree raw monotonicity). *For  $n \geq 3$ , if a merge uses an entry equal to  $\max_i d_i$ , then  $a_p(\mathbf{d}') > a_p(\mathbf{d})$  for every  $1 \leq p \leq n - 1$ .*

**Significance.** Without any normalization the profile can fall under a general merge, since the degree drops; the conjecture isolates the directed moves, those through a maximal entry, that raise every protected coefficient simultaneously, and dimension 2, where all profiles tie, is the

equality boundary. Together with Conjecture 186 this brackets how much degree normalization a raw monotonicity needs. A first decisive theorem would be hypersurface targets, merges landing on  $Y$ . In tests on the corpus described in the stress-test section, all 47,433 deposited exact comparisons for  $3 \leq n \leq 14$  passed, reverified independently on 13,724 deduplicated tests. A counterexample would be one maximal merge losing one coefficient.

**Conjecture 188** (Real-rooted protected polynomial). *For every configuration, the polynomial  $A_{\mathbf{d}}(y)$ , after removing in odd complex dimension the single factor  $y$  forced by  $a_0 = a_n = 0$ , has only simple, strictly negative real zeros.*

**Significance.** Real-rootedness would package the protected profile as a Pólya frequency sequence, an infinite hierarchy of determinantal inequalities with Conjecture 189 as its first layer, and suggests a hidden total-positivity or free-fermion structure in the protected sector. The evidence is numerical root location, the weakest evidence type in this part: all 369 deposited polynomials through dimension 12 passed scaled root tests with smallest normalized separation  $1.276 \times 10^{-3}$ , a figure reproduced independently to six decimals. A proof route is a multiplier sequence realizing the merge recursion; in odd complex dimension palindromy already forces the zero at  $-1$  of the reduced polynomial, and since the coefficients are exact integers, Sturm-sequence certification can replace the floating scans outright. A first decisive theorem would be the hypersurface family. A counterexample would be one configuration with a nonreal or repeated zero, which higher-precision arithmetic could exhibit.

**Conjecture 189** (Strict ultra-log-concavity). *For every configuration and every interior triple of positive coefficients,*

$$(a_p / \binom{n}{p})^2 > (a_{p-1} / \binom{n}{p-1})(a_{p+1} / \binom{n}{p+1}).$$

**Significance.** Binomial normalization is the strongest classical strengthening of log-concavity, the form delivered by Hodge–Riemann packages, and the conjecture asserts it strictly for the protected profile, sharp concentration for the protected-sector distribution; it is close to prior work because a geometric realization of the coefficient sequence would place it directly inside that machinery, in the manner of [103]. Conjecture 188 implies the strict form in full, a strengthening: Newton’s inequalities are strict for polynomials with simple real zeros, and the odd-dimensional reduction multiplies the normalized sequence by the strictly log-concave factor  $p(n-p)$ , so the statement stands or falls with Conjecture 188 and is retained as its first determinantal layer, with a direct intersection-theoretic proof the route that would decouple the two. A first decisive theorem would be hypersurfaces in odd dimension. In tests on the corpus described in the stress-test section, all 6,540 deposited exact inequalities through dimension 14 passed, the count and the outcomes reproduced independently exactly. A counterexample would be one flat or reversed normalized triple.

**Conjecture 190** (Degree-normalized Mahler growth). *For every merge,  $M(A_{\mathbf{d}'})/D(\mathbf{d}') > M(A_{\mathbf{d}})/D(\mathbf{d})$ , where  $M$  is the multiplicative Mahler measure, the modulus of the leading coefficient times  $\prod_{\rho} \max(1, |\rho|)$  over the zeros  $\rho$ .*

**Significance.** The Mahler measure couples coefficient growth to the placement of the zeros, and the conjecture asserts that per unit degree it grows strictly under every consolidation of equations, returning the Mahler order of Conjecture 159 on the protected polynomial in place

of the graph associahedron. Under Conjecture 188 the measure is a product over negative real zeros, but reality alone does not control how the zeros move under a merge, so the route needs interlacing or majorization of the root motion, the recorded missing step. The evidence is numerical and. A first decisive theorem would be merges within the hypersurface-adjacent family. In tests on the corpus described in the stress-test section, all 4,267 deposited merge tests through dimension 12 passed, reverified independently on 1,083 deduplicated tests. A counterexample would be one merge with nonincreasing normalized measure.

**Conjecture 191** (Spectral-radius growth with low-dimensional rigidity). *Let  $\rho(\mathbf{d})$  be the largest modulus of a zero of the reduced polynomial of Conjecture 188, the polynomial  $A_{\mathbf{d}}$  with its forced factor removed. If a merge uses a maximal entry, then  $\rho(\mathbf{d}') \geq \rho(\mathbf{d})$ , with equality for every such merge in complex dimensions  $n = 2, 3, 5$  and strict inequality in every other dimension  $n \geq 4$ .*

**Significance.** Equation consolidation is conjectured to be a monotone spectral flow of the protected index, and the rigidity pattern is exact where palindromy forces it: in dimension 2 all profiles coincide, in dimension 3 the reduced polynomial always has its zero at  $-1$ , and in dimension 5 the tested radii agree along every maximal merge, while from dimension 4 onward, dimension 5 excepted, the flow is strictly expansive. The evidence is numerical and. Palindromy alone cannot deliver the dimension-5 rigidity: the reduced cubic factors as  $y+1$  times a palindromic quadratic whose root pair varies with the coefficient ratio, so the tested equality demands a further identity among the dimension-five coefficients, and a first decisive theorem would be that identity. In tests on the corpus described in the stress-test section, all 2,102 deposited maximal-merge tests through dimension 12 matched the equality pattern, reverified independently on 617 deduplicated tests with no violation of either clause. A counterexample would be a contraction, or a strict increase in dimensions 2, 3, or 5.

## 45 Signature laws for complete intersections

**Conjecture 192** (Signature endpoint extremality). *In every even complex dimension  $n \geq 4$ ,  $S(Q) \leq S(\mathbf{d}) \leq S(Y)$ , with equality only at the corresponding endpoint. Dimension 2 is the calibration: every configuration is a  $K3$  surface with  $S = 16$ .*

**Significance.** The signature is an index, an oriented-cobordism invariant, and the divisibility obstruction for free finite group actions, so universal endpoints bound quotient and anomaly data across the landscape; the conjecture is the signature specialization of Conjecture 185 but is stated separately because the signature carries its own rigidity, verified here with equality forced only at the endpoints in every tested dimension. Since the total degree strictly decreases under every merge, the normalized law of Conjecture 193 does not by itself give the raw corridor, a correction; a first decisive theorem would be the upper endpoint from the maximal-merge law of Conjecture 194 along maximal chains. In tests on the corpus described in the stress-test section, all 8,190 deposited configurations through dimension 26 passed in exact arithmetic, the rigidity clause reverified independently on all 8,187 eligible configurations. A counterexample would be one configuration outside the corridor or a non-endpoint tie.

**Conjecture 193** (Degree-normalized signature monotonicity). *Every merge in even complex dimension satisfies  $S(\mathbf{d}')/D(\mathbf{d}') > S(\mathbf{d})/D(\mathbf{d})$ .*



**Significance.** The signature density per unit degree increases strictly under every consolidation of equations, an index-density monotone and pruning rule for quotient searches, and the signature sibling of Conjecture 186; the proof route is one-merge positivity in the L-genus characteristic series [117], the analogue for  $x/\tanh x$  of the  $\chi_y$  computation. A first decisive theorem would be merges of two quadrics. In tests on the corpus described in the stress-test section, all 302,130 deposited exact comparisons through dimension 26 passed, reverified independently on 55,665 deduplicated tests. A counterexample would be one merge with nonincreasing signature density.

**Conjecture 194** (Maximal-degree signature monotonicity). *In even complex dimension  $n \geq 4$ , a merge using a maximal entry strictly increases  $S$ .*

**Significance.** As with the profile, the raw signature can fall under an arbitrary merge, and the conjecture isolates the directed moves that always raise it, giving a strictly increasing global index along maximal-merge chains from the all-quadric configuration to the hypersurface; dimension 2 is again the equality boundary. A first decisive theorem would combine Conjecture 193 with a bound on the degree drop. In tests on the corpus described in the stress-test section, all 80,294 deposited exact tests through dimension 26 passed, reverified independently on 23,284 deduplicated tests. A counterexample would be one maximal merge lowering the signature.

**Conjecture 195** (The quadric ratio law with limit 27, resolved true). *For even  $n$  let  $\Sigma_n = S(Q)$  for the all-quadric configuration of  $n + 1$  quadric equations in complex dimension  $n$ . Then  $R_n = \Sigma_{n+2}/\Sigma_n$  is strictly increasing in  $n$ ,  $R_n < 27$ , and*

$$R_n \rightarrow 27, \quad 27 - R_n \sim \frac{27}{n}.$$

**Significance.** The all-quadric ladder is conjectured to grow by an asymptotic factor of exactly 27 every two dimensions, with a clean first-order correction, a sharp asymptotic fingerprint that calls for the dominant singularity of the generating function of  $\Sigma_n$ ; the statement carries a limit clause and an attribution, since singularity analysis is classical once the generating function is extracted. A first decisive theorem would be the limit alone. In tests on the corpus described in the stress-test section, all 29 deposited exact ratio comparisons through  $n = 60$  passed, reproduced independently with  $R_{58} = 26.55045876$  and gap  $0.44954124$  against  $27/60 = 0.45$ . A counterexample would be a ratio at least 27, a decrease, or a different correction rate. Since deposit the conjecture has been resolved true, by proof rather than by computation: Proposition 31 shows the ladder's generating function is algebraic, and the resulting exact recurrence collapses the ratio dynamics to a decreasing map interlacing the explicit rational sequence  $q_m < 27$ , which forces boundedness, monotonicity, and the  $27/n$  rate outright; matching orders in the proved recurrence sharpens the limit clause to  $R_n = 27(1 - 1/n + \frac{37}{18}n^{-2} - \frac{281}{72}n^{-3} + O(n^{-4}))$ . Nor is the mechanism particular to quadrics: Proposition 32 extends it to every repeated multidegree block and identifies the constant as  $27 = \cot^6(\pi/6)$ , the first member of the trigonometric family  $\Lambda_d = \cot^{2(d+1)}(\pi/(2(d+1)))$ .

**Proposition 31** (Resolution of the quadric ratio law). Conjecture 195 holds in full. With  $a_m = \Sigma_{2m}$  and  $a_0 = 2$ , the generating function  $A(w) = \sum_{m \geq 0} a_m w^m$  is algebraic,

$$A(w) = \frac{2}{(1+y)(1-3y)}, \quad 4w = y(1-y)^2, \quad y(0) = 0,$$

and the coefficients satisfy  $D_m a_{m+2} = (M_m - D_m) a_{m+1} + M_m a_m$  with  $D_m = (m+2)(2m+3)(14m+9)$  and  $M_m = 6(3m+2)(3m+4)(14m+23)$ .



*Proof.* For the all-quadric configuration in complex dimension  $n = 2m$  there are  $2m + 1$  defining quadrics, so the ambient dimension is  $N = 4m + 1$ , and the classical signature formula  $\sigma = [x^N](x/\tanh x)^{N+1} \prod_i \tanh(d_i x)$  for complete intersections in  $\mathbb{CP}^N$  [117] gives  $\sigma = [x^{4m+1}](x/\tanh x)^{4m+2} \tanh(2x)^{2m+1}$ . Setting  $f(x) = x \tanh(2x)/\tanh^2 x$ , an even power series with  $f(0) = 2$ , the expression under the coefficient equals  $x^{2m+1} f(x)^{2m+1}$ , so  $\sigma = [x^{2m}] f(x)^{2m+1}$ . Because  $f$  is even, replacing  $x^2$  by  $-z$  multiplies the coefficient of  $x^{2m}$  by  $(-1)^m$  and turns  $f$  into  $\psi(z) = \sqrt{z} \tan(2\sqrt{z})/\tan^2 \sqrt{z}$ , analytic on  $|z| < \pi^2/4$ ; hence  $a_m = (-1)^m \sigma = [z^m] \psi(z)^{2m+1}$ . For each small  $w > 0$  the equation  $z = w\psi(z)^2$  has exactly one root  $z_0(w)$  near the origin, and the residue form of Lagrange inversion, summing the geometric series  $\sum_m (w\psi(z)^2/z)^m$  on a small contour and picking up the simple pole at  $z_0$ , gives

$$A(w) = \sum_{m \geq 0} [z^m] \psi(z)^{2m+1} w^m = \frac{\psi(z_0)}{1 - w(\psi^2)'(z_0)}.$$

Substituting  $z = u^2$  and  $T = \tan u$  and using the double-angle formula  $\tan 2u = 2T/(1 - T^2)$ , the root equation becomes  $4w = T^2(1 - T^2)^2$  and the residue value becomes  $2/((1 + T^2)(1 - 3T^2))$ , two exact identities; with  $y = T^2$  this is the stated parametrization.

Next, the recurrence. Write  $\theta = w d/dw$ , let  $D(\theta)$  and  $M(\theta)$  denote the polynomials  $D_m$  and  $M_m$  with  $m$  replaced by the operator  $\theta$ , and put  $B_s = w^{-s}(A - \sum_{i < s} a_i w^i)$ , the generating function of the shifted sequence  $(a_{m+s})_{m \geq 0}$ . Then

$$\sum_{m \geq 0} (D_m a_{m+2} - (M_m - D_m) a_{m+1} - M_m a_m) w^m = D(\theta) B_2 - (M(\theta) - D(\theta)) B_1 - M(\theta) B_0,$$

and under the parametrization, where  $\theta$  acts as  $\frac{y(1-y)}{1-3y} \frac{d}{dy}$ , the right side is a rational function of  $y$  that vanishes identically, a finite identity checked by exact expansion. So the recurrence holds for every  $m \geq 0$ .

Finally, the three clauses. Set  $r_m = a_{m+1}/a_m = R_{2m}$  and  $q_m = M_m/D_m$ , so the recurrence reads  $r_{m+1} = q_m - 1 + q_m/r_m$ . First,  $27 - q_m = 3(126m^2 + 271m + 118)/D_m > 0$  for every  $m$ , so  $q_m < 27$ . Second, the map  $g_m(r) = q_m - 1 + q_m/r$  is strictly decreasing on  $r > 0$  and fixes  $q_m$ , and the interlacing  $q_{m-1} < r_m < q_m$  holds for all  $m \geq 1$  by induction: the base is  $q_0 = \frac{184}{9} < 22 = r_1 < \frac{518}{23} = q_1$ ; if  $r_m < q_m$ , then, because  $g_m$  is decreasing,  $r_{m+1} = g_m(r_m) > g_m(q_m) = q_m$ ; and if  $r_m > q_{m-1}$ , then  $r_{m+1} < g_m(q_{m-1}) < q_{m+1}$ , the last inequality clearing denominators to a polynomial of degree seven in  $m - 1$  with positive coefficients. The interlacing gives  $r_m < q_m < 27$  and  $r_{m+1} > q_m > r_m$ , the boundedness and monotonicity clauses; and since  $27 - q_m \sim 27/(2m)$ , the squeeze  $27 - q_m < 27 - r_m < 27 - q_{m-1}$  gives  $27 - R_n \sim 27/n$  for  $n = 2m$ , the limit clause.  $\square$

**Proposition 32** (Block ladders and their saddle constants). Let  $B = (d_1, \dots, d_s)$  be a multidegree block with  $A = \sum_j (d_j - 1)$  odd, and for  $m \geq 0$  let  $S_m(B)$  be the sign-normalized signature of the  $(2m + 1)$ -fold repetition of  $B$ , a Calabi–Yau complete intersection of complex dimension  $n_m = A(2m + 1) - 1$ . Then

$$S_m(B) = [y^{Am+(A-1)/2}] \frac{C_B(y)^{2m+1}}{1+y}, \quad C_B(y) = \prod_{j=1}^s \frac{\tan(d_j \arctan \sqrt{y})}{\sqrt{y}},$$

where the factor with entry  $d$  equals  $\sum_i (-1)^i \binom{d}{2i+1} y^i / \sum_i (-1)^i \binom{d}{2i} y^i$ , a rational function, so the generating function  $\sum_m S_m(B) w^m$  is algebraic. The series  $C_B$  has nonnegative aperiodic

coefficients; there is a unique  $y_* > 0$  below the first positive pole of  $C_B$  with  $y_* C'_B(y_*)/C_B(y_*) = A/2$ , and with  $\Lambda_B = C_B(y_*)^2/y_*^A$ ,

$$\frac{S_{m+1}(B)}{S_m(B)} = \Lambda_B \left(1 - \frac{1}{2m} + O(m^{-2})\right),$$

so the ratios are eventually strictly increasing with limit  $\Lambda_B$ , and  $\Lambda_B - S_{m+1}/S_m \sim \Lambda_B A/n_m$ . For the equal-degree block  $B = (d)$  with  $d$  even,  $y_* = \tan^2 \frac{\pi}{2(d+1)}$  and

$$\Lambda_d = \cot^{2(d+1)} \left( \frac{\pi}{2(d+1)} \right),$$

so  $\Lambda_2 = 27$  recovers Proposition 31; and for the Euler ladders with  $r$  equations of equal degree  $d$ , the analogous saddle sits at  $t_* = 1/(d+1)$  and the ratio limit is  $d^{d+1}$ , recovering the limit  $8 = 2^3$  of Proposition 16 at  $d = 2$ .

*Proof.* For  $k = 2m + 1$  repetitions there are  $sk$  defining equations of total degree  $(A + s)k$ , so the ambient dimension is  $N = (A + s)k - 1$ , and the signature formula reads  $\sigma = [x^N](x/\tanh x)^{N+1} \prod_j \tanh(d_j x)^k$ . Write the coefficient as a contour integral and substitute  $x = it$ , then  $T = \tan t$ , then  $y = T^2$ . The powers of  $i$  collect to exactly the sign  $(-1)^{n_m/2}$  that the normalization  $S = (-1)^{n_m/2} \sigma$  removes; the Jacobian  $dt = dT/(1 + T^2)$  contributes the factor  $1/(1 + y)$ ; the map  $T \mapsto T^2$  covers the  $y$ -contour twice, which cancels the half from  $dT = dy/(2\sqrt{y})$  since the integrand is even in  $T$ ; and what remains of the integrand is  $C_B(y)^k y^{-(Ak+1)/2}/(1 + y)$ . Reading off the residue at  $y = 0$  gives the extraction identity. Each factor of  $C_B$  is rational by the tangent multiple-angle formula; writing  $C_B = P/Q$  with  $P$  and  $Q$  the products of the factor numerators and denominators, the residue argument of Proposition 31, applied to the root equation  $y^A Q(y)^2 = w P(y)^2$ , shows the generating function is algebraic.

For the saddle law, note first that the coefficients of each factor are nonnegative and aperiodic, meaning that the indices of the nonzero coefficients have greatest common divisor one: this follows by induction from  $C_1 = 1$  and the tangent addition recursion  $C_{d+1} = (C_d + 1)/(1 - yC_d)$ , whose right side is a series with nonnegative coefficients in  $y$  and  $C_d$ , and both properties persist under products. Consequently  $\mu(y) = yC'_B(y)/C_B(y)$ , the average index of the coefficients of  $C_B$  when the coefficient of  $y^j$  is weighted by  $y^j$ , is strictly increasing, starts at 0, and diverges at the first positive pole of  $C_B$ , so the saddle  $y_*$  exists and is unique. The large-powers saddle-point method [118] then gives  $S_m(B) = K_B \Lambda_B^m m^{-1/2} (1 + O(1/m))$  with  $K_B > 0$ , together with a full expansion in powers of  $1/m$ , and taking the ratio of consecutive terms gives the displayed law; the dimension form follows from  $n_m = 2Am + A - 1$ .

For the equal-degree block  $B = (d)$ , write  $y = \tan^2 \theta$ , so that  $C_d = \tan(d\theta)/\tan \theta$ . The saddle equation becomes  $\sin(2d\theta) = \sin(2\theta)$ , whose first positive solution is  $\theta_* = \pi/(2(d+1))$ ; there  $d\theta_* = \pi/2 - \theta_*$ , so  $\tan(d\theta_*) = \cot \theta_*$ , hence  $C_d(y_*) = \cot^2 \theta_*$  and  $\Lambda_d = C_d(y_*)^2/y_*^{d-1} = \cot^4 \theta_*/\tan^{2(d-1)} \theta_* = \cot^{2(d+1)} \theta_*$ . For the Euler ladders, the Chern-class manipulation of Proposition 16 generalizes to  $E_{d,r} = [t^{(d-1)r-1}] \varphi_d(t)^r$  with  $\varphi_d(t) = d(1-t)^d/(1-dt)$ ; the saddle equation  $\varphi'_d/\varphi_d = (d-1)/t$  has the exact solution  $t_* = 1/(d+1)$ , and  $\varphi_d(t_*)/t_*^{d-1} = d^{d+1}$ .  $\square$

**Remark 21** (Open question). *Not counted among the conjectures: whether every signature block ladder is strictly increasing and below its saddle constant from the first ratio onward, as Conjecture 195 asserted for quadrics. Proposition 32 proves this eventually, and for the equal-degree ladders the question has since acquired exact structure. The function  $C_d$  satisfies the Riccati identity  $2y(1+y)C'_d = dyC_d^2 - (1+y)C_d + d$ , an elementary consequence of the angle*

substitution, and, verified through degree eight, equals  $d$  times a terminating Stieltjes continued fraction with positive coefficients  $\alpha_j = (d^2 - j^2)/(4j^2 - 1)$  for  $1 \leq j \leq d - 1$ . From the Riccati identity, the function  $h_d(y) = y^{d-1}/C_d(y)^2$  obeys the exact factorization

$$\frac{h'_d(y)}{h_d(y)} = \frac{d(C_d(y) - 1)(1 - yC_d(y))}{y(1 + y)C_d(y)},$$

so every positive critical point of  $h_d$  solves  $C_d(y) = 1$  or  $yC_d(y) = 1$ , the two explicit families  $y = \tan^2(k\pi/(d - 1))$  and  $y = \tan^2((2k + 1)\pi/(2(d + 1)))$ , and on the second family  $h_d(y) = y^{d+1}$ : the constant  $\Lambda_d$  is therefore the exact reciprocal of the first critical value of  $h_d$ , not a saddle estimate. Moreover  $S_m^{(d)} = d^{2m+1}b_{(d-1)m+(d-2)/2}^{(d)}$  for the single coefficient sequence  $b_n^{(d)} = [y^n] (C_d(y)/d)^{(2n+1)/(d-1)}/(1 + y)$ , and at the first critical point  $r = \tan^2(\pi/(2(d + 1)))$  the functions  $P_d(z) = rC_d(rz)$  and  $Y_d(z) = (1 + r)P_d(z)/(1 + rz)$  are probability generating functions, the first with mean exactly  $(d - 1)/2$ , in terms of which  $R_m^{(d)}/\Lambda_d = p_{m+1}/p_m$  for the central probabilities  $p_m = [z^{(d-1)m+(d-2)/2}] Y_d(z)P_d(z)^{2m}$ ; the open question for degree  $d$  says exactly that  $p_m$  decreases strictly and is strictly log-convex. The boundary is calibrated:  $b_n^{(2)} = 1, 2, 11, 62, 367, \dots$  is not a Stieltjes moment sequence, its once-shifted Hankel determinant of order three being negative and its unshifted determinants turning negative at order nine, so no moment-theoretic or fully Hankel-positive route can succeed, while the order-two inequality  $b_{n-1}b_{n+1} > b_n^2$  survives every test, through  $n = 38$  at  $d = 2$  and  $n = 23$  at  $d = 4$ . The property also persists at the opposite boundary: as  $d \rightarrow \infty$  the normalized ratios appear to tend, for fixed  $m$ , to  $e^{-2}(2m + 3)^{2m+2}/((2m + 1)^{2m+1}(2m + 2))$ , itself strictly increasing to 1, and the exact ladders at  $d = 100$  and  $d = 200$  approach that limit monotonically. In exact scans, the first twenty-five ratios of every even equal-degree ladder through  $d = 30$  and the early ratios of hundreds of mixed blocks showed no violation, and an independent recomputation of sixteen blocks at five exact ratios each found none; the Euler ladder at  $d = 2$ , whose early ratios dip before the monotone regime of Conjecture 117 sets in, shows that such global monotonicity is not automatic.

**Conjecture 196** (Dimension-wise signature gcd law). *Let the gcd range over all Calabi–Yau complete-intersection multidegrees in even complex dimension  $n$ . Then*

$$\gcd_{\mathbf{d}} S(\mathbf{d}) = \begin{cases} 2, & n \equiv 0 \pmod{4}, \\ 2^{\max(4, 1+\nu_2(n+2))}, & n \equiv 2 \pmod{4}, \end{cases}$$

where  $\nu_2$  is the 2-adic valuation.

**Significance.** This is the signature sibling of Conjecture 121’s Euler gcd law: a universal divisibility constraint on index data across each dimension, with an unexpectedly rigid 2-adic dependence on  $n + 2$  whose spike  $2^5$  at  $n = 14$  the census confirms exactly, and whose  $n \equiv 2 \pmod{4}$  floor  $2^4$  contains Rokhlin’s theorem at  $n = 2$ ; free group actions and anomaly normalizations inherit the constraint. A proof route is a 2-adic congruence in the L-genus characteristic series followed by a small family attaining the gcd. A first decisive theorem would be the divisibility direction alone. In tests on the corpus described in the stress-test section, the exact gcd over the full census matched the formula in every even dimension  $2 \leq n \leq 26$ , all thirteen values reproduced independently. A counterexample would be one even dimension with a different gcd.

## Part XI

# Spectral entropies and multibase carry fields

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The twenty conjectures below pair a spectral programme on graphs with a return to the integers. An entropy programme (Conjectures 197 to 201) reads the Rényi entropies of the trace-normalized graph Laplacian, the density-matrix reading of a graph introduced in [119], across tree and unicyclic operations, asserting that a small family of local moves is entropy-monotone at every order at once. A metric programme (Conjectures 202 to 207) poses the same functionals for the distance signless Laplacian of [120], an object whose spectral theory is extensively surveyed in [121] but whose normalized entropy appears unstudied. A carry programme (Conjectures 208 to 216) returns to the local–global aesthetic of Part I on a new object: the carry fields  $V_p(n) = v_p(\binom{2n}{n})$  at distinct primes, finite-state functionals of the base- $p$  digits by Kummer’s theorem, for which complete cross-base decoupling is conjectured at Gaussian, local, modular, large-deviation, and cumulant resolution. The calibration layer is proved where it can be: the degree identity at Rényi order two with its tree and leaf-move corollaries and the complete-graph majorization law from Proposition 33, and the exact single-base carry-chain law forms Proposition 34. The adversarial controls marking the genuine boundaries of the entropy windows are retained in place. The spectral scans are floating-point eigensolver checks, with the smallest margins near  $10^{-4}$ ; the carry statements are limit laws, with Conjecture 216, the transducer principle, the part’s boldest statement; and Conjectures 197 and 208 sit closest to [122] and [123].

For a finite graph  $G$  with  $m \geq 1$  edges, let  $L(G)$  be the combinatorial Laplacian and  $\rho_L(G) = L(G)/(2m)$  its trace normalization, a density matrix whose eigenvalues  $p_1, \dots, p_n$  are nonnegative with unit sum; for  $\alpha > 0$ ,  $\alpha \neq 1$ , the Rényi entropy is  $H_\alpha^L(G) = (1 - \alpha)^{-1} \log \sum_i p_i^\alpha$ , with the Shannon limit  $H_1^L = -\sum_i p_i \log p_i$  at  $\alpha = 1$  and min-entropy  $H_\infty^L = -\log \max_i p_i$  at  $\alpha = \infty$ , zero eigenvalues contributing nothing. For a connected  $G$ , let  $\mathcal{D}$  be the distance matrix,  $t(v)$  the transmission  $\sum_u d(u, v)$ ,  $W(G) = \frac{1}{2} \sum_v t(v)$  the Wiener index, and  $Q_D(G) = \text{diag}(t) + \mathcal{D}$  the distance signless Laplacian [120], positive definite for connected graphs on at least three vertices by the identity  $x^T Q_D x = \sum_{i < j} d(i, j)(x_i + x_j)^2$ , with trace  $2W$ ; write  $H_\alpha^Q$  for the Rényi entropy of  $Q_D/(2W)$ . On the arithmetic side, for an odd prime  $p$  put  $V_p(n) = v_p(\binom{2n}{n})$ , by Kummer’s theorem the number of carries in the base- $p$  addition of  $n + n$ , a bounded-output finite-state functional of the digits that is not  $p$ -additive, since each output depends on the incoming carry state;  $\mu_p(N)$  and  $\sigma_p^2(N)$  denote the mean and variance of  $V_p$  over uniform  $n < N$ .

**Proposition 33** (Degree and majorization calibrations). (i) For every graph with  $m$  edges and degrees  $d_i$ ,  $H_2^L(G) = -\log((2m + \sum_i d_i^2)/(2m)^2)$ ; consequently, among trees of a fixed order the path uniquely maximizes  $H_2^L$ , and if a leaf  $x$  adjacent to  $u$  is moved to a neighbour  $v$  of  $u$  with  $d(v) \geq d(u)$ , then  $H_2^L$  strictly decreases. (ii) For every connected  $n$ -vertex graph  $G$  and every  $\alpha > 0$ ,  $H_\alpha^Q(G) \leq H_\alpha^Q(K_n)$ . (iii) For a tree  $T$  on  $n \geq 3$  vertices, joining two leaves or subdividing one edge strictly increases  $H_2^L$ , and each move also strictly increases  $H_\alpha^L$  for all sufficiently small  $\alpha > 0$ ; and among trees of a fixed order the path uniquely maximizes  $H_\infty^L$ .

*Proof.* (i)  $\text{tr } L^2 = \sum_i d_i^2 + 2m$ , since the diagonal contributes  $\sum d_i^2$  and the off-diagonal entries contribute 1 for each of the  $2m$  ordered adjacent pairs, and  $H_2^L = -\log(\text{tr } L^2/(2m)^2)$ . Among trees  $m = n - 1$  is fixed, and the path is the unique tree minimizing  $\sum d_i^2$ . The leaf move changes

$\sum d_i^2$  by  $(d_u - 1)^2 + (d_v + 1)^2 - d_u^2 - d_v^2 = 2(d_v - d_u + 1) > 0$ . (ii) The all-ones vector gives  $\lambda_{\max}(Q_D) \geq \mathbf{1}^T Q_D \mathbf{1}/n = 4W/n$ , so the largest normalized eigenvalue of any connected graph is at least  $2/n$ . The normalized spectrum of  $K_n$  is  $(2/n, (n-2)/(n(n-1)), \dots)$  with a flat tail, and a probability vector whose largest entry is at least  $2/n$  majorizes every such flat-tail vector: if the top entry is  $p_1 \geq 2/n$ , the  $k$ -th partial sum exceeds the flat-tail partial sum by at least  $(p_1 - 2/n)(1 - (k-1)/(n-1)) \geq 0$  plus the excess of the remaining entries over their flat minimum. Schur concavity of Rényi entropy finishes. (iii) Write  $A = \text{tr } L^2 = 2(n-1) + \sum_i d_i^2$ , so  $A \geq 6n - 8$  with equality for the path. Joining two leaves gives  $\text{tr } L'^2 = A + 8$  on  $n$  edges, and  $H_2^L$  increases exactly when  $8(n-1)^2 < A(2n-1)$ , which holds since  $(6n-8)(2n-1) - 8(n-1)^2 = 2n(2n-3) > 0$ ; subdividing an edge gives  $\text{tr } L'^2 = A + 6$  on  $n$  edges, and the required  $6(n-1)^2 < A(2n-1)$  holds since  $(6n-8)(2n-1) - 6(n-1)^2 = 6n^2 - 10n + 2 > 0$ . For the smallest orders, the matrix-tree theorem gives  $\prod \lambda_i = n\tau$  over positive eigenvalues: subdivision raises the number of positive eigenvalues from  $n-1$  to  $n$ , and closure into a cycle of length  $c \geq 3$  keeps that number while raising the mean of  $\log(\lambda_i/2m)$  by  $(\log c - (n-1) \log \frac{n}{n-1})/(n-1) > 0$ , since  $(n-1) \log \frac{n}{n-1} < 1 < \log 3$ , so expanding  $H_\alpha$  at  $\alpha = 0$  finishes both. For min-entropy, a non-path tree has a vertex of degree  $d \geq 3$ , and the vector equal to  $d$  there,  $-1$  on its neighbours, and  $0$  elsewhere has Rayleigh quotient at least  $d+1 \geq 4$ , while  $\lambda_{\max}(P_n) = 2 + 2\cos(\pi/n) < 4$ ; equal traces  $2(n-1)$  convert this into  $H_\infty^L(P_n) > H_\infty^L(T)$ .  $\square$

**Proposition 34** (Single-base carry-chain law). For uniform base- $p$  digits, the carry sequence in the addition  $n + n$  is a two-state Markov chain with  $P(0 \rightarrow 1) = (p-1)/(2p)$ ,  $P(1 \rightarrow 1) = (p+1)/(2p)$ , stationary carry probability  $\frac{1}{2}$ , and second eigenvalue  $1/p$ ; the stationary variance per digit is  $(p+1)/(4(p-1))$ ; and over the exact block  $0 \leq n < p^k$ ,

$$\mathbb{E}V_p = \frac{k}{2} - \frac{1-p^{-k}}{2(p-1)}.$$

*Proof.* With carry in  $c$  and uniform digit  $a$ , the carry out is 1 exactly when  $2a + c \geq p$ , giving the two transition rows, whose difference makes the second eigenvalue  $(p+1)/(2p) - (p-1)/(2p) = 1/p$  and whose symmetry fixes the stationary law at  $\frac{1}{2}$ . The stationary autocovariance of the carry indicator at lag  $k$  is  $\frac{1}{4}p^{-k}$ , so the variance per digit is  $\frac{1}{4}(1 + 2\sum_{k \geq 1} p^{-k}) = (p+1)/(4(p-1))$ . Over the exact block the digits are independent and uniform, so the carry probability after  $j$  steps satisfies  $\pi_{j+1} = (p-1)/(2p) + \pi_j/p$  with  $\pi_0 = 0$ , giving  $\pi_j = (1-p^{-j})/2$ , and summing  $j = 1$  to  $k$  gives the displayed mean.  $\square$

## 46 Rényi entropies of the Laplacian density matrix

**Conjecture 197** (All-order path maximality). For every  $n \geq 3$ , every  $n$ -vertex tree  $T \neq P_n$ , and every  $\alpha > 0$ ,  $H_\alpha^L(T) < H_\alpha^L(P_n)$ .

**Significance.** The Rényi orders form a continuum of Schur-concave probes of the normalized spectrum, and the conjecture asserts that the path is the most mixed tree under every one of them simultaneously, from the support-sensitive small orders through Shannon entropy to the min-entropy of the spectral radius; different orders can favour different spectra, so this is stronger than any single-functional extremality. The order-two and min-entropy cases are proved in Proposition 33, and near order zero the matrix-tree theorem forces every tree to share its mean log-eigenvalue, so the low-order content is that the path maximizes the variance of the

logarithms of its normalized spectrum; the nearest literature is the star-minimum conjecture and Rényi results of [122], whence the attribution; a proof route is a majorization theorem along a refinement of the tree-shift poset of [124]. A first decisive theorem would be the law for caterpillars. In tests on the corpus described in the stress-test section, all 8,775 deposited comparisons through twelve vertices at nine Rényi orders and an independent scan through eleven vertices passed, with smallest margin  $2.23 \times 10^{-4}$ . A counterexample would be one tree and one order beating the path.

**Conjecture 198** (Leaf-pair closure law). *Let  $T$  be a tree on at least three vertices and  $u, v$  distinct leaves. Then  $H_\alpha^L(T + uv) > H_\alpha^L(T)$  for every  $\alpha > 0$ .*

**Significance.** Arbitrary edge addition can lower these entropies, a phenomenon exhibited already in [122], so the leaf-pair restriction carries real content: closing the unique leaf-to-leaf path into a cycle raises degrees only at two spectrally peripheral vertices and removes one bottleneck, spreading Laplacian mass at every order. The order-two case and the smallest orders are now proved in Proposition 33, promotions; a first decisive theorem on the remaining range would be the Shannon case. In tests on the corpus described in the stress-test section, all 86,982 deposited leaf-pair comparisons through twelve vertices and an independent scan of 14,252 through ten vertices passed, with smallest margin  $1.66 \times 10^{-3}$ . A counterexample would be one tree and one leaf pair with nonincreasing entropy, most plausibly with leaves on very unequal branches.

**Conjecture 199** (Edge-subdivision law). *If  $T'$  is obtained by subdividing one edge of a tree  $T$ , then  $H_\alpha^L(T') > H_\alpha^L(T)$  for every  $\alpha > 0$ .*

**Significance.** Subdivision inserts a degree-two vertex while preserving the branching skeleton, a controlled spectral refinement rather than an arbitrary enlargement, and the conjecture asserts the refined normalized spectrum is strictly more mixed at every order; the comfortable margins suggest an interlacing-plus-normalization proof. The order-two case and the smallest orders are now proved in Proposition 33, promotions; a first decisive theorem on the remaining range would be the min-entropy case. In tests on the corpus described in the stress-test section, all 70,126 deposited subdivision comparisons through twelve vertices and 11,249 independent ones passed, with smallest margin  $6.12 \times 10^{-2}$ , the largest in the entropy programme. A counterexample would be a broom subdivided at its hub failing at high order.

**Conjecture 200** (Leaf-compression window). *Let  $x$  be a leaf adjacent to  $u$  in a tree  $T$ , let  $v \neq x$  be a neighbour of  $u$  with  $d(v) \geq d(u)$ , and let  $T'$  be the tree with  $xu$  deleted and  $xv$  added. If  $T' \not\cong T$ , then  $H_\alpha^L(T') < H_\alpha^L(T)$  for all  $1/2 \leq \alpha \leq 2$ .*

**Significance.** The move concentrates local degree mass onto a no-smaller vertex, and at order two the decrease is the theorem of Proposition 33; the conjecture is that the concentration order persists down to order one half, and the lower boundary is calibrated, not cautious: an explicit order-12 tree violates the same direction at order 0.1 by  $7.65 \times 10^{-7}$ , a control reproduced independently to three digits, so the window cannot extend to all orders; no structural role is claimed for one half itself, a tested safe bound. A first decisive theorem would locate the true critical order. In tests on the corpus described in the stress-test section, all 15,385 deposited window comparisons and an independent scan through ten vertices passed, with smallest margin  $9.31 \times 10^{-4}$ . A counterexample would be one eligible move increasing entropy inside the window.



**Conjecture 201** (Unicyclic endpoint law). *Among connected unicyclic graphs on  $n \geq 4$  vertices, for every finite  $\alpha > 0$ , the graph  $S_n^+$  obtained from the star by joining two leaves uniquely minimizes  $H_\alpha^L$  and the cycle  $C_n$  uniquely maximizes it.*

**Significance.** The cycle is the maximally homogeneous unicyclic graph and  $S_n^+$  concentrates degree as far as one cycle permits, and the conjecture asserts that these degree extremes remain the entropy endpoints under every finite Rényi lens. The restriction to finite orders is a calibrated boundary found in the present verification: at  $n = 4$  the only two unicyclic graphs,  $C_4$  and  $S_4^+$ , have Laplacian spectra  $\{0, 2, 2, 4\}$  and  $\{0, 1, 3, 4\}$  with equal trace and equal largest eigenvalue, so their min-entropies tie exactly and the strict law fails at  $\alpha = \infty$ ; the control is retained in place. Near order zero the minimization half is delicate, since all triangle-cycled unicyclic graphs share their mean log-eigenvalue by the matrix-tree theorem, so uniqueness of  $S_n^+$  there rests on second-order log-spectral information, while the cycle half, with its maximal spanning-tree count, is first-order automatic. A first decisive theorem would be either endpoint at the Shannon order. In tests on the corpus described in the stress-test section, all 6,230 deposited comparisons and an independent exhaustive scan through seven vertices passed at every tested finite order. A counterexample would be a third extremal graph at some finite order.

## 47 Entropy of the distance signless Laplacian

**Conjecture 202** (Tree one-vertex extension law). *For every tree  $T$  and every  $\alpha > 0$ , attaching a pendant vertex anywhere, or subdividing any edge once, strictly increases  $H_\alpha^Q$ .*

**Significance.** Both operations enlarge the metric state space and the Wiener normalization, and in a tree every altered distance passes through a unique path, a global coherence that the conjecture converts into entropy growth at every order, preventing the Perron mode from ever dominating disproportionately. This is the first entropy law posed for the normalized  $Q_D$  spectrum, whose unnormalized spectral theory is surveyed in [121]. The smallest orders are automatic, since  $Q_D$  is positive definite and both operations raise the support count. A first decisive theorem would be the min-entropy case, a normalized spectral-radius decrease adaptable to graft-transformation tools [125]. In tests on the corpus described in the stress-test section, all 168,168 deposited comparisons through twelve vertices and an independent scan through ten vertices passed, with minimum margin  $4.44 \times 10^{-2}$ . A counterexample would be one tree, one operation, one order.

**Conjecture 203** (Low-order extension law). *For every connected graph  $G$  and every  $0 < \alpha \leq 2$ , pendant attachment and single edge subdivision strictly increase  $H_\alpha^Q$ .*

**Significance.** The general-graph shadow of Conjecture 202, with a substantive cutoff: at high orders the entropy is governed by the largest normalized eigenvalue, and metric extensions of graphs with many routes can sharpen that mode. The boundary is calibrated by retained controls: subdivision decreases  $H_5^Q$  in an explicit order-30 graph by  $1.00 \times 10^{-2}$  and pendant attachment decreases  $H_{10}^Q$  in an order-15 graph by  $2.32 \times 10^{-2}$ , both reproduced independently. Whether 2 is the exact universal cutoff is the sharpest open edge; it is an endpoint of the tested window rather than an explained critical exponent. A first decisive theorem would be the order-two case, an explicit second-moment inequality. In tests on the corpus described in the stress-test section,

all 10,884 deposited window comparisons and 7,672 independent ones passed. A counterexample at order two would destroy the clean boundary.

**Conjecture 204** (Star maximality). *For every  $n \geq 3$ , every  $n$ -vertex tree  $T \neq K_{1,n-1}$ , and every  $\alpha > 0$ ,  $H_\alpha^Q(T) < H_\alpha^Q(K_{1,n-1})$ .*

**Significance.** The metric programme reverses the Laplacian intuition: for the distance signless matrix the star, not the path, is conjectured to carry the most mixed normalized tree spectrum at every order, metric compression flattening the transmission profile after normalization by total distance. Together with Conjecture 197 this makes the two programmes genuinely different orders on trees. Since positive definiteness gives every tree  $H_{0+}^Q = \log n$ , the near-zero form of the law is the exact determinant inequality that the star maximize  $\det Q_D / (2W)^n$  among trees. A first decisive theorem would be that determinant law or the min-entropy case, a normalized spectral-radius minimization. In tests on the corpus described in the stress-test section, all 8,775 deposited comparisons through twelve vertices and an independent scan through eleven vertices passed, with smallest margin  $4.72 \times 10^{-4}$ . A counterexample would be one tree and one order beating the star, most plausibly as  $\alpha \rightarrow \infty$ .

**Conjecture 205** (Bipartite path minimality at the Shannon order, resolved false). *Among connected bipartite graphs on  $n$  vertices,  $P_n$  uniquely minimizes  $H_1^Q$ .*

**Significance.** Trace normalisation removes total scale. The claim is that the path's extreme metric anisotropy leaves the normalised spectrum most concentrated even against denser bipartite competitors. The Shannon restriction is deliberate because an all-order strengthening already fails among trees at order two. The deposited comparisons were exhaustive through seven vertices and included holdouts through thirty vertices. *Resolution.* The conjecture is false. The deposited fourteen-vertex tree has a degree-five hub carrying five leaves at diameter ten, and its Shannon entropy lies  $3.1535496 \times 10^{-5}$  below the path. This gap is certified by interval arithmetic at one hundred digits. A further exhaustive enumeration of all 3,159 unlabelled trees on fourteen vertices found a sharper witness  $T_*$  on vertices  $0, \dots, 13$  with edge set

$$\{\{0, 1\}, \{1, 2\}, \{0, 5\}, \{2, 3\}, \{3, 4\}, \{5, 6\}\} \cup \{\{6, j\} : 7 \leq j \leq 13\}.$$

Its degree multiset is  $(8, 2, 2, 2, 2, 2, 1, 1, 1, 1, 1, 1, 1, 1)$ , and a sixty-digit computation gives

$$H_1^Q(P_{14}) - H_1^Q(T_*) = 9.10088461876217 \times 10^{-5}.$$

The interval-certified tree remains the formal certificate. The sharper computation shows that it is not extremal even among trees of the same order. The path may remain minimal in narrower classes such as bounded-degree trees or caterpillars. Identifying the true bipartite minimiser is the surviving problem.

**Conjecture 206** (Balanced-biclique window maximality). *Among connected bipartite graphs on  $n$  vertices, for every  $1 \leq \alpha \leq 2$ ,  $H_\alpha^Q(G) \leq H_\alpha^Q(K_{\lfloor n/2 \rfloor, \lceil n/2 \rceil})$ , with equality only for the balanced complete bipartite graph.*

**Significance.** Completing the bipartite metric collapses all distances to two classes and balancing the sides equalizes the transmissions as far as parity permits, and the conjecture asserts this metric regularization maximizes a whole entropy window rather than one moment. A first decisive theorem would be the order-two case, where the entropy is the explicit functional  $-\log(\text{tr } Q_D^2 / (2W)^2)$  and the claim becomes an extremal inequality between transmissions and squared distances. In tests on the corpus described in the stress-test section, all 5,322 deposited window comparisons and 195 independent exhaustive ones passed, with smallest margin  $1.60 \times 10^{-3}$ . A counterexample would be one bipartite graph and one window order above the balanced biclique.

**Conjecture 207** (Unicyclic cycle maximality above the Shannon order). *Among connected unicyclic graphs on  $n \geq 4$  vertices,  $C_n$  uniquely maximizes  $H_\alpha^Q$  for every  $\alpha \geq 1$ , including  $\alpha = \infty$ .*

**Significance.** The cycle is transmission-regular, the most homogeneous unicyclic metric, and from the Shannon order upward that homogeneity is conjectured to dominate the normalized spectral shape. The lower boundary is calibrated, not cautious: at  $n = 4$  a noncycle unicyclic graph exceeds the cycle at order 0.1 by  $6.62 \times 10^{-4}$ , a retained control reproduced independently to three digits, so the law genuinely fails below some critical order. A first decisive theorem would be the Shannon case; locating the critical order  $\alpha_c(n)$  is the sharper target. In tests on the corpus described in the stress-test section, all 2,245 deposited comparisons and an independent exhaustive scan through seven vertices passed on the asserted range. A counterexample would be a noncycle maximizer at some order at least one.

## 48 Multibase carry fields

**Conjecture 208** (Cross-prime carry central limit law). *Fix distinct odd primes  $p_1, \dots, p_r$ . For  $n$  uniform on  $\{0, \dots, N-1\}$ , the vector with coordinates  $(V_{p_i}(n) - \mu_{p_i}(N)) / \sigma_{p_i}(N)$  converges in distribution to a standard  $r$ -dimensional Gaussian, with  $\mu_p(N) = \frac{1}{2} \log_p N + O_p(1)$  and  $\sigma_p^2(N) = \frac{p+1}{4(p-1)} \log_p N + O_p(1)$ .*

**Significance.** Each carry field is generated by the two-state automaton of Proposition 34 reading base- $p$  digits, and distinct primes give multiplicatively independent digit partitions of the same integers; the conjecture asserts that after normalization the only persistent dependence is within a base. It sits closest to the different-base theorem of [123] for  $q$ -additive functions, whence the attribution, but carries are state-dependent transducer outputs, not additive functions, which is exactly why that theorem does not settle it; the single-prime boundary is [126]. A first decisive theorem would be two primes via a joint transfer-operator estimate. In tests on the corpus described in the stress-test section, marginals at  $N = 2 \times 10^6$  track the proved chain law and all pairwise correlations among  $p \in \{3, 5, 7, 11\}$  lie below 0.08, independently below 0.04. A counterexample would be a persistent limiting correlation.

**Conjecture 209** (Bounded cross-prime covariance). *For distinct odd primes  $p \neq q$ ,  $\text{Cov}_{n < N}(V_p(n), V_q(n)) = O_{p,q}(1)$ , so the correlation is  $O_{p,q}(1/\log N)$ .*

**Significance.** Vanishing correlation is all the central limit law needs; bounded covariance says much more, that every logarithmically growing contribution cancels; the filtrations share no exact scale, yet Diophantine approximation supplies infinitely many near-alignments  $p^a \approx q^b$ , so boundedness requires genuine cancellation across near-resonant scales rather than mere absence

of exact resonance, a sharpening of emphasis. This is the part's crispest analytic target. A first decisive theorem would be the pair (3, 5). In tests on the corpus described in the stress-test section, deposited covariances through  $N = 2 \times 10^6$  stay below 0.24 against marginal variances of several units and growing, and an independent run reproduced the smallness. A counterexample would be logarithmic covariance growth with any coefficient, which would leave the central limit law intact but refute this sharper statement.

**Conjecture 210** (Cross-prime local limit factorization). *Fix distinct odd primes  $p_1, \dots, p_r$ . For every fixed  $C$ , uniformly for integers  $k_i$  with  $|k_i - \mu_{p_i}(N)| \leq C\sqrt{\log N}$ ,*

$$\Pr(V_{p_i}(n) = k_i \ \forall i) = (1 + o(1)) \prod_i \frac{1}{\sqrt{2\pi\sigma_{p_i}^2(N)}} \exp\left(-\frac{(k_i - \mu_{p_i}(N))^2}{2\sigma_{p_i}^2(N)}\right).$$

**Significance.** A central limit law does not determine individual lattice masses, and the conjecture asserts decoupling at one-cell resolution, the transducer analogue of the local limit theorems of the digit-sum literature. The most likely obstruction, arithmetic oscillation of the joint mass with  $N$  modulo powers of  $p_1 \cdots p_r$ , is exactly what a refutation should exhibit. A first decisive theorem would be the pair (3, 5) at the bivariate centre. The statement is a limit law; the deposited diagnostics at  $N = 2 \times 10^6$  are consistent with factorization at the centre. A counterexample would be a persistent excess or deficit at one lattice cell.

**Conjecture 211** (Modular equidistribution of carry vectors). *For distinct odd primes  $p_i$  and fixed moduli  $m_i \geq 2$ , the vector  $(V_{p_1}(n) \bmod m_1, \dots, V_{p_r}(n) \bmod m_r)$  equidistributes on  $\prod_i \mathbb{Z}/m_i\mathbb{Z}$  as  $n$  ranges uniformly below  $N \rightarrow \infty$ .*

**Significance.** This is the Fourier face of cross-base mixing: bounded-frequency obstructions that the central limit law cannot see, with a proof amounting to spectral contraction of every nontrivial product character of the carry outputs. A surviving residue bias would expose a spectral obstruction of the kind Conjecture 216 seeks to exclude, though possibly subtler than an exact common periodic factor; a sufficiently uniform form of Conjecture 210 would imply this law by summation over residue classes. A first decisive theorem would be one prime pair at modulus two. In tests on the corpus described in the stress-test section, deposited discrepancies at  $N = 2 \times 10^6$  fall below  $10^{-3}$  for moduli two and three, reproduced independently at  $2.5 \times 10^{-3}$  and better, with percent-level finite-size effects at modulus four recorded as the honest tail. A counterexample would be one residue vector with persistent bias.

**Conjecture 212** (Additive large deviations). *Let  $I_p$  be the large-deviation rate function per base- $p$  digit of the carry fraction of the chain of Proposition 34. For fixed distinct odd primes, the vector  $(V_{p_i}(n)/\log_{p_i} N)_i$  satisfies a large-deviation principle at speed  $\log N$  with rate  $J(x) = \sum_i I_{p_i}(x_i)/\log p_i$  throughout the interior of the attainable region.*

**Significance.** Central-limit independence is a quadratic statement; additive rates assert that even exponentially rare carry densities carry no leading cross-prime interaction energy, with the factors  $1/\log p_i$  converting digit counts to the common speed. The restriction to the interior is essential and deliberate: at the extreme boundary sit simultaneous near-carry-free events, where the Diophantine entanglement quantified in [127] lives, and the conjecture makes no claim there. No claim is made that the cumulant programme implies it, since exponential tilting probes digital

sets that no fixed cumulant order controls. A first decisive theorem would be an upper bound at one interior point for two primes. The statement is a limit law. A counterexample would be a rate defect at an interior point.

**Conjecture 213** (Bounded mixed cumulants). *For fixed distinct odd primes, every fixed joint cumulant of total order at least three involving at least two distinct primes remains  $O(1)$  as  $N \rightarrow \infty$ .*

**Significance.** Even single-base cumulants grow linearly in the digit length, while the carry chain’s complement symmetry, which exchanges carries and non-carries, caps every odd single-base cumulant at a bounded boundary term, a correction; the conjecture asserts that every mixed connected correlation is likewise bounded, which together with Conjecture 209 would give a connected-correlation proof scheme for the whole Gaussian decoupling rather than an endpoint statement. A first decisive theorem would be  $o(\log N)$  for one mixed third cumulant. In tests on the corpus described in the stress-test section, deposited mixed third and fourth cumulants stay of order one while marginal even cumulants grow, reproduced independently, evidence that is honest but noisy at this depth. A counterexample would be one mixed cumulant growing like  $c \log N$ .

**Conjecture 214** (Fixed-multiplier carry vectors). *Fix an integer  $a \geq 2$  and distinct primes  $p_1, \dots, p_r$  such that  $\text{Var}_{n < N} v_{p_i}(\frac{an}{n}) \rightarrow \infty$  and the accessible carry-state chain of each base- $p_i$  addition automaton is irreducible and aperiodic under independent uniform digits. Then the standardized vector of these valuations converges to a standard Gaussian vector.*

**Significance.** By Kummer’s theorem,  $v_p(\frac{an}{n})$  counts the carries in adding  $n$  and  $(a-1)n$  in base  $p$ , a transducer with more internal state than the doubling automaton, and the conjecture asserts the multibase decoupling mechanism survives the enrichment. The same-base quasi-additive framework of [128] reaches joint laws in one base; the cross-base claim is the new content, and the mixing hypothesis is stated explicitly, since variance growth alone does not guarantee the automaton regularity a clean central limit law needs, though the multiplier automata plausibly satisfy it whenever the variance diverges. A first decisive theorem would be  $a = 3$  with the pair  $(3, 5)$ . In tests on the corpus described in the stress-test section, deposited correlations for  $a \in \{3, 4, 5\}$  at  $N = 5 \times 10^5$  stay below 0.11, independently below 0.04 at  $a = 3$ . A counterexample would be a multiplier resonance with persistent correlation.

**Conjecture 215** (Balanced factorial-ratio vectors). *Let  $R(n) = \prod_j (a_j n)! / \prod_k (b_k n)!$  be integer-valued with  $\sum_j a_j = \sum_k b_k$ . For distinct primes  $p_i$  with  $\text{Var}_{n < N} v_{p_i}(R(n)) \sim c_{p_i} \log N$ ,  $c_{p_i} > 0$ , the standardized vector  $(v_{p_i}(R(n)))_i$  converges to a standard Gaussian vector.*

**Significance.** Legendre’s formula writes each valuation as a balanced combination of digit sums, a finite carry functional whose deterministic linear part cancels by balance, leaving logarithmic fluctuations; the conjecture asserts cross-prime decoupling for the entire classical family of integer factorial ratios, whose same-base divisibility theory is a central application of [128]. A first decisive theorem would be one Catalan-like ratio at two primes. In tests on the corpus described in the stress-test section, the Landau ratio  $(30n)!n!/((15n)!(10n)!(6n)!)$  at  $p \in \{7, 11\}$  shows the predicted logarithmic variance growth with correlation  $-0.016$  in an independent run at  $N = 10^5$ . A counterexample would be a ratio with an exceptional congruence structure forcing a shared automatic factor.

**Conjecture 216** (Multibase transducer principle). *Let  $q_1, \dots, q_r \geq 2$  be jointly multiplicatively independent bases, meaning  $\prod_i q_i^{a_i} = 1$  with integer exponents only when every  $a_i = 0$ , and let  $F_i(n)$  be the total output of a deterministic finite-state transducer reading the base- $q_i$  digits of  $n$ , with bounded transition outputs, such that (i) under independent uniform digits the accessible state chain is irreducible and aperiodic, (ii)  $\text{Var}_{n < N} F_i(n) \sim c_i \log N$  with  $c_i > 0$ , and (iii) no two of the centred functionals factor, up to a bounded coboundary, through a common nonconstant eventually periodic function of  $n$ . Then the standardized vector  $(F_i(n))_i$  converges to a standard Gaussian vector, and every fixed mixed cumulant involving at least two bases is  $O(1)$ .*

**Significance.** This is the structural umbrella over Conjectures 208, 214, and 215: joint multiplicative independence of the bases, mixing of each automaton, and the exclusion of a common periodic factor are conjectured to be the complete obstruction list for finite-state observables, the class where the additive theory of [123] stops. Joint independence is genuinely stronger than the pairwise form of the deposited statement, a repair: the bases 2, 3, 6 are pairwise independent yet satisfy  $\log 6 = \log 2 + \log 3$ , exactly the kind of higher relation a three-base mixed cumulant could feel; the two conclusions are also of different strengths, since a mixed cumulant can grow slowly and still vanish after Gaussian normalization. The exclusion hypothesis is necessary in spirit, since two automata in different bases can be engineered to emit the same periodic statistic, and sharpening its exact form is part of the problem; the statement is the part's boldest and is a programme conjecture. A first decisive theorem would be the two-base two-state case, essentially Conjectures 208 and 209. A counterexample satisfying the three hypotheses would identify a genuinely new obstruction beyond periodicity.

## Part XII

# Subdivision spectra, spanning-tree correlation, and information dimension

*Added to the deposit: 14 August 2026.*

The twenty conjectures below run four programmes at the common interface of spectral theory and information. A subdivision programme (Conjectures 217 to 221) asks what weak spectral universality under barycentric refinement, proved at the graph level by Knill [129, 130], in every Hodge degree for the barycentric rule by Knill [131], and for top-dimensional Laplacians of general inclusion-uniform rules by Mårte [132], does *not* see: exponentially rare high eigenvalues, organized by a factorial hierarchy of critical moment exponents. A spanning-tree programme (Conjectures 222 to 226) introduces the total correlation of the uniform-spanning-tree edge process, an invariant assembled from Kirchhoff marginals and the matrix-tree theorem in the tradition of transfer impedances [134], and conjectures its extremal graphs, led by a mesoscopic core-periphery shape principle at scale  $\sqrt{n/\log n}$ . A cyclic-entropy programme (Conjectures 227 to 231) starts beyond the prime-field entropy-doubling floor proved this year by Gavalakis, Goh, and Kontoyiannis [137] and conjectures the full inverse theory: localization at near-equality, Gaussian rigidity, a heat-kernel crossover profile at the wrap scale, and uniform many-summand monotonicity. A graphon programme (Conjectures 232 to 236) conjectures that the Shannon entropy of a sampled random-free graph, known to be subquadratic of essentially arbitrary



growth [143], is governed at scale  $n \log n$  exactly by the information dimension of the latent row space. The calibration layer is proved where it can be: block additivity and the exact cycle law for the tree invariant form Proposition 35, and the threshold-graphon entropy bound that anchors the second-order programme forms Proposition 36. The adversarial controls marking genuine boundaries are retained in place: the bipartite crossover at seventeen vertices in Conjecture 223, the separated-bump doubling penalty in Conjecture 227, the onset of pre-wrap failures in Conjecture 231, and the threshold graphon itself as the permanent negative control of Conjecture 234. All graph and spectral scans are floating-point checks. Three statements carry elevated exposure or overlap with prior work and are recorded where they occur: Conjecture 220, whose large-deviation law has no direct numerical support at reachable depth and is the part's boldest statement; Conjecture 219, which interpolates two published theorems [132, 131] and is close to prior work; and Conjecture 230, whose integer-lattice analogue is a recent theorem [142], so that only the modular content is new.

Throughout,  $\text{sd}^m K$  denotes the  $m$ -fold barycentric subdivision of a finite pure  $d$ -dimensional simplicial complex  $K$ , realized as the order complex of the face poset;  $\Delta_k$  is the unweighted Hodge Laplacian on  $k$ -simplices,  $f_k$  the number of  $k$ -simplices, and for the graph Laplacian  $L_m$  of the 1-skeleton,  $M_\alpha(m) = \text{Tr}(L_m^\alpha)/|V(\text{sd}^m K)|$ ; we write  $A_d = (d+1)!$ ,  $B_{d,0} = d!$ , and  $B_{d,k} = (d-k+1)!$  for  $1 \leq k < d$ . For a connected simple graph  $G$ ,  $\tau(G)$  is the number of spanning trees,  $T$  a uniform spanning tree, and  $R_e$  the unit-conductance effective resistance across  $e$ , so that  $\Pr(e \in T) = R_e$  by Kirchhoff's theorem; with  $h(x) = -x \log x - (1-x) \log(1-x)$ , the *spanning-tree correlation* is

$$\mathcal{C}(G) = \sum_{e \in E(G)} h(R_e) - \log \tau(G),$$

the total correlation of the edge-indicator vector of  $T$ , nonnegative and zero exactly on trees, since  $H(T) = \log \tau(G)$ ;  $S_{n,r} = K_r \vee \bar{K}_{n-r}$  denotes the complete split graph. On the cyclic side, laws  $\mu_p$  on  $\mathbb{Z}/p\mathbb{Z}$  with  $p$  prime are called *mesoscopic* when  $H(\mu_p) \rightarrow \infty$  and  $\log p - H(\mu_p) \rightarrow \infty$ ; convolution powers are written  $\mu_p^{*m}$ . A graphon  $W$  is *random-free* when it is  $\{0, 1\}$ -valued almost everywhere;  $G(n, W)$  is the sampled graph on  $n$  vertices,  $d_W(x, y) = \int_0^1 |W(x, u) - W(y, u)| du$  the neighborhood distance, and  $(\Omega_W, d_W, \rho)$  the *purified row space*, the completion of the quotient of  $[0, 1]$  by  $d_W$ -null pairs carrying the pushforward of Lebesgue measure. The measure  $\rho$  satisfies a *uniform local doubling bound* when there are  $C$  and  $\varepsilon_0 > 0$  with  $\rho(B(x, 2\varepsilon)) \leq C\rho(B(x, \varepsilon))$  for all  $x$  in the support and  $\varepsilon \leq \varepsilon_0$ , and is *exact-dimensional of dimension  $s$*  when  $\log \rho(B(x, \varepsilon))/\log \varepsilon \rightarrow s$  for  $\rho$ -almost every  $x$ .

**Proposition 35** (Block additivity and the cycle law). (i)  $\mathcal{C}$  is additive over the blocks of a connected graph, and bridges contribute zero. (ii)  $\mathcal{C}(C_n) = (n-1) \log \frac{n}{n-1}$ , strictly increasing in  $n$ ; consequently, subdividing an edge that lies on exactly one cycle strictly increases  $\mathcal{C}$ .

*Proof.* (i) A subgraph is a spanning tree of  $G$  exactly when its intersection with every block is a spanning tree of that block, so the uniform spanning tree is an independent product across blocks and  $\log \tau(G) = \sum_B \log \tau(B)$ ; each edge lies in exactly one block, so the marginal sum also splits, and a bridge has  $R_e = 1$ , contributing  $h(1) = 0$  and a factor 1 to  $\tau$ . (ii) In  $C_n$  every edge has  $R_e = (n-1)/n$  and  $\tau = n$ , so  $\mathcal{C}(C_n) = n h\left(\frac{n-1}{n}\right) - \log n = (n-1) \log \frac{n}{n-1}$ , which equals  $\log(1 + \frac{1}{n-1})^{n-1}$  and is strictly increasing because  $(1 + 1/x)^x$  is. An edge lying on exactly one cycle has a cycle  $C_\ell$  as its block; subdivision replaces that block by  $C_{\ell+1}$  and changes nothing else, so  $\mathcal{C}$  increases by  $\mathcal{C}(C_{\ell+1}) - \mathcal{C}(C_\ell) > 0$ .  $\square$

**Proposition 36** (Threshold-graphon entropy bound). For the random-free graphon  $W(x, y) = \mathbf{1}\{x + y \geq 1\}$ , every realization of  $G(n, W)$  is a threshold graph, the law of  $G(n, W)$  is supported on at most  $n! 2^{n-1}$  labeled graphs, and hence

$$H(G(n, W)) \leq n \log n - (1 - \log 2) n + O(\log n),$$

although the purified row space is  $[0, 1]$  with the metric  $|x - y|$  and Lebesgue measure, whose scale- $1/n$  quantization entropy is  $\log n$  per point with vanishing linear correction.

*Proof.* Sample  $x_1, \dots, x_n$  uniformly, and set  $y_i = |x_i - \frac{1}{2}|$  and  $s_i = \mathbf{1}\{x_i > \frac{1}{2}\}$ . Almost surely the  $y_i$  are distinct, and  $x_i + x_j \geq 1$  holds exactly when  $s_i = s_j = 1$ , or the signs differ and the plus-signed vertex has the larger  $y$ . Adding vertices in increasing order of  $y$ , each new vertex is adjacent to all earlier ones when its sign is plus and to none when its sign is minus: a creation sequence, so the realization is a threshold graph, and the labeled graph is a function of the  $y$ -ordering and the signs. The sign of the first vertex added affects no edge, so at most  $n! 2^{n-1}$  labeled graphs occur, and  $H \leq \log n! + (n - 1) \log 2 = n \log n - (1 - \log 2) n + O(\log n)$ . The row space assertion is immediate: rows of  $W$  are indicators of intervals  $[1 - x, 1]$ , so  $d_W(x, y) = |x - y|$  and  $\rho$  is Lebesgue measure, whose  $1/n$ -quantization entropy is  $\log n + o(1)$  per point.  $\square$

## 49 Spectral tails of barycentric subdivision

**Conjecture 217** (Critical moment exponent for barycentric vertex Laplacians). *Let  $K$  be a finite pure  $d$ -dimensional simplicial complex,  $d \geq 2$ , with at least one  $d$ -simplex. For every  $\alpha > 0$ ,*

$$\lim_{m \rightarrow \infty} \frac{1}{m} \log M_\alpha(m) = \max\{0, \alpha \log d! - \log(d + 1)!\},$$

*and at the critical order  $\alpha_c(d) = \log(d + 1)! / \log d!$  the moment  $M_{\alpha_c}(m)$  grows linearly in  $m$  up to constants depending on  $K$ .*

**Significance.** Weak spectral universality under barycentric refinement is a theorem [129, 130], but weak convergence coexists with divergent high moments, and this statement locates the exact divergence threshold, a discrete analogue of a multifractal moment transition. The mechanism is an incidence-age ledger: each refinement multiplies the number of chambers by  $(d + 1)!$  while the link of a surviving vertex grows by the factor  $d!$ , so the age- $a$  stratum has frequency about  $(d + 1)!^{-a}$  and degree scale  $d!^a$ , and  $\sum_a A_d^{-a} B_{d,0}^{\alpha a}$  changes behaviour exactly at  $\alpha_c$ ; the analytic content is transferring the ledger from degrees to eigenvalues, for which Schur–Horn majorization gives the lower half. In dimension two the threshold is  $\log_2 6 = 2.5849\dots$ , and the failure of the finite all-order Rényi defect that this transition forces is the recorded reason the predecessor claim was withdrawn. A first decisive theorem would be the case  $d = 2$ ,  $\alpha = 4$ . In tests on the corpus described in the stress-test section, degree-moment growth ratios at  $\alpha = 4$  reach 2.675 against the predicted  $16/6 = 2.6667$  by seven refinements of the triangle, with the same limit from a two-triangle start and from a non-manifold three-page book, and dense spectral moments through five refinements track the degree ledger. A counterexample would be a starting complex whose supercritical moment rate separates from the factorial prediction.

**Conjecture 218** (Hodge incidence-tail hierarchy). *For  $0 \leq k \leq d$  and the Hodge Laplacians  $\Delta_k$  on  $\text{sd}^m K$ , the empirical  $\alpha$ -moments of  $\Delta_k$  stay bounded for  $\alpha < \alpha_c(d, k) = \log(d + 1)! / \log B_{d,k}$ , grow polynomially at  $\alpha_c(d, k)$ , and grow at exponential rate  $\alpha \log B_{d,k} - \log(d + 1)!$  above it; for  $k = d$  all moments stay bounded.*

**Significance.** This organizes the Hodge spectrum by codimension: a persistent  $k$ -incidence stratum sees only the  $d - k + 1$  remaining directions of a chamber chain, giving the factorial  $B_{d,k} = (d - k + 1)!$ , while Hodge supersymmetry forces adjacent blocks to share their extreme nonzero eigenvalues, which is why  $k = 0$  and  $k = 1$  share the tail factor  $d!$ . The predicted tail-factor pattern in dimension three is 6, 6, 2, 1. The nearest boundary is the all-degree weak universality of [131], which sees none of this. A first decisive theorem would be boundedness for  $k = d$  together with the shared  $k = 0, 1$  growth rate in dimension two. In tests on the corpus described in the stress-test section, the tetrahedron probe gives maxima  $15 \rightarrow 75.06$  in degrees zero and one,  $10 \rightarrow 15.74$  in degree two, and  $7 \rightarrow 7.88$  at the top, exactly the predicted pattern, and in dimension two the degree-one maximum doubles per step while the top-degree maximum converges to six. A counterexample would be a Hodge degree whose tail factor depends on finer simplex type than codimension alone.

**Conjecture 219** (All-degree inclusion-uniform universality). *For every nontrivial inclusion-uniform subdivision rule  $\text{div}$  on pure  $d$ -complexes and every  $0 \leq k \leq d$ , the empirical spectral measure of  $\Delta_k(\text{div}^m K)$  converges weakly to a limit law depending only on the rule,  $d$ , and  $k$ , and not on  $K$ , apart from asymptotically negligible homological zero modes.*

**Significance.** This sits at the open product cell of two published theorems and is close to prior work on that account: Marte proves universality for the top-dimensional Laplacian of every inclusion-uniform rule [132], and Knill proves it in every Hodge degree for the barycentric rule [131]; the conjecture asserts the common generalization. The mechanism is that inclusion-uniformity makes the interior of every old chamber evolve by one fixed local rule while boundary contributions have exponentially smaller face counts; the delicate point is that lower Hodge blocks touch those boundaries directly, and the claim is that the coupling still carries vanishing spectral density. A first decisive theorem would be the edgewise subdivision in dimension two at  $k = 0$ . In tests on the corpus described in the stress-test section, the barycentric all-degree case is corroborated through five two-dimensional and two three-dimensional refinements; no non-barycentric rule has been machine-tested, and the conjecture is deposited with that gap on record. A counterexample would be an inclusion-uniform rule and a lower Hodge degree retaining positive-density memory of the initial gluing.

**Conjecture 220** (Spectral-tail large deviations from simplex age). *Fix  $0 \leq k < d$  and  $0 < \theta < 1$ , and let  $N_{m,k}(\theta)$  count eigenvalues of  $\Delta_k(\text{sd}^m K)$  that are at least  $B_{d,k}^{\theta m}$ . Then*

$$\lim_{m \rightarrow \infty} \frac{1}{m} \log \frac{N_{m,k}(\theta)}{f_k(\text{sd}^m K)} = -\theta \log(d + 1)!,$$

*uniformly for  $\theta$  in compact subintervals of  $(0, 1)$ ; for  $k = d$  there is no exponential tail.*

**Significance.** This is the full tail profile behind Conjectures 217 and 218, whose critical exponents are its Legendre data: an eigenvalue of scale  $B^{\theta m}$  should come from an incidence stratum that survived about  $\theta m$  refinements, and such strata number about  $A_d^{(1-\theta)m}$  of the  $A_d^m$  total. The hard content is a two-sided correspondence between old incidence and large eigenvalue, not a degree estimate; eigenvector delocalization across age strata is the named failure channel. This is the part’s boldest statement: at reachable depths the thresholds  $B^{\theta m}$  still sit inside the spectral bulk, so the deposited evidence is the mechanism and the moment data of Conjecture 217, not direct tail counts, and the independent recomputation confirms that its

shallow-depth tail counts are uninformative rather than confirmatory. A first decisive theorem would be the upper bound for  $k = 0$ ,  $d = 2$ . A counterexample would be a tail count with a strictly flatter rate than the age ledger at some  $\theta$ .

**Conjecture 221** (Finite-state renormalization for barycentric Hodge limits). *For each  $d \geq 2$  there exist an integer  $N(d)$ , a rational map  $R_d$  on  $\mathbb{C}^{N(d)}$ , and rational functions  $\Phi_{d,k}$ ,  $0 \leq k \leq d$ , such that the Schur complement of one barycentric refinement onto a generating set of  $N(d)$  boundary Green functions is exactly  $R_d$ , the iteration of  $R_d$  has a unique attracting fixed point on the relevant domain, and the Stieltjes transforms of all limiting Hodge laws  $\nu_{d,k}$  are the  $\Phi_{d,k}$  of that fixed point; for  $d = 1$  the scheme reduces to the known quadratic spectral decimation. The claim fails if the minimal rank of an exact boundary Schur-complement closure grows without bound under refinement.*

**Significance.** Knill identifies the one-dimensional limit through the quadratic map  $z \mapsto 4z - z^2$  and records that no higher-dimensional analogue is known and the higher-dimensional limits are unidentified [129]; a block-Jacobi formulation close in spirit appears in [133] without a finite rational recursion. The conjecture asserts finite-state closure: refinement acts on a finite vector of boundary Green functions by a rational map whose fixed point carries the spectral data, turning qualitative universality into solvable renormalization. The final sentence is the falsifiability clause, added at deposit: the statement stands or falls with the boundedness of the minimal closure rank, a quantity computable in principle at each depth. A first decisive theorem would be finite closure for  $d = 2$  at the top degree, where Mårte's Schreier-graph decimation [132] provides the strongest existing mechanism. In tests on the corpus described in the stress-test section, no machine evidence bears on closure; the conjecture is deposited as a programme statement on the strength of the one-dimensional precedent alone. A counterexample would be superlinear growth of the minimal boundary rank in any dimension.

## 50 Total correlation of uniform spanning trees

**Conjecture 222** (Complete-split extremality). *Among connected  $n$ -vertex graphs,  $\mathcal{C}(G)$  is maximized by a complete split graph  $S_{n,r}$  for some  $r = r_n$ , and*

$$r_n \sim \sqrt{\frac{n}{\log n}}, \quad \max_G \mathcal{C}(G) = n - (1 + o(1))\sqrt{n \log n}.$$

**Significance.** The invariant  $\mathcal{C}$  measures the total dependence of the determinantal edge process whose entropy theory begins with transfer impedances [134] and extends to general determinantal measures in [135]; its extremal theory appears nowhere, and the conjectured optimizer is neither sparse nor a standard dense extremal graph but a mesoscopic core of order  $\sqrt{n/\log n}$  joined to a near-independent periphery. The mechanism is the exact ledger  $\tau(S_{n,r}) = n^{r-1}r^{n-r-1}$  with core marginal  $2/n$  and cross marginal  $(n+r-1)/(nr)$ , whose small-marginal expansion gives the two-term deficit  $r \log(n/r) + n/(2r)$ , balanced exactly at the stated scale. A first decisive theorem would be optimality of split graphs among split-plus-one-edge perturbations. In tests on the corpus described in the stress-test section, the maximizer over all 853 connected seven-vertex graphs is the split family, hill climbs from random starts at twelve and sixteen vertices land exactly on the split optimum, structured core-density, periphery-matching, and two-core adversaries all fall below it at thirty and forty vertices, and exact family optimization to  $n = 10^7$  gives

$r^*/\sqrt{n/\log n} = 0.92$  and deficit ratio 1.14, both drifting toward one. A counterexample would be any connected graph exceeding every complete split value at its order.

**Conjecture 223** (Triangle-free bipartite extremality). *Among connected triangle-free  $n$ -vertex graphs,  $\mathcal{C}$  is maximized by a complete bipartite graph  $K_{r,n-r}$  with  $r$  maximizing the exact bipartite formula; the maximizing  $r$  satisfies  $r \sim \sqrt{n/\log n}$  and the maximum is  $n - (1 + o(1))\sqrt{n \log n}$ .*

**Significance.** The predecessor claim fixed the small side at two and is false: the retained control is the crossover  $\mathcal{C}(K_{3,14}) = 8.35012\dots > \mathcal{C}(K_{2,15}) = 8.31559\dots$  at  $n = 17$ , after which the optimizer migrates through the bipartite family at the same mesoscopic scale as Conjecture 222. Every edge of  $K_{r,s}$  has marginal  $(n-1)/(rs)$  and  $\tau = r^{s-1}s^{r-1}$ , so the same two-term deficit appears; the conjecture says triangle-freeness costs exactly the core edges and nothing else at leading order. A first decisive theorem would be optimality within the complete bipartite family. In tests on the corpus described in the stress-test section, both crossover values reproduce to twelve digits in an independent implementation, no triangle-free seven-vertex graph beats the family, and blowups of the five-cycle, the Petersen graph, and edge-diluted bicliques all fall short at ten to thirty vertices. A counterexample would be a triangle-free graph exceeding every complete bipartite value at its order.

**Conjecture 224** (Cycle minimum for bridgeless dependence). *Every bridgeless connected  $n$ -vertex simple graph satisfies  $\mathcal{C}(G) \geq \mathcal{C}(C_n) = (n-1) \log \frac{n}{n-1}$ , with equality only for  $C_n$ .*

**Significance.** Once bridges are forbidden, dependence cannot be diluted to zero, and the conjecture identifies the cycle, whose single constraint is to omit exactly one edge, as the unique minimizer; by Proposition 35 the value  $\mathcal{C}(C_n)$  increases to one, so the statement is a sharp universal gap for the graphic projection determinantal class rather than a spanning-tree count extremum. Block additivity makes the claim especially plausible at cut vertices, where dependence adds over cyclic blocks. A first decisive theorem would be the two-connected planar case. In tests on the corpus described in the stress-test section, every bridgeless graph in the seven-vertex atlas sits above the cycle, as do all theta graphs through forty vertices and one hundred eighty-three structured and random bridgeless graphs, including prisms, wheels, hypercubes, complete bipartite graphs, and cubic expanders through sixty vertices, with zero violations. The cactus case is now proved, a promotion: by Proposition 35, a bridgeless cactus with two or more blocks has correlation at least  $4 \log \frac{3}{2} > 1$  while every cycle stays below one, so the difficulty is concentrated in two-connected non-cycle blocks. A counterexample would be a bridgeless graph at or below the cycle value.

**Conjecture 225** (Core-periphery stability at the ceiling). *If connected graphs  $G_n$  satisfy  $\mathcal{C}(G_n) = n - (1 + o(1))\sqrt{n \log n}$ , then there are vertex sets  $A_n$  with  $|A_n| = (1 + o(1))\sqrt{n/\log n}$  such that all but  $o(n|A_n|)$  of the pairs between  $A_n$  and its complement are edges, while the complement spans  $o(n|A_n|)$  edges. No structure is asserted for  $G_n[A_n]$ .*

**Significance.** This is a stability principle for an information-theoretic extremal problem: near-extremality is asserted to force the mesoscopic core-periphery cut while deliberately leaving the core's internal graph free, because that freedom is real. The deposited calibration shows the optimized split and bipartite families, whose cores differ maximally, separated by only 0.46 at  $n = 10^6$  against a leading deficit in the thousands, so no leading-order statement can see

inside the core. The nearest boundary is the classical stability paradigm of extremal graph theory, here transposed to a determinantal entropy functional with no known antecedent. A first decisive theorem would be recovering the cut from the two-term deficit expansion under a split-plus-perturbation hypothesis. In tests on the corpus described in the stress-test section, random core densities inside an otherwise optimal split graph move  $\mathcal{C}$  by order one while breaking the cross edges moves it at scale  $\sqrt{n \log n}$ . A counterexample would be a near-extremal family with two well-separated mesoscopic cores or none.

**Conjecture 226** (Complete-multipartite reduction). *Among complete multipartite graphs on  $n$  vertices,  $\mathcal{C}$  is maximized by one part of size  $n - r$  together with  $r$  singleton parts, that is, by the complete split graph  $S_{n,r}$  for the optimal  $r$ .*

**Significance.** This compresses an optimization over integer partitions to one parameter and is the natural stepping stone toward Conjecture 222, since the multipartite class interpolates between the clique and the biclique while admitting the exact ledger  $\tau(K_{a_1, \dots, a_k}) = n^{k-2} \prod_i (n - a_i)^{a_i-1}$  and closed-form effective resistances, reducing the claim to a finite family of partition inequalities. A first decisive theorem would be that no partition with two parts of size at least two beats the split form at its own  $n$ . In tests on the corpus described in the stress-test section, every complete multipartite partition through  $n = 24$  was checked exactly at deposit, a further computation extended that to  $n = 30$ , and the independent recomputation extends it to all 89,133 partitions at  $n = 45$  and every  $n$  below, with the split form winning at every order. A counterexample would be a partition with a second macroscopic part beating every split value at some  $n$ .

## 51 Prime-cyclic entropy power and the wrap transition

**Conjecture 227** (Localization of near-minimizers on prime cycles). *Let  $p \rightarrow \infty$  through primes and let  $\mu_p$  be mesoscopic laws on  $\mathbb{Z}/p\mathbb{Z}$  with*

$$H(\mu_p * \mu_p) - H(\mu_p) \rightarrow \frac{1}{2} \log 2.$$

*Then there are affine automorphisms  $\phi_p$  of  $\mathbb{Z}/p\mathbb{Z}$  and cyclic intervals  $I_p$  of length  $o(p)$  with  $(\phi_p)_* \mu_p(I_p) \rightarrow 1$ .*

**Significance.** The floor itself is now a theorem: Gavalakis, Goh, and Kontoyiannis prove the prime-field doubling gain of  $\frac{1}{2} \log 2$  up to  $\varepsilon$  for entropies in a window away from both endpoints [137], a prime-field analogue of Tao’s torsion-free inequality, with earlier cyclic entropy inequalities in [136]. The conjecture is the corresponding inverse theorem: asymptotic equality forces the torsion geometry to vanish, leaving the measure inside a Freiman copy of the line, the rank-one case of the bounded-rank progression structure that drives the proof of the floor. A first decisive theorem would be localization under an additional log-concavity hypothesis. In tests on the corpus described in the stress-test section, localized discrete Gaussians sit on the floor to six decimal places at two primes and four widths, while the retained control, an equal mixture of two separated bumps, pays the full extra  $\frac{1}{2} \log 2$ , reproduced independently to machine precision. A counterexample would be an essentially cyclic near-minimizing family with no asymptotically full affine interval.

**Conjecture 228** (Gaussian rigidity after localization). *Under the conclusion of Conjecture 227, choose centered integer lifts  $X_p$  with variances  $\sigma_p^2 \rightarrow \infty$  and assume a uniform standardized*



$(2 + \delta)$ -moment bound. If the doubling gain tends to  $\frac{1}{2} \log 2$ , then  $(X_p - \mathbb{E}X_p)/\sigma_p \Rightarrow N(0, 1)$  and  $H(X_p) - \frac{1}{2} \log(2\pi e\sigma_p^2) \rightarrow 0$ .

**Significance.** This identifies the equality geometry, not just the constant, bridging prime-field inverse structure to continuous entropy-power rigidity. The moment hypothesis is load-bearing and the statement carries the corresponding caveat: in the continuous setting there are sequences whose doubling constant attains the Gaussian minimum asymptotically while remaining far from Gaussian, so unrestricted rigidity is false [139], and the recent equality-and-stability analysis of Shannon’s and Tao’s inequalities obtains qualitative stability precisely under mild moment control [138]. The conjecture asserts that after cyclic localization the same moment control suffices without log-concavity. A first decisive theorem would be the entropic convergence under an added log-concavity hypothesis via the discrete stability bounds of [138]. In tests on the corpus described in the stress-test section, Gaussian families achieve both limits simultaneously and no tested non-Gaussian family reaches the floor. A counterexample would be a uniformly integrable near-minimizing family with persistent lattice microstructure blocking the entropy asymptotic.

**Conjecture 229** (Heat-kernel wrap crossover, resolved false). *Let  $\mu_p$  have centered span-one integer lifts with variances  $\sigma_p^2$  and characteristic functions obeying  $|\mathbb{E} e^{i\theta(X_p - \mathbb{E}X_p)}| \leq e^{-c\theta^2\sigma_p^2}$  for  $|\theta| \leq \pi$  and a constant  $c > 0$  independent of  $p$ , and let  $\sigma_p = o(p)$ ,  $m_p \rightarrow \infty$ ,  $m_p\sigma_p^2/p^2 \rightarrow t \in (0, \infty)$ . With  $\theta_t$  the unit-circle heat kernel with Fourier coefficients  $e^{-2\pi^2 t j^2}$  and  $q_{p,t}$  its discretization,  $q_{p,t}(x) \propto \theta_t(x/p)$ ,*

$$\|\mu_p^{*m_p} - q_{p,t}\|_{\text{TV}} \rightarrow 0, \quad \log p - H(\mu_p^{*m_p}) \rightarrow \int_0^1 \theta_t(u) \log \theta_t(u) du.$$

**Significance.** This is the universal profile of the entire transition from line-like Gaussian convolution to Haar equilibrium on the cycle: at Fourier mode  $j$  the  $m$ -th power of the characteristic value concentrates to  $e^{-2\pi^2 t j^2}$  exactly when  $m\sigma^2/p^2 \rightarrow t$ , and the entropy defect becomes the relative entropy of the circle heat kernel from Haar. The comparison is with the *discretized* kernel, since a discrete law cannot converge in total variation to a continuous one; the explicit characteristic-function hypothesis replaces a vaguer local-limit clause at deposit. The nearest boundary is total-variation mixing and cutoff for cycle walks, which concerns the endpoint rather than the profile. A first decisive theorem would be the Gaussian-input case. In tests on the corpus described in the stress-test section, three prime-width pairs collapse onto the profile with total variation at machine roundoff and entropy defects matching the target integral to five decimals, and a lazy-walk input at up to 9,604,801 convolutions has total variation falling from  $4.6 \times 10^{-7}$  to  $1.0 \times 10^{-9}$  as  $p$  grows. A counterexample would be an admissible input law with a total-variation limit bounded away from zero at some  $t$ . *Resolution.* The conjecture is false as stated: the characteristic-function bound does not exclude rare macroscopic jumps, and a refuting family, a lazy-walk core of  $\lfloor \sqrt{p} \rfloor$  steps carrying jumps of size  $\lfloor p/3 \rfloor$  with probability  $\lfloor \sqrt{p} \rfloor \lfloor p/3 \rfloor^{-2}$ , satisfies every hypothesis with  $c = 1/6$  while its wrap limit is the heat kernel convolved with a compound-Poisson jump law, the mode-three coefficient tending to  $e^{-6\pi^2 t}$  against the asserted  $e^{-18\pi^2 t}$ ; an independent transform computation here reproduces the jump prediction to six decimals. A Lindeberg condition at scale  $p$ , or a uniform  $o(p)$  support bound, restores the intended law and is the recorded repair.

**Conjecture 230** (Fixed- $m$  entropy-power law on prime cycles). *For every fixed integer  $m \geq 2$  and every mesoscopic sequence  $\mu_p$ ,*

$$\liminf_{p \rightarrow \infty} [H(\mu_p^{*m}) - H(\mu_p)] \geq \frac{1}{2} \log m,$$

*and asymptotic equality under the hypotheses of Conjecture 228 forces the same Gaussian limit.*

**Significance.** The doubling case is the theorem of [137]; the conjecture is the full envelope in the number of summands, matching the continuous Gaussian law at every  $m$ , where iterating the doubling bound reaches only powers of two. The statement is close to prior work on its lattice side: on  $\mathbb{Z}$ , a growth floor of  $\frac{1}{2} \log N$  for  $N$ -fold sums of lattice random variables is a recent theorem of Castellano and Sekatski [142], so the content here is exclusively the modular statement for mesoscopic sequences on the cycle, where torsion could in principle interact with a non-power-of-two number of summands, together with the rigidity clause; sharp cyclic Rényi inequalities in [136] are the older boundary. A first decisive theorem would be  $m = 4$  via two doublings and a localization step. In tests on the corpus described in the stress-test section, six families at  $m \in \{2, 3, 4, 5, 7, 10\}$  and a three-hundred-draw adversarial mixture search at  $m = 3$  never fall below the floor, with the Gaussian sitting on it to six decimals. A counterexample would be a mesoscopic family strictly under  $\frac{1}{2} \log m$  at some fixed  $m$ .

**Conjecture 231** (Uniform pre-wrap entropy monotonicity). *Let  $\mu_p$ , after an affine automorphism, have integer log-concave lifts with variances  $\sigma_p^2$  and  $H(\mu_p) \rightarrow \infty$ , and let  $M_p \rightarrow \infty$  satisfy  $\log M_p = o(H(\mu_p))$  and  $M_p \sigma_p^2 / p^2 \rightarrow 0$ . Then, uniformly for  $1 \leq m \leq M_p$ ,*

$$H(\mu_p^{*(m+1)}) - H(\mu_p^{*m}) \geq \frac{1}{2} \log \frac{m+1}{m} - o(1/m).$$

**Significance.** Fixed- $m$  discrete entropy monotonicity for log-concave laws is known on the integers, with gain  $\frac{1}{2} \log((m+1)/m)$  up to vanishing error [140] and in higher lattice dimension [141]; the new content is uniformity in a number of summands growing all the way to the wrap scale, connecting the entropic central limit theorem quantitatively to the onset of modular wrapping. The wrap condition is not decorative: the deposited tests record finite-size failures once  $M_p \sigma_p^2 / p^2$  approaches  $10^{-2}$ , the retained boundary control. A first decisive theorem would be uniformity for  $M_p$  up to a fixed power of  $H(\mu_p)$ . In tests on the corpus described in the stress-test section, three log-concave families at  $p = 2,000,003$  hold the scaled margin nonnegative through one thousand summands at wrap ratio  $2.5 \times 10^{-8}$ , with the Gaussian family sitting at equality to the tested precision. A counterexample would be a log-concave family whose scaled margin turns negative strictly inside the pre-wrap window.

## 52 Information dimension of random-free graphons

**Conjecture 232** (Exact-dimensional entropy identity). *Let  $W$  be random-free with purified row space  $(\Omega_W, d_W, \rho)$ . If  $\rho$  is exact-dimensional of finite dimension  $s$  and satisfies a uniform local doubling bound, then*

$$\frac{H(G(n, W))}{n \log n} \longrightarrow s.$$

**Significance.** Random-free graph entropy is subquadratic of essentially arbitrary growth [143], and the neighborhood metric underlying  $(\Omega_W, d_W, \rho)$  carries the graphon’s estimable dimension theory [144]; the conjecture identifies the  $n \log n$  coefficient with the almost-sure local dimension, replacing regularity of the space by regularity of the measure. The mechanism is resolution counting: with  $n$  samples, rows are distinguishable at scale  $1/n$ , and exact dimension prices a typical latent label at  $s \log n$  nats. The known random-free graphon with merely linear entropy is consistent rather than adversarial here, since its row measure cannot be exact-dimensional of positive dimension with local doubling; producing such a verification is part of the problem. A first decisive theorem would be the identity under Ahlfors regularity. In tests on the corpus described in the stress-test section, one- and two-dimensional models and their mixtures show row-signature resolution slopes at the predicted dimensions, with the deposit explicitly recording that these are proxies for the coding mechanism and not measurements of  $H(G(n, W))$ . A counterexample would be an exact-dimensional doubling row measure whose sampled-graph entropy separates from  $sn \log n$ , for instance through cross-row redundancy of order  $n \log n$ .

**Conjecture 233** ( $L^1$  multifractal average-dimension law). *For random-free  $W$ , define  $D_\varepsilon(x) = \log(1/\rho(B(x, \varepsilon))) / \log(1/\varepsilon)$  on the purified row space. If  $\rho$  satisfies a uniform local doubling bound and  $D_\varepsilon \rightarrow d(\cdot)$  in  $L^1(\rho)$ , then*

$$\frac{H(G(n, W))}{n \log n} \longrightarrow \int d(x) d\rho(x).$$

**Significance.** This is the averaging principle behind Conjecture 232 for genuinely multifractal latent geometry: the coefficient is neither a Minkowski dimension nor an essential supremum but the mean pointwise information dimension, with the  $L^1$  hypothesis stated explicitly so the resolution heuristic can be converted to a code-length calculation rather than hidden behind uniform integrability, a repair recorded against the predecessor phrasing. The nearest boundary remains [143], whose flexibility theorem shows the coefficient cannot be universal without a hypothesis of exactly this kind. A first decisive theorem would be the two-atom case, a mixture of components of distinct dimensions. In tests on the corpus described in the stress-test section, mixture models show proxy slopes at the predicted averages of their component dimensions. A counterexample would be a doubling multifractal row measure whose entropy coefficient separates from the mean dimension.

**Conjecture 234** (Second-order graph entropy constant). *If the purified row space of a random-free graphon is a compact  $s$ -dimensional  $C^2$  metric manifold, boundary allowed, and  $\rho$  has a positive  $C^1$  density in local charts, then*

$$\kappa(W) = \lim_{n \rightarrow \infty} \frac{H(G(n, W)) - sn \log n}{n}$$

*exists and is finite.*

**Significance.** A leading coefficient is a first-order theory; this asserts a second-order constant, the graphon analogue of a quantization or coding constant, able to separate geometries of equal dimension. Crucially,  $\kappa(W)$  is *not* conjectured to equal the row-space quantization constant: the predecessor made that identification and Proposition 36 kills it, since the threshold graphon has the unit interval as row space yet its sampled entropy carries the strictly negative linear correction forced by threshold-graph combinatorics [146], the permanent negative control retained

here. The constant must therefore mix metric and observational-combinatorial contributions, and identifying its ingredients is part of the problem. A first decisive theorem would be existence for one-dimensional threshold-type row spaces. In tests on the corpus described in the stress-test section, exact enumeration of the threshold law through seven vertices gives support sizes 46, 332, 2,874, and 29,024, matching the labeled threshold-graph counts of [146] exactly, with linear correction already at  $-0.60$  per vertex against the quantization prediction of zero. A counterexample would be a smooth row geometry with oscillatory linear term.

**Conjecture 235** (Equipartition for random-free samples). *Under the hypotheses of Conjecture 232, let  $G_n \sim G(n, W)$  and  $I_n = -\log \Pr(G_n)$ . Then  $I_n/(n \log n) \rightarrow s$  in probability.*

**Significance.** Entropy is a mean; this is the corresponding Shannon–McMillan statement, placing almost every sampled graph in a typical set of size  $\exp((s + o(1))n \log n)$  despite the dependence induced by shared latent coordinates. Equipartition principles for coloured random graph models go back to [147], and graphon-entropy fluctuation work concerns estimators rather than the sample information variable [145]; neither touches the random-free dimension setting. The mechanism is that exact-dimensionality suppresses  $n \log n$ -scale variation in local coding rates, leaving fluctuations at linear order. A first decisive theorem would be the identity for step-function-free Ahlfors-regular row spaces via a second-moment computation of  $I_n$ . In tests on the corpus described in the stress-test section, no direct machine evidence exists at meaningful  $n$ , and the statement is deposited as the natural concentration companion of Conjecture 232 with that gap on record. A counterexample would be a family of exceptionally probable isomorphism types carrying positive information-scale mass.

**Conjecture 236** (Sparse-observation dimension law). *Reveal each potential edge of  $G(n, W)$  independently with probability  $q_n$ , condition on the mask, and let  $O_n$  be the revealed bits. Under the hypotheses of Conjecture 233, if  $nq_n \rightarrow \infty$ , then*

$$\frac{H(O_n \mid \text{mask})}{n \log(nq_n)} \longrightarrow \int d(x) d\rho(x).$$

**Significance.** This predicts exactly how latent geometric information degrades under missing data: each vertex is probed by about  $nq_n$  revealed neighbours, so the observable resolution moves from  $1/n$  to  $1/(nq_n)$  while the dimension coefficient survives untouched. Conditioning on the mask is essential, since otherwise the mask’s own Bernoulli entropy dominates, a repair recorded against the predecessor phrasing along with the removal of an opaque density condition. The nearest boundary is graphon estimation from partially observed data, which concerns recovery rates rather than information content. A first decisive theorem would be the one-dimensional Ahlfors-regular case at  $q_n = n^{-1/2}$ . In tests on the corpus described in the stress-test section, sparse-probe proxy slopes reproduce the dense-case dimensions after replacing the probe count, with the same proxy caveat on record. A counterexample would be an admissible reveal rate with an extra information loss of order  $n \log(nq_n)$ .

## Part XIII

# Flux torsion, quantum merge monotonicity, and effective string geometry

*Added to the deposit: 15 August 2026.*

The nineteen conjectures below extend the string-geometric programmes of Part X in five directions. A flux programme (Conjectures 237 to 240) lifts the discrete saturation and threshold laws of Conjectures 177 and 178 from three-forms to every odd degree and opens the integral torsion of the complete flux: full degree-wise shape laws for the prime-torsion multiplicities. A mirror programme (Conjectures 241 to 245) runs the merge order of Part X through genuinely quantum invariants, the mirror map, the genus-zero Gopakumar–Vafa numbers [150, 151], and the conifold radius in the flat coordinate, asserting strict monotonicity at every level. A stability programme (Conjectures 246 to 248) counts Bridgeland walls [156, 154], sharpens the  $\Gamma$ -constant of the Brill–Noether construction of [155], and asserts a tame atlas for wall arrangements in the spirit of definable Hodge theory [157]. An effective-geometry programme (Conjectures 249 to 253) posits polynomial condition numbers, in an algebraic gauge of distance to the discriminant, for Ricci-flat metrics near degeneration [158, 159], balanced-metric and Hermitian–Yang–Mills convergence [160, 163, 161, 162], matter spectral gaps [164, 165], and certified heterotic Yukawa couplings [166, 167]. A degeneration programme (Conjectures 254 and 255) closes with special-Lagrangian realization at infinite distance [152, 153] and a Laplace-tower rigidity law [168, 169]. The proved layer is Proposition 37: the classical monotonicity of degree and discriminant scale under merges, and the exact Chern floors of the five threefold configurations. Since deposit it has grown three times, by Proposition 39, the shifted torsion duality, by Proposition 30, whose complete-flux normal form resolves Conjecture 240 true and proves the interval-support clause of Conjecture 239, and by Proposition 38, the subcritical direction of Conjecture 238. One identity is recorded as an uncounted remark rather than a conjecture: the insertion-power formula of Remark 1, whose source construction [148] computes an infinite family of higher operations for the torus that could not be compared in full through the present channel, with the attribution recorded where it occurs. Conjectures 237 and 238 are stated in a random model; Conjectures 249 to 255 assert laws, the price of aiming at effective theorems in geometric analysis; Conjecture 255 is the part’s closest to prior work, since the small-eigenvalue asymptotics of the induced Kähler degeneration metrics are now a theorem of [170] and only the Ricci-flat, integrally LMHS-indexed content is claimed; and Conjecture 246 is possibly accessible from the existing wall geometry of [156].

Throughout, the flux notation of Part X is retained:  $E_n(R) = \Lambda^* R^n$ , and for  $H \in \Lambda^r \mathbb{Z}^n$  of odd degree  $r$ ,  $d_H = H \wedge -$ , with  $b_k(H; R)$  the cohomology dimensions and  $g_{n,r}$  the degree-wise profile on the Zariski-open locus of maximal wedge ranks, so that  $g_{n,3} = g_n$ . For the complete flux  $H_\Delta^{(n)}$ ,  $t_{p,k}(n)$  counts the  $p$ -primary cyclic summands of the integral cohomology in degree  $k$ , and  $T_p(n) = \sum_k t_{p,k}(n)$ . A dimension  $n$  is  $r$ -essential when  $g_{n,r}$  is not  $(1+t)$  times the profile one dimension lower, so that a form ignoring one coordinate cannot saturate. On the mirror side, configurations  $\mathbf{d}$  and merges are as in Part X;  $D(\mathbf{d}) = \prod_i d_i$ ,  $\Lambda(\mathbf{d}) = \prod_i d_i^{d_i}$ ; for the five one-parameter projective complete-intersection Calabi–Yau threefolds (5), (4, 2), (3, 3), (3, 2, 2), (2, 2, 2, 2), the mirror map  $q_{\mathbf{d}}(z) = z + O(z^2)$ , the genus-zero Gopakumar–Vafa invariants  $N_{\mathbf{d}}(\beta)$ , and the A-model Yukawa coupling  $K_{\mathbf{d}}(q) = D(\mathbf{d}) + \sum_{\beta \geq 1} N_{\mathbf{d}}(\beta) \beta^3 q^\beta / (1 - q^\beta)$  are the classical

mirror data [150], and  $q_c(\mathbf{d})$  is the smallest positive singularity of  $K_{\mathbf{d}}$ , the conifold point in the flat coordinate.

**Proposition 37** (Merge monotonicity and Chern floors). (i) Under every merge  $d_i, d_j \mapsto d_i + d_j - 1$ , the degree  $D$  strictly decreases and the scale  $\Lambda$  strictly increases; in particular the algebraic conifold point  $z_c = \Lambda^{-1}$  strictly decreases. (ii) For a complete-intersection Calabi–Yau threefold of multidegree  $\mathbf{d}$  in  $\mathbb{CP}^{k-1}$ ,  $k = \sum_i d_i$ , one has  $c_2(X) \cdot H = \frac{1}{2}(\sum_i d_i^2 - k)D$ , so the ratios  $\text{td}_2(X) \cdot H/H^3$  for the five threefold configurations (5), (4, 2), (3, 3), (3, 2, 2), (2, 2, 2, 2) are 5/6, 7/12, 1/2, 5/12, 1/3.

*Proof.* (i)  $d_i d_j - (d_i + d_j - 1) = (d_i - 1)(d_j - 1) > 0$  gives the degree drop. For  $\Lambda$ , the function  $g(x) = x \log x$  is strictly convex with  $g(1) = 0$ , and the pair  $(d_i + d_j - 1, 1)$  has the same sum as  $(d_i, d_j)$  with a strictly larger maximum, so majorization gives  $g(d_i + d_j - 1) > g(d_i) + g(d_j)$ , which is  $\log \Lambda' > \log \Lambda$ . (ii) From  $c(TX) = (1 + h)^k \prod_i (1 + d_i h)^{-1}$ , the degree-one term vanishes by the Calabi–Yau condition  $\sum_i d_i = k$ , and the degree-two term is  $\frac{1}{2}(\sum_i d_i^2 - k)h^2$ ; multiplying by  $H = h$  and integrating against the class of  $X$ , of degree  $D$ , gives the displayed value, and  $\text{td}_2 = c_2/12$  with  $H^3 = D$  gives the five ratios 50/60, 56/96, 54/108, 60/144, 64/192.  $\square$

### 53 Odd flux and torsion shapes

**Conjecture 237** (Odd-degree discrete saturation). *Fix an odd  $r \geq 3$  and a finite symmetric  $V \subset \mathbb{Z} \setminus \{0\}$  with  $\gcd(V) = 1$ , and let every coefficient of  $H_n \in \Lambda^r \mathbb{Z}^n$  be independent and uniform on  $V$ . Then  $\Pr[(b_k(H_n; \mathbb{Q}))_k = g_{n,r}] = 1 - O(e^{-cn})$  for some  $c = c(r, V) > 0$ .*

**Significance.** This lifts Conjecture 177 from three-forms to every odd degree, the full family in which wedge multiplication is a differential, against the Zariski-generic theory of [115]; the gcd normalization absorbs the scalar-invariance of rational ranks noted in Part X. The mechanism is anti-concentration for the structured minors of wedge multiplication, uniformly in the degree. A first decisive theorem would be a uniform positive lower bound at  $r = 5$ . In tests on the corpus described in the stress-test section, all sixty deposited dense  $r = 5$  trials and twenty-four independent trials at  $n = 9, 10$  hit a single profile. A counterexample would be an odd degree and a law with failure probability bounded away from zero.

**Conjecture 238** (Odd coverage-scale threshold). *Fix odd  $r \geq 3$ , include each coordinate  $r$ -subset independently with probability  $p_n$ , and give included monomials independent coefficients from a fixed finite symmetric nonzero set. Along  $r$ -essential dimensions,  $p_n n^{r-1}/\log n \rightarrow \infty$  implies saturation probability tending to one.*

**Significance.** The isolated-vertex scale of random  $r$ -uniform hypergraphs is conjectured to govern saturation in every odd degree, with the  $r$ -essential restriction excluding exactly the suspension dimensions where an unused coordinate is harmless, the parity phenomenon noted in Part X now built into the hypothesis. The zero direction is now proved in Proposition 38, a promotion, the essentiality hypothesis being exactly what an unused coordinate violates; the surviving supercritical direction is an extrapolation in the random model. In tests on the corpus described in the stress-test section, deposited  $r = 5$  sweeps at  $n = 9, 10$  rise from no hits to full saturation across the window. A counterexample would be an essential sequence saturating below the scale or failing above it.



**Proposition 38** (Odd coverage threshold, zero direction). Fix an odd  $r \geq 3$  and an  $r$ -essential dimension  $n$ . If  $p_n n^{r-1} / \log n \rightarrow 0$  then the saturation probability of Conjecture 238 tends to zero.

*Proof.* A coordinate lying in no chosen  $r$ -subset makes the profile  $(1+t)h$  with  $h$  the profile of the form read in the remaining  $n-1$  coordinates, and the argument of Proposition 28(i), with  $g_{n,r}$  in place of  $g_n$ , forces  $h = g_{n-1,r}$  and  $g_{n,r} = (1+t)g_{n-1,r}$  whenever such a form saturates; that is exactly what  $r$ -essentiality excludes. A fixed coordinate is unused with probability  $(1-p_n)^{\binom{n-1}{r-1}}$ , which is  $n^{-o(1)}$  when  $p_n n^{r-1} = o(\log n)$ , and two coordinates lie in only  $O(n^{r-2})$  common  $r$ -subsets, so the second-moment estimate of Proposition 28(ii) applies verbatim and unused coordinates exist with probability tending to one.  $\square$

**Proposition 39** (Shifted torsion duality). For every prime  $p$  and all  $n, k$ ,  $t_{p,k}(n) = t_{p,n+3-k}(n)$ .

*Proof.* Write  $u_k = b_k(H_\Delta; \mathbb{F}_p) - b_k(H_\Delta; \mathbb{Q})$ . Since  $d_{H_\Delta}$  raises degree by three, the universal coefficient theorem, applied to each residue class modulo three as an ordinary complex, gives  $u_k = t_{p,k}(n) + t_{p,k+3}(n)$ . The perfect pairing  $\Lambda^k \times \Lambda^{n-k} \rightarrow \Lambda^n \cong \mathbb{Z}$  intertwines  $d_{H_\Delta}$  with its own transpose up to sign, since  $H_\Delta \wedge \alpha \wedge \beta = \pm \alpha \wedge H_\Delta \wedge \beta$ , so over every field  $b_k = b_{n-k}$  and hence  $u_k = u_{n-k}$ . Setting  $s_k = t_{p,n+3-k}(n)$ , the identity in degree  $n-k$  reads  $s_k + s_{k+3} = u_k = t_{p,k}(n) + t_{p,k+3}(n)$ ; both sequences vanish for  $k \leq 2$ , since no differential enters below degree three, and beyond degree  $n$ , so induction in steps of three gives  $t_{p,k}(n) = s_k$ .  $\square$

**Conjecture 239** (Torsion shape law for the complete flux). For every prime  $p$  and every  $n$ , every interior positive triple of  $k \mapsto t_{p,k}(n)$  is strictly log-concave.

**Significance.** Part X's staircase, Conjecture 183, locates the onset of  $p$ -torsion; this asserts the complete degree-wise shape of the torsion once present, an integral refinement invisible to every field coefficient, in the twisted-cohomology setting whose integral theory is developed in [148]. The duality centre  $\frac{n+3}{2}$  sits half a step above Poincaré symmetry, the shift produced by the degree-three differential. The duality clause is proved in Proposition 39 and the interval-support clause in Proposition 30, which identifies the support as the full range  $2p+1 \leq k \leq n-2p+2$ ; both are now proved, and strict log-concavity is the surviving clause. In tests on the corpus described in the stress-test section, every nonzero profile through  $n = 12$  at  $p = 2, 3, 5$  passed all three clauses in an independent exact-rank recomputation, including the profiles 1, 10, 44, 10, 1 at  $n = 11$  and 1, 11, 54, 54, 11, 1 at  $n = 12$ , and a recomputation from the normal form extends the log-concavity check to every prime  $p \leq 23$  through  $n = 150$  without a failure. A counterexample would be a flat interior triple at any prime.

**Conjecture 240** (Post-onset torsion persistence, resolved true). For every prime  $p$ ,  $T_p(n+1) > T_p(n)$  for every  $n \geq 4p-1$ .

**Significance.** The staircase of Conjecture 183 never retreats: once  $p$ -torsion appears it grows strictly with the dimension, so each prime contributes a permanently expanding sector of torsion flux charge. Conditional on the totals of Conjecture 182, the  $p = 2$  case was already a theorem, now sharpened: the increments reduce to  $4^{m-1} - \frac{m}{m+1} \binom{2m}{m}$  and its even companion, both positive from  $m = 3$  with the small cases checked directly. In tests on the corpus described in the stress-test section, the independent recomputation gives  $T_2 = 1, 2, 10, 20, 66, 132$  from  $n = 7$  to 12 and  $T_3 = 1, 2$  at  $n = 11, 12$ , and the closed-form  $p = 2$  expression increases strictly

through  $n = 200$ . A counterexample would be one prime and one dimension with stagnant total torsion. *Resolution.* The conjecture is true for every prime, not only conditionally at  $p = 2$ : Proposition 30 gives  $T_p(2g + 2) = 2T_p(2g + 1)$  from the null vector in even dimension and  $T_{g+1} - 2T_g = \sum_{r \geq 1} \binom{g}{r-1} 2^{g+1-r} A_r > 0$  from Pascal's rule in odd dimension, both strict after onset. An independent recomputation confirms strict growth for every prime  $p \leq 11$  through dimension 150.

**Remark 1** (Insertion powers of the complete flux). The integral twisting theory of [148] attaches to a three-form its insertion powers, odd-degree forms driving an extended differential in the manner of Kraines' higher Massey powers [149]. Exact enumeration of the signed regular families on  $2m + 1$  labels gives, for  $m \leq 4$ , the identity that the  $m$ -th insertion power of the complete three-form is  $(-1)^{m-1}(2m + 1)^{m-1}$  times the complete  $(2m + 1)$ -form, a Cayley-like coefficient collapsing a sum of 76,545 terms at  $m = 4$ . Since [148] computes an infinite family of higher operations for the torus that could not be compared in full through the present channel, the identity is recorded here as a verified calibration observation with its attribution unresolved, and is not counted among the conjectures.

## 54 Quantum monotonicity along the merge order

**Conjecture 241** (Normalized mirror-map contraction). *For complete-intersection Calabi–Yau  $n$ -folds with  $n \geq 3$ , write  $\hat{q}_{\mathbf{d}}(x) = \Lambda(\mathbf{d}) q_{\mathbf{d}}(x/\Lambda(\mathbf{d})) = x + \sum_{m \geq 2} c_m(\mathbf{d})x^m$ . Under every merge,  $c_m(\mathbf{d}') \leq c_m(\mathbf{d})$  for every  $m \geq 2$ , with strict inequality for every  $m \geq 3$ .*

**Significance.** The mirror map is the transcendental heart of the classical mirror computations [150, 151], and the conjecture asserts that in the scale-free normalization fixed by  $\Lambda$ , whose merge growth is the proved half of Proposition 37, equation consolidation contracts every coefficient at once; the two-fold violations found in the deposited scan are excluded by the dimension hypothesis, and the  $m = 2$  equalities at  $n = 3$  fix the non-strict boundary exactly. A first decisive theorem would be the  $m = 2$  case in every dimension. In tests on the corpus described in the stress-test section, the deposited scan through dimension nine and order eight passed off dimension two, and an independent exact recomputation of all merges in dimensions three to five, 588 comparisons, found zero violations. A counterexample would be one merge and one order breaking the contraction.

**Conjecture 242** (Gopakumar–Vafa amplification). *For the five one-parameter threefold configurations and every merge edge,  $A_\beta = N_{\mathbf{d}'}(\beta)/N_{\mathbf{d}}(\beta)$  is strictly increasing in  $\beta \geq 1$ .*

**Significance.** Coefficientwise growth of curve counts under merges would already be a shape law; monotonicity of the ratio is much sharper, an ordering of the five geometries by quantum growth rate that refines degree-by-degree. The mechanism is the conifold hierarchy of Conjecture 244 propagated backward through Conjecture 243. A first decisive theorem would be the asymptotic ratio divergence from separated conifold radii. In tests on the corpus described in the stress-test section, deposited exact invariants through  $\beta = 60$  and an independent from-scratch mirror computation through  $\beta = 24$  agree and pass on all five edges, the amplification rising from 1.41 at  $\beta = 1$  to  $10^{15}$  scale at high degree on the steepest edge. A counterexample would be one edge and one degree with a falling ratio.

**Conjecture 243** (Monotone conifold approach). *For each of the five configurations,  $R_\beta = N(\beta + 1)/N(\beta)$  is strictly increasing, and  $R_\beta \rightarrow 1/q_c$ ; equivalently the invariants are strictly log-convex from degree one with ratio limit the conifold growth constant.*

**Significance.** That the conifold point governs the exponential growth rate of genus-zero invariants is folklore of the resurgence and enumerative literature; the content here is global monotonicity from the first degree, an unexpectedly rigid convexity of the full quantum series with no preasymptotic dip anywhere in the landscape’s simplest column. A first decisive theorem would be log-convexity in large degree from the conifold expansion. In tests on the corpus described in the stress-test section, deposited invariants through  $\beta = 60$  and the independent recomputation through  $\beta = 24$  are strictly log-convex for all five families, with final tested ratios still below the independently computed constants  $1/q_c = 1980.15, 641.78, 458.89, 269.61, 159.26$ . A counterexample would be one family and one degree with a falling ratio.

**Conjecture 244** (Merge monotonicity of the conifold constant). *For every merge edge among the five configurations,  $q_c(\mathbf{d}') < q_c(\mathbf{d})$ .*

**Significance.** The algebraic conifold point  $z_c = \Lambda^{-1}$  decreases under merges by Proposition 37; the conjecture asserts that the transcendental image under the mirror map preserves this order, so that consolidation strictly raises the quantum growth constant  $1/q_c$ . This is a comparison of periods across distinct Picard–Fuchs systems, not a formal consequence of the algebraic ordering. A first decisive theorem would be one edge by interval arithmetic on the hypergeometric sums; indeed the whole five-edge statement is a finite certification target for validated numerics, and should eventually be a computation rather than a conjecture. In tests on the corpus described in the stress-test section, independent sixty-digit evaluations of the five constants order strictly along every edge, with ratios of consecutive constants between 1.69 and 4.32. A counterexample would be one edge with reversed constants.

**Conjecture 245** (Conifold-normalized Yukawa monotonicity). *For every merge edge among the five configurations and every  $0 < x < 1$ ,  $K_{\mathbf{d}'}(xq_c(\mathbf{d}'))/D(\mathbf{d}') > K_{\mathbf{d}}(xq_c(\mathbf{d}))/D(\mathbf{d})$ .*

**Significance.** At the same fraction of the respective conifold radii, the quantum-corrected Yukawa coupling normalized by its classical intersection number strictly increases under consolidation: a scale-free comparison of quantum corrections across geometries in which both normalizations are essential, since the deposited adversarial control shows the raw instanton comparison reversing at every grid point. A first decisive theorem would be the small- $x$  regime from degree-one invariants alone. In tests on the corpus described in the stress-test section, all 495 deposited grid tests and an independent recomputation on all five edges at nineteen fractions passed, with the raw-instanton control failing everywhere, retained in place. A counterexample would be one edge and one fraction with a reversed normalized coupling.

## 55 Stability walls and sharp constants

**Conjecture 246** (Quadratic wall complexity). *Let  $X$  be a Picard-rank-one Calabi–Yau threefold with a constructed component  $\text{Stab}^\dagger(X)$  satisfying the support property,  $K$  a compact set in a fundamental domain for the  $\widetilde{\text{GL}}_2^+(\mathbb{R})$  action, and  $\|\cdot\|$  a Euclidean norm on the numerical charge*

lattice. Then there is  $C_K$  with  $N_K(v) \leq C_K(1 + \|v\|)^2$  for every primitive charge  $v$ , where  $N_K(v)$  counts numerical walls for  $v$  meeting  $K$ .

**Significance.** Local finiteness of walls is classical [154], and the Picard-rank-one wall geometry of [156] bounds the largest walls and proves finiteness in bounded regions. On a compact set the support property bounds each possible destabilising class by  $O_K(\|v\|)$ , which gives a polynomial count. In the usual rank-four numerical charge lattice, a naive count after quotienting by the direction of  $v$  is cubic rather than quadratic. The exponent two therefore needs genuine Picard-rank-one wall geometry and does not follow merely because a wall is a quadric. A first decisive theorem would isolate the extra geometric restriction that removes one lattice degree of freedom. A counterexample would be a charge sequence with superquadratic wall counts in a fixed compact region.

**Conjecture 247** (Sharp  $\Gamma$  for Brill–Noether stability). *For a Picard-rank-one complete-intersection Calabi–Yau threefold covered by the Brill–Noether construction of [155], with  $\varepsilon_*$  the supremum of  $\varepsilon$  for which their condition  $\text{BG3}(\varepsilon)$  holds, the infimum of  $\gamma$  for which  $\Gamma = \gamma H^2 - \text{td}_2(X)$  satisfies the  $\Gamma$ -twisted Bayer–Macrì–Toda ( $\Gamma$ -BMT) stability condition is exactly  $\max\{4/(H^3\varepsilon_*), \text{td}_2(X) \cdot H/H^3\}$ .*

**Significance.** The construction of [155] proves that  $\text{BG3}(\varepsilon)$  yields a valid  $\Gamma$  on this ray and leaves its minimization open; the conjecture asserts their bound is already the exact optimum, so the Brill–Noether route loses nothing. The sharpness direction is the exposed half: it requires destabilizing objects at every smaller  $\gamma$ , a converse the construction itself does not supply. The topological floor  $\text{td}_2 \cdot H/H^3$  is unconditional, and Proposition 37 computes it as  $5/6, 7/12, 1/2, 5/12, 1/3$  across the five configurations. A first decisive theorem would be optimality of the floor branch for the quintic. A counterexample would be a valid  $\Gamma$  strictly below the displayed maximum.

**Conjecture 248** (Tame wall atlas). *For  $X$  as in Conjecture 246 and compact  $K$ , fix a finite effective  $\mathbb{R}_{\text{an},\text{exp}}$  presentation of the central charge and the numerical wall equations. Let formula format record length, quantifier depth, polynomial degree and logarithmic coefficient height in that presentation. There is  $C = C(X, K)$  such that the union  $W_Q$  of numerical walls for primitive charges with  $\|v\| \leq Q$  has a defining formula of format at most  $Q^C$ , and  $K \setminus W_Q$  has a compatible definable stratification into  $O(Q^C)$  strata and chambers.*

**Significance.** This imports the definability paradigm of arithmetic Hodge theory [157] into Bridgeland geometry. Qualitative o-minimal definability alone supplies neither a canonical format nor a polynomial chamber bound. The statement therefore fixes the presentation and treats polynomial formula format and polynomial stratification count as separate claims. The numerical-wall formulation is essential because actual walls can accumulate. A first decisive theorem would give an effective definition and a quantitative cell decomposition for one wall family. A counterexample would be a wild wall arrangement or superpolynomial stratum growth.

## 56 Effective geometry of Calabi–Yau families

**Gauge convention.** Two positive gauges  $\delta$  and  $\delta'$  on the smooth locus of a fixed compact semialgebraic chart are *power-equivalent* if, after truncation to  $(0, 1]$ , there are positive constants

$c, C, a, b$  with  $c\delta^a \leq \delta' \leq C\delta^b$ . Every polynomial estimate below is understood up to this equivalence. The gauge  $\delta_{\text{alg}}$  is represented by a truncated norm of generators of the reduced discriminant ideal. For a family of bundle pairs,  $\delta_{\text{pair}}$  is represented by the minimum of  $\delta_{\text{alg}}$  and a fixed positive stability margin that vanishes exactly on the chosen chamber boundary.

**Conjecture 249** (Polynomial conditioning of Ricci-flat metrics). *For a projective polarised Calabi–Yau family over a compact semialgebraic chart, with  $\delta_{\text{alg}}$  the truncated norm of generators of the reduced discriminant ideal, the volume-normalised Ricci-flat metrics satisfy, for every derivative order  $\ell$ , bounds  $\|\nabla^\ell \text{Rm}\|_\infty \leq C_\ell \delta_{\text{alg}}^{-A_\ell}$ , with diameter bounded by  $C\delta_{\text{alg}}^{-A}$  and, in the noncollapsing case, injectivity radius bounded below by  $C^{-1}\delta_{\text{alg}}^A$ , the same polynomial form holding for the regularity scale in collapsing strata away from the canonical collapsing set.*

**Significance.** Diameter bounds at degeneration are now sharp in the degeneration parameter [158], and quantitative collapse is understood in important families [159]; the conjecture asserts the global effective statement, polynomial control of all curvature derivatives in an algebraic gauge, the form of estimate that certifies every numerical metric computation at once. Łojasiewicz inequalities make the gauge canonical up to the exponents. A first decisive theorem would be the curvature bound for one-parameter quintic degenerations. A counterexample would be a family with essential superpolynomial blow-up in the algebraic gauge.

**Conjecture 250** (Discriminant-effective balanced convergence). *In the setting of Conjecture 249, the level- $k$  balanced metrics satisfy  $\|g_{s,k} - g_s\|_{C^\ell(g_s)} \leq C_\ell \delta_{\text{alg}}(s)^{-A_\ell} k^{-1}$  whenever  $k \geq C_\ell \delta_{\text{alg}}(s)^{-B_\ell}$ , with the corresponding  $k^{-N-1}$  bounds after subtracting  $N$  Bergman expansion terms.*

**Significance.** On a fixed smooth fiber this is classical Bergman asymptotics; the conjecture makes the rate uniform in the family with polynomial deterioration at the discriminant, exactly the regime probed by the balanced-metric power laws observed numerically near large complex structure [160]. A first decisive theorem would be the  $C^0$  case near a conifold degeneration. A counterexample would be a family where the required level grows faster than any power of the inverse algebraic distance.

**Conjecture 251** (Discriminant-effective Hermitian–Yang–Mills convergence). *For an algebraic family of polarised Calabi–Yau threefolds with slope-stable bundles on a constant-stability chamber, with  $\delta_{\text{pair}}$  as in the gauge convention, the level- $k$  balanced-bundle approximations converge to the Hermitian–Yang–Mills connection in Coulomb gauge with  $\|A_{s,k} - A_s\|_{C^\ell} \leq C_\ell \delta_{\text{pair}}^{-A_\ell} k^{-1}$  whenever  $k \geq C_\ell \delta_{\text{pair}}^{-B_\ell}$ .*

**Significance.** Hermitian–Yang–Mills existence survives conifold transitions [161], iteration on a fixed stable bundle converges [163], and the numerical algorithms are mature [162]; the conjecture is the missing family statement, a polynomial condition number in distance to where the fiber or the stability degenerates. A first decisive theorem would be the fixed-fiber statement with explicit stability-margin dependence. A counterexample would be a chamber whose HYM approximation cost is essentially superpolynomial at the boundary.

**Conjecture 252** (Polynomial matter spectral gap). *In the setting of Conjecture 251, for a representation and form degree with locally constant bundle cohomology, the smallest positive eigenvalue of the bundle-valued Dolbeault Laplacian satisfies  $\lambda_+(s) \geq C^{-1}\delta_{\text{pair}}(s)^A$ , simultaneously for any fixed finite list of representations and degrees.*

**Significance.** Bundle-valued spectra are computable numerically [164, 165] and rigorous eigenvalue floors are an explicitly wanted certificate in that programme; the conjecture asserts the floor is polynomial in the algebraic gauge, with the constant-cohomology hypothesis essential since a jump lets an eigenvalue cross zero. A first decisive theorem would be the scalar case on quintics near a conifold. A counterexample would be an admissible chamber with essentially exponential spectral collapse.

**Conjecture 253** (Certified heterotic Yukawa convergence). *Under Conjectures 249 to 252, the level- $k$  physical heterotic Yukawa couplings satisfy  $|Y_{abc}^{(k)}/Y_{abc} - 1| \leq C \delta_{\text{pair}}^{-A} \delta_Y^{-A} k^{-1}$  for  $k \geq C \delta_{\text{pair}}^{-B} \delta_Y^{-B}$ , where  $\delta_Y$  is the power-equivalence class of a truncated algebraic gauge for the zero locus of the corresponding holomorphic coupling. An absolute error bound of the same form holds with no  $\delta_Y$  factor.*

**Significance.** Physical Yukawa couplings and even quark masses are now computed numerically from learned geometric data [166, 167], with only a-posteriori error diagnostics; the conjecture asserts the a-priori certificate, relative error polynomial in the two algebraic gauges, the  $\delta_Y$  factor being necessary because no uniform relative bound can survive a coupling crossing zero. A first decisive theorem would be the absolute bound for a single standard-embedding coupling. A counterexample would be an admissible family whose certified level grows faster than any power.

## 57 Degenerations and towers

**Conjecture 254** (Saturation-index special-Lagrangian realization). *For a semistable one-parameter Calabi–Yau threefold degeneration of type II, III, or IV with unipotent monodromy, let  $L_{\text{orb}}$  be the nilpotent-orbit lattice of asymptotically light D3 charges of common limiting phase,  $L_{\text{sat}}$  its rational saturation in integral homology, and  $I_\tau$  the exponent of  $L_{\text{sat}}/L_{\text{orb}}$ . Every primitive charge generating an extremal ray of the saturated light cone with the required positive limiting phase has a multiple  $m\gamma$ , with  $m$  dividing  $I_\tau$ , admitting a calibrated special-Lagrangian representative for all sufficiently deep parameters.*

**Significance.** Special-Lagrangian towers are constructed in Tyurin limits [152] and the type II–III–IV asymptotics are developed in [153]; the new clause is arithmetic, the divisor bound  $m \mid I_\tau$  tying calibrated existence to the saturation exponent of the monodromy-orbit lattice, so that the failure of primitive realization is quantized by lattice theory. At  $I_\tau = 1$  the statement reduces to primitive realization. Extremality of the ray is the stability surrogate: a light charge on an extremal ray admits no decomposition into non-proportional light charges of the common phase, and decay channels involving heavy charges are excluded deep in the limit by mass comparison, so the wall-crossing obstructions that block naive realization claims are switched off by hypothesis. A first decisive theorem would be the type II case with  $I_\tau = 1$ . A counterexample would be an extremal ray whose minimal calibrated multiple fails to divide the saturation exponent.

**Conjecture 255** (LMHS-controlled Laplace towers). *For a semistable one-parameter polarised Calabi–Yau degeneration at infinite Weil–Petersson distance, fix a normalisation of  $d_{\text{WP}}$  and volume-normalise the Ricci-flat metrics. The vanishing scalar Laplace spectrum admits a finite partition into tower types  $a$  with rational exponents  $\alpha_a$  such that: (i) the list of exponents depends only on the isomorphism class of the integral limiting mixed Hodge structure and the combinatorial*



*dual-intersection and essential-skeleton data. (ii) for each tower there is a positive normalizer  $N_a(t) = e^{2\alpha_a d_{WP}(t)} \ell_a(t)$ , with  $\ell_a$  slowly varying in  $e^{d_{WP}(t)}$ , such that for every fixed  $j$  the quantity  $N_a(t) \lambda_{a,j}(t)$  converges to the  $j$ -th eigenvalue of a canonical weighted Laplace operator on the collapsed base geometry. (iii) with eigenvalues labelled in increasing order at each parameter, every fixed-rank eigenvalue branch tending to zero belongs eventually to one of these towers. The limiting operator may depend on the canonical metric data of the collapsed base and is not asserted to be determined by the limiting mixed Hodge structure alone.*

**Significance.** For induced Kähler degeneration metrics, exact small-eigenvalue rates are now a theorem of [170]. General skeleton-Laplacian convergence is not. The conjecture makes three stronger claims for volume-normalised Ricci-flat metrics. It asserts a finite exhaustive tower decomposition, an arithmetic classification of the rational exponents by the integral limiting mixed Hodge structure and skeleton combinatorics, and spectral convergence within each tower. Only the exponent list is claimed to be determined by that discrete data. The slowly varying normalizer is deliberate: metric collapse is known to generate logarithmic scales, and only regular variation in the Weil–Petersson gauge is asserted. The limiting operator is allowed to retain canonical metric information from the collapsed base. A first decisive theorem would establish the finite decomposition for a type IV maximal degeneration. A counterexample would be a light eigenvalue escaping every listed tower or an exponent varying while the stated discrete data remain fixed, or a tower whose normalizer fails regular variation.

## Part XIV

# Convexity, defect, and dispersion laws for projective invariants

*Added to the deposit: 19 August 2026.*

The fifteen conjectures below take three of the oldest invariants of projective algebraic geometry, the middle cohomology of a complete intersection, the graded Betti table of an embedded surface, and the Hodge decomposition of a hypersurface, and conjecture the discrete shape of each as its defining degrees move. They form three programmes. A convexity programme (Conjectures 256 to 261) studies the degree-normalized primitive middle Betti number of smooth complete intersections on the full multidegree lattice, asserting a coherent discrete-convexity package: coordinate monotonicity and convexity, supermodular absolute interaction with log-submodular relative interaction, Schur-convexity at fixed degree sum, and a strict decrease under every codimension-raising split that preserves the adjunction coefficient. A syzygy programme (Conjectures 262 to 266) studies the two nontrivial rows of the Betti tables of  $\mathbf{P}^1 \times \mathbf{P}^1$  under Segre–Veronese embeddings, replacing global shape claims by a controlled-defect law: the nonlinear row is strictly log-concave, while the linear row can fail log-concavity at most once, only past its mode, only where the nonlinear row is already present, and only within one step of that row’s mode. A dispersion programme (Conjectures 267 to 270) studies the normalized primitive Hodge profiles of smooth hypersurfaces as the degree grows, asserting that the known drift toward the Eulerian profile is monotone in four canonical stochastic orders: likelihood-ratio flattening, central-window escape, convex-order spread, and total-variation distance. The proved layer is

five propositions: Proposition 40 proves all six convexity laws for curves from the exact formula  $B = \sum_i d_i - c - 2 + 2/D$ , Proposition 42 pins the two syzygy rows to the Hilbert series by an exact antidiagonal identity, Proposition 43 proves all four dispersion laws for surfaces from an explicit rational function of the degree, Proposition 41 proves the coordinate monotonicity and convexity laws of Conjectures 256 and 257 for hypersurfaces in every dimension from a closed form, and Proposition 44 derives Conjectures 268 and 269 from the likelihood-ratio flattening of Conjecture 267; in addition, for odd  $n$  the Calabi–Yau slice of Conjecture 261 is already a corollary of the merge monotonicity of Proposition 17. Two negative controls are retained in place: the unnormalized primitive Betti number fails Schur-convexity and has no split direction already for curves and surfaces, and coordinatewise monotone convergence to the Eulerian limit fails at dimension twelve, so the normalization of the convexity programme and the total-variation formulation of the dispersion programme are both forced. Conjectures 262 to 266 rest on the thirteen complete published tables, the entire deposited corpus, and are resting on a small evidence base; Conjecture 262 sits closest to the published unimodality conjecture of [184] and is the part’s closest to prior work; Conjectures 267 to 270 assert laws at every degree and dimension, and the Eulerian limit itself is exact in the abelian setting of [182] and asymptotic in the toric setting of [181], so only the monotone interpolation is claimed as new.

For a smooth complex complete intersection  $X^n(\mathbf{d}) \subset \mathbf{P}^{n+c}$  of dimension  $n \geq 1$ , codimension  $c \geq 1$ , and multidegree  $\mathbf{d} = (d_1, \dots, d_c)$  with every  $d_i \geq 2$ , write  $D(\mathbf{d}) = \prod_i d_i$ ; the Lefschetz theorem confines the new topology to the middle degree, and the primitive middle Betti number is  $b_n^{\text{prim}} = (-1)^n(\chi(X) - (n+1))$  with  $\chi$  evaluated exactly by the Chern-class formula  $\chi = D[H^n](1+H)^{n+c+1}/\prod_i(1+d_iH)$  [177, 178]. The invariant of the convexity programme is the *degree-normalized primitive complexity*  $B_{n,c}(\mathbf{d}) = b_n^{\text{prim}}/D(\mathbf{d})$ , and  $e_i$  denotes a coordinate step in the multidegree lattice. For the Segre–Veronese embedding of  $\mathbf{P}^1 \times \mathbf{P}^1$  by  $\mathcal{O}(a, b)$  with  $a, b \geq 2$ , in characteristic zero, the graded Betti numbers  $\beta_{p,p+q}(a, b)$  of the homogeneous coordinate ring vanish outside  $q = 0, 1, 2$ ; write  $\Delta_p^{(q)} = \beta_{p,p+q}^2 - \beta_{p-1,p-1+q}\beta_{p+1,p+1+q}$  for the log-concavity determinant at an interior index of the  $q$ th row, call  $p$  a *defect* of the row when  $\Delta_p^{(q)} < 0$ , and let  $m_q$  be the least index at which the row attains its maximum. For a smooth hypersurface  $X_d^n \subset \mathbf{P}^{n+1}$  with  $n \geq 2$ ,  $d \geq 3$ , the primitive Hodge numbers are  $h_p(n, d) = [t^{(p+1)d-(n+2)}](1+t+\dots+t^{d-2})^{n+2}$  by Griffiths’ formula [180]; write  $\pi_{n,d}$  for the normalized profile  $h_p/\sum_j h_j$  on  $\{0, \dots, n\}$ , symmetric about  $n/2$ , and  $\varepsilon_n(p) = A(n+1, p)/(n+1)!$  for the Eulerian profile, with  $A$  the Eulerian numbers. A profile  $P$  precedes  $Q$  in *convex order* when  $\mathbf{E}\varphi(P) \leq \mathbf{E}\varphi(Q)$  for every convex  $\varphi$ , and  $\|P - Q\|_{\text{TV}} = \frac{1}{2} \sum_p |P(p) - Q(p)|$ .

**Proposition 40** (Curve calibration for the convexity programme). For  $n = 1$  one has  $B_{1,c}(\mathbf{d}) = \sum_i d_i - c - 2 + 2/D$ , and all six laws of Conjectures 256 to 261 hold strictly for curves.

*Proof.* The genus formula for complete-intersection curves gives  $b_1 = D(\sum_i d_i - c - 2) + 2$ , whence the displayed  $B$ . Write  $u = 2/D$ . For Conjecture 256, a step in  $d_i$  adds 1 to the linear part and changes  $u$  by  $-2/(D(d_i+1))$ , so  $\Delta_i B = 1 - 2/(D(d_i+1)) \geq 1 - \frac{1}{3} > 0$ . For Conjecture 257, differencing again gives  $\Delta_i^2 B = 4/(D(d_i+1)(d_i+2)) > 0$ . For Conjecture 258, the linear part has vanishing mixed difference and  $\Delta_i \Delta_j (2/D) = 2/(D(d_i+1)(d_j+1)) > 0$ . For Conjecture 259, writing  $x = d_i$ ,  $y = d_j$ ,  $S = \sum_k d_k$ , a direct expansion using  $D D_{ij} = D_i D_j$  gives the exact identity

$$B(\mathbf{d} + e_i + e_j)B(\mathbf{d}) - B(\mathbf{d} + e_i)B(\mathbf{d} + e_j) = -1 + \frac{2(S + x + y - c)}{D(x+1)(y+1)},$$

and the right side is negative: with  $s_k = d_k - 1$  and  $R = \prod_{k \neq i,j} d_k$ , one has  $S + x + y - c =$

$2x + 2y - 2 + \sum' s_k$ , while  $xy(x+1)(y+1) \geq 6(x+y)$  and  $R \geq 1 + \sum' s_k$  give  $D(x+1)(y+1) \geq 6(x+y) + 6(x+y) \sum' s_k$ , so the fraction is less than 1 with room to spare. For Conjecture 260, at fixed  $\sum_i d_i$  and  $c$  the linear part is constant and a balancing transfer  $(x, y) \mapsto (x-1, y+1)$  with  $x \geq y+2$  multiplies  $D$  by  $(x-1)(y+1)/(xy) > 1$ , so  $u$  strictly decreases and  $B$  is strictly Schur-convex. For Conjecture 261, a split  $d \mapsto (a, b)$  with  $a+b = d+1$  fixes the linear part and multiplies  $D$  by  $ab/d > 1$ , since  $ab \geq 2(d-1) \geq d+1$ , so  $B$  strictly decreases.  $\square$

**Proposition 41** (Hypersurface calibration for the convexity programme). For a smooth hypersurface ( $c = 1$ ) of degree  $d \geq 2$  in  $\mathbf{P}^{n+1}$ , write  $x = d-1$ ; then  $B_{n,1}(d) = f_n(x)$  with  $f_n(x) = (x^{n+2} + (-1)^n x)/(x+1)^2$ , and  $f_n$  is strictly increasing and strictly convex on  $x \geq 1$  for every  $n \geq 1$ . In particular Conjectures 256 and 257 hold at codimension one in every dimension.

*Proof.* The Chern-class formula gives  $b_n^{\text{prim}} = (-1)^n(\chi - (n+1))$  with  $\chi = d[H^n](1+H)^{n+2}/(1+dH)$ , and a direct evaluation of the residue yields  $b_n^{\text{prim}} = ((d-1)^{n+2} + (-1)^n(d-1))/d$ , so  $B_{n,1}(d) = b_n^{\text{prim}}/d = f_n(x)$  with  $x = d-1$ ; for  $n = 1$  this is  $(d-1)(d-2)/d$ , matching Proposition 40. Differentiating,  $(x+1)^3 f'_n(x) = nx^{n+2} + (n+2)x^{n+1} + (-1)^n(1-x)$ : for odd  $n$  the last term is  $x-1 \geq 0$ , and for even  $n$  the numerator is  $x^{n+1}(nx+n+2) - (x-1) \geq x^{n+1}(nx+n+1) > 0$  since  $x-1 \leq x^{n+1}$ , so  $f'_n > 0$  on  $x \geq 1$ . A second differentiation gives  $(x+1)^4 f''_n(x) = x^n\{n(n-1)x^2 + 2(n-1)(n+2)x + (n+1)(n+2)\} + 2(-1)^n(x-2)$ , whose bracket is positive and, times  $x^n$ , dominates the sign-varying correction  $2(-1)^n(x-2)$  for every  $n \geq 1$  and  $x \geq 1$ , so  $f''_n > 0$ . Strict positivity of  $f'_n$  and  $f''_n$  gives the strict first and second integer differences of Conjectures 256 and 257 at codimension one.  $\square$

**Proposition 42** (Antidiagonal calibration for the syzygy programme). Let  $N+1 = (a+1)(b+1)$  and  $c_j = [t^j](1-t)^{N+1} \sum_{k \geq 0} (ka+1)(kb+1)t^k$ . Then  $\beta_{j-1,j}(a, b) - \beta_{j-2,j}(a, b) = (-1)^{j-1}c_j$  for every  $j \geq 2$ , the resolution has length  $N-2$ , and  $\beta_{N-2,N} = (a-1)(b-1)$ .

*Proof.* The embedding is projectively normal, so the Hilbert series of the coordinate ring is  $\sum_k \dim H^0(\mathcal{O}(ka, kb)) t^k = \sum_k (ka+1)(kb+1)t^k$ , and taking Euler characteristics of the minimal free resolution over the polynomial ring in  $N+1$  variables gives  $\sum_p (-1)^p \beta_{p,j} = c_j$  in every internal degree  $j$ . Since  $\beta_{p,j}$  vanishes unless  $j-p \in \{0, 1, 2\}$ , the degree- $j$  alternating sum collapses to the two displayed terms for  $j \geq 2$ . The coordinate ring is a normal semigroup ring, hence Cohen–Macaulay of dimension three and codimension  $N-2$ , so the resolution has length  $N-2$  and the numerator  $h(t) = (1-t)^3 \sum_k (ka+1)(kb+1)t^k$  is a polynomial whose coefficients are third differences of a quadratic sequence:  $h(t) = 1 + (N-2)t + (a-1)(b-1)t^2$  exactly. Then  $(1-t)^{N+1} \sum_k (ka+1)(kb+1)t^k = (1-t)^{N-2}h(t)$  has top coefficient  $c_N = (-1)^{N-2}(a-1)(b-1)$ , and since  $\beta_{N-1,N} = 0$  the degree- $N$  identity reads  $(-1)^{N-2}\beta_{N-2,N} = c_N$ , giving  $\beta_{N-2,N} = (a-1)(b-1)$ .  $\square$

**Proposition 43** (Surface calibration for the dispersion programme). For  $n = 2$  set  $x(d) = \pi_{2,d}(0) = \frac{(d-2)(d-3)}{6(d^2-3d+3)}$ . Then  $x(d)$  is strictly increasing in  $d \geq 3$  with  $x(d) < \frac{1}{6}$  and  $x(d) \rightarrow \frac{1}{6}$ , and consequently all four laws of Conjectures 267 to 270 hold for surfaces, strictly for  $d \geq 4$ .

*Proof.* For a smooth surface of degree  $d$  one has  $h_0 = \binom{d-1}{3}$  and  $\sum_j h_j = d^3 - 4d^2 + 6d - 3 = (d-1)(d^2 - 3d + 3)$ , giving the displayed  $x(d) = \frac{(d-2)(d-3)}{6(d^2-3d+3)}$ . Its increment has the sign of  $(2d-5)(d^2-3d+3) - (d^2-5d+6)(2d-3) = 2d^2 - 6d + 3 > 0$  for  $d \geq 3$ , so  $x$  is strictly increasing, and  $x < \frac{1}{6}$  is the inequality  $(d-2)(d-3) < d^2 - 3d + 3$ , which reduces to  $2d > 3$ . The profile is  $(x, 1-2x, x)$  and  $\varepsilon_2 = (\frac{1}{6}, \frac{4}{6}, \frac{1}{6})$ , so on three symmetric points every one of the four laws is

equivalent to the monotone increase of  $x$  below its limit  $\frac{1}{6}$ : the likelihood ratio, the central mass  $1 - 2x$ , every convex functional, and  $\|\pi_{2,d} - \varepsilon_2\|_{\text{TV}} = 2(\frac{1}{6} - x)$  are all monotone in  $x$ .  $\square$

**Proposition 44** (Dispersion consequences of likelihood-ratio flattening). Conjecture 267 implies Conjectures 268 and 269.

*Proof.* The profile  $\pi_{n,d}$  is symmetric about  $n/2$ . Conjecture 267 is the monotone-likelihood-ratio order  $\pi_{n,d} \leq_{\text{lr}} \pi_{n,d+1}$  on the common positive support, which for a symmetric profile is equivalent to the likelihood-ratio order of the centered displacement  $R = |P - n/2|$ ; since likelihood-ratio order implies first-order stochastic dominance,  $R$  under degree  $d+1$  dominates  $R$  under degree  $d$ , an outer position newly nonzero only strengthening the dominance. Each central window  $\{|P - n/2| \leq \frac{n}{2} - k\}$  is a lower set for  $R$ , so its mass decreases, which is Conjecture 268. For any convex  $\varphi$  the symmetrization  $\psi(p) = \frac{1}{2}(\varphi(p) + \varphi(n-p))$  is convex and nondecreasing in  $R$ , and symmetry gives  $\mathbf{E}\varphi(P) = \mathbf{E}\psi(P)$ ; stochastic dominance of  $R$  then yields  $\mathbf{E}\varphi(\pi_{n,d}) \leq \mathbf{E}\varphi(\pi_{n,d+1})$ , which is Conjecture 269.  $\square$

## 58 Normalized primitive cohomology of complete intersections

**Conjecture 256** (Coordinate monotonicity of normalized complexity). *For every  $n \geq 1$ ,  $c \geq 1$ , every  $\mathbf{d} \in \mathbf{Z}_{\geq 2}^c$ , and every coordinate  $i$ ,  $B_{n,c}(\mathbf{d} + e_i) > B_{n,c}(\mathbf{d})$ .*

**Significance.** Raising one defining degree multiplies the raw topological size by the growing degree factor, and the conjecture asserts that the primitive middle cohomology outruns that trivial multiplicative scale in every dimension and codimension: normalized complexity has strictly positive marginal growth on the whole degree lattice. The mechanism is transparent for curves, where Proposition 40 proves it from the exact formula, but the alternating Chern-class sum offers no visible positivity in higher dimension. The nearest related results are the explicit primitive-Betti formulas of [177] and the Euler-characteristic implications of [178], neither of which orders the degree lattice. The hypersurface case, monotonicity at every  $n$ , is now proved in Proposition 41; a first decisive theorem in higher codimension would be codimension two. In tests on the corpus described in the stress-test section, all 59,100 coordinate steps on the exhaustive grid and 13,608 steps on the extended adversarial grid passed, with no failure. A counterexample would be a multidegree at which added degree is sublinear after normalization.

**Conjecture 257** (Strict discrete convexity in each degree). *Under the same hypotheses,  $B_{n,c}(\mathbf{d} + 2e_i) - 2B_{n,c}(\mathbf{d} + e_i) + B_{n,c}(\mathbf{d}) > 0$ .*

**Significance.** Beyond positive marginal growth, the marginal growth itself accelerates: each one-variable restriction of  $B$  is strictly convex on the integer lattice, so normalized complexity bends upward in every defining degree. The mechanism cannot be coefficientwise, since the Chern-class expression is alternating; Proposition 40 proves the curve case with second difference  $4/(D(d_i + 1)(d_i + 2))$ , a shape that suggests a positive rational generating function in general, and Proposition 41 proves the hypersurface case in every dimension. The nearest related result is [177]. A first decisive theorem would be convexity for quadric-heavy multidegrees, where the formula degenerates. In tests on the corpus described in the stress-test section, all 59,100 second differences on the exhaustive grid and 13,608 on the adversarial extension were strictly positive. A counterexample would be one multidegree with a concave step, marking an unexpected cancellation in the Chern-class sum.

**Conjecture 258** (Additive supermodularity across degrees). *For  $i \neq j$ ,  $B(\mathbf{d} + e_i + e_j) - B(\mathbf{d} + e_i) - B(\mathbf{d} + e_j) + B(\mathbf{d}) > 0$ .*

**Significance.** Distinct defining equations reinforce one another: raising  $d_i$  makes a subsequent raise of  $d_j$  strictly more valuable in absolute normalized complexity, so the discrete Hessian of  $B$  is positive off the diagonal as well as on it. Together with Conjecture 257 this asserts a full multivariate discrete convexity, and the curve case, mixed difference  $2/(D(d_i + 1)(d_j + 1))$ , is proved in Proposition 40. The nearest related results are [177, 179]. A first decisive theorem would be supermodularity at codimension two, where only one mixed pair exists. In tests on the corpus described in the stress-test section, all 84,600 mixed second differences on the exhaustive grid and 17,281 on the adversarial extension were strictly positive. A counterexample would be a pair of degrees whose interaction turns negative.

**Conjecture 259** (Strict log-submodularity). *Whenever the four values are positive,  $B(\mathbf{d} + e_i + e_j) B(\mathbf{d}) < B(\mathbf{d} + e_i) B(\mathbf{d} + e_j)$ .*

**Significance.** Set against Conjecture 258, this is the striking pairing of the programme: absolute interaction of two degrees is positive while proportional interaction is negative, so  $\log B$  is strictly submodular where  $B$  is supermodular, the signature of diminishing relative returns familiar from discrete statistical mechanics and rare for a cohomological invariant. The curve case follows from the exact identity of Proposition 40, whose right side exposes the competition between the two orders. The nearest related result is [177]. A first decisive theorem would be log-submodularity for pairs of quadrics and cubics in all dimensions. In tests on the corpus described in the stress-test section, all 84,600 cross-products on the exhaustive grid and 17,281 on the adversarial extension obeyed the strict inequality. A counterexample would separate the two interaction orders at some multidegree, itself a sharp structural datum.

**Conjecture 260** (Schur-convexity at fixed degree sum). *If  $\mathbf{d}$  majorizes  $\mathbf{d}'$  in  $\mathbf{Z}_{\geq 2}^c$  with equal coordinate sums, then  $B_{n,c}(\mathbf{d}) \geq B_{n,c}(\mathbf{d}')$ , strictly unless the vectors agree up to permutation.*

**Significance.** Among complete intersections with the same dimension, codimension, and adjunction coefficient  $\sum_i d_i - (n + c + 1)$ , the balanced multidegrees minimize normalized primitive complexity and the lopsided ones maximize it: redistributing positivity unevenly enriches primitive topology even when the canonical class data are unchanged. The unnormalized statement is false already for curves, where  $b_1^{\text{prim}}(2, 4) = 18 < 20 = b_1^{\text{prim}}(3, 3)$ , a retained control showing the normalization reverses the order; the normalized curve case is proved in Proposition 40 through the Schur-concavity of the degree product. The nearest related results are the extremal Euler-characteristic theorems of [179]. A first decisive theorem would be the single balancing transfer at codimension two. In tests on the corpus described in the stress-test section, all 47,000 balancing transfers on the exhaustive grid and 15,754 on the adversarial extension strictly decreased  $B$ . A counterexample would be a transfer at which balance raises normalized complexity.

**Conjecture 261** (Adjunction-preserving splits decrease complexity). *For  $d_i = d \geq 3$  and  $a, b \geq 2$  with  $a + b = d + 1$ , replacing the degree  $d$  by the pair  $(a, b)$  gives  $B_{n,c+1}(\dots, a, b, \dots) < B_{n,c}(\dots, d, \dots)$ .*



**Significance.** The split raises the codimension by one while preserving the adjunction coefficient, so it moves between presentations that the canonical class cannot distinguish, and the conjecture orients this move: two lower-degree equations are strictly simpler per unit degree than one high-degree equation. The unnormalized statement has no fixed direction, with  $b_2^{\text{prim}}(3) = 6 > 5 = b_2^{\text{prim}}(2, 2)$  against  $b_1^{\text{prim}}(4) = 6 < 8 = b_1^{\text{prim}}(2, 3)$ , both retained as controls. The split is the inverse of the merge order of Part VII, and for odd  $n$  its Calabi–Yau slice is already a theorem: there  $B = E_n/D + (n + 1)/D$ , and both terms strictly increase under a merge, the first by Proposition 17 and the second because merging strictly decreases the degree product, so the odd-dimensional Calabi–Yau case follows outright; the even-dimensional and general-adjunction cases are the open content. The curve case is proved in Proposition 40. A first decisive theorem would be the even Calabi–Yau case, where the same two mechanisms compete. In tests on the corpus described in the stress-test section, all 147,750 admissible splits on the exhaustive grid and 346,825 on the adversarial extension passed strictly. A counterexample would be a split that concentrates rather than dilutes normalized topology.

## 59 Controlled defects in Segre–Veronese syzygies

**Conjecture 262** (Strict log-concavity of the nonlinear row). *For every  $a, b \geq 2$  and every interior index  $p$  of the positive support of the  $q = 2$  row,  $\Delta_p^{(2)} > 0$ .*

**Significance.** The nonlinear row is exactly the row that resists the Gaussian heuristics of asymptotic syzygy theory, and the conjecture asserts that it nevertheless obeys the strongest local curvature law short of real-rootedness, strict log-concavity across its whole support. The statement is stronger than the published unimodality conjecture for these rows [184], which is the part’s closest literature boundary and its attribution; the normality heuristic itself is [186], and log-concavity of syzygy rows is known to fail for small indices in other families [187], so the claim is genuinely calibrated to this surface. A first decisive theorem would be log-concavity of the three trailing entries, accessible from the tail formulas of [185]. In tests on the corpus described in the stress-test section, all 48 interior triples of the thirteen complete published tables were strictly log-concave, the tables reverified file by file against the primary dataset and pinned by the antidiagonal identity of Proposition 42; the statement rests on that small corpus. A counterexample would be the first curvature pathology of a nonlinear row.

**Conjecture 263** (The linear row is log-concave through its mode). *For every  $a, b \geq 2$  and every interior index  $p \leq m_1$  of the  $q = 1$  row,  $\Delta_p^{(1)} > 0$ .*

**Significance.** Global log-concavity of the linear row is false, with the retained control  $\beta_{14,15}^2 - \beta_{13,14}\beta_{15,16} = -3,391,704$  at  $\mathcal{O}(2, 6)$ , so the right law is one-sided: the ascending side and the mode keep strict log-concavity, and every observed failure sits deep in the descending tail. The rising side is the regime where the finite tables are most compatible with the asymptotic normality picture of [186], and the conjecture extracts its most robust local consequence. The nearest related results are [184, 185]. A first decisive theorem would be log-concavity of the first three entries at every  $(a, b)$ , where binomial formulas for the low syzygies are available. In tests on the corpus described in the stress-test section, all 102 interior triples at or before the mode passed strictly across the thirteen tables; the small-corpus limit applies. A counterexample would be a curvature reversal on the rising side, refuting the one-sided picture.



**Conjecture 264** (At most one defect in the linear row). *For every  $a, b \geq 2$ , the set  $\{p : \Delta_p^{(1)} < 0\}$  has at most one element.*

**Significance.** Unimodality permits arbitrarily many local curvature reversals; this conjecture forbids all but one, asserting that the linear row is a nearly totally positive sequence with a single controlled exception. Across the thirteen published tables, five rows have no defect and eight have exactly one, at indices 14, 16, 18, 20, 22 for  $\mathcal{O}(2, b)$ ,  $b = 6, \dots, 10$ , and 14, 16, 19 for  $\mathcal{O}(3, b)$ ,  $b = 4, 5, 6$ , an arithmetic progression pattern in the first family that any proof should explain. The nearest related results are [184, 187]. A first decisive theorem would be uniqueness of the defect for  $\mathcal{O}(2, b)$ , the family with the cleanest tail formulas [185]. In tests on the corpus described in the stress-test section, no table carried two defects; the small-corpus limit applies. A counterexample would be a linear row with two curvature reversals, breaking the controlled-defect picture.

**Conjecture 265** (Defects occur only inside the nonlinear overlap). *For every  $a, b \geq 2$ , if  $\Delta_p^{(1)} < 0$  then  $\beta_{p,p+2}(a, b) > 0$ .*

**Significance.** The conjecture couples a shape property of one row to nonvanishing in the next: the linear row can lose log-concavity only at homological positions where nonlinear syzygies are already present, so a vanishing theorem for  $K_{p,2}$  would automatically restore strict log-concavity of  $K_{p,1}$  there. All eight observed defects obey the coupling, and the rational-normal-scroll boundary  $a = 1$ , where the nonlinear row is empty, has defect-free linear rows across the eight published scroll tables, consistent with the picture from outside the stated range. The nearest related results are [184, 186]. A first decisive theorem would be defect-freeness wherever the nonlinear row vanishes, a statement about the 2-linear part alone. In tests on the corpus described in the stress-test section, the coupling held in all eight defect tables; the small-corpus limit applies. A counterexample would be a defect before the nonlinear row enters, decoupling the two rows.

**Conjecture 266** (Defect-mode locking). *For every  $a, b \geq 2$ , if the linear row has a defect at  $p_*$  then  $p_* \in \{m_2, m_2 + 1\}$ .*

**Significance.** This is the boldest statement of the syzygy programme: the unique loss of curvature in the linear row is phase-locked to the concentration point of the nonlinear row, landing at its mode or one step beyond in every observed table, exactly at  $m_2$  for  $\mathcal{O}(2, b)$  and at  $m_2$  or  $m_2 + 1$  for  $\mathcal{O}(3, b)$ . A proof would exhibit a cross-row mechanism in minimal resolutions that no current theory predicts; even a failure would be valuable, since the displacement  $p_* - m_2$  is a sharp statistic to track across embeddings. The nearest related results are [184, 186, 185]. A first decisive theorem would be the locking for  $\mathcal{O}(2, b)$ , where both rows have explicit tails. In tests on the corpus described in the stress-test section, the locking held in all eight defect tables; this statement rests on the small corpus, whose limits apply with full force. A counterexample would be a defect drifting away from the nonlinear mode.

## 60 Monotone dispersion of hypersurface Hodge profiles

**Conjecture 267** (Likelihood-ratio flattening in the degree). *For every  $n \geq 2$ ,  $d \geq 3$ , and  $0 \leq p < q \leq \lfloor n/2 \rfloor$ ,  $h_p(n, d+1) h_q(n, d) \geq h_p(n, d) h_q(n, d+1)$ .*

**Significance.** As the degree grows, outer Hodge pieces gain relative weight against pieces nearer the middle, uniformly in every pair of positions: a monotone likelihood-ratio comparison between the Hodge structures of different varieties. The coefficient vector is a convolution of discrete uniforms, but the sampling lattice moves with the degree, so no fixed-lattice total-positivity argument applies directly; the surface case is proved in Proposition 43. Equalities occur while an outer piece is still zero, so the non-strict form is the uniform statement. The nearest related results are [180, 181]. A first decisive theorem would be the outermost pair  $(0, 1)$  at every  $n$ , the geometric-genus-to-next-piece ratio. In tests on the corpus described in the stress-test section, all 71,920 cross-products on the grid and 164,502 on the extension passed. A counterexample would be a degree step at which the profile sharpens toward the center.

**Conjecture 268** (Central-window escape). *For every  $n \geq 2$ ,  $d \geq 3$ , and  $1 \leq k \leq \lfloor n/2 \rfloor$ ,  $\sum_{p=k}^{n-k} \pi_{n,d+1}(p) \leq \sum_{p=k}^{n-k} \pi_{n,d}(p)$ .*

**Significance.** Every central window of the Hodge diamond loses normalized mass monotonically as the degree grows: cohomology drains from the middle toward the outer pieces at every scale simultaneously, a position-sensitive statement strictly stronger than growth of any single spread functional. The mechanism is the same moving-lattice convolution as Conjecture 267, of which this is a consequence for symmetric profiles, proved in Proposition 44; the surface case is Proposition 43. The nearest related results are [180, 182]. A first decisive theorem would be the innermost window  $k = \lfloor n/2 \rfloor$ , the central Hodge piece alone. In tests on the corpus described in the stress-test section, all 13,050 window comparisons on the grid and 25,272 on the extension passed. A counterexample would be a window recapturing mass at some degree step.

**Conjecture 269** (Convex-order dispersion). *For every  $n \geq 2$  and  $d \geq 3$ , the profile  $\pi_{n,d}$  precedes  $\pi_{n,d+1}$  in convex order.*

**Significance.** The degree flow is a mean-preserving spread: every convex functional of the normalized Hodge position, the variance, all even central moments, every exponential moment, increases weakly at each degree step, so infinitely many numerical Hodge inequalities are packaged into one canonical stochastic order. Like Conjecture 268, this order is a consequence of Conjecture 267 for symmetric profiles (Proposition 44): the window comparisons give stochastic dominance of the centered displacement, and symmetrizing a convex statistic reduces the convex order to that dominance. On finite symmetric support the order is equivalent to the call-function comparisons  $\mathbf{E}(P - t)_+$ , which is how it is tested exactly; the surface case is Proposition 43, and the only coincident consecutive profiles in the tested range are the cubic and quartic threefolds. The nearest related results are [180, 181]. A first decisive theorem would be the variance alone, the second-moment shadow of the order. In tests on the corpus described in the stress-test section, all 1,682 degree transitions on the grid and 2,730 on the extension passed. A counterexample would be a convex statistic moving against the spread.

**Conjecture 270** (Monotone total-variation convergence to the Eulerian profile). *For every  $n \geq 2$  and  $d \geq 3$ ,  $\|\pi_{n,d+1} - \varepsilon_n\|_{\text{TV}} \leq \|\pi_{n,d} - \varepsilon_n\|_{\text{TV}}$ , with equality only at  $(n, d) = (3, 3)$ .*

**Significance.** The normalized profiles drift toward the Eulerian distribution, exactly the law of descents of a random permutation of  $n + 1$  letters; the conjecture upgrades that drift to monotone convergence in total variation from the first degree onward, a finite-degree error principle that

one verified threshold would convert into bounds at every larger degree. Convergence alone forces no monotonicity, and the coordinatewise strengthening is false, a retained control: at  $n = 12$  the  $p = 5$  mass crosses its Eulerian limit between degrees six and seven and overshoots, so the cancellation across positions asserted here is the genuine content. The Eulerian limit is exact for hypersurfaces in abelian varieties [182], asymptotic in the toric setting [181], and feeds effective arithmetic applications [183]; only the monotone interpolation is claimed. A first decisive theorem would be a one-crossing law between  $\pi_{n,d}$  and  $\varepsilon_n$ . In tests on the corpus described in the stress-test section, the distance never increased across 1,682 grid and 2,730 extension transitions, the sole equality being the cubic–quartic threefold coincidence. A counterexample would be a degree step moving the profile away from the Eulerian law.

## Part XV

# Stochastic order, curvature, and dependence laws for self-avoiding walks

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The fifteen conjectures below concern the uniform measure on  $n$ -step self-avoiding walks on the hypercubic lattice, the classical model whose most basic monotonicity questions, including whether the mean-square displacement grows with the length, remain open [194, 190]. They form four programmes. A curvature programme (Conjectures 271 to 273) asserts exact all-length curvature laws for three count sequences: the square-lattice counts  $c_n$  are strictly log-concave in steps of two with curvature relaxing monotonically to zero, sharpening the ratio analysis of Hammersley’s programme carried out by Guttmann and Jensen [188]; every radial-moment sum  $M_q(n)$  obeys the same parity log-concavity at every real order  $q \geq 0$ ; and the bridge counts  $b_n$  obey the opposite law, strict log-convexity from  $n = 5$  onward, so that walk ratios would descend to  $\mu^2$  while bridge ratios ascend to  $\mu$ . A transport programme (Conjectures 274 to 278) asserts stochastic-order growth in the length: likelihood-ratio growth of the bridge height and of the coordinate span, coordinatewise monotone couplings of the absolute endpoint vector and of the span vector under  $n \rightarrow n + 2$ , and stochastic growth of the contact count; the endpoint coupling strictly implies the open mean-square-displacement monotonicity along each parity class. A dependence programme (Conjectures 279 to 282) assigns signs to four joint laws at every fixed length: the two spans are negatively quadrant dependent, radius and terminal atmosphere positively, radius and contacts negatively, and contacts and atmosphere negatively, an extension–compaction dependence structure for a measure with no known FKG-type property. A phase-and-dimension programme (Conjectures 283 to 285) states a partition-curvature trichotomy across the collapse transition whose three constants are the established predictions of [195, 196, 197], and two strict cross-dimensional laws: the probability that a nonbacktracking walk is self-avoiding rises with the dimension at every fixed length, and every positive radial moment falls. The proved layer is three calibrations: Proposition 45 proves the extreme-order regime of both moment laws from a two-walk extremal count and a spanning-axes inequality  $c_n(d + 1) > (1 + 1/d) c_n(d)$ ; Proposition 46 computes the top four diagonals of the bridge-height array in closed form and proves the top corner of the likelihood-ratio law at every length; and Proposition 47 computes the survival probabilities  $q_{d,n}$  for  $n \leq 7$  as explicit rational functions of  $d$  and proves Conjecture 284

there in every dimension at once; and Proposition 48 proves Conjecture 285 at length two for every dimension and every positive moment order. Four negative controls are retained in place: the absolute endpoint coordinates are not negatively quadrant dependent already at  $n = 3$ , so the span law of Conjecture 279 is not a symmetry artifact; the full radial distributions cross between adjacent dimensions, so the moment formulation of Conjecture 285 is forced; bridge log-convexity fails exactly at  $n = 2$  and  $n = 4$ , so the threshold of Conjecture 273 is sharp; and the finite-size roots of the curvature diagnostic of Conjecture 283 oscillate below  $n \approx 12$ , so that statement is asymptotic only. Conjecture 271 is the part's closest to prior work: its statistic and its asymptotics are explicitly Guttman–Jensen's [188], and only the exact strict inequalities and the second-level monotonicity are proposed as new; Conjecture 283 is a consolidation, since each of its three regimes is separately the standard prediction and the new content is the uniform sequence-level formulation.

An  $n$ -step self-avoiding walk on  $\mathbf{Z}^d$  is a nearest-neighbour path  $\omega = (\omega_0, \dots, \omega_n)$ ,  $\omega_0 = 0$ , with distinct vertices;  $c_n(d)$  is their number,  $c_n = c_n(2)$ , and  $U_n$  is the uniform measure on the square-lattice walks, whose connective constant is  $\mu = \lim c_n^{1/n} \approx 2.63815853$ . Write  $\omega_n = (X_n, Y_n)$ ,  $R_n^2 = |\omega_n|^2$ ,  $M_q(n) = \sum_{\omega} |\omega_n|^q$ , and let  $S_x, S_y$  be the coordinate spans  $\max_i x_i - \min_i x_i$  and its  $y$ -analogue. The terminal atmosphere  $A_n \in \{0, 1, 2, 3\}$  is the number of unvisited neighbours of  $\omega_n$ ; since every  $(n + 1)$ -step walk extends a unique  $n$ -step walk,  $\sum_{\omega} A_n(\omega) = c_{n+1}$ , so  $\mathbf{E}[A_n] = c_{n+1}/c_n$  exactly, the identity underlying atmospheric sampling methods [201, 200]. The contact count  $C_n$  is the number of nonbonded nearest-neighbour pairs of visited vertices, with partition function  $Z_n(u) = \sum_{\omega} u^{C_n(\omega)}$  for the interacting walk [203, 204]. A bridge is a walk with  $x_0 = 0$ ,  $x_i \geq 1$  for  $i \geq 1$ , and  $x_i \leq x_n$  for all  $i$ ;  $b_n$  counts them,  $B(n, h)$  counts those of height  $x_n = h$ , and  $b_1, \dots, b_{10} = 1, 3, 7, 17, 41, 101, 251, 631, 1591, 4029$ , with  $b_m b_n \leq b_{m+n}$  and  $b_n^{1/n} \rightarrow \mu$  by Hammersley–Welsh unfolding [192]. A uniformly random  $n$ -step nonbacktracking walk on  $\mathbf{Z}^d$  is self-avoiding with probability  $q_{d,n} = c_n(d)/(2d(2d - 1)^{n-1})$  [205]. For laws  $P, Q$  on  $\mathbf{Z}$  or  $\mathbf{Z}^2$ ,  $P \leq_{\text{st}} Q$  means  $Q$  dominates  $P$  on upper sets, equivalently a coupling exists with the  $Q$ -variable coordinatewise at least the  $P$ -variable;  $\leq_{\text{lr}}$  is likelihood-ratio order, the total positivity of the two-row array; and  $(X, Y)$  is positively (negatively) quadrant dependent when  $\mathbf{P}(X \geq a, Y \geq b)$  is at least (at most)  $\mathbf{P}(X \geq a)\mathbf{P}(Y \geq b)$  for all thresholds. All finite verifications below are exact: integer enumeration, integer cross-products, and integer maximum flows for the coupling laws, with the full joint laws enumerated through  $n = 21$  ( $c_{21} = 2,408,806,028$  walks at the top length), bridges through  $n = 23$ , dimensions through  $d = 10$ , and published counts through  $n = 79$  [189].

**Proposition 45** (Extreme-order calibration for the moment laws). (i) For all  $n \geq 2$  and  $d \geq 1$ ,  $c_n(d + 1) > (1 + 1/d) c_n(d)$ . (ii) For every  $n \geq 3$  there is  $q_0(n)$  such that  $M_q(n)^2 > M_q(n - 2)M_q(n + 2)$  for all real  $q \geq q_0(n)$ . (iii) For every  $n \geq 2$  and  $d \geq 2$  there is  $q_1(n, d)$  such that  $\mathbf{E}_{d+1}[R_n^q] < \mathbf{E}_d[R_n^q]$  for all real  $q \geq q_1(n, d)$ .

*Proof.* (i) Group walks by the set of coordinate axes their steps use:  $c_n(d) = \sum_{k \geq 1} \binom{d}{k} s_{n,k}$ , where  $s_{n,k}$  counts  $n$ -step walks in  $\mathbf{Z}^k$  using all  $k$  axes. Since  $\binom{d+1}{k} / \binom{d}{k} = (d+1)/(d+1-k) \geq (d+1)/d$  for  $k \geq 1$ , strictly for  $k \geq 2$ , and  $s_{n,2} > 0$  for  $n \geq 2$ , the sum grows strictly faster than the factor  $(d+1)/d$ . (ii) Exactly  $2d$  walks, the straight ones, attain  $|\omega_n| = n$ ; every other walk has  $|\omega_n|^2 \leq n^2 - 2n + 2$ , since  $|x| + |y| \leq n$  with  $|x| \leq n - 1$  forces  $x^2 + y^2 \leq (n - 1)^2 + 1$ . Hence  $M_q(n) = 4n^q(1 + \delta_n(q))$  on the square lattice with  $0 \leq \delta_n(q) \leq \frac{c_n}{4} \theta_n^q$  and  $\theta_n = \sqrt{n^2 - 2n + 2}/n < 1$ , so

$$\frac{M_q(n)^2}{M_q(n - 2)M_q(n + 2)} \geq \left( \frac{n^2}{n^2 - 4} \right)^q \frac{1}{(1 + \delta_{n-2}(q))(1 + \delta_{n+2}(q))} \rightarrow \infty$$

as  $q \rightarrow \infty$ . (iii) Likewise  $\mathbf{E}_d[R_n^q] = (2d n^q / c_n(d))(1 + \delta)$  with  $\delta \rightarrow 0$ , so the ratio of consecutive-dimension moments tends to  $\frac{(d+1)c_n(d)}{d c_n(d+1)}$ , which is strictly below 1 by (i).  $\square$

**Proposition 46** (Top-corner calibration for the bridge programme). For all admissible  $n$ ,  $B(n, n) = 1$ ,  $B(n, n-1) = 2(n-1)$ ,  $B(n, n-2) = 2(n-2)^2$ , and  $B(n, n-3) = \frac{2}{3}(n-3)(2n^2 - 12n + 19)$ ; consequently the top two adjacent cross-products of Conjecture 274 hold at every length:

$$\begin{aligned} B(n, n-1)B(n+1, n) - B(n, n)B(n+1, n-1) &= 2(n-1)(n+1), \\ B(n, n-2)B(n+1, n-1) - B(n, n-1)B(n+1, n-2) &= \frac{4}{3}(n-2)(n-1)(n^2 - n - 3), \end{aligned}$$

both positive for  $n \geq 2$  and  $n \geq 3$  respectively.

*Proof.* A bridge of height  $n - k$  with  $v$  vertical steps and  $m$  backward horizontal steps has  $n = h + 2m + v$ , so  $k = 2m + v$ . If  $m \geq 1$  and  $v = 0$  the walk lives on the axis and a backward step revisits it; if  $m = 1$  and  $v = 1$ , the backward step must immediately follow the vertical step, since anything else backtracks or revisits the axis, after which any forward step re-enters the column just left, while placing the backward step last makes the running maximum  $n - 2$  exceed  $x_n = n - 3$ ; so all horizontal steps point forward and  $v = k$  for  $k \leq 3$ . The  $x$ -coordinate is then nondecreasing, distinct columns never collide, and within a column a vertical run is self-avoiding exactly when its signs are constant, so the count is  $\sum_r N_r 2^r$  over placements of the  $k$  vertical steps into the  $n - 1$  slots after the forced first step,  $N_r$  the number of placements with  $r$  maximal runs. For  $k = 1, 2, 3$  this gives  $2(n-1)$ ,  $4\binom{n-1}{2} - 2(n-2) = 2(n-2)^2$ , and  $2(n-3) + 8\binom{n-3}{2} + 8\binom{n-3}{3} = \frac{2}{3}(n-3)(2n^2 - 12n + 19)$ . The two displayed identities are then polynomial algebra, and their factorizations are positive on the stated ranges.  $\square$

**Proposition 47** (Survival calibration in every dimension). For all  $d \geq 2$ ,

$$\begin{aligned} 1 - q_{d,4} &= \frac{2(d-1)}{(2d-1)^3}, & 1 - q_{d,5} &= \frac{2(d-1)(4d-3)}{(2d-1)^4}, & 1 - q_{d,6} &= \frac{8(d-1)(3d^2 - 2d - 2)}{(2d-1)^5}, \\ 1 - q_{d,7} &= \frac{2(d-1)(32d^3 - 28d^2 - 46d + 41)}{(2d-1)^6}, \end{aligned}$$

and each  $q_{d,n}$ ,  $4 \leq n \leq 7$ , is strictly increasing in  $d$  on all of  $d \geq 2$ ; Conjecture 284 therefore holds for  $n \leq 7$  in every dimension.

*Proof.* A nonbacktracking walk of length at most 5 fails to be self-avoiding only through a four-cycle among steps 1–4 or steps 2–5; counting the two events and their intersection (which forces the fifth step to repeat the first) gives  $c_4(d) = 2d(2d-1)^3 - 2d(2d-2)$  and  $c_5(d) = 2d(2d-1)^4 - 2d(2d-2)(4d-3)$ , whence the first two displays. For  $n = 6, 7$ , the axis-grouping identity  $c_n(d) = \sum_k \binom{d}{k} s_{n,k}$  of Proposition 45 expresses  $c_n$  as a polynomial in  $d$  of degree  $n$  determined by exact enumeration in  $\mathbf{Z}^k$  for  $k \leq n$ ; this gives  $c_6(d) = 64d^6 - 160d^5 + 112d^4 + 20d^2 - 34d$  and  $c_7(d) = 128d^7 - 384d^6 + 352d^5 - 80d^4 + 192d^3 - 372d^2 + 166d$ , agreeing with the classical  $1/d$ -expansion polynomials of Fisher and Gaunt [207] and reverified here against direct enumeration for  $2 \leq d \leq 10$ . For each  $n \leq 7$  the difference  $q_{d+1,n} - q_{d,n}$  is a rational function whose denominator is positive and whose numerator, expanded in  $t = d - 2$ , has every coefficient nonnegative and constant term positive (for  $n = 4$  the numerator is  $32t^3 + 168t^2 + 280t + 142$ ), so the difference is strictly positive for all real  $d \geq 2$ .  $\square$

**Proposition 48** (Radial moments at length two). For  $n = 2$ , every  $d \geq 2$ , and every real  $q > 0$ ,  $\mathbf{E}_{U_2^{(d)}}[|\omega_2|^q] = 2^{q/2} + (2^q - 2^{q/2})/(2d - 1)$ , strictly decreasing in  $d$ ; in particular Conjecture 285 holds at  $n = 2$ .

*Proof.* A two-step self-avoiding walk on  $\mathbb{Z}^d$  is a nonbacktracking walk: conditional on the first step, one of the  $2d - 1$  continuations is straight, with  $|\omega_2| = 2$ , and the other  $2d - 2$  turn, with  $|\omega_2| = \sqrt{2}$ . Averaging  $|\omega_2|^q$  over the uniform choice gives  $(2^q + (2d - 2)2^{q/2})/(2d - 1) = 2^{q/2} + (2^q - 2^{q/2})/(2d - 1)$ , and since  $2^q > 2^{q/2}$  for  $q > 0$  the second term strictly decreases in  $d$ .  $\square$

## 61 Curvature laws for counts, radial moments, and bridges

**Conjecture 271** (Parity log-concavity and curvature relaxation of walk counts). For  $\rho_n = c_{n+2}c_{n-2}/c_n^2$  and every  $n \geq 2$ :  $0 < \rho_n < 1$  and  $\rho_n < \rho_{n+2} < 1$ .

**Significance.** The statement says that  $\log c_n$  is strictly concave in steps of two and that its discrete curvature relaxes monotonically to zero: the two-step ratios  $c_{n+2}/c_n$  decrease strictly along each parity class onto their limit  $\mu^2$ , so each would be a certified upper bound on  $\mu^2$  improving monotonically, complementing O'Brien's monotonicity  $c_{n+1} \geq c_n$  [193], with the ratio limit and its  $n^{-1/3}$  rate Kesten's [191]. This conjecture is the part's closest to prior work: the statistic  $\rho_n$  and its asymptotics  $\rho_n \sim 1 - 4(\gamma - 1)/n^2 = 1 - 11/(8n^2)$  are exactly the objects of Guttman and Jensen's analysis of Hammersley's programme [188] with the exponent  $\gamma = 43/32$  of [195], and monotonicity hypotheses of this type are the assumptions under which they derive the existence of critical exponents; the exact strict all- $n$  inequalities and the second-level claim that the curvature itself is monotone are the content proposed here. A first decisive theorem would be  $\rho_n < 1$  for all  $n$ , a strict two-step Fekete inequality beyond submultiplicativity. In tests against the published counts through  $n = 79$  [189], every  $\rho_n$  and every increment obeys both inequalities, the largest  $\rho_{77} = 0.99977583$ ; a violation at large  $n$  would show correction terms overturning the scaling curvature after its apparent onset.

**Conjecture 272** (Radial-moment parity log-concavity at every order). For every real  $q \geq 0$  and every  $n \geq 3$ ,  $M_q(n)^2 > M_q(n - 2)M_q(n + 2)$ .

**Significance.** At  $q = 0$  this is the concavity clause of Conjecture 271; the conjecture extends it to the entire Mellin family of endpoint-weighted counts, whose predicted growth  $M_q(n) \sim A_q \mu^n n^{\gamma-1+q\nu}$  with  $\nu = 3/4$  [195] always carries a positive polynomial exponent, so the scaling curvature  $-4(\gamma - 1 + q\nu)/n^2$  is negative at every order. Since  $c_n \leq M_q(n) \leq n^q c_n$  for  $q > 0$ , each  $M_q$  has growth constant  $\mu$ , and the conjecture would produce an infinite hierarchy of metric-weighted two-step ratios decreasing onto  $\mu^2$ , upper bounds that see endpoint geometry. Proposition 45 proves the extreme-order regime: for each  $n$  the inequality holds for all sufficiently large  $q$ , by the two-walk extremal count at radius  $n$ . The metric series themselves are enumerated to great length in [189]; no moment-sum log-concavity statement appears there or elsewhere located. A first decisive theorem would be the law at  $q = 2$ , the mean-square-displacement-weighted count. In exact tests at  $q \in \{0, 2, 4, 6, 8\}$  and on a grid of 501 orders through  $q = 50$ , all lengths  $n \leq 20$  pass, minimum ratio 1.00413; a counterexample would exhibit an order at which endpoint weighting overturns coefficient curvature.

**Conjecture 273** (Eventual strict log-convexity of bridge counts). For every  $n \geq 5$ ,  $b_n^2 < b_{n-1}b_{n+1}$ .



**Significance.** Bridges are predicted to grow as  $b_n \sim B\mu^n n^{\gamma_b-1}$  with the negative polynomial exponent  $\gamma_b - 1 = -7/16$  [199], so their logarithmic curvature  $+7/(16n^2)$  has the opposite sign to Conjecture 271: walk ratios would fall to  $\mu^2$  while bridge ratios rise to  $\mu$ , squeezing the connective constant from both sides. The conjecture upgrades that prediction to an exact inequality from the sharp threshold onward; the negative control is exact,  $b_2^2 - b_1b_3 = b_4^2 - b_3b_5 = 2$ , so log-convexity fails at  $n = 2$  and  $n = 4$  and at no other tested length. Under the conjecture the ratios  $b_{n+1}/b_n$  increase, and since  $b_n^{1/n} \rightarrow \mu$  [192] they increase onto  $\mu$ , so each ratio is a certified lower bound; supermultiplicativity [192] gives no pointwise convexity, and Kesten’s irreducible-bridge renewal structure [191] suggests the natural route, since Kaluza-type theorems derive log-convexity from renewal representations. A first decisive theorem would be log-convexity of the renewal sequence generated by irreducible bridges. In exact tests through  $n = 23$  ( $b_{23} = 862,447,881$ , extending the previously enumerated range), every inequality from  $n = 5$  holds, minimum ratio 1.00081; a violation would mark a second curvature reversal in a sequence believed monotone-regular.

## 62 Monotone transport in the length

**Conjecture 274** (Bridge height rises in likelihood-ratio order). *For every  $n \geq 1$  the bridge height satisfies  $H_n \leq_{\text{lr}} H_{n+1}$ : for all  $h_1 < h_2$ ,  $B(n, h_2)B(n+1, h_1) \leq B(n, h_1)B(n+1, h_2)$ .*

**Significance.** Likelihood-ratio order is the strongest standard stochastic order: it asserts total positivity of order two for the full array  $B(n, h)$ , implies stochastic and hazard-rate dominance, and gives a single-crossing structure for the height laws that renewal and unfolding arguments could exploit. The obstruction is that uniform reweighting across lengths need not preserve the geometric bias of an added step; no total-positivity statement appears in the bridge literature [192, 199]. Proposition 46 proves the top corner of the law at every length: the four leading diagonals are explicit polynomials and the two top adjacent cross-products are  $2(n-1)(n+1)$  and  $\frac{4}{3}(n-2)(n-1)(n^2-n-3)$ , positive always, so the conjectured order is a theorem where the array is thinnest. A first decisive theorem would be  $\text{TP}_2$  of the full array in a strip of bounded height, where transfer matrices apply. In exact integer tests of all cross-products, not only adjacent ones, for  $n \leq 22$ , every inequality holds; a counterexample would exhibit a height at which longer bridges become relatively rarer, breaking single-crossing.

**Conjecture 275** (Monotone coupling of the absolute endpoint). *For every  $n \geq 0$  there is a coupling of  $U_n$  and  $U_{n+2}$  under which  $(|X_n|, |Y_n|) \leq (|X_{n+2}|, |Y_{n+2}|)$  coordinatewise almost surely.*

**Significance.** This is the strongest of the transport laws: a single coupling forcing every increasing functional of the absolute endpoint to grow when two steps are added, in a model where even the growth of  $\mathbf{E}[R_n^2]$  is a recorded open problem [194, 190]. The conjecture strictly implies that open monotonicity along each parity class, and much more: all upper-orthant probabilities and all increasing moments grow. The parity shift respects bipartiteness, and the statement is distributional, since an individual walk can move inward over two steps. Strassen’s theorem makes each instance finitely decidable: the coupling exists if and only if every upper-set inequality holds, checked here by exact integer maximum flow. A first decisive theorem would be the marginal law  $|X_n| \leq_{\text{st}} |X_{n+2}|$ . In exact max-flow tests for every  $n \leq 19$ , the transport deficit is zero in all cases; a counterexample at some larger  $n$  would exhibit an explicit increasing endpoint event whose probability falls with length, a finite-size reversal of excluded-volume swelling and a sharp datum about the open problem.

**Conjecture 276** (Monotone coupling of the span vector). *For every  $n \geq 0$  there is a coupling under which  $(S_x, S_y)$  at length  $n$  is coordinatewise dominated by  $(S_x, S_y)$  at length  $n + 2$  almost surely.*

**Significance.** Spans grow along any fixed path, but truncating a uniform  $(n + 2)$ -step walk does not give a uniform  $n$ -step walk, so pathwise monotonicity does not transfer to the ensembles; the conjecture says the reweighting never overturns it, in the strongest product-order sense. Every increasing functional of the bounding box, the area proxy  $S_x S_y$ , escape probabilities from rectangles, and mixed span moments would then grow monotonically along parity, providing monotone finite-volume approximants for the confinement and compression ensembles studied in [208]. Because spans are pathwise monotone, an extension-biased coupling corrected by local surgeries is a plausible mechanism, unlike for the endpoint. A first decisive theorem would be stochastic growth of  $\max(S_x, S_y)$ . In exact max-flow tests for every  $n \leq 19$ , all transport deficits vanish; a counterexample would locate a box size whose escape probability falls when two steps are added.

**Conjecture 277** (Span likelihood-ratio expansion). *For every  $n \geq 0$ ,  $S_{x,n} \leq_{\text{lr}} S_{x,n+2}$ .*

**Significance.** Marginally, Conjecture 276 gives only stochastic growth of  $S_x$ ; this law upgrades it to likelihood-ratio order, total positivity of the span-count array on each parity class, controlling the odds of large against small spans and yielding single-crossing tail structure for confinement estimates. Transfer matrices organize square-lattice walks by width, so the natural target is a positivity-preserving operator whose minors encode the order, the same structure conjectured for bridges in Conjecture 274. The span-conjugate ensembles of [208] give the nearest literature, with no ordering statements in the length. A first decisive theorem would be  $\text{TP}_2$  for spans confined to a strip of width three. In exact integer tests of all cross-products for  $n \leq 19$ , every inequality holds; a counterexample would give a span value whose relative odds fall with length, breaking the single-crossing picture.

**Conjecture 278** (Stochastic growth of the contact count). *For every  $n \geq 2$ ,  $C_n \leq_{\text{st}} C_{n+1}$ .*

**Significance.** Extending a fixed walk cannot destroy contacts, but the uniform law at length  $n + 1$  reweights  $n$ -step prefixes by their atmospheres, and contact-rich walks have systematically smaller atmospheres if Conjecture 282 is right, a selection bias working against the pathwise gain; the conjecture asserts the gain always wins at every tail threshold. In particular, the normalized partition factor  $Z_n(u)/c_n$  is then nondecreasing in  $n$  for every  $u \geq 1$  and nonincreasing for  $u \leq 1$ , a monotonicity statement bearing on the attractive regime  $u > 1$ , where even the existence of the limiting free energy remains open, in contrast to the repulsive regime settled by subadditivity. The contact-scaling literature [203] and the interacting-walk estimates of [204] study the marginals; no stochastic monotonicity in length appears there. A first decisive theorem would be monotonicity of  $\mathbf{P}(C_n \geq 1)$ . In exact tail tests for all  $n \leq 20$ , every cross-multiplied inequality holds; a counterexample would exhibit a contact level whose tail probability falls with length, quantifying the atmosphere selection effect.

## 63 Dependence laws at fixed length

**Conjecture 279** (Negative quadrant dependence of the two spans). *For every  $n \geq 1$ ,  $(S_x, S_y)$  is negatively quadrant dependent under  $U_n$ :  $\mathbf{P}(S_x \geq a, S_y \geq b) \leq \mathbf{P}(S_x \geq a) \mathbf{P}(S_y \geq b)$  for all  $a, b$ .*

**Significance.** A fixed step budget makes the two axes compete: horizontal extension consumes steps and room that vertical extension needs, and the conjecture asserts the competition at every joint tail, not merely in the covariance, which is  $-2.283$  at  $n = 16$ . The retained negative control sharpens the claim: the same statement for  $(|X_n|, |Y_n|)$  is false already at  $n = 3$ , so the negative dependence is a property of accumulated geometry, not of symmetry or of the step budget alone. The uniform self-avoiding measure has no known FKG or negative-association structure, so any proof would introduce a genuinely new correlation inequality; a natural route is a negative-lattice property of the bivariate span counts after a cumulative transform. A first decisive theorem would be  $\text{Cov}(S_x, S_y) \leq 0$  for all  $n$ . In exact tests of every upper-tail determinant through  $n = 21$ , all signs are as conjectured; a counterexample would locate a scale at which two-axis extension becomes cooperative rather than competitive.

**Conjecture 280** (Positive quadrant dependence of radius and atmosphere). *For every  $n \geq 1$ ,  $(R_n^2, A_n)$  is positively quadrant dependent under  $U_n$ .*

**Significance.** The atmosphere is a microscopic escape capacity and the radius a macroscopic extension observable; the conjecture couples them at every joint tail: walks that have travelled far are locally more extendable. Through the identity  $\mathbf{E}[A_n] = c_{n+1}/c_n$ , the law ties endpoint geometry to the one-step coefficient ratio, and it would give a comparison principle for atmospheric sampling methods, whose attrition and weights are controlled by exactly this quantity [201, 200]; the atmospheric-collapse computations of [202] point the same way qualitatively, with no formal correlation statement located anywhere. Because  $A_n$  is itself an extension count, the conjectured family of inequalities is equivalent to the average atmosphere inside radial upper sets dominating the global average, a formulation open to direct extension-counting attack. A first decisive theorem would be  $\text{Cov}(R_n^2, A_n) \geq 0$  for all  $n$ , the covariance being  $+4.636$  at  $n = 16$ . In exact tests of every upper-tail determinant through  $n = 21$ , all signs hold; a counterexample would exhibit a radius beyond which far endpoints are paradoxically more caged.

**Conjecture 281** (Negative quadrant dependence of radius and contacts). *For every  $n \geq 1$ ,  $(R_n^2, C_n)$  is negatively quadrant dependent under  $U_n$ .*

**Significance.** Contacts reward folding, displacement requires extension, and the conjecture makes the opposition exact at every pair of thresholds. Its weakest consequence is already the physically central sign: differentiating the interacting-walk expectation at the athermal point gives  $\partial_\beta \mathbf{E}_\beta[R_n^2]|_{\beta=0} = \text{Cov}(R_n^2, C_n) \leq 0$ , so an infinitesimal attraction shrinks the mean-square displacement, the folklore mechanism of collapse [203, 204] for which no rigorous finite- $n$  sign result appears to be known. The full quadrant law is far stronger, coupling every increasing function of radius negatively with every increasing function of contact number. A natural strengthening to test first is whether the contact-conditioned radius laws fall in likelihood-ratio order, which would imply the quadrant law; the enumerated joint tables permit exactly that screening. A first decisive theorem would be the covariance sign alone. In exact tests of every upper-tail determinant through  $n = 21$ , all signs hold; a counterexample would put a threshold pair at which attraction and extension cooperate, a genuine surprise for the collapse picture.

**Conjecture 282** (Negative quadrant dependence of contacts and atmosphere). *For every  $n \geq 1$ ,  $(C_n, A_n)$  is negatively quadrant dependent under  $U_n$ .*

**Significance.** Contact-rich conformations are locally crowded, and the conjecture asserts that the crowding reaches the endpoint at every joint tail: attraction suppresses extendability. Its covariance consequence,  $\partial_\beta \mathbf{E}_\beta[A_n]|_{\beta=0} \leq 0$  with  $\mathbf{E}_0[A_n] = c_{n+1}/c_n$ , says that switching on attraction immediately lowers the expected number of legal extensions, the mechanism by which the collapse transition throttles growth ratios, seen qualitatively in the atmospheric-collapse study [202]. Jointly with Conjecture 278 the law forms a closed circuit: contacts grow with length, yet contact-rich prefixes are underrepresented among extendable walks, and the circuit would become a recursion scheme if both signs were theorems. A first decisive theorem would be monotone decrease of  $\mathbf{E}[A_n \mid C_n \geq m]$  in  $m$  at fixed small  $n$ , an explicitly finite statement. In exact tests of every upper-tail determinant through  $n = 21$ , all signs hold; a counterexample would exhibit a contact level whose walks are locally freer, inverting the crowding heuristic.

## 64 Phase and dimension laws

**Conjecture 283** (The collapse trichotomy of partition curvature). *Let  $K_n(u) = \log Z_{n+2}(u) - 2 \log Z_n(u) + \log Z_{n-2}(u)$  and let  $u_\theta$  be the square-lattice collapse point,  $u_\theta \approx 1.9474$  [204]. Along either parity class:  $n^2 K_n(u) \rightarrow -11/8$  for  $0 < u < u_\theta$ ;  $n^2 K_n(u_\theta) \rightarrow -4/7$ ; and  $n^{3/2} K_n(u) \rightarrow \sigma(u)$  with  $\sigma(u) > 0$  for  $u > u_\theta$ .*

**Significance.** This conjecture is a consolidation: each regime restates the standard prediction, and the new content is only the uniform sequence-level formulation. In the swollen phase the constant is  $-4(\gamma - 1)$  with  $\gamma = 43/32$  [195]; at the collapse point it is  $-4(\gamma_\theta - 1)$  with the tricritical exponent  $\gamma_\theta = 8/7$  of Duplantier and Saleur [196]; and in the collapsed phase the positive  $n^{-3/2}$  curvature is exactly the surface term  $-s(u)\sqrt{n}$  in the collapsed scaling form of Owczarek, Prellberg, and Brak [197], proved in the partially directed model by Legrand and Pétrélis [198], and studied for radius-of-gyration laws in [209]. What the trichotomy adds is a phase detector reading a single sequence: the discrete curvature of  $\log Z_n$  changes sign and scaling exponent across the transition, with no radius or specific-heat observable. The retained control is that the finite-size roots of  $K_n$  oscillate strongly below  $n \approx 12$ , so nothing monotone is claimed at finite length; the roots computed here through  $n = 19$  run 1.9843, 2.1022, 1.9697, 2.0589, 2.0016 from  $n = 15$ , consistent with slow drift toward  $u_\theta$ . A first decisive theorem would be the swollen-phase clause for the strictly self-avoiding model, or the collapsed clause for any two-dimensional undirected ensemble. A failure in any regime would force revision of the corresponding scaling form itself.

**Conjecture 284** (Self-avoidance survival rises with dimension). *For every  $n \geq 4$  and  $d \geq 2$ ,  $q_{d+1,n} > q_{d,n}$ .*

**Significance.** A nonbacktracking walk fails to be self-avoiding only by closing a loop of length at least four, and added transverse directions dilute every such pattern; the conjecture asserts that the dilution is monotone at every finite length, not merely in the  $1/d$  expansion of Fisher and Gaunt [207]. For  $n \leq 3$  survival is identically 1, so  $n = 4$  is the first nontrivial length, and Proposition 47 proves the law for all  $n \leq 7$  in closed form, in every dimension at once. Since

$q_{d,n}^{1/n} \rightarrow \mu_d/(2d-1)$ , the full conjecture implies that the normalized connective constant  $\mu_d/(2d-1)$  is nondecreasing in  $d$ , a normalized companion to the strict increase  $\mu_{d+1} > \mu_d$  that follows from the edge-addition theorem of Grimmett and Li [206], and a cross-dimensional comparison that lace-expansion combinatorics [205] seems well suited to attack, pattern by pattern. A first decisive theorem would be  $n = 8$ , where six-cycles first interact. In exact rational tests for all enumerated pairs,  $d \leq 10$  at  $n \leq 7$  and up to  $n = 14$  for  $d = 2, 3$ , every adjacent-dimension difference is positive, the smallest being  $6.3 \times 10^{-4}$ ; a counterexample would exhibit a length at which raising the dimension makes self-trapping relatively likelier.

**Conjecture 285** (All radial moments fall with dimension). *For every  $n \geq 2$ , every real  $q > 0$ , and every  $d \geq 2$ ,  $\mathbf{E}_{U_n^{(d+1)}}[|\omega_n|^q] < \mathbf{E}_{U_n^{(d)}}[|\omega_n|^q]$ .*

**Significance.** Self-avoidance relaxes as the dimension grows, so excluded-volume swelling should weaken; the conjecture asserts the weakening in every positive radial moment at every finite length, comparing uniform ensembles of the same step number across dimensions. The retained negative control forces this formulation: full stochastic dominance of the radial laws is false, the distributions crossing already at  $(d, n) = (2, 3)$ , so a Mellin-order statement is the strongest available. If the moments obey the standard scaling  $\mathbf{E}[R_n^q] \sim A_{d,q} n^{q\nu_d}$ , the conjecture is incompatible with  $\nu_{d+1} > \nu_d$ , giving conditional monotonicity of the metric exponents through their conjectured values  $\nu_2 = 3/4$  [195] down to the mean-field  $1/2$ ; no rigorous fixed-length cross-dimension comparison of displacement appears in the literature [194, 190]. Proposition 45 proves the extreme-order regime from the spanning-axes inequality, and Proposition 48 proves the length-two case for every dimension and every positive order, so the open content is bounded orders at longer lengths. A first decisive theorem there would be the  $q = 2$  case at some fixed small  $n$  for all  $d$ . In exact and gridded tests across  $d \leq 10$  and all enumerated lengths through  $n = 14$ , every adjacent-dimension difference has the conjectured sign, the smallest gap  $1.8 \times 10^{-4}$ ; a counterexample would give a moment order at which higher dimension swells the walk.

## Part XVI

# Budgeted irreversibility and the spectral geometry of relaxation

*Added to the deposit: 19 August 2026.*

The nine conjectures below concern a single organizing question of non-equilibrium dynamics: when a relaxation process is allowed only a finite budget of irreversibility, how is that budget spectrally allocated? They form four programmes. A current-fluctuation law (Conjecture 286) asserts that a reversible shortcut added to a driven ring at fixed activity strictly amplifies the standardized skewness of the winding current, against the mixing intuition, sharpening the cumulant inequalities of [217, 218] into a monotonicity law. A frontier programme (Conjectures 287 to 291) studies the linear diffusion  $dX_t = -(S + A)X_t dt + \sqrt{2S} dW_t$  with fixed Gaussian steady state, whose entropy-production rate is the exact quadratic  $\Sigma(A) = \text{tr}(A^\top S^{-1}A)$  [211], and conjectures the complete architecture of the budgeted-gap frontier: nested mode recruitment ordered by the reversible spectrum, piecewise-convex arcs with downward marginal-gain resets, exact two-mode support at small budget, a rank-two extremizer for the oscillation frequency,

and a pairwise formula for the cheapest exceptional point, the thermodynamic analogue of the nearest-defective-matrix problem [216]; the unlimited endpoint is the theorem of Lelièvre, Nier, and Pavliotis [210], and the spectral-perturbation bound of [212] and the dissipation–coherence decomposition of [213] bound the same objects, so Conjecture 290 is closely related to the latter work. A coherent-acceleration programme (Conjectures 292 and 293) transfers the frontier architecture to primitive Lindblad dynamics with detailed balance, where a stationary Hamiltonian is a skew perturbation in the Gelfand–Naimark–Segal (GNS) geometry [221] and coherent gap enhancement is the theorem of Li and Lu [219]: sequential operator-mode recruitment under the commutator-accessibility constraint, and saturation of the classical envelope exactly on  $\mathfrak{su}(2)$ -closed mode systems. A preparation-geometry programme (Conjecture 294) turns from rates to eigenvector geometry in the anomalous-relaxation setting of the Markovian Mpemba effect [223, 224]: the measure of preparations that robustly cancel the slow mode should collapse in inverse proportion to the biorthogonal condition number as a Liouvillian exceptional point is approached, quantifying the antagonism between exceptional-point acceleration [225, 226] and robustness [228]. The proved layer is three calibrations: Proposition 49 solves the two-mode frontier in closed form, identifies the second-slowest mode as the unique best infinitesimal partner of the slowest, and gives the exact pair costs used throughout; Proposition 50 proves that the classical frontier is an upper envelope for every coherently driven detailed-balance Lindbladian and that the unital qubit attains it exactly; and Proposition 51 proves the basin-collapse law exactly in the qubit model, with the sphere prefactor equal to one. Four negative findings are retained as controls, and they shape the part: the Markov analogues of the exceptional-point frontier laws were tested at optimizer level and discarded, since on a three-state ring the budgeted-gap frontier reaches its supremum only on the unidirectional boundary at infinite entropy production, on rings of four and five states the slow pair’s real part continues to improve past the exceptional point rather than pinning, on the four-state instance the slow sector turns complex strictly before the frontier’s plateau, and the fixed-trace pinning that motivated those claims is an artifact of three-state spectra; gap optimizers generically sit at defective matrices [215] and at exceptional points of Liouvillians [214], so only the budgeted, structure-revealing refinements below are claimed as new; and a proposed resonance criterion for entangling coherent acceleration was tested in exact second-order arithmetic and discarded, since in every nondegenerate product instance the best local Hamiltonian already attains the full optimum, the commutator structure forcing every entangling transition out of a simple product slow mode to carry a strictly larger spectral denominator than its local counterpart, while the claimed advantage appears only through slow-mode degeneracies outside the criterion’s own hypotheses; and a proposed capacity principle equating the number of slow modes simultaneously cancellable by unitary preparation with the maximal rank of the tangent overlap map was refuted on a generic qutrit instance, where the slow coordinate’s range over the fixed-spectrum orbit omits zero entirely, the spectral offset between the stationary state and the orbit dominating the orbit’s fluctuation, so feasibility of Mpemba preparation on an orbit is an affine range condition invisible to the tangent geometry, constructive steering protocols [229] notwithstanding, and Conjecture 294 carries that condition as an explicit hypothesis. Conjectures 287, 288, 292, and 293 assert laws for every dimension.

For a continuous-time generator  $Q$  on a finite connected graph with stationary law  $\pi$ , the activity is  $\kappa = \sum_i \pi_i \sum_{j \neq i} q_{ij}$  and the entropy-production rate is  $\sigma = \frac{1}{2} \sum_{i \neq j} (\pi_i q_{ij} - \pi_j q_{ji}) \log(\pi_i q_{ij} / \pi_j q_{ji})$ . On the homogeneous  $N$ -ring with affinity  $F$ , rates  $e^{\pm F/(2N)}$ , the winding current  $J_t$  has long-time cumulant rates  $c_m$  obtained from the tilted generator, and  $S_3 = |c_3|/c_2^{3/2}$  is its standardized skewness. For the linear diffusion above,  $S = S^\top > 0$  is diagonal with



simple spectrum  $s_1 < \dots < s_n$  after an orthogonal change of basis,  $A^\top = -A$ , the stationary covariance is the identity,  $r(A) = \min_j \operatorname{Re} \lambda_j(S + A)$ ,  $\Omega(A) = \max_j |\operatorname{Im} \lambda_j(S + A)|$ , and the frontier is  $r_*(\Sigma) = \sup\{r(A) : \Sigma(A) \leq \Sigma\}$  with  $\Sigma(A) = \sum_{i < j} a_{ij}^2(1/s_i + 1/s_j)$ ; since the real parts sum to  $\operatorname{tr} S$ , always  $r(A) \leq \bar{s} = \operatorname{tr} S/n$ , and  $\bar{s}$  is attained at finite budget [210]. A mode  $i$  is *active* for  $A$  when the  $i$ th row of  $A$  is nonzero. For a primitive Lindblad dissipator  $\mathcal{D}$  in detailed balance with respect to  $\rho_*$ , the operator  $K = -\mathcal{D}$  is positive self-adjoint in the GNS inner product with eigenmodes  $KF_j = s_j F_j$  on the traceless sector, a stationary Hamiltonian ( $[H, \rho_*] = 0$ ) acts as the skew-adjoint  $\mathcal{A}_H = -i[H, \cdot]$  with matrix elements  $a_{jk}(H)$  and coherent budget  $\Sigma_H = \sum_{j < k} |a_{jk}(H)|^2(1/s_j + 1/s_k)$ , and  $r_{\text{OU}}$  denotes the frontier of the diagonal spectrum  $\{s_j\}$ . For a density matrix  $\rho_0$  with simple spectrum,  $\mathcal{O}_{\rho_0}$  is its unitary orbit, the left Liouvillian eigenmodes  $\ell_1, \ell_2, \dots$  are ordered from slow to fast and split into real functionals,  $T_\rho : K \mapsto (\langle \ell_\alpha, i[K, \rho] \rangle)_\alpha$  is the tangent overlap map with real rank  $\operatorname{rank} T_\rho$ , and for a family  $\mathcal{L}(h)$  with simple slow eigenvalue the biorthogonal condition number is  $\kappa_{\text{eig}}(h) = \|\ell_1(h)\| \|r_1(h)\|$  under  $\langle \ell_1, r_1 \rangle = 1$ ; the  $\varepsilon$ -robust basin on an orbit  $\mathcal{O}$  is  $\mathcal{B}_\varepsilon(h) = \{\rho \in \mathcal{O} : |\langle \ell_1(h), \rho - \rho_* \rangle| \leq \varepsilon\}$  with the invariant orbit measure  $\mu_{\mathcal{O}}$ .

**Proposition 49** (Pair calibration for the frontier programme). (i) A perturbation supported on the pair  $(i, j)$  leaves  $\operatorname{span}\{e_i, e_j\}$  invariant with eigenvalues  $\frac{s_i + s_j}{2} \pm \sqrt{(\frac{s_j - s_i}{2})^2 - a_{ij}^2}$ ; the block reaches its exceptional point at cost  $\Sigma_{ij}^{\text{EP}} = \frac{(s_j - s_i)^2}{4}(1/s_i + 1/s_j)$  and generates frequency  $\Omega_{ij}(\Sigma) = (\Sigma/(1/s_i + 1/s_j) - (\frac{s_j - s_i}{2})^2)_+^{1/2}$  at cost  $\Sigma$ . (ii) The small-budget rate gain of coupling the slowest mode to mode  $j$  is  $\gamma_{1j} = s_1 s_j / ((s_j - s_1)(s_j + s_1))$  per unit entropy production, strictly decreasing in  $s_j$ : the second-slowest mode is the unique best infinitesimal partner. (iii) For  $n = 2$  the frontier is

$$r_*(\Sigma) = \frac{s_1 + s_2}{2} - \sqrt{\left(\frac{s_2 - s_1}{2}\right)^2 - \frac{\Sigma}{1/s_1 + 1/s_2}} \quad (\Sigma \leq \Sigma_{12}^{\text{EP}}), \quad r_*(\Sigma) = \bar{s} \quad (\Sigma \geq \Sigma_{12}^{\text{EP}}),$$

continuous, strictly increasing and strictly convex on the arc, with marginal gain diverging at the threshold and resetting to zero beyond it; all laws of Conjectures 287 to 291 hold for  $n = 2$ .

*Proof.* (i) is the characteristic polynomial of the  $2 \times 2$  block, which is invariant because  $A$  vanishes off the pair, together with the cost formula  $\Sigma = a_{ij}^2(1/s_i + 1/s_j)$ . For (ii), expanding the block eigenvalue gives  $r = s_1 + a^2/(s_j - s_1) + O(a^4)$  at cost  $a^2(1/s_1 + 1/s_j)$ , whence  $\gamma_{1j}$ ; as a function of  $s_j$ ,  $s_j/(s_j^2 - s_1^2)$  has derivative of the sign of  $-(s_j^2 + s_1^2) < 0$ . For (iii), when  $n = 2$  the only skew matrices are the pair blocks, so the frontier is the substitution  $a^2 = \Sigma/(1/s_1 + 1/s_2)$  into (i); the arc is the composition of the strictly convex increasing map  $\Sigma \mapsto -\sqrt{K - c\Sigma}$ , its derivative  $c/(2\sqrt{K - c\Sigma})$  diverges as  $\Sigma \uparrow \Sigma_{12}^{\text{EP}}$ , and beyond the threshold both real parts equal  $\bar{s}$ , which no feasible  $A$  can exceed. Recruitment, convexity, two-mode support, the frequency formula, and the cheapest exceptional point are then all immediate for  $n = 2$ .  $\square$

**Proposition 50** (Envelope calibration for the coherent programme). (i) For every primitive detailed-balance dissipator with traceless-sector spectrum  $\{s_j\}$  and every stationary Hamiltonian  $H$ ,

$$\operatorname{gap}(-\mathcal{D} - \mathcal{A}_H) \leq r_{\text{OU}}(\Sigma_H).$$

(ii) For the unital qubit dissipator with Bloch decay rates  $(s_1, s_2, s_3)$  and  $H = \frac{1}{2} h \cdot \sigma$ , the traceless Liouvillian spectrum equals  $-\operatorname{spec}(S + A(h))$  with  $A(h)$  the cross-product matrix of  $h$ , every  $3 \times 3$  skew matrix is realized, and the envelope is attained exactly.

*Proof.* In the GNS inner product  $K = -\mathcal{D}$  is positive self-adjoint on the traceless sector and  $\mathcal{A}_H$  is skew-adjoint and real in a  $K$ -eigenbasis; the gap of  $K + \mathcal{A}_H$  is the objective  $r$  evaluated at the particular skew matrix  $a(H)$ , whose cost is  $\Sigma_H$  by definition, and the frontier is the supremum over all skew matrices of that cost, which proves (i). For (ii), on Bloch coordinates the dissipator acts as  $\text{diag}(s_1, s_2, s_3)$  and  $-i[H, \cdot]$  acts as  $h \times r$ , an arbitrary antisymmetric matrix as  $h$  ranges over  $\mathbb{R}^3$ , so the Liouvillian restricted to the traceless sector is conjugate to  $-(S + A(h))$  and the relaxation problem coincides with the three-dimensional frontier problem.  $\square$

**Proposition 51** (Basin calibration for the preparation programme). Let  $\mu$  be the uniform measure on the unit sphere of Bloch vectors and let the slow left functional of the family  $B(h) = S + A(h)$ ,  $S = \text{diag}(s_1, s_2, s_3)$ ,  $A(h)$  the rotation block  $a_{12} = -h$ , be normalized by  $\langle \ell_1, r_1 \rangle = 1$  with  $\|r_1\| = 1$ , so  $\kappa_{\text{eig}}(h) = \|\ell_1(h)\|$ . Then for every  $0 < \varepsilon \leq \kappa_{\text{eig}}(h)$  and every  $h$  below the exceptional point  $h_c = (s_2 - s_1)/2$ ,

$$\mu\{x : |\langle \ell_1(h), x \rangle| \leq \varepsilon\} = \frac{\varepsilon}{\kappa_{\text{eig}}(h)},$$

exactly, and  $\kappa_{\text{eig}}(h) = \sqrt{h_c/2} (h_c - h)^{-1/2} (1 + o(1))$  as  $h \uparrow h_c$ , so the basin measure collapses at the exact rate  $\varepsilon \sqrt{2(h_c - h)/h_c}$ .

*Proof.* The functional  $x \mapsto \langle \ell_1, x \rangle$  is linear, and by the Archimedes projection theorem the coordinate of a uniform point of  $S^2$  along any unit direction is uniform on  $[-1, 1]$ , so the band  $|\langle \ell_1, x \rangle| \leq \varepsilon$  has measure  $\min(1, \varepsilon/\|\ell_1\|)$ , and  $\|\ell_1\| = \kappa_{\text{eig}}$  under the stated normalization. For the divergence rate, write  $d = h_c = (s_2 - s_1)/2$  and  $u = \sqrt{d^2 - h^2}$ ; the slow eigenvalue of the block  $\begin{pmatrix} s_1 & -h \\ h & s_2 \end{pmatrix}$  has right eigenvector  $r \propto (h, u - d)$  and left eigenvector  $\ell \propto (h, d - u)$ , so

$$\kappa_{\text{eig}}(h) = \frac{\|\ell\| \|r\|}{|\langle \ell, r \rangle|} = \frac{h^2 + (d - u)^2}{h^2 - (d - u)^2},$$

and as  $h \uparrow d$  one has  $u = \sqrt{2d}\sqrt{d - h} (1 + o(1))$ , numerator  $\rightarrow 2d^2$ , denominator  $= (h - d + u)(h + d - u) = 2d u (1 + o(1))$ , giving  $\kappa_{\text{eig}} = \sqrt{d/2} (d - h)^{-1/2} (1 + o(1))$ .  $\square$

## 65 A current-fluctuation law for equilibrium shortcuts

**Conjecture 286** (Reversible shortcuts amplify winding skewness). *For every even  $N \geq 4$  and affinity  $F \neq 0$ , add to the homogeneous driven  $N$ -ring a reversible antipodal chord of strength  $g$  carrying no winding increment, rescaling all rates to keep the activity fixed. Then the standardized skewness  $S_3(g)$  of the long-time winding current is strictly increasing in  $g \geq 0$ .*

**Significance.** The chord carries no affinity and no counted current, and the mixing intuition says extra reversible pathways should Gaussianize the winding statistics; the conjecture asserts the exact opposite at every chord strength: the shortcut suppresses coherent winding progress faster than it suppresses the third cumulant, so the standardized non-Gaussianity strictly grows. The standardized skewness uses the absolute third cumulant,  $S_3 = |c_3|/c_2^{3/2}$ , and is therefore even in  $F$ : reflecting the ring sends  $F \mapsto -F$  and the winding tilt  $z \mapsto -z$ , so the scaled cumulant generator obeys  $\psi_{-F}(z) = \psi_F(-z)$ , making the signed third cumulant odd in  $F$  while the variance rate  $c_2$  and hence  $S_3$  stay invariant; the law is thus symmetric under affinity reversal, and the

positive-affinity tests below cover both signs. The nearest results are the skewness and kurtosis bounds of Wampler and Barato [217], proved for unicyclic networks and known to fail multicyclic ones, and the steady-state cumulant bounds of [218]; neither orders the effect of a topological perturbation. The statement is one-dimensional in  $g$  for each  $(N, F)$ , so it is unusually exposed: a single high-precision decrease refutes it. A first decisive theorem would be the sign of  $dS_3/dg$  at  $g = 0$  for all even  $N$ . In tests with fourth-order tilted-cumulant differences across  $N \leq 10$ , affinities  $0.5 \leq F \leq 12$ , and chords through  $g = 50$ , all 420 sampled ratios exceeded one and every consecutive pair increased, the smallest ratio being 1.0000739, at the smallest chord; a counterexample would exhibit a mixing pathway that Gaussianizes a driven current, locating the boundary of the amplification mechanism.

## 66 The finite-budget frontier of linear diffusions

**Conjecture 287** (Sequential spectral recruitment). *For every diagonal  $S$  with simple spectrum there are thresholds  $0 = \Sigma_0 < \Sigma_1 \leq \Sigma_2 \leq \dots$  and a choice of maximizer  $A(\Sigma)$  of the frontier at every budget whose active-mode set is  $\{1, \dots, m\}$  with  $m$  nondecreasing in  $\Sigma$ : no mode is recruited before every slower mode.*

**Significance.** The conjecture is a spectral water-filling law with a non-Hermitian twist: at small budget all resources go to the two slowest modes, by the exact quotient of Proposition 49(ii), and the claim is that recruitment stays ordered by the reversible spectrum at every scale, so the optimizer’s invariant subspace grows by one canonical direction at a time until the theorem of [210] takes over at the ceiling. The dissipation–coherence trade-offs of [213] bound the same objects without addressing optimizer architecture. A first decisive theorem would be ordered recruitment for  $n = 3$ . In global optimizations over spectra  $(1, 2, 5)$ ,  $(1, 1.3, 7)$ ,  $(1, 2, 2.05)$ ,  $(1, 50, 50.5)$ ,  $(1, 2, 5, 8)$ , and  $(1, 2, 3, 7)$  across budgets spanning four orders of magnitude, every observed active set was an initial segment  $\{1, \dots, m\}$  with  $m$  nondecreasing, including near-degenerate and widely split spectra; a counterexample would be a spectrum whose optimum skips a slow mode, which would overturn the marginal-return mechanism itself.

**Conjecture 288** (Convex arcs with marginal-gain resets). *For simple  $S$  the frontier  $r_*(\Sigma)$  is continuous and piecewise real-analytic; on each maximal interval with constant active set it is convex; the one-sided marginal gain jumps only downward at arc boundaries; and  $r_*$  reaches the ceiling  $\bar{s}$  at a finite budget and is constant beyond.*

**Significance.** Convex arcs mean increasing returns to irreversibility inside a fixed spectral architecture, the opposite of concave resource allocation: budget should be pushed into the currently active block all the way to its structural threshold, not spread smoothly. Proposition 49(iii) proves the extreme form for  $n = 2$ , where the marginal gain diverges at the exceptional point and resets to zero, and the ceiling attainment at finite budget is the unconstrained theorem of [210]; the conjecture asserts that this convex-then-reset geometry survives every recruitment in every dimension. The nonsmooth-optimization literature knows that spectral-abscissa optimizers sit at defective matrices [215], which is where the resets live. A first decisive theorem would be convexity of the first arc for  $n = 3$ . In fine-grid frontier computations for  $(1, 2, 5)$  over fifty-two warm-started budgets, the discrete slopes increase monotonically within each of the two resolved arcs, from 0.69 toward 4.9 on the first and from 0.18 toward 1.1 on the second, with the only

slope decreases at the recruitment threshold near  $\Sigma \approx 0.38$  and at the terminal entry into the ceiling, which is reached exactly,  $8/3$  to  $10^{-6}$ , by budget  $\Sigma \approx 4.1$ ; a counterexample, a concave stretch inside an arc, would restore ordinary diminishing returns somewhere on the frontier.

**Conjecture 289** (Finite two-mode radius). *For simple  $S$  there exists  $\Sigma_1 > 0$  such that for every  $0 < \Sigma < \Sigma_1$ , every global maximizer of the frontier, up to sign and rotations inside exact degeneracies, is supported on the two slowest modes;  $\Sigma_1$  is the first budget at which the two-mode branch collides or loses optimality to a branch containing mode three.*

**Significance.** Proposition 49(ii) proves that the second-slowest mode is the unique best infinitesimal partner, so the two-mode block wins at first order; the conjecture upgrades the perturbative selection to exact support on a whole interval, the base case of the recruitment law and the natural first proof target of the programme, accessible to analytic perturbation theory plus an isolation argument. In global optimizations at spectra  $(1, 2, 5)$  and budgets through 0.16, the largest coupling outside the  $(1, 2)$  block was  $2 \times 10^{-5}$ , and for the widely split spectrum  $(1, 50, 50.5)$  the two-mode support persists through budgets in the hundreds, consistent with the block’s exceptional-point cost; a counterexample would be a spectrum where dense couplings beat the pair at arbitrarily small budget, contradicting the strict gap in the quotient  $\gamma_{1j}$ .

**Conjecture 290** (Rank-two oscillation extremizer). *For every diagonal  $S > 0$  and budget  $\Sigma$ ,*

$$\sup_{\Sigma(A) \leq \Sigma} \Omega(A) = \max_{i < j} \Omega_{ij}(\Sigma),$$

*with  $\Omega_{ij}$  the pair formula of Proposition 49(i): no coupling of three or more modes generates a larger spectral frequency than the best single rotation block.*

**Significance.** For isotropic  $S$  the statement is the equality case of the operator-norm–Frobenius-norm inequality for skew matrices; the anisotropic weighting  $1/s_i + 1/s_j$  and the detuning penalty  $(s_j - s_i)/2$  break that argument, and the conjecture asserts the rank-two concentration survives. This conjecture is the part’s closest to prior work: the dissipation–coherence bound of [213] constrains entropy production per oscillation period in the same family, and whether the envelope above is a saturated form of that bound cannot be settled from its abstract; the exact pairwise envelope and its attainment are proposed as the new content, recorded pending that paper’s full text. A first decisive theorem would be the  $n = 3$  case, a two-parameter reduction. In adversarial optimizations over dense skew couplings at spectra  $(1, 2, 5, 8)$ ,  $(1, 1.1, 1.2, 1.3)$ ,  $(1, 2, 50, 51)$ , and the five-mode  $(1, 3, 3.2, 9, 10)$  across five budgets each, the adversarially optimized dense coupling coincided with the pair envelope to eight digits in every case; a counterexample would exhibit collective oscillation beating every pairwise rotation, a genuinely new non-normal mechanism.

**Conjecture 291** (Cheapest exceptional point is a pair). *For simple  $S$ , the infimum of  $\Sigma(A)$  over antisymmetric  $A$  making  $S + A$  defective equals  $\min_{i < j} \Sigma_{ij}^{\text{EP}}$ , attained by the single block of Proposition 49(i) on a minimizing pair.*

**Significance.** This is a thermodynamic Wilkinson problem: the distance to defectivity, classically measured in unweighted norms and characterized through coalescing pseudospectral components [216], is here measured in entropy production over the stationarity-preserving antisymmetric perturbations, and the conjecture asserts that the cheapest route to an exceptional

point is always the elementary two-mode one, computable from pairwise rates. Exceptional points are often treated as delicate high-dimensional nonnormal phenomena [222]; a proof would make their minimal thermodynamic cost a two-body quantity. The retained Markov control sharpens the claim’s scope: the analogous exceptional-point frontier laws fail for jump processes, where current perturbations are not block-closed, so the pair structure is a genuinely linear-diffusive mechanism. A first decisive theorem would be the  $n = 3$  case via the discriminant variety. In adversarial minimizations of the eigenvalue separation at 90, 97, and 99.5 percent of the predicted cost across four spectra, the optimizer converged onto the pair block itself, the minimal separation matching the block’s own value  $(s_j - s_i)\sqrt{1 - \Sigma/\Sigma_{ij}^{\text{EP}}}$  to six digits and vanishing to machine precision exactly at the predicted cost; a counterexample would be a genuinely collective cheapest defectivity.

## 67 Coherent acceleration of dissipative quantum dynamics

**Conjecture 292** (Recruitment under commutator accessibility). *For a primitive detailed-balance dissipator with simple traceless spectrum, every small-budget maximizer of  $\text{gap}(-\mathcal{D} - \mathcal{A}_H)$  over stationary Hamiltonians of coherent budget  $\Sigma_H$  concentrates, as  $\Sigma_H \rightarrow 0$ , on the accessible direction maximizing the restricted quotient  $\sum_{j>1} |a_{1j}(H)|^2 / (s_j - s_1)$  over  $\Sigma_H$ ; and there is a maximizing selection whose active dissipative modes are nested and nondecreasing in the budget, with changes only at spectral collisions or accessibility branch crossings.*

**Significance.** Coherent gap enhancement is the theorem of Li and Lu [219]; the conjecture concerns the optimizer’s architecture, which their variational framework does not address. The new obstruction is algebraic: commutators with stationary Hamiltonians realize only a subspace of skew matrices, and coupling the slow mode may force side couplings elsewhere, wasting budget, so the selected partner is the accessible optimum of the same Rayleigh quotient that drives Conjecture 287, not necessarily the second-slowest mode. For maximally accessible dissipators, stationary state proportional to the identity, the constraint vanishes and the classical recruitment law applies verbatim through Proposition 50. A first decisive theorem would be the perturbative selection statement, a finite-dimensional constrained Rayleigh problem. In tests on a unital qutrit with an eight-rate Gell-Mann dissipator, the finite-budget optimizer aligned with the predicted accessible direction to cosine 1.0000 and its leading matrix element coupled the two slowest eigenoperators, with the forced side couplings an order smaller; a counterexample would exhibit accessibility overturning sparse recruitment itself rather than merely redirecting it.

**Conjecture 293** (Envelope saturation is Lie-algebraic). *For a primitive detailed-balance Lindbladian, the envelope of Proposition 50 is attained on a nontrivial budget interval if and only if the dissipative eigenmodes active in the classical optimizer close under commutation into a direct sum of three-dimensional  $\mathfrak{su}(2)$ -type modules together with spectators, so that the unrestricted skew optimizer is realized by a Hamiltonian commutator.*

**Significance.** Proposition 50 makes the classical frontier an unconditional upper bound and shows the unital qubit attains it because the Bloch cross product realizes every rotation; the conjecture characterizes exactly when that saturation persists: the active operator modes must form the adjoint representation of an  $\mathfrak{su}(2)$ , making the optimal rotation physically implementable by a stationary Hamiltonian. The interval hypothesis is the safeguard against

accidental coincidences: saturation at an isolated budget forces nothing, while saturation on a nontrivial interval pins the whole optimizer path inside the commutator image. This is a sharp structural dichotomy, coherence saturates the classical bound precisely on quantum systems that look classical in the relevant sector, and it would give the first necessary-and-sufficient criterion in the coherent-acceleration literature [219, 220]. A first decisive theorem would be necessity for a qutrit dissipator whose slow modes do not close. The unital qubit equality is verified to machine precision, and the qutrit computations of Conjecture 292, whose accessible optimum falls twenty-six percent below the unrestricted classical optimum, exhibit the conjectured strictness on a non-closing system; a counterexample would be saturation without closure, revealing an unforeseen exact-control mechanism.

## 68 The geometry of anomalous relaxation

**Conjecture 294** (Exceptional points collapse robust Mpemba basins). *Let  $\mathcal{L}(h)$  approach a second-order exceptional point  $h_c$  in its slow sector from the diagonalizable side, normalize the slow eigenvectors by  $\|r_1\| = 1$  and  $\langle \ell_1, r_1 \rangle = 1$  so that  $\kappa_{\text{eig}}(h) = \|\ell_1(h)\|$  and the basin  $\mathcal{B}_\varepsilon(h) = \{\rho \in \mathcal{O} : |\langle \ell_1(h), \rho - \rho_* \rangle| \leq \varepsilon\}$  is fixed in scale, and suppose the slow functional vanishes transversally on the fixed smooth orbit  $\mathcal{O}$ , uniformly near the exceptional point: the tangential gradient of the unit-normalized slow functional is bounded above and below by positive constants along its zero set on  $\mathcal{O}$ , and that zero set is nonempty with hypersurface measure bounded below, uniformly for  $h$  near  $h_c$ . Then there are constants  $0 < c < C$  with*

$$c \frac{\varepsilon}{\kappa_{\text{eig}}(h)} \leq \mu_{\mathcal{O}}(\mathcal{B}_\varepsilon(h)) \leq C \frac{\varepsilon}{\kappa_{\text{eig}}(h)}$$

*for all sufficiently small  $\varepsilon$  and all  $h$  near  $h_c$ : the measure of preparations that robustly cancel the slow mode collapses in exact inverse proportion to the biorthogonal condition number.*

**Significance.** Exceptional points maximize acceleration for exactly tuned preparations [225, 226], eigenvector rotation can switch Mpemba behavior with spectra pinned [227], and preparation noise is known to degrade the effect [228]; the conjecture quantifies the antagonism as a clean scaling law, the coarea mechanism being that biorthogonal normalization makes the slow functional's gradient on the orbit grow like  $\kappa_{\text{eig}}$ , so the  $\varepsilon$ -slab thins in exact proportion. Proposition 51 proves the law in the qubit model with prefactor one and identifies the divergence rate  $(h_c - h)^{-1/2}$ , so the basin closes like  $\varepsilon\sqrt{h_c - h}$ ; the open content is the general orbit, where the uniform-transversality hypothesis records exactly the remaining boundary, tangency of the limiting zero set to the orbit, at which the upper constant would fail. A first decisive theorem would be the law for generic qutrit families. In sphere samplings of the qubit model and of a four-dimensional companion, the product of basin fraction and condition number held constant within sampling error while the condition number grew fifty-fold toward the exceptional point, and the basin fraction was linear in  $\varepsilon$  throughout; a counterexample would exhibit robust exceptional-point Mpemba preparation, welcome news for experiments and fatal for the coarea picture.



## Part XVII

# Inverse problems and design laws for multivariate extremes

*Added to the deposit: 19 August 2026.*

The thirteen conjectures below concern a single organizing question of extreme-value theory: how much of the dependence structure of rare events can be recovered from partial, thinned, or redesigned observation, and at what cost in stability. They form five programmes. A geometric programme (Conjectures 295 to 298) studies the extremal coefficient  $\theta_m(\Gamma)$  of the Hüsler–Reiss family [230, 231] as a functional on variogram matrices: an isoperimetric recruitment staircase at fixed total variogram energy with a proved symmetry-breaking boundary, the regular simplex maximal for three variables and at weak dependence but strictly beaten below it by configurations carrying fewer active components, a pairwise closure law at weak dependence sharpening the consistency inequalities of Schlather and Tawn [232] into an asymptotic equivalence, a scale-elasticity bound of one half paired with entrywise monotonicity whose conjunction yields a dimension-free sparsification law for resistance variograms, strengthening the variogram-level guarantees of [236] to the extremal coefficient itself, and a positive-curvature law at complete dependence whose convex-body shadow is the Wills functional [238]. A cluster-tomography programme (Conjectures 299 and 300) turns the missing-data results for subsampled sequences [242, 243, 244] into an inverse theory: the Bernoulli-thinning response curve should identify the entire limiting cluster-size law through its generating function, and the two-parameter systematic-plus-Bernoulli response should identify the cluster support exactly up to the equivalence generated by its modular occupancy data. A topological programme (Conjectures 301 and 302) makes the topology of extreme directions an inferential target: a bottleneck-consistency law for persistent homology of intermediate-order angular samples at the rate  $b(U(n/k)) + (\log k/k)^{1/s}$ , composing the diagram-estimation rates of [251] with a finite-threshold angular bias, and a leakage trichotomy locating exactly when hidden regular variation [254, 255] is topologically visible after support filtering. A tail-tomography programme (Conjectures 303 and 304) quantifies the inverse problem of angular reconstruction from one-sided projection tails: a severe stability barrier, no Hölder inverse on measures and only a logarithmic modulus on Hölder classes, the analytic-kernel phenomenon of moment problems [259] instantiated for the fractional transforms whose injectivity dichotomy is recorded in Proposition 55 following [257, 258], and a parameter-count identifiability threshold for  $s$ -atomic angular measures at exactly  $m = sd - 1$  generic fractional projections. A design programme (Conjectures 305 to 307) redesigns block maxima themselves: Poissonized block sizes should remove the universal  $V^2/(2n)$  finite-block correction and the corresponding  $n^{-1}$  estimator bias documented in the block-maxima second-order theory [261, 262, 263], factorial-cumulant matching should extend the removal to a full ladder of orders, and exact second-order cancellation in a Hall family [264] should create a cusp at which threshold adaptation, whose logarithmic penalty away from the cusp is the theorem of Carpentier and Kim [267], cannot be uniformly oracle-optimal, with critical window  $|c| \asymp n^{-\beta/(\alpha+4\beta)}$ . The proved layer is four calibrations. Proposition 52 gives the exact Gaussian-orthant formula for  $\theta_m$  used throughout, proves the elasticity bound in the pairwise case, and proves that every collinear configuration has vanishing second curvature, the easy direction of Conjecture 298. Proposition 53 proves the cluster-level thinning algebra: the Bernoulli response formula, the systematic random-phase formula, the

identifiability of the occupancy multisets from the joint response, the recovery of the entire difference multiset from occupancy data by Möbius inversion, and the closure of occupancy equivalence under dilation. Proposition 55 proves the integer–fractional injectivity dichotomy for the projection transforms, with an explicit two-atom versus three-atom pair invisible at  $\alpha = 2$  and separated at  $\alpha = 3/2$ . Proposition 56 proves the exact Poissonization identity for Pareto seeds and constructs, for every order  $r$ , finitely supported non-Poisson integer block-size laws whose factorial cumulants match the Poisson law through order  $r$ , with the exact probability-generating-function identity showing that the finite-block error then falls to order  $n^{-r}$ . Five negative findings are retained as controls, and they shape the part. The deposited isoperimetry claim, that the regular simplex maximizes  $\theta_m$  at fixed energy for every  $m$ , is refuted for  $m \geq 4$ : the width identity of Proposition 52(iv) makes configurations with fewer active components strictly better away from weak dependence, with exact orthant-probability instances at  $m = 4$ , scale 0.12, and  $m = 5$ , scale 0.35, where the four-active configuration beats the simplex by  $3.5 \times 10^{-3}$  and is found by unrestricted optimization, so Conjecture 295 keeps the simplex law at three variables and states the recruitment staircase above. A proposed modular reconstruction claim, that the occupancy multisets over all moduli determine a finite cluster support up to translation and reflection, is refuted: the six-mark pair  $\{0, 1, 4, 10, 12, 17\}$  and  $\{0, 1, 8, 11, 13, 17\}$  of Bloom [246], the classical counterexample to Piccard’s theorem on difference multisets [247, 248], shares every occupancy multiset, by finite verification below the diameter and trivially above it, so occupancy tomography can never distinguish this pair; the repaired Conjecture 300 asks instead for determination within size five and for the two-member bound on equivalence classes, over the exhaustive verification that all ten occupancy collisions among the 6437 canonical sets of size at most six through diameter eighteen and the 3312 of size at most seven through diameter fourteen are homometric pairs, while the pair  $\{0, 1, 3, 8, 10, 14\}$  versus  $\{0, 1, 7, 9, 11, 14\}$  shows the occupancy data strictly refines the difference multiset. The subsampling formulas themselves, the Bernoulli and systematic extremal-index responses at the cluster level, are known in substance for subsampled stationary sequences [242, 243, 244] and are therefore recorded as calibration, not conjecture; what is claimed as new in Conjectures 299 and 300 is the uniform process-level transport and the inverse-theory content. A multiplicity observed at the identifiability threshold of Conjecture 304 on one random instance was exposed as an artifact of a near-coincident atom pair, ill-conditioning inflating rounding-level copies of the single true solution, and the atom-separation hypothesis of the conjecture records that boundary. A second multiplicity is genuine and refutes global uniqueness at the counting boundary itself: at three atoms in three dimensions, two of six well-separated random instances admit a second exact interior mixture at  $m = sd - 1$ , verified to forty-digit residuals, and both become unique one measurement higher, so the global clause of Conjecture 304 begins at  $m = sd$  and the counting-boundary refutation is recorded in place. Conjectures 295, 296, 297, 301, 302, and 305 to 307 assert laws for every dimension or every sample size; Conjecture 297 is closely related to the sparsification guarantees of [236], and Conjecture 307 carries one against the adaptation lower bounds of [267]. Since deposit, Conjecture 298 has been resolved true by Proposition 54.

A variogram matrix  $\Gamma$  on  $m$  sites is symmetric with zero diagonal and conditionally negative definite, so that  $\Gamma_{ij} = \|x_i - x_j\|^2$  for centered points  $x_1, \dots, x_m$  in some  $\mathbb{R}^k$ ; the Hüsler–Reiss max-stable vector with variogram  $\Gamma$  [230] has extremal coefficient  $\theta_m(\Gamma) \in [1, m]$ , the exponent function at **1**, with spectral representation  $\theta_m(\Gamma) = \mathbb{E} \max_i \exp(\langle Z, x_i \rangle - \|x_i\|^2/2)$  for a standard Gaussian  $Z$ , pairwise coefficients  $\theta_{ij} = 2\Phi(\sqrt{\Gamma_{ij}}/2)$ , energy  $\mathcal{S}(\Gamma) = \sum_{i < j} \Gamma_{ij}$ , and scale elasticity  $\eta(t) = d \log \theta_m(t\Gamma) / d \log t$ . For a stationary regularly varying sequence whose exceedances

converge to a compound-Poisson cluster process [239] with cluster size  $N$ ,  $\mathbb{E}N < \infty$ , the extremal index is  $\theta = 1/\mathbb{E}N$  [240, 241], and when the limiting cluster has a finite support  $S \subset \mathbb{Z}$ , defined up to translation, the occupancy counts are  $n_{m,r}(S) = |\{s \in S : s \equiv r \pmod{m}\}|$ , the occupancy data of  $S$  is the family of unordered multisets  $\mathcal{N}_m(S) = \{n_{m,0}(S), \dots, n_{m,m-1}(S)\}$  over all  $m \geq 2$ , two sets are occupancy equivalent when their occupancy data agree, and homometric when their difference multisets agree. A  $d$ -variate regularly varying vector has angular measure  $H$  on the simplex  $\Delta_{d-1}$ ; the fractional projection transform is  $(T_\alpha H)(a) = \int (a \cdot w)^\alpha H(dw)$  for directions  $a \in (0, \infty)^d$ ; and for block maxima,  $F$  lies in a max-domain with exponent function  $V$  and second-order rate  $A(n)$ , a block of size  $n$  has exceedance probability  $q_n(x) = \bar{F}(a_n x + b_n)$ , and a randomized block size  $N_n$  has factorial cumulants  $\kappa_r^{(F)}$  defined by  $\log \mathbb{E}(1+u)^{N_n} = \sum_{r \geq 1} \kappa_r^{(F)} u^r / r!$ , so that  $\log \mathbb{E}(1-q)^{N_n} = \sum_{r \geq 1} \kappa_r^{(F)} (-q)^r / r!$  exactly.

**Proposition 52** (Tilting calibration for the dependence programme). (i) For every admissible  $\Gamma$  realized by centered points  $x_1, \dots, x_m$ ,

$$\theta_m(\Gamma) = \sum_{i=1}^m \Phi_{m-1} \left( \left( \frac{\Gamma_{ij}}{2} \right)_{j \neq i}; \left( \frac{\Gamma_{ij} + \Gamma_{ik} - \Gamma_{jk}}{2} \right)_{j,k \neq i} \right),$$

where  $\Phi_{m-1}(\cdot, C)$  is the centered Gaussian orthant probability with covariance  $C$ . (ii) For  $m = 2$  the elasticity satisfies  $0 \leq \eta(t) \leq \sup_{x>0} x\varphi(x)/(2\Phi(x)) = 0.1472\dots < 1/2$  for all  $t > 0$ . (iii) If the points  $x_i$  are collinear, then with  $M = \max_i \langle Z, x_i \rangle$  and  $I$  the almost-surely unique argmax,  $\mathbb{E}[M^2 - \|x_I\|^2] = 0$ : every collinear configuration has vanishing second curvature. (iv) For  $3 \leq k \leq m$  let the  $k$ -active configuration place  $k$  points at the vertices of a centered regular simplex and  $m - k$  points at the origin, with energy  $S$ . Its Gaussian width is

$$W_k(m, S) = \mathbb{E} \max_i \langle Z, x_i \rangle = \sqrt{\frac{S}{m}} \frac{e_k}{\sqrt{k-1}},$$

where  $e_k$  is the expected maximum of  $k$  independent standard Gaussians; and  $e_3/\sqrt{2} = 0.59841\dots$  exceeds  $e_4/\sqrt{3} = 0.59431\dots$  and  $e_5/2 = 0.58148\dots$ , so for  $m \in \{4, 5\}$  the triangle configuration  $k = 3$  has strictly larger width than the full simplex  $k = m$ , and hence strictly larger  $\theta_m$  at fixed energy for all sufficiently small scale.

*Proof.* (i) Write  $G_i = \langle Z, x_i \rangle$  and  $\sigma_i^2 = \|x_i\|^2$ . Each summand of  $\theta_m = \sum_i \mathbb{E}[e^{G_i - \sigma_i^2/2} \mathbf{1}\{G_i - \sigma_i^2/2 \text{ maximal}\}]$  is an Esscher tilt: under the measure with density  $e^{G_i - \sigma_i^2/2}$  the vector  $Z$  is a standard Gaussian shifted by  $x_i$ , so the summand equals the probability that  $\langle Z, x_j - x_i \rangle \leq (\sigma_i^2 + \sigma_j^2 - 2\langle x_i, x_j \rangle)/2 = \Gamma_{ij}/2$  for all  $j \neq i$ , a Gaussian orthant probability with covariances  $\langle x_j - x_i, x_k - x_i \rangle = (\Gamma_{ij} + \Gamma_{ik} - \Gamma_{jk})/2$ . (ii) Here  $\theta_2(t\Gamma) = 2\Phi(x)$  with  $x = \sqrt{t}\Gamma_{12}/2$ , so  $\eta = x\varphi(x)/(2\Phi(x))$ , which is positive, vanishes at  $x \rightarrow 0$  and  $x \rightarrow \infty$ , and is bounded by  $x\varphi(x) \leq \varphi(1) = 0.2419\dots$  since  $\Phi(x) \geq 1/2$ ; the stated supremum is attained at  $x = 0.8399\dots$ . (iii) For collinear  $x_i = c_i e$  with a unit vector  $e$  and  $\zeta = \langle Z, e \rangle$ , only the two extreme coefficients  $c_{\min} < c_{\max}$  can be argmax, according to the sign of  $\zeta$ , so

$$\mathbb{E}[M^2 - \|x_I\|^2] = c_{\max}^2 \mathbb{E}[(\zeta^2 - 1)\mathbf{1}\{\zeta > 0\}] + c_{\min}^2 \mathbb{E}[(\zeta^2 - 1)\mathbf{1}\{\zeta < 0\}] = 0,$$

each term vanishing because  $\mathbb{E}[\zeta^2 \mathbf{1}\{\zeta > 0\}] = \frac{1}{2} = \Pr(\zeta > 0)$ . For (iv): a centered configuration has  $\sum_{i < j} \Gamma_{ij} = m \sum_i \|x_i\|^2$ , so the  $k$  active vertices have squared norm  $a = S/(mk)$ ; unit regular-simplex directions  $u_1, \dots, u_k$  satisfy  $\langle u_i, u_j \rangle = -1/(k-1)$  and  $\sum_i u_i = 0$ , so the Gaussian vector  $(\langle Z, u_i \rangle)_i$  has the law of  $\sqrt{k/(k-1)} (\xi_i - \bar{\xi})_i$  for independent standard  $\xi_i$ , whose expected maximum

is  $\sqrt{k/(k-1)} e_k$ ; the origin points never exceed the active maximum, which is nonnegative because the active values sum to zero; so  $W_k = \sqrt{a} \sqrt{k/(k-1)} e_k = \sqrt{S/m} e_k / \sqrt{k-1}$ . The constants are  $e_3 = \frac{3}{2}\pi^{-1/2}$  and the classical one-dimensional quadratures  $e_4 = 1.029375\dots$ ,  $e_5 = 1.162964\dots$ , giving the stated ordering with margins  $4 \times 10^{-3}$  and  $1.7 \times 10^{-2}$ ; the width dominates the expansion of Conjecture 298 as the scale falls, so the triangle configuration exceeds the simplex for small scale.  $\square$

## 69 The geometry of Hüsler–Reiss dependence

**Conjecture 295** (Isoperimetry and symmetry breaking at fixed energy). *(a) For  $m = 3$  and every  $S > 0$ , among all admissible variograms with energy  $\mathcal{S}(\Gamma) = S$ , the equidistant variogram  $\Gamma^{\text{eq}}$  with all off-diagonal entries  $S/\binom{m}{2}$ , realized by a regular triangle, uniquely maximizes  $\theta_3(\Gamma)$ , up to relabeling. (b) For every  $m \geq 4$  and every  $S$ , the maximum of  $\theta_m$  at energy  $S$  is attained by a  $k$ -active configuration of Proposition 52(iv),  $k$  components at the vertices of a centered regular simplex and  $m - k$  coincident at the centroid, for some  $3 \leq k \leq m$ , uniquely up to relabeling; the maximizing  $k$  is nondecreasing in  $S$ , equals 3 for all sufficiently small  $S$ , equals  $m$  for all sufficiently large  $S$ , and passes through every intermediate value, so the fixed-energy optimizer recruits one component at a time through a staircase of symmetry-breaking transitions.*

**Significance.** The extremal coefficient is the effective number of asymptotically independent components, and the deposited claim was that at fixed total variogram energy the most even allocation always buys the most independence; the refutation recorded in the preamble and in Proposition 52(iv) shows this fails for  $m \geq 4$ , because a centered second-moment budget rewards concentrating separation on few directions: the width identity  $W_k = \sqrt{S/m} e_k / \sqrt{k-1}$  peaks at three active components, so near complete dependence a regular triangle with the other components parked at the centroid strictly beats the simplex. What replaces the isoperimetry is a recruitment staircase inside the same one-parameter family. In exact orthant computations at  $m = 5$ , the three-active configuration is optimal among the family at scale 0.1, the four-active configuration from scale 0.2 through 0.5, beating the simplex by  $3.5 \times 10^{-3}$  at scale 0.35, and the full simplex from scale 0.7; unrestricted eighteen-parameter optimization at scale 0.35 converges to the four-active configuration to three decimals, with paired Monte Carlo over one hundred million common samples confirming its margin  $+3.4 \times 10^{-3} \pm 1.5 \times 10^{-5}$  over the simplex, while at scales 1.0 and 2.4 no candidate beat the simplex; at  $m = 4$  the single interior transition sits near scale 0.125. At  $m = 3$  adversarial optimization returns the equilateral triangle to eight digits at every scale, so (a) keeps the deposited law where it is true. The weak-dependence endpoint is forced by convexity of the pairwise deficit at large scale, and the nearest theorems, the simplex mean-width results, classical in Gaussian form [233] with a proof announced [234], fix the circumradius rather than the second moment and see no transition; the functional here is the finite-point Wills exponential [238]. A counterexample would be a scale where some configuration outside the  $k$ -active family wins, or a re-entrant staircase, either of which would break the recruitment mechanism.

**Conjecture 296** (Pairwise closure at weak dependence). *Fix an admissible  $\Gamma$  with strictly positive off-diagonal entries such that no three-or-higher-way Gaussian large-deviation saddle shares the exact rate of a contributing pairwise saddle. Then, with  $\delta_{ij}(t) = 2 - \theta_{ij}(t\Gamma) = 2[1 - \Phi(\sqrt{t\Gamma_{ij}}/2)]$ ,*

$$m - \theta_m(t\Gamma) \sim \sum_{i < j} \delta_{ij}(t), \quad t \rightarrow \infty.$$

**Significance.** At independence  $\theta_m = m$ , and the deficit  $m - \theta_m$  measures residual simultaneous dependence; the consistency inequalities of Schlather and Tawn [232] give the Bonferroni bound  $m - \theta_m \leq \sum_{i < j} \delta_{ij}$  for every max-stable law, so the conjecture asserts that for Hüsler–Reiss models the bound is asymptotically exact: pair overlaps carry Gaussian-tail rates  $t^{-1/2} e^{-t\Gamma_{ij}/8}$ , and any triple overlap should demand an additional transversal Gaussian constraint and hence a strictly larger rate, exactly as pair interactions dominate dilute systems. The genericity hypothesis is the honest boundary: a third site sitting exactly on a pairwise saddle can conceivably share its exponential rate, and the conjecture isolates rather than hides that set. The higher-order expansions known for Hüsler–Reiss maxima [235] concern convergence of Gaussian triangular arrays to the limit, not the independence asymptotics of  $\theta_m$  in the variogram scale, so the equivalence appears to be new. In exact tests on random, equilateral, and near-degenerate geometries at  $m = 3, 4$  the ratio of the two sides rose monotonically toward one across  $t = 6$  to 96, reaching 0.9215, 0.9795, 0.9988, 0.999996, 1.000000 on a random triangle, 1.000000 from  $t = 12$  on a random and a near-degenerate quadrilateral, and rising more slowly, 0.831 to 0.998, on the equilateral triangle, whose symmetric triple competition is the strongest. A counterexample would be a geometry whose triple saddle survives at pairwise rate, and would identify the exceptional set as substantive rather than technical.

**Conjecture 297** (Scale elasticity and monotone dependence). *For every admissible nonzero  $\Gamma$ : (a)  $0 \leq \eta(t) \leq \frac{1}{2}$  for all  $t > 0$ ; (b) if  $\Gamma'$  is admissible and  $\Gamma_{ij} \leq \Gamma'_{ij}$  entrywise, then  $\theta_m(\Gamma) \leq \theta_m(\Gamma')$ . Consequently, if  $L, \tilde{L}$  are connected graph Laplacians with  $(1 - \varepsilon)L \preceq \tilde{L} \preceq (1 + \varepsilon)L$  on  $\mathbf{1}^\perp$  and  $\Gamma(L)$  is the resistance variogram  $\Gamma_{ij}(L) = (e_i - e_j)^\top L^\dagger (e_i - e_j)$ , then*

$$(1 + \varepsilon)^{-1/2} \leq \frac{\theta_m(\Gamma(\tilde{L}))}{\theta_m(\Gamma(L))} \leq (1 - \varepsilon)^{-1/2}.$$

**Significance.** The two claims are a differential and an order-theoretic rigidity for the extremal coefficient, and together they convert spectral sparsification into a dimension-free guarantee for extremal dependence: the Laplacian sandwich passes to effective resistances entrywise, monotonicity passes it to  $\theta_m$ , and the elasticity bound integrates to the displayed square-root distortion, so the conclusion follows from (a) and (b) by a three-line chain recorded here. Proposition 52 proves (a) for  $m = 2$  with best constant 0.1472, far inside the conjectured bound, and the bound  $1/2$  is the log-log elasticity of  $\theta_m - 1$ , whose leading  $\sqrt{t}$  term at  $t \downarrow 0$  from Conjecture 298 has elasticity exactly  $1/2$ , whereas the elasticity of  $\theta_m$  itself,  $\eta(t) = t\theta'_m/\theta_m = (c/2)\sqrt{t} + O(t)$ , tends to zero there, so  $1/2$  is a conjectural cap and not a value forced by the complete-dependence expansion; monotonicity in the entrywise order is Slepian-type folklore for which no published statement at the level of  $\theta_m$  on the full admissible cone was found. This conjecture is close to prior work: spectral sparsification of Hüsler–Reiss graphical models is the subject of [236], which proves multiplicative variogram preservation and likelihood control under exactly this sandwich, so the new content is the transfer to the extremal coefficient with the universal exponent one half, strengthening rather than paralleling their guarantees; the graphical-model calculus itself is [231, 237]. In tests, the elasticity over eighty-three random geometries at  $m \leq 5$ , twelve scales each, and over near-degenerate and elongated shapes, never exceeded 0.334, and  $\theta_m$  was monotone on all 150 admissible entrywise-increase pairs, sixty-seven of them with non-variogram difference, and on sixty additive cone directions. A counterexample to (a) above one half would break the sparsification law at its stated exponent; one to (b) would exhibit dependence that decreases when every pair separates.



**Conjecture 298** (Positive curvature at complete dependence, resolved true). *For centered points  $x_1, \dots, x_m$  realizing  $\Gamma$ , with  $M = \max_i \langle Z, x_i \rangle$  and  $I$  the argmax, as  $t \downarrow 0$ ,*

$$\theta_m(t\Gamma) = 1 + \sqrt{t} \mathbb{E}M + t q(\Gamma) + o(t), \quad q(\Gamma) = \frac{1}{2} \mathbb{E}[M^2 - \|x_I\|^2],$$

*and  $q(\Gamma) \geq 0$  always, with  $q(\Gamma) = 0$  if and only if, after deleting non-exposed points, the convex hull of the  $x_i$  is a point or a segment.*

**Significance.** The first-order coefficient  $\mathbb{E}M$  is the Gaussian mean-width functional of the configuration, so the expansion says the departure of  $\theta_m$  from complete dependence opens at Sudakov's term and the conjecture locates the sign and the null set of the second term: a genuinely two-or-higher-dimensional exposed geometry always curves upward. The convex-body shadow is classical, the Wills functional of a body expands in intrinsic volumes with nonnegative second coefficient vanishing exactly on segments [238], but  $\theta_m$  is the finite-point analogue, its argmax form  $\mathbb{E}[M^2 - \|x_I\|^2]$  is not an intrinsic volume of the hull, and no strong-dependence expansion of Hüsler–Reiss extremal coefficients appears in the extremes literature. Proposition 52(iii) proves the null direction, every collinear configuration has  $q = 0$  by an exact sign-conditioning identity, so the open content is strict positivity off segments together with the validity of the expansion. In tests,  $q$  was computed exactly for planar configurations by polar quadrature: adversarial minimization at  $m = 3$  and  $m = 4$  drives the minimum to  $-2 \times 10^{-16}$ , machine zero, attained in both cases at collinear configurations, while thin triangles of height  $h$  over a unit base keep  $q > 0$  with  $q$  vanishing linearly in  $h$ , from  $4.3 \times 10^{-2}$  at  $h = 0.3$  to  $4.8 \times 10^{-4}$  at  $h = 0.003$ , and with  $w$  and  $q$  computed exactly by polar quadrature on a random planar triple, the finite- $t$  residual  $[\theta_3(t\Gamma) - 1 - \sqrt{t}w]/t$  rose to the exact  $q = 0.08370$  through 0.07993, 0.08183, 0.08277, 0.08324 as  $t$  fell from  $6.4 \times 10^{-3}$  to  $10^{-4}$ , the deficit shrinking like  $\sqrt{t}$ . A counterexample would be an exposed planar configuration with vanishing or negative curvature, which would refute the finite-point Wills analogy at its first nontrivial order. Since deposit the conjecture has been resolved true: Proposition 54 proves the expansion together with the nonnegativity of  $q(\Gamma)$  and the equality classification, the positivity following from a Gaussian integration by parts for the support function of the exposed hull.

## 70 Cluster inverse problems under thinning

**Proposition 53** (Thinning calibration for the cluster programme). Let the limiting cluster have size  $N$  with generating function  $G_N$  and, where stated, finite support  $S \subset \mathbb{Z}$ . At the level of the limiting compound-Poisson cluster process: (i) independent retention with probability  $p$  gives the response  $\theta(p) = \theta(1 - G_N(1 - p))/p$ ; (ii) retention of one residue class modulo  $m$  with uniform random phase gives  $\theta_m^{\text{sys}} = \theta \mathbb{E}R_m(S)$ , where  $R_m(S)$  is the number of occupied residues; (iii) the joint response  $Q_{m,S}(p) = \frac{1}{m} \sum_{r=0}^{m-1} [1 - (1 - p)^{n_{m,r}(S)}]$ , known for  $p$  in any infinite subset of  $(0, 1)$ , determines the occupancy multiset  $\mathcal{N}_m(S)$ ; (iv) the occupancy data  $\{\mathcal{N}_m(S) : m \geq 2\}$  determines the difference multiset of  $S$ ; (v) occupancy equivalence is closed under dilation:  $\mathcal{N}_m(tS) = \mathcal{N}_{m/\gcd(t,m)}(S)$  for every  $t \geq 1$ , so  $S \sim S'$  implies  $tS \sim tS'$ .

*Proof.* (i) Thinning replaces each cluster by a  $\text{Binomial}(N, p)$  survivor count  $K$ ; the surviving-cluster intensity is the original cluster intensity times  $\Pr(K > 0) = 1 - G_N(1 - p)$ , the exceedance intensity is scaled by  $\mathbb{E}K = p\mathbb{E}N$ , and the extremal index is their ratio, giving  $(1 - G_N(1 - p))/(p\mathbb{E}N)$ . (ii) With uniform phase  $U$ , the retained count is  $K = |\{s \in S : s \equiv U\}|$ , so



$\Pr(K > 0 \mid S) = R_m(S)/m$  and  $\mathbb{E}(K \mid S) = |S|/m$ ; the ratio of expectations is  $\mathbb{E}R_m(S)/\mathbb{E}|S|$ . (iii) Averaging Bernoulli thinning over the phase gives exactly  $Q_{m,S}(p)$ ; the functions  $p \mapsto 1 - (1-p)^k$  for distinct positive integers  $k$  are linearly independent, being polynomials of distinct degrees in  $1-p$ , so the coefficients of the representation of  $Q_{m,S}$  in this family, which are  $\frac{1}{m}$  times the multiplicities of the occupancy counts, are determined. (iv) From  $\mathcal{N}_m(S)$  form  $g(m) = \sum_r \binom{n_{m,r}(S)}{2}$ , the number of pairs of  $S$  congruent modulo  $m$ , and set  $g(1) = \binom{|S|}{2}$ ; if  $c_\delta$  is the number of pairs at difference  $\delta \geq 1$  then  $g(m) = \sum_{m|\delta} c_\delta$  for every  $m \geq 1$ , a finitely supported divisor sum, and Möbius inversion gives  $c_\delta = \sum_{j \geq 1} \mu(j) g(j\delta)$ . (v) Write  $g = \gcd(t, m)$ ,  $m = gm_1$ ,  $t = gt_1$ . Then  $ts \bmod m = g(t_1s \bmod m_1)$ , and since  $\gcd(t_1, m_1) = 1$  the map  $u \mapsto t_1u$  permutes the residues modulo  $m_1$ ; so the nonzero occupancy counts of  $tS$  modulo  $m$  are exactly those of  $S$  modulo  $m_1$ .  $\square$

**Proposition 54** (Resolution of the complete-dependence curvature law). Conjecture 298 holds in full: the displayed expansion is valid,  $q(\Gamma) \geq 0$  for every admissible  $\Gamma$ , and  $q(\Gamma) = 0$  if and only if the convex hull of the exposed points is a point or a segment.

*Proof.* Write  $\varepsilon = \sqrt{t}$ ,  $G_i = \langle Z, x_i \rangle$ ,  $M = \max_i G_i$  with almost-surely unique  $\arg\max I$  after deleting non-exposed and duplicate points, and  $V_\varepsilon = \max_i (\varepsilon G_i - \frac{\varepsilon^2}{2} \|x_i\|^2)$ , so  $\theta_m(t\Gamma) = \mathbb{E}e^{V_\varepsilon}$ . From  $|V_\varepsilon - \varepsilon M| \leq \frac{\varepsilon^2}{2} \max_i \|x_i\|^2$  and uniform integrability of  $e^{\varepsilon M}$  for small  $\varepsilon$ , dominated expansion gives  $\theta_m(t\Gamma) = 1 + \varepsilon \mathbb{E}M + \frac{\varepsilon^2}{2} \mathbb{E}[M^2 - \|x_I\|^2] + o(\varepsilon^2)$ , the asserted expansion with  $q(\Gamma) = \frac{1}{2} \mathbb{E}[M^2 - \|x_I\|^2]$ . Now  $M = h_K(Z)$  is the support function of  $K = \text{conv}\{x_i\}$ ; the  $x_i$  are centered, so  $0 = \frac{1}{m} \sum_i x_i$  lies in the relative interior of  $K$  and  $M \geq 0$ , while  $M$  is convex and 1-homogeneous with  $\nabla M = x_I$  almost everywhere. Gaussian integration by parts and Euler's identity  $\langle z, \nabla M \rangle = M$  give

$$\mathbb{E}[M^2] = \mathbb{E}\langle Z, M \nabla M \rangle = \mathbb{E}\|\nabla M\|^2 + \mathbb{E}[M \Delta M],$$

where  $\Delta M \geq 0$  is the curvature measure of the convex function  $M$ ; since  $M \geq 0$  the last term is nonnegative and  $\|\nabla M\|^2 = \|x_I\|^2$  almost everywhere, so  $q(\Gamma) \geq 0$ . Equality forces  $M = 0$  on  $\text{supp } \Delta M$ ; if  $\dim K \geq 2$  then 0 lies in the relative interior of  $K$ , so  $M > 0$  at every nonzero point of the affine span of  $K$ , on which the nonzero curvature measure  $\Delta M$  is supported, and the term is strictly positive. Hence  $q(\Gamma) = 0$  iff  $\dim K \leq 1$ , and for a point or a segment the curvature measure sits where  $M = 0$ , giving  $q = 0$ .  $\square$

**Conjecture 299** (Bernoulli-thinning spectroscopy of clusters). *Let  $(X_t)$  be stationary and regularly varying, with exceedances converging to a compound-Poisson cluster process with cluster size  $N$ ,  $\mathbb{E}N < \infty$ , and extremal index  $\theta = 1/\mathbb{E}N$ , under mixing and anti-clustering conditions. Retain each observation independently with probability  $p$  and compress the time axis. Then the retained sequence satisfies the same hypotheses with extremal index  $\theta(p) = \theta(1 - G_N(1-p))/p$ , uniformly for  $p$  in compact subsets of  $(0, 1]$ ; consequently the response curve  $p \mapsto \theta(p)$  identifies the entire limiting cluster-size law.*

**Significance.** The formula itself is the cluster-level algebra of Proposition 53(i), and extremal indices of randomly subsampled sequences are an established subject [244, 243, 242], so the formula is claimed as calibration, not novelty; the conjectural content is the transport theorem, that thinning commutes with the cluster limit uniformly enough in  $p$  for the curve to be an inferential object, and the spectroscopic consequence, that a one-parameter family of scalar

extremal indices inverts to the full cluster-size distribution via  $G_N(1-p) = 1 - p\theta(p)/\theta$ , for which no statement was found. Cluster-size inference is otherwise a direct estimation problem [245]; the conjecture replaces estimation of a distribution by estimation of a curve of scalars, each a standard blocks quantity. In process-level tests on a four-lag moving-maximum process with twelve million steps, the blocks estimator of the thinned sequence matched the formula at  $p = 1, 0.7, 0.4, 0.2$  with ratios 0.988, 0.989, 0.954, 0.976, the common shortfall being the finite-threshold bias of the estimator itself. A counterexample would be a process whose thinned clusters fail the predicted limit at some  $p$ , which would locate a genuine obstruction to interchanging thinning and clustering.

**Conjecture 300** (Occupancy tomography of cluster supports). *(a) Every finite  $S \subset \mathbb{Z}$  with  $|S| \leq 5$  is determined by its occupancy data up to translation and reflection. (b) For every finite size, each occupancy-equivalence class of finite integer sets contains at most two translation-reflection classes.*

**Significance.** The two-parameter thinning response determines exactly the occupancy data, by Proposition 53(iii), so this conjecture is the precise inverse theory of cluster-shape tomography, and its shape is dictated by a refutation recorded in the part preamble: full reconstruction is impossible, because Bloom’s pair  $\{0, 1, 4, 10, 12, 17\}$ ,  $\{0, 1, 8, 11, 13, 17\}$  [246], the classical counterexample to Piccard’s theorem [247], shares every occupancy multiset, by finite check through modulus seventeen and trivially beyond the diameter. Occupancy equivalence thus sits strictly between congruence and homometry: it implies homometry by Proposition 53(iv), the census found every collision homometric, ten collisions among the 6437 canonical sets of size at most six through diameter eighteen and the 3312 of size at most seven through diameter fourteen, while the homometric pair  $\{0, 1, 3, 8, 10, 14\}$ ,  $\{0, 1, 7, 9, 11, 14\}$  is occupancy-distinguished, so the refinement is strict on both sides, and dilation closure, Proposition 53(v), generates infinite equivalent families from each collision. Among Sidon sets clause (b) is a theorem: occupancy equivalence implies homometry, and Bekir and Golomb proved Bloom’s pair essentially the only homometric Sidon ambiguity [248]; the general case, and the five-mark determination (a), verified exhaustively through diameter twenty-two, are open, with the turnpike literature [249] the nearest boundary. A counterexample to (a) would be a five-mark ambiguity below every census run; one to (b) would be a triple of pairwise occupancy-equivalent, pairwise non-congruent sets.

## 71 The topology of extreme directions

**Conjecture 301** (Consistency of persistent angular homology). *Let  $X$  be  $d$ -variate regularly varying whose angular law  $H$  is supported on a compact  $C^2$   $s$ -dimensional manifold  $M$  of positive reach in the open simplex, with density bounded above and below. Let  $A_{n,k}$  be the angular components of the  $k = k_n$  largest radii,  $k \rightarrow \infty$ ,  $k/n \rightarrow 0$ , and suppose the conditional angular law beyond radius  $t$  couples to  $H$  with Hausdorff bias at most  $b(t) \rightarrow 0$ . Then for every homological degree  $q$ ,*

$$d_B(\text{Dgm}_q(\text{VR}(A_{n,k})), \text{Dgm}_q(\text{VR}(M))) = O_p\left(b(U(n/k)) + \left(\frac{\log k}{k}\right)^{1/s}\right),$$

*with  $U$  the radial quantile function and  $d_B$  the bottleneck distance.*

**Significance.** Angular inference in extremes estimates masses, faces, and linear structure; the conjecture makes loops and components of the support an estimand with a rate, composing two known machines, the stability of persistence diagrams under Hausdorff perturbation [252] and the  $(\log k/k)^{1/s}$  covering rates of manifold samples [253, 251], with a genuinely extreme-value ingredient: the angular points are selected by top- $k$  radii, are not an independent sample from  $H$ , and sit at finite threshold, displaced by  $b(U(n/k))$ . The nearest body of work, the topology of extreme sample clouds and its persistence limit theorems [250], studies the crackle of the cloud itself, not consistency for the angular support, so the statement appears new while its proof budget, a Poissonization of the order-statistics selection, is plausibly within reach; the rate also gives a threshold-selection principle, balancing radial bias against topological sampling error. In process-level tests on a ring in the three-part simplex with an explicit second-order bias  $b(t) = 0.9/t$  and two million observations, the Hausdorff distance tracked the predicted two-term sum within a factor 0.55 to 1.22 across  $k = 50$  to 3200, top- $k$  covering radii matched independent-sample covering radii, and the Vietoris–Rips diagrams of an independent  $\mathbb{F}_2$  implementation showed a single dominant loop with death 0.3123 against the analytic  $\sqrt{3}\rho = 0.3118$ , the bottleneck error falling from 0.115 at  $k = 40$  to 0.049 at  $k = 120$ . A counterexample would show the intermediate-order selection itself distorting the covering rate, the failure mode the conjecture exists to exclude.

**Conjecture 302** (A leakage trichotomy for hidden extremes). *Let  $X = R_0W_0 + R_1W_1$  with independent components:  $R_0$  regularly varying with index  $\alpha_0$ ,  $R_1$  with index  $\alpha_1 > \alpha_0$ , and angular laws  $W_0 \sim H_0$ ,  $W_1 \sim H_1$  supported on compact sets  $M_0$ ,  $M_1$  separated in the simplex,  $H_1$  as in Conjecture 301. Remove the angular neighborhood  $N_{\delta_n}(M_0)$ , rerank the surviving radii, and let  $L_n$  be the ratio of the primary to the hidden mechanism’s contribution to the surviving exceedance intensity at the hidden threshold. Let  $\Lambda_n$  be the expected number of surviving leaked primary exceedances in the reranked top- $k_n$  sample, and suppose the leaked support  $M_{\text{leak}}$  is persistence-separated from  $M_1$ ,  $d_B(\text{Dgm}_q(M_1 \cup M_{\text{leak}}), \text{Dgm}_q(M_1)) \geq c_0 > 0$ . Then: if  $\Lambda_n \rightarrow 0$ , the persistence diagrams of the surviving angular sample recover those of  $M_1$  at the rate of Conjecture 301; if  $\Lambda_n \rightarrow \infty$  they do not; and for  $\Lambda_n \rightarrow \lambda \in (0, \infty)$  the surviving leaked primaries are asymptotically Poisson- $\lambda$  and the probability of recovery within any fixed bottleneck tolerance below the hidden feature scale converges to a limit  $p(\lambda) \in (0, 1)$ , so feature detection switches on the leaked-count scale  $\Lambda_n \asymp 1$ , that is  $L_n \asymp 1/k_n$ .*

**Significance.** Hidden regular variation is a competition of tail orders on nested cones [254, 255], and the filtering primitive, removing the cone of the first-order limit and asking what remains, is exactly the detection scheme of Mitra and Resnick; the conjecture makes the competition quantitative at the level of topology, asserting that what decides visibility is not the index gap  $\alpha_1 - \alpha_0$  but the finite-threshold leakage ratio after support removal, because a handful of leaked heavy points can fill a loop or bridge components. The two-mechanism additive model makes the ratio well-defined without reference to unobservable labels, and sparse-support methods [256] are the statistical neighbors. In tests at  $\alpha_0 = 1$ ,  $\alpha_1 = 2$  with primary mass on the axes spread by a parameter  $\eta$  and hidden mass on a ring: the unfiltered top sample had three components and no loop; after filtering, at leakage 0.034 and 0.081 the ring was recovered with bottleneck error 0.090 and 0.075, while at leakage 2.9 and 120 the error rose to 0.143 and 0.156, the failure value being half the true persistence, so the diagram no longer matched the ring on either side of the crossover, exactly the predicted switch. The governing scale is the leaked count, not the intensity ratio: under the intermediate-order sampling of Conjecture 301 a fixed ratio  $L_n \rightarrow \lambda > 0$  leaks

order- $k_n$  points and floods the diagram, so the nondegenerate crossover sits at  $\Lambda_n \asymp 1$ , and the finite-sample transition seen near  $L_n \asymp 1$  is its reflection at the tested sizes. A counterexample would be a regime with  $\Lambda_n \rightarrow 0$  and persistent contamination, or recovery surviving  $\Lambda_n \rightarrow \infty$ , either of which would refute the leaked count as the governing scale.

## 72 Tail tomography and its stability

**Proposition 55** (Moment calibration for the tomography programme). (i) If  $\alpha > 0$  is not an integer, then  $T_\alpha H$  restricted to any open subset of  $(0, \infty)^d$  determines the finite measure  $H$  on  $\Delta_{d-1}$  uniquely. (ii) If  $\alpha = q \in \mathbb{N}_0$ , then  $T_q H$  depends only on the moments of  $H$  of degree at most  $q$ , and injectivity fails; explicitly, for  $d = 2$ ,  $q = 2$ , the measures

$$H_1 = \frac{1}{2}\delta_{0.2} + \frac{1}{2}\delta_{0.8}, \quad H_2 = 0.28125\delta_{0.1} + 0.4375\delta_{0.5} + 0.28125\delta_{0.9}$$

satisfy  $T_2 H_1 = T_2 H_2$  identically while  $T_{3/2} H_1 \neq T_{3/2} H_2$ .

*Proof.* (i) Since  $\mathbf{1} \cdot w = 1$  on the simplex, directions  $a = c(\mathbf{1} + z)$  with  $\|z\|_\infty < 1$  give  $(a \cdot w)^\alpha = c^\alpha(1 + z \cdot w)^\alpha$ , and  $z \mapsto \int (1 + z \cdot w)^\alpha H(dw)$  is real-analytic on  $\|z\|_\infty < 1$ , its binomial series converging uniformly. Agreement of  $T_\alpha H$  on an open set forces agreement of this analytic function everywhere on the polydisc, hence of all its Taylor coefficients at  $z = 0$ , which are  $\binom{\alpha}{k}$  times the degree- $k$  moment forms of  $H$ ; for noninteger  $\alpha$  every  $\binom{\alpha}{k} \neq 0$ , so all mixed moments agree, and moments determine a finite measure on a compact simplex by Stone–Weierstrass and Riesz representation. (ii) For integer  $q$  the binomial series terminates at degree  $q$ . Identifying  $w \in \Delta_1$  with its first coordinate,  $(a \cdot w)^2 = (a_2 + (a_1 - a_2)w)^2$  involves only  $\int w^j dH$  for  $j \leq 2$ ; both displayed measures have total mass 1, first moment  $0.028125 + 0.21875 + 0.253125 = 0.5$  and second moment  $0.0028125 + 0.109375 + 0.2278125 = 0.34$  against  $\frac{1}{2}(0.2 + 0.8) = 0.5$  and  $\frac{1}{2}(0.04 + 0.64) = 0.34$ , so  $T_2 H_1 = T_2 H_2$ ; their third moments are 0.26 and 0.2456..., so by (i) any noninteger power separates them, and at  $\alpha = 3/2$  the relative difference reaches  $2.0 \times 10^{-3}$  on the tested direction grid.  $\square$

**Conjecture 303** (The stability barrier of fractional tail tomography). *Fix noninteger  $\alpha > 0$ ,  $0 < \eta < 1$ , and the direction box  $A_\eta = \{a : \eta \leq a_i \leq \eta^{-1}\}$ . (a) There exist no  $C, \beta > 0$  with  $W_1(H_1, H_2) \leq C \|T_\alpha H_1 - T_\alpha H_2\|_{L^\infty(A_\eta)}^\beta$  for all probability measures  $H_1, H_2$  on  $\Delta_{d-1}$ . (b) On the class of angular densities lying in a Hölder ball  $\{f : \|f\|_{C^s} \leq L, f \geq c > 0\}$ , the inverse admits a logarithmic modulus of continuity:  $W_1(H_1, H_2) \lesssim (\log(1/\varepsilon))^{-\kappa_s}$  with  $\varepsilon$  the transform error, for some  $\kappa_s > 0$  depending on  $s$ , and no polynomial modulus  $\varepsilon^\beta$  is valid on the class.*

**Significance.** Proposition 55 settles identifiability, integers lose it algebraically, fractional powers keep it, the dichotomy known in substance from the theory of zonoid equivalence and diagonal Minkowski classes [257] and visible in extreme-value practice as the non-uniqueness of angular measures behind the tail pairwise dependence matrix at  $\alpha = 2$  [258]; the conjecture is the quantitative sequel, that fractional identifiability is severely ill-posed, because the kernel  $(a \cdot w)^\alpha$  is real-analytic on the box, its singular values collapse geometrically, and recovery passes through arbitrarily high moments, the phenomenon known for analytic-kernel moment problems [259] but not instantiated for one-sided projection tails of angular measures. In discretized tests at  $\alpha = 3/2$  against  $\alpha = 2$ : the  $\alpha = 2$  transform matrix had exact numerical rank three while the  $\alpha = 3/2$  matrix had effective ranks 4, 10, 15 at thresholds  $10^{-4}$ ,  $10^{-8}$ ,  $10^{-12}$  with tenth singular

value  $2.2 \times 10^{-9}$ , and zero-mass oscillating perturbations at frequencies 2 through 16 drove the ratio of transform norm to Wasserstein scale from 0.70 down to 0.048, the super-polynomial collapse the barrier predicts. A counterexample to (a), a global Hölder inverse, would make nonparametric angular tomography from tail constants routine, against all numerical evidence; a proof of (b) would explain why sparse and parametric tail models succeed where unrestricted reconstruction fails.

**Conjecture 304** (A parameter-count threshold for sparse tail tomography). *Let  $H = \sum_{r=1}^s \pi_r \delta_{w_r}$  with distinct atoms in the open simplex  $\Delta_{d-1}^\circ$ , weights  $\pi_r > 0$  summing to one, and let  $p = sd - 1$  be the parameter count. Fix noninteger  $\alpha > 0$  and observe  $y_j = (T_\alpha H)(a_j)$  at generic positive directions  $a_1, \dots, a_m$ . Then, outside a proper real-analytic exceptional set of atom configurations and designs, and quantitatively away from atom collisions and the simplex boundary: for  $m < p$  identifiability fails on a positive-dimensional fiber; for  $m = p$  the Jacobian has full rank and the fiber is finite, so the mixture is locally rigid but need not be unique; and for every  $m \geq sd$  the measurement vector determines  $H$  globally up to permutation of atoms, with local inverse modulus controlled by atom separation and boundary distance.*

**Significance.** The placement of the global threshold one measurement above the count is forced by two computations recorded in the preamble. A near-coincident atom pair first produced ten apparent interpolants at  $m = p$  that all collapsed under forty-digit Newton polishing into the single true solution, an artifact of conditioning fixed by the separation hypothesis; then a genuine multiplicity survived every polish: at  $(d, s) = (3, 3)$ , two of six well-separated random instances admit a second exact interior three-atom mixture at  $m = p = 8$ , with atoms nowhere near the truth and residuals below  $10^{-25}$ , and in both instances the ambiguity disappears at  $m = 9$ . So the counting boundary buys exactly local rigidity, the fractional analogue of the identifiability counts of multivariate Prony and super-resolution theory, which live on integer moments and have no counterpart here, and global uniqueness is conjectured to begin at  $m = sd$ , where both refuting instances become unique again. In tests at noninteger  $\alpha = 1.37$ : at  $m = p - 1$  the fiber is visibly a continuum, fifty distinct exact interior interpolants from one hundred twenty starts at  $(d, s) = (3, 2)$ ; at  $m = p$ , all six separated instances at  $(d, s) = (3, 2)$  and four of six at  $(3, 3)$  returned exactly the true mixture, the other two returning the truth plus one extra that dissolves at  $m = 9$ . A counterexample would be a separated generic instance with two exact mixtures persisting above the count, which would push the design requirement higher still.

## 73 Randomized block designs

**Proposition 56** (Design calibration for the block programme). (i) If  $\bar{F}(x) = x^{-\alpha}$  and  $N_n \sim \text{Poisson}(n)$ , then  $\Pr(M_{N_n}/n^{1/\alpha} \leq x) = e^{-x^{-\alpha}}$  exactly for every  $n$ . (ii) For any block-size law,  $\log \mathbb{E}(1 - q)^{N_n} = \sum_{r \geq 1} \kappa_r^{(F)} (-q)^r / r!$ , so matching factorial cumulants of the Poisson law through order  $r$  cancels the sampling corrections through order  $q^r$ . (iii) For every  $r \geq 2$  there is a law  $\xi$  on a finite subset of  $\{0, 1, 2, \dots\}$ , not Poisson, whose cumulants match those of  $\text{Poisson}(1)$  through order  $r$ ; the convolution  $N_n = \xi_1 + \dots + \xi_n$  then has  $\mathbb{E}N_n = n$ ,  $\text{Var } N_n = n$ ,  $\kappa_j^{(F)}(N_n) = 0$  for  $2 \leq j \leq r$ ,  $\kappa_{r+1}^{(F)}(N_n) = O(n)$ , and for the Pareto seed

$$\sup_x |\Pr(M_{N_n}/n^{1/\alpha} \leq x) - e^{-x^{-\alpha}}| = O(n^{-r})$$

uniformly on compact  $x$ -sets. For  $r = 3$  the law on  $\{0, 1, 3\}$  with weights  $(\frac{1}{3}, \frac{1}{2}, \frac{1}{6})$  and for  $r = 4$  the law on  $\{0, 1, 2, 4\}$  with weights  $(\frac{3}{8}, \frac{1}{3}, \frac{1}{4}, \frac{1}{24})$  are such laws.

*Proof.* (i) The Poisson generating function gives  $\mathbb{E}(1 - q)^{N_n} = e^{-nq}$ , and  $q = x^{-\alpha}/n$  makes the exponent exactly  $-x^{-\alpha}$ . (ii) is the definition of factorial cumulants at argument  $u = -q$ . (iii) The moment vector  $(m_1, \dots, m_r)$  of Poisson(1) is the convex combination  $\sum_{k \geq 0} e^{-1}/k! (k, k^2, \dots, k^r)$  of points on the integer moment curve, so by Carathéodory it is a convex combination of at most  $r + 1$  such points: a law  $\xi$  on at most  $r + 1$  nonnegative integers with the first  $r$  moments, hence the first  $r$  cumulants, of Poisson(1). No finitely supported law is Poisson. Cumulants add under convolution, so  $\kappa_j(N_n) = n\kappa_j(\xi)$  matches  $\kappa_j(\text{Poisson}(n))$  for  $j \leq r$ ; each factorial cumulant  $\kappa_j^{(F)}$  is a fixed linear combination of  $\kappa_1, \dots, \kappa_j$ , and Poisson( $n$ ) has  $\kappa_j^{(F)} = 0$  for  $j \geq 2$ , so  $\kappa_j^{(F)}(N_n) = 0$  for  $2 \leq j \leq r$ , while  $\kappa_{r+1}^{(F)}(N_n)$  is  $n$  times a constant depending on  $\xi$ . Because the summand factorial cumulants vanish through order  $r$ , the analytic function  $q \mapsto \log \mathbb{E}(1 - q)^\xi$  equals  $-q + O(q^{r+1})$  on  $[0, q_0]$ , so  $\log \mathbb{E}(1 - q)^{N_n} = -nq + O(nq^{r+1})$ ; with  $q = x^{-\alpha}/n$  bounded on compacts this is  $-x^{-\alpha} + O(n^{-r})$ , and exponentiating gives the uniform bound. For the displayed laws the moment checks are exact rational arithmetic: on  $\{0, 1, 3\}$  the raw moments are  $\frac{1}{2} + \frac{3}{6} = 1$ ,  $\frac{1}{2} + \frac{9}{6} = 2$ ,  $\frac{1}{2} + \frac{27}{6} = 5$ , matching Poisson(1) through order three with fourth cumulant  $0 \neq 1$ ; on  $\{0, 1, 2, 4\}$  the raw moments are 1, 2, 5, 15, matching through order four with fifth cumulant differing; so both are non-Poisson with all factorial cumulants of order at most  $r$  vanishing.  $\square$

**Conjecture 305** (Poissonized blocks remove the universal bias). *Let  $F$  lie in a max-domain of attraction with a uniform second-order expansion  $n\bar{F}(a_n x + b_n) = V(x) + A(n)B(x) + o(A(n)) + o(n^{-1})$  on compact interior sets. Then Poissonized block maxima satisfy  $\log \Pr(M_{N_n} \leq a_n x + b_n) = -V(x) - A(n)B(x) + o(A(n)) + o(n^{-1})$ , the universal  $V(x)^2/(2n)$  correction of fixed blocks being absent; and for generalized-extreme-value likelihood estimation from many independent blocks, Poissonization leaves the first-order asymptotic variance unchanged while removing the universal  $n^{-1}$  bias contribution, so that when  $A(n) = o(n^{-1})$  the estimator bias improves from order  $n^{-1}$  to the intrinsic second order.*

**Significance.** The identity behind the conjecture is elementary and exact for the Pareto seed, Proposition 56(i), the Poisson generating function exponentiating the one-observation tail with no binomial discretization error, and the object it removes is precisely the universal finite-block term documented in the second-order block-maxima theory [261, 262] and in the comparison of block maxima with threshold methods [263]; the distributional clause needs no lift beyond the seed, since  $\Pr(M_{N_n} \leq y) = \mathbb{E}F(y)^{N_n} = e^{-n\bar{F}(y)}$  exactly for Poisson exposure and every  $F$ , so the second-order expansion removes the  $V^2/(2n)$  term across the whole second-order max-domain; what is genuinely conjectured is the estimator clause, the transfer of that cancellation to generalized-extreme-value likelihood, in the framework of [260]. Maxima over random sample sizes have been studied for their dependence attractors, not as a bias-cancellation design, so the design reading appears new, and the natural objection, that a Poisson number of observations per block is threshold exceedance in disguise, is answered by the estimator clause: the procedure remains a blocks procedure with blockwise likelihood, changing only the exposure law. In estimator-level tests with three hundred thousand blocks of Pareto data, the Fréchet shape estimate from fixed blocks carried the predicted order- $1/n$  bias while the honestly Poissonized version removed it: shape bias +0.0396 against +0.0017 at  $n = 32$ , and +0.0100 against +0.0031 at  $n = 128$ , the fixed-block bias falling exactly like  $1/n$  and the Poissonized residual sitting at the finite-sample



likelihood level. A counterexample would be a second-order family where Poissonization leaves an  $n^{-1}$  term, which would locate a non-universal component in the finite-block correction.

**Conjecture 306** (The factorial-cumulant design ladder). *Fix  $r \geq 2$  and let  $N_n$  be an  $r$ -th ladder law as in Proposition 56(iii). Then in every max-domain with the second-order expansion of Conjecture 305, block maxima with exposure  $N_n$  satisfy the expansion with sampling-mechanism corrections of order  $O(n^{-r})$  in place of  $V^2/(2n)$ ; and generalized-extreme-value likelihood estimation from many independent randomized blocks retains the first-order limit law and variance of deterministic blocks while its sampling-mechanism bias falls to  $O(n^{-r}) + o(A(n))$ .*

**Significance.** The algebra is proved: Proposition 56 exhibits, for every  $r$ , integer exposure laws with variance  $n$  whose factorial cumulants vanish through order  $r$  and whose exact finite-block error for the Pareto seed is  $O(n^{-r})$ , so the ladder exists constructively, block design becoming a quadrature problem in which each matched factorial cumulant annihilates one universal error term. What remains conjectural is the statistical half: that the cancellation survives general second-order domains and passes through blockwise likelihood with unchanged first-order variance, and that the bounded-support ladder laws, whose cumulants are all  $O(n)$ , avoid the degradation that heavy exposure randomization would cause. No design principle of this kind appears in the random-sample-size or block-maxima literature. In exact tests at Pareto seed with  $\alpha = 2$  across  $n = 64$  to  $16384$ : fixed blocks gave sup-error slope  $-1.001$ , the two-point law  $n \pm \sqrt{n}$  gave  $-2.002$ , and the ladder laws of Proposition 56 gave  $-3.004$  and  $-4.004$  in fifty-digit arithmetic, each error falling by its predicted factor, eighty-three-fold and two-hundred-fifty-six-fold, per fourfold block growth. A counterexample would be a domain in which some non-universal  $n^{-j}$  term,  $j \leq r$ , survives every exposure law, which would split the finite-block error into design-cancellable and intrinsic parts.

**Conjecture 307** (An adaptation cusp at second-order cancellation). *In a smooth Hall family  $\bar{F}_\lambda(x) = x^{-\alpha}[1 + c(\lambda)x^{-\beta} + d(\lambda)x^{-2\beta} + o(x^{-2\beta})]$  with  $c(\lambda_0) = 0$ ,  $c'(\lambda_0) \neq 0$ ,  $d(\lambda_0) \neq 0$ , the oracle threshold for Hill-type estimation moves from  $k \asymp n^{2\beta/(\alpha+2\beta)}$  generically to  $k \asymp n^{4\beta/(\alpha+4\beta)}$  at cancellation, the two regimes meeting at the critical window  $|c(\lambda)| \asymp n^{-\beta/(\alpha+4\beta)}$ ; and no data-driven threshold selector is uniformly oracle-rate optimal, with the same constants, over both  $|c(\lambda)| \leq Cn^{-\beta/(\alpha+4\beta)}$  and  $|c(\lambda)| \gg n^{-\beta/(\alpha+4\beta)}$ : an adaptation penalty, at least logarithmic or a nontrivial constant, is unavoidable in the cusp.*

**Significance.** This conjecture is close to prior work: that adaptive tail estimation pays an unavoidable logarithmic-type penalty is the theorem of Carpentier and Kim [267], with the threshold-selection literature going back to [264, 266, 265]; the new content is the location and geometry of the obstruction, a cusp of measure zero in the parameter but of width  $n^{-\beta/(\alpha+4\beta)}$  in samples, where exact cancellation doubles the effective second-order exponent and the oracle itself is discontinuous in  $\lambda$  at fixed  $n$ . The window exponent is forced by a two-line balance, the revived first correction  $|c|u^{-\beta}$  matching the second correction  $u^{-2\beta}$  at the cancellation-oracle threshold, and the conjecture asserts this balance is not just heuristic but the exact boundary of uniform adaptivity. In tests, the tail-error slope doubled from  $-\beta$  to  $-2\beta$  at cancellation across two decades of thresholds, and the empirically critical  $|c|$  tracked  $n^{-\beta/(\alpha+4\beta)}$  with constant ratio  $0.500$ , the value of  $d(\lambda_0)$ , across  $n = 10^3$  to  $10^9$ . A counterexample, a selector uniformly optimal across the cusp, would be a genuinely new adaptation mechanism, evading the usual two-point lower-bound construction at its most degenerate point.

## Part XVIII

# Extremal, randomized, and inverse laws for persistent topology

*Added to the deposit: 20 August 2026.*

The twelve conjectures below concern a single organizing question of topological data analysis: which parts of a persistence computation are laws rather than artifacts, sharp constants instead of worst-case bounds, invertible summaries instead of lossy signatures, and calibrated randomness instead of parameter sensitivity. They form five programmes. An extremal-and-sampling programme (Conjectures 308 to 310) concerns degree-one Vietoris–Rips persistence: a circle isoperimetry asserting that among  $n$  points on a Euclidean circle the regular polygon uniquely maximizes the multiplicative persistence ratio, the chordal, every- $n$ , ratio-level refinement of the geodesic persistence-set extremality of Gómez and Mémoli [270]; a saturation law asserting that the classical factor two of Vietoris–Rips stability [271] is asymptotically attained by greedy landmarking of smooth convex curves; and a two-sided Johnson–Lindenstrauss threshold [272, 273] for the simultaneous preservation of the loop ratios of orthogonal hexagonal channels under Gaussian projection, whose upper half is known in substance [274, 275] and whose lower half would be the first matching bound for a topological statistic. A statistical programme (Conjectures 311 and 312) makes the geometry of Wasserstein means of diagrams [276, 277] quantitative: an optional feature of prevalence  $q$  appears in the mean at exactly  $q$  times its conditional persistence up to exponentially small matching leakage, sharpening the shrinkage phenomenon visible in the probabilistic Fréchet means of [278] and the grouping calculus of [279], and empirical means use the intended latent grouping up to  $CNm e^{-c\gamma^2/\sigma^2}$ . A Mapper programme (Conjectures 313 and 314) builds on an exact phase calculus proved here: randomizing the phase of a uniform one-dimensional cover [280] makes the survival probability of an isolated Reeb cycle the universal piecewise-linear function  $F_\alpha(\ell/h)$  with an explicit kernel, the phase-averaged Betti curve of a chain of cycles is exactly additive with the spans recoverable from its kinks, and the dithered midpoint estimator of a branch critical value is exactly unbiased with the quantization variance  $s^2/12$  of [282]; the conjectures transfer this calculus to general tame Reeb graphs with a stable inversion, and to discretized and sampled Mapper with mesh-order corrections. A tomography programme (Conjecture 315) concerns rectangle-decomposable two-parameter modules [283], for which the fibered barcode over all lines is a complete invariant [284]: a proposition proved here sharpens completeness to a single generic direction with the line parameterization retained, the deposited three-direction claim being recorded as calibration, and the conjecture asserts the quantitative sequel, that three fixed generic directions give a locally bi-Lipschitz invariant with polynomially controlled constant and that the direction profile of the matching distance is never needle-like, so uniformly random slopes see near-maximal discrepancy; it is closely related to the local metric equivalence of Vipond [285], proved with a universal constant over all lines, and the Lipschitz foliation bounds of [286] behind deterministic approximation schemes [287, 288]. A finite-summary programme (Conjectures 316 to 319) asks how small a complete or faithful summary can be: a restricted-isometry law for random Fourier sketches [289] of the persistence-weighted Gaussian kernel [290], bi-Lipschitz to the Wasserstein metric on separated bounded-cardinality classes, which the known non-embeddability theorems [291, 292] do not forbid because they require collisions or unbounded cardinality, with restricted-class embeddings known in spirit [293];

a Whitney-count law [294] asserting that  $4m + 1$  generic evaluations of the Gaussian persistence surface, whose continuum injectivity is known [295] while grid images are not injective [296], form a complete invariant of  $m$ -point diagrams; a moment law asserting that the first  $4m$  moments of the aggregate landscape [297] determine a generic  $m$ -bar diagram, a compressed instance of the finite algebraic coordinate systems of [298, 299]; and an average-case kinetic law asserting that random affine motion of  $n$  planar points produces only  $O(n^2 \log n)$  expected changes of the degree-one persistence pairing, far below the quartic worst-case count that vineyard algorithms [300, 301, 302, 303] are built to survive. The proved layer is three calibrations. Proposition 57 proves the exact survival law with kernel  $F_\alpha(x) = \min(1, \max(0, (x - 1)/(1 - \alpha)))$ , the exact additivity and kink inversion of the phase-averaged Betti curve for chains of isolated cycles, and the exact dither law for branch critical values. Proposition 58 proves that one generic direction of the parameterized fiber field determines a generic rectangle-decomposable module, by tracking the interval pairs of the fiber barcodes through the parallel family and reading each rectangle off the envelope of its track. Proposition 59 proves the exact closed form of the aggregate-landscape moments and the one-bar case of the moment law, a single bar being determined by its first two moments. Three negative findings are retained as controls, and they shape the part. The unrestricted planar version of the extremality claim is refuted: a six-point perturbation search found planar configurations whose multiplicative loop persistence exceeds the regular hexagon's by the factor 1.052, exact to the arithmetic of the boundary-matrix reduction, so the circle hypothesis of Conjecture 308 is essential and is recorded with the refuting instance; this failure of unrestricted extremality appears to be undocumented. The deposited three-direction tomography claim is true but not sharp, one direction sufficing by Proposition 58, and is recorded as calibration. A deposited majority-threshold claim for barycenter features, that an optional feature appears in the mean exactly when its prevalence exceeds one half, is refuted by the flat grouping algebra itself, the mean feature shrinking continuously with prevalence instead, and Conjecture 311 carries the corrected law. The deposited linear stable-inversion clause of Conjecture 313 is refuted at coalescing spans: the two-cycle span multisets  $\{\ell, \ell\}$  and  $\{\ell - \delta, \ell + \delta\}$  have bottleneck distance  $\delta$ , yet on the affine middle branch of the survival kernel the symmetric perturbation cancels, so the phase-averaged Betti curves differ only in  $O(\delta)$ -wide neighborhoods of the moving kinks and  $\|S_G - S_{G'}\|_{L^2(dh/h)} = \Theta(\delta^{3/2})$ , whence no bound  $\delta \leq C\eta$  can hold; the repaired clause requires the span values to be pairwise separated, near-collisions admitting only a Hölder inverse of exponent  $2/3$ , the constant  $\eta/\delta^{3/2} \rightarrow 6.51$  at  $\alpha = 0.3$ ,  $\ell = 0.5$  found by exact integration. Conjectures 309 to 312, 313, 314, and 316 to 319 assert asymptotic or every-instance laws; Conjecture 308 is closely related to [270], and Conjecture 315 is closely related to [285] and [286].

For a finite metric space  $X$ , the Vietoris–Rips complex at scale  $r$  has as simplices the subsets of diameter at most  $r$ ; homology is over the two-element field, computed here by boundary-matrix reduction of the full two-skeleton, which suffices in degree one. For a degree-one bar  $(b, d)$  with  $b > 0$  the multiplicative persistence is  $d/b$ , and  $\Pi_1^\times(X)$  is its maximum over bars, set to one if none exists;  $d_B$  and  $W_2$  are the bottleneck and two-Wasserstein distances with diagonal matching. For a regular  $n$ -gon inscribed in the unit circle the dominant degree-one bar has  $b_n = 2 \sin(\pi/n)$  and  $d_n = 2 \sin(\lceil n/3 \rceil \pi/n)$  [269, 268], so  $\Pi_1^\times = \sin(\lceil n/3 \rceil \pi/n) / \sin(\pi/n)$ , reproduced exactly by the present implementation for  $n \leq 12$ . A uniform one-dimensional Mapper cover has interval length  $h$ , overlap fraction  $\alpha \in (0, 1)$ , step  $s = h(1 - \alpha)$ , intervals  $[ks - U, ks - U + h]$ , and uniformly random phase  $U$  on  $[0, s]$ ; the Mapper graph is the simple nerve of the connected components of the interval preimages, and an isolated cycle of a tame Reeb graph is a pair of arcs joining two branch

points whose filter values differ by the span  $\ell$  (its degree-one extended-persistence bar length), with all other critical values farther than the cover scale. A rectangle-decomposable module is a finite direct sum of interval modules supported on axis-parallel rectangles; for a positive-slope direction  $v$ , the fiber field assigns to each line parallel to  $v$ , indexed by its transverse intercept  $\tau$ , the multiset of parameter intervals cut by the rectangles, and endpoints are generic when the finitely many coincidence conditions among side coordinates and crossing values fail. The Gaussian persistence surface of a diagram  $D$  is  $\rho_D(z) = \sum_{p \in D} \text{pers}(p) \exp(-\|z - p\|^2/2\sigma^2)$ ; the aggregate landscape is  $A_D(t) = \sum_i \max\{0, \min(t - b_i, d_i - t)\}$  with moments  $M_r(D) = \int t^r A_D(t) dt$ ; and for points moving affinely in the plane, a vineyard event is a time at which the multiset of (birth edge, death triangle) pairs of the degree-one reduction changes.

**Proposition 57** (Phase calculus for one-dimensional Mapper). Fix  $\alpha \in (0, 1)$  and the uniform random phase. (i) For an isolated cycle of span  $\ell$ ,

$$\Pr(\beta_1(M_{h,U}) = 1) = F_\alpha(\ell/h), \quad F_\alpha(x) = \min\left(1, \max\left(0, \frac{x-1}{1-\alpha}\right)\right),$$

and  $\beta_1 = 0$  otherwise: the survival probability is a universal piecewise-linear function of  $\ell/h$ , zero for  $h \geq \ell$  and one for  $h \leq \ell/(2-\alpha)$ . (ii) For a series chain of isolated cycles with spans  $\ell_1, \dots, \ell_r$ , the phase-averaged Betti curve satisfies  $S(h) = \mathbb{E}_U \beta_1(M_{h,U}) = \sum_j F_\alpha(\ell_j/h)$  exactly for every  $h > 0$ , and the multiset  $\{\ell_j\}$  is recovered from  $S$ : its kinks in  $h$  occur exactly at the points  $\ell_j$  and  $\ell_j/(2-\alpha)$ , and descending induction on the largest kink identifies every span with multiplicity. (iii) For an isolated branch critical value  $c$ , the midpoint estimator  $\hat{c}(U) = L(U) - s/2$ , with  $L(U)$  the lower endpoint of the first interval whose preimage has the post-branch component count, satisfies  $\hat{c} - c \sim \text{Unif}(0, s] - s/2$ , so  $\mathbb{E}\hat{c} = c$  exactly and  $\text{Var } \hat{c} = s^2/12$ .

*Proof.* (i) If some interval  $I$  lies strictly inside the open span  $(c, c + \ell)$  between the two branch values, its preimage in the cycle is two arcs, giving two nerve vertices; the neighboring intervals containing the branch values have connected preimages, and the resulting simple nerve contains exactly one cycle through the doubled vertices, while everything else is a tree by isolation, so  $\beta_1 = 1$ . If no interval lies inside the span, every interval meeting it contains a branch value, its preimage is connected through the branch point, all nerve vertices along the feature are single, and the nerve is a path, so  $\beta_1 = 0$ ; note that when two single vertices overlap inside the span their preimages meet in two arcs but the simple nerve still carries one edge. The lattice of interval left endpoints has spacing  $s$ , and an interval lies inside the span exactly when its left endpoint falls in a window of length  $(\ell - h)_+$ , so the survival probability is  $\min(1, (\ell - h)_+/s) = F_\alpha(\ell/h)$ . (ii) The survival criterion is local to each span, the criteria are functions of the same phase, and expectations add; each summand  $F_\alpha(\ell_j/h)$  is smooth in  $h$  except at  $h = \ell_j$  and  $h = \ell_j/(2-\alpha)$ , where the one-sided slopes differ, and distinct spans contribute kink locations  $\ell_j$  that are largest among all kinks of their terms, so the largest kink of  $S$  is the largest span, and subtracting its term and iterating recovers the multiset. (iii) An interval  $[lo, lo + h]$  has the post-branch count exactly when  $lo > c$ , by isolation, so  $L$  is the smallest lattice point above  $c$ ; as  $U$  runs uniformly through one period,  $L - c$  is uniform on  $(0, s]$ , and  $\hat{c} - c = (L - c) - s/2$  is centered uniform of width  $s$ .  $\square$

**Proposition 58** (One direction determines a rectangle module). Let  $M = \bigoplus_{i=1}^m \mathbf{k}_{R_i}$  be rectangle-decomposable with generic endpoints, and fix any direction  $v$  of positive slope. Then the fiber field of  $M$  along  $v$ , with the line parameterization retained, determines the multiset  $\{R_1, \dots, R_m\}$ .

*Proof.* Along the parallel family, the interval cut by a fixed rectangle has entry and exit endpoints that are piecewise-linear functions of the intercept  $\tau$ , with two possible slopes according to which side the line enters or leaves, and each rectangle's interval exists on a closed  $\tau$ -window, entering and leaving with zero length at the corner-touching lines. Two distinct rectangles cannot cut identical intervals over a  $\tau$ -interval: equality of entry functions on an interval forces the same active side and the same side coordinate, excluded by generic endpoints; equality of the pairs at a single  $\tau$  requires an entry crossing and an exit crossing at the same intercept, one further coincidence excluded by genericity. Hence the multiset-valued fiber field decomposes into uniquely tracked continuous single-valued branches, one per rectangle, distinct pairs never colliding. On its window each branch sweeps the entry point along the union of the left and bottom sides, a polyline from the top-left to the bottom-right corner, so the coordinate extremes of the traced entry and exit points recover all four sides of the rectangle, and the multiset is determined.  $\square$

**Proposition 59** (Moment calibration for aggregate landscapes). Write bar  $i$  as center  $c_i = (b_i + d_i)/2$  and half-width  $a_i = (d_i - b_i)/2$ . (i) For every  $r \geq 0$ ,

$$M_r(D) = \sum_{i=1}^m \sum_{\substack{0 \leq j \leq r \\ j \text{ even}}} \binom{r}{j} c_i^{r-j} \frac{2a_i^{j+2}}{(j+1)(j+2)}.$$

(ii) For  $m = 1$ , the first two moments determine the bar:  $M_0 = a^2$  and  $M_1 = c a^2$ , so  $a = \sqrt{M_0}$  and  $c = M_1/M_0$ ; in particular the first  $4m$  moments of Conjecture 318 suffice at  $m = 1$  with two to spare. (iii) The second distributional derivative  $A''_D = \sum_i (\delta_{b_i} - 2\delta_{c_i} + \delta_{d_i})$  is a signed measure with at most  $3m$  atoms whose moments obey  $\mu_0 = \mu_1 = 0$  and  $\mu_k = k(k-1)M_{k-2}$  for  $k \geq 2$ ; hence for a diagram with  $3m$  distinct nodes the first  $6m - 2$  aggregate moments  $M_0, \dots, M_{6m-3}$  determine the diagram, by Prony reconstruction of the nodes from  $\mu_2, \dots, \mu_{6m-1}$  and the pairing  $b_i + d_i = 2c_i$  of endpoints to centers by the weights  $+1$  and  $-2$ .

*Proof.* The aggregate landscape is the sum of the tent functions  $t \mapsto \max\{0, \min(t - b_i, d_i - t)\} = \max\{0, a_i - |t - c_i|\}$ . Substituting  $t = c_i + u$ ,

$$\int t^r \max\{0, a_i - |u|\} dt = \sum_{j=0}^r \binom{r}{j} c_i^{r-j} \int_{-a_i}^{a_i} u^j (a_i - |u|) du,$$

the odd- $j$  integrals vanish by symmetry, and for even  $j$  the integral is  $2(a_i^{j+2}/(j+1) - a_i^{j+2}/(j+2)) = 2a_i^{j+2}/((j+1)(j+2))$ . Part (ii) is the case  $r \leq 1$ :  $M_0 = a^2$  and  $M_1 = c a^2$  directly. (iii) The tent  $\max\{0, a_i - |t - c_i|\}$  has second derivative  $\delta_{b_i} - 2\delta_{c_i} + \delta_{d_i}$ , so  $A''_D = \sum_i (\delta_{b_i} - 2\delta_{c_i} + \delta_{d_i})$ ; two integrations by parts give  $\mu_k = \int t^k dA''_D = k(k-1)M_{k-2}$  for  $k \geq 2$ , with  $\mu_0 = \mu_1 = 0$  by compact support. A signed measure with  $s = 3m$  distinct atoms is determined by  $2s$  consecutive moments through the Prony–Hankel construction off a proper algebraic set; the known  $\mu_0 = \mu_1 = 0$  leave  $\mu_2, \dots, \mu_{6m-1}$ , that is  $M_0, \dots, M_{6m-3}$ , and the weight  $-2$  marks each center against  $+1$  at each endpoint, so  $b_i + d_i = 2c_i$  pairs them uniquely off a further proper algebraic set.  $\square$

## 74 Extremal and sampling laws for Vietoris–Rips persistence

**Conjecture 308** (Circle isoperimetry for multiplicative persistence). *For every  $n \geq 4$  and every set  $X$  of  $n$  distinct points on the Euclidean unit circle,*

$$\Pi_1^\times(X) \leq \frac{\sin(\lceil n/3 \rceil \pi/n)}{\sin(\pi/n)},$$

with equality exactly for the regular  $n$ -gon, up to isometries of the circle.

**Significance.** This conjecture is close to prior work: the extremality of the regular polygon among circle configurations is known in the geodesic metric through the persistence sets of Gómez and Mémoli [270], who characterize the Vietoris–Rips diagrams of  $n$ -point subsets of the circle and prove the regular polygon the unique extreme configuration in the principal cases; the present statement is the chordal refinement, every  $n$  at once, for the multiplicative ratio rather than additive persistence, over the exact regular-polygon bars of [269, 268]. The circle hypothesis is essential and its necessity is a retained control: in the unrestricted plane, a perturbation search with exact two-skeleton reduction found six-point configurations beating the regular hexagon by the factor 1.052, so no global planar isoperimetry of this type exists, a failure that appears to be undocumented. The mechanism is a two-sided balancing: equal spacing minimizes the largest chordal gap, which controls the birth from below, while distributing the third-circle chords that fill the fundamental cycle, which controls the death from above; a proof must handle the combinatorial switching of the death simplex across the chambers of the gap simplex. In adversarial tests at  $n = 4$  to 10, fifteen hundred random circular configurations per  $n$  plus local optimization from the best of them and from perturbed regular polygons never exceeded the regular value, the closest competitor reaching 0.9988 of it, while the formula itself was reproduced to  $10^{-15}$  for  $n \leq 12$ . A counterexample would exhibit irregular spacing manufacturing a more persistent loop, and would separate the chordal from the geodesic extremal theory.

**Conjecture 309** (Saturation of the stability factor two). *Let  $\Gamma \subset \mathbb{R}^2$  be a  $C^3$  strictly convex closed curve, let  $L_m$  be the first  $m$  points of a farthest-point sampling of  $\Gamma$  from a generic seed, and let  $\rho_m = d_H(L_m, \Gamma)$ . Then*

$$\lim_{m \rightarrow \infty} \frac{d_B(\text{Dgm}_1(L_m), \text{Dgm}_1(\Gamma))}{\rho_m} = 2.$$

**Significance.** Vietoris–Rips stability [271] bounds the bottleneck error of a landmark diagram by twice the Hausdorff distance, and the conjecture asserts that on the most classical objects, smooth convex loops under the canonical deterministic sampling scheme, the constant two is not slack but attained in the limit: the discrete loop cannot be born before the largest gap chord closes, the largest greedy cell has radius  $\rho_m$  and gap chord  $2\rho_m + O(\rho_m^3)$ , and the continuum loop is born at zero, so the birth discrepancy alone forces the factor, provided the death error stays of lower order, which is the open half. On the exactly regular inscribed  $m$ -gon the model computation is elementary,  $b_m/\rho_m = 2 \cos(\pi/2m) \rightarrow 2$ . No sharpness result for the stability constant under sampling was found; the neighboring literature bounds errors from above. In tests with exact reduction and greedy samples from three seeds, the ratio on the circle ran 1.990, 2.014, 2.032, 2.000 at  $m = 24$  to 128 against the analytic bar  $(0, \sqrt{3})$ , and on the ellipse with axes 1.3 and 0.7 it ran 1.955 to 2.019 at  $m = 16$  to 100 against a dense reference bar. A counterexample would be a convex curve where death error cancels birth error, revealing a curvature-dependent constant below two.

**Conjecture 310** (A two-sided projection threshold for loop ratios). *For  $q \geq 2$  let  $H_1, \dots, H_q$  be regular hexagons in mutually orthogonal coordinate two-planes of  $\mathbb{R}^{2q}$ , let  $A$  be an  $m \times 2q$  Gaussian matrix with independent  $N(0, 1/m)$  entries, and for fixed small  $\varepsilon > 0$  let  $\mathcal{E}_{q,m,\varepsilon}$  be the event that every channel ratio  $\Pi_1^\times(AH_j)/\Pi_1^\times(H_j)$  lies within  $\varepsilon$  of one. Then there are constants*



$0 < c < C$  with

$$m \leq c \varepsilon^{-2} \log q \implies \Pr(\mathcal{E}_{q,m,\varepsilon}) \rightarrow 0, \quad m \geq C \varepsilon^{-2} \log q \implies \Pr(\mathcal{E}_{q,m,\varepsilon}) \rightarrow 1,$$

as  $q \rightarrow \infty$ .

**Significance.** Johnson–Lindenstrauss projection [272] preserves persistence because it preserves all pairwise distances, the principle quantified for distance-function persistence by Sheehy [312], and upper bounds of this type are known [274, 275]; the conjecture asks for the converse scale on a purely topological observable: the restrictions of a Gaussian matrix to the  $q$  orthogonal planes are independent blocks, a single channel’s ratio distortion has a tail exponential in  $m\varepsilon^2$ , and the extreme value of  $q$  independent channels should cross from failure to success exactly when  $qe^{-cm\varepsilon^2} \asymp 1$ , the same logarithmic law that governs metric distortion and is optimal for it [273]. The lower half would be the first matching lower bound for a persistence statistic; the concern, recorded here, is that it reduces to metric Johnson–Lindenstrauss optimality because hexagon ratios encode the anisotropy of the projected plane. In tests at  $\varepsilon = 0.3$  the empirical critical dimension fit  $m^* = 5.92 \log q + 1.00$  across  $q = 4$  to 256 with maximum residual 1.4, and the transition profile at  $q = 64$  rose from 0.00 at  $m = 14$  through 0.45 at  $m = 24$  to 0.92 at  $m = 44$ . A counterexample would be a threshold at a scale other than  $\log q$ , showing that loop ratios discard enough metric information to change the dimension law.

## 75 The statistical geometry of diagram means

**Conjecture 311** (Prevalence shrinkage with exponential leakage). *Let random diagrams consist of a background diagram plus, with probability  $q$ , one signal point, the signal cluster separated from the background and the diagonal by a matching margin  $\gamma$ , with conditional perturbations centered subgaussian of scale  $\sigma \ll \gamma$ . Then the population  $W_2$  Fréchet mean has a unique signal point whose diagonal coordinate satisfies  $\bar{U} = \mathbb{E}[U \mid \text{present}] + O(e^{-c\gamma^2/\sigma^2})$  and whose distance to the diagonal satisfies*

$$\bar{V} = q \mathbb{E}[V \mid \text{present}] + O(e^{-c\gamma^2/\sigma^2}),$$

*uniformly over bounded separated model classes.*

**Significance.** When an absent diagram matches the candidate mean point to the diagonal, the flat first-order conditions give exactly  $\bar{V} = q \mathbb{E}V$ , so a rare feature does not vanish from the mean, it shrinks toward the diagonal in proportion to its prevalence; the phenomenon is visible in the probabilistic Fréchet means of [278] and the grouping calculus of [277, 279], and a deposited majority-threshold claim, appearance exactly above prevalence one half, is refuted by this algebra and retained as a control. The conjectural content is global: that the full Fréchet problem inherits the flat law with only exponentially small matching leakage, so that the persistence of a barycentric feature is interpretable as conditional strength times prevalence, and prevalence becomes estimable from calibrated mean diagrams. In tests with a real alternating-minimization mean over five hundred diagrams per prevalence, the ratio of the mean feature’s diagonal distance to  $q \mathbb{E}[V \mid \text{present}]$  was 0.98 to 1.02 across  $q = 0.1$  to 0.9, and shrinking the margin to three noise scales left the deviation at the finite-sample floor near 0.02, consistent with leakage remaining below sampling error at every tested margin. A counterexample would be polynomial-in-margin leakage, which would make mean diagrams systematically misestimate rare features.

**Conjecture 312** (Exponential stability of latent groupings). *Let  $D_1, \dots, D_N$  be independent diagrams from a latent model with  $m$  feature clusters, matching margin  $\gamma$  between clusters and to the diagonal, subgaussian scale  $\sigma$ , and prevalences bounded away from zero. Then*

$$\Pr\{\text{some optimal mean grouping differs from the latent grouping}\} \leq CNm e^{-c\gamma^2/\sigma^2},$$

*and on the complementary event the empirical  $W_2$  Fréchet mean is unique, its coordinates obey the flat projection formulas, and featurewise central-limit behavior holds at rate  $N^{-1/2}$ .*

**Significance.** The singular geometry of diagram space enters averaging only through combinatorial matching changes, and under a margin every such change requires a noise excursion of scale  $\gamma$ ; a union bound over samples and features then predicts the stated form, and the substance is that all globally optimal Wasserstein assignments respect the latent grouping whenever no local excursion occurs, upgrading the deterministic uniqueness condition of [279] to a quantitative probabilistic regime in which barycenter statistics are ordinary Euclidean statistics. No exponential grouping-stability theorem was found. In tests with two latent features at margin one, the identity-swap frequency of the optimal matching was 0 at noise scales through 0.25, then 0.0027, 0.0103, 0.0307 at scales 0.35, 0.45, 0.60, a margin-controlled transition with the observed logarithms decaying faster than  $\gamma^2/4\sigma^2$ . A counterexample would be a polynomial failure rate, forcing heavy-tailed corrections to every barycenter confidence statement.

## 76 A phase calculus for Mapper

**Conjecture 313** (Stable survival tomography of Reeb graphs). *Let  $(G, f)$  be a tame Reeb graph whose  $r$  degree-one extended-persistence bars have lengths (spans)  $\ell_1, \dots, \ell_r$ , each below the separation ( $\ell_j < \gamma$ ), and pairwise critical-value separation at least  $\gamma > 0$ , and let  $S_G(h) = \mathbb{E}_U \beta_1(M_{h,U})$  be the phase-averaged Betti curve at overlap  $\alpha$ . Then (a)  $S_G(h) = \sum_j F_\alpha(\ell_j/h)$  for every  $h < \gamma$ , with the kernel of Proposition 57; and (b) the inversion is stable away from the collision stratum: if two tame Reeb graphs with  $\gamma$ -separated features and distinct span values pairwise separated by at least  $\beta > 0$  have curves with  $\|S_G - S_{G'}\|_{L^2((0,\gamma), dh/h)} \leq \eta$ , then the equal-cardinality span multisets are within bottleneck distance  $C(\gamma, \alpha, r, \beta) \eta$  of each other; the span separation is essential, coalescing spans admitting only a Hölder inverse of exponent  $2/3$ .*

**Significance.** Proposition 57 makes the phase-averaged calculus exact for chains of isolated cycles: survival is the universal kernel  $F_\alpha$ , curves add, and kinks invert; the conjecture asserts that the calculus survives on general tame Reeb graphs below the separation scale, where nested and adjacent features can interact through shared branch points, the span multiset being read canonically as the degree-one persistence bar lengths of  $(G, f)$ , and that the kink inversion is quantitatively stable, turning the notorious cover-position sensitivity of Mapper [280] into an invertible multiscale signature: the randomness of the phase is not noise but a probe, in the spirit of randomized numerical analysis rather than of the sampling randomness studied in [281]. In tests, the empirical survival probability of an isolated cycle matched  $F_\alpha$  to three decimals at overlaps 0.30 and 0.45 across the full range of  $\ell/h$ , and the two-cycle curve matched the additive prediction within 0.0055 across twenty-eight scales spanning both kinks, the curve passing value one exactly at the smaller span. A counterexample to (a) would be an interaction term surviving separation, to (b) an instability of the kink locations under curve perturbation, either of which would locate a genuine obstruction to randomized Mapper inference. The span bound  $\ell_j < \gamma$  is

essential and is retained as a control: a feature whose span exceeds the separation saturates the survival kernel,  $F_\alpha(\ell_j/h) = 1$ , throughout the interaction-free window  $h < \gamma$ , so its length leaves no trace on  $S_G$  there; two graphs with spans 3 and 4 at  $\gamma = 1$  have identical curves on  $(0, \gamma)$ , and stable inversion requires every kink, at  $h = \ell_j$  and  $h = \ell_j/(2 - \alpha)$ , to lie inside the observed window. A second control is the collision stratum: two equal spans  $\{\ell, \ell\}$  and a symmetric split  $\{\ell - \delta, \ell + \delta\}$  have curves differing only near their moving kinks, so the  $L^2(dh/h)$  distance is of order  $\delta^{3/2}$  while the span bottleneck is  $\delta$ ; the deposited linear inversion of clause (b) is thus false without the span-separation hypothesis, and near collisions the inverse modulus is no better than  $\delta = O(\eta^{2/3})$ .

**Conjecture 314** (The dither law survives discretization). *For a Reeb graph discretized at filter mesh  $\eta = o(h)$ , or sampled densely enough that its Mapper is that of the underlying graph, the midpoint estimator of an isolated branch critical value satisfies*

$$\mathbb{E}\hat{c} - c = O(\eta), \quad \text{Var}(\hat{c}) = \frac{s^2}{12} + O(s\eta + \eta^2),$$

*uniformly over admissible discretizations, with constants depending only on the local branch geometry.*

**Significance.** Proposition 57(iii) is the continuum law: random cover translation converts the deterministic quantization bias of Mapper critical values into centered noise of exactly the dithered-quantizer variance  $s^2/12$  [282]; the conjecture asserts that nothing worse than mesh-order corrections appears on discretized or sampled objects, so that branch locations come with exact error bars, an elementary but genuinely usable uncertainty quantification that the Mapper literature does not provide. The content is that the first-split event on a discrete graph tracks the continuum event to one mesh cell, without a topological bias from components fragmenting at the sampling scale. In tests on a discretized branch graph at mesh  $10^{-3}$ , the phase-averaged bias was between  $-1.7 \times 10^{-4}$  and  $-4 \times 10^{-5}$  across three cover resolutions, of mesh order as predicted, and the variance ratio to  $s^2/12$  was 1.000 at all three. A counterexample would be a discretization scheme with an order- $s$  bias, which would identify a topological failure mode of dithering absent from classical quantization.

## 77 Rectangle tomography

**Conjecture 315** (Three-direction stability and random-slope observability). *Fix a box, a maximum count  $m$ , and  $\delta > 0$ , and consider rectangle-decomposable modules whose corner coordinates, side lengths, and pairwise separations are at least  $\delta$ , and whose corners stay at distance at least  $\delta$  from the coincidence loci of  $v_1, v_2, v_3$ , the configurations at which two rectangles cut a common fiber-interval endpoint along some  $v_j$ . (a) For generic fixed positive-slope directions  $v_1, v_2, v_3$ , the invariant  $d_3(M, N) = \max_j \sup_{L \parallel v_j} d_B(\text{Dgm}(M_L), \text{Dgm}(N_L))$  is locally bi-Lipschitz equivalent to the minimal corner displacement  $d_{\text{rect}}$ , with constants polynomial in  $m$  and  $1/\delta$ . (b) The weighted direction profile  $g(\theta)$  of the matching distance between two such modules is Lipschitz relative to its own maximum on compact slope ranges,  $\text{Lip } g \leq C \|g\|_\infty$  with  $C$  polynomial in  $m$ , the box size,  $1/\delta$ , and the slope margin, so its near-maximal set has measure bounded below and  $K$  uniformly random slopes recover a  $(1 - \varepsilon)$ -approximation of the matching distance with failure probability exponentially small in  $K$ .*

**Significance.** This conjecture is close to prior work on two counts. Local bi-Lipschitz equivalence between the matching distance, the supremum over all lines, and the interleaving distance is the theorem of Vipond [285], with a universal constant; and Lipschitz dependence of the fiber bottleneck on the line is classical in the foliation method [286] and is what deterministic approximation schemes exploit [287, 288]. What is asked here is strictly finite-direction and strictly quantitative: three fixed directions replacing the continuum of lines, with an inverse constant that is polynomial rather than combinatorial in the number of rectangles, over the identifiability base of Proposition 58, which sharpens the all-line completeness of [284] to a single generic direction and records the deposited three-direction claim as calibration, the polynomial conditioning claimed only on the transversality-margin class, since with three fixed directions a module can drift toward a direction-specific endpoint coincidence while keeping its own corners  $\delta$ -separated; and a width lower bound for the near-maximal set of the profile, which is what makes Monte Carlo approximation of the matching distance trustworthy, complementing the switch-point enumeration of [288]. In tests on separated three-rectangle modules, the ratio of corner displacement to the three-direction metric stayed below 0.48 over thirty perturbed pairs, the empirical profile Lipschitz constant stayed below 22, and the near-maximal fraction of one hundred eighty slopes never fell below 0.017. A counterexample to (a) would be a separated family with exponentially degenerating three-direction conditioning; to (b), a needle-like profile, which would doom random-direction estimates on even the simplest module class.

## 78 Finite summaries and dynamic complexity

**Conjecture 316** (A restricted isometry for persistence sketches). *On the class of diagrams with at most  $m$  points, persistence and pairwise separation at least  $\delta$ , in a box of diameter  $R$ , let  $\Phi_M$  be the random Fourier sketch of the persistence-weighted Gaussian embedding at bandwidth comparable to  $\delta^{-1}$  with  $M$  frequencies. Then there are class constants  $0 < c_0 < C_0$ , polynomial in  $m, R, 1/\delta$ , such that with probability  $1 - \eta$ ,*

$$c_0(1 - \varepsilon) W_2(D, E) \leq \|\Phi_M(D) - \Phi_M(E)\|_2 \leq C_0(1 + \varepsilon) W_2(D, E)$$

for all  $D, E$  in the class, provided  $M \geq C\varepsilon^{-2}m \log(mR/(\delta\eta))$ .

**Significance.** Diagrams famously do not embed: bounded-cardinality classes admit no bi-Lipschitz map to finite-dimensional Hilbert space in general [291], and the full space embeds in no Hilbert space at all [292]; the conjecture threads the needle those theorems leave open, because their obstructions require collisions or vanishing separation, which the class excludes, and restricted-class embeddings with controlled distortion are known in spirit [293]. The proposal is concrete: the kernel mean embedding of the persistence-weighted Gaussian kernel [290] should be bi-Lipschitz to transport on the separated class, and random Fourier features [289] should preserve that comparison at dimension nearly linear in the diagram cardinality, giving a streaming-compatible, provably Wasserstein-faithful vectorization. The polynomial dependence of the lower constant is the exposed clause: inverting Gaussian mixtures from kernel embeddings can be exponentially ill-conditioned in the number of components, and separation excludes collisions without yet excluding that decay. In tests, the ratio of kernel distance to  $W_2$  over eight hundred separated four-point pairs stayed within the two-sided band  $[0.90, 2.75]$ , and the sketch-to-kernel relative deviation halved with each fourfold increase in frequencies, from 0.10 at  $M = 16$  to 0.0087 at  $M = 1024$ , the parametric rate the concentration half requires. A counterexample would be a

separated family collapsing the lower constant, which would locate a genuinely non-Euclidean direction inside the separated class.

**Conjecture 317** (Whitney-count evaluations of persistence surfaces). *Fix  $\sigma > 0$  and the class of diagrams with exactly  $m$  points, pairwise  $\delta$ -separated and  $\delta$ -far from the diagonal, in a compact box. Choose  $q \geq 4m + 1$  evaluation points independently from any distribution with positive density on a neighborhood of the box. Then, almost surely in the evaluations, the map  $D \mapsto (\rho_D(z_1), \dots, \rho_D(z_q))$  is injective on the class, and locally bi-Lipschitz off the discriminant.*

**Significance.** The continuum Gaussian surface is injective [295], while gridded persistence images are not [296]; the conjecture asserts that the dimension-theoretically minimal number of generic point evaluations, the Whitney embedding count  $4m + 1$  for the  $2m$ -dimensional diagram manifold [294], already suffices, so that a complete summary needs no grid and no continuum, only  $4m + 1$  numbers. The mechanism is transversality of the real-analytic evaluation family, adjacent to the moment identifiability of Gaussian mixtures, with the persistence weights breaking the symmetries that make mixture recovery delicate. In tests at  $m = 3$ ,  $q = 13$ , sixty adversarial searches for well-separated pairs with equal evaluations never drove the evaluation gap below  $2.0 \times 10^{-2}$ , and local Jacobians had full rank throughout the search paths. A counterexample would be a collision pair at the Whitney count, pushing the truth toward the information-theoretic worst case and revealing hidden algebraic relations among Gaussian surface evaluations.

**Conjecture 318** (Aggregate-landscape moment determinacy). *For a Zariski-open, full-measure set of  $m$ -bar diagrams, the first  $4m$  moments  $M_0, \dots, M_{4m-1}$  of the aggregate landscape determine the diagram uniquely up to bar permutation.*

**Significance.** The full landscape is invertible [297], and finite complete coordinate systems for barcodes exist [298, 299]; the compression claimed here is that  $4m$  scalar integrals of one aggregated function suffice, through the exact closed form of Proposition 59, a constrained Prony problem in which each bar contributes a rigid two-parameter moment stream rather than a free point mass. The one-bar case is proved, two moments sufficing, and no claim of numerical robustness is made, the smallest singular values of the moment map decaying rapidly; a deposited claim that the first  $2m$  moments form a robust coordinate system was withdrawn on conditioning grounds by the source analysis and is retained here only as the observation that multistart inversion of the first  $2m$  exact moments already returned a unique reconstruction on every tested instance at  $m = 2, 3$ , so the  $4m$  threshold is conservative and possibly not sharp; the generic determinacy from  $6m - 2$  aggregate moments is unconditional, Proposition 59(iii), and the conjectured  $4m$  is the compression below that Prony benchmark. In tests, exact-moment multistart inversion at  $m = 2$  and  $m = 3$  found exactly one solution at  $2m$ ,  $3m$ , and  $4m$  moments across one hundred fifty starts each. A counterexample would be a moment symmetry among tent functions, a hidden relation that would be interesting in itself; a proof would give the smallest known complete statistic for generic barcodes.

**Conjecture 319** (Near-quadratic vineyard complexity in the mean). *Let  $n$  planar points move affinely,  $x_i(t) = a_i + tv_i$  on  $[0, 1]$ , with  $a_i, v_i$  independent standard Gaussians, and let  $N_1(n)$  be the number of times the degree-one Vietoris–Rips persistence pairing changes. Then  $\mathbb{E}N_1(n) = O(n^2 \log n)$ , and  $N_1(n) = O(n^2 \log^2 n)$  with probability  $1 - n^{-c}$ .*

**Significance.** There are quadratically many edge lengths and quartically many candidate order swaps, and vineyard-type algorithms [300, 301, 302, 303] are engineered to process whatever event sequence arrives; the conjecture asserts that on random affine motion almost all swaps are topologically silent, the pairing changing only near kinetically critical local configurations, so the true event count is governed by the sparse geometry of the moving point set rather than by the full distance arrangement, the first average-case law for dynamic persistence. In exact tests with adaptive time refinement at  $n = 6$  to 14, the mean event count grew with log-log exponent 2.36, matching the local slope 2.45 of  $n^2 \log n$  over the same range and far below the quartic candidate count, with absolute counts twenty to thirty times smaller than  $n^4/100$ . A superquadratic counterexample would show topological pairings are kinetically volatile in a way static sparsity cannot see, itself a finding with algorithmic consequences.

## Part XIX

# Temporal irreversibility and spectral selection in nonequilibrium dynamics

*Added to the deposit: 22 August 2026.*

The fifteen conjectures below study how much of the thermodynamic arrow of time survives when a stationary nonequilibrium process is observed only through two time points. For an irreducible continuous-time Markov chain the complete trajectory over an interval of length  $t$  carries a path-space entropy production  $\sigma t$ ; reducing the trajectory to its endpoints keeps only the *lag irreversibility*  $L(t)$ , the Kullback–Leibler divergence between the stationary two-time law and its time reverse, and data processing gives  $0 \leq L(t) \leq \sigma t$ . The organizing object is the *temporal sampling efficiency*  $\eta(t) = L(t)/(\sigma t) \in [0, 1]$ , the fraction of dissipation still visible at lag  $t$ . The conjectures form three programmes. A temporal-and-spectral programme (Conjectures 320 to 326) asserts monotone loss of efficiency, spectral bounds on the temporal support and the integrated budget of the arrow of time, an irreversibility-specific relaxation exponent that doubles the slowest dissipative mode, its generic coincidence with the ordinary spectral gap together with an exceptional invisible-slow-mode phenomenon, and the discrete mode selection performed by deterministic coarse-graining. A unicyclic-and-winding programme (Conjectures 327 to 331) sharpens these on a single driven cycle, where endpoint sampling loses exactly the integer winding number, giving a strict efficiency law, an amplitude cap, a weak-affinity wrapped-Brownian scaling limit, a ballistic winding-aliasing horizon, and a single Péclet crossover unifying the two. A Gaussian programme (Conjectures 332 to 334) transfers the peak-horizon, integrated-budget, and mode-doubling laws to multivariate Ornstein–Uhlenbeck processes, where every quantity is an explicit function of the drift and diffusion matrices. The proved layer is Proposition 60: the endpoint data-processing bounds, the short-time expansion  $L(t) = \sigma t + O(t^2)$  giving  $\eta(t) \rightarrow 1$ , and the exact Gaussian lag-KL formula. Conjectures 323, 325, 326, and 334 assert long-time exponential laws; Conjectures 329 to 331 are scaling-limit statements; Conjecture 324 is a genericity statement carrying its own exceptional set. The central observable is the lag irreversibility of Merino Negrete, Maldonado, and Salgado-García [304], and the closest spectral precedent is the finite-time cross-correlation asymmetry of Van Vu, Vo, and Saito [306]; the normalization by path entropy production, the temporal horizons and



integrated budgets, the doubled irreversibility exponent and its coarse-grained selection, and the winding scaling laws are what is proposed as new. Since deposit, Conjectures 323, 325, and 334, the discrete and Gaussian irreversibility-doubling and coarse-grained mode-selection laws, have been resolved true by Propositions 61 and 68 from the quadratic equivalence of lag KL with the antisymmetric endpoint flux, and Conjecture 329, the wrapped-Brownian limit, by Proposition 65. Since deposit, Conjectures 320 and 327, the efficiency-monotonicity and unicyclic strict-monotonicity laws, have been resolved false by Proposition 63: a homogeneous ten-state ring has non-monotone efficiency, rising by more than two orders of magnitude across an interior lag interval through winding aliasing, so the winding effect of Conjectures 330 and 331 produces a resonant revival rather than the proposed monotone loss. Since deposit, Conjecture 326, the hidden-mode exponential-separation law, has been resolved false by Proposition 62: a product of two driven three-cycles has an oscillatory dominant mode with exact zeros, so the coarse-to-full ratio returns to one at infinitely many times and the liminf separation exponent is zero rather than the exponent gap. Since deposit, Conjecture 331, the universal diffusive-to-ballistic Péclet crossover, has been resolved false as stated by Proposition 66: the ballistic first-crossing coefficient depends on the edge affinity, so no single Péclet function recovers the ballistic law of Conjecture 330 for every fixed affinity, though a one-variable crossover survives in the weak-edge-bias regime. Since deposit, Conjecture 328, the sharp unicyclic amplitude cap  $\sup_t \gamma L(t)/\sigma \leq \frac{1}{2}$ , has been resolved false by Proposition 64: an explicit near-decomposable seven-state ring reaches  $0.643 > \frac{1}{2}$ , so the constant  $\frac{1}{2}$  is not universal. Since deposit, Conjectures 332 and 333, the Gaussian peak-horizon  $g_A t_* \leq 1$  and spectral-area  $\int L_{OU} \leq \sigma_{OU}/g_A^2$  bounds, have likewise been resolved false by Proposition 67: a five-dimensional Jordan-block Ornstein–Uhlenbeck drift with  $g_A = 1$  has its lag-KL peak at  $t_* \approx 1.069$  and normalized area  $\approx 1.266$ , so the spectral abscissa alone cannot bound the temporal horizon or the integrated budget of a strongly nonnormal drift, and a repaired bound needs a nonnormality constant.

Let  $X_t$  be an irreducible continuous-time Markov chain on  $\{1, \dots, n\}$  with row generator  $Q = (q_{ij})$ , stationary law  $\pi > 0$ , and bidirected support ( $q_{ij} > 0 \Leftrightarrow q_{ji} > 0$ ), so the steady-state entropy production is finite. Write  $P_t = e^{tQ}$ , the stationary endpoint law  $F_t(i, j) = \pi_i P_t(i, j)$ , its time reverse  $F_t^\top$ , the lag irreversibility

$$L(t) = D_{\text{KL}}(F_t \| F_t^\top) = \sum_{i,j} F_t(i, j) \log \frac{F_t(i, j)}{F_t(j, i)},$$

and the entropy-production rate  $\sigma = \sum_{i \neq j} \pi_i q_{ij} \log(\pi_i q_{ij} / \pi_j q_{ji})$ , so that  $\eta(t) = L(t)/(\sigma t)$ . Let  $q_{ij}^* = \pi_j q_{ji} / \pi_i$  be the reversed generator,  $\bar{Q} = (Q + Q^*)/2$  the additive reversibilization,  $\gamma = \lambda_2(-\bar{Q}) > 0$  its reversible  $L^2(\pi)$  spectral gap, and  $g = \min\{-\Re \lambda : \lambda \in \text{spec}(Q) \setminus \{0\}\}$  the ordinary spectral-abscissa gap. Write  $A_t = F_t - F_t^\top = \Pi e^{tQ} - e^{tQ^\top} \Pi$  with  $\Pi = \text{diag}(\pi)$ , and let  $\alpha_{\text{irr}}$  be the smallest  $-\Re \lambda > 0$  over generator eigenmodes whose coefficient in  $A_t$  is nonzero (on generalized eigenspaces for Jordan blocks, the polynomial factors not affecting the rate). For a deterministic map  $f : \{1, \dots, n\} \rightarrow \{1, \dots, m\}$  let  $F_t^f(a, b) = \sum_{f(i)=a, f(j)=b} F_t(i, j)$ ,  $L_f(t) = D_{\text{KL}}(F_t^f \| (F_t^f)^\top)$ , and  $\alpha_{\text{irr}}(f)$  the smallest such rate surviving the block summation. For the homogeneous  $N$ -cycle with clockwise/counterclockwise rates  $p_N = e^{A/(2N)}$ ,  $q_N = e^{-A/(2N)}$  and time  $t_N = N^2 \tau / (p_N + q_N)$ , the wrapped Gaussian on  $x \in [0, 1)$  is  $w_{A, \tau}(x) = \sum_{k \in \mathbb{Z}} (2\pi \tau)^{-1/2} \exp[-(x + k - A\tau/2)^2 / (2\tau)]$ , with reverse  $w_{-A, \tau}$ ; at fixed edge affinity  $a \neq 0$  write  $p = e^{a/2}$ ,  $q = e^{-a/2}$ ,  $v = p - q$ , the Péclet number  $\text{Pe} = N(p - q)/(p + q)$ , and  $t_c(N, \cdot) = \inf\{t : \eta_N(t) \leq c\}$ . For the Ornstein–Uhlenbeck process  $dX_t = AX_t dt + \sqrt{2D} dW_t$  with  $A$  Hurwitz and  $D \succ 0$ , the stationary covariance solves  $AC + CA^\top + 2D = 0$ , the two-time law is Gaussian with cross-covariance  $Ce^{A^\top t}$ ,  $L_{OU}(t)$  is

the KL divergence to its block-swapped reverse,  $\sigma_{\text{OU}} = \text{tr}[(A + DC^{-1})^\top D^{-1}(A + DC^{-1})C]$ ,  $g_A = \min_\lambda(-\Re\lambda(A))$ ,  $H(t) = e^{At}C - Ce^{A^\top t}$ , and  $\alpha_H$  the slowest exponential rate in  $H(t)$ .

**Proposition 60** (Endpoint bounds, short-time efficiency, and the Gaussian lag KL). For every chain in the stated class,  $0 \leq L(t) \leq \sigma t$ , and as  $t \downarrow 0$  single-jump paths dominate, giving  $L(t) = \sigma t + O(t^2)$  and  $\eta(t) \rightarrow 1$ ; as  $t \rightarrow \infty$ ,  $F_t \rightarrow \pi \otimes \pi$  and  $L(t) \rightarrow 0$ . For the Ornstein–Uhlenbeck process the two-time covariance and its block swap have equal determinants, so

$$L_{\text{OU}}(t) = \frac{1}{2}[\text{tr}(\Sigma_t^{R-1}\Sigma_t) - 2d].$$

*Proof.* The endpoint map is a measurable function of the path, so data processing for KL gives  $L(t) \leq D_{\text{KL}}$  of the path law to its reverse  $= \sigma t$ , and  $L(t) \geq 0$ ; positivity of  $\sigma$  makes the interval nondegenerate. For small  $t$ ,  $P_t = I + tQ + O(t^2)$ , and the off-diagonal contribution to  $\sum F_t \log(F_t/F_t^\top)$  is  $t \sum_{i \neq j} \pi_i q_{ij} \log(\pi_i q_{ij}/\pi_j q_{ji}) + O(t^2) = \sigma t + O(t^2)$ , the diagonal terms cancelling to first order, whence  $\eta(t) \rightarrow 1$ ; irreducibility gives  $P_t \rightarrow \mathbf{1}\pi^\top$  and  $F_t \rightarrow \pi \otimes \pi$ , a symmetric limit, so  $L(t) \rightarrow 0$ . For the Gaussian case both covariances are  $2d \times 2d$  block matrices differing by a swap of the two time blocks, a similarity by the block permutation, so  $\det \Sigma_t^R = \det \Sigma_t$  and the log-determinant term of the Gaussian KL vanishes, leaving the displayed trace formula.  $\square$

## 79 Temporal irreversibility and spectral selection

**Conjecture 320** (Temporal sampling-efficiency monotonicity, resolved false). *For every finite irreducible bidirected chain with  $\sigma > 0$ , the efficiency  $\eta(t) = L(t)/(\sigma t)$  is non-increasing on  $(0, \infty)$ , and strictly decreasing for a generic nonequilibrium chain.*

**Significance.** Data processing gives only  $\eta \leq 1$ ; the conjecture asserts the far stronger statement that lengthening the observation lag can never raise the *fraction* of path irreversibility retained by two endpoints, so every coarsening of the observation clock loses thermodynamic information per unit elapsed time. The comparison is between different endpoint maps, so no single contraction theorem applies; a proof route compares a length- $t$  path to a random subdivision into blocks and seeks a conditional-relative-entropy inequality for the endpoint statistic, while a counterexample would exhibit a resonant lag at which more dissipation becomes visible. The nearest related result is the lag-irreversibility observable of [304]. A first decisive theorem would be monotonicity for three-state chains. A counterexample would be a chain and two lags with  $\eta$  strictly rising. *Resolution.* The conjecture is false: on the homogeneous 10-cycle with edge affinity 1,  $\eta$  falls to  $9.3 \times 10^{-6}$  at  $vt/N \approx 0.54$  and then rises by a factor above 130 to  $1.2 \times 10^{-3}$  at  $vt/N \approx 0.68$  before mixing erases it, verified at eighty digits (Proposition 63). The mechanism is the resonant lag the significance anticipates: near half a winding the forward and reverse wrapped endpoint laws nearly coincide and then separate again before diffusion mixes the cycle, so the winding effect of Conjectures 330 and 331 produces a revival of efficiency rather than monotone loss.

**Conjecture 321** (Two-gap-time location bound). *If  $t_*$  is any global maximizer of  $L(t)$ , then  $\gamma t_* \leq 2$ .*

**Significance.** Since  $L$  rises linearly from zero and returns to zero, it has a maximizer; the reversible part of the dynamics erases endpoint memory while the antisymmetric part supplies irreversibility, making the dimensionless product  $\gamma t_*$  a natural temporal horizon for the visible

arrow of time. A proof may combine a reversible  $L^2(\pi)$  contraction for  $P_t$  with a comparison between  $L$  and an antisymmetric  $\chi^2$  divergence, the obstacle being that KL can be sensitive to rare states, so a successful argument must avoid  $\pi_{\min}^{-1}$  prefactors. The nearest related results are the entropy-production spectral bounds of [212]. A first decisive theorem would be the bound for birth–death chains with one added shortcut. In tests on the corpus described in the stress-test section, the maximum  $\gamma t_*$  over sizes three to nine was 0.775, far inside the bound. A counterexample with  $\gamma t_* \gg 1$  would isolate arrow-of-time information latent long after reversible mixing.

**Conjecture 322** (Spectral integrated-budget bound). *For every chain in the class,  $\int_0^\infty L(t) dt \leq \frac{2\sigma}{\gamma^2}$ .*

**Significance.** The integral has the units of  $\sigma/\gamma^2$ , the unique scale built from the entropy-production rate and the reversible gap, so the conjecture caps the total area under all two-endpoint irreversibility measurements once the mixing timescale is fixed. Monotonicity alone does not imply it, since a monotone function can decay arbitrarily slowly, so genuine spectral input is required; a proof route bounds  $L$  by a weighted  $\chi^2$  norm of  $A_t$  and applies additive-reversibilization Dirichlet-form estimates. The nearest related results are [306, 212]. A first decisive theorem would be the bound restricted to reversibilizations with a spectral gap of multiplicity one. In tests on the corpus described in the stress-test section, the maximum normalized area  $\gamma^2 \int L dt / \sigma$  was 0.527. A counterexample would be a chain whose efficiency curve decays slowly enough to exceed the budget.

**Conjecture 323** (Irreversibility-gap doubling law, resolved true). *The long-time envelope of lag KL decays at twice the slowest irreversible rate:*

$$\liminf_{t \rightarrow \infty} -\frac{1}{2t} \log L(t) = \alpha_{\text{irr}},$$

*and generically  $L(t) = t^m e^{-2\alpha_{\text{irr}} t} [c(t) + o(1)]$  with  $m \geq 0$  a Jordan multiplicity and  $c$  bounded, nonnegative, and not identically zero.*

**Significance.** At long times  $F_t = \pi \otimes \pi + \delta F_t$  and KL is locally quadratic in the antisymmetric perturbation  $A_t$ , so a leading amplitude  $e^{-\alpha_{\text{irr}} t}$  in  $A_t$  doubles to  $e^{-2\alpha_{\text{irr}} t}$  in  $L$ ; the conjectural content is the clean modal characterization under degeneracy, complex modes, and cancellations. A proof route uses the Jordan–Chevalley expansion of  $e^{tQ}$  and a second-order KL expansion around  $\pi \otimes \pi$ . The nearest related result is the spectral-gap-controlled correlation-asymmetry decay of [306]. A first decisive theorem would be the diagonalizable case with a simple slowest irreversible mode. In tests on the corpus described in the stress-test section, fitting the KL tail and the antisymmetric-flux tail gave a ratio of exponents with median 1.0000 over the tested chains. A counterexample would be a chain whose lag KL decays at a rate other than twice an antisymmetric-flux rate. *Resolution.* Now proved, Proposition 61: the exponent identity holds unconditionally, and the refined law holds with power  $m = 2r$ , twice the Jordan-polynomial degree, rather than a single multiplicity, without genericity.

**Conjecture 324** (Generic equality of spectral and irreversibility gaps). *Fix a strongly connected bidirected graph and vary its positive rates. Outside a measure-zero analytic exceptional set, and excluding detailed balance,  $\alpha_{\text{irr}} = g$ ; strict inequality  $\alpha_{\text{irr}} > g$  occurs exactly when every mode at real-part gap  $g$  contributes zero to  $A_t$ .*

**Significance.** A generic eigenmode has no reason to be annihilated by the antisymmetrization  $M \mapsto \Pi M - M^\top \Pi$ , so vanishing imposes analytic constraints on the rates; the conjecture makes this precise and names a new phenomenon, a slowest state-space mode that is thermodynamically invisible, so long-lag irreversibility can serve as a spectroscopy distinguishing dissipative from merely slow modes. The nearest related results are [212, 213]. A first decisive theorem would be genericity for a fixed three-state driven graph. In tests on the corpus described in the stress-test section, a designed product of a slow reversible pair and a fast driven three-cycle realized strict  $\alpha_{\text{irr}} > g$  with the KL tail controlled by the driven factor, confirming the structured exception. A counterexample to genericity would be a positive-measure set of rates with  $\alpha_{\text{irr}} > g$ .

**Conjecture 325** (Coarse-grained irreversibility mode-selection law, resolved true). *For every deterministic partition with  $L_f \not\equiv 0$ ,*

$$\liminf_{t \rightarrow \infty} -\frac{1}{2t} \log L_f(t) = \alpha_{\text{irr}}(f).$$

**Significance.** The partition acts linearly at the probability level and can cancel whole modal coefficients before KL makes the observable quadratic, so coarse-graining does not continuously renormalize the late-time exponent but selects the slowest irreversible eigenmode the observation map fails to annihilate, predicting a discrete hierarchy of observable exponents rather than a continuum. The nearest related results are the coarse-grained dissipation inferences of [308, 309, 310]. A first decisive theorem would be the two-block partition of a driven ring. In tests on the corpus described in the stress-test section, coarse observation of a driven-ring product selected the faster irreversible mode as predicted. A counterexample would be a partition whose coarse lag KL decays at a rate absent from the selected modal set. *Resolution.* Now proved, Proposition 61.

**Proposition 61** (Irreversibility-gap doubling and coarse-grained mode selection). Let  $Q$  be an irreducible bidirected generator with stationary  $\Pi$ , endpoint law  $F_t = \Pi e^{tQ}$ , and antisymmetric part  $A_t = F_t - F_t^\top \not\equiv 0$ . For all large  $t$  there are constants  $0 < c < C$  with  $c\|A_t\|_F^2 \leq L(t) \leq C\|A_t\|_F^2$ ;  $A_t$  is a finite exponential polynomial  $\sum_k t^k e^{\lambda t} B_{\lambda,k}$ ; and with  $\alpha_{\text{irr}} = \min\{-\Re \lambda : \text{the term survives in } A_t\}$  and  $r$  the largest polynomial degree among surviving terms of real part  $-\alpha_{\text{irr}}$ ,

$$\liminf_{t \rightarrow \infty} -\frac{1}{2t} \log L(t) = \alpha_{\text{irr}}, \quad L(t) = t^{2r} e^{-2\alpha_{\text{irr}} t} [c(t) + o(1)],$$

with  $c$  a nonnegative bounded trigonometric polynomial, not identically zero. For every deterministic partition the same holds with  $L_f$  and  $\alpha_{\text{irr}}(f)$  in place of  $L$  and  $\alpha_{\text{irr}}$  whenever  $L_f \not\equiv 0$ . This proves Conjectures 323 and 325; the refined power is  $m = 2r$ , twice the Jordan-polynomial degree, and holds with no genericity hypothesis.

*Proof.* Grouping the two orderings,  $L(t) = \sum_{i < j} (F_{ij} - F_{ji}) \log(F_{ij}/F_{ji})$ , and the mean-value theorem writes each term as  $(F_{ij} - F_{ji})^2 / \xi_{ij}$  with  $\xi_{ij}$  between  $F_{ij}$  and  $F_{ji}$ . Since  $F_{ij}, F_{ji} \rightarrow \pi_i \pi_j > 0$ , the  $\xi_{ij}$  are bounded above and below for large  $t$ , giving  $c\|A_t\|_F^2 \leq L(t) \leq C\|A_t\|_F^2$ . The Jordan expansion  $e^{tQ} = \sum_k t^k e^{\lambda t} N_{\lambda,k}$  makes  $A_t$  a finite exponential polynomial; dividing by  $t^r e^{-\alpha_{\text{irr}} t}$  leaves a nonzero matrix trigonometric polynomial  $T(t)$  plus  $o(1)$ , whose Frobenius norm has positive limsup, so  $\liminf(-t^{-1} \log \|A_t\|_F) = \alpha_{\text{irr}}$  and  $L(t) = t^{2r} e^{-2\alpha_{\text{irr}} t} [c(t) + o(1)]$  with  $c(t) = \sum_{i < j} T_{ij}(t)^2 / (\pi_i \pi_j)$ . Block summation sends  $F_t$  to  $F_t^{(f)}$ , a linear image whose entries tend to  $\pi_A \pi_B > 0$ , so the identical argument applies to  $L_f$  with the block-summed flux  $A_t^{(f)}$ , giving  $\alpha_{\text{irr}}(f)$  whenever  $L_f \not\equiv 0$ .  $\square$

**Conjecture 326** (Hidden-mode exponential separation, resolved false). *If  $\alpha_{\text{irr}}(f) > \alpha_{\text{irr}}$ , then  $\liminf_{t \rightarrow \infty} -\frac{1}{2t} \log[L_f(t)/L(t)] = \alpha_{\text{irr}}(f) - \alpha_{\text{irr}}$ ; if the two exponents coincide with nonzero leading coefficients,  $L_f(t)/L(t)$  approaches a nonzero bounded envelope.*

**Significance.** This is the operational consequence of Conjectures 323 and 325: when a partition hides the slowest dissipative mode, the observed and full arrows of time separate at exactly the gap between the two irreversibility exponents, turning the mode-selection law into a measurable ratio. The nearest related results are [306, 309]. A first decisive theorem would follow from Conjectures 323 and 325 with control of the leading coefficients. A counterexample would be a bounded ratio despite distinct irreversibility exponents. *Resolution.* The conjecture is false: Proposition 62 exhibits a product of two driven three-cycles for which the coarse-to-full ratio  $L_f/L$  returns to one at infinitely many times, so the liminf separation exponent is zero rather than  $\alpha_{\text{irr}}(f) - \alpha_{\text{irr}}$ . The obstruction is that the nominally dominant slow mode is oscillatory with exact zeros; a repaired statement must use a windowed supremum, a time-averaged logarithm, or nonoscillatory leading modes.

**Proposition 62** (Failure of hidden-mode exponential separation). There is a product of two independent homogeneous driven three-cycles, a slow factor with  $(p_s, q_s) = (1, 0.01)$  and a fast factor with  $p_f + q_f = \frac{16}{15}$  and  $p_f - q_f = 0.99\sqrt{11/10}$ , for which the coarse (fast-factor) exponent  $\alpha_{\text{irr}}(f) = 1.6$  exceeds the full exponent  $\alpha_{\text{irr}} = 1.515$ , yet  $\liminf_{t \rightarrow \infty} -\frac{1}{2t} \log[L_f(t)/L(t)] = 0$ . Hence Conjecture 326 is false.

*Proof.* For a homogeneous driven three-cycle the nonzero generator eigenvalues are  $-\alpha \pm i\omega$  with  $\alpha = \frac{3}{2}(p + q)$  and  $\omega = \frac{\sqrt{3}}{2}(p - q)$ , so the antisymmetric endpoint flux is proportional to  $e^{-\alpha t} \sin(\omega t)$  and the lag irreversibility  $L(t) \asymp e^{-2\alpha t} \sin^2(\omega t)$  vanishes exactly at  $t = n\pi/\omega$ . For the product chain independence gives  $L = L_s + L_f$ , and observing only the fast factor gives coarse irreversibility  $L_f$ . At  $t_n = n\pi/\omega_s$  the slow factor's flux vanishes,  $L_s(t_n) = 0$ , while  $L_f(t_n) > 0$  because  $\omega_f/\omega_s = \sqrt{11/10}$  is irrational, so  $L_f(t_n)/L(t_n) = 1$  and  $-\frac{1}{2t_n} \log[L_f/L] = 0$ . Since  $t_n \rightarrow \infty$ , the liminf is zero, not the gap  $\alpha_{\text{irr}}(f) - \alpha_{\text{irr}} = 0.085$ ; the values are reproduced in the verification script. The defect is intrinsic: a liminf of a ratio of oscillatory quantities is destroyed by the exact zeros of the nominally dominant mode.  $\square$

## 80 Unicyclic networks and winding aliasing

**Conjecture 327** (Unicyclic strict monotonicity, resolved false). *For every finite bidirected unicyclic chain with nonzero total cycle affinity,  $\eta'(t) < 0$  for all  $t > 0$ .*

**Significance.** A single cycle carries one thermodynamic affinity, so no two affinity-bearing loops can trade dominance as the lag changes, and the only loss mechanism is temporal winding ambiguity, suggesting a total ordering in  $t$  stronger than the general Conjecture 320. A proof route uses the scalar cycle-current representation and total-positivity properties of birth–death lifts on the universal cover. The nearest related results are the finite-lag affinity bounds of [305]. A first decisive theorem would have been strictness for the homogeneous ring. A counterexample would be a ring with a genuine rising efficiency interval. *Resolution.* The conjecture is false on the homogeneous ring itself: Proposition 63 exhibits a bidirected 10-cycle whose efficiency rises by more than two orders of magnitude across an interior lag interval, confirmed at eighty digits,



so the apparent increments are genuine and not the near-decomposable cancellation artifacts once supposed.

**Proposition 63** (A homogeneous-ring counterexample to efficiency monotonicity). On the homogeneous continuous-time  $N$ -cycle with clockwise rate  $p = e^{a/2}$ , counterclockwise rate  $q = e^{-a/2}$ , uniform stationary law, and  $\sigma = (p - q)a$ , the efficiency  $\eta(t) = L(t)/(\sigma t)$  is not monotone. For  $N = 10$  and  $a = 1$ , writing  $u = (p - q)t/N$ , an independent eighty-digit Fourier diagonalization of the circulant generator gives  $\eta = 9.31 \times 10^{-6}$  at  $u = 0.54$ ,  $1.58 \times 10^{-4}$  at  $u = 0.56$ ,  $6.65 \times 10^{-4}$  at  $u = 0.60$ , and  $1.22 \times 10^{-3}$  at  $u = 0.68$ , an increase by a factor above 130. Hence Conjectures 320 and 327 both fail, the latter on the simplest possible one-cycle support.

*Proof.* The endpoint law is circulant,  $P_t(k) = \frac{1}{N} \sum_m e^{\lambda_m t} \omega^{-mk}$  with  $\omega = e^{2\pi i/N}$  and  $\lambda_m = p(\omega^m - 1) + q(\omega^{-m} - 1)$ , and  $L(t) = \sum_k P_t(k) \log(P_t(k)/P_t(-k))$ ; the displayed values follow from these exact formulas at eighty digits and are reproduced in the verification script. The mechanism is winding aliasing: near half a winding the forward and reverse wrapped endpoint means nearly coincide, collapsing  $\eta$ , then separate again before diffusion mixes the cycle, so  $\eta$  revives. The revival is far above any conditioning artifact. The first-crossing horizons of Conjectures 330 and 331 are unaffected, reflecting the same half-winding collapse.  $\square$

**Conjecture 328** (Sharp unicyclic spectral-amplitude cap, resolved false). *For every nonequilibrium unicyclic chain,  $\sup_{t>0} \frac{\gamma L(t)}{\sigma} \leq \frac{1}{2}$ .*

**Significance.** Since  $\gamma L/\sigma = (\gamma t)\eta(t)$ , this asks for the largest time-rescaled endpoint irreversibility before mixing erases directionality, an inequality that would combine one-cycle topology, the reversible gap, and entropy production with no activity or minimum-probability prefactor. The nearest related results are [305, 212]. A first decisive theorem would have been the homogeneous ring, where the ratio stays near 0.2. In tests on the corpus described in the stress-test section, well-mixed rings kept the ratio below 0.44, and the deposited constant  $\frac{1}{2}$  was proposed as potentially sharp. *Resolution.* The conjecture is false: Proposition 64 exhibits an explicit near-decomposable seven-state ring with  $\gamma L/\sigma \geq 0.643 > \frac{1}{2}$ , verified to fifty digits, so no universal cap at  $\frac{1}{2}$  holds; the correct sharp constant, if finite, remains open, and a near-decomposable regime is where it is approached.

**Proposition 64** (A near-decomposable amplitude-cap counterexample). There is a bidirected seven-state ring, with one weak edge three orders of magnitude below the others, whose reversible gap  $\gamma$ , entropy-production rate  $\sigma$ , and maximal lag irreversibility satisfy  $\gamma \sup_t L(t)/\sigma = 0.643$  to three places. Hence the sharp cap  $\frac{1}{2}$  of Conjecture 328 fails.

*Proof.* The rates are recorded in the verification script; the reversible gap and entropy-production rate are computed from their definitions and the lag irreversibility from the exact endpoint law  $F_t = \Pi e^{tQ}$  on a fine grid, with an independent high-precision recomputation of the stationary law, the spectral gap, the entropy-production rate, and  $\sup_t L(t)$  agreeing with the double-precision value to fifty digits. The resulting ratio is  $0.643 > \frac{1}{2}$ . The gap  $\gamma \approx 0.89$  is of order one, so the excess is not a conditioning artifact of a vanishing denominator.  $\square$

**Conjecture 329** (Wrapped-Brownian irreversibility limit, resolved true). *For each fixed  $A \neq 0$  and  $\tau > 0$ ,*

$$\eta_N(t_N) \longrightarrow \mathcal{E}(A, \tau) := \frac{2}{A^2 \tau} \int_0^1 w_{A, \tau}(x) \log \frac{w_{A, \tau}(x)}{w_{-A, \tau}(x)} dx,$$



locally uniformly on compacta of  $(A, \tau) \in (\mathbb{R} \setminus \{0\}) \times (0, \infty)$ .

**Significance.** Under the weak-affinity diffusive scaling the net displacement before wrapping is a Skellam difference converging to Brownian motion on the circle with drift  $A/2$  and unit variance rate, the full entropy production tends to  $A^2\tau/2$ , and the wrapped KL gives the stated normalized limit, an exact continuum law for endpoint irreversibility on a driven ring. A proof route proves a local central limit theorem for the wrapped Skellam displacement and controls the KL by density lower bounds. The nearest related results are the circulant mixing-time moments of [311]. A first decisive theorem would be the limit at a single  $(A, \tau)$ . In tests on the corpus described in the stress-test section, rings up to  $N = 128$  matched  $\mathcal{E}(A, \tau)$  to within  $5 \times 10^{-5}$ . A counterexample would be a scaling point where  $\eta_N(t_N)$  misses the wrapped-Brownian value. *Resolution.* Now proved, Proposition 65.

**Proposition 65** (Wrapped-Brownian irreversibility limit). On the homogeneous  $N$ -cycle with  $p_N = e^{A/(2N)}$ ,  $q_N = e^{-A/(2N)}$ , and  $t_N = N^2\tau/(p_N + q_N)$ , the efficiency satisfies  $\eta_N(t_N) \rightarrow \mathcal{E}(A, \tau)$  locally uniformly on compacta of  $(A, \tau) \in (\mathbb{R} \setminus \{0\}) \times (0, \infty)$ . This proves Conjecture 329.

*Proof.* The endpoint law is the wrapped difference of two independent Poisson counts; its mode- $m$  Fourier coefficient  $\exp(t_N[p_N(\omega^m - 1) + q_N(\omega^{-m} - 1)])$  converges to  $e^{-2\pi^2 m^2 \tau + i\pi m A \tau}$ , so by Gaussian Fourier-tail control the wrapped lattice density converges uniformly to the wrapped Gaussian  $w_{A, \tau}$ , which is bounded below. Continuity of the Kullback–Leibler integrand and  $\sigma_N t_N = (p_N - q_N)(A/N)t_N \rightarrow A^2\tau/2$  give  $\eta_N(t_N) \rightarrow \mathcal{E}(A, \tau)$ , with local uniformity inherited from the uniform Fourier convergence.  $\square$

**Conjecture 330** (Ballistic winding-aliasing horizon). For fixed edge affinity  $a \neq 0$  and level  $c \in (0, 1)$  there is a finite positive  $u_c(a)$  with  $|v|t_c(N, a)/N \rightarrow u_c(a)$  as  $N \rightarrow \infty$ , so fixed-bias endpoint irreversibility is lost on the ballistic winding timescale  $N/|v|$ , not the diffusive mixing timescale.

**Significance.** The path entropy production is proportional to the signed winding current, which an endpoint reveals only modulo  $N$ ; at fixed nonzero drift the uncertainty about the integer winding number becomes macroscopic once the mean displacement is of order  $N$ , producing a ballistic information horizon even though positional mixing is diffusive. A proof route compares competing winding sectors by Laplace asymptotics. The nearest related results are [305, 311]. A first decisive theorem would be the horizon for  $c = \frac{1}{2}$  on the homogeneous ring. In tests on the corpus described in the stress-test section, half-efficiency times up to  $N = 192$  scaled linearly in  $N$  with slope near one, and  $|v|t_{1/2}/N$  approached a finite constant. A counterexample would be a diffusive  $N^2$  scaling of the fixed-bias horizon.

**Conjecture 331** (Diffusive-to-ballistic irreversibility crossover, resolved false). For every fixed  $c \in (0, 1)$  there is a universal function  $\Phi_c$  with  $t_c = \frac{N^2}{p+q}\Phi_c(\text{Pe})[1+o(1)]$  in the joint large- $N$  scaling, where  $\Phi_c(z) \rightarrow \Phi_c(0) \in (0, \infty)$  as  $z \rightarrow 0$  and  $\Phi_c(z) \sim u_c/|z|$  as  $|z| \rightarrow \infty$ , so Conjectures 329 and 330 are its weak- and strong-drift limits.

**Significance.** The Péclet number is the single dimensionless group controlling the transition from diffusive to ballistic loss of winding information, and the conjecture asserts that the half-efficiency time collapses onto one crossover curve interpolating the two proved-limit regimes. A proof route matches the wrapped-Brownian and ballistic asymptotics through a uniform

winding-sector expansion. The nearest related results are [311]. A first decisive theorem would be the collapse at small Péclet. In tests on the corpus described in the stress-test section, at fixed affinity the scaled half-efficiency time across  $N = 64, 128, 192$  collapsed with a cross- $N$  spread below one percent. A counterexample would be a Péclet value where the scaled times fail to collapse. *Resolution.* The conjecture is false as stated: Proposition 66 shows the ballistic first-crossing coefficient  $u_c(a)$  of Conjecture 330 depends on the edge affinity  $a$ , so no single universal  $\Phi_c(\text{Pe})$  can have Conjecture 330 as its strong-drift limit for every fixed affinity. A genuine one-variable Péclet crossover survives in the weak-edge-bias regime  $a_N \rightarrow 0$ , which is the intended restriction.

**Proposition 66** (Affinity-dependence of the ballistic crossover coefficient). The ballistic first-crossing coefficient  $u_c(a)$  of Conjecture 330 depends nontrivially on the edge affinity  $a$ , so no single universal  $\Phi_c(\text{Pe})$  with  $\Phi_c(z) \sim u_c/|z|$  recovers Conjecture 330 for every fixed affinity. With the Skellam rate function  $I_{u,a}(d) = \frac{u}{v} J_{p,q}(dv/u)$ ,  $p = e^{a/2}$ ,  $q = e^{-a/2}$ ,  $v = p - q$ , the level- $\frac{1}{2}$  coefficient solving  $\min\{1, I_{u,a}(1-u)/(au)\} = \frac{1}{2}$  equals 0.2923, 0.2894, 0.2806, 0.2576, 0.2354 at  $a = 0.2, 0.5, 1, 2, 3$ , with  $u_{1/2}(a) \rightarrow \frac{1}{2(1+\sqrt{1/2})} = 0.2929$  as  $a \downarrow 0$  and  $u_{1/2}(a) \rightarrow 0$  as  $a \rightarrow \infty$ . Hence Conjecture 331 is false as a universal crossover covering every fixed affinity.

*Proof.* Since  $\text{Pe} \asymp |v|N/(p+q)$ , fixing  $a$  and letting  $N \rightarrow \infty$  drives  $\text{Pe} \rightarrow \infty$ , and matching  $t_c = \frac{N^2}{p+q} \Phi_c(\text{Pe})$  to the ballistic law  $t_c \sim u_c(a)N/|v|$  of Conjecture 330 forces the tail constant  $u_c$  in  $\Phi_c(z) \sim u_c/|z|$  to equal  $u_c(a)$  for every  $a$ . The fluctuation relation gives  $I_{u,a}(-u) = au$ , so  $\eta_\infty(u, a) = \min\{1, I_{u,a}(1-u)/(au)\}$ ; as  $a \downarrow 0$  the Skellam rate has the Gaussian form  $I_{u,a}(1-u) = a(1-2u)^2/(4u) + o(a)$ , giving  $u_c(a) \rightarrow 1/(2(1+\sqrt{c}))$ , while as  $a \rightarrow \infty$  backward jumps vanish and  $I_{u,a}(1-u)$  tends to the finite Poisson cost  $(1-u) \log \frac{1-u}{u} - 1 + 2u$  against the linear denominator  $au$ , forcing  $\eta_\infty \rightarrow 0$  and  $u_c(a) \rightarrow 0$ . The tabulated values are reproduced in the verification script. A single  $\Phi_c$  therefore cannot hold across fixed affinities; restricting to the weak-edge-bias regime  $a_N \rightarrow 0$  with  $\text{Pe}$  finite or  $o(N)$  restores a genuine one-variable Péclet crossover.  $\square$

## 81 Nonequilibrium Ornstein–Uhlenbeck dynamics

**Conjecture 332** (OU one-relaxation-time peak horizon, resolved false). If  $\sigma_{\text{OU}} > 0$  and  $t_*$  is a global maximizer of  $L_{\text{OU}}(t)$ , then  $g_A t_* \leq 1$ .

**Significance.** The Gaussian lag KL rises from zero as infinitesimal trajectory irreversibility becomes resolvable and falls as cross-time covariance vanishes, and the slowest drift timescale appears to cap the location of the dominant peak even under strong nonnormal transient amplification, a sharper Gaussian analogue of Conjecture 321. A proof route expresses the Gaussian KL through canonical correlations of the two-time covariance. The nearest related results are the OU dissipation decompositions of [213] and the signal-response mapping irreversibility of [307]. A first decisive theorem would be the normal (commuting) drift case. In tests on the corpus described in the stress-test section, the maximum  $g_A t_*$  over stable systems in dimensions two to five, including nonnormal drifts, was 0.774. A counterexample would be a stable OU process whose lag-KL peak lies beyond one drift-relaxation time. *Resolution.* The bound is false: Proposition 67 exhibits a five-dimensional nonnormal drift, a single Jordan block with every eigenvalue  $-1$  (so  $g_A = 1$ ) and isotropic diffusion, whose lag-KL peak lies at  $t_* = 1.069 \dots$ , giving  $g_A t_* = 1.069 > 1$ ; the earlier stress test sampled only drifts with negative-definite symmetric

part and so missed the strong nonnormal transient amplification that pushes the horizon past one drift-relaxation time, and a repaired bound must control nonnormality (a numerical-abscissa or hypocoercive constant), not the spectral abscissa alone.

**Conjecture 333** (OU spectral-area bound, resolved false). *For every stable nonequilibrium OU process,  $\int_0^\infty L_{\text{OU}}(t) dt \leq \frac{\sigma_{\text{OU}}}{g_A^2}$ .*

**Significance.** This is the Gaussian analogue of Conjecture 322 with a sharper constant, Gaussianity removing the rare-state KL amplification that forces the factor-of-two safety margin for arbitrary chains. A proof route uses Lyapunov/operator inequalities for the integrated two-time covariance. The nearest related results are [213, 307]. A first decisive theorem would be the scalar and normal-drift cases. In tests on the corpus described in the stress-test section, the maximum normalized area  $g_A^2 \int L_{\text{OU}} dt / \sigma_{\text{OU}}$  was 0.553, including strongly nonnormal drifts. A counterexample would be a stable OU process exceeding the budget. *Resolution.* The bound is false: for the same five-dimensional Jordan-block drift of Proposition 67,  $g_A = 1$  and  $\sigma_{\text{OU}} = 32$  while  $\int_0^\infty L_{\text{OU}} dt = 40.50\dots$ , so the normalized area is  $1.266 > 1$ ; as with Conjecture 332, only a nonnormality-corrected budget can hold.

**Proposition 67** (A nonnormal Ornstein–Uhlenbeck counterexample to the peak-horizon and spectral-area bounds). Let  $A = -I_5 + 2N$  with  $N$  the  $5 \times 5$  nilpotent first-superdiagonal shift and  $D = I_5$ , a Hurwitz drift with every eigenvalue  $-1$ , so  $g_A = 1$ . The stationary covariance solving  $AC + CA^\top + 2D = 0$  has integer entries, the entropy-production rate is  $\sigma_{\text{OU}} = 32$ , and the Gaussian lag KL  $L_{\text{OU}}$  of Proposition 60 attains its global maximum at  $t_* = 1.0693\dots$ , so  $g_A t_* = 1.0693\dots > 1$ , and has total area  $\int_0^\infty L_{\text{OU}} dt = 40.497\dots$ , so  $g_A^2 \int L_{\text{OU}} dt / \sigma_{\text{OU}} = 1.2655\dots > 1$ . Hence Conjectures 332 and 333 both fail: the spectral abscissa  $g_A$  alone bounds neither the peak location nor the integrated budget of the Gaussian arrow of time.

*Proof.* The covariance  $C$  is obtained by solving the Lyapunov equation; substitution confirms  $AC + CA^\top + 2I_5 = 0$  with zero residual, and  $\sigma_{\text{OU}} = \text{tr}[(A + DC^{-1})^\top D^{-1}(A + DC^{-1})C] = 32$ . Using the Jordan-polynomial matrix exponential  $e^{At}$  and the trace formula of Proposition 60,  $L_{\text{OU}}$  rises from  $L_{\text{OU}}(t) = \sigma_{\text{OU}} t + O(t^2)$ , is still increasing at  $t = 1$  ( $L'_{\text{OU}}(1) \approx 1.60 > 0$ ), peaks at  $t_* = 1.0693\dots$ , and has integral  $40.497\dots$ ; all quantities are reproduced in the verification script, and the margins (6.9% and 26.5%) far exceed any conditioning artifact.  $\square$

**Conjecture 334** (OU antisymmetric-covariance doubling law, resolved true). *For every stable OU process with  $\sigma_{\text{OU}} > 0$ ,*

$$\liminf_{t \rightarrow \infty} -\frac{1}{2t} \log L_{\text{OU}}(t) = \alpha_H,$$

*so the long-time Gaussian arrow of time decays at twice the slowest time-antisymmetric covariance mode.*

**Significance.** At long times  $\Sigma_t$  approaches  $C \oplus C$ , its difference from the time-reversed covariance is linear in  $H(t)$ , and Gaussian KL between nearby covariances is quadratic in that difference, so the exponent doubles, the Gaussian counterpart of Conjecture 323. A proof route is a matrix Taylor expansion of the Gaussian KL around  $C \oplus C$ . The nearest related results are [213]. A first decisive theorem would be the normal-drift case, where  $H$  has an explicit modal form.

In tests on the corpus described in the stress-test section, the ratio of the lag-KL tail exponent to twice the  $H(t)$  tail exponent had median 1.0000 over stable systems in dimensions two to five. A counterexample would be a decay rate other than twice an antisymmetric-covariance rate. *Resolution.* Now proved, Proposition 68.

**Proposition 68** (Gaussian antisymmetric-covariance doubling). For a stable Ornstein–Uhlenbeck process with stationary covariance  $C \succ 0$  and  $\sigma_{OU} > 0$ , the two-time covariance tends to  $C \oplus C \succ 0$ , its time-antisymmetric part is  $H(t) = e^{At}C - Ce^{A^\top t} \neq 0$ , and for all large  $t$  there are  $0 < c < C'$  with  $c\|H(t)\|_F^2 \leq L_{OU}(t) \leq C'\|H(t)\|_F^2$ . Since  $H$  is a matrix exponential polynomial with slowest surviving rate  $\alpha_H$ ,  $\liminf_{t \rightarrow \infty} -\frac{1}{2t} \log L_{OU}(t) = \alpha_H$ . This proves Conjecture 334.

*Proof.* The Gaussian Kullback–Leibler divergence between the two-time law and its time reversal is a smooth function of the covariance vanishing to second order at the positive-definite limit  $C \oplus C$ , hence is bounded above and below by  $\|H(t)\|_F^2$  near the limit;  $H(t) = e^{At}C - Ce^{A^\top t}$  is an exponential polynomial, and the  $\liminf$  argument of Proposition 61 gives the exponent  $\alpha_H$ .  $\square$

## Part XX

# Log-concavity laws for exactly solved random conformal geometry

*Added to the deposit: 22 August 2026.*

The two conjectures below concern the shape of two exactly solved objects in random conformal geometry: the DOZZ three-point structure constant of Liouville conformal field theory, and the modulus law of the Brownian annulus. Each asserts strict log-concavity of an exact special-function density along its natural spectral or scale coordinate. Such a law forces one preferred value, forbids secondary shoulders or competing saddles, and turns qualitative bootstrap and shape questions into convex-analysis problems, with globally controlled saddle-point approximations and certified one-dimensional quadrature as corollaries. Both objects are given by closed forms—the Barnes-function representation of the DOZZ constant [313] and the exact modulus transform of the Brownian annulus [314]—so both conjectures are exactly and finitely checkable on any grid of parameters, in contrast with continuum-limit shape statements that admit only finite-size surrogates. The proved layer is Proposition 69: the reduction of each curvature to the part of the closed form that carries its sign, the vanishing of the DOZZ modulus at zero momentum, and the boundary-swap symmetry of the annulus modulus. The nearest precedents are the DOZZ formula of Kupiainen, Rhodes, and Vargas [313] and the annulus modulus law of Ang, Remy, and Sun [314]; the strict log-concavity in the internal Liouville momentum and in the logarithmic annulus modulus, together with the charge chamber that makes the first hold, is what is proposed as new. Both statements are checkable on grids but not proved, and a rigorous proof requires continuous rather than gridwise control of the second derivative; the interior charge chamber of the first is necessary, its two boundaries each carrying an explicit sign reversal recorded below. Since deposit, Conjecture 336, the annulus modular log-concavity, has been resolved false by Proposition 70: at large boundary-length ratio the logarithmic-modulus curvature turns positive, so log-concavity fails in the tail, while the DOZZ law of Conjecture 335 stands.

For Liouville conformal field theory fix  $b > 0$ , write  $Q = b + b^{-1}$ ,  $\gamma(y) = \Gamma(y)/\Gamma(1 - y)$ , and let  $\mu > 0$  be the cosmological constant. The Barnes function  $\Upsilon_b$  is defined for  $0 < \Re x < Q$  by

$$\log \Upsilon_b(x) = \int_0^\infty \frac{dt}{t} \left[ \left( \frac{Q}{2} - x \right)^2 e^{-t} - \frac{\sinh^2((\frac{Q}{2} - x)t/2)}{\sinh(bt/2) \sinh(t/2b)} \right],$$

and extends to an entire function. The DOZZ structure constant for real charges  $\alpha_1, \alpha_2, \alpha_3$  with  $\bar{\alpha} = \alpha_1 + \alpha_2 + \alpha_3$  is

$$C_b(\alpha_1, \alpha_2, \alpha_3) = [\pi\mu\gamma(b^2)b^{2-2b^2}]^{(Q-\bar{\alpha})/b} \frac{\Upsilon'_b(0) \prod_{i=1}^3 \Upsilon_b(2\alpha_i)}{\Upsilon_b(\bar{\alpha} - Q) \prod_{i=1}^3 \Upsilon_b(\bar{\alpha} - 2\alpha_i)},$$

and the principal series is parameterized by internal momentum  $P \geq 0$  through  $\alpha = \frac{Q}{2} + iP$ . For an equal external charge  $a$  write  $M_a(P) = |C_b(a, a, \frac{Q}{2} + iP)|$  for the symmetric spectral modulus. For the Brownian annulus, following Ang, Remy, and Sun [314], let  $\eta$  be the Dedekind eta function, and conditional on the boundary-length ratio  $x > 0$  let  $X_\tau$  be the positive random variable with Mellin transform

$$\mathbb{E}[X_\tau^{it}] = \frac{2\pi t}{3 \sinh(2\pi t/3)} e^{-2\pi\tau t^2/3}, \quad t \in \mathbb{R},$$

and density  $\rho_\tau$ ; the modulus  $\tau > 0$  then has density  $p_x(\tau) \propto \eta(2i\tau) \rho_\tau(x)$ , the constant of proportionality depending only on  $x$ .

**Proposition 69** (Curvature reductions and boundary symmetry).  $\Upsilon_b$  is entire with simple zeros exactly at  $-b\mathbb{N}_0 - b^{-1}\mathbb{N}_0$  and at  $Q + b\mathbb{N}_0 + b^{-1}\mathbb{N}_0$ ; in particular  $\Upsilon_b(Q) = 0$ , so  $M_a(0) = 0$  and the log-concavity of  $M_a$  is a statement on  $P > 0$ . The prefactor modulus and the momentum-independent factor  $|\Upsilon_b(2a)|$  are constant in  $P$ , so

$$\frac{d^2}{dP^2} \log M_a(P) = \frac{d^2}{dP^2} \left[ \log |\Upsilon_b(Q + 2iP)| - 2 \log |\Upsilon_b(\frac{Q}{2} + iP)| - 2 \log |\Upsilon_b(2a - \frac{Q}{2} + iP)| \right].$$

For the Brownian annulus,  $\log p_x(e^s) = \log \eta(2ie^s) + \log \rho_{e^s}(x) + \kappa(x)$  with  $\kappa$  independent of  $s$ , so  $d^2 \log p_x(e^s)/ds^2$  is independent of the normalization, and  $p_{1/x}(\tau) = x^2 p_x(\tau)$ , so that curvature is invariant under  $x \mapsto 1/x$ .

*Proof.* The standard shift relations  $\Upsilon_b(x+b) = \gamma(bx)b^{1-2bx}\Upsilon_b(x)$  and  $\Upsilon_b(x+b^{-1}) = \gamma(x/b)b^{2x/b-1}\Upsilon_b(x)$  extend  $\Upsilon_b$  to an entire function with the stated zeros;  $Q = Q + 0 \cdot b + 0 \cdot b^{-1}$  is one of them, so  $\Upsilon_b(Q) = 0$  and the numerator factor  $\Upsilon_b(2\alpha_3) = \Upsilon_b(Q + 2iP)$  vanishes at  $P = 0$ , giving  $M_a(0) = 0$ . In  $C_b(a, a, \frac{Q}{2} + iP)$  the base  $\pi\mu\gamma(b^2)b^{2-2b^2}$  is real and positive for  $0 < b^2 < 1$ , so the prefactor modulus is that base raised to  $\Re[(Q - \bar{\alpha})/b] = (Q/2 - 2a)/b$ , independent of  $P$ ; for general  $b > 0$  the base or a self-duality branch factor may enter, but any such factor has modulus exponential-linear in  $P$  and is therefore annihilated by the second  $P$ -derivative, so the curvature identity holds for every  $b > 0$ ; the factors  $\Upsilon_b(2a)$  are real and  $P$ -independent, and  $\Upsilon_b(\bar{\alpha} - 2\alpha_3) = \Upsilon_b(2a - \frac{Q}{2} - iP)$  is the complex conjugate of  $\Upsilon_b(2a - \frac{Q}{2} + iP)$ , so has the same modulus; collecting the surviving factors gives the displayed identity. For the annulus,  $\eta(2i\tau)$  is real and positive for  $\tau > 0$ , the proportionality constant is  $s$ -independent, and  $\rho_\tau(x) = g_\tau(\log x)/x$  with  $g_\tau$  the even density of  $\log X_\tau$ , whence  $\rho_\tau(1/x) = x^2 \rho_\tau(x)$  and the swap invariance follows.  $\square$

## 82 Spectral and modular log-concavity

**Conjecture 335** (DOZZ spectral log-concavity). *For every  $b > 0$  and every real external charge  $a$  with  $\frac{Q}{3} \leq a < \frac{Q}{2}$ , the symmetric DOZZ spectral modulus is strictly log-concave in the internal momentum:*

$$\frac{d^2}{dP^2} \log |C_b(a, a, \frac{Q}{2} + iP)| < 0 \quad \text{for all } P > 0.$$

**Significance.** The principal-series decomposition of a symmetric four-point Liouville amplitude weights the internal momentum by  $M_a(P)^2$  times a conformal block, so strict log-concavity of  $M_a$  gives the spectral density a single mode and forbids competing saddles, the structure-constant half of a total-positivity picture for Liouville data. The leading operator-product factor already carries the sign; the conjectural content is that the Barnes-function descendant structure never overturns it. The chamber  $\frac{Q}{3} \leq a < \frac{Q}{2}$  is necessary at both ends: below  $\frac{Q}{3}$  the spectral curvature turns positive, already at  $b = 1$ ,  $a = 0.55$ , and the joint log-concavity of  $(\alpha_1, \alpha_2, \alpha_3) \mapsto \log |C_b|$  fails even at the symmetric point, a positive Hessian eigenvalue appearing at  $(1.3, 1.3, 1.3)$ , so only the one-dimensional spectral statement inside the interior chamber survives. A proof route turns the second momentum derivative of the integral representation of  $\log \Upsilon_b$  into a sign inequality for a hyperbolic-kernel integral. The nearest related result is the DOZZ formula of Kupiainen, Rhodes, and Vargas [313]. A first decisive theorem would be the self-dual point  $b = 1$ , where  $\Upsilon_1$  has an elementary form. In tests on the corpus described in the stress-test section, an independent evaluation of the exact  $\Upsilon_b$  integral over  $b \in \{0.5, 0.8, 1, 1.3\}$  and the full chamber gave strictly negative spectral curvature throughout, with the two boundary controls turning positive as required. A counterexample would be one regular point inside the chamber with nonnegative spectral curvature.

**Conjecture 336** (Brownian-annulus modular log-concavity, resolved false). *For the Brownian annulus and every boundary-length ratio  $x > 0$ , the modulus density is log-concave in the logarithmic modulus:*

$$s \mapsto \log p_x(e^s) \quad \text{is strictly concave on } \mathbb{R}.$$

**Significance.** The modulus of a random annulus is the one conformal invariant of its shape, and the conjecture asserts that after passing to the multiplicative coordinate  $s = \log \tau$  its law has a single preferred scale, with no secondary shoulder at very thin or very thick annuli. The density is the product of a Dedekind-eta conformal-ghost factor and a heat-smoothed log-ratio density, and the content is that this product preserves concavity in logarithmic time, which neither factor forces separately. A proof route combines tail control as  $\tau \rightarrow 0, \infty$  with a sign computation for the second logarithmic derivative of the exact modulus transform. The nearest related result is the annulus modulus law of Ang, Remy, and Sun [314]. A first decisive theorem would be the symmetric annulus  $x = 1$ . In tests on the corpus described in the stress-test section, an independent Fourier inversion of the exact transform, multiplied by the exact eta factor, gave strictly negative logarithmic-modulus curvature at every moderate boundary ratio tested, with the  $x \mapsto 1/x$  symmetry reproduced. A counterexample would be a boundary ratio and a modulus at which the logarithmic density is locally convex. *Resolution.* The conjecture is false at large boundary ratio. Writing  $\log X_\tau = Z + G_\tau$  with  $Z$  logistic of scale  $\frac{2}{3}$  and  $G_\tau$  Gaussian of variance  $\frac{4\pi\tau}{3}$ , the logistic right tail  $f_Z(z) \sim \frac{3}{2}e^{-3z/2}$  gives  $g_\tau(y) \sim \frac{3}{2}e^{-3y/2}e^{3\pi\tau/2}$ , so as  $x = e^y \rightarrow \infty$  the logarithmic-modulus curvature tends to a positive limit  $\mathcal{C}_\infty(\tau)$ , equal to



0.9746 at  $\tau = 0.30$ ; a finite certificate already appears at  $x = e^5$ ,  $\tau = 0.30$ , where the curvature is  $+0.380$  (Proposition 70). The deposited scan reached only moderate boundary ratios and so missed the tail; by the  $x \mapsto 1/x$  symmetry the same convexity occurs in the opposite tail.

**Proposition 70** (Failure of annulus log-concavity at large boundary ratio). For the Brownian-annulus modulus density  $p_x(\tau) \propto \eta(2i\tau)\rho_\tau(x)$ , the logarithmic-modulus curvature  $d^2 \log p_x(e^s)/ds^2$  is positive for sufficiently large boundary ratio  $x$ . At  $x = e^5$ ,  $\tau = 0.30$  an independent Fourier inversion of the exact modulus transform, multiplied by the exact eta factor, gives curvature  $+0.3798$ ; the value grows to  $+0.90$  at  $x = e^7$  and to the positive limit  $\mathcal{C}_\infty(0.30) = 0.9746$  as  $x \rightarrow \infty$ . Hence Conjecture 336 is false.

*Proof.* From the Mellin transform  $\mathbb{E}[X_\tau^{it}] = \frac{2\pi t}{3 \sinh(2\pi t/3)} e^{-2\pi t^2/3}$ ,  $\log X_\tau = Z + G_\tau$  with  $Z$  centred logistic of scale  $\frac{2}{3}$  and  $G_\tau \sim N(0, \frac{4\pi\tau}{3})$  independent. Since  $f_Z(z) = \frac{3}{8} \operatorname{sech}^2(\frac{3z}{4}) \sim \frac{3}{2} e^{-3z/2}$  as  $z \rightarrow \infty$ , Gaussian averaging gives  $g_\tau(y) \sim \frac{3}{2} e^{-3y/2} \mathbb{E}[e^{3G_\tau/2}] = \frac{3}{2} e^{-3y/2} e^{3\pi\tau/2}$  locally uniformly in  $\tau$ , and likewise after two  $\tau$ -derivatives. With  $\eta(2i\tau)$  real and  $\log \eta(2i\tau) = -\pi\tau/6 + \sum_{n \geq 1} \log(1 - e^{-4\pi\tau n})$ , and writing  $x_n = 4\pi n\tau$ ,  $q_n = e^{-x_n}$ , the curvature of  $\log p_x(e^s)$  in  $s = \log \tau$  tends as  $x \rightarrow \infty$  to  $\mathcal{C}_\infty(\tau) = \frac{4\pi}{3}\tau + \sum_{n \geq 1} (\frac{x_n q_n}{1-q_n} - \frac{x_n^2 q_n}{(1-q_n)^2}) > 0$ , in which the  $\frac{4\pi}{3}\tau$  term is the logistic-Gaussian tail contribution  $(-\frac{\pi}{6} + \frac{3\pi}{2})\tau$  and the sum is the eta  $q$ -series, which does not vanish at finite  $\tau$  and cannot be dropped. At  $\tau = 0.30$  this gives  $\mathcal{C}_\infty = 0.9746$ . The finite value at  $x = e^5$ ,  $\tau = 0.30$  is obtained by direct high-precision Fourier inversion of the exact transform, stable under refinement of the step, and is reproduced in the verification script.  $\square$

## Part XXI

# Bold conjectures in theoretical fluid dynamics

*Added to the deposit: 22 August 2026.*

The fifteen conjectures below are sharp, falsifiable statements about the small-scale and singular behaviour of the incompressible Euler, Navier–Stokes, Boussinesq, generalized surface quasi-geostrophic, and point-vortex systems. They fall in six groups: two-dimensional small-scale creation (Conjectures 337 to 339), enhanced dissipation and hypoelliptic mixing (Conjectures 340 and 341), the geometry and regularity of three-dimensional Euler and Navier–Stokes singularities (Conjectures 342 to 344), band-limited kinematics and rapid rotation (Conjectures 345 and 346), stratified and active-scalar singularities (Conjectures 347 to 349), and point-vortex collapse and non-Beltramization (Conjectures 350 and 351). Each is a determinate law with a stated mechanism, a nearest-literature boundary, a first decisive theorem, and a route to disproof, and each is deliberately bold: a proof or a counterexample would reorganize how one reads the corresponding small-scale or singular problem. The proved layer is Proposition 71: the exact three-vortex self-similar collapse anchor, the degenerate one-drift enhanced-dissipation exponent, the homogeneity-reduced generalized-SQG neck reduction, and the exact Beltrami Lamb-vector identity with Frenet dilation invariance. These are statements about continuum solutions—blow-up signatures, vanishing-viscosity and large-bandwidth limits, and long-time or asymptotic laws—so they lie and every one; the accompanying diagnostics test reduced models, exact anchors, or prerequisite scalings, never the full law. Several carry attributions where a sharp

exponent or anchor is due to earlier work: the degenerate-shear exponent of Albritton, Beekie, and Novack [322] (Conjecture 341), the Lorentz-endpoint boundary of Phuc [336] (Conjecture 344), and the self-similar three-vortex collapse of Aref [334] (Conjecture 350); Conjecture 347 is where the part's novelty is most in question, since at equal viscosity and diffusivity its critical-layer anomaly may reduce to known inviscid repeated-root asymptotics times the Kelvin diffusion factor, and only its sharp generic coefficient is claimed as new.

Throughout,  $u$  is a divergence-free velocity field on the flat torus or on  $\mathbb{R}^d$ ,  $\omega = \nabla \times u$  (the scalar curl in two dimensions) its vorticity,  $S = \frac{1}{2}(\nabla u + \nabla u^\top)$  the strain,  $\xi = \omega/|\omega|$  the vorticity direction, and  $\hat{f}(k)$  the Fourier coefficients. Objects specific to a single conjecture are defined in its statement.

**Proposition 71** (Exact anchors and reductions). (i) For three planar point vortices with circulations  $\Gamma = (\Gamma_1, \Gamma_2, \Gamma_3)$  satisfying  $\Gamma_1\Gamma_2 + \Gamma_2\Gamma_3 + \Gamma_3\Gamma_1 = 0$  and vanishing angular impulse, the motion is exactly self-similar with a finite-time collapse or expansion; the symmetric choice  $\Gamma = (1, 1, -\frac{1}{2})$  satisfies the collapse condition. (ii) For the degenerate one-drift shear generator  $X_0 = y^m \partial_x$ ,  $X_1 = \partial_y$  with dissipation  $-\nu X_1^* X_1$ , the enhanced-dissipation gap obeys  $\text{gap}(L_\nu) \asymp \nu^{m/(m+2)}$  as  $\nu \downarrow 0$ . (iii) The homogeneity-reduced neck equation  $\dot{d} = -d^{1-\beta}$ ,  $0 < \beta \leq 1$ , has  $d(t) = [\beta(T-t)]^{1/\beta}$ , so  $d(t) \sim a_\beta(T-t)^{1/\beta}$  and, under the reciprocal-curvature ansatz  $\kappa_{\max} \sim b_\beta(T-t)^{-1/\beta}$ , the product  $d\kappa_{\max}$  tends to a finite nonzero constant. (iv) A Beltrami field ( $u \times \omega \equiv 0$ ), in particular the ABC field, has identically vanishing Lamb vector; and the dimensionless pair (total curvature, scale-invariant torsion variation) of a  $C^4$  space curve is invariant under spatial dilation.

*Proof.* (i) is the classical self-similar three-vortex analysis: the collapse condition makes the moment of inertia quadratic in time with the similarity geometry fixed by vanishing angular impulse [334]; the displayed vector gives  $1 \cdot 1 + 1 \cdot (-\frac{1}{2}) + (-\frac{1}{2}) \cdot 1 = 0$ . (ii) is the degenerate-shear enhanced-dissipation estimate obtained by Fourier reduction in  $x$  and hypoelliptic balance of the  $m$ -fold commutator against diffusion [322]. (iii) integrates the separable ODE directly. (iv) is immediate:  $u \times \omega = 0$  makes the Lamb vector vanish pointwise, and total curvature  $\int \kappa ds$  and  $\int L|\partial_s \tau|/(1 + L|\tau|) ds$  are scale-free, hence invariant under  $u \mapsto \lambda u$ .  $\square$

## 83 Small-scale creation in two-dimensional Euler and Navier–Stokes

**Conjecture 337** (Bulk-enchrophy spectral front). *Let  $\omega$  be a smooth mean-zero solution of the 2D Euler equation on  $\mathbb{T}^2$  and, for  $q \in (0, 1)$ , let  $K_q(t) = \inf\{K \geq 0 : \sum_{|k| \leq K} |\hat{\omega}(k, t)|^2 \geq q \|\omega(t)\|_2^2\}$  be the radius carrying a fraction  $q$  of the enstrophy. For every  $0 < q < q' < 1$  there are constants  $C, c$  depending only on  $q, q'$  such that  $K_q(t) \leq CK_{q'}(0) \exp(c\|\omega_0\|_\infty t)$  for all  $t \geq 0$ .*

**Significance.** Pointwise vorticity gradients can grow double-exponentially [315, 316]; the conjecture asserts that a fixed *bulk* quantile of the conserved enstrophy nonetheless migrates outward only single-exponentially, separating genuine small-scale creation from transport of a macroscopic fraction of enstrophy. A proof route bounds the flux of enstrophy past a moving Fourier radius by the  $L^\infty$ -transport rate of the vorticity. The nearest results are the small-scale creation constructions of Kiselev and Šverák [315]. A first decisive theorem would be the bound for  $q$  near zero. In tests on the corpus described in the stress-test section, a pseudo-spectral Euler

run showed the bulk quantiles  $K_{90}, K_{99}$  moving far more slowly than  $\max |\nabla \omega|$ . A counterexample would be a globally regular Euler flow with genuinely bulk super-exponential spectral transfer.

**Conjecture 338** (Super-exponential gradient sparsity). *For  $\eta \in (0, 1)$  let  $G_\eta(t) = \{x : |\nabla \omega(x, t)| \geq \eta \|\nabla \omega(t)\|_\infty\}$ . If a smooth 2D Euler solution has times  $t_n \rightarrow \infty$  with  $\log \|\nabla \omega(t_n)\|_\infty / t_n \rightarrow \infty$ , then  $|G_\eta(t_n)| \rightarrow 0$  for every fixed  $\eta \in (0, 1)$ .*

**Significance.** The statement turns any super-exponential gradient-growth scenario into a geometric intermittency requirement: near-maximal gradients must occupy a vanishing area under the area-preserving Euler flow. A proof route couples the  $L^2$ -conservation of vorticity to an area estimate for level sets of  $|\nabla \omega|$ . The nearest results are again [315, 316]. A first decisive theorem would be the implication for a single fixed  $\eta$  under a hyperbolic-point growth ansatz. In tests on the corpus described in the stress-test section, a regular run showed a strong negative correlation between  $\log \max |\nabla \omega|$  and the area fraction above 90% of that maximum. A counterexample would permit macroscopically large regions of near-maximal gradient.

**Conjecture 339** (Vanishing-viscosity global branch selection). *On  $\mathbb{T}^2$  fix mean-zero vorticity and initial palinstrophy  $P(\omega_0) = P_0 > 0$ . For viscosity  $\nu > 0$  and horizon  $T$  let  $D_{\nu, T} = \nu \int_0^T \|\nabla \omega(t)\|_2^2 dt$  and  $M_{\nu, T}$  its supremum under the palinstrophy constraint. Fix  $\tau \in (0, \infty)$ , let  $\nu_j \rightarrow 0$  and  $T_j \sqrt{P_0} \rightarrow \tau$ , and take relative near-maximizers  $D_{\nu_j, T_j}(\omega_{0,j}) \geq (1 - \varepsilon_j) M_{\nu_j, T_j}$  with  $\varepsilon_j \rightarrow 0$ . With  $\ell_j = \|\omega_{0,j}\|_2 / \|\nabla \omega_{0,j}\|_2$  and  $W_j(y) = P_0^{-1/2} \omega_{0,j}(x_j + \ell_j y)$ , every concentration limit is, after torus isometries, reflection, and sign, the same nontrivial profile  $W_\tau$ ; and, when the normalized objectives are equicoercive, they  $\Gamma$ -converge at fixed  $\tau$  to a variational problem whose maximizer is unique modulo these symmetries.*

**Significance.** Finite-time enstrophy-dissipation and palinstrophy extremizers are computed numerically with several local branches [317, 318, 319]; the conjecture asserts unique selection only at the global vanishing-viscosity level, turning a family of numerical extreme events into one universal variational object. A proof route is a  $\Gamma$ -convergence and concentration-compactness analysis of the rescaled objective. The nearest results are the maximal-dissipation studies of [319]. A first decisive theorem would be uniqueness of the concentration profile at a single  $\tau$ . In tests on the corpus described in the stress-test section, a concentration-family surrogate reproduced the proposed symmetry reduction and profile collapse without performing the optimization. A counterexample would be two inequivalent global limits, establishing genuine non-universality of worst-case dissipation.

## 84 Enhanced dissipation and hypoelliptic mixing

**Conjecture 340** (Reeb-graph spectral desert). *Let  $H$  be a Morse Hamiltonian on a compact surface whose regular level sets form finitely many cells with quadratic saddles, and let  $L_\nu = -\nabla^\perp H \cdot \nabla + \nu \Delta$  on mean-zero scalars. In every fixed bounded window of imaginary frequency, as  $\nu \rightarrow 0$  the spectrum splits into slow eigenvalues  $\lambda_j(\nu) = -\nu \mu_j + o(\nu)$ , where  $\mu_j$  are the eigenvalues of the effective weighted diffusion on the Reeb graph of  $H$ , and remaining eigenvalues with  $\Re \lambda \leq -c_H \nu^{1/2}$ ; in particular no eigenvalue branch decays at a rate  $\nu^\alpha$  with  $\frac{1}{2} < \alpha < 1$ .*

**Significance.** The law combines the  $O(\nu)$  slow streamline-averaged dynamics on the Reeb graph [320] with cellular enhanced dissipation [321] into a two-band spectral statement with a forbidden intermediate range. A proof route matches a graph-diffusion resolvent to a boundary-layer estimate near the separatrices. The nearest results are [320, 321]. A first decisive theorem would be the two-band separation for a single cell with one saddle. In tests on the corpus described in the stress-test section, a coarse cellular operator robustly recovered the  $O(\nu)$  slow band while the complementary  $O(\nu^{1/2})$  band was not grid-resolvable, a limitation recorded there. A counterexample would be an eigenvalue branch in the forbidden intermediate range.

**Conjecture 341** (Hörmander-step enhanced-dissipation universality). *Consider  $\partial_t f + X_0 f = -\nu \sum_{j=1}^r X_j^* X_j f$  on a compact manifold with smooth invariant volume,  $X_0$  volume-preserving, in a one-drift equiregular Hörmander class where the missing drift directions are generated by Lie words containing exactly one  $X_0$  and repeated commutators with the diffusive fields. Let  $m$  be the maximal number of diffusive commutators needed to span the tangent space. If there is no nonconstant common first integral, then  $\text{gap}(L_\nu) \asymp \nu^{m/(m+2)}$  as  $\nu \rightarrow 0$ .*

**Significance.** Degenerate-shear models already realize the exponent  $m/(m+2)$ ; the conjecture is that bracket depth alone fixes the exponent across all one-drift equiregular geometries, a geometric universality class for enhanced dissipation. A proof route is a quantitative hypoelliptic hypocoercivity estimate keyed to the commutator depth. The nearest result, and the source of the anchor exponent, is Albritton, Beekie, and Novack [322]; the exponent itself is theirs, and only the universality across geometries is claimed as new, an attribution carried in the statement. A first decisive theorem would be a two-dimensional non-shear model with  $m = 2$ . In tests on the corpus described in the stress-test section, Fourier-reduced shear operators with depths  $m = 1, 2, 3$  recovered the exponents  $m/(m+2)$ . A counterexample would be a one-drift equiregular geometry whose exponent depends on more than the bracket depth.

## 85 Geometry and regularity of three-dimensional singularities

**Conjecture 342** (Frenet-frame completion barrier). *Suppose a Type-I Euler singularity at  $T$  is carried by a connected high-vorticity tube  $R_t$  containing a maximum-vorticity point, with radius comparable to the length  $L(t)$  of a  $C^4$  central vortex-line segment  $\gamma_t$  of uniform chord-arc constant and nonvanishing curvature, so that Frenet torsion  $\tau$  and  $\partial_s \tau$  are defined. With  $K(t) = \int \kappa ds$  and  $\text{Tor}(t) = \int L(t) |\partial_s \tau| / (1 + L(t) |\tau|) ds$ , one has  $\limsup_{t \rightarrow T} [K(t) + \text{Tor}(t)] = \infty$ .*

**Significance.** Classical geometric blow-up criteria emphasize curvature and vorticity-direction coherence; the conjecture makes torsion variation the missing Frenet obstruction inside a coherent tube, so bounded total curvature forces the scale-invariant torsion diagnostic to diverge. A proof route feeds the Beale–Kato–Majda integral [324] through the tube geometry. The nearest results are the three-dimensional Euler geometric criteria surveyed in [323]. A first decisive theorem would be the barrier under an axisymmetric-with-swirl tube model. In tests on the corpus described in the stress-test section, the dimensionless (total curvature, torsion variation) pair was confirmed dilation-invariant to relative error below  $10^{-6}$  on a modulated helix; the singular statement is conditional on a Type-I tube. A counterexample would be a bona fide Type-I tube with bounded curvature and torsion.

**Conjecture 343** (Type-I stretching coherence). *Let  $\Omega(t) = \|\omega(t)\|_\infty$ ,  $A_\lambda(t) = \{x : |\omega(x, t)| \geq \lambda\Omega(t)\}$ ,  $\alpha = \xi \cdot S\xi$ , and let  $\bar{\alpha}_\lambda, V_\lambda$  be the  $|\omega|^2$ -weighted mean and standard deviation of  $\alpha$  over  $A_\lambda$ . If a Type-I singularity satisfies  $0 < c \leq (T - t)\Omega(t) \leq C$ , then for every  $\lambda$  sufficiently close to 1 there is a sequence  $t_n \rightarrow T$  with  $(T - t_n)\bar{\alpha}_\lambda(t_n) \rightarrow 1$  and  $(T - t_n)V_\lambda(t_n) \rightarrow 0$ .*

**Significance.** The law upgrades the scalar Type-I rate to concentration of the entire local stretching-rate distribution on the intense-vorticity core, forcing an emergent coherent stretching clock in any Type-I blow-up. A proof route differentiates  $\log \Omega$  and controls the weighted variance of  $\alpha$  by the Type-I hypothesis. The nearest results are the Beale–Kato–Majda control [324] and the geometric-depletion literature in [323]. A first decisive theorem would be the mean statement  $(T - t_n)\bar{\alpha}_\lambda \rightarrow 1$  alone. In tests on the corpus described in the stress-test section, a smooth low-resolution Taylor–Green run gave a finite subcritical weighted stretching diagnostic, the singular limit being inaccessible; the statement is conditional on a Type-I singularity. A counterexample would be a Type-I blow-up whose core stretching-rate distribution stays spread.

**Conjecture 344** (Weak-Lorentz projected-Lamb regularity criterion). *Let  $u$  be a Leray–Hopf solution of the 3D Navier–Stokes equations on  $\mathbb{R}^3$ , or on  $\mathbb{T}^3$  in the Galilean frame with mean velocity zero, let  $\omega = \nabla \times u$ ,  $P$  the Leray projector, and  $\Lambda = P(u \times \omega)$  the projected Lamb vector. If  $\Lambda \in L^2((0, T); L_x^{3/2, \infty})$ , then  $u$  is smooth on  $(0, T]$ .*

**Significance.** The criterion rewards local Beltrami depletion geometrically: a solution whose projected Lamb vector is small in the weak-Lorentz critical norm cannot be singular. Its Stokes scaling lands exactly at the unresolved  $L_t^\infty L_x^{3, \infty}$  velocity endpoint, so the statement is deliberately placed one Lorentz index beyond the range known to be regular. A proof route combines a forced-Stokes estimate with the Lamb-vector structure. The nearest results are the near-Beltrami depletion of Farhat and Grujić [325] and, as the boundary anchor, the finite-index regularity of Phuc [336]:  $L_t^\infty L_x^{3, q}$  regularity is known for  $q < \infty$ , which is why the endpoint  $q = \infty$  is chosen, an attribution carried in the statement. A first decisive theorem would close the criterion under an additional smallness of the strong  $L^{3/2}$  norm. In tests on the corpus described in the stress-test section, a smooth Taylor–Green run computed a finite weak- $L^{3/2}$  Lamb quasi-norm while the exact ABC Beltrami field annihilated the Lamb vector; the criterion is an analytic implication. A counterexample would be a singular solution with finite critical projected-Lamb norm.

## 86 Band-limited kinematics and rapid rotation

**Conjecture 345** (Helicity-defect law for stretching extremals). *Let  $V_N$  be divergence-free mean-zero fields on  $\mathbb{T}^3$  with Fourier support  $|k| \leq N$ . Fix admissible kinetic energy  $K$  and enstrophy  $E$ , put relative helicity  $\rho = H/(2\sqrt{KE})$ , and let  $M_N(\rho; K, E) = \sup |\int \omega \cdot S\omega|$  over  $V_N$  at these constraints and helicity  $\rho$ . Whenever  $\rho$  and 0 are both uniformly feasible as  $N \rightarrow \infty$  and the enstrophy-to-energy ratio  $E/K$  is bounded away from its first-shell minimum, so that zero-sum stretching triads persist and  $M_N(0; K, E) > 0$  uniformly,  $M_N(\rho; K, E)/M_N(0; K, E) \rightarrow (1 - \rho^2)^{3/2}$  locally uniformly for  $\rho$  in compact subsets of  $(-1, 1)$ .*

**Significance.** The law quantifies how close-to-Beltrami alignment suppresses the strongest possible vortex stretching, through a single universal helicity-defect factor in the large-bandwidth



extremal envelope. The triad-active hypothesis is necessary: at the first-shell ratio  $E/K = 1$  every admissible field is supported on  $|k| = 1$ , which contains no zero-sum triad  $k + \ell + m = 0$ , so the stretching functional vanishes identically and the ratio is undefined; an interior ratio populates the second shell and restores  $M_N(0) > 0$ . A proof route is a constrained variational analysis of the stretching functional with the two quadratic and one helicity constraints. The nearest results are the maximum-entropy-production optimizations with energy, enstrophy, and helicity constraints in the depletion literature. A first decisive theorem would be the two-shell case. In tests on the corpus described in the stress-test section, a two-shell helical Monte Carlo held energy and enstrophy fixed while varying helicity and confirmed the suppression sign, not the exponent. A counterexample would be a different exponent, revealing multiscale geometry beyond the simple helicity defect.

**Conjecture 346** (Diophantine small-divisor rotation law). *For  $K \geq 1$  let  $\delta(K)$  be the minimum nonzero inertial-wave triad detuning among triads with maximal wavenumber at most  $K$  on an anisotropic torus, and assume the sharp arithmetic law  $\delta(K) = K^{-\tau+o(1)}$ . For fixed data of finite Sobolev regularity  $H^s$  whose Fourier coefficients obey a polynomial lower envelope  $|\hat{u}(k)| \gtrsim |k|^{-s'}$  along the wavenumbers of the triads realizing  $\delta(K_*)$  (a frequency-rich nondegeneracy, which also excludes a single inertial-wave mode, for which  $(u \cdot \nabla)u \equiv 0$ ; for  $C^\infty$  data the super-polynomial Fourier decay empties this condition, so the lower envelope forces finite regularity and only the upper bound  $E(\Omega, \nu) \lesssim \epsilon_*^{1+o(1)}$  survives without it) and  $O(1)$  times let  $E(\Omega, \nu)$  be the error between rotating 3D Navier–Stokes at rotation  $\Omega$  and its exact-resonant normal form in a fixed lower Sobolev norm, with a strictly positive data constant. In the joint regime  $K_* = (\Omega/\nu)^{1/(\tau+2)} \rightarrow \infty$  and  $\epsilon_* = (\Omega^2 \nu^\tau)^{-1/(\tau+2)} \rightarrow 0$ , one has  $E(\Omega, \nu) = \epsilon_*^{1+o(1)}$  up to Sobolev-loss, subpower, and data constants, and the  $\Omega$  power cannot be uniformly improved over tori realizing the same sharp  $\tau$ .*

**Significance.** The law ties the fluid averaging error to the sharp asymptotic exponent of the minimum nonzero triad detuning, making arithmetic properties of the torus quantitatively observable in the rate of rotational averaging and avoiding the ambiguity of a nonoptimal Diophantine lower bound. A proof route bounds near-resonant contributions by the minimum detuning through a normal-form transformation. The nearest results are the near-resonance approximations of Cheng and Sakellaris [326]. A first decisive theorem would be the upper bound  $E \leq \epsilon_*^{1-o(1)}$ . In tests on the corpus described in the stress-test section, integer triad enumeration on an irrational aspect ratio showed the expected algebraic minimum-detuning trend and fewer exact resonances than a rational comparison torus. A counterexample would be nonlinear cancellation beating the naive small-divisor balance.

## 87 Stratified and active-scalar singularities

**Conjecture 347** (Critical-Richardson viscous log anomaly). *For a fixed Kelvin ray  $(k, \eta)$ ,  $k \neq 0$ , in the linearized equally viscous and diffusive 2D Boussinesq system about Couette flow at Richardson number  $\text{Ri} = \frac{1}{4}$ , let  $\hat{\omega}_{k,\eta}(t)$  be the vorticity amplitude along the sheared frequency  $(k, \eta - kt)$ . For generic data with nonzero projection onto the repeated critical indicial mode,  $\sup_{1 \leq t \leq c(\nu k^2)^{-1/3}} |\hat{\omega}_{k,\eta}(t)|/(1+t)^{1/2} \sim C_{k,\eta} \log(1/\nu)$  as  $\nu \rightarrow 0$ , with  $C_{k,\eta} > 0$ ; for every fixed  $\text{Ri} > \frac{1}{4}$  the analogous normalized supremum is  $O(1)$ .*



**Significance.** The conjecture predicts a sharp matched-asymptotic interaction between the inviscid critical repeated-root logarithm and the viscous enhanced-dissipation cutoff, identifying  $\text{Ri} = \frac{1}{4}$  as a genuinely singular viscous asymptotic threshold rather than merely the endpoint of inviscid stability. A proof route matches the inviscid Kelvin solution to the viscous cutoff on the enhanced-dissipation time. The nearest results are the stratified Couette analyses of Yang and Lin [327] and Masmoudi, Said-Houari, and Zhao [328]. A first decisive theorem would be the reduced-ODE log growth at  $\text{Ri} = \frac{1}{4}$ . This is where the part's novelty is most in question: at equal viscosity and diffusivity the diffusion term is a scalar multiple of the identity on the Kelvin  $(\omega, \theta)$  system, so the viscous fundamental matrix factors as  $\exp(-\nu \int_0^t (k^2 + (\eta - ks)^2) ds)$  times the inviscid one and the logarithm is inherited from the inviscid critical repeated root; only the sharp generic coefficient  $C_{k,\eta}$  and the uniform viscous matching are claimed as new. In tests on the corpus described in the stress-test section, a reduced Kelvin indicial ODE with an  $e^{-\nu t^3/3}$  cutoff produced a growing logarithmic normalized peak precisely at  $\text{Ri} = \frac{1}{4}$ . A counterexample would be an  $O(1)$  normalized supremum at  $\text{Ri} = \frac{1}{4}$ .

**Conjecture 348** (Generalized-SQG neck-curvature similarity law). *Use  $u = \nabla^\perp \Lambda^{-2+\beta} \theta$  with  $0 < \beta \leq 1$ . Suppose a smooth generalized-SQG patch boundary develops a single symmetric self-similar neck singularity at  $T$ , with minimum opposing-boundary distance  $d(t)$  and maximum curvature  $\kappa_{\max}(t)$  in the same similarity core. Then  $d(t) \sim a_\beta (T-t)^{1/\beta}$ ,  $\kappa_{\max}(t) \sim b_\beta (T-t)^{-1/\beta}$ , and  $d(t)\kappa_{\max}(t) \rightarrow c_\beta \in (0, \infty)$ .*

**Significance.** The law proposes a universal reciprocal neck/curvature exponent and a nonzero dimensionless product for the single-neck class, a simple diagnostic across the Euler-to-SQG interpolation. A proof route is a self-similar reduction of the patch contour dynamics near the neck. The nearest results are the no-splash and blow-up analyses of Kiselev and Luo [329], Gancedo and Patel [330], and the self-similar profiles of Hou and collaborators [331]. A first decisive theorem would be the exponent for the reduced neck ODE. In tests on the corpus described in the stress-test section, the homogeneity-reduced neck ODE of Proposition 71 reproduced  $d \sim (T-t)^{1/\beta}$  with  $d\kappa = O(1)$ . A counterexample would be a neck singularity with a different exponent or  $d\kappa \rightarrow 0, \infty$ .

**Conjecture 349** (Boussinesq source-core co-location and alignment). *On  $\mathbb{T}^2$  let  $\theta^\circ = \theta - \langle \theta \rangle$ ,  $\ell_\theta = \|\theta^\circ\|_\infty / \|\nabla \theta\|_\infty$ ,  $A_\lambda^\omega = \{|\omega| \geq \lambda \|\omega\|_\infty\}$ , and  $A_\lambda^\theta = \{|\partial_x \theta| \geq \lambda \|\partial_x \theta\|_\infty\}$ . If a smooth inviscid Boussinesq solution develops an isolated finite-time singularity at  $x_*$ , then, intersecting both intense sets with a fixed ball isolating the core, for every  $\lambda$  close to 1 the distance between the localized sets divided by  $\ell_\theta$  tends to zero; moreover, along a sequence in the core with distinct strain eigenvalues,  $\nabla \theta$  aligns with the most compressive strain eigenvector.*

**Significance.** The law connects the exact vorticity source  $\partial_x \theta$  with geometric strain alignment in a necessary blow-up-core localization, forcing the source, the vorticity maximum, and the compressive scalar-gradient mechanism to merge at the scalar-gradient scale. A proof route tracks the co-evolution of the intense sets under the shared transport. The nearest results are the Boussinesq blow-up asymptotics of Grundy and Kay [332] and the stagnation-point analysis of [333]. A first decisive theorem would be the co-location clause alone. In tests on the corpus described in the stress-test section, a short smooth pseudo-spectral Boussinesq run showed a decreasing normalized distance between the vorticity maximum and the source maximum; the singular statement is conditional on a finite-time singularity. A counterexample would be a blow-up with separated source and vorticity cores.

## 88 Point-vortex collapse and non-Beltramization

**Conjecture 350** (Minimal-admissible point-vortex collapse core). *Fix nonzero circulations for which at least one collapse-admissible subset exists—a subset being admissible if its isolated subsystem has a nontrivial self-similar finite-time collapse—and let  $m_*$  be the least admissible cardinality. Let  $\mathcal{C}$  be the set of initial configurations undergoing a finite-time collision, with the resonance loci where two distinct minimal cores collapse simultaneously removed. Then the union of collision strata whose fastest-collapsing core contains exactly  $m_*$  vortices is relatively open and dense in  $\mathcal{C}$ , while strata with a strictly larger fastest core lie in positive-codimension substrata; and along every collision in this relatively generic set, similarity rescaling of the fastest core converges to an isolated self-similar  $m_*$ -vortex collapse with every spectator asymptotically farther than the core diameter.*

**Significance.** The law organizes the complicated planar collision set by a minimal-core universality principle: the smallest admissible collapsing subsystem generically dominates. A proof route combines the classical self-similar collapse classification with a transversality analysis of the collision set. The nearest results are the self-similar three-vortex collapse of Aref [334], whose collapse anchor is used here and is attributed in the statement, and the recent collapse analyses of Drivas and collaborators [335]. A first decisive theorem would be relative density of the  $m_* = 3$  stratum for a circulation vector admitting a unique minimal core. In tests on the corpus described in the stress-test section, the exact anchor  $\Gamma = (1, 1, -\frac{1}{2})$  perturbed by two distant spectators contracted the core to about 2.4% of its diameter while the spectator/core scale separation amplified. A counterexample would be a robust same-codimension larger fastest core coexisting with a smaller admissible core.

**Conjecture 351** (High-vorticity non-Beltramization barrier). *On  $\mathbb{R}^3$  with finite energy, or on  $\mathbb{T}^3$  in the zero-mean Galilean frame, let  $C_\lambda(t)$  be the connected component of  $\{|\omega| \geq \lambda\Omega(t)\}$  containing a maximum-vorticity point, and  $B_\lambda(t) = \int_{C_\lambda} |u \times \omega|^2 / \int_{C_\lambda} |u|^2 |\omega|^2$  when the denominator is nonzero. Any Type-I Euler singularity with  $0 < c \leq (T-t)\Omega(t) \leq C$  satisfies  $\limsup_{t \rightarrow T} B_\lambda(t) > 0$  for every  $\lambda$  sufficiently close to 1.*

**Significance.** The law makes persistent non-Beltramization a necessary local condition for a Type-I Euler singular core, ruling out asymptotically force-free high-vorticity cores as blow-up mechanisms. A proof route shows that a vanishing local Lamb ratio would deplete the nonlinearity enough to contradict the Type-I rate. The nearest results are the near-Beltrami depletion of Farhat and Grujić [325] and the Beale–Kato–Majda control [324]. A first decisive theorem would be the barrier under axisymmetry with swirl. In tests on the corpus described in the stress-test section, the ABC field gave a Lamb-vector root-mean-square about  $10^{-14}$  of a Taylor–Green comparison, confirming the diagnostic sharply detects exact Beltrami depletion; the singular statement is conditional on a Type-I singularity. A counterexample would be a Type-I core that asymptotically Beltramizes.

## Part XXII

# Metric geometry and spectral convexity in random conformal geometry

*Added to the deposit: 22 August 2026.*

The thirteen conjectures below complete the random-conformal-geometry bundle whose two exactly solved shape members—strict log-concavity of the DOZZ structure constant and of the Brownian-annulus modulus—are Conjectures 335 and 336 of Part XX. They fall in three groups: the metric geometry of Liouville quantum gravity (LQG), covering geodesic stars, the radial metric residual, metric-ball volume and boundary, space-filling SLE, and distance profiles (Conjectures 352 to 359); the spectral convexity of Liouville conformal field theory, covering monotone likelihood ratio and log-concavity of sphere and torus blocks and DOZZ-weighted spectral densities (Conjectures 360 to 363); and quantum-area nesting in conformal loop ensembles (Conjecture 364). Each is a determinate, sharp, falsifiable law with a stated mechanism, a nearest-literature boundary, and a first decisive theorem. The metric-geometry statements are continuum-limit laws (Hausdorff dimensions, scaling exponents, mixing and functional central limit theorems); the CFT statements are exact special-function shape properties whose descendant contributions are here tested only in the global ( $SL(2)$  or global-torus) block limit, a reduced-model check and a verification target. The proved layer is Proposition 72: the exact scaling triangle relating the boundary-length, roughness, and branching exponents, and the anchor reductions at the Brownian point  $\gamma = \sqrt{8/3}$ . Several statements are sharp interpolations of recognized open problems and attribute to the works whose anchors they extend.

Fix  $\gamma \in (0, 2)$ , write  $Q = \frac{2}{\gamma} + \frac{\gamma}{2}$ , let  $h$  be a Gaussian free field variant,  $\mu_h$  the  $\gamma$ -LQG area measure, and  $D_h$  the  $\gamma$ -LQG metric, with Hausdorff dimension  $d_\gamma = \dim_H(\mathbb{C}, D_h)$  and  $\xi = \gamma/d_\gamma$ ; the metric obeys Weyl scaling  $D_{h+c} = e^{\xi c} D_h$  and the area obeys  $\mu_{h+c} = e^{\gamma c} \mu_h$ , and  $d\sqrt{8/3} = 4$ . For Liouville CFT write  $b > 0$ ,  $Q_b = b + b^{-1}$ , central charge  $c = 1 + 6Q_b^2$ , principal-series weights  $\Delta(P) = \frac{Q_b^2}{4} + P^2$ , sphere OPE blocks  $\mathcal{F}_P(z)$  and torus blocks  $\mathcal{F}_{a,P}^{\text{tor}}(q)$  in the standard normalizations, and  $C_b$  for the DOZZ structure constant of Part XX. Objects specific to a single conjecture are defined in its statement.

**Proposition 72** (Scaling triangle and Brownian anchors). Set  $p_\gamma = d_\gamma/2$ ,  $H_\gamma = 2/d_\gamma$ , and  $\alpha_\gamma = 1 + 2/d_\gamma$ . Then  $p_\gamma = 1/H_\gamma = 1/(\alpha_\gamma - 1)$ , so a single scalar controls the metric boundary-length dimension, the distance-profile roughness exponent, and the perimeter branching index. At the Brownian point  $d_\gamma = 4$  these are  $2, \frac{1}{2}, \frac{3}{2}$ , the known values. The affine star spectrum  $s_k(\gamma) = 2 + \frac{3-k}{2}(d_\gamma - 2)$  satisfies  $s_1 = d_\gamma$ ,  $s_3 = 2$  for every  $d_\gamma$ , and  $s_k = 5 - k$  at  $d_\gamma = 4$ ; and  $s_k \rightarrow 2$  as  $d_\gamma \downarrow 2$ .

*Proof.* The three displayed identities are immediate from the definitions of  $p_\gamma, H_\gamma, \alpha_\gamma$ ; at  $d_\gamma = 4$  they evaluate to  $2, \frac{1}{2}, \frac{3}{2}$ . The affine spectrum is the unique affine-in- $k$  function with  $s_1 = d_\gamma$  and  $s_3 = 2$ , and substituting  $d_\gamma = 4$  gives  $5 - k$ , while  $d_\gamma \downarrow 2$  sends every fixed  $s_k$  to 2.  $\square$

## 89 Metric geometry of Liouville quantum gravity

**Conjecture 352** (Affine LQG geodesic-star spectrum). *For a point  $z$  call  $z$  a  $k$ -star point if there are  $r > 0$  and unit-speed  $D_h$ -geodesics  $\eta_1, \dots, \eta_k$  from  $z$  with disjoint interiors on  $(0, r]$ ;*

let  $S_k$  be the set of such points. With  $s_k(\gamma) = 2 + \frac{3-k}{2}(d_\gamma - 2)$ : if  $s_k(\gamma) > 0$  then almost surely  $\dim_{\mathbb{H}}^{D_h} S_k = s_k(\gamma)$ ; if  $s_k(\gamma) < 0$  then  $S_k = \emptyset$  almost surely; at the threshold  $s_k = 0$  the set has  $D_h$ -dimension at most zero.

**Significance.** The formula is the unique affine-in- $k$  spectrum fixed by the two anchors  $s_1 = d_\gamma$  (typical points, rigorous) and  $s_3 = 2$  (the three-star value, whose lower bound  $\dim S_3 \geq 2$  is proved by Bhatia, Gwynne, and Harper and whose sharpness is their conjecture), so it compresses an entire geodesic-branching spectrum into the single unknown  $d_\gamma$  and predicts a maximal multiplicity  $k_{\max} \approx 3 + 4/(d_\gamma - 2)$ . A proof route extends the Lipschitz-function lower-bound technology to upper bounds. The nearest results are the general- $\gamma$  three-star lower bound of Bhatia, Gwynne, and Harper [337] and the strong-confluence framework of Bhatia and Kavvasias [338]; the interpolation itself is the proposed content, an attribution carried in the statement. A first decisive theorem would be  $\dim S_2$  at one non-Brownian  $\gamma$ . In tests on the corpus described in the stress-test section, the formula reproduces the Brownian spectrum  $5 - k$  and the  $d_\gamma \downarrow 2$  limit. A counterexample would be a computed  $\dim S_2$  or  $\dim S_4$  off the affine line.

**Conjecture 353** (Exponential scale-mixing of the radial metric residual). *Let  $h$  be a whole-plane GFF with  $h_1(0) = 0$ ,  $h_r(0)$  its circle average, and  $X_t = \log[e^{\xi Q t - \xi h_{e^{-t}}(0)} D_h(0, \partial B(0, e^{-t}))]$ , a process stationary in  $t$  by scale covariance. Its strong-mixing coefficient satisfies  $\alpha_X(s) \leq C e^{-cs}$ , and  $\sup_t \Pr[\sup_{0 \leq u \leq 1} |X_{t+u}| > x] \leq C e^{-cx}$ ; consequently  $\sup_{0 \leq u \leq T} |X_u| = O(\log T)$  almost surely and the radial log-distance process has the Brownian functional scaling limit and law of the iterated logarithm.*

**Significance.** Stationarity of  $X_t$  is exact from Weyl scaling; the conjectural content is quantitative decorrelation across logarithmic scales and a pathwise envelope sharp enough to transfer Brownian functional and iterated-logarithm behaviour uniformly in time. A proof route seeks a spectral gap for a scale-shift Markov dynamics on the field. The nearest result is the metric-growth stationarity of Dubédat and Falconet [339]. A first decisive theorem would be polynomial mixing. In tests on the corpus described in the stress-test section, finite Liouville first-passage-percolation diagnostics matched the Weyl field-response coefficient and showed no residual growth on the tested scales. A counterexample would be polynomially decaying scale correlations or excursions of order  $\sqrt{t}$ .

**Conjecture 354** (Quantum-cone metric-ball shell functional CLT). *For a rooted LQG surface let  $V(r) = \mu_h(B_{D_h}(z, r))$  and  $Y_j^{(r)} = -\log[V(re^{-(j+1)})/V(re^{-j})]$ . There is a stationary inward-scale version  $(Y_j^{\text{cone}})$  for the rooted  $\gamma$ -quantum-cone tangent law that is exponentially  $\alpha$ -mixing with  $\mathbb{E}Y_0^{\text{cone}} = d_\gamma$  and finite positive asymptotic variance  $\sigma_\gamma^2$ , with  $\frac{1}{\sqrt{n}} \sum_{j < nt} (Y_j^{\text{cone}} - d_\gamma) \Rightarrow \sigma_\gamma B_t$  in  $D([0, 1])$ ; and for any finite-volume surface rooted at a quantum-typical point,  $(Y_j^{(r)})_{j \in J} \Rightarrow (Y_j^{\text{cone}})_{j \in J}$  as  $r \downarrow 0$ .*

**Significance.** The law upgrades the first-order volume exponent to a second-order Gaussian expansion  $\log V(r) = d_\gamma \log r + \sigma_\gamma B_{\log(1/r)} + o_{\text{Pr}}(\sqrt{\log(1/r)})$ , the metric-ball analogue of logarithmic-scale central limit theorems in multiplicative chaos. A proof route establishes exponential mixing of annular scale blocks via the GFF Markov property. The nearest result is the all-moment volume control of Ang, Falconet, and Sun [340]. A first decisive theorem would be the finite-variance central limit theorem for a single scale increment. In tests on the corpus

described in the stress-test section, finite LFPP under-resolves the exponent (slope near 2.3 against 4), a bias reported there. A counterexample would be power-law shell covariance or a non-Gaussian stable limit.

**Conjecture 355** (Pointwise sfSLE metric Hölder exponent). *Let  $\eta$  be a whole-plane space-filling  $\text{SLE}_{16/\gamma^2}$  independent of  $h$ , parameterized by LQG area,  $\mu_h(\eta([s, t])) = t - s$ . For Lebesgue-almost every  $t$ , almost surely  $\lim_{\varepsilon \downarrow 0} \log \text{diam}_{D_h} \eta([t - \varepsilon, t + \varepsilon]) / \log \varepsilon = 1/d_\gamma$ .*

**Significance.** An interval of area  $2\varepsilon$  explores, at a typical time, a cell of  $D_h$ -diameter  $\varepsilon^{1/d_\gamma}$ ; the conjecture asserts the pointwise metric exponent exists and equals the dimensional value almost everywhere, a differentiation statement for the mating-of-trees parameterization. A proof route combines the Minkowski-content identification of  $\mu_h$  with two-sided diameter control. The nearest result is the metric Hölder continuity and Minkowski content of Gwynne and Sung [341]. A first decisive theorem would be the exponent in probability at a typical time. In tests on the corpus described in the stress-test section, a discrete mated-CRT surrogate gave the reciprocal product  $\widehat{d}\widehat{H} \approx 1$  despite finite-size bias in each factor. A counterexample would be a genuine multifractal diameter spectrum at typical times.

**Conjecture 356** (Stable-CSBP perimeter universality). *For every  $\gamma \in (0, 2)$ , filled metric balls in a whole-plane or quantum-cone  $\gamma$ -LQG surface admit a canonical scalar outer-boundary length  $L_h^\gamma(r)$ , measurable from the metric-measure surface, with  $L_{h+c}^\gamma(e^{\xi c}r) = e^{\gamma c/2} L_h^\gamma(r)$ . Under reverse metric-radius exploration, after the natural normalization and excursion/Doob conditioning, the perimeter process is the conditioned/time-reversed stable continuous-state branching process of index  $\alpha_\gamma = 1 + 2/d_\gamma$ .*

**Significance.** The index is forced by self-similarity: a stable CSBP of index  $\alpha$  has mass scaling like  $\text{time}^{1/(\alpha-1)}$ , and matching the proposed perimeter dimension  $d_\gamma/2$  gives  $\alpha_\gamma = 1 + 2/d_\gamma$ , exactly  $\frac{3}{2}$  at  $d_\gamma = 4$ . The consequential claim is that a scalar perimeter closes the growth dynamics Markovianly for every  $\gamma$ , a hidden integrable skeleton. A proof route constructs the boundary length from the trace field and identifies its generator. The nearest results are the open filled-ball boundary-length problem of Gwynne and Miller [342] and the boundary-trace stationarity of [339]; the index and scalar process are the proposed content, an attribution carried in the statement. A first decisive theorem would be existence of the canonical length. In tests on the corpus described in the stress-test section, the index reduces to  $\frac{3}{2}$  at the Brownian point via Proposition 72. A counterexample would be a perimeter process that is not Markov in its scalar length.

**Conjecture 357** (Linear confluence response). *On a unit-area  $\gamma$ -LQG sphere sample  $z, x, y$  independently from  $\mu_h$ . On  $\{D_h(x, y) \leq \varepsilon\}$  let  $b$  be the terminal point of the common initial segment of the geodesics from  $z$  to  $x, y$ ,  $\Delta = \max\{D_h(b, x), D_h(b, y)\}$ , and  $C_\varepsilon = \Delta/D_h(x, y)$ . Conditioned on  $D_h(x, y) \leq \varepsilon$ ,  $C_\varepsilon$  converges in law as  $\varepsilon \downarrow 0$  to a nondegenerate  $C_\gamma \in (0, \infty)$  with finite moments in a neighbourhood of zero.*

**Significance.** Strong confluence gives only an  $\varepsilon^{1-o(1)}$  defect scale; the conjecture sharpens it to an exactly linear tangent law with a universal nondegenerate ratio, the geodesic analogue of a tangent distribution for the typical geodesic tree. A proof route seeks a scaling limit of the branch geometry at conditioned typical points. The nearest result is the strong confluence of

Bhatia and Kavvadias [338]. A first decisive theorem would be tightness of  $C_\varepsilon$  away from 0 and  $\infty$ . In tests on the corpus described in the stress-test section, a finite LFPP confluence diagnostic gave a log-log slope near 1.04 after a max/min implementation fix, a finite-grid check only. A counterexample would be  $C_\varepsilon \rightarrow 0, \infty$  or a nonlinear defect exponent.

**Conjecture 358** (Filled metric-ball outer boundary dimension). *Let  $h$  be a whole-plane GFF and  $B_{D_h}^{\infty, \bullet}(0, r)$  the filled metric ball targeted at infinity. For every deterministic  $r > 0$ , almost surely  $\dim_H^{D_h} \partial B_{D_h}^{\infty, \bullet}(0, r) = d_\gamma/2$ .*

**Significance.** The ordinary ball boundary has dimension  $d_\gamma - 1$ ; the filled outer boundary is conjectured to carry the boundary-length scaling dimension  $d_\gamma/2$ , the same scalar appearing in the perimeter index of Conjecture 356 and the roughness exponent of Conjecture 359, so three otherwise different observables encode one number. A proof route relates the outer boundary to the boundary length of Conjecture 356. The nearest results are the ordinary boundary dimension of Gwynne [343] and the open-problem discussion of Gwynne and Miller [342]. A first decisive theorem would be one rigorous non-Brownian estimate. In tests on the corpus described in the stress-test section, the formula returns 2 at  $d_\gamma = 4$  and is exactly compatible with the scaling triangle. A counterexample would be a non-Brownian dimension off  $d_\gamma/2$ .

**Conjecture 359** (Fractional-Brownian tangent of distance profiles). *Let  $\mathcal{D}$  be a finite-boundary-length  $\gamma$ -LQG disk,  $z$  sampled from  $\mu_h$ , and  $X(u) = D_h(z, \partial(u))$  the distance to the boundary point at length coordinate  $u$ . At boundary-length-almost every  $u_0$  there is  $\sigma_\gamma > 0$  with  $\varepsilon^{-H_\gamma}[X(u_0 + \varepsilon t) - X(u_0)] \Rightarrow \sigma_\gamma B_{H_\gamma}(t)$ ,  $H_\gamma = 2/d_\gamma$ , in local uniform topology, where  $B_H$  is two-sided fractional Brownian motion.*

**Significance.** The variation theorem forces the roughness exponent  $H_\gamma = 2/d_\gamma$ ; the new content is that the local tangent is Gaussian fractional Brownian motion, persistent for  $d_\gamma < 4$  and antipersistent for  $d_\gamma > 4$ , so matter coupling appears as sign-changing increment correlations in a boundary observable. A proof route upgrades the  $d_\gamma/2$ -variation to a self-similar tangent law. The nearest result is the  $d_\gamma/2$ -variation of distance profiles of Bhatia [344]. A first decisive theorem would be the Brownian case  $H = \frac{1}{2}$  at  $\gamma = \sqrt{8/3}$ . In tests on the corpus described in the stress-test section, a finite LFPP structure-function fit gave a smoother-than-Brownian exponent, reported as pre-asymptotic bias. A counterexample would be a non-Gaussian or non-fBm tangent with the correct exponent.

## 90 Spectral convexity in Liouville conformal field theory

**Conjecture 360** (Sphere spectral monotone likelihood ratio). *Fix  $b > 0$ , four equal real external charges  $Q_b/3 \leq a < Q_b/2$ , and  $0 < z_1 < z_2 \leq \frac{1}{2}$ . The OPE-normalized sphere block  $\mathcal{F}_P(z)$  is nonzero on this domain and  $E = P^2 \mapsto |\mathcal{F}_{\sqrt{E}}(z_2)|^2/|\mathcal{F}_{\sqrt{E}}(z_1)|^2$  is strictly increasing; equivalently  $\partial_E \partial_z \log |\mathcal{F}_{\sqrt{E}}(z)|^2 > 0$ .*

**Significance.** This is a total-positivity statement: moving the cross-ratio toward the centre of moduli space shifts spectral weight monotonically toward larger internal dimensions in the likelihood-ratio order, far stronger than pointwise positivity. The leading  $z^\Delta$  factor sets the sign; the conjecture is that Virasoro descendants never overturn it. A proof route controls the



descendant series through the Zamolodchikov recursion. The nearest work is the block/recursion and positivity literature; no monotone-likelihood-ratio statement in the internal momentum was located. A first decisive theorem would be the global  $SL(2)$  block, where the sign is explicit. In tests on the corpus described in the stress-test section, the global-block limit gave strictly positive discrete log-ratio slopes in every tested case; the full-block statement is tested only in that limit. A counterexample would be two internal momenta whose likelihood ratio reverses.

**Conjecture 361** (Four-point spectral log-concavity). *For  $b > 0$ ,  $Q_b/3 \leq a < Q_b/2$ ,  $0 < z \leq \frac{1}{2}$ , and  $P > 0$ , the principal-series magnitude  $I_z(P) = |C_b(a, a, \frac{Q_b}{2} + iP)|^2 |\mathcal{F}_P(z)|^2$  is positive and  $\frac{d^2}{dP^2} \log I_z(P) < 0$ .*

**Significance.** Strict log-concavity gives the four-point spectral density a unique mode and rules out shoulders or competing saddles in the interior chamber, turning bootstrap questions into convex-analysis problems. It combines the DOZZ curvature of Conjecture 335 with the block contribution, requiring only their sum to stay negative. A proof route adds the Barnes-kernel curvature of Part XX to a descendant Hessian bound. The nearest boundary is the DOZZ formula of Kupiainen, Rhodes, and Vargas [313]; the interior chamber is necessary, the broader version being falsified at  $b = 1$ ,  $a = 0.55$ . A first decisive theorem would be the  $b = 1$  case. In tests on the corpus described in the stress-test section, exact- $\Upsilon_b$  quadrature combined with the global block gave negative curvature throughout the chamber and positive curvature at the excluded sub-chamber charge; the block contribution is tested only in the global limit. A counterexample would be positive curvature at a regular interior point.

**Conjecture 362** (Torus thermal spectral monotone likelihood ratio). *Fix  $b > 0$  and a real external charge  $Q_b/3 \leq a < Q_b/2$ , and use the full torus block  $\mathcal{F}_{a,P}^{\text{tor}}(q) = q^{\Delta(P) - c/24} \mathcal{H}_{a,P}^{\text{tor}}(q)$  with  $q = e^{-2\pi\beta}$ . For  $\beta_2 > \beta_1$  the blocks are nonzero for  $P > 0$  and  $E = P^2 \mapsto |\mathcal{F}_{a,\sqrt{E}}^{\text{tor}}(q_2)|^2 / |\mathcal{F}_{a,\sqrt{E}}^{\text{tor}}(q_1)|^2$  is strictly decreasing.*

**Significance.** Making the torus colder suppresses higher internal Liouville momentum in the strongest one-dimensional likelihood-ratio sense; the bare Boltzmann factor supplies the sign and the conjecture is that the descendant series preserves it, giving a family of thermal correlation inequalities for non-rational CFT. A proof route controls the toric descendant factor  $\mathcal{H}^{\text{tor}}$ . The nearest result is the probabilistic toric-block construction of Ghosal, Remy, Sun, and Sun [345]. A first decisive theorem would be the global torus block. In tests on the corpus described in the stress-test section, the global torus-descendant proxy gave negative log-ratio slopes at every tested temperature pair and charge; the full-block statement is tested only in that limit. A counterexample would be a reversal near  $q \rightarrow 1$ , where descendant entropy is strongest.

**Conjecture 363** (Multipoint joint spectral log-concavity of the modulus). *For  $n \geq 5$ , real moduli  $\mathbf{z}$  in a positive OPE cell, equal physical external charges  $Q_b/3 \leq a < Q_b/2$ , and internal momenta  $\mathbf{P} = (P_1, \dots, P_{n-3})$  on the principal series, let  $K_{\mathbf{z}}(\mathbf{P})$  be the standard-normalized comb sewing kernel (product of DOZZ vertices and the multipoint block) and  $\Phi_{\mathbf{z}}(\mathbf{P}) = \log |K_{\mathbf{z}}(\mathbf{P})|$ . Then  $K_{\mathbf{z}} \neq 0$  and  $\nabla_{\mathbf{P}}^2 \Phi_{\mathbf{z}} < 0$  for every  $P_j > 0$ .*

**Significance.** This is the multivariate modulus analogue of Conjecture 361: a unique joint spectral maximum with no flat or competing directions in the comb channels, giving global convex control on the size of multipoint sewing contributions. A proof route seeks diagonal

dominance or an M-matrix structure in the comb graph after a coordinate change. The nearest result is the multipoint WKB analysis of Artemev and Khromov [346]. A first decisive theorem would be the five-point case at  $b = 1$ . In tests on the corpus described in the stress-test section, a five-point coupled proxy from exact  $b = 1$  DOZZ vertices and global blocks had negative-definite finite-difference Hessians on the tested interior grid; the full-block statement is tested only in that limit. A counterexample would be a positive Hessian direction of  $\log |K_{\mathbf{z}}|$  in the chamber.

## 91 Nesting in conformal loop ensembles

**Conjecture 364** (Quantum-cone nesting area-ratio functional CLT). *Let  $\kappa \in (8/3, 4) \cup (4, 8)$  with  $\gamma = \sqrt{\kappa}$  for  $\kappa < 4$  and  $\gamma = 4/\sqrt{\kappa}$  for  $\kappa > 4$ . For nested  $\text{CLE}_{\kappa}$  around the marked point of the matched  $\gamma$ -quantum-cone tangent law, with  $U_n$  the successive marked components,  $A_n = \mu_h(U_n)$ , and  $Y_n = -\log(A_{n+1}/A_n)$ : the sequence  $(Y_n)$  admits a stationary version that is exponentially  $\alpha$ -mixing with analytic scaled cumulant generating function near zero and positive asymptotic variance, so  $\frac{1}{\sqrt{n}} \sum_{j < nt} (Y_j - v_{\gamma, \kappa}) \Rightarrow \sigma_{\gamma, \kappa} B_t$  in  $D([0, 1])$ ; and finite-volume deep-nesting increments converge to this stationary law.*

**Significance.** Euclidean CLE nesting has a renewal structure in logarithmic conformal radius; the conjecture asserts the analogous Gaussian second-order law for successive ratios of genuine continuum LQG areas, which mix the radial mode, multiplicative chaos, and random boundary geometry. A proof route seeks regeneration at nesting scales with quantitative absolute continuity of area increments. The nearest results are the CLE nesting theory of Miller, Watson, and Wilson [347] and the LQG-CLE reweighting of Holden and Lehmkuehler [348]; the critical point  $\kappa = 4$  is excluded. A first decisive theorem would be the central limit theorem for the stationary chain. In tests on the corpus described in the stress-test section, a structural surrogate coupling exact CLE conformal-radius cumulants to a radial-GFF area approximation showed skewness and excess kurtosis decaying toward zero with depth. A counterexample would be long-range dependence or a non-Gaussian stable partial-sum limit.

## Part XXIII

# Hydrodynamic limits of interacting particle systems

*Added to the deposit: 22 August 2026.*

The fifteen conjectures below propose hydrodynamic scaling limits for interacting particle systems whose microscopic rules carry a new structural ingredient: an endogenous local jump index, refreshed or fatigued environments, non-Markovian renewal clocks, singular reaction surfaces, finite crossing budgets, shared common noise, odd multi-species transport, and moving or storing membranes. They fall in five groups (Conjectures 365 to 367, 368 to 370, 371 to 373, 374 to 376, and 377 to 379). Each states a microscopic model, the conjectured limiting equation, its conserved quantities, the critical scaling that makes the new mechanism survive, and a route to disproof. These are scaling-limit statements about continuum PDE or SPDE,, and every one; the diagnostics test event-count scalings, conservation, reduced linearizations, or exact ODE

reductions, never the limit theorem. The proved layer is Proposition 73: exact mass conservation of the limiting equations, the fatigue mobility-budget identity, the perturbative self-arrest residual, and the critical event-count exponents. Four statements have explicit high overlaps with prior work, since a closely related continuum object or one-particle input appeared in 2025–2026 work: the Markov-switching Robin equation of Colantoni [354] (Conjecture 374), the scalar vorticity/odd-diffusion exclusion of De Carlo, Gabrielli, and Gonçalves [355] (Conjecture 375), the critical-percolation random-walk scaling limit of Dankovic, Markering, Miller, and Yuan [357] (Conjecture 377), and the finite-reservoir Wentzell theorem of Franco, Franco, and Gonçalves [359] (Conjecture 379); for these the novel content is the exact microscopic derivation and the specific interior/two-sided/multi-species structure, not the limiting object itself.

Throughout,  $\eta$  is an exclusion configuration on the discrete torus  $\mathbb{T}_N$  or on  $\mathbb{R}$ ,  $\rho$  the limiting density, and time is diffusively scaled unless stated otherwise. Objects specific to a single conjecture are defined in its statement.

**Proposition 73** (Conservation, mobility budget, self-arrest residual, and critical counts). (i) Each limiting equation displayed below conserves the relevant total mass (for multi-species models, each color mass): integrating the gain–loss or divergence form and interchanging dummy variables cancels the flux. (ii) For the fatigue ODE  $\dot{a} = c(1 - a)^p$  with  $c = 2\rho(1 - \rho) > 0$ , the conductance reaches zero in finite time iff  $p < 1$ , with  $t_{\text{hit}} = 1/(c(1 - p))$ , while the integrated mobility  $\int_0^\infty (1 - a)^p dt = 1/c$  is finite for every  $p$ . (iii) For the occupation-deepened trap system linearized about  $\rho = \bar{\rho}$  with  $b = 0$ , the first-order perturbation solves  $\partial_t v = e^{-a\bar{\rho}t} \Delta v$ , so  $v(t) = e^{\tau(t)\Delta} v_0$  with  $\tau(t) = (1 - e^{-a\bar{\rho}t})/(a\bar{\rho})$  and  $v_\infty = e^{\Delta/(a\bar{\rho})} v_0$ , generally nonzero. (iv) A codimension-one reaction surface at generator rate  $N^\theta$  contributes at order  $N^{\theta-1}$ , critical at  $\theta = 1$ ; and a membrane of step  $N^a$ , move rate  $N^{-b}$ , and slow conductance  $\alpha/N$  has order-one Brownian motion, healing length, and slow activity per dwell simultaneously iff  $b = 2a$  and  $a = \frac{1}{2}$ .

*Proof.* (i) is the standard conservative-form cancellation. (ii) integrates  $\dot{a} = c(1 - a)^p$  directly:  $(1 - a)^{1-p}$  is linear in  $t$  for  $p \neq 1$ , giving finite  $t_{\text{hit}}$  iff  $p < 1$ , and  $\int (1 - a)^p dt = \int_0^1 da/c = 1/c$ . (iii) solves the linear nonautonomous heat equation by the time change  $\tau(t) = \int_0^t e^{-a\bar{\rho}s} ds$ . (iv) is the event-count balance:  $N^{d-1}$  surface sites times rate  $N^\theta$  times mass change  $N^{-d}$  is  $N^{\theta-1}$ ; and  $N^2 \cdot N^{-b} (N^a/N)^2 = N^{2a-b}$ , a dwell of  $N^b$  diffuses  $N^{b/2}$ , and slow ringing per dwell is  $\alpha N^{b-1}$ , jointly order one exactly at  $b = 2a = 1$ .  $\square$

## 92 Density-adaptive and refreshed exclusion hydrodynamics

**Conjecture 365** (Density-adaptive Lévy exclusion). On  $\mathbb{T}_N$  let  $r_x^N(\eta)$  be the mean occupation in a block of radius  $\ell_N$  with  $\ell_N \rightarrow \infty$ ,  $\ell_N/N \rightarrow 0$ , and  $\log N/\sqrt{\ell_N} \rightarrow 0$ , and let  $\alpha : [0, 1] \rightarrow [\alpha_-, \alpha_+] \subset (0, 2)$  be  $C^1$ . A particle at  $x$  jumps to a vacant  $y$  at rate  $N^{\alpha(r_x^N)} c(\alpha(r_x^N)) |y - x|_N^{-1-\alpha(r_x^N)}$ . For smooth profiles bounded away from 0, 1 the empirical density converges to the conservative variable-order gain–loss equation

$$\partial_t \rho(u) = \text{PV} \int_{\mathbb{T}} [K_{\alpha(\rho(v))}(u - v) \rho(v) (1 - \rho(u)) - K_{\alpha(\rho(u))}(v - u) \rho(u) (1 - \rho(v))] dv.$$

**Significance.** The spatial stable index is an endogenous local-density field, so the macroscopic operator has a state-dependent order; the stability condition  $\log N/\sqrt{\ell_N} \rightarrow 0$  is necessary because a vanishing block-density error otherwise creates an order-one multiplicative rate error through  $N^{\alpha(r)}$ . A proof route establishes the replacement lemma for the source-dependent exponent

with exponential control of the logarithmic amplification. The nearest results are the variable-order fractional kinetics of Kolokoltsov and Troeva [349] and the fractional porous-medium hydrodynamics of Cardoso, de Paula, and Gonçalves [350]. A first decisive theorem would be the case  $\alpha_+ < 1$ . In tests on the corpus described in the stress-test section, the amplification diagnostic  $\log N/\sqrt{\ell_N}$  decreased with  $N$  and lower density gave longer accepted jumps. A counterexample would be a surviving logarithmic renormalization or a different symmetrized operator.

**Conjecture 366** (Refreshing heavy traps: crossover trichotomy). *Start from a reversible heavy-trap exclusion model with static anomalous clock  $\theta_N$  and index  $\beta \in (0, 1)$ , refresh each trap at rate  $q_N$  independently of the particles, and set  $\lambda_N = q_N \theta_N$ . If  $\lambda_N \rightarrow 0$  the static fractional-kinetics limit persists; if  $\lambda_N \rightarrow \lambda \in (0, \infty)$  the hydrodynamics is governed by a model-dependent renewal/Bernstein symbol  $\psi_\lambda(s)$  with  $\psi_0(s) = cs^\beta$  and  $\psi_\lambda(s) \sim m_\lambda s$  as  $s \downarrow 0$ ; if  $\lambda_N \rightarrow \infty$  the limit is ordinary diffusion after effective-diffusivity renormalization.*

**Significance.** Refreshing a trapped environment interpolates between anomalous and diffusive transport, but the critical symbol is model-dependent rather than universally tempered-stable; the theorem-sized problem is to identify  $\psi_\lambda$  from the microscopic refreshed process. A proof route computes the one-particle resolvent of the refreshed renewal clock. The nearest results are static Bouchaud-trap fractional kinetics and dynamic-random-environment hydrodynamics. A first decisive theorem would be the  $\lambda \rightarrow 0$  persistence. In tests on the corpus described in the stress-test section, a generic heavy waiting time interrupted by exponential refresh showed the truncated mean waiting time falling monotonically with refresh rate, supporting the crossover mechanism without a fixed symbol. A counterexample would be a universal symbol independent of the refresh rule.

**Conjecture 367** (Critical sparse shortcuts add a teleportation operator). *Let nearest-neighbor SSEP run on  $\mathbb{T}_N$  at rate  $N^2$ , and add  $M_N$  macroscopic shortcut edges ringing at rate  $\kappa$ , with  $M_N/N \rightarrow \lambda/2$  and endpoint-pair measure converging to uniform. The empirical density converges to  $\partial_t \rho = D\partial_{uu}\rho + \kappa\lambda(\bar{\rho} - \rho)$ , with  $\bar{\rho} = \int_{\mathbb{T}} \rho$ ; for a nonuniform endpoint kernel the nonlocal term becomes the corresponding conservative jump operator.*

**Significance.** Each shortcut moves  $O(N^{-1})$  mass and there are  $O(N)$  of them, so their aggregate is an order-one nonlocal mean-field damping toward the spatial average, a Euclidean local-plus-teleportation hydrodynamic crossover. A proof route uses the exact local-equilibrium cancellation  $\rho(u)(1 - \rho(v)) - \rho(v)(1 - \rho(u)) = \rho(u) - \rho(v)$ . The nearest literature concerns random walks on small-world graphs and IPS on sparse graphs, not this local-plus- $O(N)$ -shortcut limit. A first decisive theorem would be the annealed rewiring case. In tests on the corpus described in the stress-test section, the first Fourier mode decayed at the predicted mean-field rate. A counterexample would be a different damping coefficient or a non-mean-field limit.

## 93 Renewal exclusion, fatigue, and boundary memory

**Conjecture 368** (Crowding-renormalized successful-reset renewal exclusion). *Let a particle of age  $a$  attempt jumps with hazard  $\beta/(a + a_0)$ ,  $0 < \beta < 1$ , resetting its age only after a successful jump. In dimension  $d \geq 2$ , linearizing about a homogeneous equilibrium  $\rho_*$ , on the clock  $t_{N,*} = N^{2/\gamma(\rho_*)}$  with  $\gamma(\rho_*) = \beta(1 - \rho_*)$  the density perturbation converges to the fractional*

diffusion  $\partial_t^{\gamma(\rho_*)} u = D(\rho_*) \Delta u$ ; a genuinely nonconstant profile admits no single global polynomial clock, requiring a variable-order-in-time renewal kernel.

**Significance.** Because a jump resets the age only on success and a vacancy is seen with probability  $1 - \rho_*$ , the successful-event hazard is thinned to  $\beta(1 - \rho_*)/(a + a_0)$ , so the hard-core constraint changes the temporal universality exponent, not merely a diffusivity. A proof route Markovizes the system with an empirical age measure and integrates the age out. The nearest results are non-Markovian exclusion models and variable-order temporal CTRW equations. A first decisive theorem would be the linearized fractional response at fixed  $\rho_*$ . In tests on the corpus described in the stress-test section, fitted tail exponents of the thinned-hazard proxy matched  $\beta(1 - \rho_*)$  at densities 0.1, 0.4, 0.7 to 0.013. A counterexample would be restoration of the isolated exponent  $\beta$  by decorrelation of the environment seen by old particles.

**Conjecture 369** (Activity-fatigued conductances). *Attach damage  $a_b(t) \in [0, 1]$  to each bond, exchanging symmetrically at rate  $N^2(1 - a_b)^p$ , with  $a_b$  increased by  $N^{-2}$  at each successful crossing (a crossing budget of order  $N^2$ ). Writing  $g(a) = (1 - a)^p$ , the damage field then evolves macroscopically as  $\partial_t a = 2g(a)\rho(1 - \rho)$ , this law derived from the increment rule rather than imposed. The coupled hydrodynamic limit is  $\partial_t \rho = \partial_u(g(a)\partial_u \rho)$ ,  $\partial_t a = 2g(a)\rho(1 - \rho)$ , with finite-time loss of ellipticity at  $p < 1$  but finite integrated mobility for every  $p$ , so some smooth nonconstant data retain nonconstant long-time limits.*

**Significance.** Activity damages the medium it flows through, and the sharp  $p = 1$  transition governs finite-time loss of ellipticity while the finite total diffusive budget can freeze non-equilibrium profiles for a broader range of  $p$ . A proof route proves finite-time hydrodynamics up to the first degeneracy, then controls the self-consistent damage field. The nearest result is the history-dependent substrate exclusion of Schulz, Kolomeisky, and Frey [351]. A first decisive theorem would be finite-time hydrodynamics up to degeneracy. In tests on the corpus described in the stress-test section, the exact fatigue ODE of Proposition 73 confirmed finite hitting iff  $p < 1$  and integrated mobility  $1/(2\rho(1 - \rho))$ . A counterexample would be complete homogenization for all nonconstant data.

**Conjecture 370** (Common heavy-tailed boundary clock: random Stieltjes boundary law). *Let a single heavy-tailed renewal process control all left-boundary updates, scaled so that  $N^{-1}R_N(N^2t) \Rightarrow \kappa_L E_t^L$  with  $E^L$  an inverse- $\gamma$ -stable clock, and analogously on the right. The bulk solves  $\partial_t \rho = \partial_{uu} \rho$ , but the boundary fluxes are singular random Stieltjes measures: in weak form  $d\langle \rho_t, G \rangle = \langle \rho_t, G'' \rangle dt + \kappa_L G(0)[\alpha - \rho_t(0)]dE_t^L + \kappa_R G(1)[\beta_R - \rho_t(1)]dE_t^R$ , remaining random after averaging over the clocks.*

**Significance.** A single shared heavy-tailed clock does not self-average—its inverse-stable limit is genuinely random—so the correct object is a random-measure-driven boundary law, not the deterministic fractional Robin condition of a naive law of large numbers. A proof route proves joint convergence of the boundary counting process and the empirical field. The nearest topics are inverse-stable time changes and fractional boundary PDE, whose ingredients are individually classical. A first decisive theorem would be the one-sided clock. In tests on the corpus described in the stress-test section, the coefficient of variation of the single renewal clock stayed near 0.67 and did not vanish, while averaging 36 clocks drove it to 0.11, the signature that forced the random-clock statement. A counterexample would be a self-averaging deterministic boundary law from one shared clock.



## 94 Catalysts, mixed clocks, and crossing budgets

**Conjecture 371** (Codimension-one nonlinear catalyst transmission law). *Let  $\Sigma_N$  approximate a smooth hypersurface  $\Sigma \subset \mathbb{T}^d$  with bulk SSEP at rate  $N^2$  and catalyst birth/death flips at generator rate  $N^\theta$  with reaction  $R(r) = b(r)(1 - r) - d(r)r$ . The critical exponent is  $\theta = 1$ , at which  $\partial_t \rho = \Delta \rho$  off  $\Sigma$  with continuous trace and derivative jump  $[\partial_n \rho]_\Sigma = -R(\rho|_\Sigma)$ ; for  $\theta < 1$  the catalyst disappears and for  $\theta > 1$  a stable zero of  $R$  pins the trace.*

**Significance.** The lower-dimensional set carries no independent conserved density; it acts as a singular nonlinear reaction measure, a codimension-one transmission law. A proof route is the surface event-count balance of Proposition 73 with a boundary replacement estimate. The nearest result is the field-road bulk/surface SSEP of Alfaro, Mourragui, and Treton [352]. A first decisive theorem would be the linear-reaction case. In tests on the corpus described in the stress-test section, the catalyst contribution scaled as  $N^{\theta-1}$ , reproducing the  $N^{-1/2}, 1, N^{1/2}$  regimes at  $\theta = \frac{1}{2}, 1, \frac{3}{2}$ . A counterexample would be a different critical exponent or a persistent independent surface density.

**Conjecture 372** (Mixed-clock two-species Maxwell–Stefan hydrodynamics). *Let species  $A$  use exponential attempt clocks and species  $B$  use i.i.d. heavy-tailed attempt clocks of index  $\gamma \in (0, 1)$ , reset after every attempt. Observed on time  $N^{2/\gamma}$  with  $A$ 's microscopic rate  $N^{2-2/\gamma}$ , both species stay hydrodynamically visible, and around a homogeneous composition the linearized law is the mixed-order Maxwell–Stefan system  $s\hat{\delta}a - \delta a_0 = -k^2(D_*\hat{\delta})_A$ ,  $s^\gamma\hat{\delta}b - s^{\gamma-1}\delta b_0 = -k^2(D_*\hat{\delta})_B$  with non-diagonal exclusion cross-diffusion  $D_*$ .*

**Significance.** The two species carry different temporal universality classes coupled by hard-core exclusion, so the relaxation modes are roots of a genuinely hybrid matrix dispersion relation, not one ordinary plus one independent fractional mode. A proof route uses subexponential-tail stability of the attempt-reset clock under geometric blocking, which leaves the index  $\gamma$  unchanged. The nearest results are two-species exclusion cross-diffusion [353] and bidirectional non-Markovian exclusion. A first decisive theorem would be the linearized dispersion relation. In tests on the corpus described in the stress-test section, fitted successful-wait exponents stayed at  $\gamma$  across vacancy probabilities and the clock-balance identity held to machine precision. A counterexample would be a blocked-clock index shift or a diagonal decoupling.

**Conjecture 373** (Finite crossing-budget membrane: endogenous stopping-time transition). *Give a critical slow bond (microscopic rate  $\alpha/N$ ) a budget  $B_N = \lfloor \kappa N \rfloor$  of successful exchanges, after which its rate is zero forever. With unsigned local activity  $A(r_-, r_+) = r_-(1 - r_+) + r_+(1 - r_-)$ , the normalized success count satisfies  $S_N(t)/N \rightarrow \alpha \int_0^t A(\rho(s, 0^-), \rho(s, 0^+)) ds$ , and the density obeys the critical Robin transmission law before the exhaustion time  $\tau = \inf\{t : \alpha \int_0^t A \geq \kappa\}$  and reflecting conditions after, so the PDE changes type at an endogenous stopping time.*

**Significance.** A cumulative successful-crossing capacity permanently switches a critical membrane, so the macroscopic boundary condition changes type at a stopping time generated by the integrated local activity—a genuinely endogenous phase transition. A proof route enlarges the slow-bond replacement theory by the normalized success count and applies a continuous-mapping theorem for the hitting time. The nearest machinery is the classical slow-bond Robin transition, which lacks the endogenous stopping mechanism. A first decisive theorem is the joint law of



large numbers up to exhaustion. In tests on the corpus described in the stress-test section, mean exhaustion times concentrated on the predicted value with standard deviation shrinking as  $N^{-1/2}$ . A counterexample would be a signed-current rather than unsigned-activity exhaustion law.

## 95 Common noise, odd transport, and self-generated arrest

**Conjecture 374** (Telegraph-modulated critical membrane). *Let a single two-state Markov chain  $Z_t \in \{0, 1\}$ , converging on diffusive time to a finite-rate telegraph process, multiply a critical slow bond's rate,  $\alpha Z_t/N$ . Conditional on  $Z$ , the density converges to the heat equation with time-dependent Robin coefficient  $\alpha Z_t$ , so the empirical density converges jointly with  $Z$  to a random hydrodynamic limit; the fast-switch regime converges to the averaged Robin law.*

**Significance.** A single finite-dimensional common noise survives a conservative hydrodynamic limit because there is no spatial averaging mechanism to replace  $Z_t$  by its mean at order-one switching. A proof route is a quenched hydrodynamic limit conditional on the telegraph path. *Priority.* The continuum Markov-switching Robin equation is already established by Colantoni [354], so the novel content is only the exclusion derivation of that object and the fast-switch averaging; the statement has this high overlap with prior work. A first decisive theorem would be the quenched limit at fixed switching rate. In tests on the corpus described in the stress-test section, a macroscopic gap proxy had substantial randomness at finite switching rate, averaging out only as switching accelerated. A counterexample would be self-averaging of the common noise in the conservative limit.

**Conjecture 375** (Multi-species chiral-vorticity odd cross-diffusion). *Use at least two conserved colors plus vacancies, with symmetric color-vacancy exchanges and plaquette rotations at rates  $c_{\pm}(\xi) = N^2 c_0 [1 \pm \varepsilon \Phi(\mathcal{N}(\xi))]$  depending only on the rotation-invariant plaquette composition  $\mathcal{N}(\xi)$ , so grand-canonical product color measures are invariant. The diffusive hydrodynamics is  $\partial_t r_i = \nabla \cdot \sum_j [D_{ij}^S(r) \nabla r_j + D_{ij}^O(r) R \nabla r_j]$  with  $R$  rotation by  $\pi/2$  and a composition-dependent odd cross matrix  $D^O$  that changes sign with  $\varepsilon$  and affects individual density equations.*

**Significance.** The genuinely new requirement is a density-changing odd cross sector,  $\nabla \cdot (D_{ij}^O(r) R \nabla r_j) = \nabla D_{ij}^O \cdot R \nabla r_j \neq 0$ , which needs composition dependence and cross-coupling, each color mass still conserved. A proof route verifies the invariant product family, then computes the first nonzero odd cross Green–Kubo coefficient perturbatively. *Priority.* The broad scalar odd/vorticity exclusion is already established by De Carlo, Gabrielli, and Gonçalves [355] (a scalar antisymmetric sector affects the current but not the density), and hard-core odd diffusion is known [356]; only the multi-species density-changing cross sector is claimed, a high overlap with prior work carried in the statement. A first decisive theorem would be a nonzero odd cross coefficient at a symmetric composition. In tests on the corpus described in the stress-test section, a transverse species-1 current reversed sign with chirality while all color counts were exactly conserved. A counterexample would be a vanishing density-changing odd cross sector for all admissible  $\Phi$ .

**Conjecture 376** (Perturbative self-generated arrest). *Attach trap depth  $E_x(t) \geq 0$  with symmetric bond rate  $N^2 \exp[-(E_x + E_{x+1})/2]$  and  $dE_x/dt = a\eta_t(x) - bE_x$ . The finite-time limit is  $\partial_t \rho = \partial_u(e^{-e} \partial_u \rho)$ ,  $\partial_t e = a\rho - be$ ; with  $b = 0$ , linearizing about  $\rho = \bar{\rho}$  gives residual  $v_{\infty} = e^{\Delta/(a\bar{\rho})} v_0$ ,*

so for small smooth mean-zero data the full nonlinear system converges to the nonconstant limit  $\rho_\infty = \bar{\rho} + \epsilon e^{\Delta/(a\bar{\rho})} v_0 + O(\epsilon^2)$ .

**Significance.** Occupation deepens the traps it sits in, and the finite total heat-time  $\tau(\infty) = 1/(a\bar{\rho})$  freezes a residual nonconstant profile—a concrete, testable arrest expansion rather than a generic-arrest heuristic. A proof route proves finite-time hydrodynamics with a positive conductance lower bound, then controls the nonlinear remainder in a weighted norm with finite time integral. The nearest result is the history-dependent substrate exclusion of [351]. A first decisive theorem is the finite-time coupled limit. In tests on the corpus described in the stress-test section, the conservative nonlinear PDE reproduced the linear residual fraction to four digits with mass error  $10^{-16}$ . A counterexample would be complete relaxation despite the finite mobility budget.

## 96 Critical geometry and moving membranes

**Conjecture 377** (Critical-percolation exclusion transfer). *Let  $(V_N, c_N, \mu_N)$  be the rescaled critical triangular-percolation graph sequence for which simple random walk converges quenched to the semigroup  $P_t^\omega$  of  $CLE_6$  Brownian motion on the limiting resistance space, with time normalization  $a_N$ . Run SSEP on  $V_N$  with the same conductances and time change. If the initial color field converges to  $\rho_0 \mu^\omega$ , then quenched  $\pi_t^N \Rightarrow \rho_t \mu^\omega$  with  $\rho_t = P_t^\omega \rho_0$ , so exclusion produces no additional transport exponent beyond the one-particle normalization.*

**Significance.** The conjecture transfers the one-particle scaling to the many-particle exclusion system on a critical random fractal where volume, resistance, and geometry are singular. A proof route abstracts the transfer step: measured-Gromov/Dirichlet-form convergence of the walk plus exclusion local ergodicity in the same random geometry. *Priority.* The one-particle scaling limit is now a theorem of Dankovic, Markering, Miller, and Yuan [357], so the remaining content is the exclusion-transfer step; the statement has this overlap with prior work. A first decisive theorem would be local ergodicity uniform in critical bottlenecks. In tests on the corpus described in the stress-test section, a random-walk proxy on the correct critical triangular model gave the expected mean-squared-displacement geometry. A counterexample would be an additional slowdown exponent from exclusion correlations despite one-particle convergence.

**Conjecture 378** (Critically mobile slow membrane). *Let a distinguished slow bond of rate  $\alpha/N$  move by microscopic distance  $N^a$  at rate  $N^{-b}$ . At the uniquely matched scaling  $a = \frac{1}{2}$ ,  $b = 1$  (step  $\sqrt{N}$ , move rate  $N^{-1}$ , conductance  $\alpha/N$ ), the membrane path converges to Brownian motion and the density converges to a nontrivial random singular-interface heat flow conditional on that Brownian path; the membrane is not annealed away, and the exact moving-Robin generator is a sub-conjecture.*

**Significance.** Three requirements—Brownian macroscopic motion, healing length equal to the next jump, and order-one critical slow activity per dwell—uniquely select the scaling, and at that scaling a random non-annealed interface survives the limit. A proof route works in the moving-defect frame over dwell intervals with a Mosco-type convergence of random time-dependent forms. The nearest result is the moving-defect SSEP of Chatterjee, Chatterjee, and Pradhan [358]. A first decisive theorem is tightness of the Brownian membrane with survival of the interface. In

tests on the corpus described in the stress-test section, the rescaled membrane variance and slow-bond attempts per dwell both stayed order one across sizes. A counterexample would be an annealed deterministic-membrane limit at the matched scaling.

**Conjecture 379** (Interior two-sided capacitor membrane). *Split a one-dimensional system into left and right SSEP bulks and an interior storage node of capacity  $\lfloor \kappa N \rfloor$ , conserving total mass, with side-to-storage rates of order  $N$ . The limit is  $\mu_t = \rho_- du + \rho_+ du + m(t)\delta_0$  with heat equations in the bulks, nonlinear vertex fluxes  $F_{\pm}(r, m) = a_{\pm}r(1 - m/\kappa) - b_{\pm}(m/\kappa)(1 - r)$ , boundary conditions  $-\partial_{n_{\pm}}\rho_{\pm} = F_{\pm}$ , and  $\dot{m} = F_- + F_+$ , conserving  $\int \rho_- + \int \rho_+ + m$ .*

**Significance.** A single atomic storage state couples two macroscopic traces and mediates delayed conservative transfer between them, an interior two-sided capacitor. A proof route couples the two bulk replacement estimates through the storage occupation. *Priority.* Franco, Franco, and Gonçalves [359] prove finite-reservoir Wentzell boundary behaviour for exclusion; C15 differs in being interior, two-sided, conservative, and shared, but the conceptual distance is small, so the statement carries a high overlap with prior work. A first decisive theorem would be the single-side reduction. In tests on the corpus described in the stress-test section, total microscopic mass was conserved exactly and the stored macroscopic mass stayed order one across sizes. A counterexample would be a vanishing or non-shared storage variable in the limit.

## Part XXIV

# Fixed-traffic circulation laws for nonequilibrium Markov chains

*Added to the deposit: 22 August 2026.*

The four conjectures below concern a controlled family of finite-state continuous-time Markov chains—fixed-traffic circulation rays—in which the stationary law, every undirected stationary edge traffic, and every escape rate are held fixed while an irreversible stationary circulation of strength  $\alpha$  is increased. The ray therefore varies time-reversal asymmetry without touching the equilibrium landscape or the local activity. Two conjectures concern visited-set statistics (Conjectures 380, 382, 383) and one the spectral geometry of the generator (Conjecture 381). Each is a determinate law with a stated mechanism, a nearest-literature boundary, a first decisive theorem, and a route to disproof. The proved layer is Proposition 74: the endpoint spectral bounds, the exact even-analytic dependence of visited-set laws on  $\alpha$ , and an explicit high-precision counterexample showing that the natural stronger companion—whole-ray monotonicity of the fast real spectral edge—is *false*, so only the endpoint bound survives there. Conjecture 381 is the strongest of the four; its own stronger global-unimodality form is also false and is recorded as a control. The visited-set laws are exact finite-state statements, tested here by exact augmented-chain propagation rather than trajectory sampling, but with the alternating-derivative hierarchy of Conjecture 380 checked only to finite order.

Let the state space be  $\{1, \dots, n\}$  with stationary law  $\pi > 0$ . For each undirected edge fix a symmetric traffic  $T_{ij} = T_{ji} > 0$  and an antisymmetric divergence-free current  $J_{ij} = -J_{ji}$ ,  $\sum_j J_{ij} = 0$ , with  $|J_{ij}| < T_{ij}$ . The circulation ray has off-diagonal rates  $L_{\alpha, ij} = (T_{ij} + \alpha J_{ij})/(2\pi_i)$ ,  $0 \leq \alpha \leq 1$ , and zero row sums; then  $\pi$  is stationary, the undirected traffic  $q_{ij}(\alpha) + q_{ji}(\alpha) = T_{ij}$

and the escape rate  $r_i = \frac{1}{2\pi_i} \sum_j T_{ij}$  are  $\alpha$ -independent. Writing  $H_\alpha = -D^{1/2}L_\alpha D^{-1/2}$  with  $D = \text{diag}(\pi)$  and restricting off the stationary mode gives  $G_\alpha = K + \alpha C$  with  $K = K^* \succ 0$  and  $C^* = -C$ . Let  $h(\alpha) = \max\{\Re \lambda : \lambda \in \sigma(-L_\alpha) \setminus \{0\}\}$  and  $g(\alpha) = \min\{\Re \lambda : \lambda \in \sigma(-L_\alpha) \setminus \{0\}\}$  be the fast and slow real spectral edges. Let  $V_t = \{X_s : 0 \leq s \leq t\}$  be the visited set,  $w_i^{(\gamma)} = \pi_i^\gamma / \sum_j \pi_j^\gamma$  escort weights, and  $U_{\gamma,t} = \sum_i w_i^{(\gamma)} \mathbf{1}_{\{\tau_i > t\}}$  the unexplored escort mass, with  $x = \alpha^2$ .

**Proposition 74** (Endpoint bounds, even dependence, and the failure of fast-edge ray monotonicity). For every ray,  $\Re \lambda(G_\alpha) \in [\lambda_{\min}(K), \lambda_{\max}(K)]$ , so  $h(\alpha) \leq h(0) = \lambda_{\max}(K)$  and  $g(\alpha) \geq g(0) = \lambda_{\min}(K)$ . Under a stationary start, path reversal maps the law of  $L_\alpha$  to that of  $L_{-\alpha}$  and fixes every visited-set statistic, so  $V_t^{(\alpha)} \stackrel{d}{=} V_t^{(-\alpha)}$  and all finite-state visited-set probabilities are even analytic functions of  $\alpha$ , whence  $x = \alpha^2$  is the natural variable. The endpoint bound does *not* extend to whole-ray monotonicity of the fast edge: there is a bidirected six-state ray on which  $h(\alpha)$  strictly increases over an interior interval by more than 0.6, verified to 40 digits, so the companion statement  $h(\beta) \leq h(\alpha)$  for  $\alpha \leq \beta$  is false.

*Proof.* For an eigenpair  $G_\alpha v = \lambda v$ ,  $\Re \lambda = v^* K v / v^* v \in [\lambda_{\min}(K), \lambda_{\max}(K)]$  by the Rayleigh bound, and at  $\alpha = 0$ ,  $G_0 = K$  attains both ends; this gives the endpoint inequalities. The adjoint identity  $L_{-\alpha} = L_\alpha^*$  in  $L^2(\pi)$  and the invariance of  $V_t$  under time reversal give the equality in law and hence even analyticity in  $\alpha$ . The counterexample is an explicit valid ray (positive rates for all  $\alpha \in [0, 1]$ ,  $\pi$  stationary to machine precision) whose fast-edge curve, recomputed by high-precision eigenvalue evaluation on a refined grid, rises across an interior interval; the increase exceeds 0.6 at 40-digit precision, far above any conditioning artifact, so it is genuine.  $\square$

## 97 Visited-set and spectral circulation laws

**Conjecture 380** (Escort unexplored-mass alternating-derivative hierarchy). *For every integer  $k \geq 1$ , integer  $r \geq 0$ , time  $t \geq 0$ , and  $0 \leq \gamma \leq 1$ , with  $x = \alpha^2$ ,*

$$(-1)^r \frac{d^r}{dx^r} \mathbb{E}_{\sqrt{x}}[U_{\gamma,t}^k] \geq 0$$

*throughout the admissible physical interval.*

**Significance.** Every moment of the unexplored escort mass is a positive combination of finite-target avoidance probabilities, and  $x = \alpha^2$  is forced as the variable by the time-reversal even-analyticity of Proposition 74; the conjecture asserts a complete alternating-derivative hierarchy in the squared circulation, unifying stationary-random-target avoidance ( $\gamma = 1$ ) with expected state-space exploration ( $\gamma = 0$ ). A proof route seeks an operator Stieltjes representation in  $x$ , or a target-avoidance identity made positive by pairing a path with its reverse. The nearest results are the nonreversible-chain hitting/variance improvements of Rey-Bellet and Spiliopoulos [361] and Huang and Mao [362]. A first decisive theorem would be the first moment at  $\gamma \in \{0, 1\}$ . In tests on the corpus described in the stress-test section, exact augmented-chain propagation on chains of sizes three to six recorded thousands of alternating-difference inequalities with no violation beyond roundoff; the infinite hierarchy is checked only to finite order. A counterexample would be a moment, order, and lag with the wrong derivative sign.

**Conjecture 381** (One-maximum principle for the slow spectral edge). *The slow real spectral edge  $g(\alpha)$  has at most one strict interior local maximum on  $(0, 1)$ .*

**Significance.** The endpoint bound  $g(\alpha) \geq g(0)$  says irreversibility cannot push the slow edge below the reversible gap, but says nothing about the shape in between; the conjecture asserts a rigid one-maximum shape imposed by the Markov positivity cone, allowing a later shallow rebound but not a second interior peak. A proof route studies branch collisions of the symmetric-plus-skew pencil  $K + \alpha C$  under the constraint that not every such pencil is a positive generator. The nearest results are the affinity/winding eigenvalue-localization bounds of Kolchinsky, Ohga, and Ito [212]. A first decisive theorem would be the birth–death-plus-one-shortcut case. The stronger global-unimodality form of this conjecture is *false*, an eight-state ray realizing an interior maximum followed by a shallow minimum and an endpoint rebound, which is retained as a control. In tests on the corpus described in the stress-test section, no ray among thousands of general and stressed families showed two interior maxima of  $g$ . A counterexample would be a ray with two strict interior maxima of the slow edge.

**Conjecture 382** (Unicyclic range stochastic acceleration). *For every bidirected chain whose undirected support is connected of cycle rank one, with  $X_0 \sim \pi$  and  $R_t = |V_t|$ : for every  $m$ ,  $t \geq 0$ , and  $0 \leq \alpha \leq \beta \leq 1$ ,*

$$\Pr_{\beta}(R_t \geq m) \geq \Pr_{\alpha}(R_t \geq m).$$

**Significance.** On a single-cycle support the whole finite-time range distribution is stochastically accelerated by circulation, first-order stochastic dominance along the ray that implies expected-range improvement and, at  $m = n$ , acceleration of cover time. A proof route uses the unique-cycle decomposition, coupling excursions into attached trees to a biased cycle traversal. The nearest results are the nonreversible mixing improvements of Chen and Hwang [360] and Hua and Chen [363]; the unicyclic restriction is substantive, the same range ordering being false on general fixed-traffic graphs and retained as a control. A first decisive theorem would be expected-range monotonicity. In tests on the corpus described in the stress-test section, exact augmented-chain propagation on unicyclic families recorded thousands of stochastic-order inequalities with no violation beyond roundoff; an earlier apparent violation was traced to comparing probabilities at  $\alpha$ -dependent times rather than a fixed lag and vanishes under the correct fixed-time comparison. A counterexample would be a unicyclic ray with a genuinely non-monotone range tail.

**Conjecture 383** (Pure-cycle stationary-mass coverage acceleration). *For a simple-cycle support with  $X_0 \sim \pi$  and  $M_t = \pi(V_t)$ : for every  $z \in [0, 1]$ ,  $t \geq 0$ , and  $0 \leq \alpha \leq \beta \leq 1$ ,*

$$\Pr_{\beta}(M_t \geq z) \geq \Pr_{\alpha}(M_t \geq z).$$

**Significance.** On a simple cycle the visited set is a contiguous arc and the divergence-free current is one-dimensional, so the fixed-traffic deformation is unusually rigid; the conjecture asserts stochastic acceleration of the stationary mass of the visited set, which for nonuniform  $\pi$  is strictly more than the range acceleration of Conjecture 382. A proof route reduces to a two-boundary arc-growth process whose clockwise and counterclockwise rates change oppositely under circulation at fixed total activity. The nearest result is the circulant maximum-entropy mixing-time extremality of Avrachenkov, Cottatellucci, Maggi, and Mao [311], whose homogeneous case is adjacent and attributed. A first decisive theorem would be the homogeneous cycle. In tests on the corpus described in the stress-test section, exact augmented-chain propagation on nonuniform cycles recorded tens of thousands of stochastic-order inequalities with no violation, cross-checked

against Gillespie simulation. A counterexample would be a cycle and mass threshold with a non-monotone coverage tail.

## Part XXV

# Cross-shadow universality

*Added to the deposit: 24 August 2026.*

The five conjectures below share a single principle. In each, two limiting observables are read off the *same* discrete randomness, often with no literal separation of the input coordinates, and are nevertheless conjectured to decouple because they probe distinct resolution, frequency, or representation-theoretic sectors. We call this *cross-shadow universality*: an Archimedean or edge shadow and an arithmetic or bulk shadow of one random object become asymptotically independent even though each is a function of all of the input. They fall in five settings. A random-matrix setting (Conjecture 384) couples the real microscopic spectrum of a symmetric integer Wigner matrix to its fixed-odd-prime Smith cokernel. An arithmetic setting (Conjecture 385) asks a bounded-alphabet random determinant to obey random-integer multiplicative laws in the growing-dimension regime. An integrable-probability setting (Conjecture 386) separates the modular charge of a Plancherel partition from its Airy edge. A representation-theoretic setting (Conjecture 387) asserts that the exponentially large unstable Fourier spectrum of a symmetric-group word measure cannot beat its best stable mode. A local–global setting (Conjecture 388) factorizes the real and fixed-prime root processes of a random Littlewood polynomial. Each is a determinate law with a stated mechanism, a nearest-literature boundary, a first decisive theorem, and a route to disproof. The proved layer is Proposition 75: the nonsingular determinant–cokernel identity, the exact Parseval identity that makes the word-measure chi-square canonical, the  $p$ -adic unit-root property of Littlewood polynomials, the summable fixed-prime singularity tilt, and Rostam’s core normalization. The marginals in every case are mature theorems; what is proposed as new is in each case the joint law, for which no antecedent was found. These are asymptotic-independence statements, so the accompanying finite-size diagnostics can only exclude gross obstructions: a small observed correlation is compatible with either a finite-size effect or genuine slow-decaying dependence, and no such computation certifies independence. Two statements are flagged as beginning with odd primes only: the  $p = 2$  paired-cokernel structure of Conjecture 384 and the collapsed reduction  $+1 \equiv -1$  of Conjecture 388 are exceptional and are deferred to secondary statements.

**Proposition 75** (Cross-shadow anchor identities). (i) For an integral matrix  $A_n$  with  $\det A_n \neq 0$ , the cokernel  $\mathbb{Z}^n/A_n\mathbb{Z}^n$  is finite of order  $|\det A_n|$ , so the Archimedean size and the total arithmetic size coincide and cannot factor; the conjectured factorization is therefore of *local* shadows only. (ii) For a conjugation-invariant word measure  $\mu_{w,N}$  on  $S_N$  with uniform reference  $U_N$ ,

$$\chi^2(\mu_{w,N} \| U_N) = \sum_{\rho \in \widehat{S_N}, \rho \neq 1} |\mathbb{E}_w \chi_\rho|^2, \quad \mathbb{E}_w \chi_\rho = \mathbb{E}[\chi_\rho(w(\sigma_1, \dots, \sigma_r))],$$

an exact Parseval identity over the nontrivial irreducibles. (iii) Every root in  $\mathbb{Q}_p$  of a Littlewood polynomial  $F_n(X) = \sum_{j=0}^n \varepsilon_j X^j$ ,  $\varepsilon_j = \pm 1$ , is a  $p$ -adic unit. (iv) The fixed-prime singularity probability of a balanced random matrix over  $\mathbb{F}_p$  is  $c_p = 1 - \prod_{j \geq 1} (1 - p^{-j}) = 1/p + O(p^{-2})$ , so



the tilt  $c_p - 1/p$  is summable over primes. (v) For a fixed  $e$ -core,  $t^{-1/2}|\text{core}_e(\lambda_t)|$  and  $(4/\pi)\Gamma_e^{\text{Ros}}$  differ only by the deterministic factor  $\pi/4$ .

*Proof.* (i) The Smith normal form writes  $A_n = U \text{diag}(d_1, \dots, d_n)V$  with  $U, V \in \text{GL}_n(\mathbb{Z})$ , so  $\mathbb{Z}^n/A_n\mathbb{Z}^n \cong \bigoplus_i \mathbb{Z}/d_i$  has order  $\prod_i |d_i| = |\det A_n|$ . (ii) Writing  $f = d\mu_{w,N}/dU_N = N!\mu_{w,N}$ , the class functions carry the orthonormal basis  $\{\chi_\rho\}$  under  $\langle a, b \rangle = \mathbb{E}_{U_N}[a\bar{b}]$ ; expanding  $f - 1 = \sum_\rho c_\rho \chi_\rho$  gives  $c_\rho = \mathbb{E}_{U_N}[f\bar{\chi}_\rho] = \mathbb{E}_w \bar{\chi}_\rho$  for  $\rho \neq \mathbf{1}$  and  $c_{\mathbf{1}} = 0$ , and Parseval gives  $\chi^2(\mu\|U) = \|f - 1\|^2 = \sum_{\rho \neq \mathbf{1}} |\mathbb{E}_w \chi_\rho|^2$ . (iii) If  $|x|_p < 1$  then  $|F_n(x)|_p = |\varepsilon_0|_p = 1$  since every higher term lies in  $p\mathbb{Z}_p$ , so  $F_n(x) \neq 0$ ; if  $|x|_p > 1$  the leading term  $\varepsilon_n x^n$  strictly dominates the rest in  $|\cdot|_p$ , so again  $F_n(x) \neq 0$ ; hence any root has  $|x|_p = 1$ . (iv) Expanding  $\prod_{j \geq 1} (1 - p^{-j}) = 1 - p^{-1} - p^{-2} + O(p^{-3})$  near its dominant terms gives  $c_p = p^{-1} + O(p^{-2})$ , and  $\sum_p (c_p - 1/p) < \infty$ . (v) is Rostam's definition of the normalized core statistic.  $\square$

**Conjecture 384** (Local spectral–Smith factorization). *Let  $A_n$  be a symmetric integral Wigner matrix with independent upper-triangular entries: off-diagonal entries have a fixed integer law  $\xi$  with  $\mathbb{E}\xi = 0$ ,  $\mathbb{E}\xi^2 = \sigma^2 > 0$ , enough moments for edge universality, and a fixed anti-concentration bound modulo every odd prime; diagonal entries are centered with the remaining Wigner hypotheses. Let  $\mathcal{E}_n$  be the rescaled real upper-edge process of  $A_n/(\sigma\sqrt{n})$  and, for an odd prime  $p$ , let  $C_{n,p} = (\text{coker } A_n[p^\infty], \delta_{n,p})$  be the  $p$ -primary cokernel with its symmetric discriminant pairing when  $\det A_n \neq 0$ , and a cemetery state otherwise, an event of probability tending to zero. Then for every fixed finite set  $P$  of odd primes,*

$$(\mathcal{E}_n, (C_{n,p})_{p \in P}) \implies \mathcal{E}_{\text{GOE}} \otimes \bigotimes_{p \in P} \mu_p^{\text{sym}},$$

with  $\mu_p^{\text{sym}}$  the universal paired-cokernel law, and the same product limit holds with any fixed finite collection of microscopic bulk and edge spectral windows in place of  $\mathcal{E}_n$ . The  $p = 2$  analogue, with the correct 2-adic paired state space, is a secondary statement.

**Significance.** The real microscopic spectrum is governed by resolvents at scale  $n^{-1}$  in the bulk and  $n^{-2/3}$  at the edge, while a fixed  $p$ -primary cokernel is governed by congruence rank and lifting through bounded powers of  $p$ ; both are functions of the same entries, and the conjecture is resolution separation without input separation. Global quantities cannot factor, since  $|\det A_n| = |\text{coker } A_n|$  on nonsingularity by Proposition 75(i); the claim is about the local shadows. The nearest results are the paired-cokernel universality of Hodges [364] and the quantitative symmetric-cokernel universality of Shen [365] on one side, and real Wigner edge and bulk universality on the other; no joint theorem couples the two shadows of one integral matrix. A first decisive theorem would be, for one odd prime and one three-point ensemble,  $(n^{2/3}(\lambda_{\max} - 2), \dim_{\mathbb{F}_p} \ker(A_n \bmod p)) \Rightarrow \text{TW}_1 \otimes \nu_p$ . In tests on the corpus described in the stress-test section, symmetric  $\{-1, 0, 1\}$  matrices show odd-prime edge/nullity correlations small and unstable in sign, while  $p = 2$  retains a persistent finite-size correlation, the reason the statement begins with odd primes. A counterexample would be a structured family of almost-singular real eigenvectors whose arithmetic also raises modular rank, a coupling channel neither marginal theory sees.

**Conjecture 385** (Random-integer multiplicative universality of determinants). *Let  $B_n$  have i.i.d. Bernoulli- $\frac{1}{2}$  entries in  $\{0, 1\}$  and set  $D_n = |\det B_n|$ , conditioned on  $D_n \geq 3$ . Then*

$$\frac{\omega(D_n) - \log \log D_n}{\sqrt{\log \log D_n}} \Rightarrow N(0, 1),$$

and simultaneously the ranked normalized logarithms of the prime factors converge in the ranked simplex,

$$\left( \frac{\log Q_{1,n}}{\log D_n}, \frac{\log Q_{2,n}}{\log D_n}, \dots \right) \Rightarrow PD(1),$$

so the normalized logarithm of the largest prime factor converges to the largest component of a Poisson–Dirichlet  $PD(1)$  partition.

**Significance.** For each fixed prime, reduction modulo  $p$  turns  $B_n$  into a balanced matrix, so  $\Pr(p \mid D_n) \rightarrow c_p = 1/p + O(p^{-2})$ ; the local tilt  $c_p - 1/p$  is summable by Proposition 75(iv), so it may shift bounded centering constants but should vanish on the Erdős–Kac and Poisson–Dirichlet scales. This is a dimension-driven arithmetic universality: the input alphabet is fixed forever while the determinant acquires  $n^2$  variables and degree  $n$ , unlike classical Erdős–Kac results that fix a polynomial and enlarge the input box. The  $\{0, 1\}$  alphabet is essential; a  $\pm 1$  determinant carries a deterministic power-of-two divisor incompatible with random-integer statistics. The nearest results are finite-field singularity universality of Maples [366] and the non-square theorem of Bary-Soroker and Kozma [367], with the magnitude theory of Tao–Vu and Nguyen–Vu [368, 369]; no Erdős–Kac or Poisson–Dirichlet law for a bounded-alphabet random determinant was found. A first decisive theorem would be a central limit theorem for  $\sum_{p \leq e^{cn}} \mathbf{1}_{p \mid D_n}$  for some fixed  $c > 0$ . In tests on the corpus described in the stress-test section, exact determinant factorizations give  $\omega(D_n)$  of order  $\log \log D_n$  with a bounded excess, and a largest-prime fraction drifting toward the Golomb–Dickman value  $0.62433\dots$ , though the factorable sizes are tiny. A counterexample would be a systematic excess of medium primes leaving the normal law intact while destroying  $PD(1)$ , or the reverse.

**Conjecture 386** (Plancherel modular-charge/Airy factorization). *Let  $\lambda_t$  have poissonized Plancherel measure with parameter  $t$ , fix  $e \geq 2$ , and let  $X^{(e)}(\lambda_t)$  be the  $e$ -abacus charge vector in the root lattice, with Rostam’s normalization  $Y_t^{(e)} = e\sqrt{\pi/2}t^{-1/4}X^{(e)}(\lambda_t)$ . Let  $\mathcal{A}_t$  be the Plancherel upper-edge process. Then*

$$(Y_t^{(e)}, \mathcal{A}_t) \Longrightarrow G_e \otimes \mathcal{A}_{\text{GUE}},$$

with  $G_e$  Rostam’s Gaussian charge limit, a centred Gaussian vector on the root-lattice hyperplane with Rostam’s covariance, and  $\mathcal{A}_{\text{GUE}}$  the Airy<sub>2</sub> edge; consequently, with  $\Gamma_e^{\text{Ros}}$  Rostam’s Gamma-sum law,

$$\left( \frac{\pi}{4\sqrt{t}} |\text{core}_e(\lambda_t)|, t^{-1/6}(\lambda_{t,1} - 2\sqrt{t}) \right) \Longrightarrow \Gamma_e^{\text{Ros}} \otimes F_2,$$

with  $F_2$  the Tracy–Widom GUE distribution and the deterministic  $\pi/4$  scaling of Proposition 75(v).

**Significance.** The Plancherel partition is a discrete-Bessel determinantal process, and the two observables probe different sectors of it: the Airy statistic lives at the turning point of the limit shape, while the  $e$ -core charge is a periodic residue imbalance accumulated across the occupied Fermi sea, neither a compactly supported local statistic nor a smooth linear statistic. The mechanism is frequency-sector separation of a high-frequency modular bulk observable from a turning-point edge observable. The nearest results are Johansson’s Airy edge asymptotics [370] and Rostam’s explicit core-size law [371]; no joint distribution or independence statement was found, and the generic independence of smooth linear statistics from the edge does not apply because the  $e$ -core is built from periodic residue classes and a quadratic charge. A first decisive

theorem would be the  $e = 2$  scalar case, joint convergence of the normalized 2-abacus charge and the longest row to  $G \otimes F_2$ , via a Fredholm determinant with a periodic charge twist and an edge projector. In tests on the corpus described in the stress-test section, exact RSK sampling of poissonized Plancherel partitions gives core/edge correlations that fluctuate around zero and shrink from  $t = 400$  to  $t = 900$ , with the  $\pi/4$  normalization reproduced. A counterexample would be a persistent nonzero covariance, or zero covariance with higher-order dependence, distinguishing periodic modular statistics from smooth bulk ones.

**Conjecture 387** (Stable-spectrum saturation of word measures). *Let  $w \in F_r$  have primitive abelianization in  $\mathbb{Z}^r$  but be non-primitive in  $F_r$ , and let  $\mu_{w,N}$  be the word measure it induces on  $S_N$  by evaluating  $w$  at  $r$  independent uniform permutations. With  $\lambda[N] = (N - |\lambda|, \lambda_1, \dots)$  and  $\chi_{\lambda[N]}$  the stable irreducible character, set*

$$\beta_{\text{st}}(w) = \inf_{\lambda \neq \emptyset} \liminf_{N \rightarrow \infty} \frac{-\log |\mathbb{E}_w \chi_{\lambda[N]}|}{\log N}, \quad -\log 0 = +\infty,$$

with  $N^{-\infty+o(1)}$  read as smaller than every power of  $N$ . Then

$$\chi^2(\mu_{w,N} \| U_N) = N^{-2\beta_{\text{st}}(w)+o(1)} \quad \text{and} \quad D_{\text{KL}}(\mu_{w,N} \| U_N) = N^{-2\beta_{\text{st}}(w)+o(1)},$$

so the aggregate contribution of all unstable irreducible representations cannot improve the polynomial exponent of non-uniformity beyond the best stable Fourier mode.

**Significance.** The exact Parseval identity of Proposition 75(ii) makes the chi-square the squared  $\ell^2$  norm of the full nontrivial Fourier spectrum, so the conjecture is a genuine global saturation principle rather than a statement about finitely many stable observables: an exponentially large unstable spectrum does not collectively beat the best stable mode. The nearest results are the word-measure and stable-invariant theory of Hanany–Puder and Puder–Shomroni [372, 373, 374], which control fixed stable characters through primitivity-type invariants; no treatment of the KL or chi-square divergence of an  $S_N$  word measure from uniform, or of the complete Fourier tail against the stable sector, was found. Two implications are bundled that stable-character asymptotics alone do not deliver: the equality of the chi-square and Kullback–Leibler exponents requires a tail bound on the likelihood ratio, since rare large values of the ratio can separate the two, and the exclusion of the unstable representations requires a bound on their aggregate Plancherel mass, since exponentially many individually negligible modes could dominate a global divergence; a proof would establish both. A first decisive theorem would be, for  $w = [x, y]x$ , the chi-square bound  $N^{-2+o(1)}$  under an explicit bound on the aggregate unstable mass, followed by the Kullback–Leibler bound from a likelihood-ratio tail estimate. In tests on the corpus described in the stress-test section, exact enumeration of  $S_N^2$  for  $w = [x, y]x$  through  $N = 6$  keeps  $N^2 \chi^2$  and  $N^2 D_{\text{KL}}$  of order one, consistent with the exponent 2, with the Parseval identity confirmed exactly. A counterexample would be a family of unstable irreducibles collectively contributing more chi-square mass than every stable character, or KL and chi-square with different exponents through rare density-ratio spikes, either defining a new asymptotic invariant of words. If the emerging stable theory identifies  $\beta_{\text{st}}(w)$  with the stable primitivity rank, the exponent becomes  $2s\pi(w)$ ; that identification is deliberately kept separate.

**Conjecture 388** (Adelic root-process factorization). *Let  $F_n(X) = \sum_{j=0}^n \varepsilon_j X^j$  with independent signs  $\varepsilon_j = \pm 1$ . Let  $\mathcal{R}_n = \sum_{r: F_n(r)=0} \delta_{n(|r|-1)}$  be the real-root process near the unit circle, and for*

an odd prime  $p$  let  $\mathcal{Z}_{p,n} = \sum_{\alpha \in \mathbb{Q}_p: F_n(\alpha)=0} m_\alpha \delta_\alpha$  be the  $p$ -adic root process on  $\mathbb{Z}_p^\times$  (every  $\mathbb{Q}_p$  root is a unit by Proposition 75(iii)). Then for every fixed finite set  $P$  of odd primes,

$$(\mathcal{R}_n, (\mathcal{Z}_{p,n})_{p \in P}) \implies \nu_\infty \otimes \bigotimes_{p \in P} \nu_p,$$

with  $\nu_\infty$  the limiting real microscopic root process and  $\nu_p$  the universal  $p$ -adic root process for the Littlewood law. The prime 2 is excluded, since modulo 2 one has  $+1 \equiv -1$  and the reduction loses all coefficient randomness.

**Significance.** For an odd prime the residues do distinguish  $+1$  from  $-1$ , so the  $p$ -adic observer can in principle reconstruct the whole sign sequence: there is no literal decomposition into independent real and  $p$ -adic digits, and the mechanism is adelic mode orthogonality. The real process near  $\pm 1$  is controlled by approximately Gaussian evaluation and derivative modes at Archimedean scale, while fixed-prime roots are governed by congruence evaluation and Hensel lifting; each is a low-complexity nonlinear functional of the same high-dimensional sign vector, and the overlap of their informative modes is conjectured to vanish. The nearest results are Michelen’s real microscopic root process [375] and the non-Archimedean mean, correlation, and universality theory of Shmueli, Caruso, and Shen [376, 377, 378], whose expected  $\mathbb{Q}_p$  root count is  $(p-1)/(p+1)$ ; no joint Archimedean/ $p$ -adic root theorem for a single random coefficient sequence was found. A first decisive theorem would be  $(\mathcal{R}_n, S_{p,n}) \Rightarrow \nu_\infty \otimes \sigma_p$  for the count  $S_{p,n}$  of simple roots modulo  $p$ , each of which Hensel-lifts uniquely. In tests on the corpus described in the stress-test section, the expected  $p$ -adic root count approaches  $(p-1)/(p+1)$  and the real-near-unit and simple-root-mod- $p$  counts show only small, slowly decaying correlations. A counterexample would be a surviving coupling at  $x = \pm 1$ , where a real root near 1 and the event  $F_n(1) \equiv 0 \pmod{p}$  share the evaluation  $F_n(1)$ ; the correct theorem might then require conditioning on finitely many shared evaluation modes.

## Part XXVI

# Entropy and rare-event laws for random algebraic and topological structures

*Added to the deposit: 29 August 2026.*

The five conjectures below place deep algebraic and topological invariants—matrix-polynomial spectra, matroid Kazhdan–Lusztig polynomials, Khovanov homology, Jacobian lattices of hyperbolic surfaces, and Heegaard Floer homology—on canonical random ensembles and assert entropy, threshold, and rare-event laws for them. They fall in five settings. A random-matrix setting (Conjecture 389) matches the zeros of a random matrix polynomial at a tropical breakpoint to a two-coefficient pencil in a double-scaling regime. A matroid setting (Conjecture 390) locates the transition from subextensive to extensive Kazhdan–Lusztig complexity of the random graphic matroid exactly at the Erdős–Rényi giant-cycle threshold. A categorified-invariant setting (Conjecture 391) asserts a deterministic convex free energy for the Khovanov bigrading of random braid closures. A random-surface setting (Conjecture 392) pairs every fixed initial segment of Jacobian successive minima with the homologically independent short-geodesic lengths through

a two-sided thin-handle equivalence. A three-manifold setting (Conjecture 393) asserts that the Heegaard Floer rank of a random Heegaard splitting grows at exactly the homological torsion rate, with no excess exponential entropy. Each is a determinate law with a stated mechanism, a nearest-literature boundary, a first decisive theorem, and a route to disproof. The proved layer is Proposition 76: the two-face tropical reduction and its spherical-ensemble pencil law, the block multiplicativity of matroid Kazhdan–Lusztig polynomials with the triviality of forests, the convexity and linear boundedness of the Khovanov free energy at finite length, the all-fixed- $k$  short-geodesic Poisson tails with their collar-capacity normalization, and the Floer–homology lower bound  $\widehat{\text{rank}} HF \geq |H_1|$  on rational homology spheres. The statements are asymptotic laws for invariants whose exact computation is infeasible at scale, so the accompanying diagnostics test exact finite calibrations, decategorified proxies, and reduced models rather than the laws themselves; where a proxy does not reach the hard mechanism, in particular the homological cancellations of Conjectures 391 and 393, that gap is recorded where it occurs.

The homological successive-length functional  $\ell_j^{\text{hom}}$  used below is defined explicitly in Conjecture 392.

**Proposition 76** (Anchor identities and reductions). (i) Let  $P_{\beta,N}(z) = \sum_{k=0}^d e^{\beta a_k} z^k G_{k,N}$  with  $G_{k,N}$  independent complex Ginibre matrices of entry variance  $1/N$ , let  $x_*$  be a tropical breakpoint of  $\Phi(x) = \max_k(a_k + kx)$  attained exactly by  $r < s$ , put  $m = s - r$  and  $\delta = \min_{k \neq r,s}(\Phi(x_*) - (a_k + kx_*)) > 0$ . After the rescaling  $z = e^{\beta x_*} \zeta$  and removal of the common scalar factor, every inactive coefficient carries the factor  $e^{-\beta \Delta_k} \leq e^{-\beta \delta}$ , and the zeros of the dominant pencil  $G_{r,N} + \zeta^m G_{s,N}$  are exactly the  $m$ -th roots of the eigenvalues of  $-G_{s,N}^{-1} G_{r,N}$ : the  $m$ -th-root lift of the complex spherical ensemble. (ii) Matroid Kazhdan–Lusztig polynomials are multiplicative under direct sums, the cycle matroid of a graph is the direct sum of the cycle matroids of its blocks, every Boolean matroid has  $P \equiv 1$ , and hence every forest has  $P_{M(G)} \equiv 1$ ; the cycle  $C_k$  contributes the uniform matroid  $U_{k-1,k}$ . (iii) For any link diagram with  $c$  crossings on  $m$  strands,  $\log Z(u, v) = \log \sum_{i,j} \dim_{\mathbb{F}_2} Kh^{i,j} e^{ui+vj}$  is convex in  $(u, v)$  and bounded by  $C(c + m)(1 + |u| + |v|)$ , so along a random braid word of length  $n$  with bounded generators,  $n^{-1} \log Z_n(u, v)$  is convex and uniformly bounded on compacta. (iv) With the Weil–Petersson short-geodesic intensity  $\lambda(\ell) = \frac{e^\ell + e^{-\ell} - 2}{2\ell}$ , the Poisson mean is  $\Lambda(\varepsilon) = \int_0^\varepsilon \lambda = \frac{\varepsilon^2}{4} + O(\varepsilon^4)$ . For each fixed  $k$ , the  $k$  shortest curves have connected complement with probability tending to one in the large-genus limit; consequently  $\varepsilon^{-2k} \Pr(\ell_k^{\text{hom}}(X_g) \leq \varepsilon) \rightarrow 1/(4^k k!)$  as  $g \rightarrow \infty$  and then  $\varepsilon \downarrow 0$ . The collar capacity satisfies  $\pi \text{cap}(\ell)/\ell \rightarrow 1$  as  $\ell \downarrow 0$ . (v) For a rational homology sphere  $Y$ ,  $\widehat{\text{rank}}_{\mathbb{F}_2} HF(Y) \geq |H_1(Y; \mathbb{Z})|$ , so the excess Floer entropy  $E = \log \widehat{\text{rank}}_{\mathbb{F}_2} HF(Y) - \log |H_1(Y; \mathbb{Z})|$  is nonnegative.

*Proof.* (i) Dividing  $P_{\beta,N}(e^{\beta x_*} \zeta)$  by  $e^{\beta \Phi(x_*)}$  gives coefficient  $e^{\beta(a_k + kx_* - \Phi(x_*))} = e^{-\beta \Delta_k}$  at  $\zeta^k$ , with  $\Delta_r = \Delta_s = 0$  and  $\Delta_k \geq \delta$  otherwise. Since  $G_{s,N}$  is almost surely invertible,  $\det(G_{r,N} + \zeta^m G_{s,N}) = \det(G_{s,N}) \prod_i (\zeta^m - \lambda_i)$  with  $\lambda_i$  the eigenvalues of  $-G_{s,N}^{-1} G_{r,N}$ ; the eigenvalues of  $G_{s,N}^{-1} G_{r,N}$  form the complex spherical ensemble [381], which is invariant under rotation, so the zeros are its  $m$ -th-root lift. (ii) Multiplicativity and  $P_{B_n} \equiv 1$  are proved by Elias, Proudfoot, and Wakefield [382]; a bridge is a coloop, so a forest’s cycle matroid is Boolean, and the block decomposition of the cycle matroid is classical. (iii)  $Z$  is a finite sum of exponentials with nonnegative integer weights, so  $\log Z$  is convex by Hölder; the cube of resolutions has  $2^c$  vertices, each carrying at most  $c + m$  circles, so  $\dim_{\mathbb{F}_2} Kh \leq 2^{2c+m}$  while  $|i| \leq c$  and  $|j| \leq 2c + m + c$ , giving the linear bound; a word of length  $n$  in fixed generators has  $c \leq Cn$  crossings. (iv)  $\lambda(\ell) = \sum_{r \geq 1} \ell^{2r-1}/(2r)!$ , so  $\Lambda(\varepsilon) = \sum_{r \geq 1} \varepsilon^{2r}/(2r(2r)!) = \frac{\varepsilon^2}{4} + \frac{\varepsilon^4}{96} + \dots$ . Mirzakhani and Petri [388] give the Poisson



law and show that, for each fixed  $k$ , the relevant short  $k$ -multicurve has connected complement with probability tending to one; its classes are therefore homologically independent. Hence  $\Pr(\ell_k^{\text{hom}} \leq \varepsilon) \sim \Pr(\text{Poisson}(\Lambda(\varepsilon)) \geq k) \sim \Lambda(\varepsilon)^k/k!$ , proving the displayed constant. The collar of a geodesic of length  $\ell$  has modulus  $\pi/\ell + O(1)$ , whence  $\pi \text{cap}(\ell)/\ell \rightarrow 1$ . (v)  $\widehat{HF}(Y)$  splits over the  $|H_1(Y; \mathbb{Z})|$   $\text{Spin}^c$  structures, and each summand has Euler characteristic  $\pm 1$  [392], hence is nonzero over  $\mathbb{F}_2$ .  $\square$

**Conjecture 389** (Double-scaling tropical shadowing). *In the setting of Proposition 76(i), suppose  $\beta_N \delta / \log N \rightarrow \infty$ . Then on every compact annular sector  $K \subset \mathbb{C}^\times$  contained in one branch of the  $m$ -th-root map and of angular width strictly less than  $2\pi/m$ , after removing a boundary layer whose limiting intensity vanishes, the zeros of the full matrix polynomial in  $K$  can be matched to the zeros of  $G_{r,N} + \zeta^m G_{s,N}$  so that the maximal matching displacement is  $o_{\mathbb{P}}(N^{-1/2})$ . Consequently, after microscopic unfolding, the selected zero point process has the same local weak limit as the corresponding single-branch lift of the complex spherical ensemble.*

**Significance.** At fixed  $N$  the limit  $\beta \rightarrow \infty$  is a perturbative reduction to the two-face pencil, but in the joint regime the exponentially small inactive coefficients compete with the  $N^{-1/2}$  microscopic spacing and the nonnormal condition numbers of the pencil, and the conjecture asserts that the tropical combinatorics still selects the local law; the matching formulation is primary, and pointwise convergence of correlation densities is a consequence only under additional uniform-integrability and density hypotheses. The nearest results are the tropical localization of matrix-polynomial eigenvalues of Noferini, Sharify, and Tisseur [379] and Akian, Gaubert, and Sharify [380], which bound moduli but carry no microscopic law, and the spherical-ensemble limit of Krishnapur [381]. The obstruction to the uniform matching is that the minimal spacing and the largest eigenvalue condition number of the random pencil are extreme statistics over  $N$  roots, and a nonnormal generalised eigenvalue can move much farther than the perturbation norm, so the threshold may need a condition-number exponent; a safer first statement matches all but  $o(N)$  well-conditioned zeros. A first decisive theorem would be the case  $d = 2$  with one exponentially suppressed coefficient at a single breakpoint, the smallest case in which the matching is not tautological, or any joint regime  $\beta_N = c \log N$  with  $c$  large. In tests on the corpus described in the stress-test section, at profile  $(-2, 0, -1, 0)$  the  $\sqrt{N}$ -scaled optimal matching displacement fell from 0.095 to 0.009 across  $N = 4$  to 20 at  $\beta = 2 \log N$  and stalled near 0.4 at  $\beta = 0.75 \log N$ , locating the double-scaling threshold. A counterexample would be a breakpoint and regime with  $\beta_N \delta / \log N \rightarrow \infty$  whose matching displacement fails to be  $o(N^{-1/2})$ , and the crossover  $\beta_N \delta \asymp c \log N$  with its possible critical kernels is the sharper open boundary.

**Conjecture 390** (Kazhdan–Lusztig entropy percolation). *Let  $G_n \sim G(n, c/n)$ , let  $M(G_n)$  be its cycle matroid,  $P_{M(G_n)}(t)$  its matroid Kazhdan–Lusztig polynomial, and  $H_n(c) = n^{-1} \log P_{M(G_n)}(1)$ . There is a deterministic  $H_{\text{KL}}: (0, \infty) \rightarrow [0, \infty)$  such that  $H_n(c) \rightarrow H_{\text{KL}}(c)$  in probability for every fixed  $c \neq 1$ , with  $H_{\text{KL}}(c) = 0$  for  $0 < c < 1$  and  $H_{\text{KL}}(c) > 0$  for  $c > 1$ ; and at criticality  $H_n(1) \rightarrow 0$  in probability.*

**Significance.** The Erdős–Rényi giant-cycle transition at  $c = 1$  is asserted to be simultaneously the transition from subextensive to extensive intersection-cohomology complexity of the random graphic matroid: by Proposition 76(ii) forests contribute nothing and the carrier of entropy is the cycle-connected material, with the cycle calibration  $k^{-1} \log P_{U_{k-1,k}}(1) \rightarrow \log 3$  showing that sparse degree-two cyclic blocks already carry positive entropy density. A proof of the



critical clause requires a quantitative upper bound on  $\log P_M(1)$  in terms of the size of the critical cyclic core, and a supercritical lower bound should use genuine direct-sum blocks or a proved comparison inequality rather than monotonicity under arbitrary minors, which is not available. The subcritical clause is close to a theorem: by Proposition 76(ii) the polynomial factors over the unicyclic components, whose total size is tight for  $c < 1$ , so the clause reduces to an exponential bound  $P_{U_{k-1,k}}(1) \leq C^k$ , which the explicit coefficients of the uniform matroids supply; the content is the critical clause and the supercritical limit and positivity. The nearest results are the matroid Kazhdan–Lusztig theory of Elias, Proudfoot, and Wakefield [382], the uniform-matroid formulas of Gedeon, Proudfoot, and Young [383], and the deletion inequality of Braden and Vysogorets [384]. A first decisive theorem would be the subcritical clause, where only bounded unicyclic complexity survives, or a lower bound for the giant cycle-connected core. In tests on the corpus described in the stress-test section, an independent implementation of the defining recursion reproduced the closed cycle formula exactly through  $k = 25$  and the entropy sequence rising toward  $\log 3$ . A counterexample would be a supercritical density with subextensive Kazhdan–Lusztig complexity, or extensive complexity below the threshold.

**Conjecture 391** (Khovanov free energy of random braid closures). *Fix  $m \geq 3$  and a finitely supported probability measure  $\mu$  on the braid group  $B_m$  whose induced walk is aperiodic on the symmetric-group quotient and whose support generates a subgroup whose image in  $B_m/Z(B_m)$ , acting on the  $m$ -punctured disc, is a nonelementary subgroup of its mapping class group. Let  $\beta_n = X_1 \cdots X_n$  with  $X_i$  i.i.d.  $\mu$ , let  $L_n = \widehat{\beta_n}$  be the oriented closure, and with normalized oriented Khovanov bigradings over  $\mathbb{F}_2$  set  $Z_n(u, v) = \sum_{i,j} \dim_{\mathbb{F}_2} Kh^{i,j}(L_n; \mathbb{F}_2) e^{ui+vj}$ . For every fixed  $(u, v) \in \mathbb{R}^2$  there is a deterministic finite  $F_\mu(u, v)$  with  $n^{-1} \log Z_n(u, v) \rightarrow F_\mu(u, v)$  almost surely and in  $L^1$ , and  $F_\mu$  is convex; in particular  $n^{-1} \log \text{rank}_{\mathbb{F}_2} Kh(L_n; \mathbb{F}_2) \rightarrow F_\mu(0, 0)$ .*

**Significance.** Finite-length convexity and linear boundedness are Proposition 76(iii); the conjectural content is the deterministic limit itself, since closure and passage to homology destroy the subadditivity available at the chain level, so no Kingman-type argument applies directly, and differentiability of  $F_\mu$ , a large-deviation principle for the bigrading, and a variational characterization are separate later questions rather than hypotheses. The nearest results are the categorified Jones polynomial of Khovanov [385], the Lyapunov theory of Alexander polynomials of random positive braids of Dunfield and Tiozzo [386], and the polynomial-time algorithm for Khovanov homology of three-braid closures of Kelomaki and Schütz [387], which makes direct testing at  $m = 3$  feasible. The existence of the limit is not a consequence of Kingman’s theorem, since the Khovanov homology of a closure has no evident submultiplicativity under multiplication of braids; the missing ingredient is a quasi-multiplicative inequality for a tangle-level norm, compared afterwards with the closed-braid partition function. A first decisive theorem would be that inequality, giving the existence of the limit at  $(u, v) = (0, 0)$  for one nondegenerate walk on  $B_3$ . In tests on the corpus described in the stress-test section, the decategorified reduced-Burau proxy at  $t = -1$  gave a stable positive determinant exponent near 0.08 on three-strand walks, a scale check that does not test the homological cancellation itself, the recorded gap. A counterexample would be almost-sure oscillation of  $n^{-1} \log \text{rank} Kh$  between distinct limit points for a nondegenerate walk.

**Conjecture 392** (Two-sided thin-handle Hodge successive-minima equivalence). *Let  $X_g$  be a closed genus- $g$  hyperbolic surface sampled from normalized Weil–Petersson volume, equip  $H^1(X_g; \mathbb{Z})$  with the Hodge  $L^2$  norm, and let  $m_j(J(X_g))$  be the successive minima of the Jacobian*

*lattice. Define*

$$\ell_j^{\text{hom}}(X_g) = \inf \left\{ L : \text{rank}_{\mathbb{Z}} \left\langle [\gamma] : \gamma \text{ primitive nonseparating simple closed geodesic, } \ell(\gamma) \leq L \right\rangle \geq j \right\}.$$

*For every fixed  $k \geq 1$  and every  $\eta > 0$ ,*

$$\lim_{\varepsilon \downarrow 0} \lim_{g \rightarrow \infty} \mathbb{P} \left( \max_{1 \leq j \leq k} \left| \frac{\pi m_j(J(X_g))^2}{\ell_j^{\text{hom}}(X_g)} - 1 \right| > \eta \mid \min\{\pi m_k(J(X_g))^2, \ell_k^{\text{hom}}(X_g)\} \leq \varepsilon \right) = 0.$$

*Consequently, with the short-geodesic law of Proposition 76(iv), in the same sequential limit*

$$\varepsilon^{-2k} \mathbb{P}(\pi m_k(J(X_g))^2 \leq \varepsilon) \longrightarrow \frac{1}{4^k k!}.$$

**Significance.** The  $k = 1$  clause is the first-minimum law as originally deposited; the all-fixed- $k$  formulation extends it to every successive minimum. The two-sided conditioning is essential: collars give Hodge upper bounds when  $k$  thin independent handles are already known, but a tail transfer must also exclude  $k$  anomalously short integral Hodge combinations on a surface without those handles. The homological rank in  $\ell_j^{\text{hom}}$  excludes separating curves and bounding pairs; two disjoint homologous curves can otherwise create only one lattice direction and change the leading electrical resistance. The capacity normalization  $\pi \text{cap}(\ell)/\ell \rightarrow 1$  fixes the factor  $\pi$  only in the shrinking window, so no fixed-window point-process transfer is asserted. The nearest results are Muetzel’s successive-minimum upper bounds from short curves [389], deterministic period-matrix degeneration, and the short-geodesic process of Mirzakhani and Petri [388]; the uniform stochastic converse in large genus is the new content. A first decisive theorem is the one-sided law conditioned on exactly  $k$  short curves whose union has connected complement. A counterexample would be a positive-probability family of Hodge-thin surfaces with fewer than  $k$  corresponding independent thin handles.

**Conjecture 393** (Floer cancellation saturation for random Heegaard splittings). *Fix  $g \geq 2$  and a finitely supported probability measure  $\mu$  on the mapping class group of the closed genus- $g$  surface whose support generates it as a semigroup. Let  $\phi_n = X_1 \cdots X_n$  with  $X_i$  i.i.d.  $\mu$ , let  $Y_n = H_g \cup_{\phi_n} H_g$  be the random Heegaard splitting, and let  $Q_n = \{b_1(Y_n) = 0\}$ . Define the excess Floer entropy  $E_n = \log \text{rank}_{\mathbb{F}_2} \widehat{HF}(Y_n) - \log |H_1(Y_n; \mathbb{Z})| \geq 0$ . Conditional on  $Q_n$ ,  $E_n/n \rightarrow 0$  in probability; equivalently, whenever the homological torsion exponent  $\alpha_\mu = \lim_n n^{-1} \log |H_1(Y_n; \mathbb{Z})|$  exists in probability on  $Q_n$ , the Heegaard Floer rank satisfies  $n^{-1} \log \text{rank}_{\mathbb{F}_2} \widehat{HF}(Y_n) \rightarrow \alpha_\mu$ .*

**Significance.** Nonnegativity of  $E_n$  on  $Q_n$  is Proposition 76(v), and the conjecture asserts that the Heegaard Floer differential cancels every additional exponential contribution, so that the Floer rank grows at exactly the homological torsion rate; this is compatible with random splittings failing to be L-spaces, since  $\text{rank } \widehat{HF}/|H_1|$  may still diverge subexponentially, and the true subexponential scale of  $E_n$  is the natural second question. The nearest results are the random Heegaard splittings of Maher [390], the random-three-manifold model of Dunfield and Thurston [391] in which  $b_1(Y_n) = 0$  holds asymptotically almost surely so the conditioning is benign, and the non-L-space expectation for random splittings recorded in the problem literature [393]. A first decisive theorem would be any subexponential upper bound on  $E_n$  for one nondegenerate walk, or the identity for a solvable subfamily. In tests on the corpus described in the stress-test section, exact integral symplectic transvection walks reproduced a positive

homological exponent near 0.19, a calibration of the abelian shadow only, which cannot witness Floer cancellation, the recorded gap. A counterexample would be a walk with  $E_n \geq cn$  on  $Q_n$  with positive probability, and such an excess exponent would itself be a new Floer-theoretic dynamical invariant.

## Part XXVII

# Random Hamiltonian cocycle entropy

*Added to the deposit: 30 August 2026.*

This part states one conjecture, a quenched entropy law for random products of Hamiltonian diffeomorphisms of a closed surface; the companion law for higher Jacobian minima is stated as the all-fixed- $k$  form of Conjecture 392. Proposition 77 records two exact in-model calibrations for the random-product statement.

**Proposition 77** (Deterministic and lazy-walk calibrations). (i) If the driving measure is the point mass  $\delta_\phi$  at a Hamiltonian diffeomorphism of a closed symplectic surface, the all-free-homotopy-class finite-bar side of Conjecture 394 is the deterministic barcode entropy of  $\phi$ , hence equals  $h_{\text{top}}(\phi)$ . (ii) More generally, fix  $p \in (0, 1)$  and  $\mu = (1 - p)\delta_{\text{id}} + p\delta_\phi$ . If  $S_n$  is the number of  $\phi$ -steps by time  $n$ , then  $\Phi_n = \phi^{S_n}$  and

$$h_{\text{top}}^{\text{a}}(\mu) = p h_{\text{top}}(\phi), \quad \lim_{\varepsilon \downarrow 0} \limsup_{n \rightarrow \infty} \frac{1}{n} \log^+ b_\varepsilon^{\text{fin}}(\Phi_n) = p h_{\text{top}}(\phi)$$

almost surely. Thus Conjecture 394 holds for this nontrivial two-atom commuting walk.

*Proof.* (i) The equality is the surface theorem of Çineli, Ginzburg, and Gürel [394]; the sequential formulation is developed in [395]. Their convention takes the direct sum over every free homotopy class. Removing the fixed finite collection of infinite bars does not change the exponential entropy. (ii) The sequence  $S_j$  visits every integer from 0 to  $S_{n-1}$ , so the random Bowen metric through time  $n$  is exactly the deterministic Bowen metric of  $\phi$  through time  $S_{n-1}$ . Also  $b_\varepsilon^{\text{fin}}(\Phi_n) = b_\varepsilon^{\text{fin}}(\phi^{S_n})$ . The strong law gives  $S_n/n \rightarrow p$ ; since every integer is eventually visited, the deterministic limsup reparametrizes by the factor  $p$ . Apply (i).  $\square$

**Conjecture 394** (Quenched random Floer barcode entropy). *Let  $(\Sigma, \omega)$  be a closed symplectic surface, let  $\mu$  be a finitely supported probability measure on  $\text{Ham}(\Sigma, \omega)$ , let  $X_1, X_2, \dots$  be i.i.d. with law  $\mu$ , and put  $\Phi_0 = \text{id}$  and  $\Phi_n = X_n \circ \dots \circ X_1$ . For  $\varepsilon > 0$ , fix coefficients  $\mathbb{F}_2$ , and let  $\mathcal{B}^{\text{all}}(\psi; \mathbb{F}_2)$  be the unpinned absolute fixed-point Floer barcode of  $(\Delta, \text{graph } \psi)$ , formed by the direct sum over every free homotopy class. For nondegenerate  $\psi$  set*

$$b_\varepsilon^{\text{fin}}(\psi) = \#\{I \in \mathcal{B}^{\text{all}}(\psi; \mathbb{F}_2) : \varepsilon < |I| < \infty\}.$$

*For degenerate  $\psi$ , use the lower-semicontinuous perturbation extension*

$$b_\varepsilon^{\text{fin}}(\psi) = \liminf_{\substack{\tilde{\psi} \rightarrow \psi \\ \tilde{\psi} \text{ nondegenerate}}} b_\varepsilon^{\text{fin}}(\tilde{\psi}),$$

where the convergence is through Hamiltonian diffeomorphisms in the  $C^\infty$  topology. Write  $\log^+ x = \log \max\{1, x\}$ . For a sample path  $\mathbf{X}$ , set

$$d_n^{\mathbf{X}}(x, y) = \max_{0 \leq j < n} d(\Phi_j x, \Phi_j y), \quad h_{\text{top}}^{\mathbf{q}}(\mu) = \lim_{\delta \downarrow 0} \limsup_{n \rightarrow \infty} \frac{1}{n} \log s_n(\mathbf{X}, \delta),$$

where  $s_n(\mathbf{X}, \delta)$  is the largest cardinality of a  $\delta$ -separated set for  $d_n^{\mathbf{X}}$ ; the last quantity is the almost-sure constant quenched random topological entropy. Then almost surely

$$\lim_{\varepsilon \downarrow 0} \limsup_{n \rightarrow \infty} \frac{1}{n} \log^+ b_\varepsilon^{\text{fin}}(\Phi_n) = h_{\text{top}}^{\mathbf{q}}(\mu).$$

**Significance.** The statement promotes the deterministic surface theorem to an i.i.d. cocycle and would make persistent Floer growth a symplectic formula for quenched orbit complexity. The all-class convention is essential: the deterministic entropy theorem can fail if Floer bars in noncontractible free homotopy classes are omitted [406]. The order of limits and the inner limsup are also essential because no quasi-multiplicative inequality for long-bar counts under random composition is known. More fundamentally, the right side sees the whole prefix path  $(\Phi_0, \dots, \Phi_{n-1})$ , whereas the left side sees fixed-point Floer data only for the endpoint  $\Phi_n$ . The nearest results are the deterministic equality and sequential barcode entropy of Çineli, Ginzburg, and Gürel [394, 395], Gong’s deterministic refinement [396], and Bogenschütz’s random entropy formalism [397]. Beyond an upper composition/volume estimate, a proof needs a random closing or crossing-energy theorem that turns typical prefix complexity into endpoint Floer bars. The lazy-walk proposition is an exact two-atom Hamiltonian calibration; no Anosov torus proxy is used. In tests on the corpus described in the stress-test section, only that calibration was checked, exhaustively through word length sixteen; no Floer barcode was computed. A counterexample could arise from endpoint cancellation or the absence of a random closing mechanism even when the path itself has positive entropy; the decisive stress test is a measure whose support contains nearly cancelling diffeomorphisms, which the commuting and lazy calibrations do not probe.

## Part XXVIII

# Cut-rank laws for graph states, forests, and random width

*Added to the deposit: 30 August 2026.*

For a finite simple graph  $G = (V, E)$  and  $S \subseteq V$ , write

$$\rho_G(S) = \text{rank}_{\mathbb{F}_2} A_G[S, V \setminus S], \quad Z_G(x, y) = \sum_{S \subseteq V} x^{|S|} y^{\rho_G(S)}.$$

For an edge-labelled graph over  $\mathbb{F}_q$ , the same symbol denotes rank over  $\mathbb{F}_q$ ; label zero means that the edge is absent. Fixing a nontrivial additive character  $\chi: \mathbb{F}_q \rightarrow \mathbb{C}^\times$ , its associated  $q$ -level graph state is

$$|G\rangle = q^{-|V|/2} \sum_{z \in \mathbb{F}_q^V} \chi\left(\sum_{u < v} A_{uv} z_u z_v\right) |z\rangle.$$

This explicit convention applies to every prime power, including even characteristic. We write  $\text{rw}$  and  $\text{bw}$  for rank-width and branch-width,  $\nu$  for maximum matching number, and  $\text{Odd}_G(A)$  for the vertices having an odd number of neighbours in  $A$ .

The fifteen conjectures below form four connected programmes: quenched and extreme cut-rank laws for dense random graph states; shape, stability, extremal, and reconstruction laws for forests; extensive width constants in random regular, sparse, and percolated graphs; and typical rigidity and orbit entropy under local complementation. The graph-state Schmidt-rank identity, local-complementation invariance, the Oum–Seymour theory of rank-width, and the known order-of-magnitude bounds for random graphs are calibrations, not new claims [398, 399, 400, 402].

**Proposition 78** (Cut-rank calibrations). (i) Across the bipartition  $S \mid V \setminus S$ , the graph state  $|G\rangle$  has Schmidt rank  $q^{\rho_G(S)}$  and entropy  $\rho_G(S) \log q$ . Binary local complementation, and its weighted finite-field analogue, preserve the whole labelled function  $S \mapsto \rho_G(S)$ . (ii) If  $F$  is a forest and a uniform random bipartition is chosen, its crossing-edge set is a uniform subset  $B \subseteq E(F)$ , every  $B$  has the same colouring-fibre size, and

$$\rho_F(S) = \nu((V(F), B)).$$

(iii) Put  $P_{q,0} = 1$ ,  $P_{q,k} = \prod_{i=1}^k (1 - q^{-i})$ , and  $P_{q,\infty} = \prod_{i \geq 1} (1 - q^{-i})$ . The limiting nullity law of a uniform  $m \times (m + s)$  matrix over  $\mathbb{F}_q$ , for fixed  $s \geq 0$ , is

$$\pi_{q,s}(d) = q^{-d(d+s)} \frac{P_{q,\infty}}{P_{q,d} P_{q,d+s}}, \quad d \geq 0.$$

*Proof.* For (i), put  $M = A_G[S, V \setminus S]$ . Diagonal unitaries remove the phases contributed by edges internal to either side, leaving the coefficient matrix

$$q^{-|V|/2} (\chi(u^\top M v))_{u \in \mathbb{F}_q^S, v \in \mathbb{F}_q^{V \setminus S}}.$$

Its rows are additive characters indexed by the image of  $M^\top$ : there are exactly  $q^{\text{rank } M}$  distinct rows, they are mutually orthogonal, and they occur with equal multiplicity. Hence the nonzero Schmidt coefficients are equal and number  $q^{\text{rank } M}$ , which gives the asserted rank and entropy. For  $q = 2$  this is the standard graph-state identity [398]. Define weighted local complementation at  $v$  with parameter  $a \in \mathbb{F}_q$  by  $A'_{jk} = A_{jk} + a A_{jv} A_{vk}$  for  $j \neq k$ , with zero diagonal, and define vertex scaling at  $v$  by multiplying all incident labels by  $b \in \mathbb{F}_q^\times$ . These are the finite-field graph operations of Bahramgiri–Beigi for arbitrary prime-power  $q$  [420]; in their odd-prime graph-state setting they describe the local Clifford action [421]. Across a fixed cut, weighted local complementation adds multiples of the  $v$ -row to the other crossing rows when  $v \in S$ , or of the  $v$ -column to the other crossing columns when  $v \notin S$ ; vertex scaling multiplies one crossing row or column by a nonzero scalar. This direct argument works over every finite field, so every cut rank is preserved over  $\mathbb{F}_q$ . On a forest, fixing one vertex colour in each connected component makes the map from the remaining vertex two-colourings to crossing-edge sets bijective; hence every crossing-edge set has exactly  $2^{\kappa(F)}$  colouring preimages. A bipartite forest’s adjacency bi-matrix has rank over  $\mathbb{F}_2$  equal to its term rank: a nonzero maximal minor has a unique perfect matching, since two perfect matchings would create a cycle. This is (ii). Finally, the exact rank count for finite rectangular matrices, divided by  $q^{m(m+s)}$  and then sent to infinity, gives (iii).  $\square$

**Conjecture 395** (Quenched balanced-cut nullity universality). *Fix a prime power  $q$  and  $s \in \{0, 1\}$ . Let  $G_{2m+s,q}$  be the loopless complete edge-labelled graph in which every unordered vertex pair receives an independent uniform label in  $\mathbb{F}_q$  (with zero interpreted as no edge), and set*

$$\mu_G(d) = \binom{2m+s}{m}^{-1} \#\{S : |S| = m, m - \rho_G(S) = d\}.$$

*Then  $\mu_G$  converges in probability, in total variation, to  $\pi_{q,s}$ .*

**Significance.** Every fixed cut matrix is uniform. Its exact finite nullity probability is

$$q^{-d(d+s)} \frac{P_{q,m} P_{q,m+s}}{P_{q,d} P_{q,d+s} P_{q,m-d}},$$

which tends to Proposition 78(iii); the content is self-averaging over the exponentially many, heavily overlapping cuts of one state. A proof would turn the random-matrix nullity law into a quenched entanglement spectrum and requires dependence-sensitive concentration, not independent sampling. The nearest results concern fixed-cut stabilizer entanglement and average cut rank, rather than the empirical law of all cuts of one realization [401]. In tests on the corpus described in the stress-test section, the finite nullity law was reproduced exactly by matrix rank computation over small fields; the self-averaging itself was not tested.

**Conjecture 396** (Dependence-sensitive extreme balanced-cut defect). *In the setting of Conjecture 395, let  $D_{\max}(G) = \max_{|S|=m} (m - \rho_G(S))$ . Then*

$$\frac{D_{\max}(G_{2m+s,q})}{\sqrt{(2m+s) \log_q 2}} \longrightarrow 1$$

*in probability.*

**Significance.** The exact rank formula gives  $\Pr\{\text{nullity} \geq d\} = q^{-d(d+s)+O_q(1)}$  uniformly for  $d = O(\sqrt{m})$ , while  $\log_q \binom{2m+s}{m} = (2m+s) \log_q 2 + O(\log m)$ ; these formulas force the displayed first-moment location. The union-bound upper estimate follows. The lower bound is a genuine de-clustering problem: overlapping cuts share quadratically many entries, complementary cuts are identical when  $s = 0$ , and rare defective cuts might occur in large dependency clusters. In tests on the corpus described in the stress-test section, binary enumeration at orders 10–18 gave mean normalized values 0.72–0.80, increasing across that range; this calibrates the scale but is not treated as independence evidence.

**Conjecture 397** (Poisson law for random graph-state distance). *For  $G \sim G(n, 1/2)$  put*

$$f(G) = \min_{\emptyset \neq A \subseteq V} (|A| + |\text{Odd}_G(A) \setminus A|)$$

*and*

$$\Lambda_n(\ell) = \sum_{a=1}^{\ell} \binom{n}{a} \Pr\{\text{Bin}(n-a, 1/2) \leq \ell - a\}.$$

*Uniformly along integer sequences for which  $\Lambda_n(\ell)$  remains in a compact subset of  $(0, \infty)$ , the number of nonempty  $A$  whose displayed weight is at most  $\ell$  converges in total variation to  $\text{Poisson}(\Lambda_n(\ell))$ . Consequently  $\Pr(f(G) > \ell) = e^{-\Lambda_n(\ell)} + o(1)$ .*



**Significance.** Here  $f(G) = \delta_{\text{loc}}(G) + 1$ , the minimum degree up to local complementation plus one. To make the code equivalence explicit, choose  $\omega \in \mathbb{F}_4 \setminus \mathbb{F}_2$  and set

$$C_G = \{x(A_G + \omega I) : x \in \mathbb{F}_2^V\} \subseteq \mathbb{F}_4^V.$$

Symmetry of  $A_G$  makes the rows trace-Hermitian orthogonal, and their  $\omega$ -diagonal entries make them independent over  $\mathbb{F}_2$ , so  $C_G$  is self-dual. The word indexed by  $A = \text{supp } x$  has support  $A \cup \text{Odd}_G(A)$ . Consequently its minimum nonzero Hamming weight is

$$\min_{\emptyset \neq A \subseteq V} |A \cup \text{Odd}_G(A)| = \delta_{\text{loc}}(G) + 1 = f(G),$$

where the equality to  $\delta_{\text{loc}}(G) + 1$  is the standard local-minimum-degree characterization [407, 408]. The mean has the equivalent exact form

$$\Lambda_n(\ell) = 2^{-n} \sum_{w=1}^{\ell} \binom{n}{w} (3^w - 1),$$

whose first-order threshold is already known. The new assertion is the Poisson finite-length law: the sharp random-linear-code analogue does not cover the symmetric graph-state generator, where  $A$ ,  $B$ , and  $A \triangle B$  create structured dependencies [409]. In tests on the corpus described in the stress-test section, the exact mean formula was checked against enumeration at small orders; the Poisson law itself was not tested.

**Conjecture 398** (Typical rigidity of the labelled cut-rank function). *For  $G \sim G(n, 1/2)$ , with probability tending to one, every graph  $H$  on the same labelled vertex set satisfying*

$$\rho_H(S) = \rho_G(S) \quad (S \subseteq V)$$

*is obtainable from  $G$  by local complementations, without permuting the vertices.*

**Significance.** The converse is standard, but universal completeness is false. A current unlabelled-orbit census finds no collision below nine vertices and respectively 1, 3, 15 colliding LC-orbit pairs at orders 9, 10, 11 [410]; structural descriptions through minimal local sets give another nearby boundary [411]. Those results are unlabelled, whereas the conjecture and the exact verifier concern one fixed labelled vertex set. In tests on the corpus described in the stress-test section, exhaustive enumeration through six vertices found exact equality between labelled cut-rank fibres and labelled local-complementation orbits, including all 760 fibres at order six. The conjecture says that the alternative isotropic representations behind structured exceptions have vanishing density. Since local-unitary equivalence preserves every cut rank, the claim would also imply generic LU-LC; no redundant LU-LC conjecture is stated separately.

**Conjecture 399** (Positive-rank-tail log-concavity). *For every connected graph  $G$ , let  $c_r(G) = \#\{S \subseteq V(G) : \rho_G(S) = r\}$  and  $R = \max_S \rho_G(S)$ . Then*

$$c_r(G)^2 \geq c_{r-1}(G)c_{r+1}(G), \quad 2 \leq r < R.$$

**Significance.** The statement deliberately begins at rank two, and that starting point is sharp. The connected graph on vertices  $0, \dots, 12$  with edge set

$$\begin{aligned} &\{(0, 8), (0, 10), (1, 2), (1, 8), (2, 7), (3, 7), (3, 12), (4, 6), \\ &\quad (4, 7), (4, 8), (5, 6), (5, 8), (7, 8), (7, 9), (7, 10), (7, 11), \\ &\quad (7, 12), (8, 9), (8, 11), (8, 12)\} \end{aligned}$$

has exact cut-rank profile  $(2, 28, 410, 1896, 3296, 2208, 352)$ , so  $c_1^2 = 784 < 820 = c_0 c_2$ . Thus rank-one log-concavity is false, not merely unproved. Conditioning on  $|S|$  fails already at order four. In tests on the corpus described in the stress-test section, the positive tail passed all 1,252 nonempty graph-atlas graphs on at most seven vertices and a separate seeded sample of 8,887 connected random graphs of orders 8–13. A proof would expose a previously unknown negative-dependence or injection principle for graph-state entanglement multiplicities.

**Conjecture 400** (Full log-concavity for percolated forests). *For a forest  $F$ , define*

$$a_j(F) = \#\{B \subseteq E(F) : \nu((V(F), B)) = j\}.$$

*Then  $(a_0(F), \dots, a_{\nu(F)}(F))$  is log-concave.*

**Significance.** By Proposition 78(ii), these coefficients are, up to a constant colouring fibre, the cut-entanglement law of a forest graph state. The rooted recursion

$$A_T = \prod_i (A_i + 2B_i), \quad B_T = z \left( 2^k \prod_i (A_i + B_i) - \prod_i (A_i + 2B_i) \right), \quad H_T = A_T + B_T,$$

with  $A_{\text{leaf}} = 1$  and  $B_{\text{leaf}} = 0$ , computes the polynomial without enumerating edge sets. In tests on the corpus described in the stress-test section, the recursion was used to check exact log-concavity on every unlabelled tree through order 20; the checked-in default verifier runs a smaller reproducible range. The boundary is real: real-rootedness fails at seven vertices for  $(1, 13, 30, 20)$ , while ultra-log-concavity fails for the 13-vertex graph6 tree `LqD?I?@0??g??@` with profile  $(1, 75, 432, 1060, 1360, 912, 256)$ .

**Conjecture 401** (Hurwitz stability of the percolated-matching polynomial). *For every forest  $F$ , every zero of*

$$H_F(z) = \sum_{B \subseteq E(F)} z^{\nu((V(F), B))}$$

*has strictly negative real part.*

**Significance.**  $H_F$  is neither the classical matching polynomial nor the maximal-matching polynomial: its coefficients count all edge subsets according to the maximum matching number they induce. The preceding two-state recursion gives a forest-specific mechanism, distinct from the Hochster-strand Hurwitz conjecture in Part VII. In tests on the corpus described in the stress-test section, the Routh conditions were certified exactly on every unlabelled tree through order 20, on paths through order 80, and on several large structured families; the checked-in default verifier uses exact coefficients with a smaller numerical root scan. There can be no uniform root-modulus margin, since  $H_{S_n}(z) = 1 + (2^{n-1} - 1)z$  has a negative root tending to zero.

**Conjecture 402** (Path–star stochastic extremality under bond noise). *If  $T_p$  retains each edge of an  $n$ -vertex tree independently with probability  $p \in (0, 1)$ , then for every integer  $k$ ,*

$$\Pr\{\nu((S_n)_p) \geq k\} \leq \Pr\{\nu(T_p) \geq k\} \leq \Pr\{\nu((P_n)_p) \geq k\}.$$

*For  $n \geq 4$ , equality at every threshold characterizes the star and path, respectively, up to isomorphism.*

**Significance.** The star lower bound is elementary; the substantive assertion is the stochastic path upper bound and its equality rigidity. Classical results make the path and star extremal for the ordinary Hosoya index [412], but no known theorem gives the all-threshold law for the maximum matching of an edge-percolated tree. The coefficientwise form,  $N_{S_n,k,r} \leq N_{T,k,r} \leq N_{P_n,k,r}$  for the numbers  $N_{T,k,r}$  of  $r$ -edge subsets with matching number at least  $k$ , implies the stochastic order for every  $p$  at once and holds on every tree through twelve vertices in independent recomputation; it is the natural proof target. The star half is immediate: every tree on  $n$  vertices retains at least one edge with the same probability  $1 - (1 - p)^{n-1}$ , and the star has matching number at most one, so the content is the path half and the two uniqueness clauses. In tests on the corpus described in the stress-test section, exact Bernstein-basis comparison through order 17 checked every probability  $p$  simultaneously; the default verifier runs through order 11. A proof should couple the rooted matching recursions or identify tree transformations preserving stochastic order.

**Conjecture 403** (Fixed-leaf broom minimum). *For  $2 \leq L \leq n - 1$ , obtain  $B_{n,L}$  from  $K_{1,L}$  by subdividing one pendant edge  $n - L - 1$  times. For every fixed  $p \in (0, 1)$ , among all  $n$ -vertex trees  $T$  with exactly  $L$  leaves,*

$$\nu((B_{n,L})_p) \leq_{\text{st}} \nu(T_p)$$

*and equality in distribution holds if and only if  $T \cong B_{n,L}$ .*

**Significance.** The endpoint classes  $L = 2$  and  $L = n - 1$  are unique; the claim begins at  $3 \leq L \leq n - 2$ . The same broom is already known to minimize the ordinary Hosoya index at fixed leaf count [413, 414]; the new content is stochastic domination of the maximum matching after bond percolation. The coefficientwise form  $N_{B_{n,L},k,r} \leq N_{T,k,r}$  likewise holds on every tree through twelve vertices in independent recomputation, and generic collisions of the whole percolated matching distribution between nonisomorphic trees begin at ten vertices, so the uniqueness clause cannot be derived from the distribution determining the tree. In tests on the corpus described in the stress-test section, coefficientwise checks through order 17 found neither an inequality failure nor another equality case.

**Conjecture 404** (Typical tree reconstruction from the entanglement enumerator). *Let  $T_n$  be a uniform labelled tree on  $[n]$ . With probability tending to one, every tree  $T'$  satisfying  $Z_{T'}(x, y) = Z_{T_n}(x, y)$  is isomorphic to  $T_n$ .*

**Significance.** Universal reconstruction is false and is not asserted. The first collision occurs at order 15: the nonisomorphic graph6 trees `NpCa?C@?a??@?@0???G` and `NpCc?D??G?_@0???g??` have the same  $Z$ . In tests on the corpus described in the stress-test section, exhaustive enumeration found this to be the only collision class among all 7,741 unlabelled 15-vertex trees, with none through order 14. The claim asks whether this exceptional sparsity persists asymptotically; no

further unrooted collision was found through order 18. The decisive obstruction is a Schwenk-type pair of nonisomorphic rooted limbs with identical rooted transfer data for  $Z$  [415]: such limbs could be substituted in arbitrary hosts, and fringe-subtree laws would then threaten the typical claim [416]. No rooted transfer collision was found through order 16. A proof must rule out substitutable limbs or show that their occurrence is negligible. The consequence is typical identifiability, not an algorithmic inversion claim.

**Conjecture 405** (Rank-width density of random regular graph states). *For each fixed  $d \geq 3$ , if  $G_{n,d}$  is a uniform simple  $d$ -regular graph and  $n$  tends through values with  $dn$  even, there is a constant  $\kappa_d \in (0, 1/3]$  such that*

$$\frac{\text{rw}(G_{n,d})}{n} \longrightarrow \kappa_d$$

*in probability.*

**Significance.** Expansion gives linear lower bounds and the universal decomposition bound gives the upper endpoint, but neither yields convergence. The proposed mechanism is a limit for the minimum balanced cut-rank profile of the configuration model, plus recursive balanced decompositions losing only  $o(n)$ . In tests on the corpus described in the stress-test section, exact dynamic programming on small cubic graphs gave mean widths 2.45, 2.60, 2.90, 3.00 at orders 8, 10, 12, 14; the checked-in verifier uses a smaller seeded sample and reports its own values. Recent average-case complexity results for random regular graph states do not establish a normalized rank-width limit [417].

**Conjecture 406** (Cubic branch-width density and kernel transfer). *As  $N \rightarrow \infty$  through even integers, there is a constant  $\beta_3 > 0$  such that for a uniform simple cubic graph  $H_N$ ,*

$$\frac{\text{bw}(H_N)}{N} \longrightarrow \beta_3$$

*in probability. In addition, let  $M_N$  be a cubic configuration multigraph on  $N$  vertices and, conditional on its realization, let  $H_N^*$  be any simple graph obtained using  $r_N = o(N)$  edge deletions and insertions. Form  $S_N$  by subdividing all but  $b_N = o(N)$  edges of  $H_N^*$  at least once, then applying arbitrary further subdivisions and taking arbitrary one-vertex sums with trees. Uniformly over these choices,*

$$\text{rw}(S_N) = \text{bw}(H_N^*) + o(N).$$

*Consequently, if  $\varepsilon \downarrow 0$ ,  $\varepsilon^3 n \rightarrow \infty$ , and  $G_n \sim G(n, (1 + \varepsilon)/n)$ , then*

$$\frac{\text{rw}(G_n)}{\varepsilon^3 n} \longrightarrow \frac{4}{3}\beta_3$$

*in probability.*

**Significance.** Oum's incidence-graph theorem gives rank-width within one of branch-width after subdividing every edge once [418]; the explicit transfer clause is the missing robustness statement for unreplaced edges, repairs, repeated subdivision, and pendant trees. The factor  $4/3$  is the cubic-kernel vertex density. Existing weakly supercritical results determine only the  $\Theta(\varepsilon^3 n)$  scale [403]. In tests on the corpus described in the stress-test section, only small cubic rank-width values were computed; the density and the transfer clause were not tested.

**Conjecture 407** (Rank-width surface tension for off-critical percolated grids). *For fixed  $p \in [0, 1] \setminus \{1/2\}$ , let  $Q_L(p)$  be bond percolation on the  $L \times L$  square grid. There is a deterministic  $\tau_{\text{rw}}(p)$  such that*

$$\frac{\text{rw}(Q_L(p))}{L} \longrightarrow \tau_{\text{rw}}(p)$$

*in probability, with  $\tau_{\text{rw}}(p) = 0$  for  $p < 1/2$ ,  $\tau_{\text{rw}}(p) > 0$  for  $p > 1/2$ , and  $\tau_{\text{rw}}(1) = 1$ .*

**Significance.** Below criticality all components are logarithmic. Above it, percolated-grid tree-width is linear [405], and since the percolated grid is planar the theorem of Fomin, Oum and Thilikos that a planar graph of rank-width  $k$  has tree-width at most  $72k - 2$  [404] makes its rank-width linear as well; so the sign of  $\tau_{\text{rw}}(p)$  on both sides of  $1/2$  is a consequence of the two cited theorems, and the content is the existence of the limit and its value. At  $p = 1$ , Jelínek’s exact formula  $\text{rw}(Q_L) = L - 1$  fixes the normalization [419]. The new content is existence of a surface-tension constant, plausibly through almost-subadditive rectangle gluing; the critical point is deliberately left open. In tests on the corpus described in the stress-test section, no grid experiment was run; the statement rests on the cited theorems and the exact normalization at  $p = 1$ .

**Conjecture 408** (Saturation of local-complementation orbit entropy). *Let  $\mathcal{O}_{\text{LC}}(G)$  be the set of labelled graphs obtainable from  $G$  by local complementations. For  $G \sim G(n, 1/2)$ ,*

$$\frac{1}{n} \log_2 |\mathcal{O}_{\text{LC}}(G)| \longrightarrow \log_2 3$$

*in probability.*

**Significance.** Bahramgiri and Beigi express the binary orbit size as  $\varepsilon(L)/\lambda(L)$ , where  $\varepsilon(L)$  counts Eulerian vectors of the isotropic system and  $\lambda(L)$  is its index, yielding the universal bound  $|\mathcal{O}_{\text{LC}}(G)| \leq 3^n$  [420]. The conjecture therefore asserts two separate mechanisms:  $\varepsilon(L) = 3^{n-o(n)}$  and  $\lambda(L) = 2^{o(n)}$ . In tests on the corpus described in the stress-test section, seeded exact enumeration gave mean normalized logarithms 1.285, 1.320, 1.376, 1.383 at orders 7, 8, 9, 10, with sample maxima reaching 1.471, versus  $\log_2 3 = 1.585$ . Existing orbit maps reach small orders [422]; a proof would establish maximal labelled-representative entropy, not by itself an algorithmic lower bound.

**Conjecture 409** (Rank-width density curve for sparse Erdős–Rényi graph states). *For every fixed  $c > 0$  there is a deterministic  $\kappa_{\text{ER}}(c)$  such that*

$$\frac{\text{rw}(G(n, c/n))}{n} \longrightarrow \kappa_{\text{ER}}(c)$$

*in probability, where  $\kappa_{\text{ER}}(c) = 0$  for  $c \leq 1$  and  $\kappa_{\text{ER}}(c) > 0$  for  $c > 1$ .*

**Significance.** The qualitative phases are known: subcritical components have bounded rank-width, critical components are sublinear, and every fixed supercritical density has linear rank-width [402, 403]. The new claim is existence of the normalized expectation limit; standard bounded-difference concentration is then expected to give convergence in probability. Its proposed mechanism passes to the two-core and kernel, proves a limit for balanced cut-rank optimization under the local weak degree law, and shows that attached trees do not alter the leading width.

Conjecture 406 predicts the boundary asymptotic as  $c \downarrow 1$  but does not imply the fixed- $c$  limit. In tests on the corpus described in the stress-test section, no sparse Erdős–Rényi width computation was run; the statement rests on the cited phase results.

## Part XXIX

# Arithmetic games, prime continuants and Hessian shadows

*Added to the deposit: 2 September 2026.*

This part states fifteen conjectures in four groups. The first four concern the arithmetic of the continuants of a Lebesgue-typical real number: how often a convergent denominator is prime, how prime denominators at neighbouring indices repel or attract, which squarefree and totient densities the denominators obey, and how their prime divisor counts fluctuate. Three concern Hausdorff dimensions of continued-fraction sets cut out by a primality condition on the continuants, by digit progressions, and by a residue condition on the continuants. Five concern zero-sum games whose payoff matrices are Legendre-symbol Hankel and Toeplitz boxes or random Hankel and Toeplitz matrices; for the Legendre half boxes the game value is, for many primes, exactly a ratio of coordinates of the fundamental unit of a real quadratic field. Three concern point counts of plane quartics, their Hessian curves and their dual curves over finite fields. The proved layer is Proposition 79. Throughout,  $x \in (0, 1)$  is irrational with continued fraction  $x = [0; a_1, a_2, \dots]$ , convergents  $p_n/q_n$ , and  $\mu$  is Lebesgue measure; for an odd prime  $p$ ,  $\chi_p$  is the Legendre symbol and we write  $m = (p - 1)/2$ ,

$$H_p = (\chi_p(i + j))_{1 \leq i, j \leq m}, \quad T_p = (\chi_p(i - j))_{1 \leq i, j \leq m},$$

and  $v(A)$  for the value of the zero-sum game with payoff matrix  $A$  to the row player, both players using mixed strategies. For  $p \equiv 1 \pmod{4}$ ,  $\varepsilon_p > 1$  denotes the fundamental unit of  $\mathbb{Q}(\sqrt{p})$ ; it has norm  $-1$ . The Dedekind function is  $\Psi(k) = k \prod_{\ell|k} (1 + 1/\ell)$  and the Lévy constant is  $L = \pi^2/(12 \log 2)$ , so that  $\log q_n(x) \sim nL$  for  $\mu$ -almost every  $x$  [423].

**Proposition 79** (Local laws, symmetric games, exact Legendre values and quartic identities). (i) For every integer  $k \geq 2$  and  $\mu$ -almost every  $x$ , the set  $\{n : k \mid q_n(x)\}$  has density  $1/\Psi(k)$ . For every prime  $p$  and almost every  $x$ , the set  $\{n : p \nmid q_n(x)q_{n+1}(x)\}$  has density  $(p - 1)/(p + 1)$ , and the set  $\{n : p \mid q_n(x), p \mid q_{n+2}(x)\}$  has density  $g_p/(p + 1)$ , where  $g_p = \sum_{j \geq 1} \log_2(1 + 1/(jp(jp + 2)))$  is the Gauss measure of  $\{x : p \mid a_1(x)\}$ . (ii) If  $A$  is a circulant matrix, then  $v(A)$  equals the mean of the entries of  $A$ ; if  $A$  is antisymmetric, then  $v(A) = 0$ ; and for any sequence  $(\xi_k)$  the Toeplitz matrix  $(\xi_{i-j})_{1 \leq i, j \leq m}$  and the Hankel matrix  $(\xi_{i+j-m-1})_{1 \leq i, j \leq m}$  have the same value. (iii) For  $p \in \{17, 41, 73, 89, 97, 113, 137, 193, 233, 241, 281, 313, 337, 409, 433, 449, 457, 521, 569, 601, 617\}$  one has  $v(H_p) = -y/x$ , where  $\varepsilon_p = x + y\sqrt{p}$ ; for  $p \in \{5, 13\}$  one has  $v(H_p) = -y/x$  with  $\varepsilon_p = (x + y\sqrt{p})/2$ , namely  $v(H_5) = -1$  and  $v(H_{13}) = -1/3$ ; for  $p = 401$  the unit is  $\varepsilon_p^5$  and for  $p = 577$  it is  $\varepsilon_p^7$ . For  $p \in \{5, 13, 29, 37, 61, 101, 109, 157, 181, 277, 349, 373, 421, 541, 613\}$  one has  $v(T_p) = +y/x$ , where  $\varepsilon_p^3 = x + y\sqrt{p}$ ; for  $p = 229$  the unit is  $\varepsilon_p^9$ . In every listed case  $x + y\sqrt{p}$  has norm  $-1$ , and an optimal strategy common to both players has full support  $\{1, \dots, m\}$  for  $T_p$  and support  $\{1, \dots, m - 1\}$  for  $H_p$ . (iv) Let  $q$  be an odd prime power and  $F \in \mathbb{F}_q[X, Y, Z]$  a



quartic form defining a smooth curve. Then

$$\sum_{P \in \mathbb{P}^2(\mathbb{F}_q)} \chi_q(F(P)) = q t(F),$$

where  $\chi_q$  is the quadratic character of  $\mathbb{F}_q$  and  $t(F)$  is the trace of the Frobenius on the  $E_7$  root lattice of the degree-two del Pezzo surface  $w^2 = F$ ; in particular  $t(F) \in \mathbb{Z}$  and  $|t(F)| \leq 7$ . (v) Let  $C$  be a smooth plane quartic over  $\mathbb{F}_q$ ,  $q$  a power of a prime larger than three so that the Gauss map is separable and  $C$  is reflexive, all of whose flexes are ordinary and whose 28 bitangents are ordinary (no hyperflex). Write  $b_s$  for the number of  $\mathbb{F}_q$ -rational bitangents whose two points of tangency are  $\mathbb{F}_q$ -rational and  $b_c$  for the number whose two points of tangency are swapped by the Frobenius. Then the dual curve satisfies  $\#C^\vee(\mathbb{F}_q) = \#C(\mathbb{F}_q) - b_s + b_c$ . (vi) Let  $W$  be a standard Brownian motion on  $[0, 2]$  and, for a step function  $f \geq 0$  on  $[0, 1]$  with  $\int_0^1 f = 1$ , let  $I_f(y) = \int_0^1 f(x) dW(x + y)$  for  $y \in [0, 1]$ , a continuous function of  $y$ . For  $R \geq 1$  let  $V_R$  be the supremum of  $\inf_{y \in [0, 1]} I_f(y)$  over such  $f$  with  $\|f\|_2 \leq R$ . Then  $V_R$  is a random variable, nondecreasing in  $R$ , and almost surely

$$\inf_{y \in [0, 1]} (W(y + 1) - W(y)) \leq V_R \leq \sup_{x \in [0, 1]} (W(x + 1) - W(x)),$$

so that  $V = \lim_{R \rightarrow \infty} V_R$  exists and is almost surely finite.

*Proof.* (i) Write  $r_n = q_n/q_{n-1} \in \mathbb{P}^1(\mathbb{F}_p)$  for the projective class of the column  $(q_{n-1}, q_n)$  modulo  $p$ ; since consecutive continuants are coprime,  $r_n$  is well defined, and the recursion  $q_{n+1} = a_{n+1}q_n + q_{n-1}$  gives  $r_{n+1} = a_{n+1} + 1/r_n$ . Each map  $r \mapsto a + 1/r$  is a bijection of  $\mathbb{P}^1(\mathbb{F}_p)$ , so the product of the Gauss measure with the uniform measure on  $\mathbb{P}^1(\mathbb{F}_p)$  is invariant under the skew product  $(x, r) \mapsto (Gx, a_1(x) + 1/r)$ ,  $G$  the Gauss map; this skew product is ergodic, which is the content of the equidistribution theorems of Moeckel [424] and Jager and Liardet [425] for  $(q_{n-1}, q_n)$  modulo  $p$ . The event  $p \mid q_n$  is  $r_n = 0$ , of stationary probability  $1/(p + 1)$ , and for a general modulus  $k$  the same argument on  $\mathbb{P}^1(\mathbb{Z}/k)$ , which has  $\Psi(k)$  points, gives  $1/\Psi(k)$ ; the ergodicity of the skew product for composite  $k$  is part of the cited theorems. Since  $r_n = 0$  forces  $r_{n+1} = \infty$ , the events  $p \mid q_n$  and  $p \mid q_{n+1}$  are disjoint, whence the density  $1 - 2/(p + 1)$ . Finally  $r_n = 0$  forces  $r_{n+2} = a_{n+2}$ , so  $p \mid q_n$  and  $p \mid q_{n+2}$  together mean  $r_n = 0$  and  $p \mid a_1(G^{n+1}x)$ ; the integrand  $\mathbf{1}_{r=0} \mathbf{1}_{p|a_2(x)}$  has product-measure integral  $g_p/(p + 1)$ , and the ergodic theorem for the skew product gives the density. (ii) Every row and every column of a circulant matrix is a permutation of the first, so the uniform strategy of either player makes every pure reply of the other player yield the mean entry; both guarantees coincide. Since  $v(-A^\top) = -v(A)$  for every  $A$ , if  $A^\top = -A$  then  $v(A) = v(-A^\top) = -v(A)$ , so  $v(A) = 0$ . Reversing the order of the columns turns  $(\xi_{i-j})$  into  $(\xi_{i+j-m-1})$  and does not change the value. (iii) Each matrix is symmetric. For the listed  $p$ , the linear system  $\sum_i x_i A_{ij} = v$  for  $j$  in the stated support,  $\sum_i x_i = 1$ ,  $x_i = 0$  off the support, has a unique rational solution, which is nonnegative and satisfies  $\sum_i x_i A_{ij} = v$  for every column  $j$ , including the column  $m$  omitted from the support of  $H_p$ . Thus  $x$  guarantees  $v$  to the row player, and by symmetry of  $A$  the same vector guarantees  $v$  to the column player, so  $v(A) = v$ . The rational numbers  $v$  so obtained are the stated unit coordinate ratios; the exact rational arithmetic is reproduced in the stress-test section. (iv) Let  $X$  be the surface  $w^2 = F(X, Y, Z)$  in the weighted projective space  $\mathbb{P}(1, 1, 1, 2)$ . Since  $F$  is smooth,  $X$  is a smooth del Pezzo surface of degree two, the double cover of the plane branched along  $C = \{F = 0\}$ , and  $\#X(\mathbb{F}_q) = \sum_P (1 + \chi_q(F(P)))$  because  $F(\lambda P) = \lambda^4 F(P)$ . For a smooth rational surface Weil's formula gives  $\#X(\mathbb{F}_q) = q^2 + q \operatorname{tr}(\operatorname{Frob} \mid \operatorname{Pic} \bar{X}) + 1$  [448]. The Picard lattice has rank eight and

splits orthogonally as  $\mathbb{Z}K_X \oplus K_X^\perp$  with  $K_X^\perp \cong E_7(-1)$ , the Frobenius fixing  $K_X$  and acting on  $K_X^\perp$  through the Weyl group  $W(E_7)$  [449]. Hence  $\text{tr} = 1 + t(F)$  and the character sum equals  $\#X(\mathbb{F}_q) - (q^2 + q + 1) = qt(F)$ ; an orthogonal transformation of a rank-seven lattice has trace at most seven in absolute value. (v) The Gauss map  $C \rightarrow C^\vee$ ,  $P \mapsto T_PC$ , is the normalisation of  $C^\vee$  and is defined over  $\mathbb{F}_q$ . Under the ordinarieness hypotheses the singular points of  $C^\vee$  are the 24 cusps, images of the flexes, each with one branch, and the 28 nodes, images of the bitangents, each with two branches through the two points of tangency. Away from the nodes the Gauss map is a bijection between  $\mathbb{F}_q$ -points, and a rational point of  $C$  maps to a rational point of  $C^\vee$ . A rational node whose branches are rational receives two rational points of  $C$ , a rational node with conjugate branches receives none, and a non-rational node receives no rational point since its branches are not rational. Counting gives the identity. (vi) For  $f = \sum_i f_i \mathbf{1}_{[x_i, x_{i+1})}$  one has  $I_f(y) = \sum_i f_i(W(x_{i+1} + y) - W(x_i + y))$ , continuous in  $y$ , and

$$\int_0^1 I_f(y) dy = \sum_i f_i \left( \int_{x_i+1}^{x_{i+1}+1} W - \int_{x_i}^{x_{i+1}} W \right) = \int_0^1 f(x)(W(x+1) - W(x)) dx \leq \sup_{x \in [0,1]} (W(x+1) - W(x)),$$

since  $f \geq 0$  integrates to one. The infimum of  $I_f$  over  $y$  is at most its average, which gives the upper bound uniformly in  $f$  and  $R$ . The density  $f \equiv 1$  is admissible for every  $R \geq 1$  and gives  $I_f(y) = W(y+1) - W(y)$ , which gives the lower bound. Since  $I_f(y)$  depends continuously on the breakpoints and values of  $f$ , uniformly in  $y$ , and a density of norm exactly  $R$  is a limit of densities of smaller norm, the supremum is unchanged when restricted to step functions with rational breakpoints and values, so  $V_R$  is measurable; it is nondecreasing in  $R$  and bounded, hence convergent.  $\square$

**Conjecture 410** (Prime continuant law). (i)  $\mu\{x : q_N(x) \text{ prime}\} = (2 \log 2 + o(1))/N$  as  $N \rightarrow \infty$ . (ii) For  $x$  uniformly distributed, the point process  $\sum_{n: q_n(x) \text{ prime}} \delta_{\log n - T}$  converges in law, as  $T \rightarrow \infty$ , to the Poisson process on  $\mathbb{R}$  of intensity  $2 \log 2$ . (iii) The count  $\#\{n \leq N : q_n(x) \text{ prime}\}$ , for  $x$  uniformly distributed, has Gaussian fluctuations of variance  $(2 \log 2 + o(1)) \log N$ .

**Significance.** The almost-sure count,  $\#\{n \leq N : q_n(x) \text{ prime}\} = (2 \log 2 + o(1)) \log N$  for  $\mu$ -almost every  $x$ , which was clause (i) of the statement as first deposited, is a theorem of Bykovskii [426], who proved that the number of prime denominators  $q_n \leq X$  is  $(\log 4 + o(1)) \log \log X$  for almost every  $x$ ; by Lévy's theorem  $\log \log q_N = \log N + O(1)$ , which is the same statement, and the numerators  $p_n(x) = q_{n-1}(Gx)$  follow through the Gauss shift  $G$ , which preserves null sets. The conjecture retains the marginal, Poisson and Gaussian clauses, which refine that theorem. The constant is forced by two exact inputs and one heuristic. By Proposition 79(i) the local probability that  $p$  does not divide  $q_n$  is  $p/(p+1)$  rather than  $1 - 1/p$ , so the Hardy–Littlewood correction to the probability  $1/\log q_n$  of primality is  $\prod_p \frac{p/(p+1)}{1-1/p} = \prod_p \frac{p^2}{p^2-1} = \zeta(2)$ ; by Lévy's theorem  $\log q_n \sim n\pi^2/(12 \log 2)$ ; and the product  $\zeta(2) \cdot 12 \log 2 / \pi^2 = 2 \log 2$  cancels every  $\pi$ , which is the signature of the mechanism. The Poisson clause records that the  $O(1/n)$ -scale repulsion between neighbouring indices (Conjecture 411) is invisible at the logarithmic scale. The nearest results are the theorem of Erdős and Mahler that some irrationals have every  $q_n$  prime [427], the equidistribution of continuants modulo a fixed integer [424, 425], and the theorem of Lévy on the growth of  $q_n$  [423], and the almost-sure count of Bykovskii [426], whose expected value  $2 \log 2 \cdot \log N$  is below two for  $N \leq 4$  and reaches ten only near  $N = 1400$ ; no antecedent was found for the marginal, Poisson and Gaussian clauses; the Erdős–Mahler construction itself follows from Dirichlet's theorem applied to the progression

$aq_{n-1} + q_{n-2}$ . A first decisive theorem would be the lower bound  $\mu\{x : q_N(x) \text{ prime}\} \gg 1/N$ , which requires primes in the values of the continuant polynomials with the correct order of magnitude. In tests on the corpus described in the stress-test section, the empirical probability that  $q_n$  is prime, multiplied by  $n$ , was within one percent of  $2 \log 2$  for  $100 \leq n \leq 200$  over  $1.38 \times 10^5$  uniformly random  $x$ , and the counts for  $p_n$  agreed to within two percent. A counterexample would be a constant other than  $2 \log 2$ , which would signal a failure of the independence of primality from the digit dynamics, or a non-Poisson limit, which would reveal long-range correlations among prime continuants.

**Conjecture 411** (Prime-pair correlations along the convergents). (i)  $\mu\{x : q_n(x), q_{n+1}(x) \text{ both prime}\} = (24 \log^2 2 / \pi^2 + o(1)) / n^2$ , that is, primality at consecutive indices is negatively correlated by the exact factor  $6/\pi^2$  relative to the product of the two marginal probabilities of Conjecture 410. (ii) The same asymptotic holds for  $\mu\{x : p_n(x), q_n(x) \text{ both prime}\}$ . (iii) At lag two the correlation factor is

$$C_2 = \prod_p \left( 1 - \frac{1}{p^2} + \frac{g_p(p+1)}{p^2} \right) = 0.91957 \dots,$$

with  $g_p$  as in Proposition 79(i), so that  $\mu\{x : q_n, q_{n+2} \text{ both prime}\} \sim C_2 (2 \log 2)^2 / n^2$ . (iv) For every fixed lag  $h \geq 1$ ,  $\mu\{x : q_n(x), q_{n+h}(x) \text{ both prime}\} \ll_h n^{-2}$ ; consequently, for almost every  $x$  and every fixed  $h \geq 1$ , only finitely many  $n$  have  $q_n(x)$  and  $q_{n+h}(x)$  both prime, and only finitely many  $n$  have  $p_n(x)$  and  $q_n(x)$  both prime.

**Significance.** The factor  $6/\pi^2$  is exact in the model: consecutive continuants are coprime, so the local event that  $p$  divides neither has probability  $(p-1)/(p+1)$  by Proposition 79(i), and  $\prod_p \frac{(p-1)/(p+1)}{(p/(p+1))^2} = \prod_p (1 - p^{-2})$ . At lag two the return to the residue class zero passes through the digit  $a_{n+2}$ , so the Gauss–Kuzmin law enters through  $g_p$  and the constant  $C_2$  is transcendental-looking rather than a zeta value; the two constants are different in kind and both are testable. Clause (iv) is Borel–Cantelli applied to its own pair bound, of which (i)–(iii) are the cases  $h = 1, 2$  and the numerator-denominator pair; the bound for  $h \geq 3$  follows the same local mechanism with the lag- $h$  transition of the projective state, and it says that a typical real number has only finitely many convergents whose numerator and denominator are both prime, a statement of a different flavour from the Erdős–Mahler construction [427]. The nearest results are the pair equidistribution of [425] and the twin-prime heuristics of Hardy and Littlewood for polynomial pairs; no prior treatment of correlations of primality is known along the convergents of a typical number. A first decisive theorem would be the upper bound  $\mu\{x : q_n, q_{n+1} \text{ both prime}\} \ll n^{-2}$  by a sieve on the continuant recursion, which already gives (iv). In tests on the corpus described in the stress-test section, the lag-one pair counts over  $1.38 \times 10^5$  random  $x$  for  $60 \leq n \leq 200$  showed the predicted deficit of thirty-nine percent relative to independence, with a sampling error of about two percent that is of the same size as the residual deviation, the numerator–denominator pairs behaved likewise, and the lag-two counts agreed with  $C_2$  to within five percent. A counterexample would be a lag-one factor different from  $6/\pi^2$ , which would mean that primality of  $q_n$  and of  $q_{n+1}$  is correlated beyond the local obstruction of coprimality.

**Conjecture 412** (Squarefree and totient densities of continuants). For  $\mu$ -almost every  $x$ ,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \#\{n \leq N : q_n(x) \text{ squarefree}\} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n \leq N} \frac{\varphi(q_n(x))}{q_n(x)} = \prod_p \left( 1 - \frac{1}{p(p+1)} \right) = 0.70445 \dots,$$

and more generally, for every multiplicative  $f$  that is bounded on prime powers and satisfies  $\sum_p \sum_{k \geq 1} |f(p^k) - f(p^{k-1})|/\Psi(p^k) < \infty$ , a class containing the squarefree indicator and  $\varphi(n)/n$ ,  $N^{-1} \sum_{n \leq N} f(q_n(x))$  converges to  $\prod_p (1 + \sum_{k \geq 1} (f(p^k) - f(p^{k-1}))/\Psi(p^k))$ .

**Significance.** For integers the squarefree density and the mean of  $\varphi(n)/n$  coincide at  $6/\pi^2$ ; along the continuants of a typical number both are governed by the projective-line local law  $1/\Psi(p^k)$  of Proposition 79(i) and coincide again, at a constant that is not a zeta value. The general clause is the Delange-type mean value theorem transported to the orbit of a skew product. What is proved is each finite-modulus density; what is conjectured is the uniformity in the modulus that allows the sieve over all primes to be closed, which is the same gap that separates Proposition 79(i) from Conjecture 410. The nearest results are the mean value theorems of Delange and Wirsing for multiplicative functions and the continuant equidistribution of [425]; no antecedent was found for arithmetic mean values along the convergents of a typical real number. A first decisive theorem would be a large sieve inequality for the sequence  $(q_n(x))_{n \leq N}$  uniform in moduli up to  $N^\delta$ . In tests on the corpus described in the stress-test section, the density of  $n \leq 200$  with  $q_n(x)$  free of squares of primes below  $10^4$  was 0.70425 over  $1.38 \times 10^5$  random  $x$ , against the truncated product 0.70445, while  $6/\pi^2 = 0.60793$  is excluded. A counterexample would be a density other than the product, which would show that the large prime squares dividing continuants do not follow the local law.

**Conjecture 413** (Erdős–Kac law along the convergents). *For  $\mu$ -almost every  $x$  and every real  $t$ ,*

$$\lim_{N \rightarrow \infty} \frac{1}{N} \# \left\{ n \leq N : \frac{\omega(q_n(x)) - \log n}{\sqrt{\log n}} \leq t \right\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^t e^{-u^2/2} du,$$

where  $\omega$  counts distinct prime divisors; the centring  $\log n$  may be replaced by  $\sum_{p \leq q_n} 1/(p+1)$ , which differs from  $\log \log q_n$  by a bounded amount.

**Significance.** Along the integers the Erdős–Kac theorem [428] rests on the Turán–Kubilius inequality, which needs the divisibility densities uniformly in primes up to a power of the size of the integer. Along the convergents of a fixed typical  $x$  the densities are known prime by prime (Proposition 79(i)) but not uniformly, and the local law  $1/(p+1)$  shifts the mean by the convergent constant  $\sum_p (1/p - 1/(p+1))$  without changing the variance, so the Gaussian shape is the same while the centring is subtly different. The nearest results are the classical theorem and its versions along polynomial values and along shifted primes; no antecedent was found along continued-fraction denominators of a typical real number. A first decisive theorem would be a second-moment bound  $\sum_{n \leq N} (\omega(q_n) - \log n)^2 \ll N \log N$  for almost every  $x$ . In tests on the corpus described in the stress-test section, only the finite-modulus local laws of Proposition 79(i) were checked, so the Gaussian law itself remains unverified numerically. A counterexample would be a non-Gaussian limit or a variance other than  $\log n$ , which would mean that the prime factors of successive continuants are correlated across many primes at once.

**Conjecture 414** (Dimension of the prime-continuant sets). (i) *The set  $\{x \in (0, 1) : q_n(x) \text{ is prime for every } n \geq 1\}$  has Hausdorff dimension exactly  $1/2$ .* (ii) *The set  $\{x \in (0, 1) : q_n(x) \text{ is composite for every } n \geq 1\}$  has Hausdorff dimension 1.*

**Significance.** Erdős and Mahler proved that the set in (i) is nonempty and, by the same construction, uncountable [427]. The set is not the set of numbers whose partial quotients are

prime, whose dimension was determined exactly by Das and Simmons [430]; a prime condition on the denominators  $q_n$  couples every digit to the whole past. Its dimension is governed by a non-stationary digit restriction: given the past, the admissible next digits  $a$  are those with  $aq_{n-1} + q_{n-2}$  prime, a set of relative density about  $1/(\log a + \log q_{n-1})$ , which is a sparse but infinite digit set whose Dirichlet series has abscissa  $1/2$  at every stage and whose density decays as  $q_{n-1}$  grows, so the local dimension tends to  $1/2$  from above. In (ii) the constraint at level  $n$  removes a set of measure about  $(2 \log 2)/n$ , and the surviving measure decays polynomially, which is the regime in which full dimension is expected. The nearest results are Good's theorem that  $\{x : a_n(x) \geq N \ \forall n\}$  has dimension tending to  $1/2$  [429] and the dimension theory of infinite iterated function systems [432]; no antecedent was found for the dimension of either prime-constrained set. Both clauses are expected to be provable with these tools: the upper bound in (i) from the Brun–Titchmarsh inequality applied level by level, the lower bound from Good's construction with digit ranges growing so fast that only the prime number theorem in a fixed progression is needed, and (ii) from a non-autonomous iterated function system with finitely many digits; the statements are recorded because no such proof is published and because the rate at which truncated versions of the set in (i) approach dimension one half is open. A first decisive theorem would be either bound in (i). In tests on the corpus described in the stress-test section, no numerical estimate of these dimensions was attempted; the only checked inputs are the prime densities of Conjecture 410. A counterexample would be a dimension strictly between  $1/2$  and  $1$  in (i), which would show that the prime-progression constraint retains a positive entropy at every scale.

**Conjecture 415** (Digit sets in arithmetic progressions). *For integers  $1 \leq a \leq m$  let  $E_{a,m}$  be the set of  $x \in (0, 1)$  all of whose partial quotients lie in  $a + m\mathbb{N}_0 = \{a, a + m, a + 2m, \dots\}$ . Then, uniformly for  $a/m$  in any compact subinterval of  $(0, 1]$ ,*

$$\dim_{\mathrm{H}} E_{a,m} = \frac{1}{2} + \frac{1}{2m} - \frac{\log m}{2m^2} - \frac{\psi_{\Gamma}(a/m)}{2m^2} + O\left(\frac{\log^2 m}{m^3}\right),$$

where  $\psi_{\Gamma}$  is the digamma function; in particular  $\dim_{\mathrm{H}} E_{m,m} = \frac{1}{2} + \frac{1}{2m} - \frac{\log m}{2m^2} + \frac{\gamma}{2m^2} + O(m^{-3} \log^2 m)$  with Euler's constant  $\gamma$ .

**Significance.** The pressure of the digit set  $a + m\mathbb{N}_0$  at  $s = \frac{1}{2} + \delta$  is governed by the Hurwitz zeta value  $m^{-1-2\delta} \zeta(1+2\delta, a/m)$ , whose Laurent expansion  $1/(2\delta) - \psi_{\Gamma}(a/m) + O(\delta)$  produces the logarithm and the digamma term when the pressure equation is inverted; the operator corrections beyond the leading rank-one part enter only at the next order. The statement is therefore a precise asymptotic law with all three constants explicit, and the digamma term distinguishes residue classes of the same density; the digits are restricted to one residue class throughout, which is unrelated to the occurrence of arithmetic progressions inside a digit sequence. The nearest results are Hensley's expansion  $1 - 6/(\pi^2 N) - 72 \log N/(\pi^4 N^2) + O(N^{-2})$  for digits at most  $N$  [431], Good's law  $\dim \rightarrow 1/2$  for digits at least  $N$  [429], and the rigorous computations of Jenkinson and Pollicott for finite digit sets [433]; the work of Nakajima, Takahasi and Wang on numbers whose partial quotients contain arbitrarily long arithmetic progressions [438] concerns a different digit constraint, and no antecedent was found for a fixed progression. A first decisive theorem would be the two-term law  $\frac{1}{2} + \frac{1}{2m} + O(m^{-2} \log m)$  by the Mauldin–Urbański pressure formalism [432]. In tests on the corpus described in the stress-test section, Chebyshev collocation of the transfer operator gave dimensions for  $m \leq 384$  and  $a/m \in \{1, \frac{1}{2}, \frac{1}{4}\}$  whose residuals against the four-term expansion were below  $0.18/m^2$  throughout and below  $0.02/m^2$  for  $a/m = \frac{1}{4}$ , while



omitting the digamma term leaves residuals of size  $0.98/m^2$  at  $a/m = \frac{1}{2}$ . A counterexample would be a residue class whose dimension violates the digamma term, which would mean the pressure is not controlled by the Hurwitz zeta function of the progression.

**Conjecture 416** (Continuant avoidance dimension). *For a prime  $p$  let  $A_p = \{x \in (0, 1) : p \nmid q_n(x) \text{ for every } n \geq 1\}$ . Then*

$$\dim_{\mathbb{H}} A_p = 1 - \frac{6 \log 2}{\pi^2} \cdot \frac{1}{p} + O\left(\frac{1}{p^2}\right),$$

and  $\mu\{x : p \nmid q_n(x) \text{ for all } n \leq N\} = \exp(-(\rho_p + o(1))N)$  with  $\rho_p = (p+1)^{-1}(1 + O(1/p))$ .

**Significance.** Forbidding one of the  $p+1$  states of the projective-line chain of Proposition 79(i) lowers the pressure at  $s = 1$  by the stationary mass  $1/(p+1)$  of that state to first order, and dividing by the Lyapunov exponent  $\pi^2/(6 \log 2)$  of the Gauss map converts the pressure drop into the dimension drop; the coefficient  $6 \log 2/\pi^2$  is the reciprocal of the entropy of the Gauss measure, which equals the Lyapunov exponent by Rokhlin's formula. The set  $A_p$  is the set of real numbers all of whose convergents have denominator prime to  $p$ , a natural object in  $p$ -adic approximation. The nearest results are the equidistribution of continuants modulo  $p$  [425, 424] and the thermodynamic formalism of the Gauss map [432]; no antecedent was found for the dimension of  $A_p$ . For each fixed  $p$  the escape rate  $\rho_p$  and the dimension drop are governed by the spectral perturbation theory of transfer operators with small holes on  $[0, 1] \times \mathbb{P}^1(\mathbb{F}_p)$ , so the open content is the uniformity in  $p$  that produces the constant  $6 \log 2/\pi^2$  and the size of the  $O(p^{-2})$  term; a first decisive theorem would be  $\rho_p(p+1) \rightarrow 1$ . In tests on the corpus described in the stress-test section, collocation gave  $(1 - \dim A_p)(p+1) = 0.581, 0.532, 0.499, 0.482, 0.469$  for  $p = 7, 11, 17, 23, 31$ , decreasing towards  $6 \log 2/\pi^2 = 0.4214$  with a correction of size  $3.3/p$ , and direct simulation gave escape rates  $\rho_p(p+1) = 1.10, 1.07, 1.04, 1.03$  for  $p = 41, 61, 97, 197$ . A counterexample would be a limit constant other than  $6 \log 2/\pi^2$ , which would mean that the forbidden state is entered in clusters at a rate not captured by its stationary mass.

**Conjecture 417** (Unit law for the Legendre half-box games). *Let  $p \equiv 1 \pmod{4}$  be prime. (i) The linear system  $\sum_{i=1}^{m-1} x_i \chi_p(i+j) = v$  ( $1 \leq j \leq m-1$ ),  $\sum_{i < m} x_i = 1$ , is nonsingular and its value is  $v = -y/x$  for a unit  $x + y\sqrt{p}$  of norm  $-1$  in  $\mathbb{Z}[\sqrt{p}]$  (a solution of  $x^2 - py^2 = -1$ ) or, when  $p \equiv 5 \pmod{8}$ , possibly a unit  $(x + y\sqrt{p})/2$  of norm  $-1$  in the maximal order; the unit is an odd power  $\varepsilon_p^e$  of the fundamental unit, and  $e \neq 1$  only if  $p$  is of the form  $k^2 + 1$  or  $k^2 + 4$ . (ii) The full linear system  $\sum_{i=1}^m x_i \chi_p(i-j) = v$  ( $1 \leq j \leq m$ ),  $\sum_i x_i = 1$ , has value  $v = +y/x$  with  $x + y\sqrt{p} = \varepsilon_p^{3e}$  when  $p \equiv 5 \pmod{8}$ , and  $v = -y/x$  with the same unit as in (i) when  $p \equiv 1 \pmod{8}$ , in which case the solution always has a negative coordinate. (iii) Whenever the solution in (i) (respectively (ii)) is nonnegative it is the unique optimal strategy of both players and  $v(H_p)$  (respectively  $v(T_p)$ ) equals the unit ratio exactly; this happens for a set of primes  $p \equiv 1 \pmod{8}$  of positive relative density in case (i) and for a set of primes  $p \equiv 5 \pmod{8}$  of positive relative density in case (ii). (iv)  $\sqrt{p}v(H_p) \rightarrow -1$  as  $p \rightarrow \infty$  through primes  $p \equiv 1 \pmod{8}$ , and  $\sqrt{p}v(T_p) \rightarrow +1$  as  $p \rightarrow \infty$  through primes  $p \equiv 5 \pmod{8}$ ; in the unit regime of (iii) this is automatic, since  $py^2/x^2 = 1 + 1/x^2$  or  $1 + 4/x^2$ .*

**Significance.** A zero-sum game with payoffs  $\pm 1$  has a rational value, and the conjecture says that for the half-box Legendre matrices this rational number is a coordinate ratio  $y/x$  of a unit of  $\mathbb{Q}(\sqrt{p})$ , so that  $v^2 p - 1 = 1/x^2$  or  $4/x^2$  is a reciprocal square. The mechanism is the



Pell-type structure of the Legendre Hankel and Toeplitz half matrices: their determinants were conjectured by Chapman and proved by Vsemirnov to be expressible through the fundamental unit [434, 439, 440]. For a symmetric nonsingular block  $A$  the indifference value is  $v = 1/(\mathbf{1}^\top A^{-1} \mathbf{1})$ , the ratio of  $\det A$  to the determinant of  $A$  bordered by a row and a column of ones, so clauses (i) and (ii) are equivalent to an evaluation of the bordered Legendre determinants, which the known evaluations do not cover. The game-theoretic content is clause (iii): when the unit strategy is a probability vector, minimax selects it, so the value of an arithmetically defined game is a unit ratio, and the values converge to  $\mp 1/\sqrt{p}$  at the rate  $1/(2x^2\sqrt{p})$  dictated by the regulator. The nearest results are the evaluations of Legendre determinants in [434, 439, 440]; no antecedent was found connecting matrix games to units of quadratic fields. A first decisive theorem would be clause (i) for one infinite family, which is a statement about the inverse of a Legendre Hankel matrix. In tests on the corpus described in the stress-test section, exact rational solution of both systems for every prime  $p \equiv 1 \pmod{4}$  below 1000 produced a unit of norm  $-1$  in every case, the fundamental unit or its cube except at  $p \in \{229, 257, 401, 577\}$  where higher odd powers occur; the unit strategy of (i) was nonnegative for twenty-one of the twenty-four primes  $p \equiv 1 \pmod{8}$  below 600, the exceptions being 257, 353 and 593, and that of (ii) for sixteen of the twenty-eight primes  $p \equiv 5 \pmod{8}$  below 620, giving the exact values of Proposition 79(iii), and linear programming with primal and dual certificates gave  $|\sqrt{p}v(H_p) + 1| < 0.01$  for every  $p \equiv 1 \pmod{8}$  below 700 and  $|\sqrt{p}v(T_p) - 1| < 0.06$  for every  $p \equiv 5 \pmod{8}$  below 600; these certificates bear on clauses (i) to (iii) at the tested primes only, and the density clause of (iii) and the limits of (iv) are supported by nothing beyond the quoted ranges. A counterexample would be a prime for which the linear system value is not a unit ratio, which would separate the game from the Pell structure of the determinants.

**Conjecture 418** (Least odd residue law in the complementary classes). *Let  $\ell(p)$  denote the least odd prime that is a quadratic residue modulo  $p$ . (i) As  $p \rightarrow \infty$  through primes  $p \equiv 5 \pmod{8}$  with  $\ell(p) \in \{3, 5\}$ ,  $\sqrt{p}v(H_p) \rightarrow -1$ ; as  $p \rightarrow \infty$  through primes  $p \equiv 1 \pmod{8}$  with  $\ell(p) \in \{3, 5\}$ ,  $\sqrt{p}v(T_p) \rightarrow +1$ . (ii) For every prime  $\ell \geq 7$ , along primes  $p \equiv 5 \pmod{8}$  with  $\ell(p) = \ell$  one has  $\limsup \sqrt{p}v(H_p) < 0$  and  $\liminf \sqrt{p}v(H_p) > -1$ , and along primes  $p \equiv 1 \pmod{8}$  with  $\ell(p) = \ell$  one has  $\liminf \sqrt{p}v(T_p) > 0$  and  $\limsup \sqrt{p}v(T_p) < 1$ ; consequently neither  $\sqrt{p}v(H_p)$  over  $p \equiv 5 \pmod{8}$  nor  $\sqrt{p}v(T_p)$  over  $p \equiv 1 \pmod{8}$  converges.*

**Significance.** In the two classes complementary to those of Conjecture 417(iv), namely the Hankel game for  $p \equiv 5 \pmod{8}$  and the Toeplitz game for  $p \equiv 1 \pmod{8}$ , the unit strategy has negative coordinates and the optimal strategies must exploit a different structure: multiplication by a small quadratic residue  $\ell$  preserves the Legendre symbol and maps the box  $[1, m/\ell]$  into  $[1, m]$ , giving the players a dilation symmetry whose reach is measured by  $\ell(p)$ . The conjecture records that the value is then governed by the least odd residue, an invariant of the prime that is classically tied to the least quadratic non-residue and to Linnik's problem. The nearest results are the bounds on the least quadratic residue and non-residue of Vinogradov and Burgess as presented in [441]; no antecedent was found for a game value governed by this invariant. A first decisive theorem would be (i) for  $\ell(p) = 3$ , which requires an explicit near-optimal strategy built from the dilation by three. In tests on the corpus described in the stress-test section, values with primal and dual certificates for all  $p \equiv 5 \pmod{8}$  below 1400 gave  $\sqrt{p}v(H_p)$  between  $-1.02$  and  $-0.995$  when  $\ell(p) = 3$ , between  $-1.11$  and  $-0.80$  when  $\ell(p) = 5$ , between  $-0.81$  and  $-0.68$  when  $\ell(p) = 7$ , and between  $-0.55$  and  $-0.44$  when  $\ell(p) \in \{11, 13, 17\}$ ; the Toeplitz values for  $p \equiv 1 \pmod{8}$  below 700 lay in  $[0.82, 0.97]$  for  $\ell(p) = 3$ , in  $[0.80, 0.95]$  for  $\ell(p) = 5$ , in  $[0.67, 0.79]$  for

$\ell(p) = 7$  and in  $[0.34, 0.42]$  for  $\ell(p) \geq 11$ ; the class  $\ell(p) = 5$  is the least settled, with the widest spread around the conjectured limit. A counterexample would be convergence of  $\sqrt{p} v(H_p)$  to  $-1$  along primes with  $\ell(p) \geq 7$ , which would mean that the least residue is only a finite-size effect.

**Conjecture 419** (Brownian shift games). *Let  $(\xi_k)_{k \in \mathbb{Z}}$  be independent random variables with mean zero, variance one and finite fourth moment, and let  $K^{(m)} = (\xi_{i+j})_{1 \leq i, j \leq m}$ ; by Proposition 79(ii) the Toeplitz game  $(\xi_{i-j})_{1 \leq i, j \leq m}$  has the same value as the Hankel game of the shifted sequence, so it suffices to treat  $K^{(m)}$ . Then  $\sqrt{m} v(K^{(m)})$  converges in law to a nondegenerate symmetric limit  $V$  that does not depend on the law of  $\xi$ , and  $V$  is the value of the continuum game*

$$V = \lim_{R \rightarrow \infty} \sup_f \inf_{y \in [0, 1]} \int_0^1 f(x) dW(x + y),$$

where  $W$  is a standard Brownian motion on  $[0, 2]$  and the supremum runs over step-function probability densities  $f$  on  $[0, 1]$  with  $\|f\|_2 \leq R$ , as in Proposition 79(vi), which proves that the capped values increase to an almost surely finite limit  $V$ ; the assertion includes that  $V$  is nondegenerate and symmetric. In contrast, for independent  $\xi$  arranged in a circulant matrix  $\sqrt{m} v$  converges to a standard Gaussian, and for a full matrix of independent entries  $m v$  is tight.

**Significance.** A Toeplitz or Hankel game has only  $2m - 1$  random bits, its rows are shifts of a single random walk, and the payoff of a mixed row strategy against a pure column is a Riemann–Stieltjes sum of the strategy against that walk; Donsker’s theorem then suggests a continuum limit at the scale  $m^{-1/2}$ , one order larger than the  $1/m$  scale of full random games proved by Cosson and Massoulié [445] after the experiments of Faris and Maier [443] and the bound of Jonasson [444]. The circulant contrast is Proposition 79(ii). The nearest results are those three papers on full random games and the spectral theory of random Toeplitz matrices; no antecedent was found for structured random games. The cap is needed to make the continuum game well posed: the probability densities in  $L^2$  are not a compact class, and Donsker’s theorem for a fixed  $f$  does not justify exchanging the random-matrix limit with an optimisation over all densities, so the game is defined through the capped values, whose convergence to a finite limit is Proposition 79(vi); the conjectural content is the identification of the discrete limit with  $V$ , its universality, nondegeneracy and symmetry. A first decisive theorem would be tightness of  $\sqrt{m} v(K^{(m)})$  together with a lower bound  $\mathbb{E}|v(K^{(m)})| \gg m^{-1/2}$ . In tests on the corpus described in the stress-test section, linear programming with primal and dual certificates for two hundred games per size gave standard deviations of  $\sqrt{m} v(K^{(m)})$  equal to 0.83, 0.76, 0.87, 0.86, 0.92 for  $m = 25, 50, 100, 200, 400$  with  $\pm 1$  entries and 0.81, 0.80, 0.95 for  $m = 100, 200, 400$  with Gaussian entries, against a flat 1.4 to 1.7 for  $m v$  in the full  $\pm 1$  ensemble; a symmetric Toeplitz variant gave 1.15 to 1.19. A counterexample would be a dependence of the limit on the entry law, or a scale other than  $m^{-1/2}$ , which would place structured games outside the Donsker regime.

**Conjecture 420** (Hessian big monodromy over finite fields). *For a smooth plane quartic  $C$  over  $\mathbb{F}_q$  let  $H_C$  be its Hessian sextic and write  $a_C = (q + 1 - \#C(\mathbb{F}_q))/\sqrt{q}$ ,  $a_H = (q + 1 - \#H_C(\mathbb{F}_q))/\sqrt{q}$ . As  $q \rightarrow \infty$  through prime powers with characteristic at least five, for  $C$  uniformly random among smooth quartics with smooth Hessian, the pair  $(a_C, a_H)$  converges in law to  $(\text{tr } U_6, \text{tr } U_{20})$  with  $U_6 \in \text{USp}(6)$  and  $U_{20} \in \text{USp}(20)$  independent and Haar distributed; the number of  $\mathbb{F}_q$ -rational flexes converges to the fixed-point count of a uniform element of the symmetric group  $S_{24}$ , independently of the pair. The same holds for smooth curves of every degree  $d \geq 4$  in characteristic larger than  $d$ , with the Hessian of degree  $3(d - 2)$ , genus  $(3d - 7)(3d - 8)/2$ , and the flexes numbering  $3d(d - 2)$ .*

**Significance.** The Hessian of a random quartic is a random sextic drawn from a fourteen-dimensional subfamily of the twenty-seven-dimensional space of sextics, so the Katz–Sarnak equidistribution for the full family of sextics [447] does not apply; the conjecture asserts that this special family nevertheless has the full symplectic monodromy, and that the monodromy of the pair  $(C, H_C)$  is the product  $\mathrm{Sp}(6) \times \mathrm{Sp}(20)$ , even though the two curves share the 24 flexes; for cubics the Hessian construction does create a correspondence, the Hesse pencil, so the restriction to  $d \geq 4$  is essential. The joint statement with the flexes is the more accessible part, because the monodromy of the flexes is the full symmetric group by Harris [446] and a connected group cannot map onto a finite one. The nearest results are Katz–Sarnak for families of plane curves [447] and Harris on the Galois groups of flexes and bitangents [446]; the monodromy of the Hessian family appears not to have been studied. A first decisive theorem would be the irreducibility of the geometric monodromy of the Hessian family, which is a degeneration argument in the space of quartics. In tests on the corpus described in the stress-test section, point counts of 150 random quartics and their Hessians over  $\mathbb{F}_{41}$  gave  $\mathbb{E}a_H^2 = 1.06$ ,  $\mathbb{E}a_C a_H = 0.06$ ,  $\mathbb{E}a_C^2 a_H^2 = 0.85$  and  $\mathbb{E}a_H^4 = 2.60$ , against 1, 0, 1, 3 for the independent symplectic law, all within one standard error of the sample, while rational singular points of the Hessian appeared in a fraction  $0.04 \approx 1/q$  of the curves. A counterexample would be a nonzero limiting correlation between  $a_C$  and  $a_H$ , which would exhibit a hidden algebraic relation between a curve and its Hessian beyond the flexes.

**Conjecture 421** (Sato–Tate independence of a curve and its Hessian). *Let  $C$  be a smooth plane quartic over  $\mathbb{Q}$  whose Hessian  $H_C$  is smooth and such that the Mumford–Tate groups of the Jacobians of  $C$  and of  $H_C$  are  $\mathrm{GSp}(6)$  and  $\mathrm{GSp}(20)$ , with no nonzero homomorphism between the two Jacobians over  $\overline{\mathbb{Q}}$ . Write  $a_p(C) = p + 1 - \#C(\mathbb{F}_p)$ . Then, as  $p \rightarrow \infty$  through primes of good reduction, the pairs  $(a_p(C)/\sqrt{p}, a_p(H_C)/\sqrt{p})$  are equidistributed with respect to the product of the trace laws of  $\mathrm{USp}(6)$  and  $\mathrm{USp}(20)$ .*

**Significance.** This is the arithmetic counterpart of Conjecture 420: for a single curve over  $\mathbb{Q}$  the Sato–Tate conjecture in the sense of Serre [450] predicts equidistribution of each normalised trace, and the joint statement adds that the Hessian construction introduces no correlation between the Frobenius classes of a genus-three and a genus-ten curve that share their 24 flexes. Under the hypotheses the Sato–Tate group of the product abelian variety is expected to be  $\mathrm{USp}(6) \times \mathrm{USp}(20)$ , so the conjecture follows from the generalised Sato–Tate conjecture for the product; the content is the identification of that group for the Hessian pair, which no known result provides. This places the statement inside the product theory: the Mumford–Tate conjecture is stable under products [435] and implies the algebraic Sato–Tate conjecture of Banaszak and Kedlaya [437, 436], and the lemma that full individual Mumford–Tate groups with no homomorphism between the Jacobians force the product group is the step the hypotheses are designed to supply; the conjecture is thus a concrete instance of that theory rather than a new independence principle, and its content is the verification of the hypotheses for the Hessian pair. The nearest results are the proved Sato–Tate law for genus one and the classification of Sato–Tate groups in low genus [450]; no antecedent was found for the Hessian pair. A first decisive theorem would be the absence of a nontrivial  $\overline{\mathbb{Q}}$ -homomorphism between the two Jacobians for one explicit quartic. In tests on the corpus described in the stress-test section, two random integral quartics with 200 primes of good reduction below 1300 gave  $\mathbb{E}a_H^2 = 1.12, 0.97$ ,  $\mathbb{E}a_C a_H = 0.08, 0.08$  and  $\mathbb{E}a_C^2 a_H^2 = 0.89, 0.82$ , consistent with the product law within the sampling error of 0.1. A counterexample would be a positive limiting correlation, which would force a common isogeny

factor between the two Jacobians for a generic quartic.

**Conjecture 422** (Tangency law and dual-curve point counts). *As  $q \rightarrow \infty$  with characteristic at least five, for a random smooth plane quartic over  $\mathbb{F}_q$  the triple  $(r, b_s, b_c)$ , where  $r$  is the number of  $\mathbb{F}_q$ -rational bitangents and  $b_s, b_c$  are as in Proposition 79(v), converges in law to  $(F, B, F - B)$ , where  $F$  is the number of fixed points of a Haar-random element of  $\mathrm{Sp}(6, \mathbb{F}_2)$  on the 28 odd theta characteristics and, conditionally on  $F$ ,  $B$  is binomial with parameters  $F$  and  $\frac{1}{2}$ . Consequently  $\#C^\vee(\mathbb{F}_q) - \#C(\mathbb{F}_q)$  converges in law to  $F - 2B$ , a symmetric integer law with variance one. Jointly, the number of rational bitangents whose two lines on the del Pezzo double cover are rational converges to the count of fixed weights of a Haar-random element of  $W(E_7)$  in its minuscule 56-point action, with  $t(F)$  of Proposition 79(iv) converging to the reflection-representation trace of the same element.*

**Significance.** The bitangents of a quartic have monodromy  $\mathrm{Sp}(6, \mathbb{F}_2)$  by Harris [446], which fixes the law of  $r$ ; the new content is the refinement by the two points of tangency, whose monodromy is claimed to be the full extension by  $(\mathbb{Z}/2)^{28}$ , so that a rational bitangent has rational or conjugate tangency points with independent fair probabilities. The mechanism is the hyperflex divisor in the space of quartics: a small loop around it exchanges the two tangency points of one bitangent while fixing everything else, so the kernel of the monodromy on tangency points contains one transposition and, by transitivity, all of them. The consequence for the dual curve is a clean limiting law for  $\#C^\vee(\mathbb{F}_q) - \#C(\mathbb{F}_q)$  through Proposition 79(v), and the line statistic separates the  $W(E_7)$  action on lines of the double cover from the tangency action. The nearest results are Harris’s monodromy theorems [446] and the classical geometry of bitangents and degree-two del Pezzo surfaces [449]; no antecedent was found for the tangency-point law or for point counts of dual quartics. The hyperflex-loop argument together with Deligne’s equidistribution theorem is expected to prove the statement once it is checked that no classical relation among the tangency discriminants cuts the kernel to a proper subgroup; the statement is recorded because that monodromy computation is not in the literature. In tests on the corpus described in the stress-test section, 150 random quartics over  $\mathbb{F}_{41}$  gave  $\mathbb{E}r = 1.03$ ,  $\mathbb{E}(b_s - b_c)^2 = 1.08$  and, among curves with a rational bitangent, a proportion  $0.39 \pm 0.05$  with all tangency pairs rational against 0.39 predicted from the empirical law of  $r$ , while the character sum of Proposition 79(iv) was divisible by  $q$  for every curve and had  $\mathbb{E}(t^2) = 0.97$  against 1. A counterexample would be an excess of rational tangency pairs, which would mean that the tangency monodromy is a proper subgroup of the wreath extension.

**Conjecture 423** (Positivity of the quarter-box Legendre games). *For a prime  $p \equiv 3 \pmod{4}$  let  $m = \lfloor p/4 \rfloor$  and  $Q_p = (\chi_p(i + j))_{1 \leq i, j \leq m}$ . Then  $v(Q_p) > 0$  for every prime  $p \equiv 3 \pmod{4}$  with  $p \geq 7$ , and there are constants  $0 < c_1 < c_2 < \infty$  with  $c_1 \leq \sqrt{m} v(Q_p) \leq c_2$  for all such  $p$ , while  $\sqrt{m} v(Q_p)$  does not converge. For  $p \equiv 1 \pmod{4}$  the sign of  $v(Q_p)$  is positive and negative for infinitely many  $p$  each.*

**Significance.** For  $p \equiv 3 \pmod{4}$  the Legendre sums over initial intervals are positive by Dirichlet’s class number formula,  $\sum_{n < p/2} \chi_p(n) = (2 - \chi_p(2))h(-p)$ , and the quarter-interval sums are class-number multiples as well [441, 442]; the row player of the quarter-box game can therefore start from strategies whose payoffs are interval sums of  $\chi_p$ , and the conjecture asserts that the optimal value inherits the sign of the class number at the scale  $\sqrt{p}/m \asymp m^{-1/2}$ . The class number explains the sign but not the size: over the tested primes the normalised

value varies by a factor below three while  $h(-p)/\sqrt{p}$  varies by a factor fifteen, and the two are negatively correlated, with correlation about  $-0.46$ , so the absence of a limit must come from the finer structure of the character sums. For  $p \equiv 1 \pmod{4}$  there is no sign bias and both signs occur. The nearest results are the classical positivity theorems for character sums collected by Berndt [442]; no antecedent was found for their game-theoretic consequence. A first decisive theorem would be  $v(Q_p) \geq 0$  from an explicit row strategy with all interval payoffs nonnegative. In tests on the corpus described in the stress-test section, values with primal and dual certificates for all 87 primes  $p \equiv 3 \pmod{4}$  between 100 and 1200 were positive, with  $\sqrt{m} v(Q_p)$  between 0.45 and 1.29 and mean 0.89, while for the 84 primes  $p \equiv 1 \pmod{4}$  in the same range 39 values were negative. A counterexample would be a prime  $p \equiv 3 \pmod{4}$  with  $v(Q_p) \leq 0$ , which would show that the column player can neutralise the class-number bias.

**Conjecture 424** (Vanishing law for the half box in the class 3 modulo 4). *For primes  $p \equiv 3 \pmod{4}$  one has  $\sqrt{p} v(H_p) \rightarrow 0$ ; more precisely  $v(H_p) = O(p^{-3/2})$  along  $p \equiv 7 \pmod{8}$ , while  $p^{3/2} v(H_p)$  is not bounded along  $p \equiv 3 \pmod{8}$ . Moreover  $v(H_p) \leq 0$  for every  $p \equiv 7 \pmod{8}$ , while along  $p \equiv 3 \pmod{8}$  the value is positive for infinitely many  $p$  and negative for infinitely many  $p$ .*

**Significance.** When  $\chi_p(-1) = -1$  the Toeplitz half box  $T_p$  is antisymmetric and has value zero by Proposition 79(ii), and the Hankel half box is an arithmetic perturbation of it: reversing the columns of  $H_p$  produces the Toeplitz matrix  $(\chi_p(2)\chi_p(2(i-j)+1))$ , whose symbol differs from the antisymmetric one by a shift. The conjecture says the perturbation is felt only at the order  $p^{-3/2}$ , a full power of  $p$  below the unit scale of Conjecture 417, and that the sign of  $\chi_p(2)$  decides whether the perturbation is bounded at that order; why a sign in the symbol should change the magnitude by a power of  $p$  is not explained by the reversal identity alone, and the unboundedness clause rests on a spread of one order of magnitude over eighteen primes. The nearest results are the antisymmetry observation and the Legendre determinant evaluations for  $p \equiv 3 \pmod{4}$  of Vsemirnov [439]; no antecedent was found for the value of this game. A first decisive theorem would be the upper bound  $|v(H_p)| \ll p^{-1}$  from an explicit strategy. The statement as first deposited asserted  $v(H_p) \leq 0$  for every  $p \equiv 3 \pmod{4}$  with  $p > 43$ ; this was refuted at  $p = 547$ , where the value is positive, with an exact rational primal and dual certificate on supports of size 156,  $v(H_{547}) = 1.0289 \times 10^{-4}$  and  $p^{3/2} v(H_p) = 1.32$ , and the corrected sign clause is retained. In tests on the corpus described in the stress-test section, linear programming with primal and dual certificates for all 86 primes  $p \equiv 3 \pmod{4}$  below 1000 gave nonpositive values throughout the class 7 (mod 8), with  $p^{3/2}|v(H_p)|$  at most 3.6, attained at  $p = 631$ , and value zero at  $p \in \{7, 23, 47, 71\}$ ; in the class 3 (mod 8) the value is positive exactly at  $p \in \{43, 547, 691, 907\}$ , with  $p^{3/2} v(H_p)$  between 1.3 and 2.0 there, vanishes at  $p \in \{11, 19\}$ , and is negative otherwise, with  $p^{3/2} v(H_p)$  as low as  $-10.7$  at  $p = 251$ ; the value at  $p = 547$  is certified in exact arithmetic. A counterexample would be a prime  $p \equiv 7 \pmod{8}$  with  $|v(H_p)| > p^{-1}$  or with positive value, which would mean the perturbation of antisymmetry acts at the leading order.



## Part XXX

# Forking without stable witnesses: quantum modules over graphs

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The stable forking conjecture of Kim and Pillay [451, 452] asserted that in a simple theory every instance of forking is witnessed by a stable formula: if  $p \in S(B)$  forks over  $A \subseteq B$ , then some  $\varphi(x, b) \in p$  with  $\varphi(x, y)$  stable forks over  $A$ . Freitag and Mutchnik have refuted it [453]: their counterexample is an infinite-dimensional module over the division ring of fractions of the quantum algebra of the random bipartite graph, a supersimple theory of infinite rank in which forking is characterised by global linear disjointness of closures and in which the noncommutativity of the scalars encodes the order property in the intersections of closed subspaces. This part states fifteen conjectures that follow the mechanism of that example in three directions: which parts of the stable forking conjecture survive, how the model theory of a graph transfers to the module built from it, and what the scalar division rings themselves look like. The proved layer is Proposition 80; its last three clauses isolate the mechanism, namely that one hidden vertex in a formula leaves it stable while two hidden vertices transport the edge relation of the graph into the vector sort, and that the theory of the counterexample is not one-based.

Throughout,  $k$  is a countable field of characteristic different from two,  $q = -1$ , and  $\Gamma$  is a countable graph with edge relation  $E$ . The *quantum algebra* of  $\Gamma$  is the  $k$ -algebra

$$A_q(\Gamma) = k\langle x_v : v \in \Gamma \rangle / (x_u x_v + x_v x_u \ (uEv), \ x_u x_v - x_v x_u \ (u \neq v, \neg uEv)),$$

so that generators at adjacent vertices anticommute and all other pairs commute; the index  $q$  records that the same constructions make sense for any root of unity or any  $q \in k^\times$  once the edges are oriented.  $D_q(\Gamma)$  is its division ring of fractions (Proposition 80(i)). For  $1 \leq n \leq \omega$ ,  $M_q^{(n)}(\Gamma)$  is the two-sorted structure with a vertex sort carrying  $E$ , a vector sort  $V$  carrying the abelian group structure of a left  $D_q(\Gamma)$ -vector space of dimension  $n$  together with a unary function  $w \mapsto \lambda w$  for each  $\lambda \in k$ , and the function  $(v, w) \mapsto x_v w$  from the vertex sort times  $V$  to  $V$ ;  $M_q(\Gamma)$  is  $M_q^{(\omega)}(\Gamma)$  and  $T_q(\Gamma)$  is its complete theory. The scalars of  $D_q(\Gamma)$  are not a sort; the scalars named by the language are the  $k$ -linear combinations of products of generators at named vertices. We write  $\Gamma_{\text{bip}}$  for the random bipartite graph with its bipartition named and  $\Gamma_{\text{rand}}$  for the random graph. A theory has *stable forking* if it satisfies the conclusion of the stable forking conjecture for all sets  $A \subseteq B$ , and *stable forking over models* if it satisfies it whenever  $A$  is a model. For  $k \geq 1$ , a formula  $\varphi(x; y_1, \dots, y_k)$  has the  *$k$ -independence property* if there are tuples  $a_i^j$  ( $i < \omega$ ,  $1 \leq j \leq k$ ) and  $b_S$  ( $S \subseteq \omega^k$ ) with  $\varphi(b_S; a_{i_1}^1, \dots, a_{i_k}^k)$  if and only if  $(i_1, \dots, i_k) \in S$  [461, 462]; a formula  $\varphi(x; y)$  is  *$k$ -dependent* if no partition of  $y$  into  $k$  blocks gives it the  $k$ -independence property. In a simple theory a formula is stable if and only if it is 1-dependent. When a formula  $\varphi(\bar{x}, \bar{b})$  with parameters  $\bar{b}$  is called stable, all of its parameters, vertices included, are displayed as variables and  $\varphi(\bar{x}; \bar{y})$  is required to have no order property; this is the notion of the stable forking conjecture. Where vertex parameters are instead treated as named constants, as in Proposition 80(iv) and Conjecture 436, this is said explicitly. The counterexample of Freitag and Mutchnik lives on a module over  $D_q(\Gamma_{\text{bip}})$ ; the conjectures below concern the structures  $M_q(\Gamma)$  in the language just fixed, which is expected to be interdefinable



with theirs but is defined here independently, and the statements about the counterexample itself are stated for  $T_q(\Gamma_{\text{bip}})$ .

**Proposition 80** (Quantum algebras of graphs and their modules). (i) Fix a linear order on  $\Gamma$ . The ordered monomials  $x_{v_1}^{a_1} \cdots x_{v_n}^{a_n}$  with  $v_1 < \cdots < v_n$  form a  $k$ -basis of  $A_q(\Gamma)$ ;  $A_q(\Gamma)$  is a domain graded by the exponent vector in  $\mathbb{N}^{(\Gamma)}$ , an iterated Ore extension of  $k$  when  $\Gamma$  is finite, and a directed union of such algebras in general; it satisfies the left and right Ore conditions, so  $D_q(\Gamma)$  exists. An induced-subgraph embedding  $\Gamma' \rightarrow \Gamma$  induces embeddings  $A_q(\Gamma') \rightarrow A_q(\Gamma)$  and  $D_q(\Gamma') \rightarrow D_q(\Gamma)$ , and every automorphism of  $\Gamma$  extends to an automorphism of  $D_q(\Gamma)$ ; in particular  $D_q(\Gamma')$  embeds in  $D_q(\Gamma_{\text{rand}})$  for every countable  $\Gamma'$ . (ii) An element of  $A_q(\Gamma)$  is central if and only if each of its monomials is central, and the monomial  $x^a = \prod_v x_v^{a_v}$  is central if and only if  $\sum_{u: uEv} a_u$  is even for every vertex  $v$ , that is, the exponent vector reduced modulo two lies in the kernel of the adjacency matrix over  $\mathbb{F}_2$ . Hence for finite  $\Gamma$  of odd order the centre contains a monomial with some odd exponent, while for  $\Gamma_{\text{rand}}$  and  $\Gamma_{\text{bip}}$  the centre of  $A_q(\Gamma)$  is the polynomial ring  $k[x_v^2 : v \in \Gamma]$ . (iii) If the adjacency matrix of  $\Gamma$  has infinite rank over  $\mathbb{F}_2$ , in particular for  $\Gamma_{\text{rand}}$  and  $\Gamma_{\text{bip}}$ , then  $D_q(\Gamma)$  satisfies no polynomial identity and is infinite-dimensional over its centre; if moreover  $k$  has positive characteristic, then the theory of the division ring  $D_q(\Gamma)$  in the ring language is not simple. (iv) The reduct of any model of  $T_q(\Gamma)$  to the vector sort with each  $x_v$  ( $v \in \Gamma$ ) as a unary function symbol, without the vertex sort, is a module over  $A_q(\Gamma)$ ; its theory is stable and every formula in it is equivalent to a Boolean combination of positive primitive formulas. Consequently every formula of  $T_q(\Gamma)$  in which the vertex variables are replaced by named constants and no vertex quantifier occurs is stable. Moreover the module reducts of  $M_q^{(n)}(\Gamma)$  for  $1 \leq n \leq \omega$  are all elementarily equivalent, so infinite dimensionality over  $D_q(\Gamma)$  is not expressible in the module reduct. (v) Let  $\psi(w; u)$  be the formula  $\exists v (w = x_v u)$  and  $\theta(w; u)$  the formula  $\exists v \exists v' (w = x_v x_{v'} u)$ , both with vector variables  $w, u$  and no parameters. Then  $\psi$  has no order pattern of length three in any model of  $T_q(\Gamma)$ , hence is stable, for every  $\Gamma$ . For distinct vertices  $a, c$  and any nonzero vector  $b$ ,  $\theta(x_a b, -x_c^{-1} b)$  holds if and only if  $aEc$ . Consequently, if  $\Gamma$  contains for every  $n$  distinct vertices  $a_1, \dots, a_n, c_1, \dots, c_n$  with  $a_i Ec_j$  if and only if  $i \leq j$ , as  $\Gamma_{\text{bip}}$  and  $\Gamma_{\text{rand}}$  do, then  $\theta$  has the order property in  $T_q(\Gamma)$ ; and if  $E$  has the independence property on distinct vertices of  $\Gamma$ , then so does  $\theta$  in  $T_q(\Gamma)$ . (vi) A supersimple theory without stable forking is not one-based. In particular the theory of the counterexample of Freitag and Mutchnik is not one-based.

*Proof.* (i) For finite  $\Gamma = \{v_1 < \cdots < v_n\}$ ,  $A_q(\Gamma)$  is the iterated Ore extension  $k[x_{v_1}][x_{v_2}; \sigma_2] \cdots [x_{v_n}; \sigma_n]$  where  $\sigma_j$  multiplies  $x_{v_i}$  by  $q$  or  $1$  according to adjacency; the Poincaré–Birkhoff–Witt basis, the domain property and the Noetherian property follow [468], and a Noetherian domain is Ore; the defining relations are symmetric in each pair, so no orientation of the edges is needed, and they are homogeneous, so the algebra is graded by the exponent vector. For countable  $\Gamma$  the algebra is the directed union of the algebras of its finite induced subgraphs, each embedded in the next by the basis description, and a directed union of Ore domains is an Ore domain because the Ore condition concerns two elements at a time. The functoriality and the automorphism extension follow from the universal property of Ore localisation [469], and the last claim from the universality of the random graph. (ii) Multiplying a monomial  $x^a$  by  $x_v$  on the left and on the right gives  $x_v x^a = (-1)^{\sum_{uEv} a_u} x^a x_v$ , so  $x_v$  commutes with a linear combination of distinct monomials if and only if it commutes with each, and  $x^a$  is central if and only if every such exponent sum is even. An alternating form on an odd-dimensional space over  $\mathbb{F}_2$  is singular, which gives the odd-order clause. For the random graph, given  $a$  with some odd entry  $a_u$ , let

$S$  be the set of vertices with odd exponent; the extension property gives a vertex  $v$  adjacent to  $u$  and to no other element of  $S$ , so  $\sum_{u'E_v} a_{u'}$  is odd; for the random bipartite graph take  $v$  on the side opposite to  $u$ , adjacent to  $u$  and to no other vertex of  $S$  on the side of  $u$ , and note that  $v$  is adjacent to no vertex on its own side. Thus the central monomials are exactly those with all exponents even. (iii) If the adjacency form has rank  $2r$  on a finite induced subgraph  $H$ , a symplectic basis of the form on  $\mathbb{F}_2^H$  gives  $2r$  monomials  $y_1, z_1, \dots, y_r, z_r$  in the generators of  $H$ , with  $y_i z_i = -z_i y_i$  and all other pairs commuting, whose squares are central; the subalgebra they generate is a quantum affine space with these relations, a domain whose division ring of fractions has centre  $k(y_i^2, z_i^2 : i \leq r)$  and, by the basis of monomials with exponents 0 or 1, dimension  $4^r$  over it, the centre of the division ring of fractions being the field of fractions of the centre by Posner's theorem, so it is a tensor product of  $r$  quaternion algebras over its centre, of polynomial identity degree  $2^r$ . A ring satisfying a polynomial identity of degree  $d$  passes it to every subring, and a division ring finite-dimensional over its centre satisfies one, so unbounded  $r$  excludes both [468]. In positive characteristic, a division ring whose theory is  $\text{NTP}_2$ , in particular one whose theory is simple, is finite-dimensional over its centre by a theorem of Milliet [472], which gives the last claim. (iv) The reduct is a module over the ring  $A_q(\Gamma)$  in the usual language of modules; the Baur–Monk theorem gives elimination of quantifiers down to positive primitive formulas and stability [466, 467]. A formula of  $T_q(\Gamma)$  with vertex parameters and without vertex quantifiers is a formula of this reduct. For the elementary equivalence, two modules are elementarily equivalent when the indices  $|\varphi(M)/\psi(M)|$  agree for all pairs of positive primitive formulas  $\psi \rightarrow \varphi$  in one free variable, each index counted as 1, a finite number or  $\infty$  [467]. A positive primitive subgroup is invariant under every endomorphism of the module, and for  $M = D_q(\Gamma)^{(n)}$  the right multiplications by column-finite matrices over  $D_q(\Gamma)$  are endomorphisms which carry any nonzero vector to any vector; hence the positive primitive subgroups of  $M$  in one variable are 0 and  $M$ . Positive primitive formulas commute with direct sums, so  $\varphi(D_q(\Gamma)^{(n)}) = 0$  if and only if  $\varphi(D_q(\Gamma)) = 0$ , and all indices agree across  $n$ , each being 1 or  $\infty$ . (v) We use that scalars act faithfully on nonzero vectors:  $db = d'b$  with  $b \neq 0$  forces  $d = d'$ , and  $x_v$  is invertible in  $D_q(\Gamma)$ . First a lemma: if  $x_s x_t^{-1} = x_{s'} x_{t'}^{-1}$  in  $D_q(\Gamma)$  for vertices  $s, t, s', t'$ , then  $(s, t) = (s', t')$  or  $s = t$  and  $s' = t'$ . Indeed  $x_s x_t^{-1} x_{t'} = x_{s'}$ , and  $x_t^{-1} x_{t'} = \varepsilon x_{t'} x_t^{-1}$  with  $\varepsilon = \pm 1$ , so  $\varepsilon x_s x_{t'} = x_{s'} x_t$  in  $A_q(\Gamma)$ ; both sides are homogeneous of degree one in each of their generators, so  $\{s, t'\} = \{s', t\}$  as multisets, which is the claim. Now suppose  $w_1, w_2, w_3, u_1, u_2, u_3$  satisfy  $\psi(w_i; u_j)$  exactly when  $i \leq j$ . From  $\psi(w_1; u_j)$  there are vertices  $v_j$  with  $w_1 = x_{v_j} u_j$ . If  $w_1 = 0$  then every  $u_j$  is 0 and  $\psi(w_3; u_2)$  and  $\psi(w_3; u_3)$  have the same truth value, a contradiction; so  $u_j = x_{v_j}^{-1} w_1$  with  $w_1 \neq 0$ . From  $\psi(w_2; u_2)$  and  $\psi(w_2; u_3)$  there are  $v', v''$  with  $x_{v'} x_{v_2}^{-1} = x_{v''} x_{v_3}^{-1}$ . By the lemma either  $v_2 = v_3$ , so  $u_2 = u_3$  and again  $\psi(w_3; u_2)$ ,  $\psi(w_3; u_3)$  agree, or  $v' = v_2$ , so  $w_2 = w_1$  and  $\psi(w_2; u_1)$  holds, a contradiction in both cases. A formula with no order pattern of length three is stable. The facts used, namely that each  $x_v$  acts bijectively on  $V$ , that scalars act faithfully on nonzero vectors, the lemma and the sign identity below, are expressed by sentences with vertex quantifiers that hold in  $M_q(\Gamma)$ , so they hold in every model. For  $\theta$ :  $\theta(x_a b, -x_c^{-1} b)$  says that  $x_a = -x_v x_{v'} x_c^{-1}$  for some  $v, v'$ , that is,  $x_a x_c = -x_v x_{v'}$  in  $A_q(\Gamma)$ ; by the grading  $\{v, v'\} = \{a, c\}$ , and  $(v, v') = (a, c)$  would give  $2x_a x_c = 0$ , impossible in a domain of characteristic different from two, so  $(v, v') = (c, a)$  and the condition is  $x_a x_c = -x_c x_a$ , which holds exactly when  $aEc$ . Given the vertices  $a_i, c_j$ , the vectors  $w_i = x_{a_i} b$  and  $u_j = -x_{c_j}^{-1} b$  satisfy  $\theta(w_i; u_j)$  exactly when  $i \leq j$ ; the independence property transfers in the same way. In  $\Gamma_{\text{bip}}$  and  $\Gamma_{\text{rand}}$  the required configurations exist by the extension axioms, with the  $a_i$  and the  $c_j$  on opposite sides in the bipartite case. (vi) Supersimple theories eliminate hyperimaginaries [455], and a one-based simple theory that

eliminates hyperimaginaries has stable forking [471]. Freitag and Mutchnik prove that their theory is supersimple and lacks stable forking [453].  $\square$

**Conjecture 425** (Finite-rank stable forking). *Every supersimple theory of finite  $SU$ -rank has stable forking.*

**Significance.** The counterexample of Freitag and Mutchnik has infinite rank, and they searched for one there because the finite-rank case is tightly constrained [453]: Mutchnik proved that stability of forking over a base in a finite-rank supersimple theory is decided by finitely many pregeometries in each rank [454], and the one-based case is a theorem in every rank, since supersimple theories eliminate hyperimaginaries [455] and one-based simple theories with elimination of hyperimaginaries have stable forking [471]. The conjecture is the surviving half of the original statement, and its mechanism is that noncommutative scalars of the kind that encode the order property in closed subspaces force infinitely many independent coordinates, hence infinite rank. The nearest results are the one-based theorem just cited and the pregeometry reduction, whose author records that stability of forking over a base is open even for finite-rank supersimple theories; the statement is that recognised open case, raised again as an open question by Freitag and Mutchnik in the paper that refutes the general conjecture [453], and recorded here as an inherited problem rather than a new one, because the refutation makes it the sharp remaining form of the original conjecture. A first decisive theorem would be the rank-two case without countable categoricity, the countably categorical case for elements of rank two being a theorem of Peretz [456]. In tests on the corpus described in the stress-test section, no computation bears on this statement. A counterexample would be a finite-rank supersimple theory, necessarily not one-based, in which a forbidden pregeometry of Mutchnik’s finite list embeds in a rank-one type.

**Conjecture 426** (Rank-one stable forking). *Let  $T$  be simple,  $A \subseteq B$ , and  $p \in S(B)$  a type of  $SU$ -rank one that forks over  $A$ . Then some stable formula in  $p$  forks over  $A$ . Granted Conjecture 433(i), the rank bound is sharp: the types there have  $SU$ -rank two and fork over their base without any stable forking formula.*

**Significance.** This localises the stable forking question from theories to types and identifies the rank of the forking type as the parameter. The type-local statement cannot hold for all finite ranks: the vector  $w = x_y x_{y'} b + m$  of Conjecture 433(i) is interalgebraic over  $Mb$  with the pair  $y, y'$  of hidden vertices, because by the grading the pairs  $(v, v')$  with  $w - m = x_v x_{v'} b$  are orderings of  $\{y, y'\}$ , so its type over  $Mb$  has  $SU$ -rank two, and the conjecture there says that it has no stable forking witness. What survives is the rank-one case, where a single hidden coordinate is available and the formulas that pin it down have one hidden vertex, which Conjecture 436(i) predicts to be the stable situation and Proposition 80(v) proves in the simplest case. The mechanism is that a rank-one forking type over  $B$  is coordinatised by a single independent parameter over the base, and a formula in the type that depends on that parameter alone has no room for the exchange of two hidden coordinates that produces the order property. The nearest results are the stable forking theorem of Peretz for elements of  $SU$ -rank two in supersimple countably categorical theories [456] and the finite-rank reductions of Mutchnik [454]; no antecedent was found for the type-local statement in an arbitrary simple theory. A first decisive theorem would be the case where  $p \restriction A$  has  $SU$ -rank two, the smallest case in which the forking extension is not algebraic. In tests on the corpus described in the stress-test section, no computation bears on this statement.

A counterexample would be a rank-one type forking over a set with only unstable witnesses, which would show that even a single hidden coordinate can carry the order property.

**Conjecture 427** (Stable forking for countably categorical simple theories). *Every  $\aleph_0$ -categorical simple theory has stable forking.*

**Significance.** The module counterexample is far from countably categorical: the type of a pair of vectors records a noncommutative rational scalar, and there are infinitely many of these. Countable categoricity bounds the number of types and, in the simple case, is expected to force finite rank on homogeneous structures [457]; the conjecture asserts that it also removes the room in which forking escapes stable witnesses. The statement is a surviving special case of the original programme rather than a consequence of the counterexample; it is among the questions Freitag and Mutchnik leave open [453], and it is recorded here as an inherited problem. The nearest results are the theorem of Koponen that binary simple homogeneous structures are supersimple of finite rank [474] and its extension to all simple theories with quantifier elimination in a finite relational language, which are of finite rank and one-based [457] and hence have stable forking by the theorem of Kim [471], the stable forking theorem of Peretz for elements of rank two in supersimple countably categorical theories [456], the lowness results of Palacín for countably categorical simple theories under a strong form of stable forking [470], and the example of Mirabi of a countably categorical simple theory in which stable formulas do not compose [458], which shows that countable categoricity does not protect stability of formulas under composition. No antecedent was found in print for the full countably categorical case. A first decisive theorem would be the supersimple countably categorical case. In tests on the corpus described in the stress-test section, no computation bears on this statement. A counterexample would be a countably categorical simple theory of infinite rank, or of finite rank but not one-based, in which some forking type has only unstable witnesses.

**Conjecture 428** (The counterexample is two-dependent). (i) *For every countable graph  $\Gamma$  and every  $k \geq 1$ ,  $T_q(\Gamma)$  is  $k$ -dependent if and only if  $\text{Th}(\Gamma)$  is  $k$ -dependent.* (ii) *In particular  $T_q(\Gamma_{\text{bip}})$  and  $T_q(\Gamma_{\text{rand}})$  are 2-dependent and not NIP, so the stable forking conjecture fails in a 2-dependent supersimple theory.*

**Significance.** In a simple theory a formula is stable exactly when it is NIP, so the refuted conjecture concerns the first level of the  $k$ -dependence hierarchy of Shelah [461, 462]; the conjecture places the counterexample at the second level. The graph is interpretable in the module, so  $k$ -dependence passes from the module to the graph, and the content is the converse. The mechanism is that the action  $(v, w) \mapsto x_v w$  is bilinear in the sense relevant to  $k$ -dependence: the module structure over a fixed algebra is stable, the graph on the vertex sort is  $k$ -dependent, and the only interaction is through a map that is linear in the vector and depends on one vertex at a time, which is the shape of the bilinear forms shown to be 2-dependent in [462]; the edge relation is recovered only by composing two such actions, and composition stays within the second level. A theory is  $k$ -dependent if none of its formulas, under any partition of the variables into an object block and  $k$  parameter blocks, has the  $k$ -independence property. The statement is at the level of theories because  $k$ -dependence of a single formula depends on how its parameters are partitioned and can be trivialised by coding parameters into one imaginary; the formula  $\theta$  of Proposition 80(v) has the independence property, but the combinatorics it transports is the binary edge relation, which is 2-dependent. The nearest results are the 2-dependence of bilinear

forms and of the random bipartite graph [462] and the transfer of NIP through interpretations; no antecedent was found for the dependence rank of the counterexample. A first decisive theorem would be clause (ii) for  $\Gamma_{\text{bip}}$ . In tests on the corpus described in the stress-test section, no computation bears on this statement. A counterexample would be a graph whose quantum module is at a strictly higher level of the hierarchy than the graph, which would show that the action of the generators encodes a hypergraph of higher arity.

**Conjecture 429** (Transfer of dividing lines from graphs to quantum modules). *For every countable graph  $\Gamma$  the following hold. (i)  $T_q(\Gamma)$  is simple if and only if  $\text{Th}(\Gamma)$  is simple, and supersimple if and only if  $\text{Th}(\Gamma)$  is supersimple. (ii)  $T_q(\Gamma)$  is NSOP<sub>1</sub> if and only if  $\text{Th}(\Gamma)$  is NSOP<sub>1</sub>. (iii)  $T_q(\Gamma)$  is stable if and only if  $\text{Th}(\Gamma)$  is stable. (iv) If  $\Gamma$  is infinite and  $\text{Th}(\Gamma)$  is supersimple of SU-rank one, then  $T_q(\Gamma)$  is supersimple of SU-rank  $\omega$ , the type of a vector outside the span of the parameters having rank exactly  $\omega$ .*

**Significance.** The graph is interpretable in  $M_q(\Gamma)$ , so each dividing line passes from the module to the graph; the content is the converse, that the module adds no complexity of its own beyond the graph that parametrises its scalars. For the random bipartite graph the simplicity direction is the theorem of Freitag and Mutchnik [453], proved by verifying the Kim–Pillay axioms for global linear disjointness. The mechanism for a general graph is a quantifier elimination up to closure in which the only unbounded quantification is over vertices, so that every combinatorial configuration in the module is coordinatised by a configuration in the graph. Clause (iv) records the rank: a vector  $x_{y_1} \cdots x_{y_n} b$  forks successively over  $b$ , then  $by_1$ , and so on, giving forking chains of every finite length, while once a vector lies in the span of finitely many parameters only finitely many hidden vertices remain, which bounds the rank of the forking extension by a finite number. The nearest results are the cited theorem, the Kim–Pillay characterisation of simple theories [451], and the Kaplan–Ramsey theory of Kim-independence [460]; no antecedent was found for the transfer principle. Clause (ii) has content only for graphs whose theory is NSOP<sub>1</sub> and not simple, and no such graph is recorded here; the clause is stated because the transfer mechanism does not distinguish the two cases. A first decisive theorem would be clause (iii), since for a stable graph the closure operator should be stationary and the module should be a stable expansion of a module. In tests on the corpus described in the stress-test section, only the algebraic layer was checked: the Poincaré–Birkhoff–Witt normal form and the commutation rule of Proposition 80(ii) were verified by exact rewriting on random products in quantum algebras of random graphs with up to eight vertices, and the two computations behind Proposition 80(v) were repeated by exact enumeration in the algebras of random graphs. A counterexample would be a stable graph whose quantum module has the order property, which would locate a source of instability in the module independent of the graph.

**Conjecture 430** (Stable forking dividing line for quantum modules). *Let  $\Gamma$  be a countable graph with simple theory. Then  $T_q(\Gamma)$  has stable forking if and only if  $\text{Th}(\Gamma)$  is stable. Moreover, when  $\text{Th}(\Gamma)$  is unstable the failure is witnessed by the type of a single vector over a set consisting of finitely many vectors and finitely many vertices.*

**Significance.** Together with Conjecture 429(iii) this says that in quantum modules stable forking and stability coincide; the direction from stability of the graph to stable forking follows from Conjecture 429(iii), since every stable theory has stable forking, so the content is the converse: a module over the quantum algebra of a graph either is stable outright or fails the



stable forking property. The mechanism is the one identified by Freitag and Mutchnik and isolated in Proposition 80(v): a formula that hides one vertex, such as  $\exists v (w = x_v u)$ , is stable, whereas a formula that hides two, such as  $\exists v \exists v' (w = x_v x_{v'} u)$ , evaluates the edge relation of the graph on the sign of a product of two generators, so that every formula expressing membership of a vector in a span with two hidden scalars inherits the order property of the graph [453]. For a stable graph no such transport is available, and Proposition 80(iv) shows that the formulas without vertex quantifiers and with vertex parameters named are stable, the vertex parameters of a stable graph entering only through graph formulas. The nearest results are the counterexample for the random bipartite graph and the stable forking theorem for one-based simple theories with elimination of hyperimaginaries [471]; no antecedent was found for the dichotomy. A first decisive theorem would be the failure of stable forking for  $\Gamma_{\text{rand}}$  by the same proof as for  $\Gamma_{\text{bip}}$ . In tests on the corpus described in the stress-test section, only the rewriting checks described under Conjecture 429 were run, and no computation bears on the dichotomy itself. A counterexample would be an unstable simple graph whose quantum module has stable forking, which would identify a graph property strictly between stability and simplicity that suffices for stable witnesses.

**Conjecture 431** (Global linear disjointness is forking in every quantum module). *Let  $\Gamma$  be a countable graph with simple theory, and work in a monster model of  $T_q(\Gamma)$  with vertex sort  $\Gamma^*$ . For a set  $A$  write  $A_\Gamma$  for its vertex part,  $A_V$  for its vector part,  $\langle A \rangle$  for the  $D_q(\Gamma^*)$ -span of  $A_V$ , and  $\text{cl}(A)$  for the pair consisting of the vertex part of  $\text{acl}(A)$  and the  $D_q(\text{acl}(A)_\Gamma)$ -span of  $A_V$ . Then for all  $A, B, C$ ,*

$$A \vdash_C B \iff \langle AC \rangle \cap \langle BC \rangle = \langle C \rangle \quad \text{and} \quad \text{cl}(AC)_\Gamma \vdash_{\text{cl}(C)_\Gamma} \text{cl}(BC)_\Gamma \quad \text{in } \text{Th}(\Gamma),$$

where the last independence is forking independence in  $\text{Th}(\Gamma)$ ; and  $T_q(\Gamma)$  has quantifier elimination up to closure, in the sense that for tuples  $\bar{a}, \bar{a}'$  and a set  $C = \text{cl}(C)$ ,  $\text{tp}(\bar{a}/C) = \text{tp}(\bar{a}'/C)$  if and only if there is an isomorphism  $\text{cl}(\bar{a}C) \rightarrow \text{cl}(\bar{a}'C)$  of two-sorted structures over  $C$  carrying  $\bar{a}$  to  $\bar{a}'$  whose vertex part is an elementary map of  $\Gamma^*$ .

**Significance.** This is the forking calculus behind Conjectures 429 and 430, and it is the statement that Freitag and Mutchnik proved for the random bipartite graph [453]. The vertex clause is needed: for a vertex  $y$  and a vector  $b$ , the sets  $\{y\}$  and  $\{x_y b, b\}$  have linearly disjoint spans and no vertices in common, yet  $y$  is definable from  $x_y b$  and  $b$ , and the clause detects this through the algebraic closure of the second set. The mechanism is that linear disjointness of spans over a division ring should satisfy the extension, symmetry and transitivity axioms of Kim and Pillay [451] by linear algebra, that the independence theorem should be inherited from the graph, and that stationarity should fail exactly as it fails in the graph. Since the span is over the division ring of the monster model, the closure of a set is not definable but only a union of definable sets, and the conjecture records that this is nevertheless the right closure for forking. The two closures differ: for a vertex  $y$  outside  $\text{acl}(b)_\Gamma$  the vector  $x_y b$  lies in  $\langle b \rangle$  but not in  $\text{cl}(b)_V$ ; the span governs forking and the local closure, a countable substructure, governs types. The nearest results are the cited theorem and the classical characterisation of forking in modules over a fixed ring by positive primitive formulas [467]; the random bipartite instance is their theorem, the statement is proposed as its extension to arbitrary simple graphs, and no antecedent was found for general graphs. A first decisive theorem would be the case of a finite graph, where the algebra is Noetherian and the independence theorem is vacuous. In tests on the corpus described in the stress-test section, only the rewriting checks described under Conjecture 429 were run, and no computation bears on the forking calculus itself. A counterexample would be a graph for



which forking in the module is strictly finer than linear disjointness, which would mean that the module carries definable structure not visible in the closure operator.

**Conjecture 432** (Where one-basedness fails in the counterexample). *In  $T_q(\Gamma_{\text{bip}})$ : (i) for every tuple of vectors  $\bar{w}$  and every boundedly closed set  $C$  there is a finite tuple  $\bar{y}$  of vertices of  $\text{acl}(\bar{w}C)$  such that the canonical base of  $\text{tp}(\bar{w}/C)$  is contained in  $\text{bdd}(\bar{w}\bar{y})$ ; (ii) the theory is CM-trivial, so that it is 1-ample but not 2-ample.*

**Significance.** Proposition 80(vi) shows that the counterexample is not one-based, although both of its ingredients are: a vector space over a division ring is modular, and the random bipartite graph has trivial forking. The conjecture locates the failure. Clause (i) says that it is entirely due to the hidden vertices: once the finitely many vertices carried by the scalars relating  $\bar{w}$  to  $C$  are named, the canonical base is bounded over the realisation, as in a module. Clause (ii) says that the failure is of the mildest kind in the ample hierarchy, as expected of a structure assembled from a trivial geometry and a modular one without a field. The mechanism is that the canonical base of the type of a vector over a closed set is the smallest subspace of  $C$  meeting its span together with the vertices of the scalars, and the vertices are algebraic over the vector and the subspace. The theorem of Kim on one-based theories is a statement about theories, not types, so it does not constrain the rank of the types without stable witnesses, which by Conjecture 433(i) can be two. The nearest results are the theorem of Kim that one-based simple theories with elimination of hyperimaginaries have stable forking [471], its refinement under weak elimination of hyperimaginaries [473], and the counterexample itself; no antecedent was found on the geometry of the counterexample. A first decisive theorem would be the computation of the canonical base of the type of a single vector over a closed set. In tests on the corpus described in the stress-test section, no computation bears on this statement. A counterexample to (i) would be a type whose canonical base needs infinitely many vertices; a counterexample to (ii) would be a 2-ample configuration in the module, which would give the counterexample a geometry richer than either of its ingredients.

**Conjecture 433** (Failure of stable forking over models). (i) *In  $T_q(\Gamma_{\text{bip}})$  let  $M$  be a model,  $b$  a vector outside  $\langle M \rangle$ ,  $m \in M$  a vector, and  $y, y'$  distinct vertices not in  $\text{acl}(Mb)$ . Then the type of  $w = x_y x_{y'} b + m$  over  $Mb$  forks over  $M$  and contains no stable formula that forks over  $M$ ; in particular  $T_q(\Gamma_{\text{bip}})$  does not have stable forking over models.* (ii) *For every countable graph  $\Gamma$  whose theory is NSOP<sub>1</sub> and unstable,  $T_q(\Gamma)$  fails the weak stable Kim-forking property: some type over a set  $B$  containing a model  $M$  Kim-forks over  $M$  while no stable formula over  $B$  in the type Kim-forks over  $M$ . In particular the proposal that every NSOP<sub>1</sub> theory has the weak stable Kim-forking property fails already for simple theories.*

**Significance.** The refuted conjecture had a weaker companion in which the base is a model, and for NSOP<sub>1</sub> theories Bossut isolated the weak stable Kim-forking property as the analogue and proposed that every NSOP<sub>1</sub> theory has it [459]; clause (ii) contradicts that proposal. The mechanism is Proposition 80(v) and does not use the base. The vector  $w$  lies in the span of  $M$  and  $b$  and not in  $\langle M \rangle$ , so its type forks over  $M$ . Two hidden vertices are essential: the type of  $x_y b + m$  over  $Mb$  also forks over  $M$ , but it contains the formula  $\exists v (w - m = x_v b)$ , which divides over  $M$  along a sequence of vectors independent over  $M$  and which Conjecture 436(i) predicts to be stable, Proposition 80(v) proving the case without the additive parameter  $m$ ; so, granted that conjecture, that type does not witness the failure. With two hidden vertices, every

formula in the type that pins  $w - m$  to a set of scalar multiples of  $b$  whose scalars are products of two generators evaluates the edge relation between the two vertices on the sign of the product, and thus has the order property of  $\Gamma_{\text{bip}}$ ; this holds whether or not the formula requires the two vertices to be adjacent, since the complement of the edge relation has the order property as well, and the variant  $\exists v \exists v' (v \neq v' \wedge \neg v E v' \wedge w - m = x_v x_{v'} b)$  evaluates  $\neg(aEc)$  on the vectors  $x_a b, x_c^{-1} b$ . Formulas with fewer hidden vertices are satisfied by vectors independent from  $b$  over  $M$ . The conjecture records that this argument is insensitive to the base being a model, and its NSOP<sub>1</sub> clause extends it along Conjecture 429(ii). Clause (ii), even for  $\Gamma_{\text{bip}}$  alone, would refute the published conjecture that every NSOP<sub>1</sub> theory has stable Kim-forking over models, so it requires a proof that classifies every stable formula in the type rather than an extension of clause (i) along Conjecture 429(ii). The nearest results are the counterexample [453], the weak stable Kim-forking property [459], and the theory of Kim-forking over models [460]; no antecedent was found on the base-dependence of the failure. A first decisive theorem would be clause (i), which requires only the characterisation of forking already proved for  $\Gamma_{\text{bip}}$  and a description of the formulas in the type. In tests on the corpus described in the stress-test section, the sign computation  $x_a x_c = -x_c x_a$  if and only if  $aEc$  behind the two-vertex mechanism was verified by exact enumeration in the algebras of random graphs, and no computation bears on the forking statement itself. A counterexample would be a stable formula in the type of  $w$  over  $Mb$  that forks over  $M$ , which would show that some formula with two hidden vertices avoids the sign mechanism.

**Conjecture 434** (Decidability and effective forking). *For  $k = \mathbb{Q}$ , the theories  $T_q(\Gamma_{\text{bip}})$  and  $T_q(\Gamma_{\text{rand}})$  are decidable; given a formula  $\varphi(x, \bar{y}, \bar{z})$  and tuples  $\bar{a}, \bar{b}$  from the computable presentation of the prime model, it is decidable whether  $\varphi(x, \bar{a}, \bar{b})$  forks over  $\bar{a}$ ; and both theories admit a computable normal form for types over finite sets, namely the isomorphism type of the closure of the tuple over the closure of the parameters in the sense of Conjecture 431.*

**Significance.** Decidability follows the pattern of the random graph, whose theory is decidable because it is countably categorical and axiomatised by extension axioms: the module theory is not countably categorical, but its types over finite sets are expected to be coordinatised by finite data, namely a finite induced subgraph and a finite-dimensional module over the quantum algebra of that subgraph, and equality of noncommutative rational scalars in the Ore localisation is expected to be decidable by computing common right multiples with Gröbner bases in the algebra, which is of solvable type [468]. Effective forking then reduces to computing intersections of finitely generated subspaces. The nearest results are decidability of the random graph, the word problem in quantum affine spaces [468], and the quantifier elimination up to closure of [453]; against them stand Macintyre’s theorem that the word problem for division rings is undecidable in general [475] and the result of Itai and Zilber that the full theory of a quantum torus at generic  $q$  interprets arithmetic [476], so decidability depends on the scalars entering only through the vertex sort; no antecedent was found on the decidability of the counterexample. The effective forking clause depends on the forking calculus of Conjecture 431 and on the prime model of Conjecture 439. A first decisive theorem would be a recursive axiomatisation of  $T_q(\Gamma_{\text{bip}})$ , which with the completeness of the theory gives decidability. In tests on the corpus described in the stress-test section, the normal form algorithm for the quantum algebra was implemented and checked to be confluent on random inputs, which is the algebraic half of the claim. A counterexample would be an undecidable  $T_q(\Gamma_{\text{bip}})$ , which would require an encoding of arithmetic in the module not present in either ingredient.

**Conjecture 435** (The theories of the scalar division rings are not simple). *For every field  $k$  of characteristic different from two and every countable graph  $\Gamma$  whose adjacency form has infinite rank over  $\mathbb{F}_2$ , the theory of the division ring  $D_q(\Gamma)$  in the ring language is not simple. In particular the theory of the scalars of the counterexample is not simple although the theory of the module is.*

**Significance.** Proposition 80(iii) proves the positive-characteristic case from Milliet’s theorem that  $\text{NTP}_2$  division rings of positive characteristic are finite-dimensional over their centres [472], and the theorem of Pillay, Scanlon and Wagner proves that no supersimple division ring is noncommutative [463]; the conjecture is the characteristic-zero case, where no such theorem is known. Its interest is the contrast it records: the module is simple, and even supersimple, only because the division ring is not a sort, so that the tree property of the scalars is invisible to the module. The mechanism is that the commutation graph of the division ring, the graph on a generating set in which two elements are joined when they fail to commute, supports an interpretation of  $\Gamma$  together with a valuation-like filtration by degree in the generators, and an infinite definable value group has the strict order property, which is incompatible with simplicity. The nearest results are the two theorems cited, the stable case [464], and the question, still open, whether every simple division ring is commutative [463]; no antecedent was found on the quantum division rings of graphs. A first decisive theorem would be the case of an infinite matching, where the division ring is an infinite tensor product of quaternion algebras over a rational function field. In tests on the corpus described in the stress-test section, only the algebraic input to Proposition 80(ii) was checked: on random graphs, odd powers of adjacent generators anticommute and even powers are central, and the central monomials are exactly those whose exponent vector lies in the kernel of the adjacency form modulo two. A counterexample would be a simple noncommutative division ring of characteristic zero, which would answer the open question in the negative.

**Conjecture 436** (Stable formulas and hidden vertices). *Let  $\Gamma$  be a countable graph with simple theory. Vertex parameters are treated throughout this statement as named constants, formulas are in vector variables, and stability is asserted for every partition of the vector variables into an object block and a parameter block. (i) Every formula of  $T_q(\Gamma)$  with at most one vertex quantifier is stable. (ii) For  $n \geq 2$  and integers  $a_1, \dots, a_n \geq 1$  of which at least two are odd, the formula  $\exists v_1 \dots \exists v_n (v_i \neq v_j \ (i < j) \wedge w = x_{v_1}^{a_1} \dots x_{v_n}^{a_n} u)$  has the order property in  $T_q(\Gamma)$  whenever  $\Gamma$  contains the half-graph configurations of Proposition 80(v). (iii) If  $\text{Th}(\Gamma)$  is unstable, then a formula is stable if and only if it is equivalent to a Boolean combination of formulas of the form  $\exists v (\chi(v, \bar{c}) \wedge \rho(\bar{w}, \bar{u}, x_v, \bar{x}_{\bar{c}}))$ , where  $\chi$  is a formula of the graph with vertex parameters  $\bar{c}$  and  $\rho$  is a positive primitive formula of the module reduct in which  $x_v$  and the generators at  $\bar{c}$  are the only scalars.*

**Significance.** This is the precise form of the observation of Freitag and Mutchnik that non-commutativity encodes the order property in the intersections of closed subspaces [453], and it extends Proposition 80(v), which proves clause (i) for the formula  $\exists v (w = x_v u)$  and clause (ii) for  $n = 2, a_1 = a_2 = 1$ . The mechanism of (ii) is the sign: with two odd exponents the vertices  $a$  and  $c$  of a half graph can be placed at the two odd positions in either order, and the two placements differ by the sign  $(-1)^{[aEc]}$ , so membership of  $x_a^{a_r} x_c^{a_s} u$  in the family evaluates the edge relation. With one hidden vertex no such exchange is available, which is the content of (i), and (iii) asserts that Boolean combinations of one-vertex formulas exhaust the stable part. Stability with named

vertex constants is implied by stability with the vertices displayed as variables, so the instability clause (ii) and the unstable side of (iii) transfer to the notion of the stable forking conjecture, which is how Conjectures 433 and 437 use them. The restriction to named vertex constants is necessary: with a vertex parameter  $z$  allowed to vary, the one-quantifier formula  $\exists v (w = x_v x_z x_v u)$  evaluates  $\neg(aEc)$  on  $w = x_a^2 b$ ,  $u = x_c^{-1} b$ ,  $z = c$ , because  $x_v x_z x_v = (-1)^{[vEz]} x_v^2 x_z$ , and so has the order property. The argument for (i) does not extend to two hidden vertices with a single odd exponent: the formula  $\exists v \exists v' (w = x_v^2 x_{v'} u)$  admits an order pattern of length three on six pairwise non-adjacent vertices, by an exchange between the even and the odd positions recorded in the verification script, so whether it is stable, and hence whether (iii) requires it to reduce to one-vertex formulas, is open. The characterisation would give a syntactic test for stable witnesses and explains the counterexample structurally. The nearest results are the Baur–Monk theorem for modules [466, 467] and the counterexample; no antecedent was found for a characterisation of the stable formulas of the counterexample. A first decisive theorem would be clause (ii) in full, which requires only the sign computation of Proposition 80(v) carried out with fixed auxiliary vertices non-adjacent to the half graph. In tests on the corpus described in the stress-test section, the sign computation behind (ii) and the length-three pattern for exponents two and one were verified by exact enumeration, and span formulas with fixed scalars, the quantifier-free case of (i), were seen to admit no half-graph of length three in finite modules over quantum algebras of small graphs, a sanity check of the mechanism and not evidence of stability; nothing in the programme tests (i) with a vertex quantifier or (iii). A counterexample to (iii) would be a stable formula with two hidden vertices not reducible to one, which would show that the stable part of the module is larger than its one-vertex fragment.

**Conjecture 437** (Nonforking extensions do not restore stable witnesses). (i) *Let  $T$  be simple. If for every  $A \subseteq B$  and every  $p \in S(B)$  forking over  $A$  there are  $C \supseteq B$  and a nonforking extension  $r \in S(C)$  of  $p$  containing a formula  $\varphi(x, c)$  with  $\varphi(x; y)$  stable that forks over  $A$ , then  $T$  has stable forking; the converse holds trivially.* (ii) *In  $T_q(\Gamma_{\text{bip}})$ , with  $M, b, m, y, y'$  and  $w = x_y x_{y'} b + m$  as in Conjecture 433(i), no nonforking extension of the type of  $w$  over  $Mb$  contains a stable formula that forks over  $M$ .*

**Significance.** A natural repair of the refuted conjecture is to allow the stable witness to appear after the parameters needed for it are added, provided the type is extended without forking. In the counterexample this repair is unavailable: the parameters needed for a stable witness are the hidden vertices  $y, y'$ , and by the vertex clause of Conjecture 431 any extension of the type of  $w$  to a set containing  $y$ , or any vertex algebraic over  $Mby$ , forks over  $Mb$ , because  $y$  and  $y'$  are algebraic over  $w, b$  and  $m$ . Clause (ii) records this, and clause (i) asserts that the phenomenon is general: the instability of the witnesses is intrinsic to the forking type and cannot be traded for parameters. One direction of (i) is trivial, since a stable witness in  $p$  lies in every extension. The nearest results are the refuted conjecture [451, 452] and its weak forms over models; no antecedent was found for the nonforking extension form. A first decisive theorem would be clause (ii), which follows from Conjectures 431 and 436(iii). In tests on the corpus described in the stress-test section, no computation bears on this statement. A counterexample to (i) would be a simple theory without stable forking in which every forking type acquires a stable witness after a nonforking extension, which would make the weakened conjecture a genuinely new dividing line.

**Conjecture 438** (Definable groups in quantum modules). *Every group interpretable in  $T_q(\Gamma_{\text{bip}})$  or in  $T_q(\Gamma_{\text{rand}})$  is abelian-by-finite, and every infinite such group has a definable subgroup of*

*finite index that is definably isomorphic to a quotient of a definable subgroup of  $V^n$  by a definable subgroup, for some  $n$ .*

**Significance.** The two ingredients of the counterexample interpret no non-abelian groups: the random graph interprets no infinite group at all, since its forking is trivial while the generic elements  $g, h, gh$  of an infinite interpretable group are pairwise independent and dependent, and a module interprets only abelian groups up to finite index. The conjecture asserts that the interaction through the quantum scalars creates no new groups, in particular no definable copy of a noncommutative multiplicative group of scalars, which is consistent with the scalars not forming a sort. The risk is that the vector sort might interpret a piece of  $D_q(\Gamma)^\times$ , for instance the group of scalars of bounded degree acting on a line, and the conjecture asserts that the degree filtration is not definable. The mechanism is the coordinatisation of definable sets by closures, which are vector spaces with vertex labels, and the expectation that a definable set of scalars of unbounded degree would interpret arithmetic in the length of products, against Conjecture 434. Conjecture 432 would not imply this even if proved, since CM-triviality does not by itself bound interpretable groups; a group-configuration analysis in the closure geometry is required. The nearest results are the description of groups definable in modules and in one-based theories [465] and the absence of definable groups in the random graph; no antecedent was found for groups in the counterexample. A first decisive theorem would be that every definable group is contained in a finite union of cosets of a definable subgroup of  $V^n$  after finitely many vertex parameters are named. In tests on the corpus described in the stress-test section, no computation bears on this statement. A counterexample would be an interpretable non-abelian group, which would give a definable trace of the scalar division ring inside the module.

**Conjecture 439** (Countable models of the counterexample). *For  $k = \mathbb{Q}$ , the structures  $M_q^{(n)}(\Gamma_{\text{bip}})$  for  $1 \leq n \leq \omega$  are pairwise elementarily equivalent and pairwise non-isomorphic, and every countable model of  $T_q(\Gamma_{\text{bip}})$  is isomorphic to one of them. In particular  $T_q(\Gamma_{\text{bip}})$  has exactly  $\aleph_0$  countable models, the prime model is  $M_q^{(1)}(\Gamma_{\text{bip}})$ , and the theory is not  $\aleph_0$ -categorical although its vertex sort is.*

**Significance.** Infinite dimensionality over the division ring is a property of the structure  $M_q(\Gamma_{\text{bip}})$  and not of its theory: linear dependence quantifies over scalars of unbounded complexity, and Proposition 80(iv) proves that the module reducts of the  $M_q^{(n)}(\Gamma_{\text{bip}})$  are elementarily equivalent. The conjecture asserts the same for the two-sorted structures, where the vertex quantifiers must be handled through the extension axioms of the graph, and adds the classification. The converse half uses elementarity: a linear dependence among vectors of a countable model with scalars from a larger model is expressed by a formula with vertex quantifiers, so it is witnessed by scalars from the model itself; for each nonzero element of the algebra with named vertices, acting bijectively on  $V$  is a sentence with vertex quantifiers true in  $M_q(\Gamma_{\text{bip}})$ , so the vector sort of every model is a vector space over the division ring and not merely a module over the algebra; and the countable model of the random bipartite graph is unique. The failure of countable categoricity is the reason Conjecture 427 does not contradict the counterexample. The nearest results are the countable categoricity of the random bipartite graph, the classification of models of a vector space over a fixed division ring by dimension, and Proposition 80(iv); no antecedent was found for the countable models of the counterexample. A first decisive theorem would be the elementary equivalence  $M_q^{(1)}(\Gamma_{\text{bip}}) \equiv M_q^{(\omega)}(\Gamma_{\text{bip}})$  in the two-sorted language. In



tests on the corpus described in the stress-test section, no computation bears on this statement. A counterexample would be a countable model with a nonstandard scalar, a vector in the closure of another that is not a definable scalar multiple of it, which would show that the closure is not determined by the vertex sort of the model.

## Stress tests and independent recomputation

Each statement was tested three ways: against the literature, against computation well beyond its original range, and against an independent reimplementaion.

*Layer 1: literature.* Five independent searches combed OEIS, arXiv, and the standard references for prior art, producing the novelty labels and the attributions used throughout. Each search was run at neighbourhood depth: defining objects, derived sequences, OEIS comments, and the abstracts and bodies of near-neighbour papers read in full rather than skimmed from search summaries—the depth at which, for instance, OEIS A088054’s intersection comment (Conjecture 21(ii)) and the second half of Lillie’s abstract (the primorial companion) are found, neither being visible in the defining sequences or in a search snippet. This layer also fixes two attributions used above: the  $k \geq 1$  Stern list of Conjecture 12 is OEIS A060003 verbatim, with 1493 the largest known Stern prime, and the primorial-twin statement is Lillie’s [5].

*Layer 2: scripted refutation.* Every falsifiable-by-instance statement was pushed at least a decade past its original bound: exception hunts across  $(10^8, 10^9]$  for Conjectures 5, 12, 17, 22; the uniformity of Conjecture 8 stressed on  $d \leq 6000$ ; the statistical laws re-tested at  $4 \times 10^9$  (Conjectures 1, 4, 9, 11, and 13, and the quintuplet calibration count reported with Conjecture 8); the recovery clause of Conjecture 3 tested on fresh moduli  $q \in (3000, 6000]$ . No statement moved. The same layer settles one statement negatively: a completeness list for Fibonacci–Lucas twins is destroyed by OEIS A080327’s catalogued index 148091 (conditionally on the probable-prime classifications of  $F_{148091}$  and  $L_{148091}$  alike, neither being a proven prime), which is why Conjecture 20 conjectures finiteness without a completeness clause and carries that index as its calibration clause.

Each statement was also verified at large scale: the Conjecture 10 profile across 150 shifts and the moment law for its constants  $C(d)$  (derived mean 2.7456 and sd 1.6840 against measured 2.7434 and 1.6726); the cubic family moments and uniformity of Conjecture 15 (294 constants, 57 count profiles); the triplet and sexy-pair races of Conjectures 18 and 19 at  $10^9$  and the Stern lanes of Conjecture 12 on 2,500 samples at  $10^8$ ; the chain of Conjecture 24 over five decades to  $10^7$ ; the null race of Conjecture 14 at  $10^7$ ; the Fibonacci–Lucas joint scan to  $p \leq 10^4$ ; factorial twins to  $n = 700$ ; the CRT-exact factorization of Conjecture 6 through a joint period of  $1.2 \times 10^8$ , with the scan extended to  $n = 6000$ ; the pinned average  $G(H)$  of Conjecture 2 computed exactly to  $H = 3000$ ; the boundary trichotomy of Conjecture 22 with two new lanes counted at  $10^6$ ; the least-summand law of Conjecture 5 on 2,000 samples; the covariance kernels of Conjectures 8(ii) and 16(ii) evaluated over 870 and 1,225 pairs, with the moving-window randomization of Conjecture 8(ii) over 2000 windows (empirical correlation matrix matching the predicted kernel entrywise at 0.86); the orientation-resolved twin-member profile of Conjecture 17 on 150 samples; the race autocorrelation and running-maximum measurements of Conjecture 9 at  $10^9$  against simulated nulls; the singular-series waiting-time clause of Conjecture 11(iii) (regression coefficient  $-0.466$  against the predicted  $-\frac{1}{2}$ ); the stratified experiment of Conjecture 3 on 80 moduli; the multibase quotient tests of Conjecture 7 on 664,577 primes; and the weighted drift and



internal null lane of Conjecture 4 on 400 fresh samples, together with the cousin-race predictions of the contamination calculus, Conjecture 1(v), at  $10^9$  (leadership log-densities 0.99 and 0.92 on the predicted sides). In each case the prediction was derived before the corresponding data were taken.

*Layer 3: independent replication.* Constants were recomputed from their definitions, and sequences recounted, by implementations sharing no code with the primary computation and working from the bare statements alone. All recomputed constants landed inside the stated truncation error, and all recounts agreed. The same layer reproduced the Fibonacci deficit recorded at Conjecture 20 and the ordering anomaly of Remark 1, confirming that both are properties of the primes rather than of a single implementation.

One methodological conclusion is worth recording. The mathematical failure modes in this subject are not statistical but *algebraic*: factorizations and congruence collapses living on density-zero or positive-density-but-structured subsequences—the cubes obstructing  $n = p + k^3$  (Theorem 1), the composite- $k$  factorization of  $D_k$  (Conjecture 23(i)), the  $k = 4$  parity branch of Conjecture 25, the inadmissible naive chain of Conjecture 24. Probabilistic sanity checks, however extensive, integrate over density and cannot see such families; only structural tests—testing a claimed exception census on special subsequences, for instance—find them. A verification that only re-runs an existing count at a larger bound is blind to every one of them.

*Part IV.* The Part IV verification programme follows the same standard. The rank formula of Proposition 3 was checked against direct linear algebra over  $\mathbb{F}_2$ ,  $\mathbb{F}_3$ , and  $\mathbb{F}_5$  on eighteen thousand random pairs of a graph and an element before the proof was written. The irreducibility claim of Conjecture 57 was verified by exact factorization over  $\mathbb{Z}$  for all 996 connected graphs on at most seven vertices. The recovery statements of Conjectures 58, 65, and 66 were tested on every known collision class of  $C_G$ : the 119 classes among the 1252 graphs on at most seven vertices, a randomized eight-vertex corpus, the unique class among all 3159 trees on fourteen vertices, refound independently by exhaustive search, and the first unicyclic collision at ten vertices, with  $Q_G$ ,  $\gamma$ , and  $i$  agreeing in every class. The rigidity statements were scanned exhaustively on at most seven vertices, with six thousand random eight-vertex graphs added for Conjecture 59 and the scan for Conjecture 61 restricted to graphs with at least four edges, and the class-restricted statements were tested on six-thousand-graph random holdouts for each of Conjectures 62, 63, and 64, spanning 2747, 3192, and 5598 distinct binary specializations respectively, together with all 656 binary classes of ten-vertex unicyclic graphs for Conjecture 63. The log-concavity and support statements of Conjectures 67 to 71 were verified on the full atlas of graphs on at most seven vertices and twenty-five hundred random graphs on eight to ten vertices, with the eight-vertex counterexample to the second-order slice strengthening reconfirmed by the independent implementation. The tree programme of Conjectures 72 to 76 was verified exhaustively over all trees through fourteen vertices, for Conjectures 73 and 74 with  $q \in \{2, 3, 4, 5\}$  through twelve vertices,  $q \in \{2, 3\}$  at thirteen, and  $q = 2$  at fourteen, and the sharpness failure at  $(n, q) = (4, 2)$  recorded at Conjecture 73 was found in this verification rather than assumed. A seven-vertex pair refutes Conjecture 63, below the ten-to-thirteen-vertex range of those holdouts. An exhaustive rescan locates it as the unique violating class among trees and connected unicyclic graphs through nine vertices, and the tree case was reverified exhaustively through fourteen vertices.

*Part V.* The Part V verification programme follows the same standard, on an independently generated corpus of all 87 unlabelled posets on at most five points, over  $\mathbb{F}_2$  and  $\mathbb{F}_3$  fully and  $\mathbb{F}_5$  to dimension six, cross-checked against the deposited enumerators. The calibration identity of Proposition 6 held with no exception. The determination statements were tested on every

adjoint-collision class: the adjoint-to-Kirillov determination of Conjecture 77 held across 64 classes, and the lower-central factor vector,  $\dim \text{Der}$ , and  $\dim \text{Cent}$  of Conjectures 78, 80, and 82 were constant on every class. The characteristic statements of Conjectures 83 and 84 were confirmed by agreement of the attained adjoint-rank and Kirillov-rank sets across  $\mathbb{F}_2$ ,  $\mathbb{F}_3$ , and  $\mathbb{F}_5$  in every case, the two-field rigidity of Conjecture 85 forced equality at  $q = 5$  across 16 three-field collision classes, and the adjoint field monotonicity of Conjecture 86 held as first-order dominance in all 221 field-pair tests while its likelihood-ratio strengthening failed. The centre-one statements were tested on all 43 centre-one records, where the Kirillov enumerator determined the adjoint enumerator,  $R_{P,q}$  had nonnegative integer coefficients, and  $R_{P,q}$  had interval support and was unimodal throughout this corpus, with log-concavity failing four times. The unimodality was subsequently refuted by a seven-point poset outside the corpus, and Conjecture 90 is recorded as resolved false. The forest statements were tested on the forest class, where the Kirillov enumerator determined the adjoint enumerator across 44 forest-collision classes, the Kirillov support filled every even rank up to  $b_K$ , the even-rank sequence was unimodal, and the bound  $b_K \leq 2b_A$  held everywhere while the unrestricted bound failed fourteen times off the forest class. The unitriangular strict log-concavity of Conjecture 95 was confirmed for  $\text{ut}_4$ ,  $\text{ut}_5$ , and  $\text{ut}_6$  over  $\mathbb{F}_2$ , for  $\text{ut}_4$  and  $\text{ut}_5$  over  $\mathbb{F}_3$ , and for  $\text{ut}_4$  over  $\mathbb{F}_5$ , and the incidence anticorrelation of Conjecture 96 gave negative covariance on every nonabelian poset and exactly zero on all sixteen abelian ones. Two statements were not independently recomputed: the cohomology recoveries of Conjectures 79 and 81 rest on the deposited computation of  $H^2$  and  $H^3$  (ordinary) and  $H^1$ ,  $H^2$ , and  $H^3$  (adjoint), our reverification having not recomputed Lie-algebra cohomology. Three statements, Conjectures 79, 81, and 85, are supported only by the restricted tests recorded with each, and their general forms should not be read as carrying evidence comparable to the other conjectures. Two further statements, Conjectures 87 and 90, are false as stated and are recorded as resolved false with the seven-point counterexamples given in their resolutions.

*Part VI.* The Part VI verification programme follows the same standard, on an arithmetic cache of every integer through 20,000,000 with segmented exact scans and an odd-only prime-neighbour sieve to  $4 \times 10^9$ , cross-checked by an independent deterministic Miller–Rabin implementation sharing no code with the primary scans. The compass frequencies of Conjectures 97 to 102 were measured over the polygonal centres through  $4 \times 10^9$ , the local-sieve model of Conjecture 97 correlating with the order-level bias at 0.896 across  $3 \leq s \leq 100$  and every promoted direction replicated on a holdout above  $10^9$ . The radical order-pattern laws of Conjectures 103 and 104 were computed exactly through  $10^9$ , the chi-square against uniformity rising from 0.45 at length four to about  $2.1 \times 10^7$  at length five and the pattern ratio settling at 13.31, with the length-four and length-five cell counts reproduced by a second implementation. Proposition 9 now proves existence of every fixed-length conditional density, leaving nonuniformity open at each length at least five. Proposition 10 proves that triangular and hexagonal centres have identical residue laws and local compass frequencies at every finite sieve depth. The order constants of Conjectures 105 to 109 were measured over  $2 \times 10^7$  triples across seven disjoint blocks, and the five-term totient barrier of Conjecture 107, since resolved false by G. Martin’s theorem, showed no run below  $4 \times 10^9$  in an exact segmented scan, while its abundancy analogue fell within the same scan to an increasing five-term run at  $n = 36,721,681$ . The shock statements of Conjectures 110 to 115 were checked over millions of steps per family against the exact jump identities of Proposition 7, the lag-one covariances negative in all 22 tested families. Proposition 11 now proves Conjectures 110 to 112 outright, with explicit six-prime witnesses reaching  $J_{3,n} = -20$ ,  $J_{4,n} = -30$ , and  $J_{3,n} = +21$ , beyond every scanned extreme, and Conjecture 115 follows from the

same identities through the multivariate Erdős–Kac theorem of [173]. Proposition 12 evaluates the truncated lag-one covariance of the central-binomial slice at exactly  $-\frac{5}{3}$  for every cutoff at least 3, confirmed numerically to five decimals at  $2 \times 10^6$ , the full covariance standing at  $-1.47$  and drifting toward the constant. Proposition 14 now proves the  $k = 2$  case of Conjecture 114. The boundary law of Conjecture 116 matched its truncated Euler sum at correlation 0.998 across the 39 gaps with at least 500 samples, with block-level correlations from 0.986 to 0.997. Six of the twenty state a bias as a numerical band rather than a derived constant, and Conjectures 102 and 113 extrapolate in a parameter rather than the index, so their general forms should be read with the corresponding caution.

*Part VII.* The Part VII verification programme rests on exact integer computation throughout: truncated integer power series for the Chern-class evaluation, gcd–lcm inclusion-exclusion for the Milnor–Orlik formula,  $\mathbf{F}_2$  Gaussian elimination for induced clique-complex homology with exact fraction-free Routh–Hurwitz determinants, and signed-orbit counting for the exterior actions, every calibration reproduced by a second implementation sharing no code with the primary scans, which recovered the quintic  $-200$ , the K3 value 24, the Poincaré-sphere 0, the empty-graph strand  $6z^2 + 8z^3 + 3z^4$ , and the  $-I_4$  profile  $(1, 0, 6, 0, 1)$ . The Euler laws of Conjectures 117 to 121 were tested on the complete configuration census through dimension 30, the merges numbering 1,568,731, the ladder exact through  $n = 200$ , and the gcd law reverified independently through dimension 26. The cap extrema of Conjectures 122 and 123 are complete exact-cap censuses through caps 50 and 40 with the  $A = 12$  transition value 222 reverified, and the scale laws of Conjectures 125 and 126 cover 52,593 tuples at scales through 10, with independent random rechecks one scale beyond. The strand laws of Conjectures 127 to 131 were tested on 34,867 graphs, exhaustive over labeled graphs through six vertices with 250 deterministic holdouts at each order seven to ten, giving 50,454 strands, and reverified independently on an exhaustive labeled corpus through five vertices. The monodromy laws of Conjectures 132 to 136 cover all 57,758 orientation-preserving signed cycle types through dimension 21, with every type through dimension 11 and the minimum 160 at dimension 12 reverified independently against Proposition 15. Six adversarial controls are retained in place, each refuting a tempting strengthening. Two of the twenty, the limit clause of Conjecture 117 and Conjecture 135, assert limits, and Conjectures 120 and 124 lie close to classical methods, so those statements should be read with the corresponding caution. Since deposit, Conjectures 117 and 118 have been resolved true and Conjecture 135 resolved false, the Landau floor of Proposition 19 fixing the compression exponent at 2 while the deposited minima respect the floor with room to spare. Proposition 18 now identifies every scale sequence as a sum of interior Ehrhart counts of  $N - 1$  integral weighted hypersimplices, proving Conjecture 125 and the convexity clause of Conjecture 126, with an independent exact stress of 24,950,000 scale inequalities on 25,000 random tuples finding no violation of the surviving log-concavity clause. Proposition 16 proves all three ladder clauses with the sharpening  $8 - Q_{n+1}/Q_n \sim 4/n$ , and its exact recurrence reproduces the deposited ladder through  $n = 200$ . Proposition 17 proves every normalised merge inequality by a positive coefficient identity, independently checked through dimension 18.

*Part VIII.* The Part VIII verification programme is exact wherever its statements are: integer Chern-coefficient evaluation for the ambient products, integer expansion of Göttsche’s product, exact age histograms, exact subset-sum functionals with rational arithmetic, and  $\mathbf{F}_2$  homology of matroid independence complexes with exact rational Routh tests, every calibration reproduced by a second implementation sharing no code with the primary scans, which recovered the quintic, K3, and tetraquadric values, the  $\text{Hilb}^2(\text{K3})$  Betti row  $(1, 23, 276, 23, 1)$ , and the moment

identities of Proposition 20 in exact rational arithmetic. The dominance and refinement laws of Conjectures 137 to 140 were tested on 42,691 moves, 42,861 splits, and 24,680 disjoint split squares through dimension 22, reverified independently through dimension 18 with the four dimension-five violations of the unqualified Conjecture 140 reproduced exactly. The profile laws of Conjectures 141 to 144 rest on 179,200, 53,070, and 48,384 exact inequalities for  $b \leq 30$  and  $n \leq 80$ , with the complete kurtosis phase diagram of Conjecture 143 reverified independently through  $n = 50$ . The age laws of Conjectures 145 to 148 combine an exact census of 450,999 admissible weight multisets with  $r \leq 60$ , reverified on 58,393 multisets with  $r \leq 40$ , with seeded Monte Carlo in four regimes whose lattice-corrected Kolmogorov distances 0.0398 to 0.0162 were independently recomputed as 0.0344 and 0.0178. The entropy laws of Conjectures 149 to 152 were tested on 16,200 random and 48,512 exact integer spectra, the 222 equality cases all two-valued, with an independent exhaustive recount through  $d = 7$  finding 122 two-valued equality cases. Propositions 21 and 22 now prove Conjectures 151 and 152. The former matroid sample did not contain the two graphic certificates now built into the exact checker. Proposition 23 refutes Conjectures 153, 154, and 156, while Proposition 24 refutes Conjecture 155. The limit clauses of Conjectures 139 and 143 and the random-model statements of Conjectures 146 to 148 are recorded where they occur, and Conjecture 145 carries an attribution, so those statements should be read with the corresponding caution.

*Part IX.* The Part IX verification programme is exact wherever its statements are: tubings enumerated exhaustively with integer  $f$ -to- $h$  conversion,  $\mathbf{F}_2$  homology of flag complexes by integer rank, sector functionals by deterministic convolution over exactly enumerated sectors, and boundary maps assembled as integer matrices, while root locations, Mahler measures, and pseudodeterminants are floating-point spectral computations, every calibration reproduced by a second implementation sharing no code with the primary scans, which recovered the Narayana and Eulerian endpoint  $h$ -polynomials, the matrix-tree value  $\text{pdet} = 5^4$  for the complete graph on five vertices, and both identities of Proposition 25, the sine product exactly and the moment integral to  $10^{-6}$ . The toric laws of Conjectures 157 to 161 were tested on 878 deposited root sets, 3,621 connected deletions, 3,956 edge additions with minimum Mahler increment 0.0263 and exact integer tail margins, and 1,857 tree tests, reverified independently on all 771 labelled connected graphs through five vertices, 2,971 deletions, 3,227 edge additions, and every labelled tree through six vertices. The McKay laws of Conjectures 162 to 166 were tested on 3,135 deposited profiles, 2,905 of them at  $d \geq 5$ , reverified by an independent exhaustive census of 379 distinct-weight profiles at  $p \in \{7, 11, 13\}$  that reproduced the two repeated-weight controls and the two-crest  $\beta_3$  control exactly. The torsion laws of Conjectures 167, 168, and 171 rest on 40,932, 6,822, and 20,466 admissible deposited triples with extreme curvatures 0.0079, 0.0770, and  $-0.0076$ , reverified independently at five  $(p, d)$  pairs with the  $s = -2$  control curvature  $+0.4980$  reproduced to six decimals over the exactly verified centering identity; the parabola of Conjecture 169 matched four exact dynamic-programming regimes with central root-mean-square errors 0.027 to 0.038, and the decoupling of Conjecture 170 was checked on three seeded ensembles of up to 60,480 character samples whose empirical correlations below  $5 \times 10^{-16}$  reproduce the exact zero forced by the involution, the informative diagnostics being the variances and fourth moments, which moved toward Gaussian values. The entropy laws of Conjectures 172 to 176 were tested on a corpus of 130 spheres and 225 stellar pairs supplying 262, 392, 130, 900, and 450 comparisons in statement order, with minimum log-concavity margins 51.8 and 17.8 and minimum stellar increment 1.42, reverified independently through dimension four. The iterated-limit clause of Conjecture 169 and the random-model clause of Conjecture 170 are recorded where they occur,

and Conjectures 157, 161, 164, 169, and 170 are close to prior work, so those statements should be read with the corresponding caution.

*Part X.* The Part X verification programme is exact wherever its statements are: characteristic-two and finite-field cohomology by exact elimination, integral torsion by Smith normal form through dimension nine, and every  $\chi_y$ -coefficient, signature, merge inequality, ratio, and gcd in exact integer or rational arithmetic, while characteristic-zero ranks beyond dimension nine are dual large-prime proxies and the root, Mahler, and radius laws are floating-point spectral checks, every calibration reproduced by a second implementation sharing no code with the primary scans, which recomputed the  $\chi_y$ -profiles from the scaled Hirzebruch series with the signature recovered as  $\chi_{y=1}$  and reproduced the K3 profile (2, 20, 2), the quintic profile, and the dimension-two signature 16. The support obstructions of Proposition 28 were checked independently: the deposited profile admits at most one unused coordinate through  $n = 120$ , the rescaling criterion was confirmed on all 1,073 spanning supports through six coordinates with no exception, and the retained coefficient control was recomputed in exact rational arithmetic, its two profiles separated exactly by the vanishing Pfaffian. Proposition 27 now pins the subcritical side of the sparse threshold at every fixed  $c < 2$  by a second-moment coverage argument. The flux laws of Conjectures 177 to 184 rest on 144 dense trials, 960 sparse trials, 2,800 finite-field samples, eight explicit witnesses, and complete-flux scans for  $4 \leq n \leq 13$  with sixteen exact Smith trials, reverified independently with 71 of 72 dense hits, all eight witnesses, exhaustive support minimality at  $n = 5, 6$ , sparse transition rates  $0/36, 5/36, 34/36$  at  $c = 1, 4, 10$ , the characteristic-three excesses 2, 4, 28 at  $n = 11, 12, 13$ , the exact torsion staircase through  $n = 9$ , and twelve universal-coefficient torsion counts with an order-four factor confirming the non-elementary control. Proposition 30 was then checked against those scans from scratch: an independent brute-force computation of the integral cohomology, using no part of the normal form, reproduced its predictions in every degree at  $p = 2, 3, 5$  through  $n = 13$ , and at  $n = 15$  in degree nine returned rational rank 3003, 911 even invariant factors, and exactly one divisible by four, the summand  $\mathbb{Z}/4$  refuting Conjecture 183. The profile laws of Conjectures 185 to 191 rest on 680 configurations with 7,124, 110,777, 47,433, and 6,540 exact and 369, 4,267, and 2,102 numerical deposited tests through dimension 14, reverified independently with matching configuration and inequality counts, zero violations, and the smallest root separation  $1.276 \times 10^{-3}$  reproduced to six decimals. The signature laws of Conjectures 192 to 196 rest on 8,190 configurations, 302,130 and 80,294 exact merge tests, 29 exact ratios through  $n = 60$ , and all thirteen even-dimensional gcDs, reverified independently in full, including the  $n = 14$  spike and  $R_{58} = 26.55045876$ . The random models of Conjectures 177, 178, 180, and 184, the beyond-computation statement of Conjecture 181, the numerical evidence of Conjectures 188, 190, and 191, and the attributions of Conjectures 184, 189, and 195 are recorded where they occur, so those statements should be read with the corresponding caution. Since deposit, Conjecture 195 has been resolved true by the proof of Proposition 31, whose generating function and recurrence are certified by exact symbolic identities and cross-checked against the exact signature ladder through  $n = 60$ ; a further exact scan through  $n = 10,000$ , redundant given the proof, confirmed what it guarantees. The block generalization of Proposition 32 was verified by exact extraction against the Hirzebruch signatures at fourteen block-repetition pairs through dimension 14, with the saddle constants  $\Lambda_4$  and  $\Lambda_6$  matched to ten digits and sixteen blocks at five exact ratios each showing no violation of the open monotonicity question.

*Part XI.* The Part XI spectral scans are floating-point eigensolver checks, on exhaustively generated corpora: all trees through twelve vertices at nine Rényi orders, all unicyclic graphs and all connected bipartite graphs through seven vertices with holdouts through thirty vertices, and



every eligible local move on each corpus member, giving 8,775 path and 8,775 star comparisons with smallest margins  $2.23 \times 10^{-4}$  and  $4.72 \times 10^{-4}$ , 86,982 leaf-pair, 70,126 subdivision, and 15,385 leaf-compression comparisons with smallest margins  $1.66 \times 10^{-3}$ ,  $6.12 \times 10^{-2}$ , and  $9.31 \times 10^{-4}$ , and 6,230 unicyclic, 168,168 tree-extension, 10,884 general-extension, 465 bipartite-minimum, 5,322 biclique-window, and 2,245 cycle-maximum comparisons, with zero violations on the asserted ranges. A second implementation sharing no code with the deposited scans rebuilt both entropy functionals from scratch and reproduced zero violations in 3,031 comparisons each for the path and star laws through eleven vertices, 14,252 leaf-pair and 11,249 subdivision comparisons, all tree extensions through ten vertices, 7,672 general-extension window comparisons, and exhaustive unicyclic and bipartite scans through seven vertices, and reproduced the deposited boundary controls to the digit: the leaf-move excess  $7.65 \times 10^{-7}$  and the unicyclic excess  $6.62 \times 10^{-4}$  at order 0.1, the order-five and order-ten subdivision decreases  $1.00 \times 10^{-2}$  and  $2.02 \times 10^{-3}$ , and the order-ten pendant decrease  $2.32 \times 10^{-2}$ ; the same verification found the exact four-vertex min-entropy tie now recorded in Conjecture 201. The carry programme is exact integer arithmetic throughout, valuations computed from Legendre digit sums: the exact block-mean formula of Proposition 34 was verified exhaustively for  $p = 3$  through  $k = 8$ ,  $p = 5$  through  $k = 6$ , and  $p = 7$  through  $k = 5$ , and the deposited diagnostics at two million integers, marginal means and variances tracking the chain law, all pairwise correlations among  $p \in \{3, 5, 7, 11\}$  below 0.08, covariances below 0.24, modular discrepancies below  $10^{-3}$  at moduli two and three, mixed third and fourth cumulants of order one, and multiplier correlations below 0.11 at half a million integers, were reproduced by an independent run at one million integers with variance slopes 0.4766, 0.3400, 0.3146, and 0.2774 against the limits  $1/2$ ,  $3/8$ ,  $1/3$ , and  $3/10$  for  $p = 3, 5, 7, 11$ , correlations below 0.04, modulus-two discrepancy  $2.5 \times 10^{-3}$ , multiplier-three correlations below 0.04, and Landau-ratio variance slopes 0.339 and 0.332 at  $p = 7, 11$  with cross-correlation  $-0.016$ . The nine carry statements are limit laws, Conjecture 216 is a programme statement, and the attributions of Conjectures 197 and 208 are recorded where they occur, so those statements should be read with the corresponding caution.

*Part XII.* The Part XII graph and spectral scans are floating-point checks. The subdivision programme was calibrated on degree and spectral moments of iterated barycentric refinements: the deposited scans run seven refinement depths in dimension two and a tetrahedral Hodge probe in dimension three, and the independent recomputation, a from-scratch implementation of subdivision as the order complex of the face poset, reproduces the supercritical moment ratio 2.675 against the predicted  $16/6$  at depth seven from three different starting complexes including a non-manifold book, the two-dimensional Hodge pattern with degree-one maxima doubling and top-degree maxima converging to six, and the three-dimensional probe maxima 75.0645, 15.7366, and 7.8794 to the deposited digits; the tail counts of Conjecture 220 are uninformative at reachable depth, and that statement rests on mechanism alone. The spanning-tree programme is exact arithmetic on closed formulas cross-checked against direct pseudoinverse and matrix-tree computation to  $1.1 \times 10^{-13}$ : the deposited scans cover the connected atlas through seven vertices, all complete multipartite partitions through twenty-four, family optimization through  $n = 10^7$ , and seeded edge-toggle climbs through twelve vertices; the independent recomputation reproduces the atlas maximizers and the bridgeless cycle minimum, reproduces the seventeen-vertex bipartite crossover values 8.315588550473 and 8.350120011590 to twelve digits, extends the exhaustive multipartite check to all 89,133 partitions at  $n = 45$ , adds all theta graphs through forty vertices and one hundred eighty-three structured and random bridgeless graphs with zero violations, runs core-density, periphery-matching, and two-core adversaries at thirty and forty vertices, all



falling below the split optimum, and finds hill climbs from random starts at twelve and sixteen vertices landing exactly on the split value, with the family scales  $r^*/\sqrt{n/\log n} = 0.92$  and deficit ratio 1.14 at  $n = 10^7$  drifting toward one. Within the split family the scale is now derived, a promotion: at  $r = c\sqrt{n/\log n}$  the deficit is  $((c + 1/c)/2 + o(1))\sqrt{n \log n}$ , uniquely minimized at  $c = 1$ , so the family clauses hold and the content is the global reduction. The cyclic programme is genuine cyclic convolution throughout: the deposited scans use fast-Fourier convolution at three primes, and the independent recomputation, with its own transform code, puts discrete Gaussians on the doubling floor to  $10^{-6}$  at two primes and four widths, reproduces the full extra half-log-two penalty of the separated-bump control, holds the fixed- $m$  floor across six families and a three-hundred-draw adversarial mixture search with minimum excess zero, matches the wrap-crossover total variation at machine roundoff for Gaussian inputs and at  $4.6 \times 10^{-7}$  falling to  $1.0 \times 10^{-9}$  for a lazy-walk input at up to 9,604,801 convolutions with entropy defects matching the heat-kernel target to five decimals, and holds the uniform monotonicity margin nonnegative through one thousand summands at  $p = 2,000,003$  and wrap ratio  $2.5 \times 10^{-8}$ . The graphon programme's deposited simulations are row-signature resolution proxies and are recorded as such, not as measurements of sampled-graph entropy; the independent recomputation adds an exact enumeration of the threshold-graphon law through seven vertices, whose support sizes 46, 332, 2,874, and 29,024 match the labeled threshold-graph counts exactly and whose linear entropy correction  $-0.60$  per vertex confirms the negative control of Proposition 36. Conjectures 220, 221, and 235 are deposited on mechanism with their evidence gaps on record, and the attributions of Conjectures 219 and 230 are recorded where they occur, so those statements should be read with the corresponding caution.

*Part XIII.* The Part XIII flux and torsion scans are exact arithmetic throughout. The deposited computation ran sixty dense degree-five sign trials and sparse sweeps at  $n = 9, 10$ , exact torsion profiles through  $n = 11$ , exact Gopakumar–Vafa invariants through degree sixty on all five merge edges with 495 conifold-fraction Yukawa tests and the raw-instanton adversarial control failing at every grid point, a 1,900-comparison mirror-map scan through dimension nine, and the insertion-power enumeration through  $m = 4$ , a signed sum over 76,545 regular families. The independent recomputation, sharing no code with it, reproduced the degree-five saturation on twenty-four fresh trials at  $n = 9, 10$ ; rebuilt the complete-flux torsion from exact modular and rational ranks through  $n = 12$  at  $p = 2, 3, 5$ , confirming interval support, shifted duality, and strict log-concavity on every nonzero profile, including 1, 10, 44, 10, 1 at  $n = 11$  and 1, 11, 54, 54, 11, 1 at  $n = 12$ , with totals  $T_2 = 1, 2, 10, 20, 66, 132$  and  $T_3 = 1, 2$ , and verified the closed-form two-primary expression to strict growth through  $n = 200$ ; recomputed the torsion profiles from the normal form of Proposition 30, finding interval support exactly  $2p + 1$  to  $n - 2p + 2$  and no log-concavity failure for every prime  $p \leq 23$  through  $n = 150$ , and strict growth of  $T_p$  for every  $p \leq 11$  over the same range; reimplemented the full hypergeometric mirror computation from scratch, matching the five known degree-one invariants 2875, 1280, 1053, 720, 512 exactly, finding zero log-convexity or amplification violations through degree twenty-four, evaluating the five conifold constants to sixty digits as  $1/q_c = 1980.146, 641.778, 458.890, 269.607, 159.255$  with strict ordering along every edge, and passing the normalized Yukawa comparison at nineteen fractions per edge; verified the five Chern floors of Proposition 37 analytically; ran an independent 588-comparison mirror-map contraction scan in dimensions three to five with zero violations; and reran the deposited insertion enumeration with all coefficients matching. Conjectures 249 to 255 assert laws, the attributions of Remark 1 and Conjectures 246 and 255 are recorded where they occur, and the raw-instanton control of Conjecture 245 is retained in

place, so those statements should be read with the corresponding caution.

*Part XIV.* The Part XIV verification programme is exact throughout: integer Chern-class series for every complete-intersection Euler characteristic, the thirteen published Betti tables of  $\mathbf{P}^1 \times \mathbf{P}^1$  reverified file by file against the primary dataset, and integer convolution for every hypersurface Hodge vector, all comparisons in exact rational arithmetic, the calibrations recovering the quintic  $-200$ , the K3 profile  $(1, 19, 1)$ , and the quintic-threefold profile  $(1, 101, 101, 1)$ . The convexity laws of Conjectures 256 to 261 were tested on the exhaustive grid of dimensions through ten, codimensions through four, and degrees through seven, 482,150 comparisons, and on an adversarial extension of 424,357 further comparisons with dimensions through sixteen and degrees through sixty, with no failure. The tables behind Conjectures 262 to 266 were pinned by the antidiagonal identity of Proposition 42 in every internal degree, with projective dimensions, quadric counts, and the canonical counts  $(a-1)(b-1)$  all matching; the forty-eight nonlinear and one hundred two pre-mode linear triples were strictly log-concave, the eight defects locked as stated, and the eight rational-normal-scroll tables at  $a = 1$ , outside the stated range, were defect-free with empty nonlinear rows, consistent with Conjecture 265 from the boundary. The dispersion laws of Conjectures 267 to 270 were tested on the grid of dimensions through thirty and degrees through sixty, 88,334 comparisons, and on an extension through dimension eighty-three and degree eighty, 195,234 further comparisons, with no failure, the coordinatewise overshoot control at dimension twelve and the cubic-quartic coincidence reproduced exactly, and the total-variation distances falling to  $2 \times 10^{-3}$  by degree three hundred. The small-corpus caveats of Conjectures 262 to 266 and the beyond-any-finite-check statements of Conjectures 267 to 270 are recorded where they occur, so those statements should be read with the corresponding caution.

*Part XV.* The Part XV verification programme is exact throughout and independent of the deposited candidate scans: fresh enumerators for square-lattice walks, bridges, and  $d$ -dimensional counts, agreeing with the deposited data files entry by entry wherever they overlap, with every inequality checked in integer arithmetic and both coupling laws certified by exact integer maximum flows via Strassen's theorem. The curvature law of Conjecture 271 was checked against the published counts through  $n = 79$ , largest ratio  $\rho_{77} = 0.99977583$ ; the joint laws of Conjectures 272 and 275 to 282 on complete enumerations through  $n = 21$ , with  $c_{21} = 2,408,806,028$  walks at the top length and five exact joint tables per length; the bridge laws of Conjectures 273 and 274 through  $n = 23$ ,  $b_{23} = 862,447,881$ , extending the deposited range by seven lengths; and the dimension laws of Conjectures 284 and 285 in exact rational arithmetic across  $2 \leq d \leq 10$ , through  $n = 14$  at  $d \leq 3$ , smallest survival gap  $6.3 \times 10^{-4}$  and smallest moment gap  $1.8 \times 10^{-4}$ . The calibration propositions were verified both symbolically and against the enumerations: the four bridge diagonals and both corner minors at every length through  $n = 23$ , the survival polynomials against eleven dimensions with the axis-grouping expansion overdetermined by three data points, and the monotonicity numerators expanded with every coefficient nonnegative. All four negative controls reproduce exactly: the endpoint-coordinate quadrant violation at  $n = 3$ , the radial-law crossings between adjacent dimensions, the bridge equalities  $b_2^2 - b_1 b_3 = b_4^2 - b_3 b_5 = 2$ , and the oscillating curvature roots, which run 2.40, 1.88, 1.80, 2.15, 2.14, 1.82, 2.48 from  $n = 4$  before settling near 2.0 through  $n = 19$ . The attribution and consolidation notes of Conjectures 271 and 283 are recorded where they occur.

*Part XVI.* The Part XVI verification programme replaced the deposited sanity probes by optimizer-level computation throughout. The skewness law of Conjecture 286 was checked with exact tilted-cumulant perturbation theory, Drazin-inverse formulas with no finite differences,

across rings of up to twelve states, affinities  $0.25 \leq F \leq 12$ , and chords  $0.002 \leq g \leq 64$ , all 420 ratios exceeding one and increasing, the smallest 1.0000739 at the smallest chord, after finite-difference artifacts at ten states were exposed and eliminated. The frontier laws of Conjectures 287 to 291 were tested by multistart global optimization over full antisymmetric couplings, not random sampling: recruitment stayed ordered on six spectra including near-degenerate and widely split ones, the two-mode support held with off-block leakage below  $2 \times 10^{-5}$ , the frontier arcs accelerated with a slope reset at recruitment and hit the trace ceiling at finite budget to  $10^{-6}$ , no dense coupling beat the pair envelope for the frequency in adversarial maximization across four spectra and five budgets per spectrum, and adversarial minimization of the eigenvalue separation at 99.5 percent of the predicted cheapest cost left separations above  $10^{-2}$  while the predicted pair coalesces to machine precision. The envelope correspondence of Proposition 50 holds to  $3.6 \times 10^{-15}$  on the qubit, the qutrit optimizer of Conjecture 292 aligned with the accessibility-restricted prediction to cosine 1.0000, and the basin law of Proposition 51 reproduced the exact  $\varepsilon/\kappa_{\text{eig}}$  measure within sampling error at every tested distance from the exceptional point, with the scaled condition number matching  $\sqrt{h_c}/2$  to three digits. The four optimizer-level negative findings, the failure of exceptional-point pinning on rings of four and five states, the infinite-budget escape of the three-state frontier, the vacuity of the entangling-resonance criterion on nondegenerate products, and the orbit-range obstruction that voids the tangent-rank capacity principle, exhibited on a generic qutrit whose slow coordinate ranges over  $[0.18, 0.51]$  and never vanishes, are recorded in the part preamble as the controls that fixed its scope.

*Part XVII.* The Part XVII verification programme combined exact orthant-probability computation, process-level simulation, exhaustive enumeration, and high-precision arithmetic, with every optimizer-level claim adjudicated against integration noise by paired Monte Carlo on common random samples. The staircase of Conjecture 295 was mapped by exact orthant sums, the three-active, four-active, and full-simplex phases of the five-variable family separated at scales near 0.15 and 0.6, unrestricted eighteen-parameter optimization at scale 0.35 converging to the four-active configuration with paired-Monte-Carlo margin  $+3.4 \times 10^{-3} \pm 1.5 \times 10^{-5}$  over the simplex on one hundred million common samples, and two earlier apparent exceedances at the same scales exposed as noise by the same adjudicator. The closure ratios of Conjecture 296 rose monotonically to 1.000000 on random geometries and to 0.998 on the equilateral one; the elasticity of Conjecture 297 never exceeded 0.334 over eighty-three geometries, twelve scales each, and degenerate shapes, against the conjectured bound one half and the proved pairwise constant 0.1472; and the curvature of Conjecture 298 was driven to machine zero exactly on collinear configurations by exact polar quadrature, with the expansion residual converging to the exact  $q$  like  $\sqrt{t}$ . The thinning formulas were tested on genuine moving-maximum processes of twelve million steps, the Bernoulli response matching to ratios 0.95 to 0.99 and the systematic resonance reproducing  $\theta \mathbb{E}R_m$  at every modulus including the exact one-third resonance; the occupancy censuses enumerated 6437 canonical sets of size at most six through diameter eighteen and 3312 of size at most seven through diameter fourteen, found ten collisions, every one a homometric pair, verified Bloom’s pair occupancy-equivalent at every modulus, and separated the homometric pair  $\{0, 1, 3, 8, 10, 14\}$ ,  $\{0, 1, 7, 9, 11, 14\}$  by occupancy at modulus five. The angular-topology laws were tested with an independent  $\mathbb{F}_2$  Vietoris–Rips implementation: the Hausdorff distance of top- $k$  angular samples tracked the two-term rate within factors 0.55 to 1.22 across  $k = 50$  to 3200, the ring diagrams showed a single dominant loop with death 0.3123 against the analytic 0.3118 and bottleneck error falling from 0.115 to 0.049, and the leakage crossover

switched recovery exactly across ratio one, bottleneck error 0.075 to 0.090 below against 0.143 to 0.156 above. The tomography dichotomy held to  $4.5 \times 10^{-16}$  at the integer power against  $2.0 \times 10^{-3}$  at the fractional one, with effective ranks three against fifteen at tolerance  $10^{-12}$  and oscillating-witness ratios falling from 0.70 to 0.048; the identifiability threshold showed a fifty-solution continuum one measurement below the count, forty-digit Newton polishing collapsed a spurious ten-fold multiplicity at a near-degenerate instance into the single truth, exposed the two genuine second mixtures at three atoms, and dissolved both one measurement higher. The design ladder was verified in fifty-digit arithmetic with sup-error slopes  $-1.001$ ,  $-2.002$ ,  $-3.004$ ,  $-4.004$  for deterministic, two-point, and the exact rational three-point and four-point ladder laws, the Fréchet shape bias fell from  $+0.0396$  to  $+0.0017$  under honest Poissonization at three hundred thousand blocks, and the cancellation cusp doubled the tail-error slope from  $-0.845$  to  $-1.600$  with the critical window tracking  $n^{-\beta/(\alpha+4\beta)}$  at constant ratio 0.500 across six decades.

*Part XVIII.* The Part XVIII verification programme was built on an independent implementation stack: Vietoris–Rips persistence in degree one by boundary-matrix reduction of the full two-skeleton over the two-element field, exact bottleneck distances by threshold search with augmenting matchings, exact two-Wasserstein matchings with diagonal slots, a one-dimensional Mapper with breadth-first nerve construction, interval tracking for rectangle-decomposable fiber fields, and adaptive time refinement for vineyard events. The regular-polygon ratio formula was reproduced to  $10^{-15}$  through  $n = 12$ ; fifteen hundred random circle configurations per  $n$  with local optimization at  $n = 4$  to 10 never exceeded the regular value, the closest competitor reaching 0.9988 of it, while the unrestricted planar control search exceeded the hexagon at the factor 1.052, fixing the circle hypothesis of Conjecture 308 and refuting the deposited planar claim. The greedy saturation ratio ran 1.990, 2.014, 2.032, 2.000 at  $m = 24$  to 128 on the circle against the analytic bar and 1.955 to 2.019 on the ellipse, and the projection threshold fit  $m^* = 5.92 \log q + 1.00$  across  $q = 4$  to 256 with maximum residual 1.4, the  $q = 64$  transition rising from 0.00 at  $m = 14$  through 0.45 at  $m = 24$  to 0.92 at  $m = 44$ . A true alternating-minimization Fréchet mean over five hundred diagrams per prevalence tracked the flat shrinkage law at ratios 0.98 to 1.02 across prevalences 0.1 to 0.9, margin shrinkage to three noise scales leaving the deviation at the 0.02 sampling floor, and the grouping-swap frequency ran 0, 0, 0.0027, 0.0103, 0.0307 at noise scales 0.15 to 0.60 under margin one, the observed logarithms decaying faster than  $\gamma^2/4\sigma^2$ . The Mapper phase calculus was confronted with four hundred phases per scale: empirical survival matched the proved kernel to three decimals at overlaps 0.30 and 0.45, a first candidate kernel with onset at  $\ell/h = \alpha$  being refuted by the same phase data before the span criterion fixed the onset at one; two-cycle additivity held within 0.0055 over twenty-eight scales spanning both kinks, the curve passing value one exactly at the smaller span; and the discretized dither estimator at mesh  $10^{-3}$  showed bias between  $-1.7 \times 10^{-4}$  and  $-4 \times 10^{-5}$  with variance ratio 1.000 to  $s^2/12$  at all three cover resolutions. One-direction tracking reconstructed one hundred sixteen of one hundred twenty random rectangle modules exactly, with zero corner error among successes, the four failures diagnosed as artifacts of the greedy tracker at coincidental crossings, none a collision of fiber fields; corner displacement stayed below 0.48 of the three-direction metric over thirty perturbed pairs with median 0.26, the empirical profile Lipschitz constant stayed below 22.1, and the near-maximal slope fraction never fell below 0.017. Kernel distance ran two-sidedly against transport in  $[0.90, 2.75]$  over eight hundred separated pairs, with sketch deviation halving at each fourfold frequency increase from 0.10 at  $M = 16$  to 0.0087 at  $M = 1024$ ; sixty adversarial collision searches at the Whitney count kept the evaluation gap above  $2.0 \times 10^{-2}$  with full-rank Jacobians throughout; exact-moment multistart inversion found a unique diagram at  $2m$ ,  $3m$ ,

and  $4m$  moments for  $m = 2$  and  $m = 3$  across one hundred fifty starts each; and exact vineyard counts at  $n = 6$  to  $14$  grew at log-log exponent  $2.36$  against the local slope  $2.45$  of  $n^2 \log n$ , twenty to thirty times below the quartic count.

*Part XX.* The Part XX verification programme evaluates two exact closed forms independently of the source computation. The DOZZ modulus of Conjecture 335 was recomputed from the Barnes integral representation of  $\Upsilon_b$  by a real-integrand quadrature sharing no code with the deposited scan, reducing the spectral curvature to the three surviving modulus factors of Proposition 69: over  $b \in \{0.5, 0.8, 1, 1.3\}$ , external charges filling  $Q/3 \leq a < Q/2$ , and momenta from  $P = 0.15$  to  $3.5$ , every discrete second derivative of  $\log M_a$  was strictly negative, the least negative maximum being about  $-5.6$ . Both boundary controls reproduced: the sub-chamber charge  $a = 0.55$  at  $b = 1$  gave a positive maximum curvature near  $+1.78$ , and the full three-charge Hessian of  $\log |C_1|$  at  $(1.3, 1.3, 1.3)$  carried a large positive eigenvalue, of order two hundred against two negative eigenvalues near  $-54$ , so the one-dimensional interior-chamber statement is the surviving one. The Brownian-annulus density of Conjecture 336 was rebuilt by numerically inverting the exact Mellin transform of  $\log X_\tau$  and multiplying by the exact Dedekind-eta factor, with the second logarithmic-modulus derivative of the product evaluated on a grid  $\tau \in [0.02, 30]$  for boundary ratios  $x$  from  $0.05$  to  $20$ : the maximum curvature stayed negative across this moderate range, from about  $-0.43$  to  $-0.58$ , with the  $x \mapsto 1/x$  symmetry to the evaluated precision; but extending the range exposes the large-boundary-ratio tail, where the curvature turns positive— $+0.380$  at  $x = e^5$ ,  $\tau = 0.30$ , tending to the positive limit  $\mathcal{C}_\infty(0.30) = 0.9746$ —so Conjecture 336 is resolved false, its deposited scan having reached only the moderate ratios. The DOZZ law of Conjecture 335 remains checkable but unproved, a rigorous proof requiring continuous rather than gridwise curvature control.

*Part XXI.* The Part XXI diagnostics are reduced-model, exact-anchor, and prerequisite-scaling checks, never evaluations of the singular laws themselves. Independent recomputation confirmed the exact anchors of Proposition 71: the three-vortex collapse condition holds exactly for  $\Gamma = (1, 1, -\frac{1}{2})$  and integration of the planar system from a vanishing-angular-impulse configuration contracted the core to about  $2.4\%$  of its initial diameter while the spectator-to-core scale separation amplified; the Fourier-reduced degenerate-shear generator returned enhanced-dissipation exponents  $0.32, 0.53, 0.63$  for bracket depths  $m = 1, 2, 3$  against  $m/(m+2) = 0.33, 0.50, 0.60$ ; the homogeneity-reduced generalized-SQG neck ODE reproduced  $d(t) \sim (T-t)^{1/\beta}$  to three digits with  $d\kappa_{\max}$  of order one; and the scale-invariant Frenet pair of a modulated helix was dilation-invariant in total curvature to relative error  $5 \times 10^{-15}$ . For the two-dimensional Euler laws a pseudo-spectral run showed the bulk enstrophy quantiles moving far more slowly than the pointwise gradient and a strong negative correlation between  $\log \max |\nabla \omega|$  and the near-maximal-gradient area; for the exact-Beltrami controls the ABC field annihilated the Lamb vector to root-mean-square about  $10^{-14}$  of a Taylor–Green comparison. Every conjecture in the part is, several as conditional on an unproven Type-I singularity, and the attributions of Conjectures 341, 344, and 350 and the elevated novelty caveat of Conjecture 347 are recorded where they occur, so those statements should be read with the corresponding caution.

*Part XXII.* The Part XXII metric-geometry statements are continuum-limit laws; the diagnostics are algebraic anchors and finite Liouville first-passage-percolation or mated-CRT surrogates whose finite-size biases are reported rather than hidden, and the field-theory statements are tested in the exact global-block limit only. Independent recomputation confirmed the algebraic layer of Proposition 72: the exponents  $p_\gamma = d_\gamma/2$ ,  $H_\gamma = 2/d_\gamma$ ,  $\alpha_\gamma = 1 + 2/d_\gamma$  satisfy  $p_\gamma = 1/H_\gamma = 1/(\alpha_\gamma - 1)$  and reduce to  $2, \frac{1}{2}, \frac{3}{2}$  at  $d_\gamma = 4$ , and the affine star spectrum hits



$s_1 = d_\gamma$ ,  $s_3 = 2$ , and  $5 - k$  at the Brownian point. For the field-theory laws an independent evaluation of the exact global  $SL(2)$  sphere block gave strictly positive discrete likelihood-ratio slopes (Conjecture 360); exact- $\Upsilon_b$  DOZZ quadrature at  $b = 1$  combined with the global block gave negative four-point curvature throughout the interior chamber and positive curvature at the excluded charge  $a = 0.55$  (Conjecture 361); the global torus-descendant proxy gave negative thermal log-ratio slopes (Conjecture 362); and a five-point coupled DOZZ/global-block proxy had negative-definite finite-difference Hessians (Conjecture 363). Every conjecture in the part is, the field-theory statements with their descendant contributions tested only in the global-block limit, and the attributions of Conjectures 352 and 356 are recorded where they occur, so those statements should be read with the corresponding caution.

*Part XXIII.* The Part XXIII statements are hydrodynamic scaling limits; the diagnostics test event-count scalings, conservation, reduced linearizations, and exact ODE reductions only. Independent recomputation confirmed the reductions of Proposition 73: the fatigue ODE has finite hitting time iff  $p < 1$  and integrated mobility  $1/(2\rho(1 - \rho))$  for every  $p$ ; the occupation-trap linearization has explicit residual  $v_\infty = e^{\Delta/(a\bar{\rho})}v_0$ , matched by the conservative nonlinear finite-difference solver to four digits with mass error  $10^{-16}$ ; the crowding-thinned successful-reset hazard has tail exponent  $\beta(1 - \rho)$ , fitted to 0.013 at densities 0.1, 0.4, 0.7; the codimension-one catalyst contribution scales as  $N^{\theta-1}$ , reproducing the subcritical, critical, and supercritical regimes at  $\theta = \frac{1}{2}, 1, \frac{3}{2}$ ; the matched moving-membrane scaling keeps the rescaled membrane variance and slow-bond attempts per dwell order one; and the crossing-budget exhaustion time concentrates on its predicted value with  $N^{-1/2}$  fluctuations. The single shared boundary clock had coefficient of variation near 0.67 that did not vanish, forcing the random-clock form of Conjecture 370, and the odd cross-current of Conjecture 375 reversed with chirality while all color masses were conserved. Every conjecture in the part is, and the high overlaps with prior work of Conjectures 374, 375, 377, and 379 against [354, 355, 357, 359] are recorded where they occur, so those statements should be read with the corresponding caution.

*Part XXIV.* The Part XXIV visited-set statements are exact finite-state laws tested by exact augmented-chain  $(X_t, V_t)$  propagation with the sparse matrix exponential, not by trajectory sampling. Independent recomputation confirmed the proved layer of Proposition 74: the ray invariants (stationarity, traffic, and escape rates constant to  $10^{-15}$ ), the endpoint spectral bounds, and the even-analytic dependence of visited-set laws on  $\alpha$ . The alternating-derivative hierarchy of Conjecture 380 was checked over moments one to four, difference orders one to four, escort exponents  $\gamma \in \{0, \frac{1}{2}, 1\}$ , and several lags on chains of sizes three to six with no violation beyond roundoff; the unicyclic range dominance of Conjecture 382 and the cycle coverage dominance of Conjecture 383 were checked exhaustively over range and mass thresholds with no violation, the latter cross-checked against Gillespie simulation, after an earlier apparent violation was traced to comparing probabilities at  $\alpha$ -dependent times and eliminated by a fixed-time comparison; and no ray among thousands of general and stressed families exhibited two interior maxima of the slow edge (Conjecture 381). The refuted companions are retained as controls: a fresh high-precision search reproduced a bidirected six-state ray on which the fast edge  $h(\alpha)$  strictly increases by more than 0.6 at 40-digit precision, refuting whole-ray fast-edge monotonicity, and an eight-state ray realizes an interior maximum of the slow edge followed by a shallow minimum and endpoint rebound, refuting its global unimodality. Conjecture 380 is checked only to finite derivative order and Conjecture 381 is the part's strongest statement.

*Part XXV.* The Part XXV statements are asymptotic-independence and universality laws, so their diagnostics can only exclude gross finite-size obstructions and cannot certify independence;



the exact identities of Proposition 75 are, however, checkable outright. Independent recomputation, sharing no code with the source diagnostics, confirmed the marginal anchors and the exact identities: the  $\{0, 1\}$  singularity probability over  $\mathbb{F}_p$  approaches  $c_p = 1 - \prod_{j \geq 1} (1 - p^{-j})$  with the summable tilt  $c_p - 1/p = O(p^{-2})$ ; the exact Parseval identity  $\chi^2(\mu_{w,N} \| U_N) = \sum_{\rho \neq \mathbf{1}} |\mathbb{E}_w \chi_\rho|^2$  was verified against a from-scratch enumeration of  $S_N^2$  for  $w = [x, y]x$  through  $N = 6$ , whose  $N^2 \chi^2$  and  $N^2 D_{\text{KL}}$  stay of order one, consistent with the exponent two; the  $p$ -adic root count of a Littlewood polynomial approaches  $(p - 1)/(p + 1)$ ; and Rostam's  $\pi/4$  core normalization was reproduced from exact RSK sampling of poissonized Plancherel partitions. The cross observables behaved as the independence conjectures require: odd-prime spectral-edge and cokernel-nullity correlations were small and sign-unstable while the  $p = 2$  correlation persisted, Plancherel core and edge correlations fluctuated around zero and shrank with  $t$ , and real-near-unit and simple-root-modulo- $p$  correlations were small and slowly decaying, the last the weakest decorrelation and Conjecture 388 correspondingly the least confident. These are falsification diagnostics, not proofs, and the finite-size correlations are consistent with either transient constraints or genuine slow-decaying dependence.

*Part XXVI.* The Part XXVI statements are asymptotic laws for invariants whose exact computation is infeasible at scale, so the diagnostics test exact finite calibrations and reduced models, not the laws themselves. Independent recomputation, sharing no code with the source diagnostics, confirmed every proved anchor of Proposition 76: a from-scratch implementation of the defining Kazhdan–Lusztig recursion reproduced the closed cycle formula  $P_{U_{k-1,k}}(1) = \sum_i \binom{k}{i} \binom{k-i-2}{i} / (i+1)$  exactly for  $3 \leq k \leq 25$ , with  $k^{-1} \log P_{U_{k-1,k}}(1)$  rising toward  $\log 3$ ; the Mirzakhani–Petri intensity integrates to  $\Lambda(\varepsilon) = \varepsilon^2/4 + \varepsilon^4/96 + \dots$ , reproducing the Poisson-tail constants  $1/(4^k k!)$  through  $k = 3$ , and the collar normalization  $\pi \text{cap}(\ell)/\ell \rightarrow 1$ ; the tropical matching experiment at profile  $(-2, 0, -1, 0)$  gave  $\sqrt{N}$ -scaled displacements falling from 0.095 to 0.009 across  $N = 4$  to 20 at  $\beta = 2 \log N$  and stalling near 0.4 at  $\beta = 0.75 \log N$ , locating the double-scaling threshold; and the decategorified proxies, the reduced-Burau determinant exponent near 0.08 on three-strand walks and the integral symplectic transvection homology exponent near 0.19, reproduced their calibration scales. The recorded gaps stand: the Burau and symplectic proxies test decategorified shadows only and cannot witness the homological cancellations of Conjectures 391 and 393, and no finite computation reaches the Weil–Petersson or double-scaling limits themselves.

*Part XXVII.* The Part XXVII verifier exhaustively checks the no-skipped-iterate identity for every zero–one word through length sixteen and separately checks the Bernoulli rate in a seeded experiment. This exactly calibrates the lazy Hamiltonian walk of Proposition 77; it computes no Floer barcode and gives no evidence for the noncommuting endpoint-versus-prefix mechanism.

*Part XXVIII.* The Part XXVIII statements were tested against exact counterexamples where stronger forms fail: a universal reconstruction claim and a statement already published for vertex minors were excluded, rank-width was corrected to branch-width in the cubic-kernel transfer, and ultra-log-concavity and rank-one log-concavity strengthenings were replaced by explicit negative controls. The checked-in verifier reproduces the finite matrix-rank calibration, labelled cut-rank/LC agreement through six vertices, the connected 13-vertex rank-one counterexample, exact small-tree shape and extremal checks, the order-fifteen  $Z_T$  collision, small cubic rank-width values, and seeded orbit growth. Exact rooted recurrences and Routh tests were run through order twenty. An independent recomputation with integer Bernstein coefficient comparisons on every unlabelled tree through eleven vertices, exact Routh–Hurwitz determinants on all 5,446 trees through fourteen vertices, and exact cut ranks over  $\mathbb{F}_2$  on all 996 connected graphs

through seven vertices found no failure of Conjectures 399 to 403 and no extra equality class in Conjectures 402 and 403; a second recomputation on all 986 trees through twelve vertices confirmed the coefficientwise Bernstein forms of Conjectures 402 and 403, which are stronger than the stated stochastic inequalities, and a third recomputation reproduced the positive-tail log-concavity of Conjecture 399 on all 996 connected graphs through seven vertices and the forest checks through twelve vertices. Neither layer is presented as evidence for the asymptotic random-width limits.

*Part XXIX.* The Part XXIX statements mix exact identities with asymptotic laws, and the verification programme separates the two. The exact layer was recomputed from scratch: the local divisibility laws of Proposition 79(i) by simulation of  $1.4 \times 10^5$  random numbers to index 200, the Legendre game values of Proposition 79(iii) by exact rational solution of the indifference systems with certification that the solution is a nonnegative vector making every column indifferent, the character-sum identity of Proposition 79(iv) on random smooth quartics over  $\mathbb{F}_{13}$ ,  $\mathbb{F}_{23}$  and  $\mathbb{F}_{41}$ , and the dual-curve identity of Proposition 79(v) by counting rational tangent lines. The asymptotic layer was calibrated as follows. Prime continuants were tested by Miller–Rabin primality of  $q_n$  and  $p_n$  for  $n \leq 200$  along  $1.38 \times 10^5$  uniformly random numbers, giving the constant  $2 \log 2$  to one percent and the lag-one and lag-two pair constants to two and five percent. Dimensions were computed by Chebyshev collocation of the transfer operator, on  $[0, 1/a]$  for the progression digit sets with  $m \leq 384$  and on  $[0, 1] \times \mathbb{P}^1(\mathbb{F}_p)$  for continuant avoidance with  $p \leq 31$ , cross-checked by the escape rates of direct simulation for  $p \leq 197$ . Game values were computed by linear programming with primal and dual certificates, for all primes below 1400 in the Legendre half and quarter boxes and for two hundred random matrices per size in the Toeplitz, Hankel and full ensembles; the unit identification of Conjecture 417(i)–(ii) was checked in exact arithmetic for every prime  $p \equiv 1 \pmod{4}$  below 1000. Quartic point counts, Hessian point counts, flexes, bitangents and tangency types were enumerated over  $\mathbb{F}_{13}$ ,  $\mathbb{F}_{23}$  and  $\mathbb{F}_{41}$  for several hundred random smooth quartics, and over primes up to 1300 for fixed integral quartics. These computations calibrate the finite anchors and exclude gross finite-size obstructions; they do not test the limits themselves. The sign clause of Conjecture 424 as first deposited was refuted by an exact certificate at  $p = 547$ , obtained by locating the optimal supports with linear programming and then solving the indifference systems over the rationals; the certificate is checked in exact arithmetic by the verification script, whose four-prime sign block is a calibration only, and the sign scan was extended to all primes  $p \equiv 3 \pmod{4}$  below 1000. A second independent recomputation solved the quarter and half boxes for the sixteen primes  $p \equiv 3 \pmod{4}$  between 1000 and 1260: every quarter-box value was positive with  $\sqrt{m} v(Q_p)$  between 0.53 and 1.29, and every half-box value was negative, with  $p^{3/2} v(H_p)$  reaching  $-16.1$  at  $p = 1091$  in the class  $3 \pmod{8}$ ; a third recomputation below 500 reproduced the quarter-box positivity and the half-box signs by class modulo 8.

*Part XXX.* The Part XXX statements are model-theoretic and their asymptotic content is not accessible to computation; the verification programme covers the proved algebraic layer of Proposition 80 only. Exact rewriting to the Poincaré–Birkhoff–Witt normal form was implemented for the quantum algebra of a graph and checked to be associative on random triples in the algebras of random graphs with up to eight vertices, the defining relations and the commutation rule  $x_v x^a = (-1)^{\sum_{u \in v} a_u} x^a x_v$  were verified on random monomials, the centrality criterion was verified against the kernel of the adjacency form modulo two, the singularity of that form in odd order and its nonsingularity with positive frequency in even order were confirmed, the extension witness behind the trivial centre of the random graph was observed with frequency increasing

to 0.995 on random graphs with forty vertices, odd powers of adjacent generators were seen to anticommute and even powers to be central, span formulas with fixed scalars were seen, as a sanity check of the mechanism and not as evidence of stability, to admit no half-graph of length three on random configurations in a finite module, and the two computations behind Proposition 80(v) were repeated by exact enumeration in the algebras of random graphs: the relation  $x_a x_c = -x_v x_{v'}$  holds for some pair  $(v, v')$  exactly when  $a$  and  $c$  are adjacent, the relation  $\pm x_s x_{t'} = x_{s'} x_t$  forces  $\{s, t'\} = \{s', t\}$ , and the explicit exponent vectors giving an order pattern of length three for the formula with exponents two and one were confirmed. Nothing in this programme tests the conjectures about forking, stability, decidability or countable models.

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